

IUCN AND TRAFFIC ANALYSES

of the proposals to amend the CITES Appendices at the
20TH MEETING OF THE CONFERENCE OF THE PARTIES

24 NOVEMBER - 05 DECEMBER 2025



TRAFFIC

IUCN/TRAFFIC analyses of the proposals to amend the CITES Appendices at the 20TH MEETING OF THE CONFERENCE OF THE PARTIES

Samarkand, Uzbekistan
24th November – 5th December 2025



Prepared by IUCN Biodiversity Assessment and Knowledge Team and Species Survival Commission and TRAFFIC

TRAFFIC



The European Union

This project was undertaken with the financial support of:
Ce projet a été réalisé avec l'appui financier de :



Environment and
Climate Change Canada

Environnement et
Changement climatique Canada

Canada – Environment and Climate Change Canada



Ministry of Agriculture, Nature and
Food Quality of the Netherlands

Netherlands – Ministry of
Agriculture, Nature and Food Quality



Fondation Franklinia



NPWS

An tSeirbhís Páirceanna
Náisiúnta agus Fiadhúlra
National Parks and Wildlife
Service

Ireland - National Parks and Wildlife Service



Federal Ministry
for the Environment, Climate Action,
Nature Conservation and Nuclear Safety

Germany – Federal Ministry for the
Environment, Nature Conservation, Nuclear
Safety and Consumer Protection (BMUV)



Gouvernement Princier
PRINCIPAUTÉ DE MONACO

Monaco – Ministry of Foreign
Affairs and Cooperation



SWEDISH
ENVIRONMENTAL
PROTECTION
AGENCY

Sweden – Scientific Authority of
CITES, Naturvårdsverket – Swedish
Environmental Protection Agency



Spain - Ministry for Ecological Transition
and the Demographic Challenge



Department
for Environment
Food & Rural Affairs

United Kingdom of Great Britain and
Northern Ireland - Department for
Environment, Food and Rural Affairs



WWF International



Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

Federal Department of Home Affairs FDHA
**Federal Food Safety and
Veterinary Office FSVO**

Switzerland – Federal Food Safety and Veterinary
Office, Federal Department of Home Affairs

Production of the 2025 IUCN/TRAFFIC Analyses of the Proposals to Amend the CITES Appendices was made possible through the support of:

- **The European Union**
- **Canada** – Environment and Climate Change Canada
- **Fondation Franklinia**
- **Germany** – Federal Ministry for the Environment, Nature Conservation, Nuclear Safety and Consumer Protection (BMUV)
- **Ireland** - National Parks and Wildlife Service
- **Monaco** – Ministry of Foreign Affairs and Cooperation
- **Netherlands** – Ministry of Agriculture, Nature and Food Quality
- **Spain** - Ministry for Ecological Transition and the Demographic Challenge
- **Sweden** – Scientific Authority of CITES, Naturvårdsverket – Swedish Environmental Protection Agency
- **Switzerland** – Federal Food Safety and Veterinary Office, Federal Department of Home Affairs
- **United Kingdom of Great Britain and Northern Ireland** - Department for Environment, Food and Rural Affairs
- **WWF International**

This publication does not necessarily reflect the views of any of the project's donors.

The International Union for Conservation of Nature (IUCN) is a membership Union uniquely composed of both government and civil society organisations. By harnessing the experience, resources and reach of its more than 1,400 Member organisations and the input of some 17,000 experts, IUCN is the global authority on the status of the natural world and the measures needed to safeguard it.

The IUCN Species Survival Commission (SSC), the largest of IUCN's seven commissions, has over 10,000 volunteer experts recruited through its network of 186 groups (including Specialist Groups, Task Forces and groups focusing solely on Red List assessments). These experts contribute to the assessment of species, development of conservation strategies, and implementation of policies that aim to halt biodiversity loss. Biodiversity loss is one of the world's most pressing crises, with many species' populations declining to critical levels. SSC is dedicated to halting this decline in biodiversity and to provide an unmatched source of information and advice to influence conservation outcomes, as well as contribute to international conventions and agreements dealing with biodiversity conservation.

TRAFFIC works to ensure that trade in wild species is legal and sustainable, for the benefit of people and the planet. With 50 years of expertise, TRAFFIC is a trusted advisor and partner to governments, businesses, law enforcement, and communities worldwide—tackling the root causes of illegal trade in wild species.

Citation: IUCN and TRAFFIC (2025). IUCN/TRAFFIC Analyses of the Proposals to Amend the CITES Appendices. Prepared by IUCN Biodiversity Assessment and Knowledge Team and TRAFFIC for the Twentieth Meeting of the Conference of the Parties to CITES. IUCN – International Union for Conservation of Nature, Gland, Switzerland.

The designations of geographical entities in this document and the presentation of the material do not imply the expression of any opinion whatsoever on the part of IUCN or TRAFFIC concerning the legal status of any country or area, or of its authorities, or concerning the delimitation of its frontiers or boundaries.

CONTENTS

FOREWORD	iv
INTRODUCTION	v
ACKNOWLEDGEMENTS AND CREDITS	vi
METHODS	vii
CoP20 Prop. 1: Deletion of Bontebok <i>Damaliscus pygargus pygargus</i> from Appendix II	1
CoP20 Prop. 2: Inclusion of Dorcas Gazelle <i>Gazella dorcas</i> in Appendix II	5
CoP20 Prop. 3: Amendment of the existing annotation to the listing of Saiga Antelope <i>Saiga tatarica</i> in Appendix II	10
CoP20 Prop. 4: Deletion of Giraffe <i>Giraffa camelopardalis</i> populations of Angola, Botswana, Eswatini, Malawi, Mozambique, Namibia, South Africa and Zimbabwe from Appendix II	21
CoP20 Prop. 5: Inclusion of Okapi <i>Okapia johnstoni</i> in Appendix I	30
CoP20 Prop. 6: Inclusion of Striped Hyaena <i>Hyaena hyaena</i> in Appendix I	34
CoP20 Prop. 7: Transfer of Guadalupe Fur Seal <i>Arctocephalus townsendi</i> from Appendix I to Appendix II	38
CoP20 Prop. 8: Deletion of Caribbean Monk Seal <i>Monachus tropicalis</i> from Appendix I	43
CoP20 Prop. 9: Amendment of the Annotation of the Population of Southern White Rhinoceros <i>Ceratotherium simum simum</i> of Namibia Listed in Appendix II	47
CoP20 Prop. 10: Transfer of the population of South-Western Black Rhinoceros <i>Diceros bicornis bicornis</i> of Namibia from Appendix I to Appendix II with an annotation	58
CoP20 Prop. 11: Inclusion of Hoffman's Two-toed Sloth <i>Choloepus hoffmanni</i> and Linné's Two-toed Sloth <i>Choloepus didactylus</i> in Appendix II	67
CoP20 Prop. 12: Transfer of Golden-Bellied Mangabey <i>Cercocebus chrysogaster</i> from Appendix II to Appendix I	76
CoP20 Prop. 13: Allow Namibia to trade in registered government-owned stocks of raw ivory and achieve full Appendix-II status for its elephant population under specified conditions	80
CoP20 Prop. 14: Amend the existing annotation for the populations of African Elephant <i>Loxodonta africana</i> in Botswana, Namibia, South Africa, and Zimbabwe	88
CoP20 Prop. 15: Inclusion of hornbill genera <i>Ceratogymna</i> and <i>Bycanistes</i> in Appendix II	95
CoP20 Prop. 16: Transfer of White-Backed Vulture <i>Gyps africanus</i> and Ruppell's Vulture <i>Gyps rueppelli</i> from Appendix II to Appendix I	104
CoP20 Prop. 17: Transfer of Peregrine Falcon <i>Falco peregrinus</i> from Appendix I to Appendix II	111
CoP20 Prop. 18: Inclusion of Great-billed Seed-finch <i>Sporophila maximiliani</i> in Appendix I and <i>S. angolensis</i> , <i>S. atrirostris</i> , <i>S. crassirostris</i> , <i>S. funerea</i> and <i>S. nuttingi</i> in Appendix II	117
CoP20 Prop. 19: Inclusion of Giant Hispaniolan Galliwasp <i>Caribicus warreni</i> in Appendix I	124
CoP20 Prop. 20: Inclusion of Mount Elliot Leaf-tailed Gecko <i>Phyllurus amnicola</i> in Appendix II	127
CoP20 Prop. 21: Inclusion of the Ringed Thin-tail Gecko <i>Phyllurus caudiannulatus</i> in Appendix II	133
CoP20 Prop. 22: Transfer of the genus <i>Amblyrhynchus</i> spp. represented by the Galápagos Marine Iguana <i>Amblyrhynchus cristatus</i> from Appendix II to Appendix I	139
CoP20 Prop. 23: Transfer of Galápagos land iguanas <i>Conolophus</i> spp. from Appendix II to Appendix I	144

CoP20 Prop. 24: Inclusion of Bale Mountain Adder <i>Bitis harenni</i> and Ethiopian Mountains Adder <i>Bitis parviocula</i> in Appendix I	149
CoP20 Prop. 25: Inclusion of the Rock Rattlesnake <i>Crotalus lepidus</i> and Central Plateau Pygmy Rattlesnake <i>Crotalus ravus</i> in Appendix II and the genera <i>Crotalus</i> and <i>Sistrurus</i> in Appendix II as lookalikes	154
CoP20 Prop. 26: Transfer of Home's Hinge-back Tortoise <i>Kinixys homeana</i> from Appendix II to Appendix I	165
CoP20 Prop. 27: Inclusion of four species of water frogs in Appendix II: <i>Pelophylax epeiroticus</i> , <i>P. lessonae</i> , <i>P. ridibundus</i> and <i>P. shqipericus</i>	171
CoP20 Prop. 28: Transfer of Oceanic Whitetip Shark <i>Carcharhinus longimanus</i> from Appendix II to Appendix I	180
CoP20 Prop. 29: Inclusion of Tope (<i>Galeorhinus galeus</i>) and Smoothhounds (genus <i>Mustelus</i>) in Appendix II	189
CoP20 Prop. 30: Transfer of manta and devil rays Family Mobulidae (Mobulids) from Appendix II to Appendix I	202
CoP20 Prop. 31: Transfer of Whale Shark <i>Rhincodon typus</i> from Appendix II to Appendix I	219
CoP20 Prop. 32: Retain giant guitarfishes <i>Glaucostegus</i> spp. in Appendix II with a zero-export quota for wild specimens traded for commercial purposes	226
CoP20 Prop. 33: Retain wedgefishes Family Rhinidae in Appendix II with a zero-export quota for wild specimens traded for commercial purposes	238
CoP20 Prop. 34: Inclusion of Gulper Sharks of the Family Centrophoridae in Appendix II	255
CoP20 Prop. 35: Inclusion of eels <i>Anguilla</i> species in Appendix II	266
CoP20 Prop. 36: Inclusion of Sea Cucumbers <i>Actinopyga echinites</i> , <i>A. lecanora</i> , <i>A. mauritiana</i> , <i>A. miliaris</i> , <i>A. palauensis</i> and <i>A. varians</i> in Appendix II	279
CoP20 Prop. 37: Inclusion of Golden Sandfish <i>Holothuria lessona</i> in Appendix II	293
CoP20 Prop. 38: Inclusion of <i>Grammostola rosea</i> and the following fourteen tarantula species in Appendix II	300
CoP20 Prop. 39: Inclusion of South African Abalone <i>Haliotis midae</i> in Appendix II with an annotation "dried form only"	306
CoP20 Prop. 40: Amend annotation #3 to exempt finished products packaged and ready for retail trade of thin-sliced roots derived from artificially propagated plants of <i>Panax quinquefolius</i>	313
CoP20 Prop. 41: Inclusion of Chilean Palm <i>Jubaea chilensis</i> in Appendix I	319
CoP20 Prop. 42: Inclusion of species <i>Beaucarnea hookeri</i> and <i>Beaucarnea glassiana</i> in Appendix II, as part of the listing of the <i>Beaucarnea</i> spp.	325
CoP20 Prop. 43: Inclusion of <i>Commiphora wightii</i> in Appendix II	329
CoP20 Prop. 44: Transfer of Cycad Spurge <i>Euphorbia bupleurifolia</i> from Appendix II to Appendix I	337
CoP20 Prop. 45: Deletion from Appendix II of <i>Afzelia bipindensis</i> populations located in Central Africa: Cameroon, Central Africa Republic, Congo, Democratic Republic of the Congo, Equatorial Guinea and Gabon	342
CoP20 Prop. 46: Transfer of Brazilwood <i>Paubrasilia echinata</i> from Appendix II to Appendix I	348
CoP20 Prop. 47: Deletion from Appendix II of populations of <i>Pterocarpus soyauxii</i> located in Central Africa: Angola, Cameroon, the Central African Republic, Congo, the Democratic Republic of the Congo, Equatorial Guinea and Gabon	356

CoP20 Prop. 48: Inclusion of species <i>Aloe bergeriana</i> , <i>Aloe jeppeae</i> , <i>Aloe subspicata</i> , and <i>Aloe welwitschii</i> in Appendix II, as part of the listing of the <i>Aloe</i> spp.	364
CoP20 Prop. 49: Transfer of Parlature's Podocarp <i>Podocarpus parlaturei</i> from Appendix I to Appendix II	368
CoP20 Prop. 50: Transfer of <i>Avonia quinaria</i> from Appendix II to Appendix I	373
CoP20 Prop. 51: Amend annotation #4	380

FOREWORD

Unsustainable and illegal trade are driving declines in many species, and biological resource use overall is a threat to 45,840 species assessed as threatened on The IUCN Red List of Threatened Species. This said, the IPBES Assessment of Sustainable Use of Wild Species found that billions of people rely on over 50,000 wild species. Striking the balance between supporting sustainable use of wild species, on the one hand, while abating unsustainable use, on the other, is therefore crucial to achievement of global goals for nature like the Sustainable Development Goals and the Kunming-Montreal Global Biodiversity Framework.

CITES plays a vital role in balancing international trade and species conservation. It is a global agreement ensuring that trade in wild animals and plants does not threaten their survival. Originating from a 1963 IUCN resolution, it came into force on 1 July 1975. For CITES to be effective, decisions—especially regarding species listings in the Appendices, must be based on the best available scientific and technical evidence. This is particularly the case when deciding whether or not to include species in the CITES Appendices, transfer species between Appendix I and II, or remove them from the Appendices altogether. To assist Parties in ensuring that such decisions are evidence-based, IUCN and TRAFFIC undertake technical reviews of the proposals to amend the CITES Appendices for each of the Conference of the Parties (CoPs). It is with great pleasure that IUCN and TRAFFIC now produce the Analyses of the Proposals for CITES CoP20, which will take place in Samarkand, Uzbekistan, in 2025. We would like to thank the team in TRAFFIC and IUCN for producing such a helpful document, synthesising a great volume of complex underlying information, in a very short time.

Information on the extinction risk, threats to, and biology of species was collected from IUCN's Species Survival Commission Specialist Group network, TRAFFIC's network of information sources on trade, and the broader scientific community, and used to evaluate the proposals and the information provided by proponents, with respect to the CITES listing criteria. The resulting document therefore brings together data provided from across a broad range of expertise, which we are confident will be of great assistance to the Parties in their consideration of the proposals.

The Analyses for CoP20 will not only provide an assessment of whether or not each proposal meets the criteria specified by CITES, but also summarises any additional considerations that may be relevant to the decision on whether or not to adopt the proposal. These include, for example, any potential implementation challenges, benefits or risks that may be associated with the adoption of the proposal.

CITES has a key role to play in the next decade. Wise, evidence-based decisions that are true to the Convention's aim of ensuring that international trade is not a threat to wild species will be needed, in alignment with the CITES Strategic Vision 2021–2030, which aims to ensure that all international trade in wild fauna and flora is legal and sustainable by 2030, contributing to halting biodiversity loss and achieving the Global Biodiversity Framework and the Sustainable Development Goals.

Dr. Jon Paul Rodríguez
Chair, IUCN Species Survival
Commission

Dr. Thomas Brooks
Chief Scientist, IUCN

Richard Scobey
Executive Director, TRAFFIC

INTRODUCTION

CITES (the Convention on International Trade in Endangered Species of Wild Fauna and Flora) was opened for signature in Washington DC on 3rd March 1973, and to date has 185 Parties from across the world. If CITES is to remain a credible instrument for conserving species affected by trade, the decisions of the Parties must be based on the best available scientific and technical information. Recognizing this, IUCN and TRAFFIC have undertaken technical reviews of the proposals to amend the CITES Appendices submitted to the Twentieth Meeting of the Conference of the Parties to CITES (CoP20).

The Analyses - as these technical reviews are known - aim to provide as objective an assessment as possible of each amendment proposal against the requirements of the Convention, as agreed by Parties and laid out in the listing criteria elaborated in Resolution Conf. 9.24 (Rev. CoP17) and other relevant Resolutions and Decisions.

For each of The Analyses, a “Summary” section presents a synthesis of available information taken from each proposal’s Supporting Statement and other sources, and a separate “Analysis” paragraph provides an assessment of whether or not the proposal is considered to meet the pertinent criteria in Resolution Conf. 9.24 (Rev. CoP17) or other relevant CITES Resolutions and Decisions. In response to feedback from Parties, an additional paragraph is included for some proposals to summarise any “Additional considerations” that may be relevant to the decision on whether or not to adopt the proposal (for example, implementation challenges and potential risks or benefits for the conservation of the species concerned). Information used to compile these sections is provided in the “Summary of available information” section. Only information from sources other than the Supporting Statement is referenced in this section, and for brevity, these references are not repeated in the “Summary”, “Analysis” or “Additional considerations” sections. See the Methods section for more information.

Although draft versions of the “Summary”, “Analysis” and “Other considerations” sections were shared with relevant experts, this was to ensure the conclusions were drawn from the best available information, therefore, the conclusions drawn do not necessarily reflect the opinions of the reviewers.

The Analyses aim to highlight relevant information on which the Parties can base their decisions. They are produced with a limited budget under severe time constraints and are not exhaustive. Where proposals include many different taxa it is not possible to address them all in detail and there will invariably be omissions and differences of interpretation. We have nevertheless tried to ensure that the document is factual and objective, and consistent in how the criteria have been interpreted and applied across the proposals.

The Analyses were completed and made available online on 19th September 2025 to allow CITES Parties and other stakeholders sufficient time to consider the information in advance of the Conference of the Parties, which convenes for the twentieth time on 24 November 2025 in Uzbekistan. The “Summary”, “Analysis” and “Additional considerations” sections will be translated into French and Spanish and made available online. Printed versions of these sections will be made available to Parties at CoP20.

ACKNOWLEDGEMENTS AND CREDITS

We again acknowledge the generous support of all the project's donors.

Many individuals and institutions contributed to the compilation of the Analyses. Those to whom we would first like to extend our thanks are the reviewers (listed below), many of them members of the IUCN Species Survival Commission, TRAFFIC staff as well as the many other scientists and experts who have volunteered their time, data, contact list and expertise to this process. For this, we are immensely grateful.

The Analyses team was made up of: Paola Mosig Reidl, Tessa Cooper, Kathy Rock, James Stevens, Rachele Stoppoloni, and Amy Woolloff (TRAFFIC), Oliver Tallowin (IUCN), and several consultants including Thomasina Oldfield, Laura Cimo, Sara Oldfield, Steven Broad, and Martin Jenkins. Richard Scoby, Richard Jenkins, Melanie Heath and Sarah Baker Ferguson are thanked for their valuable input in reviewing The Analyses. Francesca Marcolini designed the cover. Ellie Smith and George Chilcott are thanked for their patience and administrative assistance. Craig Hilton-Taylor is thanked for his invaluable assistance with the IUCN Red List. Antony Bagott provided assistance with seizure and incident data, and Graham Laidler prepared trade record data ready for analysis. Oliver Tallowin led the fundraising for this project, without which it would not have been possible to undertake. All other colleagues within TRAFFIC and IUCN are thanked for their support and good humour, sugar, and caffeine. Paola Mosig Reidl and Oliver Tallowin were responsible for overseeing the project.

We are very grateful to the reviewers who contributed their valuable time to this project. Reviewers were not asked to comment on IUCN/TRAFFIC's conclusion of whether each proposal met the relevant criteria (in the "Analysis" paragraph), for which IUCN and TRAFFIC take sole responsibility.

The reviewers were:

W. Baker (Proposal 41), D. Balfour (9, 10), S. Baruch-Mordo (9, 10, 13, 14), BirdLife International (15, 18), A. Borzee (27), Botanic Gardens Conservation International (45, 47), A. Botha (16), C. Bowden (16), N.B. Brindavanam (43), M. Brown (4), M. Burgener (28, 29, 30, 31, 32, 33, 34, 36, 37, 39), H.K. Chen (45, 46, 47, 49), S. Chng (18, 20, 21), T.J. Colston (24), C. Conand (36, 37), S. Cordero (41), A.R. Cortes (17), V. Crook (35), S. Dloniak (6), N. Dulvy (29, 30, 31, 32, 33, 34), C. Echeverria (41), F.R. Elorriaga-Verplancken (7), S. Ferreira (9, 10), E. Fletcher (40), F.E. Fonturbel (41), G. Gentile (23), B. Goettsch (44, 50), M. Gollock (35), C.J. Hernandez-Camacho (7), S.J. Inchaustegui (19), A. Jacobson (6), J. Janssen (20, 21), Q. Kagembe (6), K. Kaifu (35), A. Krivosheyeva (3), L. Luiselli (26), A. MacLeod (22), D. Mahonghol (5, 12, 45, 47), F. Maisels (12), L. Majure (44), D. Mallon (1, 2, 3), E.J. Milner-Gulland (3), N. Moraes-Barros (11), L. Musing (24, 27, 35), D. Newton (4, 9, 10, 13, 14, 42, 44, 48, 50), K. Noboa (7, 11, 19, 22, 23, 25, 38), N. Okes (39), B. Okita-Ouma (13, 14), S. Oldfield (46, 49), J.F. Ovalle (41), A. Patzelt (43), S. Pierce (31), M. Pool (11), A.C. Premoli (49), M.P. Quiroga (49), C. Rigby (28, 29, 30), G. Sant (28, 29, 30, 31, 32, 33, 34, 36, 37, 39), H. Shiraishi (35), R. Slotow (13, 14), S. Spawls (24), S. Spear (25), D. Stanton (5), M. Superina (11), A. Timoshyna (40, 43), N. Uys (44, 50), J. Valdes (18), S. von Meibom (3), M. Wardjomto (2, 16), R. West (38), R. Wilks (1, 17), O. Wright (4, 9, 10, 15, 36, 37), S. Zuther (3).

We would also like to thank Daniele and Richard Devitre for French translation, and Wendy Byrnes for Spanish translation.

METHODS

Criteria

The taxa included in the proposals were assessed according to the criteria in Res. Conf. 9.24 (Rev. CoP17). All relevant criteria were addressed regardless of those explicitly cited in the proposal. Where multi-taxon proposals included many taxa with a large number of these proposed as lookalikes, we focused on assessing those identified as traded in significant volumes against all relevant criteria.

For proposals regarding commercially exploited aquatic species, the general guidelines listed in the footnote to Annex 5 of Res. Conf. 9.24 (Rev. CoP17) with respect to the application of the definition of “decline” were taken into account when species were assessed against criterion 2aA for inclusion of species in Appendix II (refer to doc. AC25 Inf.10 for a detailed explanation of TRAFFIC’s application of listing criteria to commercially exploited aquatic species).

Information sources

To evaluate the proposals against the CITES listing criteria, information on the status and biology of species has been collected from IUCN’s Species Survival Commission (SSC) Specialist Group network, peer reviewed and grey literature, news reports and experts in the broader scientific community. TRAFFIC has drawn on its own expert network and information sources to determine the nature and scale of any trade. Where possible, currencies were converted to USD with the average exchange rates of August 2025 sourced from xe.com unless otherwise specified.

Various databases were consulted to assess reported legal and illegal trade for all taxa proposed for listing. These include the CITES Trade Database, the EU-TWIX database, the TRAFFIC Wildlife Trade Information System (WiTIS), and the United States Fish and Wildlife Service’s (USFWS) Law Enforcement Management Information System (LEMIS).

CITES Trade Database

CITES trade data were downloaded on the 2nd of July 2025, covering at least the last decade of reported trade for the taxa of interest. The last most complete year for which data exist for the majority Parties is 2023, however, some Parties have reported trade for 2024 prior to the 31st of October 2025 deadline (Res. Conf. 11.17 [Rev. CoP18]). Data were downloaded in Comparative Tabulation format. In some cases, the terms “whole organism equivalents” and “parts and derivatives” are referred to in the Analyses. Whole organism equivalents are an aggregation of term codes that can be reasonably equated to a whole individual. These include the term codes for bodies, live, skeletons, skins, and skulls (codes BOD, LIV, SKE, SKI, and SKU) where they are reported by number (see CITES Notif. 2021-044-A1 for more information on term codes). Parts and derivatives are the remaining term codes reported by number as well as other units. Sources are sometimes combined for brevity as “wild sources” and “captive sources”. Wild sources include source codes for wild, ranched, and unknown or unreported (codes “W”, “R”, “U” and unreported), and captive sources include captive-born and captive-bred or artificially propagated (codes “C”, “F”, “A” and “D”). We reported only direct imports and exports when summarising total volumes of commodities in trade to avoid duplication of records, unless otherwise specified.

LEMIS Database

LEMIS data used in the Analyses included trade reports from 2008 to 2023, reported by the USA. Data are based on USFWS Form 3-177 (Rev. 10/2017) that all wildlife importers and exporters must declare with the US Fish and Wildlife Service; this includes information on trade in CITES and non-CITES listed species. These data include records for imports, exports, and transits, as well as information regarding the outcome of the reported trade. This can be cleared, i.e. permitted trade, or refused, i.e. not permitted for trade. Records referred to as “cleared” in the Analyses refer to records that have been cleared under the Action as well as the final Disposition. Some records that are refused may subsequently be seized, abandoned, re-exported, or cleared. The reason for refusal cannot be inferred from the data. Sources are sometimes combined for brevity as “wild sources” and “captive sources”. Wild sources include source codes for wild, ranched, and unknown or unreported (codes “W”, “R”, “U” and unreported), and captive sources include captive-born and captive-bred or artificially propagated (codes “C”, “F”, “A” and “D”).

LEMIS data was obtained through a previous Freedom of Information Act request to the US Government for data in the period 2008–2020, which included taxonomic data, description of the commodity, dates, action and disposition, purpose, source, quantity and unit, the type of trade (import/export/transit), country of origin, and the trading partner. A second Freedom of Information Act request resulted in data being provided for the period 2021-2023 but only in relation to Hunting Trophies and Personal.

EU-TWIX

EU-TWIX is the Europe Trade in Wildlife Information Exchange database, managed by TRAFFIC on behalf of the participating countries. The EU-TWIX system provides users with a mailing list which facilitates the rapid exchange of information, expertise, and experience on wildlife trade enforcement. EU-TWIX also includes access to a dedicated website containing useful resources to support enforcement efforts as well as a database of wildlife seizures. The EU-TWIX database contains centralised data on seizures and offences reported by all 27 EU Member States, Bosnia and Herzegovina, Iceland, Montenegro, North Macedonia, Switzerland, Ukraine, and the UK. Access to this database is restricted to wildlife law enforcement and management authorities in Europe. Data for use in the Analyses were downloaded with prior authorisation from the relevant countries on 26th February 2025, and included seizures made in the years 2014-2024.

TRAFFIC Databases

The Wildlife Trade Information System (WiTIS) is a database managed by TRAFFIC containing information on incidents involving wildlife seizures, poaching, and law enforcement actions, in addition to market monitoring and actionable information. This information is sourced predominantly from open-source media reports as well as some CITES Management Authorities, government agencies, customs organisations, social media platforms and NGOs. The whole database was consulted for the Analyses, and data have been inputted from 2010 onwards. Nonconfidential records held in this database are publicly available via the TRAFFIC Wildlife Trade Portal. Due to the nature of this database and illegal trade, it should not be inferred that there is a direct correlation between WiTIS incident data and the overall illegal wildlife trade, or that information across locations, species or time is consistent in the database.

The references for these data sources are as follows:

CITES (2025). CITES Trade Database, UNEP World Conservation Monitoring Centre, Cambridge, UK.

EU-TWIX (2025). Trade in Wildlife Information Exchange.

LEMIS (2025). United States Fish and Wildlife Service Law Enforcement Management Information System.

TRAFFIC (2025). Wildlife Trade Information System [database].

Deletion of Bontebok *Damaliscus pygargus pygargus* from Appendix II

Proponent: South Africa

Summary: The Bontebok *Damaliscus pygargus pygargus* is an antelope endemic to South Africa. A second subspecies, the Blesbok *Damaliscus pygargus phillipsi* is widespread and numerous in southern Africa, occurring in Eswatini, Namibia and South Africa. Hybridisation between the two is known to occur. *Damaliscus pygargus pygargus* was assessed as Vulnerable in 2015 on the IUCN Red List. Having been nearly extirpated in the 1930s through hunting, by 2021 the population had recovered from approximately 17 to over 10,000. Its natural habitat, renosterveld and grassy fynbos endemic to the Western Cape province of South Africa, is severely limited and fragmented, constraining population growth to fewer than 2,500 individuals within the natural range. Extensive management has resulted in more than 70% of the population now occurring outside this area in the Eastern Cape, Free State, Limpopo, North-West, and Northern Cape provinces of South Africa, predominantly on private land.

Reported legal trade occurs in the form of hunting trophies, as well as horns and skins traded as part of trophy hunts or as byproducts of legal offtake and culling. Nearly 80% of individuals involved in international trade come from wild sources, although there are also captive-bred and ranched specimens. Between 1975 and 2024, CITES exports averaged approximately 120 individuals per year, with no evidence of illegal trade. According to LEMIS, under 50 individuals per year were imported into the USA between 2008 and 2023. Less than 2% of trophies exported are from the native range in the Western Cape. The subspecies was considered for initial review under the Review of Significant Trade process but not selected for inclusion on the basis that adequate management controls were in place in South Africa.

Threats to *D. p. pygargus* include loss of natural habitat, low genetic diversity, and the risks of hybridisation with *D. pygargus phillipsi*. Intensive management and investment in the subspecies throughout the country, both by the government and private individuals, mitigate these risks through a national subspecies-specific biodiversity management plan, a management plan for Bontebok National Park, as well as a national genetic database and testing of translocated and hunted individuals to guard against hybridisation. A metapopulation management plan is still lacking.

Analysis: *Damaliscus pygargus pygargus* numbers in South Africa have increased from tens in the 1930s to possibly around 10,000 individuals in 2021. Its natural range would have limited the population to around 2,500 individuals but the taxon has been introduced into areas beyond its indigenous range in the country for the purposes of game ranching, tourist viewing, and trophy hunting. Significant national management efforts are in place, and international trade is limited.

Annex 4 of Res. Conf. 9.24 (Rev. CoP17) provides guidance on the removal of species from Appendix II: if *Damaliscus pygargus pygargus* were removed from Appendix II, **it seems unlikely that regulation of trade would be needed to avoid this taxon from becoming eligible for inclusion in Appendix I in the near future, nor to ensure that the harvest of specimens from the wild is not reducing the wild population to a level at which its survival might be threatened.** The taxon has not been subject to recommendations under the Review of Significant Trade.

Blesbok *D. p. phillipsi*, the other subspecies, is not listed in the Appendices, therefore **deleting *D. p. pygargus* would remove any implementation issues that might arise due to one subspecies being listed in the Appendices and the other not.**

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

First included in Appendix I on 1975. Transferred to Appendix II in 1981.

Taxonomy

Damaliscus pygargus pygargus (Pallas, 1767). Scientific synonym includes *Damaliscus pygarcus ssp. dorcas* (Pallas, 1766).

CITES Standard Reference: Wilson, D.E. and Reeder, D.M. (Eds.). 2005. Mammal species of the world, a taxonomic and geographic reference. 3rd Edition, The Johns Hopkins University Press, Baltimore, Maryland. 2, 142pp.

Range

South Africa

Extant (resident): Western Cape

Extant and Introduced: Northern Cape Province, Eastern Cape Province, Free State, North-West Province

IUCN Global Category

Vulnerable B1ab(ii,iii)+2ab(ii,iii); D1 (assessed 2015, ver. 3.1)

Biological and trade criteria for retention in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)

A) Trade regulation needed to prevent future inclusion in Appendix I

B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences

The wild population of *D. p. pygargus* dropped to approximately 17 individuals in the late 1930's due to overhunting and habitat loss. Bontebok National Park was then declared to protect the remaining individuals (SANBI, 2013), and intensive conservation efforts re-established populations both within and outside of the natural distribution range (NDR). *With the intensive management of the subspecies currently, it is unlikely that population numbers will drop so low again (Mallon in litt., 2025). The total national population is estimated around 10,000 animals, with almost 2,500 of these within the NDR and extended NDR (Scientific Authority of South Africa, 2024).* Current estimates for *D. p. pygargus* on private land outside of the Western Cape are conservatively: almost 4,000 animals on 71 properties in the Eastern Cape, over 1,300 animals on 50 properties in the Free State, between 1,250 and 2,500 animals on 115 properties in the Northern Cape, 30 animals in North West on three properties and four animals on one property in Limpopo. Sixty four percent of the total *D. p. pygargus* population on private land can be considered wild based on the wildness framework described in Child et al. (2019). Extrapolating this to the national population, it is estimated that around 6,200 *D. p. pygargus* on private land within and outside of the natural distribution range could be considered wild. *Mallon in litt. (2025) states that the existence of many populations outside the indigenous range has a positive impact on subspecies viability, with national legislation and a Biodiversity Management Plan in place to mitigate threats, including hybridisation.*

Given the limited available habitat for the subspecies within the Western Cape and the possible impacts of climate change on the current available habitat for the subspecies, well-managed populations outside of the natural distribution range of the subspecies are considered of high conservation value. *The Guidelines for Using the IUCN Red List Categories and Criteria, makes provision for including benign introductions within the same ecoregion to contribute to subspecies conservation (IUCN Standards and Petitions Committee, 2024). Damaliscus pygargus pygargus numbers outside of the NDR are increasing, while those within the NDR are stable (Cowell and Birss, 2017). The extent of occurrence of the subspecies within the natural range is estimated at ~8,800 km² and the current observed area of occupancy is only around 600 km². Including all known Bontebok-containing areas within both the natural and benign introduction ranges yield almost 1,500 km² of observed occupancy.*

Legal trade occurs in the form of hunting trophies, and *horns and skins are traded as part of trophy hunts or as byproducts of legal offtake and culling. Individuals in international trade are from captive breeding, ranched, and wild sources (Radloff et al., 2017). Since the species was first listed in CITES in 1975, reported exports included at least 5,678 individuals (i.e., whole organism equivalents: bodies, live, skeletons, skulls, or trophies), with trophies making up 92% of these. Skins, skin pieces, hooves, leather products, horns, and other body parts made up the remainder of exports. Most exports were from wild sources (W, 77%), and most were hunting trophies (H, 91%). South Africa accounted for 98% of exports, although 17 wild exports, mainly in the form of trophies, were exported from other countries, including France, the USA, Denmark, Spain, and Austria. The USA imported 59% of all exports, with Spain and Mexico accounting for 6% and 4%, respectively. Exports have slowly increased in numbers over time since 1975, peaking in 1998-1999, and 2011-2012. Between 2010 and 2023, almost 2,000 *D. p. pygargus* were hunted by international clients (DFFE Professional Hunter's (PH) Register), including animals on game farms and hunting*

ranches, some of which may be considered 'non-wild' and thus not included in IUCN Red List Assessments (Mallon, in litt., 2025). According to LEMIS, at least 922 whole organism equivalents, 876 of these trophies from wild sources, were imported into the USA between 2008 and 2023. Less than 2% of trophies exported were from the Western Cape (Scientific Authority of South Africa, 2024).

Two Non-Detriment Finding (NDF) reports have been produced for the subspecies, in 2015 and 2024. The latest, in 2024, concluded that trade is considered a low threat to the subspecies survival (Scientific Authority of South Africa, 2024).

The Proponents indicate that sufficient safeguards are in place in the form of national and provincial legislation, a robust permit system, and a nationally implemented management plan to ensure that deletion from Appendix II of CITES will not lead to overexploitation and detrimental trade. SANBI is currently in the process of developing a national online subspecies population database, the South African Wildlife Population System (SAWPS), to monitor and report on population trends on a provincial and national basis.

Mallon in litt. (2025) states that the IUCN Antelope Specialist Group does not foresee any negative impacts from deleting *D. p. pygargus* from Appendix II.

Retention in Appendix II to improve control of other listed species

A) Specimens in trade resemble those of species listed in Appendix II under Res. Conf. 9.24 (Rev. CoP17) Annex 2 a or listed in Appendix I

Damaliscus pygargus pygargus and *D. p. phillipsi* are closely related antelope species native to South Africa. *D. p. pygargus* is darker brown with a distinctive, almost purple-black, coloration on its flanks, head, and upper legs. *D. p. phillipsi* is lighter brown with a brown patch dividing the white blaze on its face. While both have white bellies and legs, the bontebok's horns are black, whereas the blesbok's horns are yellow-brown at the front. The two subspecies can easily be distinguished from one another. *D. p. phillipsi* is not included in any CITES Appendix.

Additional Information

Threats

The main threats to *D. p. pygargus* include limited optimal habitat within its natural distribution range; small, fragmented populations in the absence of a coordinated meta-population management plan to conserve allelic diversity and maintain genetic diversity within and between subpopulations; low genetic variation in the natural distribution range; and risk of hybridisation with the closely related *D. p. phillipsi*.

Naturally, *D. p. pygargus* is found within the Renosterveld and grassy patches of the fynbos biome, including vegetation types which are Critically Endangered and Endangered. More than 80% of the subspecies' original range has been modified for agriculture. However, breeding efforts in the early 1900's have successfully introduced *D. p. pygargus* to suitable habitats within the Eastern Cape, Free State, Limpopo, North-West, and Northern Cape provinces. *Intensive vegetation management within the NDR is needed to ensure further natural habitat loss does not occur* (Cowell and Birss, 2017).

The historical population low of 17 individuals, and subsequent fragmentation of habitat and populations, may have contributed to low levels of genetic diversity, leading to population vulnerability and inbreeding risks. In addition, hybridisation is possible between *D. p. pygargus* and *D. p. phillipsi* and has occurred both due to the human-mediated expansion of *D. p. pygargus*' range, with translocation onto properties with or near to *D. p. phillipsi* populations, or intentional hybridisation for hunting purposes. This threat is being addressed through the development and implementation of provincial and industry protocols requiring individual DNA profiles, using a verified microsatellite marker set, prior to any animals being translocated or exported as a hunting trophy.

D. p. pygargus is vulnerable to copper deficiency and high parasite infestations. *An additional threat identified in the NDF (2024) is low economic incentives or funding models for the private sector to get involved in D. p. pygargus conservation, since the majority of the population is outside of the NDR and requires ongoing management* (Scientific Authority of South Africa, 2024).

Conservation, management and legislation

The Proponents indicate that sufficient safeguards are in place in the form of national and provincial legislation, a robust permit system, and a nationally implemented management plan to ensure that deletion from Appendix II of CITES will not lead to overexploitation and detrimental trade. SANBI is currently in the process of developing a national online subspecies population database, the South African Wildlife Population System (SAWPS), to monitor and report on population trends on a provincial and national basis.

D. p. pygargus is managed through a mix of government and private initiatives under national legislation and a subspecies-specific Biodiversity Management Plan. In addition, a management plan exists for Bontebok National Park, originally established as a refuge for the subspecies within its natural habitat. Metapopulation management

is partially in place, with monitoring of the genetics of translocated animals and management of hybridisation risk through provincial and industry protocols requiring individual DNA profiles prior to translocation or export.

The subspecies is found in several protected areas within and outside of the NDR, including three national parks and one nature reserve within the NDR and two national parks of the NDR. The Western Cape Bontebok Conservation, Translocation, and Utilization Policy (BCTUP) provides support for conservation and sustainable use, metapopulation management, habitat restoration, and threat mitigation. Phenotype testing and microsatellite marker testing occur to identify hybrid individuals, with purity certificates issued for 100% D. p. pygargus, and microchips fitted prior to translocation (Cowell and Birss, 2017).

An NDF assessment in 2015 indicated that legal and international trade in live animals and trophies poses a moderate risk to the subspecies survival. However, with improved management and monitoring this risk will be mitigated. The Bontebok BMP fulfils part of these requirements, with a metapopulation management plan still to come (Cowell and Birss, 2017). The most recent NDF assessment (2024), concluded that legal local and international trade in live animals and the export of hunting trophies poses a low risk to the survival of this subspecies, while international regulations on the import of hunting trophies remains a constraint for improved conservation outcomes (Scientific Authority of South Africa, 2024).

In addition to protection under national legislation in the form of the South African Constitution, National Environmental Management Biodiversity Act 10 of 2004 (NEM:BA), National Environmental Management: Protected Areas Act 57 of 2003 (NEM:PAA), Threatened or Protected Species Regulations 2007 (ToPS), all providing protection for the subspecies and its habitat; the subspecies is currently protected by the Convention on Biological Diversity and the World Heritage Convention, with D. p. pygargus occurring on Robben Island World Heritage Site and within the Cape Floral Region Protected Areas World Heritage Site. Provincial legislation protecting D. p. pygargus is also in place in all nine provinces of South Africa (Cowell and Birss, 2017).

No national offtake quotas are in place, but provincial offtakes are monitored and regulated (Scientific Authority of South Africa, 2024). Legal offtake primarily occurs for ecological or biodiversity reasons, as well as trophy hunting and live game sales. Ecotourism occurs, and non-trophy products are sourced from the trophy hunting industry, natural deaths, or from animals culled or hunted for meat. Most hunting occurs outside of the NDR.

Artificial Propagation/captive breeding

Due to habitat loss in the subspecies' natural range, more than 70% of the total *D. p. pygargus* population exists outside of its natural and extended natural distribution range (7,339-8,589 individuals). Population sizes on private properties are sustainable, and 64% are considered "wild" based on the wildness framework described in Child *et al.* (2019). Approximately 6% of exports reported to CITES between 2014 and 2023 were from captive or ranched sources. Several zoos and captive breeding facilities within South Africa keep and breed *D. p. pygargus*. The subspecies is easy to keep and breed in captivity. No data on the number of animals kept in zoos and breeding facilities are available. Within the private sector approximately 38 populations have less than ten animals (approximately 227 animals in total) and can be considered captive based on the wildness framework described in Child *et al.* (2019). At the species level, there are 279 individuals in different institutions worldwide, although it is unclear to which subspecies these belong (ZIMS, 2025).

References

- Child, M.F., Selier, S.A.J., Radloff, F.G.T., Taylor, A., Hoffmann, M., Nel, L., Power, J., Birss, C., Okes, N.C., Peel, M.J., Mallon, D. and Davies-Mostert, H. (2019) A framework to measure the wildness of managed large vertebrate populations. *Conservation Biology*.
- CITES (2025) List of Species. Available <https://cites.org/eng/common/reg/cb/species.html>. Accessed on 29 July 2025.
- Cowell, C.R. and Birss, C. (2017) *Biodiversity Management Plan for The Bontebok (Damaliscus pygargus pygargus) in South Africa*. Unpublished Report, jointly developed by South African National Parks, CapeNature and the National Department of Environmental Affairs. Version 1.0
- IUCN Standards and Petitions Committee (2024) Guidelines for Using the IUCN Red List Categories and Criteria. Version 16. Prepared by the Standards and Petitions Committee. Downloadable from <https://www.iucnredlist.org/documents/RedListGuidelines.pdf>
- Mallon, D. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK. IUCN SSC Antelope Specialist Group.
- Radloff, F., Birss, C., Cowell, C., Peinke, D., Dalton, D. and Kotze, A. (2017) *Damaliscus pygargus ssp. pygargus*. The IUCN Red List of Threatened Species 2017: e.T6236A50185717. <https://dx.doi.org/10.2305/IUCN.UK.2017-2.RLTS.T6236A50185717.en>. Accessed on 28 July 2025.
- SANPARKS (2013) *Bontebok National Park Management Plan for the period 2013 – 2023*.
- Scientific Authority of South Africa (2024) *Non-detriment finding for Damaliscus pygargus pygargus (Bontebok)*. Issued by the Scientific Authority of South Africa.
- Species360 Zoological Information Management System (ZIMS) (2025) zims.Species360.org.

Inclusion of Dorcas Gazelle *Gazella dorcas* in Appendix II

Proponents: Benin, Burkina Faso, Mali, Mauritania, Niger, Nigeria, Senegal, Sudan, Tunisia

Summary: The Dorcas Gazelle *Gazella dorcas* is a small antelope found throughout the Sahelo-Sahara region, where it is the most common antelope species, as well as in Israel and Jordan, and into the Horn of Africa. The species went extinct in Senegal, where it has since been reintroduced, and is possibly extinct in Nigeria and Burkina Faso. It occurs in a wide variety of arid and sub-arid habitats, with some local migration to exploit seasonal vegetation. The species has a variety of natural predators and is important in the ecosystem as a seed disperser.

In 2016, the species was assessed as Vulnerable on the IUCN Red List, based on evidence of a population decrease of over 30% over three generations (15 years), due to increasing hunting pressure and habitat degradation caused by livestock overgrazing and drought. Overall, while *G. dorcas* remains present across a broad geographic range (area of habitat ~10 million km²), its subpopulations are fragmented and declining in many areas. Its habitat is declining, particularly in West Africa. In extensive parts of its range including in the proponent countries, numbers are considered to be low or very low. It is much more numerous in Niger and especially Chad with one protected area in that country possibly holding as many as 50,000 animals.

G. dorcas is widely hunted, predominantly illegally, within its range for various purposes, including sport trophies, meat, and body parts used in decoration and traditional medicine. These items are commonly sold in local markets and herbalist shops.

The species was listed in Appendix III by Tunisia in 1976 and by Algeria in 2008. According to CITES trade data, over 4,000 whole organism equivalents of *G. dorcas* were directly exported for international trade between 2000 and 2022. Around half of the exports were live animals sourced from the wild with the remainder captive-bred; the majority were for commercial purposes. Nearly 80% of live exports were from Sudan and Niger for most of the remainder, with the UAE, Saudi Arabia, and Qatar as key importers.

While illegal trade may not significantly impact the global population, it poses a threat to smaller or isolated, national populations. Trade in horns, meat, skins, and live animals (mostly young) occurs mainly locally, but also to some extent internationally, with some seizures reported in the UAE. *G. dorcas* faces a range of pressures, with uncontrolled hunting being the most significant driver of population decline across its range. In addition to hunting, the species is affected by habitat loss and degradation due to expanding agriculture, drought, and overgrazing and by livestock and predation by domestic dogs. In some regions, killing of large numbers by militia groups and visiting hunting parties, has been reported.

Over 5,000 *G. dorcas* individuals are held *ex situ* across 21 institutions in 12 countries, including Israel, Qatar, Spain, and the USA. Reintroduction efforts have taken place in Morocco, and national restoration strategies exist in both Morocco and Tunisia. The species occurs in multiple protected areas across its range, including national parks and reserves in Algeria, Egypt, Tunisia, Mauritania, Chad, Niger, and Morocco.

Analysis: *Gazella dorcas* has a wide range across North Africa and parts of the Middle East, with a current global population estimate of more than 50,000 individuals. As such the wild population is not small, nor is it geographically restricted. While there have been local declines or in some cases extirpations, overall, the population was estimated to have reduced by 30% over three generations to 2016. The species is widely hunted for various purposes, with the resulting items commonly being sold in local markets nationally. International trade is occurring, with legal trade predominantly in live individuals, but its scale is not uniform across all range States, and recent information suggests it is not a major threat. The current level of trade does not appear to be negatively impacting the species

to the extent that it requires international regulation, so the species **does not meet the criteria for listing in Appendix II**.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

Listed on Appendix III by Algeria, 12 February 2008

Listed on Appendix III by Tunisia, 22 April 1976

Taxonomy

Gazella dorcas (Linnaeus, 1758).

Range

Native: Algeria; Chad; Djibouti; Egypt (Sinai); Eritrea; Ethiopia; Israel; Jordan; Libya; Mali; Mauritania; Morocco; Niger; Somalia; Sudan; Tunisia; Western Sahara

Possibly extinct: Nigeria

Regionally extinct: Senegal

IUCN Global Category

Vulnerable A2cd (assessed 2016, ver. 3.1)

Biological and trade criteria for inclusion in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)

A) Trade regulation needed to prevent future inclusion in Appendix I

B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences

G. dorcas is wide-ranging, occurring throughout the Sahelo-Sahara region, with some local population fragmentation and extirpations in recent decades, including in Senegal and possibly Nigeria. The species was assessed in 2016 as Vulnerable on the IUCN Red List due to an estimated population decrease of over 30% over three generations (15 years, 2001–2016).

There has been an estimated 86% reduction in range over the last 200 years (CMS Secretariat, 2023). The current area of habitat is approximately 10 million km² (Ridley et al. in prep.). Population status varies widely by country, with low to very low population numbers in parts of its range, including in proponent countries, and higher population numbers in Chad and Niger (Mallon in litt., 2025). Recent population estimates are listed in Table 1, which indicate that the total wild population numbers more than 50,000 individuals. More than 5,000 additional individuals held ex-situ (CMS Secretariat, 2023). This is significantly higher than previous estimates of 10,000–40,000 individuals in 1999–2001 (East, 1999; Mallon and Kingswood, 2001).

Table 1. *Gazella dorcas* national population estimates.

Country	Population Estimate	Source
Algeria	<1,000	Scholte and Haschim, 2013
Burkina Faso	No recent reliable population estimates	
Chad	7,700–18,000 in one survey area, up to 40,000–50,000 in total in the main reserve	Wacher <i>et al.</i> , 2023; CMS Secretariat, 2023
Djibouti	4,000	Kunzel <i>et al.</i> , 2000
Egypt	1,000–2,000	Saleh, 2001
Eritrea	Unknown	
Ethiopia	Several hundred	Scholte and Haschim, 2013
Israel	>2,000	Israel CITES MA consultation
Jordan	Unknown	
Libya	<1,000	Scholte and Haschim, 2013
Mali	2,000–2,500	East, 1997
Mauritania	No recent reliable population estimates	
Morocco	800–2,000/500–1,500	Cuzin <i>et al.</i> , 2007/ Sahelo-Saharan Antelopes, 2005

Country	Population Estimate	Source
	4,000 captive	ANEF, 2022
Niger	3,100	Wacher <i>et al.</i> , 2008
Nigeria	Thought locally extinct	Lafontaine <i>et al.</i> , 2005
Senegal	31 captive-bred individuals	Abaigar <i>et al.</i> , 2009
Somalia	Unknown	
Sudan	Unknown	
Tunisia	<1,000	Scholte and Haschim, 2013
Western Sahara	Several hundred	Sahelo-Saharan Antelopes, 2005

National trade

The species is nationally hunted for various purposes such as trophies for sport hunters, wild meat sold in local markets, and for its head, horns, and skin, which are used in decoration, traditional medicine, and the manufacture of goods. These items are commonly found in herbalist shops, where they are sold alongside herbs, oils, and other traditional remedies. In Morocco, market surveys revealed the presence of numerous parts of this species, including five whole animals, 15 skins, and 48 heads or horns, with one study estimating that 137 specimens represented around 98 individuals. In Libya, hunting has been extensive, with approximately 100 individuals reported killed in 2012 and 125 in 2015, based on expert questionnaires and local interviews. Long-term data from Libya also show that 372 individuals were recorded as hunted between 1964 and 2016, nearly half of which occurred before 1990.

International trade

Analysis of the CITES trade data between 2000 and 2022 show a total direct trade of 4,012 whole organism equivalents (WOE) (including bodies, live, skins, skulls skeleton and trophies) as reported by exporters, and 1,228 as reported by importers. WOE exporter-reported trade was almost entirely in live individuals (97%, 3,921), originating from wild (49%, 1,962) and captive-bred sources (C, 48%, 1,930), for commercial purposes (T, 53% 2,120). Wild-sourced WOE trade was predominantly exported by the Sudan (78%, 1,527) and to a lesser extent by Niger (18%, 358) and imported by 37 countries, including United Arab Emirates (39%, 757), Saudi Arabia (16%, 318) and Qatar (10%, 195). According to the LEMIS database, 2008-2022, the USA imported a total of nine wild-sourced *G. dorcas*, predominantly comprising trophies (six), and almost all for hunting purposes (five).

Illegal trade

Illegal trade is unlikely to have a significant impact on the global population, although illegal hunting may have a larger impact on smaller national populations (Mallon in litt., 2025). International trade is not listed as a major threat by the IUCN (2017) or by the CMS Secretariat (2023).

National and international illegal trade exists for parts like horns, meat, and skins, as well as whole dead animals and live animals in many countries. TRAFFIC's Wildlife Trade Portal contained three incidents of seizures involving *G. dorcas*, wherein one dead individual was seized in the United Arab Emirates in 2009, two dead individuals seized in the United Arab Emirates originating in Egypt in 2010, and 20 dead individuals, seven skins and 28 horns were seized in Niger in 2020. Illegal trade is known to occur locally, across borders, and on social media platforms, with frequent cross-border movement (CMS Secretariat, 2023), and wild meat is known to be transported across borders (Mallon in litt., 2025). Based on an examination of the CITES Annual Illegal Trade Database (AITR) on 1st July 2025 (CITES Secretariat, 2025), five seizures were reported between 2016 and 2023, including three seizures of live *G. dorcas* involving seven live specimens reported by Niger. Uses for these products include hunting trophies and wild meat, its head, skin, and horns used for decoration, and parts used in traditional medicines. Illegal capture for the pet trade is also known, although it is unclear whether this trade is international.

Inclusion in Appendix II to improve control of other listed species

A) Specimens in trade resemble those of species listed in Appendix II under Res. Conf. 9.24 (Rev. CoP17) Annex 2 a or listed in Appendix I

More research is needed on whether confusion with Appendix-I listed Slender-Horned Gazelle *Gazella leptoceros* occurs in the traded forms of both species.

B) Compelling other reasons to ensure that effective control of trade in currently listed species is achieved

Not applicable.

Additional Information

Threats

The most significant threat to *G. dorcas* is uncontrolled hunting, mainly for subsistence. While traditional hunting has impacted local populations, modern hunting with firearms and vehicles is considered to have had a substantial effect on population numbers, enabling hunters to target large numbers of individuals with greater efficiency. This form of motorised hunting is widely recognised as the primary driver of population declines across the species' range.

In addition to hunting, *G. dorcas* faces multiple pressures, including drought, habitat loss, and degradation due to expanding agriculture. Overgrazing by livestock, particularly sheep and goats, further reduces available habitat and food resources. In Eritrea, for example, the species' occurrence probability decreases with increasing livestock density and agricultural land use. Competition with domestic animals and the capture of young gazelles for the pet trade are also significant threats, particularly in countries like Djibouti.

The species is often kept as a pet, and hunting by militia groups and visiting hunting parties has been reported in several regions, including the Sahel, Egypt, and Morocco. Illegal trade in gazelle parts and live animals is widespread, occurring both in physical markets and online. This includes the sale of horns, meat, skins, and whole animals. *G. dorcas* is also used in traditional medicine and decoration, with market surveys in Morocco indicating that trade volumes may have a significant negative impact on wild populations. In some areas, illegal hunting and predation by domestic dogs further exacerbate the species' vulnerability.

Conservation, management and legislation

One population management program is run by the World Association of Zoos and Aquariums (WAZA), and one in collaboration between the European Association of Zoos and Aquariums and the Spanish National Research Council (CSIC) (IUCN SSC Antelope Specialist Group, 2017; CMS Secretariat, 2023; Mallon in litt., 2025). Population monitoring has occurred in Egypt, Morocco, Senegal, and Tunisia.

G. dorcas is protected or partially protected by national law in Algeria, Burkina Faso, Egypt, Ethiopia, Libya, Mali, Morocco, Nigeria, Somalia, the Sudan, and Tunisia. It is also protected internationally under the Convention on Migratory Species (CMS). *In addition, G. dorcas falls under the CMS Secretariat Sahelo-Saharan Megafaunal Action Plan (2023) and previously the Action Plan for Conservation and Restoration of Sahelo-Saharan Antelopes (UNEP/CMS, 1999). Reintroductions have taken place in Morocco, and there are national strategies to restore the species in Morocco and Tunisia (CMS Secretariat, 2023).*

G. dorcas occurs in multiple protected areas throughout its range, including M'Sabih Talâa Reserve and several smaller reserves (Morocco); Tassili and Ahaggar National Parks (Algeria); Bou-Hedma, Sidi Toui, Dghoumes, Oued Dekouk and Djebil National Parks (Tunisia); Elba National Park and Saint Catherine Protectorate (Egypt); Banc d'Arguin National Park (Mauritania); Ouadi Rimé-Ouadi Achim Reserve (Chad); Termit Tin Toumma National Nature Reserve (Niger) (Scholte and Hashim, 2013).

Artificial Propagation/captive breeding

Captive herds occur in multiple countries, with collections of more than 500 at the Royal farms of Bouznika and Douyiet and, more recently, R'Mila Royal Reserve and Souss-Massa National Park in Morocco, *totalling more than 4,000 individuals*, with the aim of production for reintroduction, as well as on wildlife farms in the Sudan, the Middle East, and Algeria (CMS Secretariat, 2023). *There are more than 5,000 ex-situ individuals in 21 holding institutions across 12 countries worldwide, including in Israel, Qatar, Spain, and the USA (IUCN SSC Antelope Specialist Group, 2017; CMS Secretariat, 2023; Mallon in litt., 2025; ZIMS, 2025). The species survives and reproduces well in captivity.*

Implementation challenges (including similar species)

Other gazelle species are mentioned in the proposal but *not thought to be an implementation challenge (Mallon in litt., 2025). The CMS Secretariat (2023) notes the possibility of confusion of pale individuals of G. dorcas with the Slender-Horned Gazelle Gazella leptoceros (CITES Appendix I). Given the lack of available data, conducting NDFs could be challenging for some range States.*

References

- Abaigar, T., Youm, B., Niaga, M., Ensenyat, C. and Cano, M. (2009) The role of Senegal in the recovery of the Sahelo-Saharan antelope species: the case of the reintroduction of Dorcas Gazelle. *Gnusletter* 28(1):6-8.
- CMS Secretariat (2023) *Sahelo-Saharan Megafauna Action Plan*. UNEP/CMS/COP14/Inf.29.2.1 13 June 2023. Available cms.cop14.inf.29.2.1.SSMF.AP.e.0.pdf
- CITES Secretariat (2025) *Notification to the Parties No. 2025/102: Proposals to amend Appendices I and II. Provisional assessments by the Secretariat. Convention on International Trade in Endangered Species of Wild Fauna and Flora*. <https://cites.org/sites/default/files/notifications/E-Notif-2025-102.pdf>
- East, R. (compiler) (1999) *African Antelope Database 1998*. IUCN, Gland, Switzerland and Cambridge, UK.

- IUCN SSC Antelope Specialist Group (2017) *Gazella dorcas*. *The IUCN Red List of Threatened Species 2017*: e.T8969A50186334. <https://dx.doi.org/10.2305/IUCN.UK.2017-2.RLTS.T8969A50186334.en>. Accessed on 28 July 2025.
- Lafontaine, R.-M., Beudels-Jamar, R.C., Devillers, P. and Wacher, T. (2005). *Gazella dorcas*. In: Beudels, R.C., Devillers, P., Lafontaine, R.-M., Devillers-Terschuren, J. and Beudels, M.-O. (eds), *Sahelo-Saharan Antelopes. Status and Perspectives*. Report on the conservation status of the six Sahelo-Saharan Antelopes. CMS SSA Concerted Action. 2nd edition. CMS Technical Series Publication N°11, 2005. UNEP/CMS Secretariat, Bonn, Germany.
- Mallon, D. (in litt.) (2025). *Re: Email to prospective IUCN delegates to CITES CoP20*. IUCN SSC Antelope Specialist Group.
- Mallon, D.P. and Kingswood, S.C. (2001). *Antelopes*. Part 4: North Africa, the Middle East, and Asia. Global Survey and Regional Action Plans. IUCN, Gland, Switzerland.
- Ridley, F. *et al.* (manuscript in preparation) Advancing understanding of global threat abatement potential for terrestrial vertebrates.
- Scholte, P. and Hashim, I. (2013) *Gazella dorcas*. In: J. S. Kingdon and M. Hoffmann (eds), *The Mammals of Africa*. VI. *Pigs, Hippopotamuses, Chevrotain, Giraffes, Deer, and Bovids*. Bloomsbury Publishing, London, UK.
- Species360 Zoological Information Management System (ZIMS) (2025). Available zims.Species360.org.
- Wacher, T., Amin, R., Newby, J., Hatcha, M. H., Abeye, K., Ali, H., Bourtchiakbé, S.Z. and Banlongar, F. N. (2023) Gazelle–livestock interactions and impact of water resource development in the Ouadi Rimé–Ouadi Achim Reserve, Chad. *Oryx* 57(2):205-215.

Amendment of the existing annotation to the listing of Saiga Antelope *Saiga tatarica* in Appendix II

Proponent: Kazakhstan

Summary: *Saiga tatarica*, is one of two saiga species recognized under currently adopted CITES nomenclature. It is a nomadic herding antelope that inhabits open dry steppe grasslands and semi-arid deserts across Central Asia. There are four distinct populations, three largely in Kazakhstan, where over 98% of the global population resides, and one in the Russian Federation (in the Pre-Caspian region). Two of the populations largely found in Kazakhstan have transboundary elements, part of one population migrating seasonally into the trans-Volga region of the Russian Federation and a very small number of another population migrating into Uzbekistan. The second species, *Saiga borealis*, occurs only in Mongolia.

Saiga tatarica populations were reduced from millions to low thousands of individuals by excessive hunting for meat, hides and horn in the 19th and early 20th Centuries. The primary international market is for horns from male animals (only males are horned), long used in traditional Asian medicinal practices. Under protective measures, the species recovered with numbers reaching the hundreds of thousands by the 1950s, but these gains were partly reversed when Soviet regulatory systems and border restrictions collapsed in the early 1990s. Licensed exports of saiga horn to East and Southeast Asia were replaced by largely unregulated hunting and trade. Regulated hunting and trade were resumed alongside the listing of saiga in CITES Appendix II in 1995, but legal exports were stopped within the next decade. At CoP18 in 2019, instead of a proposed Appendix-I listing, a zero quota for wild specimens traded for commercial purposes of both saiga species, was adopted as an annotation (A2) to their listings in Appendix II. The proponent now seeks to amend the annotation for *Saiga tatarica* so that the zero quota does not apply to specimens from the population of Kazakhstan, which holds over 98% of the global population.

Although legal hunting and horn export of *Saiga tatarica* were halted at the end of the 1990s populations of the species continued to suffer large fluctuations, largely caused by disease outbreaks and poaching. Most notably, in 2015 a bacterial infection caused the mortality of over 200,000 individuals in Kazakhstan within a three-week period (at that time more than half of the global population). The species has also shown vulnerability to the effects of extreme climatic events and is constrained by loss or degradation of steppe habitats due to conversion for agriculture and overgrazing and disruption of seasonal migration by development of linear infrastructure (railways, roads, pipelines, and border fences). Nevertheless, due to high fecundity, populations can quickly rebound and the latest estimate was that the Kazakhstan population numbered over 2.8 million individuals in April 2024 and was still growing rapidly (at around 40% per year).

Past trade volumes appear to have significantly exceeded actual consumption levels and large stockpiles accumulated in some destination countries. A CITES review in 2006 estimated at least 130 tonnes of horn to be held outside the range States, with potential to meet demand for 10-15 years. Most legal trade movements over the past two decades have been between these non-range countries, mostly declared as pre-Convention, therefore by inference from stockpiles accumulated legally before 1995. Nevertheless, some illegal trade from Kazakhstan and, to a lesser extent, the Russian Federation continues, with seizures totalling 1-2 tonnes of horn each year recorded in these countries during the period 2022-2024. However, recent reports for both countries indicate that poaching remains at a stable level, not posing significant threat to the population size.

Kazakhstan has taken comprehensive conservation measures for the species and is actively engaged in related international initiatives, including those coordinated under the CMS Memorandum of Understanding concerning Conservation, Restoration and Sustainable Use of the Saiga Antelope (Saiga MoU), developed in collaboration with CITES. A hunting moratorium, in place since 1999, was temporarily relaxed during the autumn-winter period of 2023/24 when limited take was permitted on an experimental basis in light of increasing conflict with farming. Approximately 43,000 animals were removed, the horns from which were stored, and the meat and hides processed for domestic sale. Recent reports indicate that a further limited harvest of males was permitted in June 2025 and

that a more extensive harvest is planned for autumn 2025, possibly targeting females only, with the goal to reduce populations by up to 20% altogether during the year.

Kazakhstan indicates that its saiga management operations are state-controlled and that annual quotas for international trade will be established based on results of population monitoring employing a long-established and credible methodology. Control measures, including hunter registration, field marking and inventory systems, have been developed to reduce the potential for laundering of horn from poached animals into the stockpile derived from legal hunting and collection of horn from natural mortality. For trade control, a marking and chain of custody system is designed to ensure that exported horns have a legal origin and that illegally harvested horns cannot be used in legal trade. A stockpile of 42,161 horns from the experimental harvest in 2023/2024 (likely weighing 5-8 tonnes) has reportedly already been marked with barcode labels and is kept under government control. Future exports are to be managed through a single government-appointed organisation and it is planned that some horn stock will be retained prior to export for market stability to buffer any restrictions that might have to be introduced if population data indicate concern. If the current zero quota is lifted, Kazakhstan proposes that its horn marking and container tagging system be adopted by the CITES Parties as a common standard for international trade control.

In the main countries that imported saiga horn in the past, domestic trade and sale remains legal, though subject to stockpile declarations and regulation. CITES Decision 19.213 and the 2021-2025 work plan under the Saiga MoU called for involved countries to adopt internal market controls for saiga parts, including registration of stockpiles, labelling of parts and products, and registration of manufacturers and traders. Although most relevant governments have responded to related CITES Secretariat requests for information on stockpile controls, it appears that inventory control and monitoring is in most cases not comprehensive or regularly updated and there is a lack of transparency about actual volumes of horns now being held.

Analysis: Under the proposed amendment of the annotation for *Saiga tatarica*, all exports of specimens of this species from Kazakhstan would be subject to Appendix-II regulation. The zero quota for wild specimens traded for commercial purposes would continue to apply for specimens from *S. tatarica* populations in the Russian Federation and Uzbekistan, and to specimens of *S. borealis*, from Mongolia. There are no specific guidelines for assessing proposals to change annotations of this nature, but it seems appropriate to ensure that satisfactory precautionary measures, as detailed in Annex 4 of Res. Conf. 9.24 (Rev. CoP17), would be in place. Kazakhstan has not proposed an export quota or other special measure as an integrated part of its proposal, so the Parties have to consider whether they are satisfied that Kazakhstan will effectively implement the requirements of the Convention, particularly Article IV and enforcement controls.

Population data necessary to support Article IV implementation are already being collected on an annual basis and the planned centralized management system clearly has potential to ensure that harvest limits could be effectively managed and their impacts assessed in a timely manner. However, the proponents make no specific statement about how overall future harvest limits will be calculated, despite expert guidance on setting sustainable hunting quotas being referenced in the national saiga conservation and management strategy. Considering the on-going threat of disease outbreak and the findings of published population modelling, some experts have expressed concern that a 20% annual cull mooted in Kazakhstan's report to a meeting of Saiga MoU Parties in 2025 may be excessive.

The inventory control and marking systems in use have strong potential to mitigate risks that saiga horn illegally harvested from within Kazakhstan or from other countries, might be infiltrated into legal trade flows to end markets. There is some risk that a resumed supply of legal horn might increase the potential for legally held stockpiles in destination countries to provide cover for parallel illegal trade from range States. Overall, this appears to be a minimal risk and current evidence indicates that poaching and illegal trade levels are not a significant concern. Furthermore, almost all the global population of saiga would be subject to the management of a single country, Kazakhstan. Even greater security would be provided by stronger complementary control and monitoring of stockpiles in all involved countries, as already recommended by the Parties.

With some clarification of how existing guidance on setting precautionary harvest levels will be applied, and reassurance about trade control cooperation with market countries, the proposal to exclude this population from the current zero-quota annotation appears to be proportionate to the risks to the species in Kazakhstan.

Other considerations: In the SS, Kazakhstan notes it is considering an alternative approach to saiga horn trade management, with only finished processed products allowed for export. Considering the nature of end uses, with saiga horn used for direct dispensing by pharmacists and manufacture of patent medicines, the viability of such an approach is questionable.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

Saiga tatarica (Linnaeus, 1766)

Wilson and Reeder (2005), the current CITES Standard Reference for saiga recognises two species, both commonly referred to as Saiga Antelopes – Saiga tatarica and Saiga borealis – and both are included in the CITES Appendices. This proposal relates to S. tatarica only.

CITES Background

Saiga tatarica was listed in Appendix II in 1995. At that time the Mongolian population was considered a subspecies Saiga t. mongolica and included in that listing. However, the subsequent adoption of Wilson and Reeder (2005) as the CITES Standard Taxonomic Reference for mammals, resulted in the splitting of this taxon into S. borealis (the population in Mongolia previously referred to as the subspecies S. t. mongolica) and S. tatarica, a division now widely recognised to have been in error but enshrined in CITES taxonomy until a new reference is adopted.

Currently, both Saiga tatarica and Saiga borealis are listed in CITES Appendix II. At CoP18 a proposal was submitted to transfer Saiga tatarica from Appendix II to Appendix I (it seems the scope of application to only one of the two currently recognised saiga species was an error made by the proponent). This proposal was not agreed by the Parties, but a zero quota for wild specimens traded for commercial purposes was adopted at CoP18 as an annotation (A2) to the Appendix II listings of both saiga species.

The CITES Parties have adopted a series of Decisions related to saiga conservation and trade at previous CoPs, including on co-operation with the Convention on Migratory Species and calls for enhancement of stockpile control and other trade management measures. Decisions 19.213-19.217 are currently in force and progress on their implementation is reported in CoP20 Doc. 85.1 (Rev. 1). Among other issues, this document raises concern that saiga horn inventory control and monitoring in relevant countries is in most cases not comprehensive or regularly updated and there is a lack of transparency about actual volumes of horns now being held. Further action is proposed after CoP20.

Range

Extant: Kazakhstan, Russian Federation, Uzbekistan

Possible: Turkmenistan

Extinct: China, Mongolia, Ukraine

IUCN Global Category

Unlike the nomenclature adopted by CITES, taxonomy used for the IUCN Red List treats saiga as a single species Saiga tatarica with two subspecies. Separate assessments have been made for the species and each of the subspecies. The nominate subspecies S. tatarica tatarica is equivalent to the species S. tatarica under CITES taxonomy.

Saiga tatarica (sensu CITES) was most recently assessed by IUCN (as S. t. tatarica) in August 2023 as Near Threatened A4bcde, a significant improvement on earlier assessments as Critically Endangered (2008) and Endangered (2018) in light of recent population recovery. However, this latest assessment included an important caveat that, because of multiple unpredictable factors, and the uncertain potential interplay between them, an assessment as Endangered, Vulnerable, or Least Concern was equally plausible (IUCN SSC Antelope Specialist Group, 2023).

The second subspecies recognised by IUCN, *S. tatarica mongolica* (treated as the separate species *S. borealis* under CITES and therefore not within the scope of this proposal) was most recently assessed in February 2018 as Endangered A4abde (IUCN SSC Antelope Specialist Group, 2018).

Population in Kazakhstan

Over the past two centuries, *Saiga tatarica* population numbers have fluctuated widely owing to over-exploitation, disease, infrastructure, weather and climate and predation. Numbers were reduced to at most a few thousand by the early 20th Century, before introduction of hunting control measures by the Soviet Union (Bekenov et al., 1998).

Saiga tatarica has four main populations. Three of them are primarily located in Kazakhstan, while the fourth is in the northwestern Pre-Caspian region of the Russian Federation. The three populations predominantly found in Kazakhstan are:

- the Ustyurt population in southwestern Kazakhstan, transboundary with Uzbekistan. *Historically, this population underwent extensive migrations, reaching Turkmenistan. However, today, due to the population's small size and significant barriers to migration, potential climate-related range shifts barriers to movement within Kazakhstan and at the border, these Steppe Saigas mainly remain in the northern escarpment of the Ustyurt Plateau in Kazakhstan (Milner-Gulland in litt., 2025; Zuther in litt., 2025);*
- the Ural population, which is mainly found in northwestern Kazakhstan, but is transboundary with the Russian Federation, where it occurs close to the border with Kazakhstan in the Trans-Volga region; and
- the Betpak-Dala population in central Kazakhstan, which now exhibits the largest seasonal migrations and occupies the largest range (some individuals occasionally cross into territory of the Russian Federation).

By the 1950s -1960s, the *Saiga tatarica* population in Kazakhstan was fairly stable. The Betpak-Dala population was the largest, followed by the Ustyurt population, while the Ural population was the smallest. The highest recorded population level during the Soviet era was 1.2 million individuals in 1974. The *Saiga tatarica* population in Kazakhstan began to decline in the early 1990s, as political and economic changes accompanying the dissolution of the USSR led to weakened conservation systems and severe economic hardships. Meanwhile, the opening of international borders expanded opportunities for the trade in saiga horns, both legal and illegal. This negative trend persisted until the early 2000s, reaching a historic low of an estimated 21,000 individuals in 2003.

In response to this decline, a wide range of conservation measures were implemented, stabilizing and subsequently increasing the *Saiga tatarica* population in Kazakhstan. Between 2003 and 2010, the population increased 4.6 times, reaching 97,300. Despite a 54% mortality event in the Ural population, the overall trend in 2011 was positive and further population growth continued until 2015, reaching 295,470 individuals. However, in 2015 the Betpak-Dala population sharply declined by 85% (from 242,500 to 36,200) due to an outbreak of bacterial disease and by 2016, the total *Saiga tatarica* population in Kazakhstan had fallen to 75,700 individuals. From 2016 to 2024, the population showed consistent annual growth. Based on the aerial survey of *Saiga tatarica* in Kazakhstan conducted in April 2024, the total population in the country was estimated at 2,833,600 individuals (annual growth 48.0%).

According to the April 2024 survey, the numbers for the main *Saiga tatarica* populations in Kazakhstan were estimated as follows:

- Ustyurt population – 63,600 (increase of 60% on 2023 estimate);
- Ural population – 1,620,000 (increase of 43% on 2023 estimate); and
- Betpak-Dala population – 1,150,000 (increase of 54% on 2023 estimate).

For comparison, a summary of information on saiga populations in other countries:

Uzbekistan: recent reports indicate that only a few hundred individuals of the Ustyurt population of *Saiga tatarica* reside in or migrate to and from Uzbekistan (Milner-Gulland et al., 2020);

Russian Federation:

- The north-western Pre-Caspian population of *S. tatarica* in the Russian Federation is isolated from the populations primarily in Kazakhstan, occurring north of the Caucasus Mountains and west of the Volga River in the Republic of Kalmykia and Astrakhan province (Milner-Gulland et al., 2020). This population was estimated through satellite survey in June 2022 at 26,000-28,000 individuals, and expert opinion was that numbers reached a little under 40,000 in June 2024 and increasing (Lushchekina and Karimova, 2025).
- The number of *Saiga tatarica* of the Ural population migrating in summer from Kazakhstan into the Trans-Volga region of the Russian Federation has grown over the past decade in line with the overall recovery of

this population, increasing from a few animals prior to 2015, up to 28,000-33,000 individuals estimated to have crossed the border in the summer of 2021 (Smelyansky et al., 2023).

Mongolia: the latest population estimate for *S. borealis*, found exclusively in Mongolia was 15,540 based on a census completed in November 2023, an increase of 12% on a 2022 estimate using the same methodology (Chimeddorj, 2025).

International trade context

Although saiga meat and hides are also commercialised in Kazakhstan for the domestic market, the primary international market is for the horns from male animals, which have long been valued for use in traditional Asian medicinal practices. Trade in saiga horns for this market has been documented over many centuries and it was the primary driver of serious over-harvesting of the species in the Russian Empire in the 19th and early 20th Centuries, followed by eventual stabilization of harvest and export levels in the Soviet Union by the 1950s (Bekenov et al., 1998; Milner-Gulland et al., 2020).

Review of Soviet Union hunting data indicated that horn exports from Kazakhstan alone totalled over 400 t between 1970-1989, averaging over 20 t a year (Milner-Gulland, 1994). Official records indicate that licensed exports declined in the early 1990s, but other evidence suggests that unreported and likely illegal trade flourished during the first half of the 1990s as Soviet Union regulatory systems collapsed (von Meibom et al., 2010). After the listing of saiga in CITES Appendix II in 1995, another 49 t of horn, averaging around 5 t a year, was declared as legally exported directly from Kazakhstan and the Russian Federation during the decade 1995-2004. The primary importers were China (including Hong Kong SAR of China), Singapore, Japan and Malaysia, with levels of re-export of horn by some of these countries also reported during this period. Legal exports from range States were halted by 2004, but seizure evidence and market surveys indicated that significant levels of illegal trade persisted during the remainder of that decade (von Meibom et al., 2010).

The historical accumulation of large stockpiles of saiga horn in some market countries suggests that import volumes were exceeding actual consumption levels over an extended period. A compilation of national records by the CITES Secretariat in a report for CoP14 concluded that a total of 130 t of saiga horn were being held in China, Japan, Malaysia, the Republic of Korea and Singapore in 2006. The same report noted an estimate of annual consumption in China of 6-10 t of saiga horn per year, suggesting that the then registered stockpiles in that country may last 10-15 years if consumption levels continued at the same level (CITES CoP14 Doc 56).

Reviews by the CITES Secretariat of recent legal saiga horn trade have indicated that some trade flows continued during the period 2006-2022 between countries where stockpiles have been reported in the past. Total annual trade volumes varied from 1-3 t, a relatively low level compared to trade reported in earlier years (AC32 Doc. 35; SC78 Doc. 68.2). With respect to more recent illegal trade flows, a CITES Secretariat review of evidence from Annual Illegal Trade Reports concluded that numbers of saiga horn seizures had declined during the period 2016-2022, the majority involving manufactured products, rather than unprocessed saiga horn (SC78 Doc 68.2). However, seizure records in the TRAFFIC WiTIS database for the years 2022-2024 demonstrate on-going confiscations of saiga horn, mostly in Kazakhstan and the Russian Federation and many reported as destined for China, totalling between 1-2 t per year. Furthermore, a recent study of online trade in saiga horn on Russian language websites revealed regular one-off offers for sale (Roberts et al., 2022). Although quantities offered per transaction were small, such trade may be channelled through small regular postal shipments that may be difficult to detect.

Comparison of market prices over time could provide important insights to saiga horn market trends. Prices paid to local collectors/hunters in range States were reported to have increased from around USD20-30/kg in the early 1990s to USD120-180/kg just over a decade later in 2006 (von Meibom et al., 2010). Prices sought by retailers in Malaysia and Singapore in 2006 averaged USD850/kg and USD880/kg respectively (von Meibom et al., 2010). A follow-up survey in Malaysia in 2018 noted a very wide range of retail prices being sought by traders (Gomez and Krishnasamy, 2019). Recent prices in China quoted in court verdicts (2014-2020) and online advertisements (2021-2024) averaged USD940/kg, with no clear upward trend (Xu Ling in litt., 2025). Additional price data may provide useful insights into the level of horn availability and demand in end markets.

Experimental re-opening of harvest in Kazakhstan

Recently, in light of the increasing size of the Saiga tatarica population and to mitigate reported social tensions with farmers and to pilot mechanisms for the sustainable use of the species, a decision was made by the government of Kazakhstan to conduct limited harvesting on an experimental basis during Autumn/Winter 2023/2024. The culling was carried out by the State Enterprise Okhotzooptom with the involvement of hunters and local communities. The horns derived from this harvest were marked and stored (see details under Precautionary Measures below). Meat and hides were processed for domestic sale.

The removal of approximately 43,000 Saiga tatarica from the Ural and Betpak Dala populations was carried out by Okhotzooptom from 10 October to 10 December 2023, was paused during the critical Saiga mating period, and then

continued from 10 January to 10 February 2024. The feasibility study for this activity was prepared by Zhangir Khan West Kazakhstan Agrarian and Technical University. Concerns about the quality of this document were raised among the expert community both in Kazakhstan and internationally (Convention on Migratory Species, 2025b).

According to recent media reports, under continuing pressure from farmers claiming extensive damage from the growing saiga population, it was agreed to reopen hunting again in 2025 (Ermagambetova, 2025; Zharbulova, 2025). A limited harvest of males was carried out in July and a more extensive cull of up to 20% of the Ural and Betpak-Dala populations is reportedly planned to take place during September-November 2025, perhaps mainly targeting females (Krivosheyeva in litt., 2025; Mallon in litt., 2025). The volume of additional horn stockpiles likely to be accumulated during this renewed hunting in 2025 is currently unknown.

Precautionary Measures

Species likely to be in demand for trade, but its management is such that the CoP will be satisfied with:

- A) Implementation by the range States of the requirements of the Convention, in particular Article IV; and**
- B) appropriate enforcement controls and compliance with the requirements of the Convention**

Kazakhstan proposes amendment to the listing of *Saiga tatarica* in Appendix II, so that the zero quota for commercial trade adopted by the Parties at CoP18 does not apply to wild specimens from its national population of the species. The SS states that the requirements of the Convention would be met by a species-specific management system for saiga derivatives, ensuring sustainable use of populations in accordance with IUCN recommendations on the sustainable use of wild living resources and strict control to prevent illegal or non-Kazakhstan-derived products from entering the market.

Kazakhstan outlines four sources for saiga horn that might be exported: future recoveries from natural mortality handed in to authorities in return for low level rewards; a currently registered stockpile accumulated during the experimental harvest during the autumn/winter of 2023/2024 (*and now presumably also from the renewed 2025 harvest*); future production from regulated hunting channelled through a government regulator; and future production from captive breeding.

In terms of Article IV implementation, Kazakhstan indicates that its harvest and trade system will be state-controlled and that annual quotas for international trade will be established. The SS states that export quotas will be set on the basis of population monitoring results and at levels aimed to ensure market stability. Surpluses will be stored under the stewardship of a state-appointed National Trade Operator in Kazakhstan. The accumulated stockpile of saiga horns will be used as reserve in cases of uncontrolled population decline due to natural causes, such as disease, harsh winters, etc., until the populations recover to a level that allows for the resumption of their sustainable use.

Monitoring of *Saiga tatarica* population numbers has been conducted since the mid-20th century. The methodology for estimating population size has been improved several times with the advent of more suitable technologies. The primary method of population monitoring is aerial surveys, which are conducted annually in April and funded by the state budget through a specialized organisation, Okhotzooptom, commissioned by the Committee of Forestry and Wildlife of the Ministry of Ecology and Natural Resources of the Republic of Kazakhstan. In addition to aerial surveys, supplementary ground-based monitoring is carried out during the conduct of scientific research and conservation activities by staff of specially protected natural areas and hunting areas within saiga habitats. Data are systematically transmitted to the Committee of Forestry and Wildlife. To standardize the saiga monitoring methodology, the "Methodological Recommendations for Aerial Surveys of Saiga in Kazakhstan" were developed and approved by the Scientific and Technical Council of the Committee of Forestry and Wildlife on March 17, 2014.

The SS does not spell out a specific formulation for setting annual harvest or export quotas, though Kazakhstan has indicated in a report to the 2025 meeting of Parties to the Saiga MoU that an on-going annual cull of 20% of the population is under consideration (Government of Kazakhstan, 2025) and a recent media report indicates that this is the threshold set for the 2025 cull (Ermagambetova, 2025). Some experts have raised concern that this level of take would however likely be excessive, considering past population modelling and recommendations of a study for the Convention on Migratory Species on sustainable use of saiga (Milner-Gulland, 1994; Milner-Gulland et al., 2020).

In terms of trade regulation, Kazakhstan reports that it is developing a domestic management system to ensure that any stored and traded saiga horns have a legal origin and that illegally harvested horns cannot be used in legal trade. Any *Saiga tatarica* obtained from natural mortality, confiscation, legal harvesting or captive breeding are to be managed by the National Trade Operator, which will hold the exclusive right to conduct international trade in saiga derivatives. A two-tier control system is envisaged: initial field marking of horns using secure barcode labels issued to registered collectors and hunters, documented photographically and inventoried through a mobile phone application; followed by central storage in sealed and tagged containers prior to export. The proponent also mentions that an alternative export policy is under consideration, under which only finished products manufactured from saiga horn would be permitted.

The marking and registration system described by Kazakhstan is likely to involve significant complexity, such as keeping track of label issuance, local marking, and managing security and consolidation of horns from distributed locations. The extent to which such systems have already been developed and tested is unknown.

As of 2007, Kazakhstan reported to the CITES Secretariat that there were no stockpiles of horns or other parts of saigas being held in the country (CoP14 Doc 56). Regular seizures have been reported since that time, but it has been reported by Kazakhstan that confiscated horns are destroyed under court order (CoP20 Doc 85.1 (Rev 1)). With respect to new stockpiles accumulated very recently, Kazakhstan reports that the 42,161 horns derived from population control in 2023/2024 (likely weighing between 5-8 t) have already been marked with barcode labels and are kept under the control of Okhotzooptom until this function is transferred to the National Operator. It is yet to be confirmed how much additional horn will be accumulated during the reopened hunting season in 2025 and whether this stock will be subject to the same marking protocol. In the event that the current restriction on commercial exports is lifted, Kazakhstan proposes that the use of its horn marking and container tagging system be adopted by the CITES Parties as a common standard for international trade control.

In terms of enforcement, Kazakhstan indicates that spot checks of the field marking and stockpile control systems would be carried out under the auspices of the Committee of Forestry and Wildlife of the Ministry of Ecology and Natural Resources. Border control is conducted by the State Revenue Committee of the Ministry of Finance. X-ray scanners, video surveillance systems, and sniffer dogs, are actively used to detect smuggling.

Additional Information

Threats

Excessive hunting for horns and meat was the primary cause of saiga population decreases in the early 20th century. Poaching for the horns in the late 1980s and early 1990s following the break-up of the former Soviet Union also caused serious reductions in saiga numbers and disrupted reproduction owing to skewing of the sex ratio caused by selective hunting of males (Milner-Gulland et al., 2003). Poaching levels later decreased and recent national reports indicate that this is not a significant current concern. In Kazakhstan, 168 poaching incidents were reported during 2021-2024. In the Russian Federation, no incidents were reported in the North-West pre-Caspian region since 2019, but five individuals from the Ural population were reportedly poached in the Saratov region in 2023 (Convention on Migratory Species, 2025b).

Saiga populations are vulnerable to extreme climatic events, both summer droughts and severe winters. Observations during the 1980s suggested that mortality during particularly harsh winters for some Saiga populations could exceed 50%, though these populations recovered over the next 2-3 years (Bekenov et al., 1998). Research and modelling to predict potential future impacts of climate change on populations of saiga and other steppe species have been identified as a priority conservation action (Convention on Migratory Species, 2025b).

The main cause of recent population fluctuations has been disease outbreaks that caused mass mortality events (MME). Most notably, in 2015 an outbreak of bacterial disease resulted in the death of more than 200,000 Saiga tatarica of the Betpak-dala population in Central Kazakhstan, at that time representing a loss of over 60% of the global population (Kock et al., 2018; Milner-Gulland et al., 2020). Similar MMEs have been documented back to the mid-1950s, apparently caused by a range of diseases, including foot-and-mouth disease, brucellosis, other bacterial infections. The species' unusually high birth rate, however, means that populations can recover rapidly from such events (Bekenov et al., 1998).

Other threats noted in the IUCN Red List assessment for the species include loss or degradation of steppe habitats due to conversion for agriculture and overgrazing, competition with livestock have affected some parts of the range and disruption of seasonal migration by development of linear infrastructure (railways, roads, pipelines, and a border fence) (IUCN SSC Antelope Specialist Group, 2023).

Conservation, management and legislation

Entering into force in 2006, the Memorandum of Understanding concerning Conservation, Restoration and Sustainable Use of the Saiga Antelope (Saiga MoU) developed under the auspices of the Convention on the Conservation of Migratory Species of Wild Animals (CMS), in collaboration with CITES, provides a non-legally binding instrument for the conservation of Saiga, which has been signed by all range States. Implementation of the Saiga MoU is managed under a Medium-Term International Work Programme (MTIWP) updated on a five-yearly basis, most recently for the period 2025-2030, which includes improving and increasing population monitoring; improving protected area networks; reducing poaching; creating alternative livelihoods; captive breeding; and raising awareness at all levels (Convention on Migratory Species, 2025a).

In light of positive signs of population recovery during the 2010s, a comprehensive exploration of the potential of future exploitation of the Saiga, and the pre-conditions which need to be fulfilled to ensure that such use would be sustainable in practice was prepared for the Convention on the Conservation of Migratory Species of Wild Animals in 2020 (Milner-Gulland et al., 2020). It included chapters on the theory of sustainable harvesting with application to

Saiga, potential management structures for Saiga harvesting, population-specific considerations, an examination of consumer market considerations and information on the international trade in Saiga products and its status under CITES.

Kazakhstan's primary conservation legislation is the *Law of the Republic of Kazakhstan No. 593 "On the Protection, Reproduction, and Use of Wildlife"*, adopted on July 9, 2004. Species subject to the highest level of protection are those listed in the country's national Red Book. Subordinate regulations establish the legal framework for a wide range of conservation measures, including rules and procedures for the export, import, and re-export of such species, ensuring strict control over their movement across borders and preventing illegal trade.

A recent review by TRAFFIC of Kazakhstan's laws on the use and trade of wildlife identified several aspects in need of strengthening, though some of the identified gaps concerned the control of captive breeding operations, which are not directly applicable to this proposal. This report also noted potential complications with enforcement of CITES trade controls between Member States of the Customs Union of the Eurasian Economic Union (Armenia, Belarus, Kazakhstan, Kyrgyzstan and Russian Federation), since goods can be moved freely between them (Tokubek uulu et al., 2024).

Saiga tatarica is not listed in Kazakhstan's national Red Book. It has always been and remains classified under law as a game species. However, to address the decline in *Saiga tatarica* populations, a moratorium on hunting was introduced in Kazakhstan in 1999 with exceptions for scientific purposes (On Measures for the Protection and Reproduction of the Saiga Resolution of the Government of the Republic of Kazakhstan dated 29 April 1999 No. 527). The moratorium on saiga harvesting in Kazakhstan has been extended six times: in 2001, 2005, 2012, 2015, 2020, and 2023. Furthermore, a resolution of the Government of the Republic of Kazakhstan dated July 25, 2012, No. 969, prohibited the use of saigas, their parts, and derivatives across the entire territory of Kazakhstan. *The legal basis for the saiga culls authorised in 2023/2024 and 2025 is a provision under national law 'On the Protection, Reproduction and Use of Wildlife' (dated 9 July 2004, No. 593), which provides for authorisation of population control in the context of disease outbreaks, agricultural damage and other socio-economic circumstances (Krivosheyeva in litt., 2025).*

Over the past five years, the government of Kazakhstan reports that it has invested more than USD18 million in programs for the protection and restoration of this species. *With support from the CMS and CITES Secretariats. Kazakhstan developed a national strategy for saiga in 2023, intended to provide guidance on conservation and sustainable management over the next years, while considering the needs of all stakeholders, in particular local communities living in the Saiga range (Convention on Migratory Species, 2023).*

Regulatory provisions in several traditional destination countries for Saiga have been reported to the CITES Secretariat in response to Decisions adopted by the Parties, most recently summarised in a report to CoP20 (CoP20 Doc 85.1 Rev 1).

- *China reported that historic stockpiles of Saiga specimens must be registered, packaged and stored at designated locations after verification of their legal origin. Stockpiles accumulated from law enforcement activities are subject to full process supervision, from confiscation to transfer and storage. The sale, purchase, and utilization of saiga specimens required prior approval.*
- *Malaysia reported that S. tatarica has been imported only by dealers in Peninsular Malaysia, primarily for use in Traditional Chinese Medicine (TCM), and all businesses in Peninsular Malaysia involving wildlife and its parts or derivatives must be licensed. These dealers are required to maintain detailed transaction records, which are regularly inspected by enforcement officers to ensure traceability. Malaysia also uses an online licensing system to monitor all licensed dealers and their trade activities. In 2024, a national inventory of saiga stockpiles was developed in collaboration with TRAFFIC to support enforcement and ensure legal stockpile regulation.*
- *Singapore reported that domestic trade and consumption of Saiga horn are allowed only for pre-Convention or legally imported specimens with valid CITES permits and traders are required to keep and maintain records of their stockpiles. Traders were required to declare their pre-Convention stockpiles of Saiga parts when it was listed in Appendix II. Saiga horn stockpiles are reportedly regularly monitored through inspections, surveys with TCM shops, and an automatic stock card module integrated with the CITES e-permit platform.*

Captive breeding

Small numbers of saiga have been kept in zoos in Europe and the USA, and in rescue centres in Central Asia, but the species is difficult to maintain in captivity (IUCN SSC Antelope Specialist Group, 2023). In Kazakhstan, Saiga tatarica are kept in captivity at two locations in semiwild conditions. In 2024, one facility was holding just over 1,000 animals and the other around 500 animals (Convention on Migratory Species, 2025b). A comprehensive review of the status of and goals for captive breeding of saiga was held in 2017 (Saiga Conservation Alliance, 2017). Although

production from captive animals is noted in the SS as one of the options for sourcing horn for export, this appears to be unlikely at any significant scale in the foreseeable future.

Potential implementation challenges (including similar species)

While Saiga tatarica and Saiga borealis show morphological differences, products in trade including their horns cannot be differentiated by non-experts. Both may be represented in stockpiles held outside the species' range States, the current source of legal trade, although it is likely that the vast majority of the horns in stockpiles are S. tatarica.

Recent media reports have speculated about a change of legal status of Saiga tatarica in the Russian Federation to allow for control of numbers of animals of the Ural population migrating into the Trans-Volga region (Korotov-Maybach, 2025). If this happens in future, management coordination between Kazakhstan and the Russian Federation would be important. However, despite a move to remove saiga other than the north-western Pre-Caspian population from the national Red Book, the species remains subject to a strict prohibition on hunting and trade under the Criminal Code of the Russian Federation (Krivosheyeva in litt., 2025).

Potential benefits and risks of the change in annotation

The participation of local community members in legal hunting and recovery of horn from natural mortality has potential to mitigate to some extent the conflict between growing and sustained larger saiga populations, and agricultural livelihoods. Significant elements of the national saiga strategy focus on delivery of local community benefits (Convention on Migratory Species, 2023) and related guidance on community-based wildlife management has been developed (Zuther et al., 2024). Kazakhstan's national saiga conservation and management strategy includes recommendations to enhance local participation in decision-making and ensure the sharing of benefits from any saiga utilization, which should be encouraged. From information currently available, it is not clear to what extent mechanisms for benefit sharing and stakeholder involvement in management decision-making are being integrated into government planning. Failure to address such issues may exacerbate conflict and increase the risk of poaching and illegal trade.

Without resumed legal supply, the saiga horn stockpiles in destination countries will eventually be depleted and incentives for illegal trade may increase without renewed supply or reduced demand. This risk is exacerbated by conflict between people in rural areas and growing saiga populations, which will continue if conservation targets are met. It is suggested by the proponent that allowing international commercial trade from the Kazakhstan population will help meet the existing demand for Saiga horns, thereby reducing poaching pressure on Saiga populations in all range countries, contributing to their conservation and sustainable use at a global level. Whether this proves to be the case depends on a range of factors, including how end markets react to the provision of new legal supply and how this in turn affects incentives for poaching and illegal trade. The longevity of stockpiles in traditional destination countries and relatively low levels of poaching and illegal trade in recent years could suggest that consumption levels for current end uses are rather stable. A particular situation to avoid is that any renewed legal horn exports are priced at such a high level that investment in illegal supply, even with associated risks, might be at significant advantage.

Other considerations:

In the supporting statement Kazakhstan notes that it is considering an alternative approach to saiga horn trade management, with only finished processed products allowed for export, but this does not appear to be the primary intention of the proposal at this time and considering the nature of end uses, with saiga used for direct dispensing by pharmacists and manufacture of patent medicines, the viability of such an approach is questionable.

References

- Bekenov, A.B., Grachev, I.A. and Milner-Gulland, E.J. (1998) The ecology and management of the Saiga Antelope in Kazakhstan. *Mammal Review* 28(1):1–52. <https://doi.org/10.1046/j.1365-2907.1998.281024.x>
- Chimeddorj, B. (2025) Mongolian saiga population hits 15,540. *Saiga News* 2024/2025 30:19. Available at: https://www.saigaresourcecentre.com/sites/default/files/2025-03/saiga_news-en_30_web_1.pdf
- Convention on Migratory Species (2023) Strategy for the conservation and management of saiga in Kazakhstan. Consultative Meeting "Identifying solutions to human-wildlife conflict involving Saiga Antelopes in Kazakhstan." Part 2: Astana Meeting. Astana, Kazakhstan, 25 May – 27 May 2023. UNEP/CMS/SCMA/Outcome 2. Available at: https://www.cms.int/sites/default/files/document/cms_scm_astana_outcome2_strategy_en_0.pdf
- Convention on Migratory Species (2025a) Medium-Term International Work Programme for the Saiga Antelope (2025-2030). Fifth Meeting of the Signatories to the Memorandum of Understanding Concerning Conservation, Restoration and Sustainable Use of the Saiga Antelope. Astana, Kazakhstan, 12 – 14 March 2025. UNEP/CMS/Saiga/MOS5/Outcome 2. Available at: https://www.cms.int/iosea-turtles/sites/default/files/document/cms_saiga_mos5_outcome%202_mtiwp_e.pdf
- Convention on Migratory Species (2025b) Overview report on saiga conservation status and saiga MOU implementation. Fifth Meeting of the Signatories to the Memorandum of Understanding Concerning

- Conservation, Restoration and Sustainable Use of the Saiga Antelope. Astana, Kazakhstan, 12 – 14 March 2025. UNEP/CMS/Saiga/MOS5/Outcome 1.
- Ermagambetova, D. (2025) Saigas have begun to be shot in the West Kazakhstan region. Kursiv News Website. Available at: <https://kz.kursiv.media/2025-07-04/dnrm-caygakov-nachali-otstrelivat-v-zko/>. Accessed on 31 July 2025
- Gomez, L. and Krishnasamy, K. (2019) A rapid assessment of the trade in Saiga Antelope horn in Peninsular Malaysia. *TRAFFIC Bulletin* 31(1):35-38. Available at: https://www.traffic.org/site/assets/files/12036/bulletin-31_1-final-web.pdf
- Government of Kazakhstan (2025) Planning of Sustainable use of Saiga in Kazakhstan. Information document for the Fifth Meeting of the Signatories to the Memorandum of Understanding Concerning Conservation, Restoration and Sustainable Use of the Saiga Antelope. Astana, Kazakhstan, 12 – 14 March 2025. UNEP/CMS/Saiga/MOS5/Inf.20. Available at: https://www.cms.int/sites/default/files/document/cms_saiga_mos5_inf.20_sustainable-use-kazakhstan_ru.pdf
- IUCN SSC Antelope Specialist Group (2018) *Saiga tatarica* ssp. *mongolica*. *The IUCN Red List of Threatened Species* 2018: e.T19833A50194613. <https://dx.doi.org/10.2305/IUCN.UK.2018-2.RLTS.T19833A50194613.en>. Accessed on 18 July 2025.
- IUCN SSC Antelope Specialist Group (2023) *Saiga tatarica* ssp. *tatarica*. *The IUCN Red List of Threatened Species* 2023: e.T19834A244065239. <https://dx.doi.org/10.2305/IUCN.UK.2023-1.RLTS.T19834A244065239.en>. Accessed on 15 July 2025.
- Kock, R.A., Orynbayev, M., Robinson, S., Zuther, S., Singh, N.J., Beauvais, W., Morgan, E.R., Kerimbayev, A., Khomenko, S., Martineau, H.M., Rystaeva, R., Omarova, Z., Wolfs, S., Hawotte, F., Radoux, J. and Milner-Gulland, E.J. (2018) Saigas on the brink: Multidisciplinary analysis of the factors influencing mass mortality events. *Science Advances* 4(1):eaao2314. <https://doi.org/10.1126/sciadv.aao2314>
- Korotov-Maybach, S. (2025) Saiga antelopes migrating from Kazakhstan will be excluded from the Red Book of the Saratov Region. 164.RU Website. Available at: <https://164.ru/text/animals/2025/07/15/75726092/>. Accessed on 4 August 2025.
- Krivosheyeva, A. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Lushchekina, A. and Karimova, T. (2025) Current status of the Saiga population in the north-west pre-Caspian region, Russian Federation. *Saiga News* 2024/2025 30:23-24. Available at: https://www.saigaresourcecentre.com/sites/default/files/2025-03/saiga_news-en_30_web_1.pdf
- Mallon, D. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Milner-Gulland, E.J. (1994) A Population Model for the Management of the Saiga Antelope. *Journal of Applied Ecology* 31(1):25-39. <https://doi.org/10.2307/2404596>
- Milner-Gulland, E.J. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Milner-Gulland, E.J., Bukreeva, O.M., Coulson, T., Lushchekina, A.A., Kholodova, M.V., Bekenov, A.B. and Grachev, I.A. (2003) Reproductive collapse in saiga antelope harems. *Nature* 422(135). <https://doi.org/10.1038/422135a>
- Milner-Gulland, E.J., Hughes, P., Bykova, E., Buuveibaatar, B., Chimeddorj, B., Karimova, T.Yu., Lushchekina, A.A., Salemgareyev, A., von Meibom, S. and Zuther, S. (2020) The sustainable use of Saiga Antelopes: Perspectives and Prospects. Fourth Meeting of the Signatories to the Memorandum of Understanding Concerning Conservation, Restoration and Sustainable Use of the Saiga Antelope. Russian Federation, Online, 28 – 29 September 2021. UNEP/CMS/Saiga/MOS4/Doc.7/Rev.1. Available at: https://www.cms.int/sites/default/files/document/unep-cms_saiga_mos4_doc.7_rev.1_sustainable-use-saiga-antelopes_e.pdf
- Roberts, D.L., Mun, K. and Milner-Gulland, E.J. (2022) A systematic survey of online trade: trade in Saiga antelope horn on Russian-language websites. *Oryx* 56(3):352-359. doi:10.1017/S0030605320001313
- Saiga Conservation Alliance (2017) Meeting report: International workshop on captive breeding for saiga conservation, 28 – 29 August 2017, held at the Institute of Ecology and Evolution of the Russian Academy of Sciences. Available at: <https://www.saigaresourcecentre.com/literature/captive-breeding-workshop-meeting-report-2017>
- Smelyansky, I., Kirilyuk, V. and Titova, S. (2023) Saigas are returning to the Russian Trans-Volga region. *Saiga News* 2022/2023 28:23-25. Available at: https://www.saigaresourcecentre.com/sites/default/files/2023-01/saiga_news-en_28_small.pdf
- Tokubek uulu, B., Kurmanov, S. and Turdubekov, U. (2024) A review of the laws related to the use and trade of wildlife in four Central Asian countries. *TRAFFIC*. Cambridge, UK. https://www.traffic.org/site/assets/files/23053/en_legal_analysis-central_asia.pdf
- von Meibom, S., Vaisman, A., Neo Liang, S.H., Ng, J. and Xu, H. (2010) Saiga Antelope Trade: Global Trends with a Focus on South-east Asia. *TRAFFIC project report to the CITES Secretariat*. *TRAFFIC Europe*. https://www.traffic.org/site/assets/files/4118/saiga_antelope_trade.pdf
- Wilson, D. and Reeder, D. (editors) (2005) *Mammal Species of the World. A Taxonomic and Geographic Reference* (3rd ed). *Johns Hopkins University Press* 2:142.
- Xu Ling (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.

- Zharbulova, N. (2025) Saiga antelopes to be shot again in Kazakhstan. Kursiv News Website. Available at: <https://kz.kursiv.media/2025-06-28/zhrb-v-kazakhstane-razreshat-otstrel-saigakov/>. Accessed on 31 July 2025.
- Zuther, S. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Zuther, S., Michel, S., Roe, D., Kubanychbekov, Z., Karimov, Kh., Sklyarenko, S.L. and Ward, S. (2024) Potential for Community-based Wildlife Management of CAMI Species. Report to the Federal Agency for Nature Conservation (BfN) and the Secretariat of the Convention on the Conservation of Migratory Species of Wild Animals (CMS). https://www.cms.int/sites/default/files/publication/Study_CBWM_final_for%20posting_en_complete.pdf

Deletion of Giraffe *Giraffa camelopardalis* populations of Angola, Botswana, Eswatini, Malawi, Mozambique, Namibia, South Africa and Zimbabwe from Appendix II

Proponents: Namibia, South Africa, United Republic of Tanzania, Zimbabwe

Summary: The Giraffe *Giraffa camelopardalis* is the world's tallest land mammal. It remains widespread across Southern and Eastern Africa, with smaller, isolated populations in West and Central Africa. In 2016, based on evidence of 36-40% declines over three generations (30 years), the IUCN Red List classified the Giraffe as Vulnerable with the best available range-wide estimates indicating a total population in 1985 of around 152,000-163,000 individuals (106,000-114,000 mature), declining to around 100,000 in 2015 (68,000 mature). The main factors responsible for the observed declines were habitat loss, illegal hunting, civil unrest and ecological changes. The presence and severity of those threats, and the conservation strategies used to manage Giraffe populations showed large regional variations. More recently increases, with notable regional variations, have taken place across all parts of the range with the 2025 population estimated at 141,000.

The proponents refer to alternative taxonomies being proposed and supported by researchers within the scientific community, which treat the subspecies *G. c. giraffa* (southern giraffe) as a separate species (*G. giraffa*) with two separate subspecies (*G. g. angolensis* and *G. g. giraffa*). However, they do not explicitly recommend the adoption of an updated nomenclature standard reference for giraffes and no updates to the taxonomy or nomenclature of the giraffe species have been recommended by the Animals Committee for consideration by the Parties at CoP20. This Proposal should therefore be understood to refer to the national populations of *Giraffa camelopardalis*, following current CITES standard nomenclature.

The species was included in Appendix II in 2019 after CoP18. At that time a number of countries (Botswana, the Democratic Republic of the Congo, Eswatini, Namibia, South Africa, United Republic of Tanzania, Zambia and Zimbabwe) took out reservations on the listing. The proponents now seek deletion of the populations of *G. camelopardalis* in Angola, Botswana, Eswatini, Malawi, Mozambique, Namibia, South Africa and Zimbabwe from Appendix II. The combined current (2025) population estimate for these eight range States is 68,000, or just under half of the global population. This represents a five-fold increase for southern African populations that were estimated to number under 15,000 in the 1970s.

Giraffes are traded both internationally and domestically. Reported trade in wild-sourced live Giraffes since the species was listed in Appendix II has been predominately from South Africa and Namibia, with 948 live specimens exported between 2019 and 2023. There is also a significant market for trophies from hunting, with a range of products being traded from hunted Giraffe exported from South Africa, Namibia and Zimbabwe. CITES reported trade from all other Giraffe range States since the 2019 listing has been negligible, comprising one trophy from Zambia (though three were reported by importers) and a small quantity of scientific specimens from Uganda and South Sudan (a non-Party). A record of 43 live individuals exported from Niger is likely a reporting error.

There are no indications of illegal killing of Giraffes for international trade; where poaching does take place this is reportedly for local use.

Analysis: The Giraffe populations referred to in this proposal do not appear to meet the criteria for inclusion in Appendix II. They do not have a restricted range nor a small population, are increasing and there is **no indication that regulation of international trade is required to prevent them becoming eligible for inclusion in Appendix I in the near future, nor is harvest for trade reducing them to a level at which their survival might be threatened.**

Adoption of this proposal would result in a split-listing of *Giraffa camelopardalis* by geographical population, with some remaining in the Appendices and others excluded. According to Annex 3 of Res. Conf. 9.24 (Rev. CoP17), "*split-listings that place some populations of a species in the Appendices and the rest outside the Appendices should not normally be permitted*" due to potential enforcement

challenges, although there are a few precedents for such split-listing in the Appendices. The only nearby Giraffe population – that in Zambia – is not part of the proposal and would remain in the Appendices if the proposal were accepted. However, reported Zambian exports are negligible and there are no indications of cross-border trade in Giraffes or Giraffe products between Zambia and the adjacent co-proponent Giraffe range States (Angola, Botswana, Namibia, Malawi, Mozambique and Zimbabwe). **It seems unlikely that such a split-listing would create significant enforcement problems.**

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

Giraffa camelopardalis was included in Appendix II at CITES CoP18 entering into effect in November 2019.

Taxonomy

Giraffa camelopardalis (Linnaeus, 1758)

The proponents refer to alternative taxonomies being proposed and supported by researchers within the scientific community. The IUCN SSC Giraffe and Okapi Specialist Group (GOSG) has recently undertaken a comprehensive taxonomic review of the genus Giraffa to clarify species and subspecies designations critical to conservation planning and Red List assessments. Following an extensive review of published literature, original datasets, and expert consultation, the IUCN SSC GOSG now recognises four distinct giraffe species and their constituent subspecies (IUCN SSC Giraffe and Okapi Specialist Group Taxonomic Task Force, 2025):

- 1) *Northern giraffe* (*Giraffa camelopardalis*)
 - *G. c. peralta* (*West African giraffe*)
 - *G. c. antiquorum* (*Kordofan giraffe*)
 - *G. c. camelopardalis* (*Nubian giraffe*)
- 2) *Reticulated giraffe* (*Giraffa reticulata*)
- 3) *Masai giraffe* (*Giraffa tippelskirchi*)
 - *G. t. tippelskirchi* (*Masai giraffe sensu stricto*)
 - *G. t. thornicrofti* (*Luangwa/Thornicroft's giraffe*)
- 4) *Southern giraffe* (*Giraffa giraffa*)
 - *G. g. giraffa* (*South African giraffe*)
 - *G. g. angolensis* (*Angolan giraffe*)

*The proponents do not explicitly recommend the adoption of an updated nomenclature standard reference for Giraffes and a Secretariat document on nomenclature notes on CoP20 proposals states that no updates to the taxonomy or nomenclature of the Giraffe species have been recommended by the Animals Committee for consideration by the Parties at CoP20 (CoP20 Inf. 6). CITES currently recognises one species of Giraffe *Giraffa camelopardalis* and therefore this analysis will be conducted on that basis, although reference will also be made to the four new species which will reflect how the listing proposal might be affected, if adopted.*

Range of *G. camelopardalis* (Muller et al., 2018). Populations proposed for deletion are underlined.

Extant: Angola; Botswana; Cameroon; Central African Republic; Chad; The Democratic Republic of the Congo; Ethiopia; Kenya; Mozambique; Namibia; Niger; Somalia; South Africa; South Sudan; United Republic of Tanzania; Uganda; Zambia; Zimbabwe

Extant & Introduced: Eswatini; Rwanda

Possibly Extinct: Mali

Extinct: Eritrea; Guinea; Mauritania; Nigeria; Senegal

Extinct & Vagrant: Burkina Faso; Malawi

IUCN Global Category

Giraffa camelopardalis (sensu CITES) Vulnerable A2acd (assessed 2016, ver 3.1)

*The proposed separate species Southern Giraffe (*G. giraffa*) made up largely of the national populations proposed for deletion from Appendix II, has not yet been assessed on the IUCN Red List, but it was suggested in the State of the Giraffe 2025 report that it should be reclassified in future as Least Concern (Marneweck et al., 2025).*

Biological and trade criteria for retention in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)**A) Trade regulation needed to prevent future inclusion in Appendix I****B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences**

This proposal concerns the populations of *Giraffa camelopardalis* in Angola, Botswana, Eswatini, Malawi, Mozambique, Namibia, South Africa and Zimbabwe (all currently classified as the subspecies *G. g. giraffa*). The most recent estimates of the size of these populations are listed in Table 1. Together they total just over 68,000 animals. South Africa is the stronghold for the species with an estimated population of 29,536 individuals, with around 16,000 individuals occurring within state protected areas of which approximately 13,009 occur within national parks. The giraffe population within South Africa's Kruger National Park alone is estimated at 12,270 (10,345 – 14,554) individuals as at the end of 2023. *Zambia's population of G. g. giraffa, which is not included in the proposal, numbers around 800 with an increasing population trend (Marneweck et al., 2025).*

Table 1. Population estimates of *Giraffa camelopardalis* included in the proposal (population estimates calculated from Marneweck et al., 2025).

Country	Population Estimate
Angola	574
Botswana	13,064
Eswatini	416
Malawi	37
Mozambique	1108
Namibia	14,095
South Africa	29,536
Zimbabwe	9201

Population trends

The best available estimates indicate a total population in 1985 of 151,702-163,452 *Giraffa camelopardalis* (106,191-114,416 mature individuals), and in 2015 a total population of 97,562 *Giraffa camelopardalis* (68,293 mature individuals) (Muller et al., 2018). However, the State of Giraffe 2025 report, provides an updated overview of giraffe conservation efforts, population estimates, and distribution across Africa, by proposed new giraffe species. For the purpose of this analysis, population estimates are presented following the previous taxonomy and the new taxonomy. The report estimates that in 2025, there are 140,701 *Giraffa camelopardalis* (Table 2). This increase can be attributed to a combination of factors. Significant progress has been made through improved data collection methods specific to *Giraffa camelopardalis*, an increase in the frequency of giraffe-specific surveys, and strengthened collaborations for data sharing which have led to more accurate population estimates and enabled more strategic conservation actions and national planning. The growth in southern African Giraffe populations is partially due to better survey coverage and the inclusion of additional private land data. Similarly, for South Africa specifically, the increase is likely an artifact of more accurate population estimates based on the inclusion of small, private sites, rather than solely a true increase (Marneweck et al., 2025).

In the last IUCN Red List Assessment (2016), it was noted that some *Giraffa camelopardalis* populations are stable or increasing, while others are declining, and each population is subject to pressure by threats specific to their local country or region.

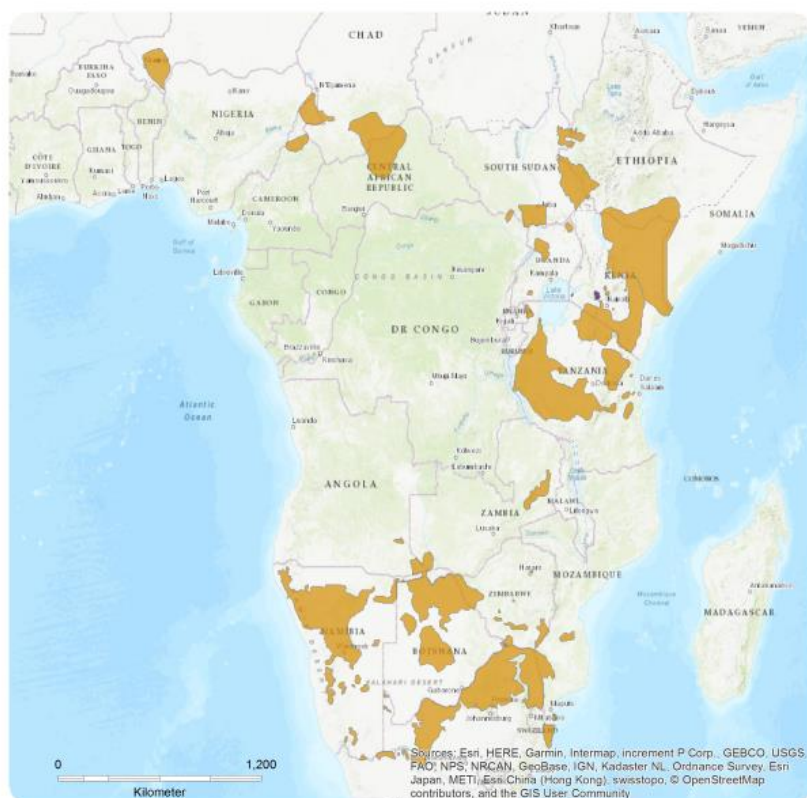


Figure 1. Range map for *Giraffa camelopardalis* (Muller et al., 2018)

Table 2. Overview of the status of *Giraffa camelopardalis* subspecies and proposed species.

Proposed New Species	Current Subspecies	Region of Africa	Range States (those proposed for removal from Appendices <u>underlined</u>)	Historic Population Estimate (Yr) ¹	CoP18 Population Estimate (Yr) ²	Proposed New Species 2025 Population Estimate ³	Current Subspecies 2025 Population Estimate (National population estimates) ⁴
Northern	<i>G. c. peralta</i>	West	Niger	50 (1990s)	607 (2017)	7,037	669 (Niger)
	<i>G. c. antiquorum</i>	Northern/Central	Cameroon, Chad, Central African Republic, the Democratic Republic of the Congo, South Sudan	13,704 (1975-1986)	1,942 (2018)		2,391 (Cameroon: 617, Central African Republic: 10, Chad: 1,657, the Democratic Republic of the Congo: 106, South Sudan: 1)
	<i>G. c. camelopardalis</i>	Northern/Eastern	South Sudan, Ethiopia	20,577 (1979-1982)	650 (2018)		3,977 (Ethiopia: 475, Kenya: 1,281, Uganda: 1,986, South Sudan: 235)
	<i>G. c. rothschildi</i>	Eastern	Uganda, Kenya	36,000-47,750 (1990s)	2,098 (2018)		
Reticulated	<i>G. c. reticulata</i>	Eastern	Kenya, Ethiopia, Somalia	1,330 (1960s)	15,785 (2018)	20,901	20,901 (Ethiopia: 101, Kenya: 20,746, Somalia: 54)
Masai	<i>G. c. tippelskirchi</i>	Eastern	Kenya, United Republic of Tanzania	66,449 (1977-1980)	31,611 (2015)	43,926	43,162 (Kenya: 14,179, Rwanda: 163, United Republic of Tanzania: 28,820)
	<i>G. c. thomicrofti</i>	Eastern	Zambia	600 (1983)	600 (2018)		764 (Zambia)
Southern	<i>G. c. angolensis</i>	Southern	<u>Angola</u> , <u>Botswana</u> , <u>Namibia</u>	5,000 (1970-2004)	14,748 (2018)	68,837	15,663 (Angola: 181, Botswana: 1,587, Namibia: 13,895)
	<i>G. c. giraffa</i>	Southern	<u>Angola</u> , <u>Botswana</u> , <u>Eswatini</u> , <u>Malawi</u> , <u>Mozambique</u> , <u>Namibia</u> , <u>South Africa</u> , <u>Zambia</u> , <u>Zimbabwe</u>	8,000 (1979)	21,387 (2016)		53,174 (Angola: 393, Botswana: 11,477, Eswatini: 416, Malawi: 37, Mozambique: 1,108, Namibia: 200, South Africa: 29,536, Zambia: 806, Zimbabwe: 9,201)
						140,701	

International trade

The SS states that giraffe products are widely traded locally and internationally, with nearly every part of the animal being used, including bones for curios and carvings, and meat increasingly sold commercially. South Africa accounts for around 94% of all international trade in giraffe specimens, with over 99% of trade originating in southern Africa (Botswana, Namibia, South Africa, Zambia and Zimbabwe). Live giraffes are exported from South Africa for various purposes, including introduction into zoos and breeding programmes, as well as reintroductions into nearby range States (including Malawi and Mozambique). Between 2019 and 2022, approximately 410 live giraffes were exported internationally from South Africa. The USA and Europe are the major markets for giraffe trophies. Between 2019 and 2022, approximately 2,064 trophies were exported from South Africa, mainly to the USA (63%) and the EU (21%), averaging 516 animals per year.

Live

Since the species was listed in Appendix II in 2019, records from the CITES Trade Database (2025) showed that 991 live, wild-sourced *Giraffa camelopardalis* were exported (Figure 2). Between 2019 and 2023, South Africa reported exporting 747 individuals, Namibia 201, and Niger 43.

¹ IUCN and TRAFFIC, 2019² IUCN and TRAFFIC, 2019³ Marneweck et al., 2025⁴ Marneweck et al., 2025

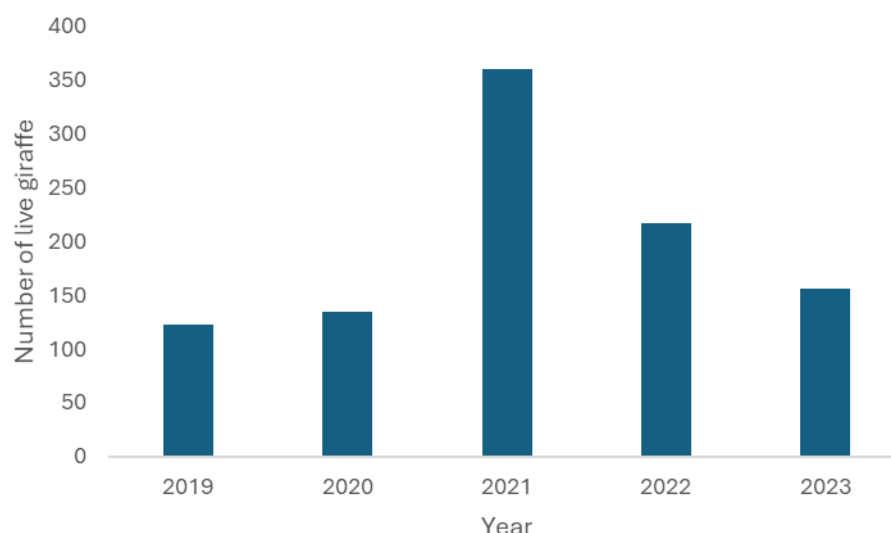


Figure 2. Number of live *Giraffa camelopardalis* exported from the wild between 2019 and 2023, as reported by exporters (CITES Trade Database 2025).

South Africa reported having exported a total of 747 individuals:

- 253 for the purpose of reintroduction or reintroduction into the wild: to Mozambique (n=130), Malawi (n=110), Macau SAR of China (n=12) and Kuwait (n=1). Although it's not clear whether the exports to Macau SAR of China and Kuwait for this purpose are a reporting error.
- 300 for commercial purposes into: Kuwait (n=110), Armenia (n=45), USA (n=45), Guatemala (n=30), Germany (n=12), United Arab Emirates (n=12), Thailand (n=12), Qatar (n=7), Saudi Arabia (n=6), Egypt (n=6), Unknown (n=6), Islamic Republic of Iran (n=3) and Turkey (n=3).
- 193 for zoos into: Thailand (n=78), Brazil (n=36), United Arab Emirates (n=27), Libya (n=18), Viet Nam (n=12), Kuwait (n=6), Egypt (n=6), Turkey (n=3), Uzbekistan (n=3), Islamic Republic of Iran (n=3), Iraq (n=3) and Malaysia (n=1).
- One for breeding in captivity to the Russian Federation.

Namibia reported having exported a total of 201 individuals:

- 197 for the purpose of reintroduction or reintroduction into the wild to: Zambia (n=100), Angola (n=89) and the Democratic Republic of the Congo (n=8).
- Four for commercial purposes to Angola.

Niger reported having exported a total of 43 live individuals in 2021. Brown in litt. (2025) noted that this record might be an error in the CITES permit process as they were not aware of this occurring. They were all for commercial purposes to France.

Trophies

The SS states that within southern Africa, most hunting trophies between 2019 and 2023 originated from South Africa (75%), Namibia (16%) and Zimbabwe (8%). Between 2010 and 2023 approximately 3,598 giraffes were hunted within South Africa by international clients, averaging 257 ± 90 animals per year. It is estimated that an additional 300 giraffe are hunted by local hunters or culled each year. In total, this equates to between 1.6% and 2.2% of the total population and the proponents consider this to be sustainable. *The basis for determining levels of sustainable offtake for this species is unknown, however South Africa published an NDF for Giraffe in 2022 stating that 2-3% is considered sustainable (South African Scientific Authority, 2022).* In Namibia and Zimbabwe, offtakes by international hunters equate to <1% of the respective national populations. Hunting mainly takes place on wildlife ranches, private game reserves and wildlife management areas.

The SS states that when hunted, every part of the animal is utilised to produce products for sale, which are marketed locally and internationally (Table 3). The long bones and scapula are mainly used for carvings and artwork, while the metatarsals and metacarpals are used for the cutting of knife handles. Approximately eight pieces can be cut per bone to make knife handles and an estimated five to 17 bones per giraffe can on average be used for carving and the manufacturing of curios. As such, South Africa could potentially export between 2,335 and 10,999 bones (and/or bone curios) per annum, sourced from the estimated ~600 giraffe sustainably

hunted or culled per year. This falls within the range exported per year and is considered sustainable. Exports of giraffe specimens are thus byproducts of a sustainable hunting industry in South Africa.

Table 3. Numbers of specimens of *Giraffa camelopardalis* by product and country, sourced from the wild, as reported by exporters and importers between 2019 and 2023, excluding scientific specimens (CITES Trade Database, 2025).

Product	Quantity in trade reported by exporter (reported by importer)					
	South Africa	Namibia	Zimbabwe	Botswana	Zambia	Mozambique
Bones	5,046 (5,706)	24 (114)	73 (18)	5		
Trophies	1,058 (1,422)	476 (288)	176 (208)	3 (6)	1 (3)	
Skin pieces	17 (552)	16 (29)	27 (1,077)			12
Skulls	194 (428)	19 (10)	22 (16)			
Skins	74 (141)	67 (16)	57 (7)			
Teeth	100 (100)	40				
Leather products (small)	44 (121)	4	34			
Bone carvings	52 (17,797)	2 (8)	(5)			
Tails	30 (20)	7 (5)	3			
Leather products (large)	2 (73)	6 (3)	22 (40)			
Feet	5 (176)	4 (9)	17 (12)			
Bone pieces	(5,669)	23	(4)	(8)		
Hair products	(5)	2				
Rug	(19)	1	1 (1)			
Derivatives	1					
Genitalia			1			
Skeletons	(1)		1			
Carvings	(1)		1			
Bodies	(62)					
Fur product (small)	(21)		(10)			
Meat	(1)					
Plates	(1)					

Scientific specimens from populations not proposed for deletion were 11 scientific specimens from South Sudan and 0.002 kg from Uganda reported by importer, and 483 g from Uganda, reported by exporter (CITES Trade Database, 2025).

When expanding the search of the CITES Trade Database to include all terms for all purposes of *Giraffa camelopardalis*, both the number of specimens and the number of kg, as reported by the exporter, there did not appear to be any exports from *Giraffa camelopardalis* range States that have populations not included in this proposal, in any significant quantity. Therefore, currently there appears that there is no significant trade in *Giraffa camelopardalis* parts that would not be distinguishable from populations not in this proposal, and therefore the risk of *Giraffa camelopardalis* parts being laundered from range States not in this proposal into range States of populations to be deleted from Appendix II, seems unlikely.

There were no significant quantities of parts (either in number of specimens or by weight) reported by exporters from *Giraffa camelopardalis* range States other than those proposed for deletion from the Appendices.

Illegal trade

The SS states that there is no evidence of illegal killing of *Giraffa camelopardalis* for the international trade and the illegal killing for domestic use is considered small to negligible. It is not clear whether this is in relation to the populations in the proposal or all range States of *Giraffa camelopardalis*.

Illegal hunting for meat takes place in several parts of the species' range (Muller et al., 2018).

Domestic vs international trade

Dunn et al. (2021) notes patterns for domestic use and international trade of *Giraffa camelopardalis*; across Central, East and West Africa, local level markets were considered to be the primary sinks for *Giraffa camelopardalis* products and limited trade across porous borders was primarily recorded in Kenya. In contrast they found that the trade of *Giraffa camelopardalis* in Southern Africa occurs domestically (local users) as well as internationally.

Retention in Appendix II to improve control of other listed species**A) Specimens in trade resemble those of species listed in Appendix II under Res. Conf. 9.24 (Rev. CoP17) Annex 2 a or listed in Appendix I**

If the *Giraffa camelopardalis* populations of Angola, Botswana, Eswatini, Malawi, Mozambique, Namibia, South Africa and Zimbabwe were removed from Appendix II there is the potential for enforcement challenges. While the SS notes that the various subspecies of giraffe can be distinguished on the shape and colour of their spot pattern, the population of Zambia is not included in this proposal. Live and trophy specimens from Zambia's population of giraffe bordering the Southern African countries would not be distinguishable from the other populations. Even under the new proposed taxonomy, the Zambian population would contain the "Southern Giraffe" and "Masai Giraffe" which are geographically separate; the "Southern Giraffe" in Zambia would be indistinguishable (unlike "Masai Giraffe") from the populations proposed to be deleted from Appendix II, due to having the same spot patterns.

The SS states that the probability of a border official encountering bones (which are not distinguishable) of any other population of giraffe is highly improbable as there is a sufficient supply of products sourced from the abundant legal, local giraffes.

Zambia has only exported one trophy in 2022 and one specimen in 2023 suggesting that trade from populations of *Giraffa camelopardalis* from range States that would remain in Appendix II are not significant.

Another issue could be the transboundary movement of Giraffe between Zambia and neighbouring countries where crossing the border would result in the regulatory regime for the animal changing (if it was taken for international trade). Fennessy in litt. (2025) reported that transboundary movement of the same species between countries does occur across the continent, however, the transboundary movement of giraffe (proposed Southern giraffe) from Zambia does not seem to occur. They also reported that there are no natural movements of southern giraffe between Zambia and Zimbabwe, and that if southern giraffe in Zambia did naturally move transboundary, it would potentially be to and from Namibia, although this has not been reported.

Additional Information**Threats**

Four major threats to *Giraffa camelopardalis* can be identified, although the severity and presence of these threats varies by region and population: (1) habitat loss (through deforestation, land use conversion, expansion of agricultural activities and human population growth) (2) civil unrest (ethnic violence, rebel militias, paramilitary and military operations), (3) illegal hunting (poaching), and (4) ecological changes (mining activity, habitat conversion to agriculture, climate-induced processes) (Muller, et al., 2018).

In Southern Africa, the main perceived threats are habitat loss and conversion of land for human development, and illegal hunting (Muller et al., 2018).

In West Africa, the main threats are habitat loss due to increasing human populations and human-wildlife conflict (Muller et al., 2018).

In Eastern and Central Africa, the main threats are habitat loss through rapid conversion of land for farming and increasing human populations, drought, illegal hunting for meat and hide, and armed conflict throughout unstable regions (Muller et al., 2018).

Conservation, management and legislation

The SS states that *Giraffa camelopardalis* is a protected species in Angola, Botswana, Eswatini, Namibia, South Africa and Zimbabwe (they are also protected in Malawi (DNPW, 2023)).

In 2021, Dunn et al. (2021) reported that hunting was prohibited in most range States except for Namibia, South Africa and Zimbabwe with a permit, and although hunting with a permit was allowed in Zambia, hunting was not currently active then.

Implementation challenges

While the populations (Angola, Botswana, Eswatini, Malawi, Mozambique, Namibia, South Africa and Zimbabwe) proposed to be deleted from the Appendix-II listing of *Giraffa camelopardalis* have increasing populations sizes and may in future be recognised under CITES as a separate "Southern Giraffe" species, under proposed nomenclature the Zambian population would be split between geographically separate Southern Giraffe and Masai Giraffe. Zambia's population has not been proposed for deletion from Appendix II.

The SS reports that the "Southern Giraffe" can be identified by its star-shaped and various shades of brown patches, however, while whole giraffes are readily identifiable, smaller parts and derivatives may be more cryptic. There is also currently no method of easily identifying giraffe bone to the species level in some of the forms in

which it is traded. Therefore, if trade in giraffe bones became prominent, it may not be possible to distinguish between populations in the Appendices and those deleted from the Appendices. *However, it is possible to distinguish the four giraffe species by their cranial morphology (Kargopoulos et al., 2024).*

Potential benefit(s) of deletion for trade regulation

The SS states that the inclusion of *Giraffa camelopardalis* in Appendix II has imposed a considerable administrative burden on range States, requiring them to issue export permits despite having entered formal reservations against the listing. It is noted that the human and financial resources dedicated to processing these permits in countries where conservation budgets are already strained could arguably be more effectively allocated to high-priority conservation efforts and regulation of other less charismatic and often neglected species.

Other comments

*Annex 3 on Special Cases in Res. Conf. 9.24 (Rev. CoP17) states that "Split-listings that place some populations of a species in the Appendices, and the rest outside the Appendices, should normally not be permitted." This proposal would result in such a case and there are a few examples of where the Parties have already accepted such split-listings. Annex 3 also states that "When split-listing does occur, this should generally be on the basis of national or regional populations, rather than subspecies." This proposal proposes the removal of some national populations with the remaining populations of *Giraffa camelopardalis* in Appendix II.*

References

- Bertola, L.D., Quinn, L., Hanghøj, K., Garcia-Erill, G., Rasmussen, M.S., Balboa, R.F., Meisner, J., Bøggild, T., Wang, X., Lin, L., Nursyifa, C., Liu, X., Li, Z., Chege, M., Moodley, Y., Brüniche-Olsen, A., Kuja, J., Schubert, M., Agaba, M., Santander, C.G., Sinding, M.-H.S., Muwanika, V., Masembe, C., Siegismund, H.R., Moltke, I., Albrechtsen, A. and Heller, R. (2024) Giraffe lineages are shaped by major ancient admixture events. *Current Biology* 34:1576-1586.e5. <https://doi.org/10.1016/j.cub.2024.02.051>
- Coimbra, R.T.F., Winter, S., Kumar, V., Koepfli, K.-P., Gooley, R.M., Dobrynin, P., Fennessy, J. and Janke, A. (2021) Whole-genome analysis of giraffe supports four distinct species. *Current Biology* 31(13):2929-2938.e5.
- Coimbra, R.T.F., Winter, S., Muneza, A., Fennessy, S., Otiende, M., Mijele, D., Masiaine, S., Stacy-Dawes, J., Fennessy, J. and Janke, A. (2023) Genomic analysis reveals limited hybridization among three giraffe species in Kenya. *BMC Biology* 21(1):215
- DNPW (2023) Malawi Giraffe Conservation Action Plan 2024-2028. Department of National Parks and Wildlife, Lilongwe, Malawi.
- Dunn, M.E., Ruppert, K., Glikman, J.A., O'Connor, D., Fennessy, S., Fennessy, J. and Verissimo, D. (2021) Investigating the international and pan-African trade in giraffe parts and derivatives. *Conservation Science and Practice* 3(5):p.e390.
- Fennessy, J. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Fennessy, J., Bidon, T., Reuss, F., Kumar, V., Elkan, P., Nilsson, M.A., Vamberger, M., Fritz, U. and Janke, A. (2016) Multi-locus analyses reveal four giraffe species instead of one. *Current Biology* 26(18):2543–2549
- IUCN and TRAFFIC (2019) IUCN/TRAFFIC Analyses of the Proposals to Amend the CITES Appendices. Prepared by IUCN Global Species Programme and TRAFFIC for the Eighteenth Meeting of the Conference of the Parties to CITES. IUCN – International Union for Conservation of Nature, Gland, Switzerland.
- IUCN SSC Giraffe and Okapi Specialist Group Taxonomic Task Force (2025) An evaluation of the taxonomic status of giraffe (*Giraffa* spp.). IUCN SSC Giraffe and Okapi Specialist Group. Windhoek, Namibia.
- Kargopoulos, N., Marugán-Lobón, J., Chinsamy, A., Agwanda, B.R., Brown, M.B., Fennessy, S., Ferguson, S., Hoffman, R., Lala, F., Muneza, A., Mwebi, O., Otiende, M., Petzold, A., Winter, S., Zabeirou, A.R.M. and Fennessy, J. (2024) Heads up – four *Giraffa* species have distinct cranial morphology. *PLOS One* 19(12):e0315043.
- Marneweck, C.J., Brown, M.B., Ekandjo, P., Fennessy, S., Hoffman, R., Kipchumba, A., Muneza, A., Otten, F. and Fennessy, J. (2025) State of Giraffe 2025: An update from the Giraffe Africa Database (GAD). Giraffe Conservation Foundation, Windhoek, Namibia.
- Muller, Z., Bercovitch, F., Brand, R., Brown, D., Brown, M., Bolger, D., Carter, K., Deacon, F., Doherty, J.B., Fennessy, J., Fennessy, S., Hussein, A.A., Lee, D., Marais, A., Strauss, M., Tutchings, A. and Wube, T. (2018) *Giraffa camelopardalis* (amended version of 2016 assessment). *The IUCN Red List of Threatened Species* 2018: e.T9194A136266699. <https://dx.doi.org/10.2305/IUCN.UK.2016-3.RLTS.T9194A136266699.en>. Accessed on 1 August 2025
- South African Scientific Authority (2022) Non-detriment finding assessment for *Giraffa camelopardalis giraffa* (South African giraffe). Reference Number: Gif_cam_gir_Jul202: https://www.wrsa.co.za/wp-content/uploads/2024/10/Giraffe_Oct-2024-1.pdf. Accessed 29 August 2025
- Winter, S., Fennessy, J. and Janke, A. (2018) Limited introgression supports division of giraffe into four species. *Ecology and Evolution* 8(20):10156–10165.

Inclusion of Okapi *Okapia johnstoni* in Appendix I

Proponent: Democratic Republic of the Congo

Summary: The Okapi *Okapia johnstoni* is a large, solitary member of the giraffe family that is now endemic to the Democratic Republic of the Congo but may have also occurred in Uganda in the past. The species prefers mixed primary and secondary forests where dense plant cover provides camouflage and a variety of plant species for consumption.

Estimates of population size are limited but in 2013, it was determined there could be 35,000-50,000 individuals. However, this estimate should be viewed with caution as it was calculated using patchily distributed dung-based surveys. In 2015, it was determined that the species had an area of occupancy of just over 14,000 km², although this is likely to be an underestimate. This species was assessed globally as Endangered in 2015 on the IUCN Red List. The justification was that the species had been decreasing since at least 1995 and was projected to continue decreasing. The rate of decline was estimated to have exceeded 50% over the past three generations (24 years). Dung surveys in 2018 in the Okapi Wildlife Reserve, the best protected area in its range, showed dung densities decreased by 60% since 1995.

The drivers of declines are illegal hunting, including by illegal armed groups in and around key protected areas and artisanal mining degrading the species' habitat. Armed groups have threatened conservation efforts including attacking the headquarters of the Okapi Wildlife Reserve, killing several people as well as 14 Okapis present at the centre, likely in retaliation for recent engagements by rangers that disrupted poaching and mining activities.

The species is totally protected in the Democratic Republic of the Congo, prohibiting the killing, capture, or commercial export. A small legal trade in captive-bred Okapi has been permitted that helps to fund conservation infrastructure. Illegal hunting for its meat, skin and oil is considered a threat to the species. Its oil is believed to have medicinal properties. There is some evidence that illegal trade occurs across the border into Uganda in Okapi products such as skins and oil, but the extent of this trade is unknown. Anecdotal reports suggest there is a trade to Southeast Asia and the Middle East, but no firm evidence is available. Likewise, an anecdotal report mentions a domestic market has opened up for the trade in Okapi oil in the Democratic Republic of the Congo, involving processing in South Africa, but this has not been corroborated.

Analysis: Information on the status of *Okapia johnstoni* is scarce. The species does not have a small population, nor a restricted area of distribution. However, acknowledging limited data sources, declines in population size have been inferred based on dung counts which would indicate that the species **has undergone a marked recent rate of decline** in its population size in the wild over the last three generations **meeting the biological criteria** in Res. Conf. 9.24 (Rev CoP17) Annex 1, paragraph C.

In the Democratic Republic of the Congo, the species cannot be exported for commercial purposes and therefore any international trade is illegal. Small numbers of animals bred in captivity have been exported legally from the Democratic Republic of the Congo to fund Okapi conservation efforts. Evidence of illegal trade is limited and, therefore **there is insufficient information to determine whether the species is affected by international trade, so it is not possible to assess whether the species overall qualifies for inclusion in Appendix I.**

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

Okapia johnstoni (P.L. Sclater, 1901)

Range

Extant: The Democratic Republic of the Congo

Extinct: Uganda

IUCN Global Category

Endangered A2abcd+4abcd (assessed 2015, ver 3.1)

Biological criteria for inclusion in Appendix I**A) Small wild population**

The population size of *Okapia johnstoni* was estimated to be 35,000-50,000 individuals in 2013. However, according to Mallon et al. (2015), this figure is better regarded as a “guestimate” as it relies on the extrapolation of a limited number of patchily distributed, dung-based surveys. In 2015, numbers were believed to be low and decreasing (Kümpel et al., 2015). The population estimates should be treated cautiously as they come from line transects conducted in the Okapi Wildlife Reserve and Maiko National Park, primarily using dung counts with challenges identified around seasonality, specific survey locations and the large uncertainty over Okapi dung decay rates. Furthermore, the data from Maiko National Park are over 30 years old.

B) Restricted area of distribution

Okapia johnstoni is endemic to the Democratic Republic of the Congo (although there are old reports of its presence in the adjoining Semliki forest of western Uganda, however, this has so far not been confirmed in the present day). The species distribution extends across parts of central, northern and eastern Democratic Republic of the Congo. North and east of the Congo River, the species range from Maiko Forest north to the Ituri Forest, then west through the Rubi, Tele and Ebola river basins, extending north towards the Ubangi River. Okapi have a much smaller range to the west and south of the Congo River, extending from the west bank of the Lomami River west to the upper Lomela and Tshuapa basins (Mallon et al., 2015).

The species recorded presence is concentrated in and around protected areas, mainly reflecting survey effort. The remoteness and inaccessibility of much of the species’ habitat make field work logistically difficult, and insecurity in the Democratic Republic of the Congo over the past two decades has further restricted survey activity. As a consequence, extensive parts of potential Okapi range are poorly studied. Furthermore, the species is secretive and its occurrence can easily go undetected, especially at low densities (Mallon et al., 2015).

Since 1980, expansion of human settlement, deforestation and forest degradation have eliminated important portions of the species range, in particular in the southern and eastern Ituri Forest, where the species was at one time abundant (Mallon et al., 2015).

The extent of occurrence (EOO) of the species is 383,190 km², but this includes unsuitable habitat such as degraded forest, swamp forest and urban areas. Excluding these gives an area of 244,405 km² for the suitable or hypothesised range. The area of occupancy (AOO) is 14,112 km², based on a grid of 5.6 x 5.6 km, the size used by most reported surveys and 450 (3.5%) of 12,764 grid squares with confirmed presence. The AOO figure is however considered likely to be a substantial underestimate, surveys have been conducted in only 1,994 (15.6%) out of 12,764 grid squares so far (Mallon et al., 2015).

C) Decline in number of wild individuals

In 2015, an IUCN Red List Assessment categorised *Okapia johnstoni* as Endangered. The justification was that it had been decreasing since at least 1995 and was projected to continue decreasing. The rate of decline was estimated to have exceeded 50% over three generations (24 years). These rates of declines were estimated based on figures from dung transect surveys in the Okapi Wildlife Reserve which suggested the dung density had decreased 43% between 1995-2007, with some reports suggesting that the population was continuing to decrease. The Okapi Wildlife Reserve was reported to be the best protected site and therefore it was inferred that this decrease was at least equalled in other parts of the species’ range.

A second set of transects was conducted in 2005-2007 using an improved spatial design with the same systematic survey conducted in 2010-2011. The 2010-2011 surveys were implemented during the three-month dry season (to avoid the impact of rainfall on dung decay), and an overall comparison between the two surveys showed an increase in dung density, a comparison between transects undertaken at the same time of year (the dry season only), in the same area of the reserve, showed no significant difference in the species’ dung density (Mallon et al., 2015).

Further dung surveys were conducted in 2018 in the Okapi Wildlife Reserve, which showed a statistically significant decrease in dung density by more than two-thirds since the 2011 surveys and a decrease in the area occupied by the species as they were mainly found in the south of the core area and south of it, whereas in 2011, their distribution covered almost the entire core and most of the east (Madidi et al., 2019). Changes in dung density are difficult to interpret, but there was clearly much lower dung density in 2018, compared to 2011, while estimates for 1995 and

2006 varied. Few dung were recorded in 2006 and 2018 (37 and 54 respectively), and therefore the accuracy in these years might be unreliable. It was also noted that the change in dung density between 1995 and 2018 (23 years) was a reduction by 60% (Madidi et al., 2019).

Trade criteria for inclusion in Appendix I

The species is or may be affected by trade

Illegal trade

Okapia johnstoni are hunted for meat and skins. Skins are used for drums, chairs and belts but in most areas, these are opportunistic uses when the animal happens to be caught (Kümpel et al., 2015).

Anecdotal reports suggest that *Okapi* are being killed and trafficked to Uganda and markets in Southeast Asia and the Middle East with their skins sold as decorations and their meat, bones and fat being targeted for purported medicinal value (Vasquez, 2022).

Between 2016 and 2024, WITIS recorded 16 skins to have been seized in the Democratic Republic of the Congo (n=8) and Uganda (n=8). 150 kg of *Okapi* meat were seized in the Democratic Republic of the Congo in 2016 in one incident, and two bottles of *Okapi* oil were seized in Uganda in 2016, in one incident, and *Okapi* oil was seized in the Democratic Republic of the Congo in 2022. The *Okapi* oil was always seized alongside skin.

It has also been reported that a market for *Okapi* oil, for unknown and unproven medical purposes has opened up. It is suspected that the fat or bone marrow is smuggled in tiny vials to South Africa where it is processed and sent back to the Democratic Republic of the Congo in the form of ready-to-use products. There is only evidence of the *Okapi* oil being used in the Democratic Republic of the Congo (Starling, 2024).

The SS states that the NGO Conserv Congo has estimated that since 2019, the products of ten *Okapis* cross the border every month.

Legal trade

Historically, export of *Okapia johnstoni* to zoos generated income for capture stations, with almost all captive *Okapi* coming from the *Okapi* capture station at Epulu (Kümpel et al., 2015).

Analysis of LEMIS indicates that there has been very little trade into the USA in *Okapi* taken from the wild between 2011 and 2019. In 2011, one wild specimen was exported for scientific purposes, originating from the Democratic Republic of the Congo and exported to the United Kingdom of Great Britain and Northern Ireland (UK), and in 2019 four wild specimens were imported for scientific purposes, originating from the Democratic Republic of the Congo. Five specimens from animals bred in captivity were exported to the UK in 2011. A further six live specimens were exported for zoos and one specimen was imported for zoos which came from animals bred in captivity.

Additional Information

Threats

The SS states that *Okapia johnstoni* can coexist with small-scale, low-level human occupation of the forest, but disappear in areas of active settlement or disturbance, and the major threat to this species is habitat loss due to logging and human settlement, including illegal occupation of protected areas.

Approximately one-third of the species known distribution is likely to be at risk by major incursions during the first quarter of this century. Areas at high risk include the southeastern Ituri Forest, the Kisangani area, Rubi-Tele, and the western and eastern limits of the species' range in the Ebola River basin and Virunga-Hoyo region respectively (Mallon et al., 2015).

The most prominent current threat to *Okapi* is the presence of illegal armed groups in and around key protected areas. These groups prevent effective conservation action, even surveys and monitoring in most sites, and engage in and facilitate elephant poaching, bushmeat hunting, illegal mining (gold, coltan and diamonds), illegal logging, charcoal production and agricultural encroachment. In a notorious incident in June 2012, armed rebels attacked the RFO HQ and killed seven people and all 14 captive *Okapi* (Mallon et al., 2015).

A recent threat analysis by Tatoutchoup (2025) found that artisanal mining is responsible for 98% of the *Okapi* population reduction since 2009, due to the destruction of habitat.

Conservation, management and legislation

National

Okapia johnstoni is listed on the Law (LOI N° 82-002 DU 28 MAI 1982 Portant réglementation de la chasse) as totally protected meaning the *Okapi* cannot be killed or captured. Furthermore, the Ministerial Decree (No. 020/CAB/MIN/ECN-EF/2006) of 20 May 2006, which approves the list of protected animal species in the

Democratic Republic of the Congo, classifying the species as a “totally protected” species. This means no animals can be exported for commercial purposes.

The species is a national symbol, appearing on the insignia of the Institut Congolais pour la Conservation de la Nature (ICCN) and on Congolese banknotes (Mallon, et al. 2015).

The Okapi Wildlife Reserve (14,000 km²) and Maiko National Park (10,800 km²) support significant populations, but numbers in both sites have declined owing to the threats listed above. Strengthening protection of these two protected areas is the single most important means to ensure long-term survival of the species (Mallon et al., 2015).

International

It is not clear whether the Okapi is protected or managed by any international agreements. The seizures of Okapi in Uganda suggests there could be some form of agreement.

Artificial Propagation/captive breeding

According to the Zoological Information Management System (ZIMS), at total of 185 Okapia johnstoni are currently held in 64 institutions across 16 countries worldwide. There is one population management programme in place for Okapia johnstoni, managed through the World Association of Zoos and Aquariums (WAZA) (ZIMS, 2025).

Potential benefit(s) of listing for trade regulation

Listing in Appendix I may enhance the establishment of cross-border enforcement (Democratic Republic of the Congo – Uganda), seizure monitoring systems and conducting awareness campaigns.

References

- Hart, J.A. (2013) *Okapia johnstoni*. In: J.S. Kingdon and M. Hoffmann (eds), *The Mammals of Africa. Volume VI: Pigs, Hippopotamuses, Chevrotain, Giraffes, Deer and Bovids*, pp. 110-115. Bloomsbury Publishing, London.
- Kümpel, N.F., Quinn, A., Queslin, E., Grange, S., Mallon, D. and Mapilanga, J.J. (2015) Okapi (*Okapia johnstoni*): conservation strategy and status review. Gland, Switzerland: IUCN and Institut Congolais pour la Conservation de la Nature, 58.
- Madidi, J., Maisels, F., Kahindo, F., Nsafusa, B., Angemito, T., Manala, M., Ngohe, O. and Liengola, I. (2019) Inventaires des Grands Mammifères et de l'Impact Humaine, Réserve de Faune à Okapis, 2018. Rapport technique No : 01/BION/RFO/2019 p. 40. WCS DRC, Kinshasa, DRC.
- Mallon, D., Kümpel, N., Quinn, A., Shurter, S., Lukas, J., Hart, J.A., Mapilanga, J., Beyers, R. and Maisels, F. (2015) *Okapia johnstoni*. The IUCN Red List of Threatened Species 2015: e.T15188A51140517. <https://dx.doi.org/10.2305/IUCN.UK.2015-4.RLTS.T15188A51140517.en>. Accessed on 23 July 2025.
- Starling, M. (2024) We know how many okapi live in zoos. In the wild? It's complicated. Mongabay. <https://news.mongabay.com/2024/09/we-know-how-many-okapi-live-in-zoos-in-the-wild-its-complicated/>. Accessed on 1 August 2025.
- Tatoutchoup, D. (2025) Okapi survival threats: a population reconstruction and threat analysis. *African Journal of Ecology* 63:e70032
- Vasquez, D. (2022) A united front against okapi trafficking. Wildlife Conservation Network. <https://wildnet.org/a-united-front-against-okapi-traffic/>. Accessed on 1 August 2025.
- Species360 Zoological Information Management System (ZIMS). (2025). *Okapia johnstoni*. zims.Species360.org. Accessed on 27 August 2025.

Inclusion of Striped Hyena *Hyaena hyaena* in Appendix I

Proponents: Israel and Tajikistan

Summary: The Striped Hyena *Hyaena hyaena* is one of four extant species within the family Hyenidae. The species has an extensive, yet patchy distribution, extending from East and North Africa, through the Middle East and Arabian Peninsula, the Caucasus and Central Asia to the Indian subcontinent. It is found in 37 range countries, and potentially present in a further ten.

The species was globally assessed as Near Threatened in 2014 on the IUCN Red List. Estimates of the species' population size are limited. In 1998, the global population estimate was between 5,000 and 9,999 mature individuals with a decreasing population trend. A current preliminary estimate based on all available empirical data suggests that the global population is greater than 15,000 individuals. Population trend data across the species' range are also limited; however, observed declines reported in a few range States have been attributed to human-induced mortality, particularly through persecution such as poisoning, collisions with vehicles, reduced availability of natural and domestic carrion sources, and shifts in livestock management practices. The species has been extirpated from parts of its range, including in Bangladesh, and there are large areas of possibly extant or unknown range due to a paucity of data.

Although there appears to be limited legal trade in the species sourced from the wild (38 specimens mostly as hunting trophies, with a few skins and live animals reported over the past ten years whilst the species has been listed in Appendix III), there are also reports of illegal trade involving opportunistic transactions in parts and live animals. However, international trade does not currently appear to be a significant driver of the species' population decrease.

Analysis: *Hyaena hyaena* does not appear to have a small population, nor a restricted area of distribution. While the species' population may be decreasing, it is not possible to determine the extent of the reduction, and it is **unlikely that it would meet the biological criteria for an Appendix I** listing in accordance with Annex 1 paragraph C. Given limited evidence of significant international trade, **the species does not appear to meet the criteria for inclusion in Appendix I** in Res. Conf. 9.24 (Rev. CoP17).

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

Hyaena hyaena (Linnaeus, 1758).

Range

Extant: Afghanistan; Algeria; Benin; Burkina Faso; Cameroon; Chad; Djibouti; Egypt; Eritrea; Ethiopia; India; Islamic Republic of Iran; Iraq; Israel; Jordan; Kenya; Lebanon; Libya; Mali; Mauritania; Morocco; Nepal; Niger; Nigeria; Oman; Pakistan; Saudi Arabia; Senegal; Somalia; the Sudan; Syrian Arab Republic; Tajikistan; Turkmenistan; United Republic of Tanzania; Tunisia; Türkiye; Western Sahara; Yemen (Dloniak in litt., 2025; Jacobson in litt., 2025).

Presence uncertain: Armenia; Azerbaijan; Central African Republic; Georgia; Ghana; Guinea; South Sudan; Uganda; Uzbekistan (Dloniak in litt., 2025; Jacobson in litt., 2025).

IUCN Global Category

Near Threatened (assessed 2014, ver 3.1)

Biological criteria for inclusion in Appendix I

A) Small wild population

In 1998, the global population of the *Hyaena hyaena* was estimated to be between 5,000 and 9,999 mature individuals, with the population trend decreasing. Population estimates vary across regions, with some areas having critically low numbers. However, population estimates are challenging due to the species' elusive nature,

nocturnal habits, solitary behaviour, low densities, infrequent sightings and the challenges of conducting surveys, and there are limited density estimates across the species range, therefore it has not been possible to model a global population size with robust statistical methods (Dloniak in litt., 2025; Jacobson in litt., 2025). A current preliminary estimate based on all available empirical data suggests that the global population is greater than 15,000 individuals (Dloniak in litt., 2025; Jacobson in litt., 2025). The population estimated in 1998 representing an underestimate (Dloniak in litt., 2025; Jacobson in litt., 2025).

There is no historic baseline distribution or range map (i.e., pre-industrial) or population estimate for *Hyaena hyaena* (Jacobson in litt., 2025).

B) Restricted area of distribution

The species has a very large, but patchy distribution, extending throughout Africa including much of east and northeast Africa south to central United Republic of Tanzania, and north including the Sahel, as well as throughout the Middle East and Arabian Peninsula, Turkey, Central Asia, and the Indian subcontinent.

The range map from the 2008 Red List assessment indicates a distribution of >23,000,000 km² (originally reported in Hoefer and Mills (1998)). The Hyena Distribution Mapping Project's current range map suggests ~15,000,000 km² of extant range or about 54% of prior range (or ~65% of the IUCN range) (Jacobson in litt., 2025). However, the 1998 range map was based on a small number of data points.

The species occupies a broad range of habitats across arid and semi-arid habitats. In some parts of its range, the species occurs near human settlements, where it may scavenge for food, demonstrating a degree of tolerance to anthropogenic environments.

The Hyena Distribution Mapping Project collects records of *Hyaena hyaena* occurrence covering more than 40 range countries. The project has no recent (post-2010) reliable data from five range countries (Azerbaijan, Ghana, Nigeria, South Sudan and Uganda) but the species may still be present. The project has no data from five range countries (Bangladesh, Kuwait, Qatar, Russian Federation and the United Arab Emirates) and the species is likely extirpated. This species' presence in Togo and the Côte d'Ivoire remains unconfirmed. The species may be extirpated from Armenia and Georgia as well (Jacobson in litt., 2025).

C) Decline in number of wild individuals

When last assessed in 2014, the population trend was reported to be decreasing, with a continuing decrease of mature individuals. Hoefer and Mills (1998) found that the species was already extinct in many localities and that populations were generally declining throughout its range.

There is limited information on the extent of these decreases, for example in Nepal (Bhandari et al., 2021), with some countries reporting stable or increasing population sizes, such as Israel (Hadad et al., 2023).

Trade criteria for inclusion in Appendix I

The species is or may be affected by trade

Legal trade

The SS states that in many countries, commercial trade in the species is prohibited due to their low numbers, with only limited uses permitted, primarily for scientific and education purposes. Although it was reported that permits are required in some countries to export *Hyaena hyaena*, enforcement of these regulations was reported to be weak. In countries where populations are healthy, sport hunting is possible, such as in the United Republic of Tanzania. In Morocco, the species is commercially hunted for use in traditional medicine, with various parts being used (especially the brain) and can fetch a high price.

Tunisia listed the species in Appendix III in 1976 but removed it in 1977 and Pakistan listed it again in Appendix III in 2014. Since then, reported international trade in the species appears to be limited. Between 2014 and 2023, a total of 38 *Hyaena hyaena* wild-sourced whole organism equivalent individuals were directly exported for commercial and hunting purposes, comprising 29 trophies, four live and four skins, as reported by exporters (Figure 1).

The United Republic of Tanzania was the primary exporter of trophies (n= 25), followed by South Africa (n=3) and Lebanon (n=1). The USA was the primary importer of trophies (n=11), followed by the Russian Federation (n=5) and Germany (n=4). Sudan exported all four live specimens to the Russian Federation in 2014, and Cameroon exported the four skins to France in 2021.

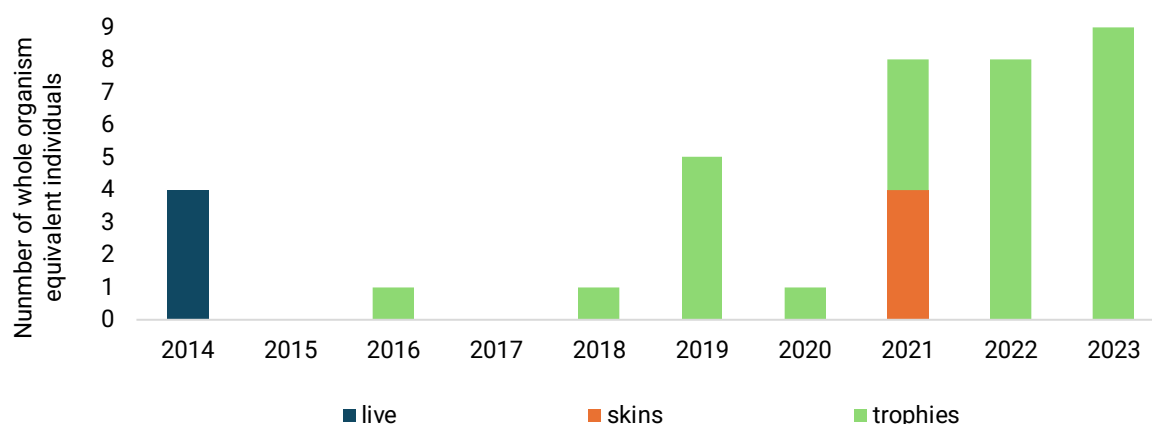


Figure 1: Direct trade in wild-sourced whole organism equivalent *Hyaena hyaena* individuals for commercial and hunting purposes, between 2014 and 2023, as reported by exporters. Source: CITES Trade Database, 2025.

A further 24 live captive-bred (23) and captive-born (1) Striped Hyaena were directly exported during this period, for zoological (18) and commercial (6) purposes. This trade was predominantly exported by Germany (5), Bahrain (4) and the Syrian Arab Republic (3), Cuba (3) and Spain (3), and mainly imported by the Philippines (4), Indonesia (4), the Dominican Republic (3) and Ukraine (3).

Illegal trade

The SS states that there is illegal trade in skins and body parts for use in traditional medicine, *but this appears to be for local use rather than international trade* (AbiSaid and Dloniak, 2015).

The SS indicates there is some evidence of Striped Hyaena cubs being traded or requested as pets, *but this appears to be in only a few individuals over a long period of time*. The trade is often clandestine and difficult to monitor, and therefore its impact on wild populations is difficult to determine.

Additional Information

Threats

The SS notes that the primary threats to the species include habitat loss, human-wildlife conflict, direct persecution (especially poisoning) and illegal trade. Decreasing natural and domestic sources of carrion due to decreasing populations of other carnivores and their prey, and changes in livestock practices have also been implicated. They are particularly vulnerable to accidental or targeted poisoning due to feeding on carrion.

AbiSaid and Dloniak (2015) report that there is also illegal trade in skins, and body parts for use in traditional medicine, and they are often kept in cages for display purposes. The species is commercially hunted in Morocco for use in traditional medicine, with various parts being used (especially the brain), and these can fetch very high prices.

The Hyena Distribution Mapping Project is not focused on international trade or illegal wildlife trade, but it does have a general sense of the published and grey literature regarding hyaenas, including the relative importance of various threats to the species discussed in the publications, and indicates that international illegal wildlife trade is not frequently mentioned or ranked as highly threatening (Jacobson in litt., 2025).

The project does collect occurrence data and any data that included a mortality was included. Roadkill is the most common type of mortality in the data set (~50%) with persecution and killing making up ~25%. It was noted that wildlife trafficking records are not likely to show up in this type of data and in many cases, if an animal was captured by people (and then either killed or rescued & released), the project did not track if the animal had been trafficked or moved to that location. In summary, it does not seem that illegal wildlife trade poses a major threat to the species (Jacobson in litt., 2025).

Conservation, management and legislation

National

Hyaena hyaena is protected under national laws in Tajikistan and many range countries although these laws vary in their effectiveness and enforcement, with some countries having stronger protections than others. National laws protecting the Striped Hyaena include hunting bans, habitat protection measures, and penalties for illegal trade. However, the effectiveness of these laws depends on the capacity and commitment of wildlife protection agencies to enforce them.

International

Hyaena hyaena was listed in Appendix III of CITES by Pakistan in 2014 (AbiSaid and Dloniak, 2015).

Artificial Propagation/captive breeding

According to the Zoological Information Management System (ZIMS), at total of 199 *Hyaena hyaena* are currently held in 66 institutions across 33 countries worldwide. There is one population management programme in place for *Hyaena hyaena*, managed through the European Association of Zoos & Aquaria (EAZA) (ZIMS, 2025).

References

- AbiSaid, M. and Dloniak, S.M.D. (2015) *Hyaena hyaena*. *The IUCN Red List of Threatened Species* 2015: e.T10274A45195080. <https://dx.doi.org/10.2305/IUCN.UK.2015-2.RLTS.T10274A45195080.en>. Accessed on 01 August 25.
- Dloniak, S. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Hofer, H. and Mills, M.G.L. (1998) Population size, threats and conservation status of hyaenas. In: Mills, M.G.L. and Hofer, H. (eds), *Hyaenas. Status Survey and Conservation Action Plan*, pp. 64-79. IUCN/SSC Hyaena Specialist Group. IUCN, Gland, Switzerland and Cambridge, UK.
- Jacobson, A. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Species360 Zoological Information Management System (ZIMS) (2025) *Hyaena hyaena*. zims.Species360.org Accessed on 27 August 2025.

Transfer of Guadalupe Fur Seal *Arctocephalus townsendi* from Appendix I to Appendix II

Proponents: Mexico and the United States of America

Summary: The Guadalupe Fur Seal *Arctocephalus townsendi* is a pinniped species native to the North Pacific. Historically found across the Channel Islands off the southwestern coast of the USA, and Guadalupe Island, the San Benito Archipelago, Cedros Island, and the Revillagigedo Archipelago off the western coast of Mexico. Once thought extinct due to intense hunting by the fur trade during the 18th and 19th centuries, the species was rediscovered on Guadalupe Island in 1954. Since then, strong conservation measures implemented by Mexico and the USA have enabled a notable population recovery.

From a remnant population of an estimated 200–500 individuals in the 1950s, *A. townsendi* has rebounded significantly. The population increased from 34,000–44,000 individuals in 2013, to around 74,000 individuals (~64,000 on Guadalupe Island and ~10,000 on San Benito Island) in 2022–2024, with continuing annual growth and a potential for recolonisation of historical sites. Nevertheless, the population remains below historical levels of up to 200,000 individuals.

The primary and only well-established breeding colony is located on Guadalupe Island, which hosts over 99% of all births. Recolonisation of former breeding sites remains slow and limited by ecological and demographic constraints. Reliance on a single breeding site increases the species' vulnerability to localised threats such as disease outbreaks, pollution, marine debris entanglement, invasive species, extreme weather events, and environmental variability (e.g. El Niño). Despite these vulnerabilities, the Guadalupe Fur Seal was globally assessed as Least Concern on the IUCN Red List in 2014. The species' distribution range is increasing, and sightings have been recorded as far north as Alaska in recent years, likely due to shifts in ocean temperatures.

Arctocephalus townsendi was included in CITES Appendix I due to its historic overexploitation and critical conservation status. International movement in this species has been extremely limited and primarily restricted to scientific specimens; only 40 transactions have been reported by direct exporters since 1975, with no recent evidence of commercial trade or any indications that trade currently poses a threat to the species, although this is unsurprising since the species is listed in Appendix I. Similarly, there is little evidence of illegal international trade. Although the USA suggests that some illegal trade may be occurring, there is no publicly available evidence to confirm this.

National legislation in the two range States provides strict protection. In Mexico, hunting and trade of *A. townsendi* have been banned since 1933, and the species is listed as a national conservation priority and a national Conservation Action Programme is in place. In the USA, the species is listed under the Endangered Species Act and the Marine Mammal Protection Act, and additional protection is granted by the Fur Seal Trade Act, which prohibits the capture, transport, importation, or possession of fur seals. These strong domestic frameworks are sufficient to ensure continued protection and monitoring. These measures would remain in place in the event of the transfer to Appendix II and therefore it is not anticipated that trade, legal or illegal would be stimulated by this.

Analysis: The population of *Arctocephalus townsendi* has recovered from a few hundred individuals in the 1950s to a population of approximately 74,000 in 2022–2024. It does not have a restricted area of distribution and its range may be expanding due to climatic changes, though it occurs at very few breeding locations. International trade has been negligible and limited primarily to scientific specimens, and the transfer of the species to Appendix II is not expected to increase demand nor create enforcement challenges, particularly as other species in the genus are listed in Appendix II. Robust national legislation is already in place that prohibits the harvest and trade of the species, ensuring that domestic controls are sufficient to enforce Appendix II provisions. The **species no longer meets the biological or trade criteria for inclusion in Appendix I and precautionary safeguards are in place**. This proposal was supported by the Animals Committee at its 33rd meeting.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

Arctocephalus townsendi was first included in Appendix II in 1975. In 1977, the entire genus was listed in Appendix II, and in 1979, the species was transferred to Appendix I, where it remains to this day.

Taxonomy

Arctocephalus townsendi (Merriam, 1897).

Synonym: *Arctocephalus philippii townsendi*

CITES Standard Reference: Wilson, D.E. and Reeder, D.M. (Eds.). 2005. *Mammal species of the world, a taxonomic and geographic reference*. 3rd Edition, The Johns Hopkins University Press, Baltimore, Maryland. 2, 142pp.

Range

Mexico and the USA

IUCN Global Category

Least Concern (assessed 2014, ver. 3.1)

Biological criteria for inclusion in Appendix I

A) Small wild population

The population of *A. townsendi* was estimated at 34,000-44,000 individuals in 2013. The numbers continue to increase, and the proponents indicate the species has colonized a new breeding area. *However, the species still occurs at very few sites: the main breeding colony on Guadalupe Island, a transitional or mixed colony at the San Benito Archipelago (Pacific Ocean), and three haul-out sites in the Gulf of California (García-Aguilar et al., 2018; Hernández-Camacho and Trites 2018; Elorriaga-Verplancken et al., 2021; Gutiérrez-Osuna et al., 2022; Gálvez et al., 2022). The most recent estimate for the Guadalupe Island colony is approximately 57,199 to 72,631 individuals (Juárez-Ruiz et al., 2022). For the San Benito Island colony, the estimate is around 10,000 individuals (Serrano-Rodríguez, 2023; Vallarino-Orozco, 2024). In total, this amounts to about 74,000 individuals, which is still significantly below the estimated 200,000 individuals that existed prior to commercial exploitation (Hubbs, 1979).*

B) Restricted area of distribution

The Guadalupe Fur Seal is distributed in temperate North Pacific waters along the west coast of North America from central Mexico/Gulf of California to southern British Columbia, Canada, with rare sightings in Alaska.

A. townsendi inhabits the interface between marine and terrestrial ecosystems, as it feeds on a high diversity of prey in coastal and oceanic waters but rests, breeds and moults in terrestrial areas, mainly on Guadalupe Island. Guadalupe Island is part of the Guadalupe Island Biosphere Reserve, whose marine and terrestrial zones represent a total area of 4,770 km². The reserve has a core zone and a buffer zone, which represent 5% and 95% of the total area, respectively. The core zone consists of Guadalupe Island, which is volcanic and has an area of 254 km²

The species breeds in Mexico on Guadalupe Island, where 99% of births take place and in recent years, few births of Guadalupe Fur Seals have occurred in San Benito Archipelago and on San Miguel Island in Mexico, comprising less than 1% of the total annual births. The San Benito Archipelago has become an important recolonization site. *However, this has not yet developed into a stable breeding colony despite more than 30 years of continuous presence. This site is predominantly composed of juvenile males (approximately 90%), with only about 60 births reported annually by 2018 (Hernández-Camacho and Trites, 2018).*

Historical evidence suggests that the main breeding colonies of the Guadalupe Fur Seal once included the Channel Islands, Guadalupe Island, the San Benito Archipelago, Cedros Island, and the Revillagigedo Archipelago (Hubbs, 1956; Le Boeuf and Bonnell, 1980; Gutiérrez-Osuna et al., 2022). The fact that only one breeding colony is currently established makes the species highly vulnerable to localized threats such as disease outbreaks, oil spills, extreme weather events, or natural disasters (Elorriaga-Verplancken et al., 2021; Hernandez, in litt., 2025).

The prolonged failure to establish a breeding colony at the San Benito Archipelago, despite persistent recolonization, indicates that the species' recovery is still partial and that limiting ecological or demographic factors remain. Sexual coercion behaviours toward pups have been documented at this site, potentially affecting the viability of emerging reproductive aggregations (Vallarino-Orozco et al., 2025). Additionally, factors such as climate change and environmental variability may influence foraging grounds and prey availability (Elorriaga-Verplancken et al., 2016;

Amador-Capitanachi et al., 2020) and ultimately impact reproductive success due to negative effects on the body mass of pups (Gálvez et al., 2020) and their survival (Gálvez et al., 2023; Hernandez, in litt., 2025).

It appears that *Arctocephalus townsendi* has been present in Farallón de San Ignacio Island, Sinaloa, since 2014 and in January 2020, 492 individuals were counted, increasing to 771 in November, most of them juveniles. Also in 2020, an incipient non-breeding colony was established on Islote de las Ánimas (Bahía de la Paz, Baja California Sur), where various counts were made throughout the year and a maximum of 119 juvenile individuals were observed in August. Evidence shows that since 2009, the species has also been observed in San Pedro Mártir Island, Sonora.

Recent sightings of the species at sites further north suggest that its current range is expanding, although these sightings are still within the limits of its historical range, *except for incipient colonies recently formed in the Gulf of California. Guadalupe Fur Seals have recently been observed as far north as Alaska, which is most likely associated with the occurrence of warmer waters (Pace et al., 2022). The species population continues to increase and rebound; therefore, it is likely that it will extend its breeding range and foraging grounds. Continued monitoring of A. townsendi is essential to obtain a better understanding of population dynamics and to identify potential local and regional threats (García-Aguilar et al., 2018). Furthermore, although the species is recovering, prolonged warming conditions or oceanographic abnormalities can pose a threat to its populations (Gálvez et al., 2020).*

C) Decline in number of wild individuals

The historical population size is unknown but has been estimated at up to 200,000 individuals prior to human influence. In the 18th and 19th centuries, the Guadalupe Fur Seal was intensively hunted by the fur industry, to the extent that it was believed to be extinct by the 1920s. In 1954, a small colony was found in Mexico in *Guadalupe Island* and protection of the species was put in place.

Since the 1950s, the species has experienced a recovery from an estimated population of 200-500. Growth began at rates of 1 to 2% per year and increased at a faster rate by the 1970s. *From 1984 to 2013, the Guadalupe Fur Seal population increased at an average annual growth rate of 5.9% (range 4.1–7.7%), and by 2013 the abundance was estimated between 34,000 and 44,000 individuals (García-Aguilar et al., 2018) with an estimated annual growth rate of 10-11%.*

The most recent estimate for the Guadalupe Fur Seal, as previously indicated, is around 57,199-72,631 individuals at an average annual growth rate of 8.4% (range 8.1–8.8%) (Juárez-Ruiz et al., 2022). The carrying capacity of the island has not yet been reached and the population continues to increase and is slowly spreading, with minor and temporary declines associated with El Niño events, posing a negative effect, especially in pups (Gálvez et al., 2020, 2023).

Trade criteria for inclusion in Appendix I

The species is or may be affected by trade

According to the CITES Trade Database, 25 transactions were recorded for *A. townsendi* between 1975 and 2024, most of them for scientific purposes (S; 52%); commercial trade in the species (30% of the transactions) occurred before 1993 or involved pre-Convention specimens (i.e., bone carvings).

The full CITES Trade Database includes at least 40 transactions reported by direct exporters (i.e., not re-exporters) since the first records dating back to 1983. In the last 10 years (i.e., 2014–2023), only 10 transactions have taken place, none of which were for commercial purposes, except for two pre-Convention specimens traded between non-range States.

Almost all records in the CITES trade data have been for scientific specimens. Very small quantities have been recorded as for commercial purposes (all of re-exports some of which of pre-convention items): number of carvings and garments (possibly three) and specimens and derivatives (around 60 kg) in the period that the species has been in the Appendices.

Specifically, in the last 10 years Mexico has issued only two CITES permits for the export of biological samples (source code W) for scientific purposes to the USA.

Regarding illegal trade, according to Mexico's Law Enforcement Authority (PROFEPA), in the last 15 years there have been no records of seizures at the national level or attempts to export specimens at ports, airports or borders. The USA indicates that some illegal trade in Guadalupe Fur Seals or specimens may occur, but information on this activity is not available.

*EU-TWIX and WITIS register no records of illegal trade of *Arctocephalus townsendi* between 2014 to 2023 and 2010 to 2025, respectively. As per LEMIS, for the period between 2008 to 2021, it includes six transactions from and to the USA; five involving imports for scientific purposes, which were cleared, and an additional one for personal purposes,*

which was refused. These transactions took place between 2010 and 2012, and all involved specimens, except for the refused one, which was a skull.

There is no national or international market for Guadalupe Fur Seals that might threaten its wild populations. Most records of international trade involve specimens for scientific purposes. Given the prohibitions on the species use by both Mexico and the USA, no commercial trade in the species is foreseen in the future.

Precautionary Measures

The species is not in demand in international trade, nor is its transfer to Appendix II likely to stimulate trade in, or cause enforcement problems for, any other species included in Appendix I: taking into account that trade is scarce, it involves mainly scientific specimens (with a high probability of proper identification at the species level) and that the other species of the genus are listed in Appendix II, no increase in demand or enforcement problems are expected.

In 1933, Mexico established a permanent ban on hunting species of fur seals and elephant seals in all waters under federal jurisdiction of the Mexican Republic. In the USA, the species is protected under the Endangered Species Act (ESA) and the Marine Mammal Protection Act (MMPA), as well as certain state laws (in California, Oregon, and Washington). Additional protection is granted by the Fur Seal Trade Act, which prohibits the capture, transport, importation, or possession of fur seals and sea otters, and the Lacey Act, which prohibits trade in wildlife, fish and plants that have been illegally taken, possessed, transported or sold. These measures would remain in place in the event of the transfer to Appendix II and, therefore, it is not anticipated that trade, legal or illegal would be stimulated by this.

The Precautionary measures in Annex 4 of Res. Conf. 9.24 (Rev. CoP17) appear to be met.

Export quota or other special measure

There are no current quotas in place for this species.

Additional Information

Threats

The main threats faced by the species are related to the effects of human activity in the areas surrounding Guadalupe Island and San Benito Archipelago:

- Pollution of the marine environment is a constant threat; an example is the ingestion of plastic pollutants. The vulnerability of *A. townsendi* to fuel/oil spills due to its fur type should also be noted.
- Signs of net entanglement were observed in 12 adult males, 3 subadults and 2 adult females during the 2013-2016 breeding seasons at Punta Sur, Guadalupe Island.
- The presence of alien invasive species on Guadalupe Island has had an impact on the health of the colony, as *Leptospira*, a bacterial spirochete that causes leptospirosis, has been detected. *A. townsendi* is also vulnerable to other pathogens such as parasites and bacteria (i.e., *Klebsiella* sp., *Pseudomonas* sp., *Proteus* sp. and *Enterobacteria* sp.).
- The increase in sea surface temperature during meteorological/oceanographic events such as El Niño and its effect on prey availability.

The fact that only one breeding colony is currently well established makes the species highly vulnerable to localized threats such as disease outbreaks, oil spills, factors such as climate change, natural disasters and environmental variability (e.g., El Niño-Southern Oscillation events), which may influence prey availability and ultimately impact reproductive success (Elorriaga-Verplancken et al., 2016).

Conservation, management and legislation

The General Wildlife Act also specifies that no marine mammal specimen may be subject to extractive exploitation for subsistence or commercial purposes., and the import, export and re-export of marine mammal specimens is prohibited, except for scientific research. The species is also included in Mexico's "List of priority species and populations for conservation", which implies promoting the development of projects for its conservation and recovery, along with that of the ecosystems, habitats and species with which it is associated. Thus, Mexico has a Conservation Action Programme for the Guadalupe Fur Seal, developed with input from government, academia and civil society. It outlines population status, threats, and conservation measures.

Mexico and the USA also have marine mammal stranding response protocols, while Canada's regulations prohibit disturbance and include management provisions.

Artificial Propagation/captive breeding

No captive breeding facilities have been reported in the three range States, *nor have any been found in the literature.*

Implementation challenges (including similar species)

All other species in the genus *Arctocephalus* are currently listed in Appendix II, supporting consistency in enforcement and identification practices. Enforcement challenges are unlikely.

Other comments

Between the 29th and 33rd meetings of the Animals Committee (AC29 and AC33, respectively), Mexico and the USA conducted a periodic review of *Arctocephalus townsendi*, in accordance with Resolution Conf. 14.8 (Rev. CoP19) *Periodic Review of species included in Appendices I and II*. The results of the review were presented to AC33 in Geneva in 2024, which supported the submission of a proposal to transfer the Guadalupe Fur Seal from Appendix I to Appendix II.

References

- Amador-Capitanachi, M.J., Moreno-Sánchez, X.G., Ventura-Domínguez, P.D., Juárez-Ruiz, A., González-Rodríguez, E., Gálvez, C., Norris, T. and Elorriaga-Verplancken, F. R. (2020) Ecological implications of unprecedented warm water anomalies on interannual prey preferences and foraging areas of Guadalupe fur seals. *Marine Mammal Science* 36(4):1254-1270. DOI: 10.1111/mms.12718
- Elorriaga-Verplancken, F.R., Norris, T., Acevedo-Whitehouse, K., Gálvez, C., Amador-Capitanachi, M.J. and Landa-Garza, M. (2021) Guadalupe fur seal (*Arctocephalus philippii townsendi*, Brunner 2004). In: Heckel, G. and Schramm, Y. (eds.), *Ecology and Conservation of Pinnipeds in Latin America*. Springer, Cham. https://doi.org/10.1007/978-3-030-63177-2_5
- Elorriaga-Verplancken, F.R., Sierra-Rodríguez, G.E., Rosales-Nanduca, H., Acevedo-Whitehouse, K and Sandoval-Sierra, J. (2016) Impact of the 2015 El Niño-Southern Oscillation on the Abundance and Foraging Habits of Guadalupe Fur Seals and California Sea Lions from the San Benito Archipelago, Mexico. *PLoS ONE* 11(5):e0155034. <https://doi.org/10.1371/journal.pone.0155034>
- Gálvez, C., Raverty, S., Galván-Magaña F., Field, C. L. and Elorriaga-Verplancken F.R. (2023) Mortality in an off-shore predator critical habitat during the Northeast Pacific marine heatwave. *Frontiers in Marine Science*. 10:1202533. DOI: 10.3389/fmars.2023.1202533
- Gálvez, C., Pardo, M. and Elorriaga-Verplancken, F.R. (2020) Impacts of extreme ocean warming on the early development of a marine top predator: The Guadalupe fur seal. *Progress in Oceanography* 180:102220. DOI: 10.1016/j.pocean.2019.102220
- García-Aguilar, M.C., Elorriaga-Verplancken, F.R., Rosales-Nanduca, H. and Schramm, Y. (2018) Population status of the Guadalupe Fur Seal (*Arctocephalus townsendi*). *Journal of Mammalogy* 99(6):1522–1528. <https://doi.org/10.1093/jmammal/gyy132>.
- Gutiérrez-Osuna, M.C., Díaz-Gaxiola, J.M., Trites, A.W. and Hernández-Camacho, C.J. (2022) New colony of Guadalupe Fur Seals (*Arctocephalus townsendi*) discovered on El Farallón de San Ignacio Island, Gulf of California. *Marine Mammal Science*. <https://doi.org/10.1111/mms.12844> I.1
- Hernández-Camacho, C.J. and Trites, A. (2018) Population viability analysis of Guadalupe Fur Seals (*Arctocephalus townsendi*). *Endangered Species Research* 37:255-267.
- Hernandez, C. (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK. IUCN SSC Pinniped Specialist Group
- Hubbs, C.L. (1979) Guadalupe Fur Seal. FAO Fisheries No. 5, Rome, Italy.
- Juárez-Ruiz, A., Pardo, M.A., Hernández-Montoya, J.C., Elorriaga-Verplancken, F.R., Milanés-Salinas, M.A., Norris, T., Beier, E. and Heckel, G. (2022) Guadalupe Fur Seal pup production predicted from annual variations of sea surface temperature in the southern California Current Ecosystem. *ICES Journal of Marine Science* 79(5):1637–1648. <https://doi.org/10.1093/icesjms/fsac097>
- Pace, C.N., Webber, M.A., Boege Tobin, D.D., Pemberton, S., Belovarac, J. and y Goertz, C.E.C. (2022) Observations of a Guadalupe Fur Seal (*Arctocephalus townsendi*) in Cook Inlet, Alaska. *Aquatic Mammals* 48(6):603–610.
- Serrano Rodríguez, Y.T. (2023). Caracterización de la colonia de lobos finos de Guadalupe (*Arctocephalus townsendi*) en el Archipiélago San Benito, México, durante el periodo reproductivo 2022. Tesis, Universidad Jorge Tadeo Lozano, Colombia.
- Vallarino-Orozco, G., Burgoin-Diaz, I.M., Elorriaga-Verplancken, F.R., Norris, T., Trites, A.W, and Hernández-Camacho, C.J. (2025) Abduction and sexual coercion of Guadalupe Fur Seal pups. *Marine Mammal Science* 41(2). <https://doi.org/10.1111/mms.13189>.
- Vallarino-Orozco, G. (2024) ¿Es la colonia de lobo fino de Guadalupe (*Arctocephalus townsendi*) del Archipiélago San Benito, México, un sitio de reproducción? Tesis maestría. Centro Interdisciplinario de Ciencias Marinas. Instituto Politécnico Nacional.

Deletion of Caribbean Monk Seal *Monachus tropicalis* from Appendix I

Proponents: Mexico and the United States of America

Summary: The genus *Monachus*, at that time regarded as comprising three species (the Mediterranean Monk Seal *M. monachus*, the Hawaiian Monk Seal *M. schauinslandi* and the Caribbean Monk Seal *M. tropicalis*) was included in Appendix I in 1975.

The Caribbean Monk Seal, first described in 1492 during the voyages of Christopher Columbus to the Americas, was found in the Caribbean Sea and part of the Gulf of Mexico, from the south of Florida to Colombia and Venezuela. Demand, primarily for skins and oil from blubber resulted in overharvesting. The species was last reliably reported in 1952 and was classified as Extinct by IUCN in 1986.

The Caribbean Monk Seal was reviewed as part of the Periodic Review process, and the Animals Committee agreed with the recommendation to delete it as an extinct species from Appendix I. Were this proposal to be accepted, the Animals Committee and Nomenclature Specialist have recommended including a note in the Appendices that reads "*Monachus* spp. (except '*Monachus tropicalis*' which is extinct and was removed from the Appendices on [insert date])".

However, the Animals Committee, at its 33rd meeting, agreed to submit for adoption by CoP20 a proposed nomenclature update concerning Monk Seals (see document CoP20 Doc.110). If adopted this will split *Monachus* into two genera; *Neomonachus*, comprising the new combinations *N. schauinslandi* and *N. tropicalis*, and *Monachus*, retaining *M. monachus* unchanged. These references would come into effect 90 days after CoP20 at the same time as the amendments to the Appendices. Were both the listing proposal and the revised taxonomy to be adopted the amendment to the listing in the appendices could be one of the following options:

- *Monachus* spp. and *Neomonachus* spp. (except '*Neomonachus tropicalis*' which is extinct and was removed from the Appendices on [insert date]), or
- *Monachus monachus* and *Neomonachus* spp. (except '*Neomonachus tropicalis*' which is extinct and was removed from the Appendices on [insert date])

There has been very little trade in the other two extant species of *Monachus* almost all in scientific specimens of the extant species. The threats that affected the species (hunting for blubber and skins) no longer exist for most seal populations.

Analysis: The Caribbean Monk Seal was last seen before the species was included in the Appendices as part of a genus listing for Monk Seals. It was declared Extinct in 1986 on the IUCN Red List.

The Precautionary measures of Annex 4 in Res. Conf. 9.24 (Rev. CoP17) allows for the deletion of extinct species from Appendix I subject to the provisions of paragraph D, although this paragraph refers to species that are regarded as possibly extinct rather than extinct. Regardless, there is no reason to assume that the species would be affected by trade in the extremely unlikely event of its rediscovery. Even though the Caribbean Monk Seal may resemble the extant species of monk seal, amending the appendices to note that the extinct taxon is not in the Appendices, as is proposed, would ensure that there was no potential for one of the extant species to be exported as the Caribbean Monk Seal and therefore **deletion would not cause difficulties in implementing the Convention or complicate the Appendices.**

In the case that the new standard reference for this taxon is adopted, **amendment of the appendices could be one of the following options:**

- *Monachus* spp. and *Neomonachus* spp. (except '*Neomonachus tropicalis*' which is extinct and was removed from the Appendices on [insert date]), or
- *Monachus monachus* and *Neomonachus* spp. (except '*Neomonachus tropicalis*' which is extinct and was removed from the Appendices on [insert date])

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

The genus *Monachus* has been split into *Monachus* and *Neomonachus*, with *Monachus tropicalis* being reclassified as *Neomonachus tropicalis*.

Range

(Extinct) Antigua and Barbuda; Bahamas; Belize; Costa Rica; Cuba; Dominica; Dominican Republic; Guadeloupe; Haiti; Honduras; Jamaica; Mexico; Nicaragua; Puerto Rico; USA. (Lowry, 2015).

IUCN Global Category

Extinct (Assessed 2014, Lowry 2015)

Biological criteria for inclusion in Appendix I

- A) **Small wild population (i) Population or habitat decline; (ii) small sub-populations; (iii) concentrated geographically during one or more life-history phases; (iv) large population fluctuations; (v) high vulnerability**

The population size in the 17th century may have been between 233,000 and 338,000 individuals distributed among 13 colonies. By the end of the 19th century it had become rare and the last photographic record of the species in the wild dates back to 1900. The last verifiable report of a sighting was in 1952. Searches by air in 1973 carried out in areas in the Gulf of Mexico and the Caribbean found no evidence of their occurrence, and it was concluded that even if a few seals were still alive, there would be little likelihood of the species recovering.

Further searches and interviews in 1986 in Mexico concluded that the species was no longer found in the Mexican Caribbean. Despite extensive searches, the Caribbean monk seal has not been verifiably sighted since 1952. In 1986, the IUCN Pinniped Specialist Group classified the species on the IUCN Red List as Extinct.

Affected by Trade.

One record of export of *Monachus tropicalis* in 2009 involved the transfer of 6 pre-Convention specimens for scientific purposes from the USA to Germany.

There has been very little trade in the other two extant species of *Monachus* (83 records of trade between 1978 and 2022, mainly for scientific purposes), both of which are listed in Appendix I.

Most reported exports have been in scientific specimens of the extant species (see Table 1). The threats that affected the species (hunting for blubber and skins) no longer exist for the majority of seal populations.

Table 1: Reported exports (including re-exports) of *Monachus* species between 1975 and 2024 (CITES Trade Database. Downloaded 28 July 25).

Taxon	Purpose	Term	Unit	Source	Total
<i>Monachus monachus</i>	Education	bodies	(blank)	0	2
	Scientific	skin pieces	(blank)	W	4
		skins	(blank)	W	5
		specimens	kg	W	1
		specimens	(blank)	W	232
	(blank)	skulls	(blank)	(blank)	2
<i>Monachus schauinslandi</i>	Personal	specimens	(blank)	W	46
	Scientific	skulls	(blank)	W	1
		specimens	(blank)	W	591
		teeth	(blank)	W	33
	Commercial	specimens	(blank)	W	10
	(blank)	specimens	(blank)	W	57
<i>Monachus spp.</i>	Circus or travelling exhibition	skins	(blank)	0	1
	Commercial	carvings	(blank)	0	1
<i>Monachus tropicalis</i>	Scientific	specimens	(blank)	0	6

Precautionary measures

Annex 4 of Res. Conf. 9.24 (Rev. CoP17), paragraph D: "Species that are regarded as possibly extinct should not be deleted from the Appendices if:

"1. they may be affected by trade in the event of their rediscovery,"

The threats that affected the species (hunting for blubber and skins) no longer exist for the majority of seal populations. Current threats that affect other seal populations mainly include factors such as climate change (impact on potential prey), competition with fisheries, gillnets, etc.

Exhaustive and appropriate surveys have been carried out leading to the IUCN categorisation of the species as Extinct (Lowry, 2015) and there is little likelihood of their rediscovery.

"2. they resemble extant species included in the Appendices"

There are two similar species listed in CITES Appendix I: *Monachus monachus* and *M. schauinslandi*. The three species can be distinguished from one another based on their morphology, mainly by bone features, size and, in some cases, by pelage colouring.

"3. their deletion would cause difficulties implementing the Convention"

Deletion of the species is not expected to imply any problems regarding implementation: ever since the Convention first came into force, there has been only one international movement of *M. tropicalis*, for scientific purposes; on account of the fact that the species became extinct before CITES entered into force (1975), all specimens meet the provisions of Resolution Conf. 13.6 (Rev. CoP18) to be considered pre-Convention, therefore, the exemptions provided in Article VII of the Convention text would be applicable. Very little trade has been recorded for other monk seal species.

Were the species to be deleted from the Appendices the proposed amendment noting the exclusion of the taxon because it is extinct would ensure that the species remaining in the Appendices could not be traded under the name of the extinct taxon.

"4. their removal would complicate the interpretation of the Appendices"

The species was listed in the Appendices in 1975 on inclusion of all species within the genus *Monachus* (at that time, *M. monachus*, *M. tropicalis*, and *M. schauinslandi*). Following the recommendations derived from the 33rd Meeting of the Animals Committee, Mexico consulted with the Nomenclature Specialist and the joint recommendation is to include a note in the Appendices that reads "*Monachus* spp. (except '*Monachus tropicalis*' which is extinct and was removed from the Appendices on [insert date])".

However, as noted in the SS and Document CoP20 Doc 110 Annex 3 a new standard reference is proposed, Scheel et al. (2014), that splits the genus Monachus into Monachus and Neomonachus, and the recommendation to list the new combinations N. schauinslandi and N. tropicalis, retaining the former names of these species as synonyms. If adopted, these references would come into effect 90 days after CoP20 at the same time at the amendments to the Appendices. Were both the listing proposal and the revised taxonomy to be adopted the amendment to the listing in the Appendices could be one of the following options:

- *Monachus spp. and Neomonachus spp. (except 'Neomonachus tropicalis' which is extinct and was removed from the Appendices on [insert date]), or*
- *Monachus monachus and Neomonachus spp. (except 'Neomonachus tropicalis' which is extinct and was removed from the Appendices on [insert date])*

An amendment noting that the Caribbean Monk Seal taxon is extinct would prevent the use of the name to export either of the extant species remaining in Appendix I.

Additional Information

Artificial Propagation/captive breeding

According to a review published in 2004, at least 18 individuals were kept in captivity on eight separate occasions; breeding was unsuccessful and survival was low (living from 1 week to 2 years). There are no further records of captive-breeding programmes for the Caribbean Monk Seal prior to its extinction.

Similar species.

Both extant species of Monk Seal have been listed in Appendix I since 1975. These are:

- Mediterranean Monk Seal (*Monachus monachus*). Turkey and Greece
- Hawaiian Monk Seal (*Monachus schauinslandi*): only found in the Hawaiian archipelago, US.

References

- Lowry, L. (2015) *Neomonachus tropicalis*. *The IUCN Red List of Threatened Species* 2015: e.T13655A45228171. <https://dx.doi.org/10.2305/IUCN.UK.2015-2.RLTS.T13655A45228171.en>. Accessed on 13 July 2025.
- Scheel, D.-M., Slater, G.J., Kolokotronis, S.-O., Potter, C.W., Rotstein, D.S., Tsangaras, K., Greenwood, A.D. and Helegen, K.M. (2014) Biogeography and taxonomy of extinct and endangered monk seals illuminated by ancient DNA and skull morphology. *ZooKeys* 409:1-33. DOI: 10.3897/ZOOKEYS.409.6244.

Amendment of the Annotation of the Population of Southern White Rhinoceros *Ceratotherium simum simum* of Namibia Listed in Appendix II

Proponent: Namibia

Summary: The Southern White Rhinoceros *Ceratotherium simum simum* is one of two subspecies of White Rhinoceros (the other being the Northern White Rhinoceros *Ceratotherium simum cottoni*, likely to be extinct in the wild).

The Rhinocerotidae family was included in Appendix I in 1977. In 1995, the South African population of *Ceratotherium simum simum* was transferred to Appendix II. In 2005, the Eswatini (then Swaziland) population was also transferred to Appendix II, and in 2023, the Namibian population was transferred to Appendix II. The current listing states: *"Only the populations of Eswatini, Namibia and South Africa; all other populations are included in Appendix I. The populations of Eswatini and South Africa are included in Appendix II for the exclusive purpose of allowing international trade in live animals to appropriate and acceptable destinations and hunting trophies. The population of Namibia is included in Appendix II for the exclusive purpose of allowing international trade in live animals for in-situ conservation only, and only within the natural and historical range of Ceratotherium simum in Africa. All other specimens shall be deemed to be specimens of species included in Appendix I and the trade in them shall be regulated accordingly."*

The current proposal is to amend the annotation of the population of *Ceratotherium simum simum* of Namibia listed in Appendix II, *"For the exclusive purpose of allowing international trade in:*

- a) *Live animals for in-situ conservation only;*
- b) *Hunting trophies; and*
- c) *Trade in rhino horn stocks owned by the Government and the Private Landowners originating in the State (excluding seizure rhinoceros horn and rhinoceros horns of unknown origin); subject to the following:*
 - i) *Only stocks registered with the Government;*
 - ii) *Only horns with RHODIS certificates;*
 - iii) *Only to trading partners that have been verified by the Secretariat, in consultation with the Standing Committee, to have sufficient national legislation and domestic trade controls; and*
 - iv) *Not before the Secretariat has verified prospective importing countries and the registered stocks.*

All other specimens shall be deemed to be specimens of species included in Appendix I, and the trade in them shall be regulated accordingly."

At the end of the 19th century the Southern White Rhinoceros was on the brink of extinction with one small population of 20-50 individuals remaining in South Africa. After years of protection and translocations, the subspecies had grown to an estimated 21,316 individuals by the end of 2012. However, following an increase in poaching, estimated numbers declined to 18,064 by the end of 2017 (a ~15% reduction). After further decreases, the population increased from 15,940 in 2021 to 16,799 in 2022, reaching 17,462 in 2023. By the end of 2024 there were estimated to be 15,750 individuals on the continent, the decrease being mostly because of an increase in poaching in South Africa.

In Namibia, from an initial reintroduction of 16 individuals in 1975, the White Rhino population grew by 6.7% annually to reach 1,500 by 2024. Currently, 1,161 (ca 80% of the total) are privately owned across roughly 74 populations, while 339 remain under state ownership in four protected areas.

Between 2022 and 2024 43 animals were legally hunted for trophies, equivalent to about 0.9% of the Namibian population. Between 2022 and 2024, 124 live individuals have been exported by Namibia. Domestic consumptive use of White Rhino and trade in its horn and other products is currently not

permitted in Namibia. Adult males are sometimes dehorned for safety or for management to deter poaching.

The greatest threat to the global population of White Rhino is poaching for the horn, with most losses occurring in South Africa. Since 2013 around 160 have been poached in Namibia; unrecovered horns are presumed to have entered illegal trade. In response to poaching risks, private landowners contend with high security costs to protect their rhino herds. The proponents note that despite current income generated from tourism, trophy hunting and live sales, the income often falls short to cover the costs, with owners reducing or disposing of their rhino herds, sometimes turning to other forms of land use and reducing the area of habitat available for conservation. The proponents also note that there are 12 million hectares of freehold land potentially suitable for White Rhino that remains largely unused as it is not financially viable for landowners to keep them. It is argued that the creation of financial incentives from rhino custodianship could encourage landowners to shift from other land uses to conservation, with a wide range of benefits.

It is not clear whether all privately held stockpiles are currently registered with the Government, nor how the stockpiles are currently secured. Approximately 70% of government-managed horns are genetically profiled and microchipped, with data integrated into a database (RhODIS). In the private sector, 80% of horns have been sampled, although only legally acquired historic horns are included under current permit frameworks. In addition to currently held stockpiles, horn from future dehorning would also be traded. The SS notes using an adaptive “pilot” framework for this but does not provide details.

The proponents set out in the annotation and in the supporting statement how management measures will be put in place to control supply, including permit control, marking of animals/products, customs and border control and how law enforcement is implemented. Information on how private landowners will engage in this process is lacking. There is also limited specific information on how the sale and exports of horn would be organised.

Namibia has also proposed the once-off export of its stockpile of rhino horn in Prop 10 relating to its population of Black Rhino. The supporting statement for Prop 10 includes a table detailing current rhinoceros horn stockpiles held by Namibia (currently almost 4.7 tonnes as of June 2025), but it does not state whether any of this is White Rhino horn.

Analysis: The Namibian population of the Southern White Rhino is already in Appendix II. There are no specific guidelines for assessing proposals to change annotations of this nature, but it seems appropriate to ensure that satisfactory precautionary measures, as detailed in Annex 4 of Res. Conf. 9.24 (Rev. CoP17), would be in place. Namibia has not proposed an export quota within the annotation, but has proposed several special measures to be approved by the Parties as an integrated part of its proposal through an annotation to the listing (Annex 4 A.2.a iii)). The key question is to consider whether the safeguards provided by this annotation, together with management measures and enforcement controls described by Namibia might adequately mitigate risks for populations of *Ceratotherium simum simum* or other rhinoceros species.

The special measures detailed in the proposed annotation address the origin of horn to be exported (registered stockpiles of legally acquired and individually identifiable horn of known origin) and a mechanism for approving any future trading partners. However, it is not stated what stockpile of horn of *Ceratotherium simum simum* exist in Namibia already and what quantity might be expected to accumulate in the future including from dehorning. The supporting statement refers to an initial cap on exports of 200 kg of horn per year, with review of impacts and suspension or revision of quotas apparently being assumed as a matter of government prerogative. There is no indication of how long this phase is envisaged. As this is not an integral part of the proposal contained in the annotation, and as noted by the proponents, this quantity could change, and the trade would be subject only to implementation of Article IV of the Convention.

Despite Namibia's successful conservation efforts and detailed provision of information on enforcement controls and compliance, the Parties may seek more explicit reference to such management measures within an annotation.

The meaning and criteria for “sufficient national legislation and domestic trade controls” in prospective importing countries and the standards and process for verifying importing countries are not elaborated. Nor is there any indication in the proposed annotation or supporting statement of how future sales of rhino horn might be managed on supply or demand sides to minimize risks of perverse outcomes. This is particularly important, because currently there are no legal end markets for rhinoceros horn, yet illegal trade persists. **It is unclear how the sale of small quantities of horn from *Ceratotherium simum simum* could be expected to affect the wider rhinoceros horn market that currently involves illegal trade of several tonnes of horn each year.** Furthermore, while the CITES Secretariat is identified as playing a significant role in verifying the suitability of proposed trading partners, it is not clear how and with what resources it would undertake this work.

Regarding allowing the international trade in hunting trophies, the number of trophies exported is currently low, comprising a small proportion of the national population and the subspecies is well managed. Therefore, **satisfactory precautionary measures are likely to be met in relation to hunting trophies.**

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

The entire rhinoceros family Rhinocerotidae was included in Appendix I of CITES in 1977. In 1995, the South African population of *Ceratotherium simum simum* was transferred to Appendix II. *In 2005, the Eswatini (then Swaziland) population was also transferred to Appendix II, and in 2023, the Namibian population was transferred to Appendix II. The current listing states, “Only the populations of Eswatini, Namibia and South Africa; all other populations are included in Appendix I. The populations of Eswatini and South Africa are included in Appendix II for the exclusive purpose of allowing international trade in live animals to appropriate and acceptable destinations and hunting trophies. The population of Namibia is included in Appendix II for the exclusive purpose of allowing international trade in live animals for in-situ conservation only, and only within the natural and historical range of *Ceratotherium simum* in Africa. All other specimens shall be deemed to be specimens of species included in Appendix I and the trade in them shall be regulated accordingly.”*

Taxonomy

Ceratotherium simum ssp. simum (Burchell, 1817) is one of two extant subspecies of *Ceratotherium simum*. The other subspecies being the Northern White Rhinoceros *C. s. cottoni*, which is likely extinct in the wild (Emslie, 2020a). As *C. s. cottoni* is possibly extinct in the wild, evidence and data provided will focus on *Ceratotherium simum simum*.

Population size and range

Range for *Ceratotherium simum simum*

Extant: South Africa

Extant & Reintroduced (resident): Botswana; Eswatini; Mozambique; Namibia; Zimbabwe

Extant & Assisted Colonisation (resident): Kenya; Uganda; Zambia

Presence Uncertain & Assisted Colonisation: Côte d'Ivoire; Senegal

Table 1. Population estimates for *C. s. simum* within range States in 2024. *indicates estimates provided in 2023.

Country	Population
Botswana	320
Côte d'Ivoire	1*
The Democratic Republic of the Congo	31*
Eswatini	62
Kenya	1,041
Mozambique	40*
Namibia	1,500
Rwanda	37
Senegal	3*

Country	Population
South Africa	12,082
Uganda	43*
Zambia	54
Zimbabwe	536
Total	15,750

IUCN Global Category

Ceratotherium simum simum Near Threatened (assessed 2020, ver 3.1)

Summary of population, trade and rhino horn in Namibia

Population

The SS states that Namibia holds the second-largest *Ceratotherium simum simum* population globally, after South Africa. As of 2024, population estimates stand at 1,500 *Ceratotherium simum simum*, based on helicopter counts, data from dehorning exercises, and reports from private owners to the Ministry of Environment, Forestry and Tourism. Records are verified through sales, exports, hunting permits, and farm inspections.

From an initial reintroduction of 16 animals in 1975, Namibia's *Ceratotherium simum simum* population rose to 1,500 by 2024. Between 2005 and 2024, the overall growth rate stood at 6.7% annually, including imports from South Africa (31 *Ceratotherium simum simum* in 2022, 18 in 2023 and none in 2024). Namibia has also exported some live specimens to the USA (6 in 2022, 5 in 2023, and 39 in 2024), to Zambia (0, 9, and 32 respectively), to Angola (0, 0, and 5), and to South Africa (0, 0, and 28) (IUCN ARSG in litt., 2025).

Presently, 1,161 *Ceratotherium simum simum* are privately owned across roughly 74 populations, while 339 remain under state ownership in four protected areas.

Table 2. Population estimates for *Ceratotherium simum simum* in Namibia (Ferreira et al., 2025).

Year	Southern White rhinoceros <i>Ceratotherium simum simum</i>
2005	293
2007	370
2010	469
2015	822
2017	975
2021	1,234
2024	1,500

Captive population

At the end of 2024, there were 1,299 *Ceratotherium simum simum* individuals worldwide in zoos and other facilities beyond Africa (Ferreira et al., 2025).

Legal trade

The SS states that domestic consumptive use of *Ceratotherium simum simum* is currently not permitted in Namibia, nor is any rhino horn commerce. Between 2022 and 2024, 43 *Ceratotherium simum simum* were hunted for trophies in Namibia, averaging about 14 per year - equivalent to just 0.9% of the national population, well below annual recruitment rates. Adult males are sometimes dehorned for safety or management, limiting potential trophy hunts. Namibia has also exported 124 *Ceratotherium simum simum* since 2022 to Angola, South Africa, Zambia and the USA (see above for details).

Tourism revenue related to rhinos is difficult to isolate and does not typically increase with higher rhino stocking levels. Because security costs and poaching risks rise steeply with larger populations, many properties keep minimal numbers of rhinos or none at all.

Illegal trade

Between 2021 and 2023, Namibia made ten seizures of rhino horn, involving specimens equivalent to 19 whole horns (hereafter horns for brevity), weighing 51 kg. Namibia was not implicated in any seizures made elsewhere. In comparison, between 2014 and 2020, Namibia made 58 seizures, involving 122 horns, weighing 335 kg. Namibia was implicated in eight seizures made elsewhere, involving 32 horns, weighing 81 kg (see Threats section below for comparison to other range States (Table 5)). It should be noted that the species of rhino was not reported (Ferreira et al., 2025).

The SS states that around 162 white rhinos have been poached in Namibia since 2013, and their horns are presumed to have entered the illicit market.

Precautionary Measures

Species likely to be in demand for trade, but its management is such that the CoP will be satisfied with:

- A) **Implementation by the range States of the requirements of the Convention, in particular Article IV; and**
- B) **appropriate enforcement controls and compliance with the requirements of the Convention**

Export quota or other special measure

Although the proposal makes no reference to the quantity of rhinoceros horn to be traded, the SS notes that the Government of Namibia proposes an initial, closely monitored phase, capped at 200 kg of horn per year, *but there is no indication of how long this phase is envisaged to last nor what the current stockpile of White Rhino horn is.*

Namibia has also proposed the export of its stockpile of rhino horn in Prop 10 relating to its population of Black Rhino. The supporting statement for Prop 10 includes a table detailing current rhinoceros horn stockpiles held by Namibia (almost 4.7 tonnes as of June 2025), but it does not include a breakdown by species.

The SS states that Namibia's existing permitting systems, DNA tracking, stringent penalties, and an updated management strategy provide robust defences. These measures will be enhanced if legal horn trade proceeds, ensuring that all rhino-derived products remain traceable and lawfully sourced.

The SS lacks information on the specific mechanism by which the proposed trade would be conducted, the meaning and criteria for "sufficient national legislation and domestic trade controls" in both Namibia and prospective importing countries and the standards and process for verifying importing countries are not detailed.

The SS provides the following details on how the horn trade will be controlled:

Supply side

Domestic controls

The SS states that in Namibia *Ceratotherium simum simum* is classified as Specially Protected Game, all aspects of ownership and transport require permits. Namibia's legal framework includes robust enforcement mechanisms and harsh penalties designed to suppress illegal trade and safeguard sustainable use. Authorities would observe poaching rates, rhino population trends, and trade volumes. If negative consequences arise—such as an increase in illegal offtake—the government stands prepared to suspend or revise quotas. This adaptive approach mitigates risk while testing the potential for a transformative conservation strategy.

International controls

Permit control: the SS states that in Namibia, the Ministry of Environment, Forestry and Tourism permit office at Windhoek issues all permits relating to *Ceratotherium simum simum* and their parts or derivatives. No competencies are delegated to local or regional authorities. Only Namibian registered game dealers are allowed to capture and trade wild animals. In the case of hunting, only Namibian registered professional hunters and operators/outfitters are allowed to conduct hunting. The Directorate of Veterinary Services in Windhoek issues all veterinary permits. A strictly applied permit system thus already exists and the control of translocations, trade in live animals and hunting and trade in hunting trophies would continue to be strictly enforced.

Marking of animals/products: the SS states that it is common practice, in the interest of facilitating control and law enforcement, that any animal traded is required to be DNA profiled and microchipped. Hunting trophies are also required to be sampled for DNA profiling. DNA samples are submitted to the Rhino DNA Index System (RhODIS) database. All horns are additionally allocated and marked with a national serial number, and the paperwork is traceable. *It would be beneficial to know who is responsible for marking the horns, where they are stored, and what security measures are in place, particularly for privately held stockpiles.*

Customs and border control: the SS states that Namibian Customs Officers check CITES, veterinary and transit permits. Where necessary, they refer to the Namibian Police or district veterinary officer.

Law enforcement: the SS states that law enforcement is a joint effort by the Ministry of Environment, Forestry and Tourism, the Protected Resources Division of the Namibian Police and the Customs Service. Law enforcement agencies rely primarily on information and well-established informer networks exist and are maintained.

The SS notes that Namibia acknowledges concerns that legal horn trade might provide cover for illicit supplies. To counter this, each horn (or part thereof) must carry a DNA certificate (via the RhODIS system), microchip ID, and the relevant CITES documents. Annual audits and specialized enforcement units will help detect discrepancies. This design improves traceability and reduces incentives to launder illegal horn.

Namibia provides a leading example of traceability and control. The country uses an electronic database that monitors both state-held and privately held stockpiles. Approximately 70% of government-managed horns are genetically profiled and microchipped, with data integrated into the RhODIS. In the private sector, 80% of horns have been sampled, although only legally acquired historic horns are included under current permit frameworks. This approach enhances accountability and acts as a deterrent to illegal activity, though it depends on regular updates and cooperation from private custodians (Ferreira et al., 2025).

Namibia has developed strong enforcement and intelligence networks that rely on contributions from multiple government bodies. However, these systems often depend on donor support and joint staffing arrangements, which could be vulnerable to future institutional or financial shifts (Ferreira et al., 2025).

Namibia's Conservancy Rhino Ranger Incentive Programme provides one of the clearest examples of community-led rhino protection. Rangers are drawn directly from local communities and supported with training, stipends, and recognition. This programme strengthens local responsibility and builds capacity for rhino monitoring at the landscape level (Ferreira et al., 2025).

Namibia has launched a dedicated Environmental Crime Court in Otjiwarongo in 2025 following the success of temporary wildlife courts. This specialised court handles high-profile wildlife cases more efficiently and is supported by inter-agency coordination mechanisms (Ferreira et al., 2025).

Namibia's Blue Rhino Task Team combines investigators from the police, Ministry of Environment, and other security agencies. The task team is supported by shared vehicles, intelligence systems, and donor contributions, enabling rapid response and data-driven decision-making (Ferreira et al., 2025).

Demand side

The SS provides sparse information on how the demand side of the trade in rhinoceros horn would be managed, noting that there will only be trade to trading partners that have been verified by the Secretariat, in consultation with the Standing Committee, to have sufficient national legislations and domestic trade controls, and not before the Secretariat has verified prospective importing countries and the registered stockpiles.

There are currently no legal end markets for rhinoceros horn, so if trade were to be allowed it is not clear what measures might be in place to prevent horn from illegal sources entering the trade chain.

The SS states that an estimated eight tonnes (equivalent to at most 1,500 rhinos) of horn are currently trafficked illegally each year. However, for the last five years, detected African rhino poaching mortalities range between 500-600 per year. Assuming two horns per rhino, weighing an average of 2.57 kg per horn for African species, would mean that in the range of 2.5 - 3 tonnes of horn are currently produced from poached rhino (between 500-600 detected poached rhinos per year over the last 5 years), noting that not all poached rhinos are detected and seizures reduce the volumes entering trade (Ferreira et al., 2025).

Additional Information

Global population and trends

*Once widespread in the bushveld areas of southern Africa south of the Zambezi River, *Ceratotherium simum simum* was on the brink of extinction by the end of the 19th century having been reduced to just one small population of approximately 20–50 animals in KwaZulu-Natal, South Africa. However, by the end of 2012, after years of protection and many translocations, the subspecies had grown to an estimated 21,316 wild animals. Following poaching increases, estimated numbers declined about 15% to 18,064 by the end of 2017 with bootstrapped 90% confidence levels of 17,212 to 18,915. South Africa remains the stronghold for this subspecies (~86.5%) conserving an estimated 18,933 individuals by the end of 2012 (Emslie, 2020b).*

*Smaller reintroduced subpopulations occur within former range States in Botswana, Namibia, Eswatini, and Zimbabwe; subpopulations of free-ranging *Ceratotherium simum simum* have also been established outside their historical range in Kenya, Zambia and more recently Uganda, although Uganda is a former *Ceratotherium simum cottoni* range State and a ~3,500 year old White Rhino subfossil indicates at one stage Kenya was also once a White Rhino range State.*

*The *Ceratotherium simum simum* population increased from 15,940 in 2021 to 16,799 in 2022, showing a 5.4% growth (Table 3). In 2023, the population reached 17,462 reflecting a reported growth of 3.9% from the previous year. By the end of 2024, there were 15,750 *Ceratotherium simum simum* living in 13 countries on the continent, a similar number to the end of 2021 when there were 15,940. Between 2023 and 2024, *Ceratotherium simum simum* had declined by 11.2%. The decline is mostly a consequence of losses experienced in South Africa, where the population declined to 14,074 *Ceratotherium simum simum* in 2023, a decline of 15.3%. There are three main reasons for the recorded decline. First, 420 *Ceratotherium simum simum* were poached, with 47.1% of those losses occurring in Hluhluwe-iMfolozi Park in South Africa's KwaZulu Natal province. This followed a shift in poaching pressure after a*

dehorning program was implemented in the Kruger National Park. Despite the dehorning, 88 *Ceratotherium simum* – most were dehorned – were still reported poached in the Kruger National Park during 2024. In response to the poaching pressure, Hluhluwe-iMfolozi Park also began dehorning. Second, some provinces did not update their population estimates after 2023. Third, there is uncertainty about whether distance sampling methods overestimated the population size in Hluhluwe-iMfolozi Park (Ferreira et al., 2025).

Table 3. Global population estimates of *Ceratotherium simum* on the continent.

Year	Number of individuals
2005	14,543
2007	17,470
2010	20,161
2015	20,375
2017	18,064
2021	15,940
2024	15,750

Eswatini also reported a substantial decline from 2023 to 2024. The sixty-two *Ceratotherium simum* reported at the end of 2024 reflects a 38.4% decline from the numbers reported at the end of 2021. These relate to challenges linked to environmental conditions and a sudden inability to effectively manage rhinos and competing herbivores due to new regulations. These regulations only allow Eswatini registered aerial support to operate in the country and have disenabled the use of affordable non-Eswatini registered wildlife air services. Implementing cost-effective various biological management interventions is thus limited (Ferreira et al., 2025).

During 2024, globally 26.1% of *Ceratotherium simum* individuals were living in semi-wild conditions, with 63.7% of all *Ceratotherium simum* on the continent owned and managed by non-state entities – mostly private but there is some community ownership (Ferreira et al., 2025).

Country populations and trends in Africa

South Africa had the most *Ceratotherium simum* (76.7%), followed by Namibia (9.5%), Kenya (6.6%), Zimbabwe (3.4%) and Botswana (2.0%) (Table 4). The other eight range States (Côte d'Ivoire, the Democratic Republic of the Congo, Eswatini, Mozambique, Rwanda, Senegal, Uganda and Zambia) each had fewer than 100 *Ceratotherium simum* and in total made up 1.7% of the continental population. Note that the populations in Côte d'Ivoire, Kenya, and Senegal are outside the species' natural range and are thus novel elements in the ecosystems in which they live. The populations in the Democratic Republic of the Congo, Rwanda and Uganda are within or on the very edge of the contemporary range of Northern white rhinos (*Ceratotherium simum cottoni*) and can be considered as equivalents (Ferreira et al., 2025).

Despite the above changes and challenges, there are a total of 336 localities with *Ceratotherium simum*, most of which are in South Africa (290), Kenya (13), Botswana (12) and Zimbabwe (7). Of interest is that Zambia reported six localities, four more than previously reported, while Namibia reported five localities (Ferreira et al., 2025).

Table 4. Population estimates for *Ceratotherium simum* within range States reported at Conferences of the Parties since before the poaching of rhinoceroses escalated in 2007. *indicates estimates provided in 2023.

Range State	Year	Southern White rhinoceros <i>Ceratotherium simum</i>
Angola	2021	3
	2024	0
Botswana	2005	99
	2007	106
	2010	135
	2015	239
	2017	452
	2021	242
	2024	320
Côte d'Ivoire	2017	1
	2021	1
	2024	1*
The Democratic Republic of the Congo	2005	-
	2007	-
	2010	-
	2015	-
	2017	-

Range State	Year	Southern White rhinoceros <i>Ceratotherium simum simum</i>
<i>Eswatini</i>	2021	20
	2024	31*
	2005	75
	2007	89
	2010	88
	2015	76
	2017	66
	2021	98
	2024	62
<i>Kenya</i>	2005	234
	2007	303
	2010	361
	2015	441
	2017	510
	2021	871
	2024	1,041
<i>Mozambique</i>	2005	7
	2007	9
	2010	6
	2015	29
	2017	29
	2021	14
	2024	40*
<i>Namibia</i>	2005	293
	2007	370
	2010	469
	2015	822
	2017	975
	2021	1,234
	2024	1,500
<i>Rwanda</i>	2005	-
	2007	-
	2010	-
	2015	-
	2017	-
	2021	30
	2024	37
<i>Senegal</i>	2017	3
	2021	3
	2024	3*
<i>South Africa</i>	2005	13,521
	2007	16,273
	2010	18,796
	2015	18,413
	2017	15,625
	2021	12,968
	2024	12,082
<i>Uganda</i>	2005	4
	2007	6
	2010	9
	2015	15
	2017	22
	2021	35
	2024	43*
<i>Zambia</i>	2005	2
	2007	1
	2010	7
	2015	10
	2017	14
	2021	8
	2024	54

Range State	Year	Southern White rhinoceros <i>Ceratotherium simum simum</i>
Zimbabwe	2005	308
	2007	313
	2010	290
	2015	330
	2017	367
	2021	417
	2024	536
Africa	2005	14,543
	2007	17,470
	2010	20,161
	2015	20,375
	2017	18,064
	2021	15,940
	2024	15,750

Illegal trade

Global

Concerns regarding the impact of poaching and illegal trade on the extinction risk of wild rhino populations are well-founded. *There was an average annual decline of approximately 514 Ceratotherium simum simum since 2015, there was little decline between 2021 and 2024 (Table 4).*

Threats

Global

The main threat to the population is illegal hunting (poaching) to supply the illegal international rhino horn trade. It is estimated that currently around 95% of rhino horn sourced in Africa for end user illegal markets in SE Asia are from this source. Rhino horn has traditionally had two main uses: use in traditional Chinese medicine, and ornamental use. Recently rhino horn has become a highly prized material for making carved expensive high-status items such as bowls and bangles. In the past it was also used to produce ornately carved handles for ceremonial daggers (jambiyas) worn in Yemen and some Middle East countries. Historically rhino horn was also used in traditional Chinese medicine (as a fever reducer). However, most recently it appears to be shavings from carvings that are illegally sold to the medicinal market at lower prices than worked items. Until recently, at the continental species level, poaching of Ceratotherium simum simum has not had a serious impact on overall numbers of Ceratotherium simum simum in Africa, with poaching losses in parts of the range being surpassed by encouraging growth rates in others. However, following declines in the largest subpopulation (Greater Kruger) continental numbers declined by an estimated 15% from 2012 to 2015. While poaching levels increased dramatically from 2007 to 2014, numbers of Ceratotherium simum simum reported poached have declined in each of the four years every year since, and half year reported poaching in 2019 in the major range State, South Africa indicates this trend is possibly continuing (Emslie, 2020b).

The significantly increased poaching since 2007 has greatly increased protection costs and risks to investment and staff. This has resulted in declining live sale prices and reduced incentives. Some private owners in South Africa have disposed of their rhino. If this worrying trend continues this threatens to possibly reverse the expansion of range and has the potential to also significantly reduce conservation budgets (due to declining live sales at lower prices). However, so far there appears to be a consolidation of Ceratotherium simum simum into larger subpopulations (Emslie, 2020b).

Namibia

The SS states that the gravest threat is illegal killing and the black-market trade in rhino horns. Poaching has risen since 2012, requiring Namibia to intensify anti-poaching strategies, interagency coordination, and stricter penalties. A secondary but critical threat is the growing cost of security, which compels many private owners to relinquish their rhino populations, *although there is no published information that formally documents or substantiates this trend*. Climate change and prolonged droughts force landowners to supplement feed, a substantial cost which many private farmers are unable to carry. As these private herds disappear, large tracts of potential rhino habitat are no longer occupied, and if no longer generating revenue, the habitat may be lost. Existing revenue streams—tourism, limited trophy hunting, and live sales—do not cover all security costs. At the same time, diminishing government budgets and uncertain donor funding put Namibia's anti-poaching measures at risk, particularly during global recessions and economically uncertain times such as pandemics. Trade restrictions also hamper potential revenue generation.

Between 2021 and 2023, Namibia made 10 seizures of rhino horns, involving 19 horns, weighing 51 kg. Namibia was not implicated in any seizures made elsewhere. In comparison, between 2014 and 2020, Namibia made 58 seizures,

involving 122 horns, weighing 335 kg. Namibia was implicated in eight seizures made elsewhere, involving 32 horns, weighing 81 kg. It should be noted that the species of rhino was not reported (Ferreira et al., 2025).

Table 5. Total numbers of seizures of rhino horn specimens by Party, and their reported or estimated weight and equivalent number of whole horns. Data are presented for the most recent three-year period since the last CoP19 report (2021-2023) and for the preceeding seven years in the analysed dataset (2014-2020). The first three columns summarise the seizures made by each Party (Seizures In (SZs In)), and the last three columns summarise the seizures made elsewhere which implicate each Party, by naming it as the country of origin, transit or destination in the reported trade chain (Seizures Out (SZs Out)). The top ten Parties are listed, sorting from highest to lowest based on their total weight (Wt) involvement in 2021-2023 (Ferreira et al., 2025).

Country	2021-2023						2014-2020					
	Szs In (No.)	Horns In (No.)	Wt In (kg)	Szs Out (No.)	Horns Out (No.)	Wt Out (kg)	Szs In (No.)	Horns In (No.)	Wt In (kg)	Szs Out (No.)	Horns Out (No.)	Wt Out (kg)
South Africa	65	442	1216	8	112	267	241	1308	3076	63	440	1064
Malaysia	1	11	29	8	167	410	5	131	335	10	245	658
Viet Nam	10	74	205	9	31	83	40	364	940	36	309	697
Qatar	4	51	123	1	4	12	2	7	20	20	150	357
UAE	0	0	0	4	38	103	4	25	71	12	130	337
Singapore	2	21	36	1	14	26	3	14	37	4	14	34
Lao PDR	1	2	4	2	28	58	3	10	27	7	13	28
Zimbabwe	2	7	14	1	18	46	8	30	45	2	5	7
Namibia	10	19	51	0	0	0	58	122	335	8	32	81
Mozambique	3	19	34	1	4	10	20	204	428	23	153	385

Around 162 white rhinos have been poached in Namibia since 2013, and their horns are presumed to have entered the illicit market.

Conservation, management and legislation

Legislation

National

The SS states that in Namibia, *Ceratotherium simum simum* is classified as “Specially Protected” under the Nature Conservation Ordinance (Ordinance 4 of 1975, as amended). All activities—hunting, capture, possession, transport, import, export, and re-export—are subject to permits. Horns and other parts are deemed “Controlled Wildlife Products” under the Controlled Wildlife Products and Trade Act (Act 9 of 2008, as amended). Convictions for illegal trade in these products can lead to fines up to NAD25,000,000 (around USD1,780,000) and/or 25 years of imprisonment. Importantly, these laws are indeed enforced across Namibia.

Under the Animal Health Act (Act 1 of 2011), veterinary permits are required for importing or moving raw wildlife products, including rhino horn, across veterinary cordon fences.

International

Namibia's *Ceratotherium simum simum* population is currently included in CITES Appendix II, strictly for in-situ live trade within Africa's historical rhinoceros range. All other specimens remain regulated as Appendix I, subject to comprehensive CITES safeguards.

Management

Namibia's White Rhinoceros Management Strategy (2022) employs a metapopulation approach, promoting genetic flow through translocations and introductions. Basic guidelines—like exchanging at least one breeding individual per generation—are exceeded by current efforts. Reintroduction to unoccupied but suitable areas also helps build up populations.

Trophy hunting (often termed “conservation hunting”) is acknowledged as an important revenue source, while live sales boost private-sector engagement. Meanwhile, increasingly sophisticated, globally networked poachers necessitate upgraded security and intelligence measures. Horn stockpile management (e.g., annual horn re-registration) helps avert theft and illegal leakage. *It is not currently clear whether private landowners are required to mark and register their horn stockpiles, and whether these are transferred to the Government of Namibia for safekeeping and therefore further clarification on how the government ensures the security and oversight of privately held stockpiles would strengthen the overall transparency of the system (IUCN ARSG in litt., 2025).*

The Ministry of Environment, Forestry and Tourism spearheads rhino monitoring and anti-poaching patrols nationwide. Key tools include aerial surveys, waterhole counts, SMART patrolling, and camera surveillance. In Etosha National Park, helicopter block counts occur bi-annually. In private reserves, mandatory microchipping, DNA sampling (RhODIS database), and permit checks ensure tight population oversight.

Artificial Propagation/captive breeding

The SS states that no formal captive breeding operations exist in Namibia.

Implementation challenges (including similar species)

The SS notes that parts of horn are not readily distinguishable from the horns of other species of rhinoceros *which is a concern as there is a potential for illegal horn from other countries to be laundered into the legal trade chain*. All the horn pieces will be required to be marked as described already.

Potential risk of a transfer from Appendix I to II

The SS states that critics of legalising trade argue that illegal demand may escalate under legal trade, particularly if the removal of legal restrictions broadens the consumer base and removes the assumed stigma associated with purchasing an illegal commodity. *Consensus on legalising the trade in rhinoceros horn has not been achieved, with arguments both for and against the trade in rhinoceros horn. The idea that stockpiles and dehorning can meet market demand for rhino horn carries some uncertainty (IUCN ARSG in litt., 2025). In IUCN's latest contemporary strategic framework for African rhino conservation 2025-2035, six strategic themes are identified for achieving thriving rhinos valued by people, with one of the themes focussed on understanding and influencing rhino horn markets. Ultimately, there is a risk that even allowing one sale of rhinoceros horn could increase demand, and it should be noted that while Namibia may have the funds after the sale to protect their population, other range States may not have the finances to combat the potential ensuing demand (Balfour et al., 2025).*

The SS does not state whether all privately held stockpiles are currently registered with the Government, nor how such stockpiles are secured given their distributions across various private holders nationwide.

Potential benefit(s) for trade regulation of a transfer from Appendix I to II

As highlighted above, Namibia is successfully managing its *Ceratotherium simum simum* population, however this comes at a significant cost, with an increasing financial burden caused by the increasing threat of illegal killing, proposing that this threat can be addressed through the sustainable use of *Ceratotherium simum simum*. Not only does protecting *Ceratotherium simum simum* come at a cost, but the SS notes that if this cost becomes too high private landowners might abandon *Ceratotherium simum simum* ownership, reducing the habitat and protections of the subspecies. The proceeds from the sale of rhinoceros horn may therefore help to contribute towards these costs.

Other comments

The SS does not state how government-generated revenue from Ceratotherium simum simum under state care will be managed, and whether the funds will be allocated to the Game Products Trust Fund or another transparent and accountable mechanism.

The SS notes using an adaptive "pilot" framework which is positive, however, the SS does not outline how trade would be adjusted or halted if unintended consequences arise from the sale of rhino horn. It would be beneficial to have independent oversight or a verification mechanism.

References

- Balfour, D. Ferreira, S.M., Gaymer, J., Lewis, C., Mafumo, H., Makoma, K., Mgoola, W., Reuben, M., Shaw, J.A. and Uri-Khob, S. (2025) African rhino conservation 2025–2035: A contemporary strategic framework. IUCN.
- Emslie, R. (2020a) *Ceratotherium simum ssp. cottoni*. The IUCN Red List of Threatened Species 2020: e.T4183A45813838. <https://dx.doi.org/10.2305/IUCN.UK.2020-1.RLTS.T4183A45813838.en>. Accessed on 31 July 2025.
- Emslie, R. (2020b) *Ceratotherium simum ssp. simum*. The IUCN Red List of Threatened Species 2020: e.T39317A45814320. <https://dx.doi.org/10.2305/IUCN.UK.2020-1.RLTS.T39317A45814320.en>. Accessed on 31 July 2025.
- Ferreira, S.M., Baruch-Mordo, S., Laidler, G., Stoppoloni, R., Sangalakula, L., Talukdar, B. and Balfour, D. (2025) The African and Asian rhinoceroses – status, conservation and trade: A report from the IUCN Species Survival Commission (IUCN/SSC) African and Asian Rhino Specialist Groups and TRAFFIC to the CITES Secretariat pursuant to Resolution Conf. 9.14 (Rev. CoP15). CoP20 Doc. 84, Annex 3, CITES Secretariat, Geneva, Switzerland.
- IUCN African Rhino Specialist Group (ARSG) (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.

Transfer of the population of South-Western Black Rhinoceros *Diceros bicornis bicornis* of Namibia from Appendix I to Appendix II with an annotation

Proponent: Namibia

Summary: The South-Western Black Rhinoceros *Diceros bicornis bicornis* is one of three extant subspecies of Black Rhinoceros (the others being the Eastern Black Rhino *Diceros bicornis michaeli* and the South-Eastern Black Rhino *Diceros bicornis minor*). The Black Rhinoceros has been included in Appendix I (under the family listing for Rhinocerotidae) since 1977.

The proposal is to transfer the population of *Diceros bicornis bicornis* of Namibia from Appendix I to Appendix II, with the following annotation:

"For the exclusive purpose of allowing trade in registered rhinoceros horn, whole or pieces subject to the following:

- i) only registered Government-owned stock, originating in the State (excluding seized rhinoceros horn and rhinoceros horns of unknown origin);*
- ii) only horns with RHODIS certificates;*
- iii) only to trading partners that have been verified by the Secretariat, in consultation with the Standing Committee, to have sufficient national legislation and domestic trade controls;*
- iv) not before the Secretariat has verified prospective importing countries and the registered stocks; and*
- v) the proceeds of trade are used exclusively for rhinoceros conservation and community development programmes within or adjacent to the rhinoceros range.*

All other specimens shall be deemed to be specimens of species included in Appendix I and the trade in them shall be regulated accordingly."

The Black Rhino was once relatively abundant in Namibia but was reduced to an estimated population of fewer than 100 by the 1960s. Numbers have recovered in recent years, increasing from around 1000 in 2004 to just over 2000 in 2024, with the country now home to over 80% of the global population. The increase is attributed to range expansion through the translocation of individuals to more secure areas, horn removal at times when poaching was a threat, close protection and monitoring of existing populations and the creation of new populations. In recent years numbers have stabilised, attributed to the impacts of drought. In Namibia, Black Rhinos are property of the State but occur in areas within different land tenure systems as part of a custodianship scheme.

Namibia has an annual hunting quota of five adult males, with the funds generated being used for conservation, although between 2009 and 2024 only 20 individuals were hunted for trophies. Between 2014 and 2024, no live individuals were exported. Domestic consumptive use of Black Rhino and trade in its horn and other products is currently not permitted in Namibia.

The proponents state that one of the most significant threats to wild rhino populations is the unsustainable cost of anti-poaching efforts. They propose that legal trade could replace illegal trade, benefiting the conservation of rhino populations, provided user demand can be met.

According to the supporting statement the proposed trade is a once-off sale of horn stockpiles although the quantity is not specified in the proposed annotation. The supporting statement includes a table detailing current rhinoceros horn stockpiles held by Namibia (almost 4.7 tonnes as of June 2025), but it does not include a breakdown by species. Namibia has also submitted a proposal to allow trade in White Rhino horn (Prop 9).

The proponents set out in the annotation and in the supporting statement what management measures will be put in place to control the supply of Black Rhino including permit control, marking of animals/products, customs and border control along with details of enforcement measures. Little specific information is presented on the measures that will be put in place to control the trade of horn

from the demand side, given that many potential destinations currently have domestic legislation that prohibits rhino horn trade.

Analysis: Namibia's population of Black Rhinoceros *Diceros bicornis bicornis* does not have a restricted range. Although its population is still small, it has increased steadily from historic lows and has now stabilized at around 2,100. The **subspecies does not appear to meet the biological criteria for Appendix I** in accordance with Annex 1 of Res. Conf. 9.24 (Rev. CoP17).

When considering the transfer of a taxon from Appendix I to Appendix II, the Resolution calls for the precautionary measures set out in its Annex 4 to be met, in this case, it appears, through an annotation forming an integrated part of the proposal. This addresses: the origin of horns to be exported (registered stockpiles of legally acquired and individually identifiable horns of known origin); a mechanism for approving any future trading partners; and the manner in which proceeds from trade will be used (for rhinoceros conservation and community development programmes within or adjacent to the rhinoceros range). The proponent has also described in the supporting statement a range of national management measures and enforcement controls aimed to mitigate risks for populations of other rhinoceros species.

With respect to the special measure annotation proposed, there is no reference made to the one-off nature of trade from registered stockpiles referred to in the supporting statement, nor the amount of horn proposed for export. In terms of future trade levels, as it stands the proposal amounts to unqualified implementation of Article IV of the Convention. **Despite Namibia's successful conservation efforts and detailed provision of information on enforcement controls and compliance, the Parties may seek more explicit reference in an annotation to the export limitation explained in the supporting statement.**

The meaning and criteria for "sufficient national legislation and domestic trade controls" in prospective importing countries and the standards and process for verifying importing countries are not elaborated. Nor is there any indication in the proposed annotation or supporting statement of how future sales of rhino horn might be managed on supply or demand sides to minimize risks of perverse outcomes. This is particularly important, because currently there are no legal end markets for rhinoceros horn, yet illegal trade persists. It is unclear how a one-off sale of Black Rhinoceros horn would affect demand in the future: opening a legal market may increase poaching pressure in other range States with fewer resources to address it. Furthermore, while the CITES Secretariat is identified as playing a significant role in verifying the suitability of proposed trading partners, it is not clear how and with what resources it would undertake this work. **Overall, insufficient precautionary measures have been detailed.**

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

The Rhinocerotidae family was listed in CITES Appendix I in 1977. *Diceros bicornis ssp. bicornis* is therefore also listed in CITES Appendix I.

Taxonomy

Diceros bicornis ssp. bicornis (Linnaeus, 1758) is one of three extant subspecies of *Diceros bicornis*.

Population size and Range (Table 1)

Range for *Diceros bicornis bicornis*

Extant: Angola and Namibia

Extant & Reintroduced: South Africa

Extinct: Botswana

Table 1. Population estimates for *Diceros bicornis bicornis* within range States as of the end of 2024 (Ferreira et al., 2025), with marginally revised continental totals for the end of 2012, 2015 and 2017 (Emslie, 2020).

Country	Population at end of 2024 (end of 2017)
Namibia	2,097 (1,857)
South Africa	500 (331)
Angola	-
Total 2024	2,597
Total 2017	2,188
Total 2015	2,212
Total 2012	1,968

IUCN Global Category

Near Threatened (assessed 2020, ver 3.1)

Biological criteria for inclusion in Appendix I**A) Small wild population**

See section C.

B) Restricted area of distribution**Namibia**

The *Diceros bicornis bicornis* population in Namibia is currently distributed throughout the country, in secure areas such as protected areas, communal land and private land as part of an innovative custodianship programme. Since 1993, the Namibian Government has made robust efforts to expand the ranges of the *Diceros bicornis bicornis* in the country within their former or historical range through translocation to private farms under the custodian programmes, which resulted in increases in population growth. Almost 17% of the land surface of Namibia has been placed in proclaimed protected areas, while 48% of the land surface in Namibia falls under commercial land, which is potential habitat available for black rhinoceros. Additionally, this land is also used for wildlife production comprising communal conservancies and freehold land used for wildlife production and tourism.

The potential range for *Diceros bicornis bicornis* is not restricted to a single habitat type and they can be found in savannahs and even dry forests.

C) Decline in number of wild individuals**Namibia**

The SS states that records from the 1700s indicate that *Diceros bicornis bicornis* was abundant at the time, but during the second half of the 19th century, *Diceros bicornis bicornis* was virtually exterminated by hunting in the southern parts of the country. By 1967, the distribution of *Diceros bicornis bicornis* in Namibia was restricted to the north-western area, north of the Hoanib River, along the Huab and Ugab Rivers and in the Etosha National Park. At that time, the national population was estimated at 90 *Diceros bicornis bicornis* (25 in the northwestern area north of the Hoanib River, 17 south of the Hoanib River, and 48 in the Etosha National Park).

The SS states that Namibia holds more than a third of all *Diceros bicornis bicornis* remaining globally, and is the stronghold of the *Diceros bicornis bicornis*, whose historical range also included southern Angola. In 2024, the *Diceros bicornis bicornis* population in Namibia was estimated at 2,098. This estimate was obtained from aerial surveys, based on standardised techniques and other monitoring activities. Over the years, *Diceros bicornis bicornis* ranges in the country have been greatly expanding, whereby 55% are now found in protected areas, 38% on freehold farms, while 7% are found on communal conservancies.

The SS states that the Namibian *Diceros bicornis bicornis* population has undergone continuous growth from just under 1,021 in 2004 to over 2,098 in 2024 (Figure 1). The stable trend identifies that the population is viable in all respects despite a decrease experienced in 2016. Since 2021, the Namibian population has changed by -1.2% per annum with Namibia noting substantial mortalities linked to drought conditions since 2015/2016 which affected *Diceros bicornis bicornis* in north-western Namibia (Ferreira et al., 2025). The SS states that the increase in number of *Diceros bicornis bicornis* before this period can be accounted for by rigorous efforts taken by the government in protecting and managing the *Diceros bicornis bicornis* population in the country. This included expansion of range through translocation of animals to more secure areas (Etosha National Park, Waterberg Plateau Park), horn removal (dehorning) at times when illegal killing was a threat, close protection and monitoring of the existing populations, and the creation of several new populations. Although all *Diceros bicornis bicornis* in Namibia belong to the State, the population occurs within different land tenure systems (protected areas, communal land and on private land as part of an innovative custodianship scheme). Etosha National Park initially served as the major donor population for translocations into other areas (custodians, conservancies and

other protected areas). In some areas, the population has reached densities that preclude maximum growth rate, and in some cases leads to losses through fighting and other intra-specific interactions.

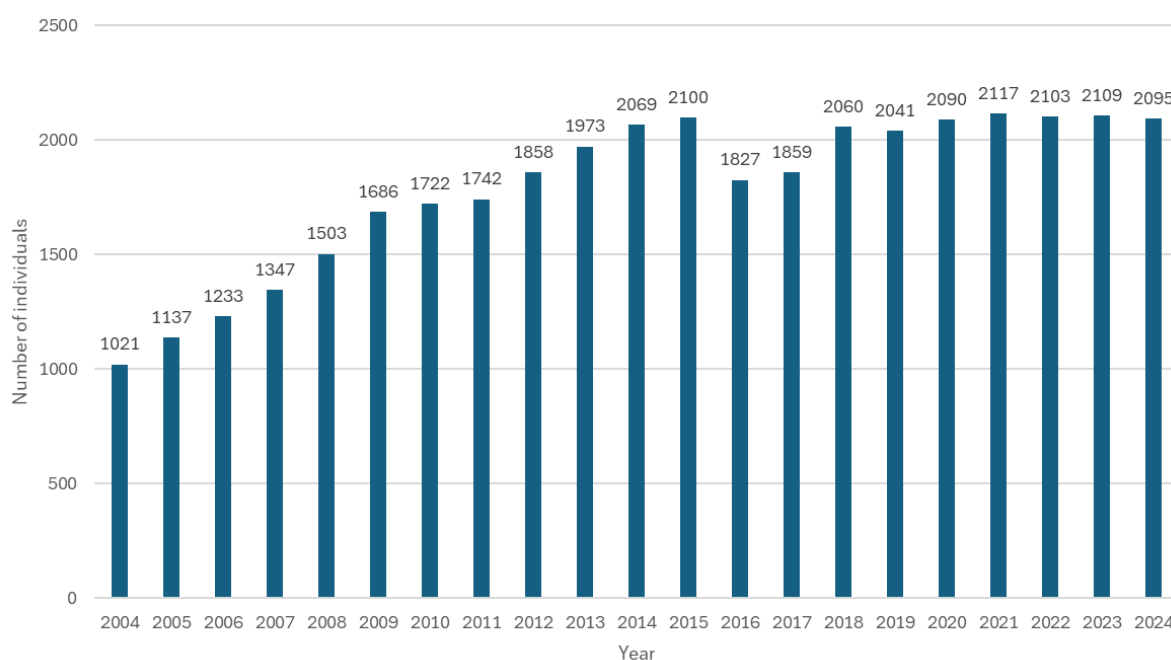


Figure 1. Trend in the number of *Diceros bicornis bicornis* in Namibia, between 2004 and 2024.

Trade criteria for inclusion in Appendix I

The species is or may be affected by trade

This proposal is to transfer the population of *Diceros bicornis bicornis* of Namibia from Appendix I to Appendix II, for the exclusive purpose of allowing trade in registered rhinoceros horn, whole or pieces, only from Government-owned stockpiles, originating in the State (excluding seized rhinoceros horn and horns of unknown origin). The proponents state that transferring the population to Appendix II will enable Namibia to trade once-off in raw horn with the funds from the sale being used to enhance the conservation of the species and its habitat.

Legal trade

The SS states that domestic consumptive use of *Diceros bicornis bicornis* and trade in its horn and other products is currently not permitted in Namibia. *CITES Parties approved the annual hunting of five adult male black rhinos each in Namibia and South Africa in 2004 (Resolution Conf. 13.5) (Ferreira et al., 2025).* This quota generates funds for conservation.

Between 2009 and 2024, twenty (20) *Diceros bicornis bicornis* were hunted for trophies in Namibia (Figure 2). Utilization of *Diceros bicornis bicornis* (trophy hunting) is undertaken within the framework of existing CITES provisions. Animals for trophy hunts are individually identified based on selected criteria. Only surplus adult male animals are considered for trophy hunting. Preference is given to hunt post-reproductive males (+25 years of age). However, occasionally other males could be targeted, where they have fought excessively, repeatedly broken out of certain areas, or disrupted the existing social structure (e.g. by killing or displacing other individuals). *The hunting market in Namibia has recovered after disruptions to international travel during the COVID-19 pandemic in 2020 (Ferreira et al., 2025).*

In Namibia, annual hunting rates range from 0.05% to 0.18% of the country's Diceros bicornis bicornis population. Average prices for hunting Diceros bicornis bicornis varied, fetching USD234,000, USD250,000 and USD170,000 during 2022, 2023 and 2024 respectively in Namibia. Namibia reported hunting revenues of USD270,000 to USD300,000 per rhino during 2018 to 2021. A collation of trophy hunting nominal fees noted USD300,299 per rhino in 2018. The recent prices for hunts suggest that revenue from trophy hunting of Diceros bicornis bicornis is reducing. It is noted as an important form of revenue to support conservation costs (Ferreira et al., 2025).

Evaluation of hunting in South Africa and Namibia demonstrated that limited, targeted hunting – particularly of older, non-breeding males – can contribute to other conservation management goals such as mitigating demographic imbalances, reducing aggression-related mortality, and enhancing genetic diversity. Economically, hunting generates significant revenue for conservation, especially in the private context (Ferreira et al., 2025; t Sas-Rolfes et al., 2022).

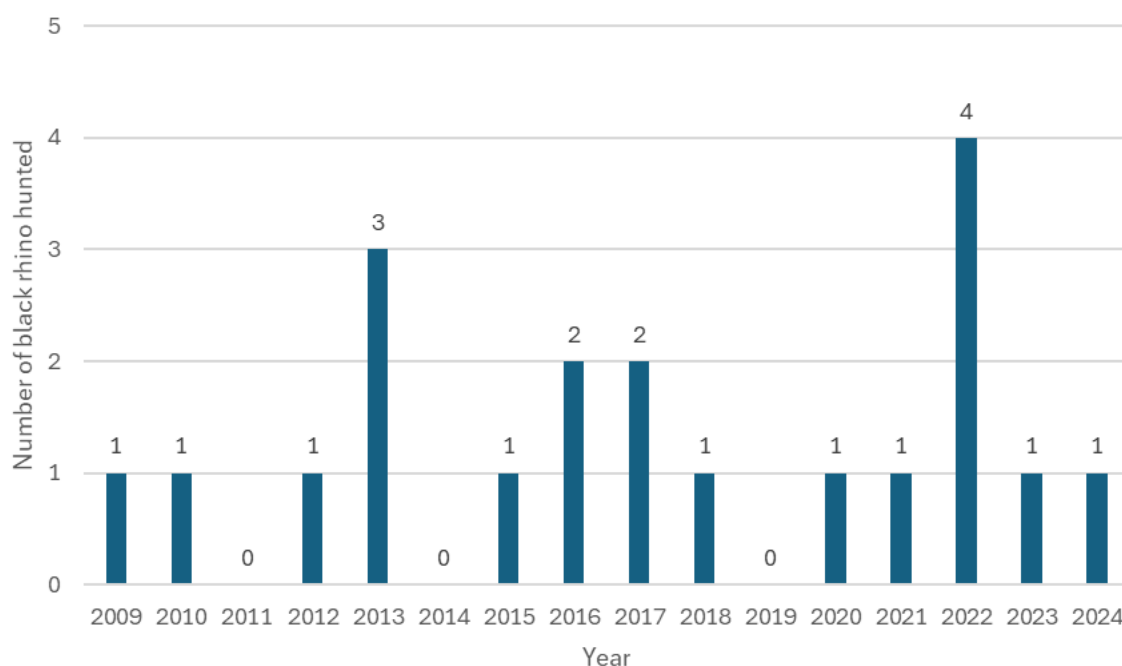


Figure 2. Number of *Diceros bicornis bicornis* hunted as trophies between 2009 and 2024.

According to the CITES Trade Database no live *Diceros bicornis bicornis* were exported from Namibia between 2014 to 2024.

Illegal Trade

Between 2013 and 2024, 637 *Diceros bicornis bicornis* were poached in Namibia (Figure 3).

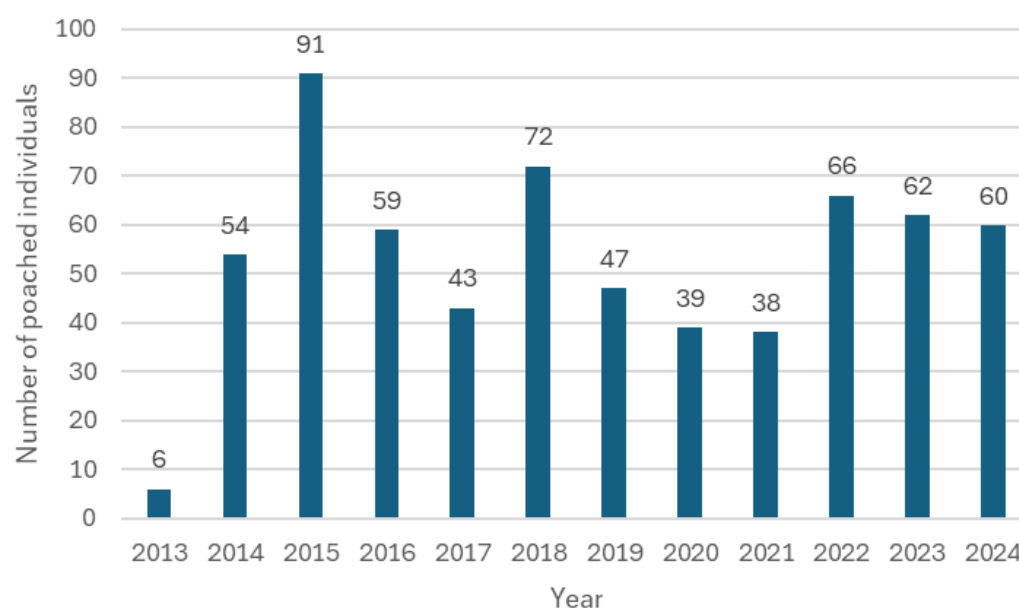


Figure 3. Number of *Diceros bicornis bicornis* poached between 2013 and 2024.

Between 2021 and 2023, Namibia made ten seizures of rhino horns, involving specimens equivalent to 19 horns (hereafter horns for brevity), weighing 51 kg. Namibia was not implicated in any seizures made elsewhere. In comparison, between 2014 and 2020, Namibia made 58 seizures, involving 122 horns, weighing 335 kg. Namibia was implicated in eight seizures made elsewhere, involving 32 horns, weighing 81 kg. It should be noted that the species of rhino was not reported (Ferreira et al., 2025).

Rhino horn stockpiles

The SS states that as of June 2025, there are 1,131 whole horns, weighing 1,606.67 kg, and 3,243 horn pieces, weighing 3,003.77 kg (equivalent of 1,111 horns) which have originated from natural mortalities or from

management related practices are held in stockpiles. Twenty-eight whole horns, weighing 48.97 kg, and six horn pieces, weighing 10.9 kg originating from horns recovered through seizures are also held in stockpiles. This results in a total stockpile weighing 4,670.31 kg. *These figures are not differentiated by species in the SS, and it is not clear what the quantity of horn from Black Rhinos is. It should be noted that the presence of only 28 whole horns held in stockpiles appears low in the context of the number of rhinos poached between 2013 and 2024 (n=637) suggesting a significant number may have entered trade illegally.*

The SS states that all seized or confiscated horns are separately stored, with some horn being held on behalf of the Protected Resources Sub-Division of the Namibian Police (PRD) as evidence for pending court proceedings. Growing horn stockpiles represent major management, administrative and security problems. Namibia maintains a detailed computer database of all specimens in storage, with source documentation, and all specimens are marked to make them individually recognizable.

Precautionary Measures

Species likely to be in demand for trade, but its management is such that the CoP will be satisfied with:

- A) Implementation by the range States of the requirements of the Convention, in particular Article IV; and**
- B) appropriate enforcement controls and compliance with the requirements of the Convention**

Export quota or other special measure

Although the proposal makes no reference to the quantity of Diceros bicornis bicornis horn to be traded, the SS notes that the scope of the proposal is limited to a once-off trade in the Diceros bicornis bicornis horn stockpile, owned by the Government (~ 4,600 kg), which they believe to be a precautionary measure. The SS notes that extensive monitoring systems are in place and all forms of utilisation and trade are subject to strict permit control, with significant penalties applied to infractions. It is not clear from the proposal that this stockpile relates to Black Rhino horn only.

Supply side

From the supply side, the annotation and stipulations set out in the control measures ensure that strong management measures are likely to be put in place to control the supply of *Diceros bicornis bicornis* horn.

Domestic controls

The SS states that in Namibia, *Diceros bicornis bicornis* is classified as Specially Protected Game (wild animals) under the Nature Conservation Ordinance 4 of 1975 as amended, and any rhino (or any portion thereof), as well as any product derived from *Diceros bicornis bicornis* is classified as a Controlled Wildlife Product under the Controlled Wildlife Products and Trade Act (Act 9 of 2008) as amended. This means that permits are required to be in possession of *Diceros bicornis bicornis* or their parts, and for utilization, movement, imports and exports. The SS notes that ample control measures are in place to ensure sustainable use and management of the *Diceros bicornis bicornis* population, and preventing illegal trade from impacting on the national population. However, all *Diceros bicornis bicornis* belongs to the State and Namibia.

International controls

Permit control: the SS states that The Ministry of Environment, Forestry and Tourism permit office at Windhoek issues all permits relating to *Diceros bicornis bicornis* and their parts or derivatives. No competencies are delegated to local or regional authorities. In the case of hunting, only Namibian registered professional hunters and operators/outfitters are allowed to conduct hunting. The Directorate of Veterinary Services in Windhoek issues all veterinary permits. A strictly applied permit system thus already exists, hunting and trade in hunting trophies would continue to be strictly enforced.

Marking of animals/products: the SS states that marking of animals/products is common practice already, and in the interest of facilitating control and law enforcement, hunting trophies are required to be sampled for DNA profiling. DNA samples are submitted to University of Pretoria for the inclusion in the RhODIS database. All horns are additionally allocated and marked with National serial numbers, and paperwork is traceable.

Customs and border control: the SS states that Namibian Customs Officers check CITES, veterinary and transit permits. Where necessary, they refer to the Namibian Police or district veterinary officer.

Law enforcement: the SS states that law enforcement is a joint effort by the Ministry of Environment, Forestry and Tourism, the Protected Resources Division of the Namibian Police and the Customs Service. Law enforcement agencies rely primarily on information and well-established informer networks exist and are maintained.

Namibia provides a leading example of traceability and control. The country uses an electronic database that monitors both state-held and privately held stockpiles. Approximately 70% of government-managed horns are genetically profiled and microchipped, with data integrated into the Rhino DNA Index System (RhODIS). In the private

sector, 80% of horns have been sampled, although only legally acquired historic horns are included under current permit frameworks. This approach enhances accountability and acts as a deterrent to illegal activity, though it depends on regular updates and cooperation from private custodians (Ferreira et al., 2025).

Namibia has developed strong enforcement and intelligence networks that rely on contributions from multiple government bodies. However, these systems often depend on donor support and joint staffing arrangements, which could be vulnerable to future institutional or financial shifts (Ferreira et al., 2025).

Namibia's Conservancy Rhino Ranger Incentive Programme provides one of the clearest examples of community-led rhino protection. Rangers are drawn directly from local communities and supported with training, stipends, and recognition. This programme strengthens local responsibility and builds capacity for rhino monitoring at the landscape level (Ferreira et al., 2025).

Namibia has launched a dedicated Environmental Crime Court in Otjiwarongo in 2025 following the success of temporary wildlife courts. This specialised court handles high-profile wildlife cases more efficiently and is supported by inter-agency coordination mechanisms (Ferreira et al., 2025).

Namibia's Blue Rhino Task Team combines investigators from the police, Ministry of Environment, and other security agencies. The task team is supported by shared vehicles, intelligence systems, and donor contributions, enabling rapid response and data-driven decision-making (Ferreira et al., 2025).

Demand side

The SS provides sparse information on how the demand side of the trade in rhinoceros horn will be managed, noting that there will only be trade to trading partners that have been verified by the Secretariat, in consultation with the Standing Committee, to have sufficient national legislations and domestic trade controls, and not before the Secretariat has verified prospective importing countries and the registered stocks.

There are currently no legal end markets for rhinoceros horn, so if trade were to be allowed it is not clear what measures might be in place to prevent horn from illegal sources entering the trade chain.

The SS states that an estimated eight tonnes (equivalent to at most 1,500 rhinos) of horn are currently trafficked illegally each year. However, for the last five years, detected African rhino poaching mortalities range between 500-600 per year. Assuming two horns per rhino, weighing an average of 2.57 kg per horn for African species, would mean that in the range of 2.5 - 3 tonnes of horn are currently produced from poached rhino (between 500-600 detected poached rhinos per year over the last five years), noting that not all poached rhinos are detected and seizures reduce the volumes entering trade (Ferreira et al., 2025).

Additional Information

Global Distribution

Today putative South-western Black Rhino or *Diceros bicornis bicornis* range includes Namibia, southern Angola, western Botswana, and southwestern and southeastern South Africa (up to the Kei River), although today they occur only in Namibia (the stronghold) and South Africa with a sighting of one animal in Angola and unconfirmed reports of possibly another three animals (Emslie, 2020). In 2020, the minimum known area of potential rhino habitat was just under 50,000 km², with 41 subpopulations present (Emslie, 2020).

Global Population

Numbers of *Diceros bicornis bicornis* had increased 380% from an estimated 456 to an estimated 2,188 in three-generations (1973–2017) based on the IUCN SSC African Rhino Specialist Group Red List historical numbers database. The lowest modelled change over a three-generation moving window up to five years into the future was a 373% increase with the highest a 414% increase. Numbers have also increased by an estimated 339% over two generations (1988–2017) and by an estimated 78% over one-generation (2002–17). This is in contrast to the other two surviving subspecies/genetic management clusters whose numbers declined significantly up to the mid-1990s before then increasing afterwards (Emslie, 2020).

At the end of 2022, the population of *Diceros bicornis bicornis* in Africa was 2,636 individuals reflecting an increase from 2021. However, by the end of 2023, the number had decreased to 2,583, marking a reduction of 2.0%. This population reduction was particularly notable in Namibia, the primary range State for the subspecies, where conservationists had previously reported stability or slight increases in numbers. The decline was attributed to increasing poaching pressures, drought and natural mortality rates exceeding birth rates (Ferreira et al., 2025).

By the end of 2024, there were 2,597 *Diceros bicornis bicornis* living in two countries namely Namibia (80.8%) and South Africa (19.2%). These rhinos were distributed between eight populations in Namibia and seven in South Africa. Overall, *Diceros bicornis bicornis* did not change much over the reporting period with a weighted growth of 0.5% (95% Ci: -3.8% to 4.8%) since 2023. Namibia continued to report a decline, but growth in South Africa compensated

for those losses. Note: authorities in Namibia make use of sample counts for a key population and thus report estimates with confidence intervals. The other populations make use of identifying individual rhinos to obtain estimates. Considering this, the overall estimate for *Diceros bicornis bicornis* on the continent had a 95% confidence interval ranging from 2,555 to 2,757. This was not a substantial improvement since 2021 with annual growth since then of 0.5% (95% CI: -1.3% to 2.2%) (Ferreira et al., 2025).

Since 2021, the Namibian population changed by -1.2% (95% CI: -2.6% to 0.3%) and the South African population by 6.8% (95% CI: 6.0% to 7.6%) per annum with the overall change 0.5% (95% CI: -1.3% to 3.7%). At the end of 2024, 98.2% *Diceros bicornis bicornis* lived in wild ranging conditions, State (60.2% of individual rhinos) and partnership management models (36.0%) were protecting *Diceros bicornis bicornis*. The median area available to *Diceros bicornis bicornis* at a locality was 680 km² but population sizes were generally small with a median of 28 individuals.

Global Trade

It has been estimated that around 95% of horn sourced in Africa for illegal markets in Asia in 2016 and 2017 came from poached animals in the wild. In some cases, horn for illegal markets has been stolen or illegally sold from horn stockpiles/private trophies/museum exhibits. Sport hunting of very small numbers of *Diceros bicornis bicornis* occurs - primarily in Namibia with permits only issued by the Ministry of Environment and Tourism in special cases. Profits from any hunt go into a Game Product Trust fund and can be used for approved projects in rhino areas (Emslie, 2020).

Live specimens for translocation are invariably taken from natural habitat, but this forms part of routine biological management of the species to meet genetic and demographic goals. This is because many established wild subpopulations require regular removals to prevent "overstocking" and maintain rapid population growth rates. Maintaining rapid underlying population growth rates also minimises loss of genetic diversity over time due to genetic drift (Emslie, 2020). Namibia has not exported any live Black Rhino between 2014 and 2023.

Threats

Global

Diceros bicornis bicornis face a variety of threats. The main threat to the species is illegal hunting (poaching) to supply the illegal international rhino horn trade. It is estimated that currently around 95% of rhino horn sourced in Africa for end user illegal markets in SE Asia are from this source. Rhino horn has traditionally had two main uses: use in Chinese medicine, and ornamental use (Emslie, 2020). Although in Viet Nam, rhino horn has also been used for gifting and medicinal purposes (USAID Vietnam, 2020). Recently rhino horn has become a highly prized material for making carved expensive high-status items such as bowls and bangles. In the past it was also used to produce ornately carved handles for ceremonial daggers (jambiyas) worn in Yemen and some Middle East countries. Historically rhino horn was also used in traditional Chinese medicine (as a fever reducer). However, most recently it appears to be shavings from carvings that are illegally sold to the medicinal market at lower prices than worked items. While *Diceros bicornis bicornis* numbers continue to increase at a continental level - poaching has slowed overall growth (Emslie, 2020).

Drought has affected Namibian populations of *Diceros bicornis bicornis* in recent years but there has been no clear impact on population in South Africa (Ferreira et al., 2025).

Conservation, management and legislation

Information related to Namibia indicates good progress regarding enforcement, having reported a relatively high number of arrests that span across all levels of poaching-related offences. Namibia's centralised wildlife crime database tracks court proceedings, bail hearings, and repeat offenders. This enables prosecutors to build stronger cases and monitor progress. The integration of case information into enforcement strategies has improved accountability, though results still depend on judicial capacity (Ferreira et al., 2025).

Artificial Propagation/captive breeding

The SS states that no captive breeding operations have been registered in Namibia.

Implementation challenges (including similar species)

The SS notes that parts of horn are not readily distinguishable from the horns of other species of rhinoceros. The proponent notes its belief that risks would be minimised by allowing only a once-off trade of the horn stockpile, but this seems at odds with the statement for Prop. 9 that an on-going export quota for horn of *Ceratotherium simum* is envisaged. All the horn pieces will be required to be marked as described already.

It should also be noted that horns of *Diceros bicornis bicornis* in the Namibian population would not be distinguishable from the South African population which could result in laundering of horns.

Potential risk(s) of a transfer from Appendix I to II

The SS states that critics of legalising trade argue that illegal demand may escalate under legal trade, particularly if the removal of legal restrictions broadens the consumer base and removes the assumed stigma associated with purchasing an illegal commodity. *Consensus on legalising the trade in rhinoceros horn has not been achieved, with arguments both for and against the trade. In IUCN's latest contemporary strategic framework for African rhino conservation 2025-2035, six strategic themes are identified for achieving thriving rhinos valued by people, with one of the themes focussed on understanding and influencing rhino horn markets. Ultimately, there is a risk that even allowing once off sale of rhinoceros horn could increase demand, and it should be noted that while Namibia may have the funds after the sale to protect their population, other range States may not have the finances to combat the potential ensuing demand (Balfour et al., 2025).*

Potential benefit(s) for trade regulation of a transfer from Appendix I to II

As highlighted above, Namibia is successfully managing its *Diceros bicornis bicornis* population, however this comes at a significant cost, with an increasing financial burden caused by the increasing threat of illegal killing, proposing that this threat can be addressed through the sustainable use of black rhinoceros. The SS states that funds from the sale of horns are desperately needed to support Namibia's conservation efforts. The funds generated would be used to maintain or improve the conservation management programmes on which various rhinoceros species depend. The SS states that law enforcement including anti-poaching and intelligence activities are extremely expensive, and unlikely on their own to succeed in the long term.

The SS states that proceeds from the trade are used exclusively for rhinoceros conservation and community development programmes within or adjacent to rhinoceros range and therefore this trade may therefore support rhinoceros conservation efforts.

References

- Balfour, D. Ferreira, S.M., Gaymer, J., Lewis, C., Mafumo, H., Makoma, K., Mgoola, W., Reuben, M., Shaw, J.A. and Uri-Khob, S. (2025) African rhino conservation 2025–2035: A contemporary strategic framework. IUCN.
- Emslie, R. (2020) *Diceros bicornis* ssp. *bicornis*. *The IUCN Red List of Threatened Species* 2020: e.T39318A45814419. <https://dx.doi.org/10.2305/IUCN.UK.2020-1.RLTS.T39318A45814419.en>. Accessed on 2 August 2025.
- Ferreira, S.M., Baruch-Mordo, S., Laidler, G., Stoppoloni, R., Sangalakula, L., Talukdar, B. and Balfour, D. (2025) The African and Asian rhinoceroses – status, conservation and trade: A report from the IUCN Species Survival Commission (IUCN/SSC) African and Asian Rhino Specialist Groups and TRAFFIC to the CITES Secretariat pursuant to Resolution Conf. 9.14 (Rev. CoP15). CoP20 Doc. 84, Annex 3, CITES Secretariat, Geneva, Switzerland.
- 't Sas-Rolfes M., Emslie R., Adcock K. and Knight M. (2022) Legal hunting for conservation of highly threatened species: The case of African rhinos. *Conservation Letters* 15:e12877.
- USAID Vietnam (2020) Research Study on Consumer Demand for Elephant, Rhino and Pangolin Parts and Products in Vietnam.

Inclusion of Hoffman's Two-toed Sloth *Choloepus hoffmanni* and Linné's Two-toed Sloth *Choloepus didactylus* in Appendix II

Proponents: Brazil, Costa Rica, Panama

Summary: Hoffmann's Two-toed Sloth *Choloepus hoffmanni* and Linné's Two-toed Sloth *Choloepus didactylus* are arboreal mammals, widely distributed in Central and South America which together comprise the family Megalonychidae. There are four other sloth species, all in the family Bradypodidae. Two of these (*Bradypus pygmaeus* and *B. variegatus*) were included in Appendix II in 1983.

Choloepus hoffmanni is distributed from Honduras in Central America through northwestern South America to Bolivia and western Brazil, while *C. didactylus* occurs in the northern Amazon forest from Brazil to Ecuador. Both species are forest-dependent, nocturnal, and exhibit cryptic behaviour, making assessment of population status challenging. Both were globally assessed as Least Concern on the IUCN Red List in 2022. Some subpopulations have been assessed as threatened at the national level, as evidence suggests that local populations, particularly of *C. hoffmanni* in Colombia, Bolivia, and parts of Central America, are experiencing decreases mainly due to ongoing habitat loss and fragmentation caused by land use change and forest fires. *C. hoffmanni* is proposed for inclusion in Appendix II accordance with Article II, paragraph 2(a) of the Convention, and *C. didactylus* for 'lookalike' reasons in accordance with paragraph 2(b).

There are no global population size estimates for either species, but both are reported to be locally common. Population density estimates for *C. hoffmanni* range from 0.2 to 1.5 individuals per hectare in various habitats across Colombia and Panama. *C. didactylus* has been recorded at densities between 0.13 and 0.9 individuals per hectare in the Brazilian Amazon and French Guiana. Recent field observations in central-southern Amazonia indicate either a significant southward range expansion for *C. didactylus* or, more likely, that the species has long been present there but hitherto unrecorded. Both species are affected by habitat loss, deforestation for agriculture and cattle ranching, forest fires, logging, and urban expansion. Despite these pressures, they exhibit some adaptability to disturbed landscapes, with *C. hoffmanni* in particular occurring in modified habitats.

Given the extensive ranges of both species, which run to many tens of millions of hectares, if these species occur in even a fraction of this area at densities comparable to those observed in studies (0.13 to 1.5 per hectare), then the global populations of both could be very large.

Both species are subject to some domestic use, for food and as medicines, but there are no indications that this has a significant impact on wild populations. Sloths in international markets are almost exclusively traded as live animals for the exotic pet trade. *Choloepus hoffmanni* was included in Appendix III by Costa Rica from 1976 to 2019 during which time 39 live individuals were reported as exported or re-exported. During this period, importing countries reported the import of 80 individuals (almost all to the USA). Most traded individuals were recorded to be wild-sourced, with a small number in trade reported as captive-bred (from the USA). US LEMIS data report somewhat higher levels of imports of *C. didactylus* with some 1340 wild-sourced individuals reported between 2008 and 2020, virtually all of these from Guyana. Legal trade among other countries is poorly documented, and no comprehensive global trade data are available. There is no evidence of commercial-scale captive breeding for either species.

Some illegal trade has also been documented, although it appears largely domestic in nature with animals reportedly taken for sale as pets or for display to tourists. From 2015 to 2019, over 1,100 sloths were seized in Colombia, though species-level identification was generally absent (both *Choloepus* species and *Bradypus variegatus* occur there). There is some evidence of illegal export using fraudulent permits. Between 2021 and 2025, around 170 sloths were reported as illegally exported from Peru (where both *Choloepus* species and *Bradypus variegatus* also occur) to countries including the United Arab Emirates, Czechia and Malaysia. Animals can reach relatively high prices (several hundred USD) on international markets.

CITES-reported trade in the Appendix-II listed *Bradypus variegatus* has been at a very low level since the species was included in the Appendices in 1983, with only 15 live animals recorded in trade, all except two imported into the USA (the second listed species, *B. pygmaeus*, is an isolated island taxon from Panama split off from *B. variegatus* in 2001 and has not appeared in trade).

Historically, distinguishing between the two species has been difficult due to overlap in their external morphology. The most apparent external differences are in pelage colour (darker in *C. didactylus*), body size and skull morphology. However, these traits are most evident in adults and are therefore of limited use when young animals are in trade.

Analysis: Both species of *Choloepus* sloths have extensive ranges in the Americas, can adapt to modified habitats and occur at moderate densities in parts of their range (up to 1.5 individuals per ha in the case of *C. hoffmanni*). Both species are currently assessed as Least Concern on the IUCN Red List and likely have large or very large populations, which are however thought to be declining overall, principally owing to large-scale habitat conversion. Both are known to have been traded legally and illegally as live animals for the exotic pet trade, predominantly for domestic markets, but reported numbers involved are small and are unlikely to have significant impacts on the wild population.

Based on such relatively low international trade levels, there is no indication that regulation of trade is required to avoid these species becoming eligible for inclusion in Appendix I in the near future, nor to ensure that harvest is not reducing the wild population to a level at which the survival of either of them might be threatened by continued harvesting or other influences. On this basis **neither species meets the criteria for inclusion in Appendix II set out in Annex 2a or b** of Res. Conf. 9.24 (Rev. CoP17).

Summary of Available Information Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

Choloepus hoffmanni was included in Appendix III by Costa Rica from 1976 to 2019, after which it was delisted. *Choloepus didactylus* has never been listed in CITES Appendices.

Taxonomy

Choloepus hoffmanni (Peters, 1858) and *Choloepus didactylus* (Linnaeus, 1758)

CITES standard reference: Wilson, D.E. and Reeder, D.M. (Eds.) (2005) *Mammal species of the world, a taxonomic and geographic reference*. 3rd Edition, The Johns Hopkins University Press, Baltimore, Maryland. 2, 142pp.

Range

Choloepus hoffmanni: Bolivia, Brazil, Colombia, Costa Rica, Ecuador, Honduras, Nicaragua, Panama, Peru, Venezuela.

Choloepus didactylus: Brazil, Colombia, Ecuador, French Guiana, Guyana, Peru, Suriname, Venezuela.

IUCN Global Category

Choloepus hoffmanni – Least Concern (assessed 2022, ver. 3.1)

Choloepus didactylus – Least Concern (assessed 2022, ver. 3.1)

Biological and trade criteria for inclusion in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)

B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences

Choloepus hoffmanni Hoffmann's Two-toed Sloth:

There are two distinct subpopulations: the northern subpopulation, which extends southwards from the north of Honduras to northwest Ecuador and western Venezuela; and the southern population, which extends from north-central Peru to the far west of Brazil (southwest of Amazonas, Rondônia, Acre and Mato Grosso) to the north of Bolivia.

The species occurs from sea level to 3,300 m asl in Costa Rica; up to 1,925 m asl in Panama; and up to 1,150 m asl in the southern Andes of Venezuela. In Colombia, it is found in the biogeographical regions of the Andean

zone, Caribbean, and Chocó, specifically in the departments of Cauca, Chocó, Cundinamarca, Nariño, Quindío, Sucre, Valle del Cauca, and Santanderes, from sea level up to 3,200 m asl.

According to the proponents, the species produce one individual per litter, and pregnancy lasts between 340-377 days. *Additional information provided by members of the IUCN SSC Anteater, Sloth and Armadillo Specialist Group notes females give birth to a single offspring about every 15 months. The young are usually dependent on their mothers for more than 200 days, which suggests that population recovery from any impact will be slow (Moraes-Barros et al., in litt., 2025).*

Population densities range from 1.05 individuals per hectare on Barro Colorado Island, Panama, to 0.3–1.5 animals per hectare in the Andean region of Colombia, and 0.2–0.83 individuals per hectare in the lowlands of northern Colombia. Biomass estimates per hectare are 3.5 kg for adults and 0.1 kg for juveniles. In Panama, *C. hoffmanni* represents between 12.3% and 14.7% of the estimated mammal biomass.

Although no global population estimate exists, these local density studies—along with the species' wide distribution and habitat flexibility—suggest a relatively large total population size. Its generalist diet and use of a wide range of forest types, including secondary and disturbed habitats, support this assumption (Chiarello et al., 2022; Moreno and Plese, 2006; Rivas et al., 2023).

The global population trend is assessed as decreasing. Some subpopulations, especially those in Colombia, Central America, Bolivia, and Brazil, are experiencing drastic declines due to habitat loss.

*According to the IUCN Red List assessment habitat degradation in Honduras is linked to extensive cattle ranching and agriculture (Chiarello et al., 2022). In Bolivia, large wildfires used to clear land for agriculture result in direct mortality through burning or suffocation, as well as indirect effects through habitat loss. In Brazil, known occurrence records mainly come from the deforestation arc in the Amazon, which is responsible for 75% of deforestation in the biome. It is estimated that habitat loss due to the expansion of pasturelands and soybean crops has amounted to 28% over three generations (i.e. 42 years). In Colombia, for instance, the UNAU Foundation reports frequent rescues of *C. hoffmanni* individuals from semi-urban areas, likely due to sloths dispersing from lost habitats or cases related to the destruction of remaining patches of natural forest (Moreno and Plese, 2006).*

Despite the above, Choloepus hoffmanni appears more resilient than species like Bradypus variegatus (listed in Appendix II), owing to its larger home ranges, greater mobility, nocturnal and solitary behaviour, and relatively aggressive nature. As a result, it is more adaptable to habitat alteration. Individuals of this species have been observed in a diversity of habitats, frequently isolated and disturbed, such as small patches of secondary forest (less than 10 ha) near urban centres, and in patches of remnant tropical dry forest in within cattle ranches (Moreno and Plese, 2006).

The IUCN assessment highlights its capacity to persist in human-modified landscapes, which likely buffers its global population against localised pressures (Chiarello et al., 2022).

Choloepus hoffmanni is listed as Least Concern due to its wide distribution, presumed large global population, and occurrence in a number of protected areas. Some subpopulations could potentially be assessed as near threatened or vulnerable. For example, In Costa Rica, the species is classified as threatened, and in Brazil, as near threatened, while in Honduras it is considered to be endangered.



Figure 1. Hoffmann's Two-toed Sloth *Choloepus hoffmanni* Distribution Map (IUCN (International Union for Conservation of Nature) 2022. *Choloepus hoffmanni*. The IUCN Red List of Threatened Species. Version 2025-1).

Choloepus didactylus

The Linné's Two-toed Sloth *inhabits tropical moist lowland and montane forests in Brazil, Colombia, Ecuador, French Guiana, Guyana, Peru, Suriname and the Bolivian Republic of Venezuela.*

Regarding reproduction in the species, females start breeding at around 3 years of age, and males at approximately 4.5 years of age. Gestation lasts about 10 months, but estimates are quite variable. A single offspring is born per litter, which is weaned at 3–5 months of age (Moraes-Barros et al., in litt., 2025).

In French Guiana, densities of C. didactylus have been recorded at 0.9 animals per hectare (Taube et al., 1999). In the Brazilian Amazon, densities range from 0.13 individuals per hectare near Manaus to 0.88 animals per hectare in flooded forests such as the Mamirauá Reserve (Queiroz, 1995; Chiarello, 2008).

These values suggest that the species can be locally common, particularly in intact habitats, and indicate the potential for large population sizes across its broad range (Chiarello et al., 2022).

Bezerra-Castro et al. (2024) documented new records of Choloepus didactylus extending up to 820 km south of its current geographic distribution. These records, and others found in literature for the central-southern Amazon, evidence a southern expansion of about 30% of the known species' area of occurrence, indicating either adaptation to new habitats or historically underestimated distribution.

Such records reinforce the assumption of a wide and potentially underestimated distribution, which further supports the IUCN's Least Concern assessment (Bezerra-Castro et al., 2024).

Although there is no comprehensive global population assessment, Choloepus didactylus is listed as Least Concern by the IUCN due to its wide distribution, presumed large population and its occurrence in several protected areas. The species faces various threats, especially habitat loss due to deforestation driven by agriculture, logging, mining, and urban expansion -particularly in Suriname- and habitat loss in Brazil has reached an estimated 9.4% of its extent of occurrence due to the expansion of pasturelands and soybean cultivation. Wildfires also pose a threat. However, the species is unlikely to be declining fast enough to qualify for listing in a threatened category (Chiarello et al., 2022).

Its tolerance for a range of forest types and presence in multiple protected areas further suggest that the species does not appear to be under significant pressure at the global level (Chiarello et al., 2022).

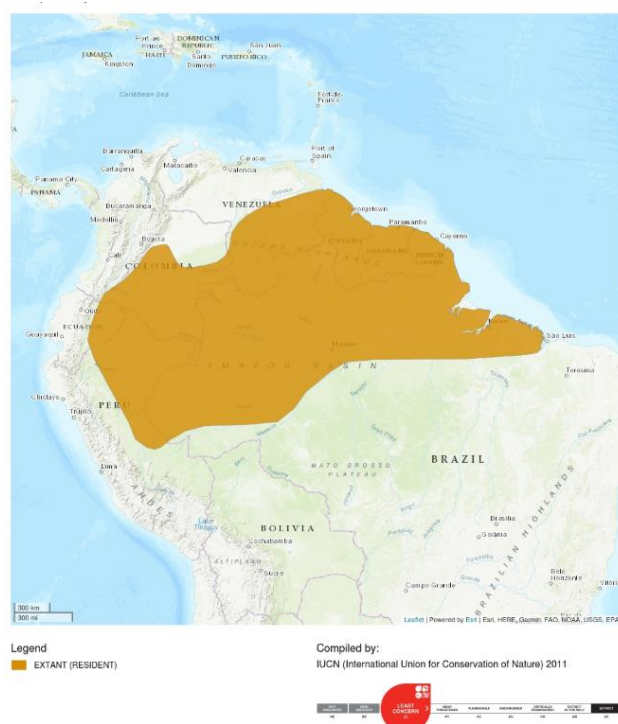


Figure 2. Linné's Two-toed Sloth *Choloepus didactylus* Distribution Map IUCN (International Union for Conservation of Nature) 2011. *Choloepus didactylus*. The IUCN Red List of Threatened Species. Version 2025-1).

Trade

Choloepus hoffmanni is used for food, as well as for medicinal, magical/religious, and ornamental purposes, and is also sold as a pet. Legal international trade exists but remains at low levels, with Honduras, Panama, Nicaragua, and Costa Rica as the main exporters, and the USA, Japan, Hong Kong SAR of China, Bahamas, and Norway as the main importers. Exports include live animals, hair, skin, skulls, specimens, tails, bones, skeletons, and powders. The SS states that the primary concern is illegal trade for both national and international pet markets. The SS also indicates that this illegal trade occurs in nearly all range countries, and that specimens are also smuggled to non-range countries such as Mexico, the USA, and various countries in Europe, Asia, and the Middle East. Both national and international illegal trade have increased over the past decade.

Legal trade

Choloepus hoffmanni was included in CITES Appendix III by Costa Rica from 1976 to 2019, when the species was removed from the Appendices. The CITES Trade Database show that from 1985 to 2019, CITES Parties recorded the export of 39 live specimens of *C. hoffmanni*, however, importing countries recorded 80 imported specimens. As *C. didactylus* has never been listed in the CITES Appendices, no trade data are available from the CITES Trade Database.

If we consider only direct trade records (i.e. excluding re-exports) of C. hoffmanni, from 1976 to 2019, exporters reported trading 24 live individuals, whereas importers reported receiving 75. Of these, only 10 individuals recorded by exporters were wild-sourced, compared to 28 recorded by importers. Of the total number of live specimens recorded in direct trade, only four of those recorded by exporters were for commercial purposes, as were 61 of those reported by importers.

The main exporters, as reported by exporters, were Guyana, Honduras, Panama, Nicaragua, and Costa Rica; however, Guyana is not a range State of *C. hoffmanni*. The main importing countries are the USA, Japan, Hong Kong SAR of China, the Bahamas, and Norway.

Analysis of LEMIS data from 2008 to 2020 show imports to the USA in Choloepus didactylus predominantly comprised 1,360 whole organism equivalent (WOE, comprising live animals, trophies, skulls, skins, skeletons, and bodies) items, specifically in live individuals (1,356, 98%). The live trade being predominantly in wild-sourced individuals for commercial purposes (1,346), with six of these being refused entry. Almost all of the trade in live individuals for commercial purposes, including the six refused entry, was exported directly by Guyana (1,312), with the remainder originating in Guyana (20) or non-range States (14, potentially representing reporting errors). This trade mostly steadily increased from 2008 to 2017, showing sharp increases in 2018 and 2020 (Figure 3).

Over the same 13-year period, 2008-2020, imports of WOE *Choloepus hoffmanni* items was much fewer. In total, only 14 live individuals for were imported, predominantly comprising 12 wild-sourced individuals for zoological (11) and personnel (1) purposes. One live captive-bred individual for commercial purposes imported from Canada was refused entry. This highlights the non-commercial nature of trade for this species into the USA.

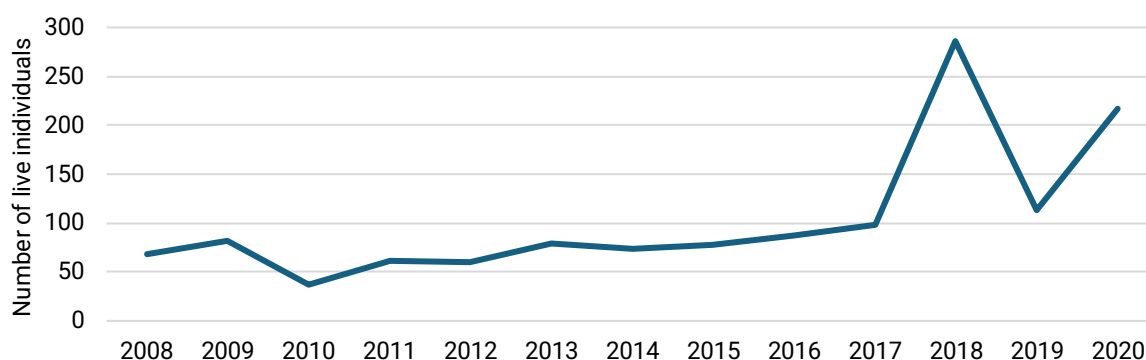


Figure 3. Imports to the USA of live wild-sourced *C. didactylus* for commercial purposes. Source: LEMIS.

The proponents note that due to challenges in species identification, some specimens in trade may have been mislabelled, a recurring issue noted by experts, as zoos often identified two-fingered sloths by their country of export (Moraes-Barros et al., in litt., 2025). For instance, animals shipped through Guyana were frequently recorded as *C. didactylus* regardless of their true origin (Moraes-Barros et al., in litt., 2025). According to the proponents, in 2024 and 2025, India imported 85 individuals of *C. didactylus* from Malaysia, Guyana, the United Arab Emirates, and Czechia.

Illegal trade

According to the proponents, sloths are poached in Central and South America. Wild-caught individuals, especially offspring, are sold as pets to tourists in Colombia and Honduras. This illegal trade is apparently a cause for concern due to its impact on the wild population. In some countries (e.g., Colombia), the illegal trade of sloths has been identified as a significant threat for conservation due to the large number of individuals being harvested from the wild, the extremely high mortality rates occurring prior to these animals arriving at pet markets, and low rates of survival in sloths being kept as pets. It is purported that high numbers of sloths were captured and sold in Colombia and Brazil, prior to 2015. It is estimated that 90% of all animals taken from the wild die while being hunted or during transport.

There are no seizure records for these species in EU-TWIX between 2014 and 2023. Meanwhile, data from WITIS between 2010 and 2025 includes one record of an incident involving one *C. hoffmanni* individual seized in a truck in Colombia in 2024, as well as one *Choloepus* spp. individual (no species identified) seized in a property in Ecuador. There were no records of incidents relating to international trade.

The IFAW report “Wildlife Crime in Hispanic America: An Analysis of Seizures and Poaching Incidents in 18 Countries (2017–2022)” documents the seizure of 20 live sloths in 10 incidents over six years, including only one *Choloepus hoffmanni*, three *Bradypus variegatus*, and 16 unidentified specimens. Most incidents were related to the local pet trade or use in tourist attractions, primarily in Amazonian regions such as Loreto, Madre de Dios, and San Martín. These numbers are very low and relate to domestic trade for pets or tourism, rather than international transactions.

Additional documented cases include the seizure of eight *Choloepus hoffmanni* individuals in a private residence in Tuxtla Gutiérrez, Mexico, in 2019; several rescues and seizures in Colombia between 2021 and 2024, including cases in Nariño, Valle del Cauca, and Cartagena; and the killing and attempted online sale of a *Choloepus didactylus* in Sucumbíos, Ecuador, in 2021 (Robin des Bois, 2019; 2021a; 2022; 2024).

A case of attempted international trade occurred in 2023 at Iquitos International Airport, Peru, where 30 *Choloepus didactylus* individuals were intercepted before being shipped to the United Arab Emirates. Eleven of them died due to poor transport conditions. Investigations linked the case to a company with a history of wildlife exports and raised concerns over the misuse of permits (Robin des Bois, 2023).

There is some evidence of these species being sold online⁵, but most Google results refer to sloths displayed in zoos or to informative or conservation-related websites. Nonetheless, demand for live baby sloths as pets appears

⁵ <https://www.exoticanimalsforsale.net/sloth-for-sale.asp>

to be increasing, particularly in the USA, where media outlets have recently highlighted the trend of keeping sloths as pets⁶.

According to information shared by the IUCN SSC Anteater, Sloth and Armadillo Specialist Group (Moraes-Barros et al., in litt., 2025), there is additional evidence of international trade in sloths, including reports of shipments to countries such as Malaysia, Thailand, the United Arab Emirates, and Czechia. While some of these cases are documented in media reports and public investigations, many do not include information at the species level, and they tend to group incidents involving sloths of both the genus *Choloepus* and *Bradypus*. However, the specialist group noted that most exported sloths belong to *Choloepus* spp., as they are easier to maintain in captivity. They noted challenges in accessing trade and seizure data from many countries, suggesting that the true extent of trade is likely underreported.

Seizures of sloths have been recorded in several Latin American countries, especially within the Amazon region. In Peru, most confiscations involved live *Choloepus* individuals, with 57 live sloths seized between 2018 and 2022. In Colombia, 1,165 sloths were seized between 2015 and 2019, although the species were not identified in official reports. In one department alone, records show 5 live *C. hoffmanni* seized, 153 voluntarily surrendered, and 104 found and rescued (Moraes-Barros et al., in litt., 2025).

The Anteater, Sloth and Armadillo Specialist Group also referred to recent high-profile cases involving illegal exports from Peru. In most cases, information at a species level is not available. These included nine sloths exported to Malaysia, as well as confirmed trafficking operations involving both *Choloepus* and *Bradypus* species. According to Ojo Público (2025), between 2021 and 2023, 70 wild-caught sloths were exported for USD28,500, and an additional 97 wild-caught sloths were exported in 2024–2025 for USD96,000. All of these exports used fraudulent permits obtained through corruption. There have also been reports of illegal sloth trade in Brazil and the USA, including seizures in areas outside the species' natural range.

Moreno and Plese (2006) report that sloth traffickers in Colombia prefer to capture offspring rather than adult sloths; in doing so, they often kill the mother. Many of these young sloths fail to adapt to captivity or die from stress-related causes before reaching their destination (Nuwer, 2013). As a result, the true impact of sloth trafficking is greater than what is reflected in official trade figures (Moraes-Barros et al., in litt., 2025). Members of the IUCN SSC Anteater, Sloth and Armadillo Specialist Group consulted for this analysis estimate that for each sloth that reaches the pet market in another country, at least nine others die due to stress and the challenges of adapting to a captive diet. Taken together, this suggests that for each sloth that reaches the pet market, there could be at least 20 losses to the wild populations (Moraes-Barros et al., in litt., 2025). However, these poached animals are frequently sold along roadsides to Colombian travellers moving between the interior and the coast for vacations, indicating that this trade is predominantly directed at the domestic market rather than international.

Inclusion in Appendix II to improve control of other listed species

A) Specimens in trade resemble those of species listed in Appendix II under Res. Conf. 9.24 (Rev. CoP17) Annex 2 a or listed in Appendix I

Choloepus hoffmanni and *C. didactylus* are the only currently recognised species in their genus and share the same basic morphology of two claws on the forelimbs and three on the hindlimbs, making them indistinguishable at the species level without close inspection (Mossman and Duke, 1973). *C. hoffmanni* is sympatric with the two-toed sloth *Bradypus variegatus* in most of its range, but sympatry with the three-toed sloth, *C. didactylus*, is limited to small portions of western South America between latitudes 0°S and 12°S. Their geographic distributions otherwise differ, with *C. hoffmanni* found from Central America into the northwestern Amazon, and *C. didactylus* occurring more broadly across the northern Amazon basin, including the Guiana Shield, northeastern Brazil, and extending westward into Colombia, Ecuador and Peru (Wilson and Reeder, 2005). The most apparent external distinction between the two *Choloepus* species lies in pelage coloration: *C. didactylus* typically exhibits darker pigmentation on the snout, hands and feet, whereas *C. hoffmanni* tends to have a lighter coat (Naples, 1982).

Historically, identification of two-toed sloth species has been difficult due to the large amount of overlap of their external morphology. The most reliable way of distinguishing between species is by body size and skull morphology. *C. didactylus* is, on average, larger (head body length: 540–880 mm; weight: 4–11 kg) than *C. hoffmanni*, although there is considerable overlap in the average ranges of the two species. It should be noted that these measurements apply to adult individuals, which is a significant limitation in the context of illegal trade given that specimens found in legal trade are usually juveniles.

⁶ <https://www.thesprucepets.com/pet-sloths-1239558>, <https://www.unep.org/news-and-stories/story/saving-sloths>, <https://spendonpet.com/baby-sloth-cost/>, <https://statestale.com/new-york/can-you-own-a-sloth-in-new-york/>, <https://resources.pangovet.com/pet-lifestyle/general/do-sloths-make-great-pets/>, <https://legalclarity.org/is-it-legal-to-own-a-sloth-as-a-pet/>, <https://petfools.com/small-pets/dos-sloths-make-good-pets-sm1/>, <https://geozoo.org/sloths-as-pets-everything-you-need-to-know/>

Additional Information

Threats

Both species face very similar threats mainly related to habitat loss.

Some subpopulations of *Choloepus hoffmanni*, particularly those in Colombia, Central America, Bolivia, and Brazil, are experiencing population decreases due to severe habitat degradation and fragmentation. In Honduras, this degradation is largely driven by the expansion of cattle ranching and agriculture. In Bolivia, large-scale wildfires often used to clear land for crops, pose a significant threat by directly killing animals or indirectly affecting them through the destruction of their habitat.

In Brazil, the few known records of *C. hoffmanni* occur in the so-called “arc of deforestation” in the Amazon, which accounts for most of the forest loss in the region. Within the species’ extent of occurrence, habitat loss has been estimated at 28% over three generations, primarily due to the expansion of pasturelands and soybean cultivation.

These sloths are also hunted by indigenous communities, and wild-caught individuals, particularly juveniles, are commonly sold as pets to tourists, a practice that is increasing and raising concern for its impact on wild populations. Individuals living in small and isolated forest patches are also vulnerable to predation by domestic dogs and other terrestrial animals.

In agroforestry systems in Costa Rica, sloths are exposed to pesticides, and electrocution has become an increasing threat in urban areas due to power line infrastructure.

C. didactylus is also affected by habitat loss resulting from deforestation linked to large-scale agriculture, logging, mining activities, and urban expansion, particularly in Suriname. In Brazil, habitat loss within its extent of occurrence (EOO) due to the expansion of pasturelands and soybean cultivation has been estimated at 9.4%. Wildfires also pose a threat, either by directly killing individuals or through the degradation of their habitat (Chiarello et al., 2022). Climate change has been identified as an additional threat. In Suriname, the drought in 2019 led to a significant mortality of sloths (Moraes-Barros et al., in litt., 2025).

The impact of the illegal pet trade, which occurs across the species’ range, is hard to quantify. Due to their cryptic, arboreal habits and cultural taboos, sloths are not heavily targeted for bushmeat and are likely hunted only opportunistically. Illegal trafficking, however, appears to be increasing in countries like Bolivia, where individuals are captured and sold as pets, despite national legislation prohibiting wildlife ownership (Chiarello et al., 2022).

Conservation, management and legislation

Range States have several national laws that regulate the use, trade and protection of sloth species. In Costa Rica, for example, wildlife removal and trade are restricted to authorized management sites for conservation or research purposes. Panama requires prior authorization for the use and transport of wildlife species. Colombia has strict laws prohibiting unauthorized capture, possession, and trade of wildlife, with severe penalties for violations. Ecuador regulates the use of wild flora and fauna through the Ministry of the Environment. Peru prohibits wildlife use without authorization. Bolivia bans keeping wild animals as pets. Brazil treats wildlife as state property, prohibiting capture, trade, and export of wild specimens except from authorized facilities; exports of captive-bred animals are permitted but no commercial breeding facilities for *Choloepus* species currently exist.

In Colombia, *in situ* and *ex situ* conservation strategies have been implemented to protect sloth populations facing threats and illegal use, including national strategies for illegal trade control and conservation programs. In Suriname, sloths are fully protected: *it is prohibited to keep them (or their parts), trade in them, or capture them sloths (Moraes-Barros et al., in litt., 2025).*

Most of the Choloepus spp. range countries state that export of wild-sourced specimens is not allowed. There is some concern that sloths illegally captured in range countries that have stricter legislation are transported across borders to neighbouring countries (such as Guyana in the case of C. didactylus) and subsequently exported as if they were legal to destinations in North America, Europe, and Asia (Moraes-Barros in litt., 2025).

There is no specific information available on population monitoring or international control measures.

Artificial Propagation/captive breeding

Following consultation with the IUCN SSC Anteater, Sloth and Armadillo Specialist Group, no dedicated sloth breeding or reproduction centres are currently known to exist.

According to the Zoological Information Management System (ZIMS), a total of 183 C. hoffmanni individuals are currently held in 54 institutions across 12 countries worldwide. These institutions include zoos, rescue and rehabilitation centres, and other facilities that house sloths, but they are not necessarily specialised in breeding or reproduction. The majority of these are located in the USA (37 institutions housing 65 sloths) and Costa Rica (four

institutions housing 93 sloths) (Species 360, 2025). However, according to the IUCN SSC Anteater, Sloth and Armadillo Specialist Group, these figures represent an underestimate, as only a minority of rescue and rehabilitation centres, as well as institutions not accredited by AZA, EAZA or similar associations, use ZIMS (Moraes-Barros et al., in litt., 2025).

Regarding *C. didactylus*, 620 individuals are held by 270 institutions across 43 countries. The majority of these are in the USA, which holds 189 individuals. Additional populations in Latin America are found in Mexico, Colombia, Brazil and French Guiana, with many others distributed throughout various European countries and a few in Asia and South Africa.

Both AZA and EAZA manage population programmes for the two *Choloepus* species: AZA through a Species Survival Plan (SSP) and EAZA through a European Studbook (ESB).

Implementation challenges (including similar species)

Implementation challenges are those implicit in any species regulated by CITES. These include having sufficient and up-to-date information on species populations to develop reliable Non-Detriment Findings and strengthening enforcement efforts to ensure compliance with CITES regulations.

Potential benefit(s) of listing for trade regulation

Inclusion of a species in Appendix II would introduce requirements for monitoring, data collection, trade records, and enforcement cooperation. It would also require range States to make NDFs before authorising exports, which is a particularly relevant measure for Guyana as it is both the country of origin, and the main exporter of most shipments recorded as entering the USA. These processes could provide more reliable information on trade levels and sources.

References

- Bezerra-Castro, A., Melo-Dias, M., Santos, T.P., Custódio, R.J., Rodrigues, D. de J. and Alves-Rosa. (2024) Lost in Amazon: Evidencing the southern extension of the geographic distribution of Linné's two-toed sloth (*Choloepus didactylus*). *Biotropica*. <https://doi.org/10.1111/btp.13353>
- Chiarello, A., Plese, T., De Thoisy, B., Pool, M., Aliaga-Rossel, E., Santos, P. and Moraes-Barros, N. (2022) *Choloepus didactylus*. The IUCN Red List of Threatened Species 2022: e.T4777A210443323. <https://dx.doi.org/10.2305/IUCN.UK.2022-2.RLTS.T4777A210443323.en>.
- IFAW (2023) *Wildlife Crime in Hispanic America: An Analysis of Seizures and Poaching Incidents in 18 Countries (2017–2022)*. International Fund for Animal Welfare.
- Moreno, S. and Plese, T. (2006) The illegal traffic in sloths and threats to their survival in Colombia. *Edentata* 7:10–18.
- Moraes-Barros, N., Superina, M., Aguilar Borbón, A., Turcios Casco, M. and Meritt Jr., D.A. (2025) in litt. to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Mossman, H.W. and Duke, K.L. (1973) *Comparative Morphology of the Mammalian Ovary*. University of Wisconsin Press. <https://pmc.ncbi.nlm.nih.gov/articles/PMC1271607/>
- Naples, V.L. (1982) *Cranial osteology and function in the tree sloths, Bradypus and Choloepus*. *American Museum Novitates*, No. 2739: 1–41. <https://digitallibrary.amnh.org/items/fbffc7b-8ef6-4b21-96c9-098b657687d9>
- Nuwer, R. (2013) Sloths Are Number One on the List of Illegally Traded Pets from Colombia. *Smithsonian magazine*. Available at: <https://www.smithsonianmag.com/smart-news/sloths-are-number-one-on-the-list-of-illegally-traded-pets-from-colombia-84585357/>
- Ojo Público (2025) *Depredadores del oriente: así operaba la mafia del tráfico de animales*. Available at: <https://ojo-publico.com/5808/depredadores-del-oriente-asi-operaba-la-mafia-traffic-animales>
- Robin des Bois (2019) *On the Trail*, No. 26. Available at: https://robindexbois.org/wp-content/uploads/ON_THE_TRAIL_26.pdf
- Robin des Bois (2021a) *On the Trail*, No. 32. Available at: https://robindexbois.org/wp-content/uploads/ON_THE_TRAIL_32.pdf
- Robin des Bois (2021b) *On the Trail*, No. 33. Available at: https://robindexbois.org/wp-content/uploads/ON_THE_TRAIL_33.pdf
- Robin des Bois (2022) *On the Trail*, No. 37. Available at: https://robindexbois.org/wp-content/uploads/ON_THE_TRAIL_37.pdf
- Robin des Bois (2023) *On the Trail*, No. 42. Available at: https://robindexbois.org/wp-content/uploads/ON_THE_TRAIL_42.pdf
- Robin des Bois (2024) *On the Trail*, No. 44. Available at: https://robindexbois.org/wp-content/uploads/ON_THE_TRAIL_44.pdf
- Species360 Zoological Information Management System (ZIMS) (2025) zims.Species360.org
- Wilson, D.E. and Reeder, D.M. (eds.) (2005) *Mammal Species of the World: A Taxonomic and Geographic Reference*. 3rd ed. Baltimore: Johns Hopkins University Press.

Transfer of Golden-Bellied Mangabey *Cercocebus chrysogaster* from Appendix II to Appendix I

Proponent: Democratic Republic of the Congo

Summary: The Golden-Bellied Mangabey *Cercocebus chrysogaster* is the largest of seven *Cercocebus* mangabey species, endemic to the Central Congo basin of the Democratic Republic of the Congo. Its total known range covers 80,000 km² split into two smaller subpopulations. The species, which was considered a subspecies of *Cercocebus agilis* until 2007, occurs in variety of forest habitats at an elevation of 300-500 m, and is largely terrestrial. Little is known of its social structure; fecundity is thought to be low, with one infant produced per year and sexual maturity reached at four to five.

C. chrysogaster was categorised as Endangered in the IUCN Red List in 2020, with a population estimated at 10,000-30,000, considered likely to have declined by 40% over two generations and projected to be decreasing at a rate of 50% over 30 years (three generations, 1999-2029). The species is primarily threatened by a combination of high hunting pressure for wild meat and habitat loss and degradation, with an estimated 30% loss of habitat over the last 20 years, owing to logging and land conversion. Wild meat trade is reported to be a major threat, with the species frequently recorded in market and roadside sales but decreasing availability reported in recent years.

The species was included in CITES Appendix II in 1977, along with the rest of the order Primates. Some legal trade has been reported over the past ten years, with 146 live individuals from wild sources exported from the Democratic Republic of the Congo to countries including Ghana, Thailand, and the United Arab Emirates. There are reports of ongoing illegal international trade in small numbers. No national management is in place, the species is not explicitly protected by national law, and very little population monitoring occurs. Minimal captive breeding occurs, with little success globally.

Analysis: According to the most recent assessments the Golden-bellied Mangabey does not have a small population, nor does it have a restricted range. It has been harvested extensively for the wild meat, for national consumption, with reports of reduced availability indicating a decreasing population, associated with overharvest and habitat loss. In the absence of greater local conservation action, this decrease is projected to reach levels that could be approaching the guideline level for a marked recent decline.

The species has been reported in international trade as live animals in small numbers (around 15 per year in the past ten years), but it seems highly unlikely that this has a significant impact on the wild population compared with harvest for local consumption and loss of suitable habitat. Overall, it therefore **seems unlikely that the species meets the criteria for inclusion in Appendix I** in Res. Conf. 9.24 (Rev. CoP17).

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

Cercocebus chrysogaster was first included in CITES Appendix II in 1977.

Taxonomy

Cercocebus chrysogaster (Lydekker, 1900)

The species has alternately been assigned as a subspecies of Cercocebus agilis or Cercocebus galeritus but retained as a separate species by Groves (2005) and elevated at CoP14 in 2007.

Range

Endemic to the Democratic Republic of the Congo

IUCN Global Category

Endangered A4cd (assessed 2020, ver. 3.1)

Biological criteria for inclusion in Appendix I**A) Small wild population**

No population estimate is available, but the population trend was reported to be decreasing in 2020. *C. chrysogaster* was reported to be locally abundant, with groups of more than 100 individuals, in the western subpopulation, but eastern subpopulations had fewer than 10 individuals per group. *C. chrysogaster* breeds annually, with one infant produced, reaching maturity at four to five years of age. With low reproductive rates and low population densities, this species is highly vulnerable to external factors. *One expert roughly estimates 10,000-30,000 individuals in the wild (Edward McLester, 2024 quoted in Kimbrough, 2024), with other experts estimating an even smaller population size (Maisels in litt., 2025).*

B) Restricted area of distribution

The species is endemic to the Democratic Republic of the Congo, with the total area of occupancy approximately 80,000 km², split into two subpopulations with limited dispersal between these, *and fragmentation within them (Dempsey et al., 2024)*. The subpopulations have a western area of 68,000 km² and an eastern range of around 12,000 km², separated by 300 km. Habitat loss and deforestation are occurring, and over the last 20 years at least 32% of the species' potential habitat has been lost due to logging and land conversion, resulting in a marked decline in range area and occupancy. *Global Forest Watch data indicate contiguous forested areas throughout the range of the species, with the Mai-Ndombe region losing 6.1% and the Sankuru region 7.1% between 2002 and 2024. The current known Area of Habitat for this species is 70,735 km² (Ridley et al., in prep).* Some local extirpations have been reported, with a declining range size, and disappearance from forest within two decades of intensive logging. Thirty per cent of the remaining forest habitat has been allocated to forest concessions.

C) Decline in number of wild individuals

McLester et al. (2025) indicate that the species is decreasing across its range. Over the past 20 years, the species has disappeared from a cumulative 39% of its total range, and population decreases have been recorded in 55% of the eastern forest block. The total decline between 1999 and 2019 is estimated at 40% of the population, and it is suspected that the decline over a 30-year period (three generations for this species) will exceed 50%. Interviews with hunters and market surveys indicate a serious decrease in wild populations, with a decrease in 80% of villages, and 25% of villages reporting the species disappearing altogether between 2002 and 2012. Surveys of wild meat markets in 2015 showed a decrease in *C. chrysogaster* presence between 2010 and 2015. In response to recent population decreases, the species was included in the IUCN Primate Specialist Group's list of the 25 most endangered primates in the world in 2022 and 2024.

Trade criteria for inclusion in Appendix I**The species is or may be affected by trade**National

C. chrysogaster are hunted in large numbers for human consumption and the wild meat trade throughout its range. *C. chrysogaster* was previously reported to be the species most commonly found in wild meat carcass counts conducted at roadsides or in villages. In the western block of the species' range it accounted for 20-37% of all wild meat recorded in three surveys between 2001 and 2007, and in the eastern block the species accounted for around 10% of all wild meat sold in a major market in a 2012 survey. *Multiple sources now indicate lower abundance of C. chrysogaster in wild meat trade (e.g., Hart and Thompson, 2020).*

Wild-caught individuals are frequently seen in captivity in both the western and eastern regions of the species range. The species is also regularly seen for sale as pets throughout their range, including in the capital Kinshasa and central markets of the Kapula block.

Legal trade

The Democratic Republic of the Congo reported exporting 146 live, wild-sourced individuals for commercial purposes between 2015 and 2024 (CITES Trade Database). This trade was predominantly imported by Ghana (24%, 35), the United Arab Emirates (21%, 30) and Thailand (21%, 30). An additional 45 live captive-bred specimens were exported for zoological purposes by Syrian Arab Republic (21), Mozambique (20) and the USA (4). Only one trade shipment in this species was recorded in the LEMIS database, 2008-2020, comprising four live captive-bred individuals imported by the USA from Hungary for zoological purposes in 2014.

Illegal trade

The SS notes seizures comprising 12 individuals in 2020, from the Zambia and Zimbabwe border and "regular seizures taking place at national and international level". *There is evidence of capture of this species for the pet trade in the Democratic Republic of the Congo (Inogwabini and Thompson, 2013), as well as the sale of infants on social media in Kazakhstan, Uzbekistan, and China (McLester et al., 2025). Dempsey et al. (2024) considered the illegal pet trade to likely pose a substantial threat to the species. However, the CITES Annual Illegal Trade Database*

(AITR) only includes one reported seizure that could be related to *C. chrysogaster*: the seizure of four *Cercocebus* spp. skulls reported by the UK in 2017 (CITES Secretariat, 2025).

Additional Information

Threats

C. chrysogaster is killed for the local and commercial wild meat trade, recorded in high numbers at roadside and village markets throughout its range. Estimates vary from 20-73% of wild meat in markets between 2001 and 2007, to 10% in 2012. Effective controls are not currently in place to prevent this. *Even within the sole protected area within the species' range, wild meat hunting is occurring and linked to population decreases. High carcass numbers appear to indicate the hunting of entire family groups, but numbers of carcasses in markets are declining* (Hart and Thompson, 2020).

Ongoing logging of natural forests results in the loss of C. chrysogaster habitat, reducing range occupancy of the species and leading to some local extirpations. Both large- and small-scale logging operations are widespread throughout the species' range (Hart and Thompson, 2020).

Capture of individuals for the pet trade also occurs throughout the species' range, sometimes linked to or arising alongside the wild meat trade, with Kinshasa reported as a destination. Reported numbers in this trade have decreased in recent years (Hart and Thompson, 2020). The pet trade is thought to be growing internationally, with live individuals from wild sources exported globally for commercial purposes, and evidence of illegal trade, including posts on social media. *Cercocebus chrysogaster* is persecuted as an agricultural pest in some areas of the Democratic Republic of the Congo.

Conservation, management and legislation

The species is listed in Appendix II of CITES and in Class B of the African Convention on the Conservation of Nature and Natural Resources, under which the species "may be hunted, killed, captured, or collected under special authorization granted by the competent authority". *C. chrysogaster* is not currently identified in the list of nationally protected species in the Democratic Republic of the Congo under the 1982 Hunting Law and the related interministerial decree No. 006/CAB/MIN/EDD/2020 and No. CAB/MIN/FINANCES/2020/069 of July 24, 2020. *However, until 2005, C. chrysogaster was thought to be a subspecies of C. agilis, which is listed as partially protected under this law* (Maisels in litt., 2025; Arrêté Ministériel N° 020/CAB/MIN/ ECN-EF/ 2006). *C. chrysogaster* is known to occur in only a single protected area, Salonga National Park. *There is no information on dedicated management plans, nor on any population monitoring projects. In 2018, the Democratic Republic of the Congo set an export quota for 300 live C. chrysogaster, however, there are no current quotas set for the species. The CITES trade data show only 29 live wild-sourced individuals were directly exported as reported by the Democratic Republic of the Congo in 2018.*

Artificial Propagation/captive breeding

Globally, the captive population of *C. chrysogaster* in zoos is very small and breeding in captivity is negligible. *Currently, nine zoos and animal parks globally hold C. chrysogaster populations, with a total population of 24 individuals* (McLester et al., 2025; ZIMS, 2025). Breeding success has been low, with a neonatal mortality rate of 23%.

Implementation challenges (including similar species)

Agile Mangabey Cercocebus agilis co-occurs in the Democratic Republic of the Congo (Maisels et al., 2020). All species of the genus *Cercocebus* have light-coloured eyelids; however, *C. chrysogaster* is clearly distinguished from other *Cercocebus* species by its unique fur colouration. *Distinguishing wild meat between species may be challenging* (Maisels in litt., 2025).

Potential risk(s) of a transfer from Appendix II to I

None identified.

Other comments

Very little recent information is available on population numbers or trends. C. chrysogaster is not protected under the Democratic Republic of the Congo legislation as of 2020 (McLester et al., 2024), *however, partial protection may be granted under the former taxonomic umbrella of C. agilis* (Maisels in litt., 2025).

References

- CITES (2025). List of Species. Available [List of species | CITES](#). Accessed 21 July 2025.
- CITES Secretariat (2025) *Notification to the Parties No. 2025/102: Proposals to amend Appendices I and II. Provisional assessments by the Secretariat. Convention on International Trade in Endangered Species of Wild Fauna and Flora.* <https://cites.org/sites/default/files/notifications/E-Notif-2025-102.pdf>
- Dempsey, A., Fernández, D., McCabe, G., Abernethy, K., Abwe, E. E., Gonedelé Bi, S., Kivai, S. M., Ngoubangoye, B., Maisels, F., Matsuda, Goodwin, R., McGraw, W. S., McLester, E., ter Meulen, T., Oates, J. F., Paddock, C. L.,

- Savvantoglou, A. and Wiafe, E. D. (2024) *Cercocebus* and *Mandrillus* conservation action plan 2024-2028. Gland, Switzerland: IUCN.
- Global Forest Watch (2025) Tree cover loss in the Democratic Republic of the Congo. Available <https://gfw.global/3ltZpp9>. Accessed on 15 July 2025.
- Hart, J.A. and Thompson, J. (2020) *Cercocebus chrysogaster*. The IUCN Red List of Threatened Species 2020: e.T4207A17956177. <https://dx.doi.org/10.2305/IUCN.UK.2020-2.RLTS.T4207A17956177.en>
- Inogwabini, B.-I. and Thompson, J.A.M. (2013) The Golden-bellied Mangabey *Cercocebus chrysogaster* (Primates: Cercopithecidae): distribution and conservation status. *Journal of Threatened Taxa* 5(7):4069–4075. <http://dx.doi.org/10.11609/JoTT.o3185.4069-75>
- Kimbrough, L. (2024) DRC's golden-bellied mangabeys: A little-known but much-threatened monkey. Mongabey. Available [DRC's golden-bellied mangabeys: A little-known but much-threatened monkey](#) Accessed 21 July 2025.
- Maisels, F., Hicks, T.C., Hart, J. and Shah, N. (2020) *Cercocebus agilis* (amended version of 2019 assessment). The IUCN Red List of Threatened Species 2020: e.T136615A167735266. <https://dx.doi.org/10.2305/IUCN.UK.2020-1.RLTS.T136615A167735266.en>. Accessed on 21 July 2025.
- Maisels, F. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK. Received 29 July 2025. Wildlife Conservation Society.
- McLester, E., Altherr, S.E., Chanterreau, F., Ho, I., Mpo, H., Myers Thompson, J.A. and Hart, J.A. (2024). Golden-Bellied Mangabey *Cercocebus chrysogaster* Lydekker 1900. Mittermeier, R.A, Reuter, K.E., Rylands, A.B., Ang, A., Jerusalinsky, L., Nash, S.D., Schwitzer, C., Ratsimbazafy, J. and Humle, T., (eds.), *Primates in Peril: The World's 25 Most Endangered Primates 2023–2025*, pp. 35-39. IUCN SSC Primate Specialist Group, International Primatological Society, Re:wild, Washington, DC.
- Ridley, F. *et al.* (manuscript in preparation) Advancing understanding of global threat abatement potential for terrestrial vertebrates.
- Species360 Zoological Information Management System (ZIMS) (2025) Available zims.Species360.org.
- TRAFFIC International (2025) Wildlife Trade Portal. Available at www.wildlifetradeportal.org. Accessed 21 July 2025.

To allow Namibia to trade in registered stocks of raw ivory (whole tusks and pieces) of Namibian origin, owned by the Government of the Republic of Namibia, for commercial purposes with trading partners that have been verified by the CITES Secretariat as having sufficient national legislation and domestic trade controls. This ensures that ivory imported from Namibia will not be re-exported and will be managed following all requirements of Resolution Conf. 10.10 concerning domestic manufacturing and trade. Furthermore, to enable Namibia to achieve full Appendix II status for its elephants, as provided for in Article IV of the Convention, thereby permitting the regulated and legal trade in Namibian elephant products, including ivory.

Proponent: Namibia

Summary: The African Elephant *Loxodonta africana* population of Namibia was transferred from Appendix I to Appendix II in 1997 at the same time as the population of Botswana and Zimbabwe, with South Africa's population similarly transferred in 2000. These transfers were subject to detailed conditions that were further modified during subsequent meetings of the Conference of the Parties and are at present expressed in Annotation A10. This annotation is the subject of another amendment proposal at CoP20 (Prop 14), for which Namibia is also a proponent.

Namibia's elephant population increased from 7,500 in 1995 to over 25,000 at the end of 2024, with its distribution expanding because of population growth and increasing habitat availability.

The proposal is *"To allow Namibia to trade in registered stocks of raw ivory (whole tusks and pieces) of Namibian origin, owned by the Government of the Republic of Namibia, for commercial purposes with trading partners that have been verified by the CITES Secretariat as having sufficient national legislation and domestic trade controls. This ensures that ivory imported from Namibia will not be re-exported and will be managed following all requirements of Resolution Conf. 10.10 concerning domestic manufacturing and trade. Furthermore, to enable Namibia to achieve full Appendix II status for its elephants, as provided for in Article IV of the Convention, thereby permitting the regulated and legal trade in Namibian elephant products, including ivory."*

It is not clear how Namibia anticipates this proposal being applied in the Appendices, if adopted. A "full Appendix II status" can be interpreted as removal of the annotation that currently applies to Namibia's population of elephants allowing the trade in any parts and derivatives in accordance with Article IV. Currently, under the listing of its population in Appendix II and under existing annotation A10, Namibia is permitted to export hunting trophies, live animals for *in-situ* conservation, hides, hair and leather goods and non-commercial ekipas. Namibia has also exported live animals to zoos under Appendix I (see Analysis for Prop 14 for discussion of live trade from Namibia).

The most significant impact of the proposal would be to allow trade in raw ivory. The Proponent states that it intends to trade in an existing stock of 46,268.30 kg of registered whole tusks and pieces of Namibian origin, owned by the Government of the Republic of Namibia, for commercial purposes. This ivory stock is stated to have been accumulated only from elephants that died of natural causes and through management practices, though a detailed breakdown in the supporting statement of stockpiles held in Namibia infers that this total includes some ivory of unknown origin for which no documentation is available. It is unclear from the supporting statement whether further trade in subsequently accumulated stocks is anticipated after current stocks have been exported.

All revenue generated from ivory sales would be paid into the Game Products Trust Fund and used exclusively for elephant conservation—including monitoring, research, law enforcement, and other management expenses—as well as for community conservation and development programmes.

The stated intention is to trade with partners that have been verified by the CITES Secretariat to have sufficient national legislation and domestic trade controls to ensure that ivory imported from Namibia will not be re-exported and will be managed according to all requirements of Res. Conf. 10.10 (Rev. CoP19) on *Trade in elephant specimens* concerning domestic manufacturing and trade. Few details

are provided as to how the proposed legal trade would take place. No potential trading partners are named and historically prominent markets either now have trade bans in force or very restrictive markets allowing only sale of ivory legally imported many years ago.

Namibia's management measures are described and there is a comprehensive conservation and management plan for elephants to 2031.

Analysis: Assuming the intention is to remove the current annotation of the listing of Namibia's African Elephant population in Appendix II, it would mean that all specimens of *Loxodonta africana* from Namibia would be subject to standard Appendix II regulation. Since the transfer of Namibia's population from Appendix I to II, trade has only been permitted under specific conditions, with raw ivory exports limited in terms of quantity and time. The greatest change of the removal of the annotation would be allowing commercial trade in ivory and ivory products under Appendix II conditions. Resolutions Conf. 10.10 (Rev. CoP19) on *Trade in elephants* and Conf. 11.20 (Rev. CoP18) on *Definition of the term 'appropriate and acceptable destinations'* would still guide any trade, the latter specifically applying to export of live elephants.

There are no specific guidelines for assessing proposals to change annotations of this nature, but it seems appropriate to ensure that satisfactory precautionary measures, as detailed in Annex 4 of Res. Conf. 9.24 (Rev. CoP17) are addressed. No such measures are integrated into the proposed amendment through an annotation (some are referred to in the supporting statement but these would not be binding) therefore Parties must be satisfied with: implementation by Namibia of the requirements of the Convention, in particular Article IV; and appropriate enforcement controls and compliance with the requirements of the Convention for both Namibia and any trading partners.

Namibia has provided details on precautions that it would implement for trade of legal (and unknown) ivory stockpiles and a comprehensive management plan to 2031 is in place. Namibia has demonstrated its ability to manage its elephant populations and stresses contribution that sale of ivory stocks could make to their conservation and associated community conservation and development programmes.

However, few details are provided as to how the proposed legal trade would be carried out and controlled; for example, it is not specified which countries might permit legal imports and on what terms they might open currently closed or highly regulated markets, how and by whom the proposed licensing of traders would be implemented, or how trade would be monitored (including in end-user markets) to avoid laundering. While the CITES Secretariat is identified as playing a significant role, it is not clear how and with what resources it would undertake this work.

With respect to the **proposed trade in ivory, there is significant uncertainty about potential impacts of the release of a legal supply on the continuing global illegal trade in ivory**. Considering the historical and ongoing impacts of trade on the conservation of elephant species and the risks associated with this proposal, it is **not possible to conclude that this proposal includes satisfactory Precautionary measures**, as detailed in Annex 4 of the Resolution.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

Loxodonta africana

*The CITES Animals Committee agreed (at AC33 in 2024) to submit for adoption by CoP20 a proposed nomenclature update concerning *Loxodonta africana*, contained in Annex 3 to CoP20 Doc. 110 on Standard Nomenclature. If adopted, this would split the current listing of African elephant into two species, the African forest elephant *L. cyclotis* and the African savanna elephant *L. africana*. However, this would not affect the population covered by this proposal, which would remain *L. africana*.*

Range: proposal relates to the population of Namibia

IUCN Global Category

As a single African species *Loxodonta africana* Vulnerable A2a (assessed 2008, ver 3.1)

New IUCN assessments of the split taxon.

- African Savanna Elephant *Loxodonta africana* Endangered A2bd, assessed 2020 (Gobush et al., 2022)
- African Forest Elephant *Loxodonta cyclotis* Critically Endangered A2abd, assessed 2020 (Gobush et al., 2021)

Biological criteria for inclusion in Appendix I

A) Small wild population and C) Decline in number of wild individuals

Namibia's elephant population has increased from just over 7,500 in 1995 to over 25,664 at end 2024 based on data from the Kavango Zambezi Transfrontier Conservation Area (KAZA TFCA) elephant survey, together with a separate survey conducted in Etosha National Park and north-western Namibia (Kunene, Omusati, and Erongo).

Table 1. Estimated elephant populations in Namibia at the end of 2024 (Source: Ministry of Environment and Tourism, reproduced from SS)

Population Estimate	25,664 (total)
Etosha National Park	3,266
Kavango Zambezi regions	12,345
Khaudum NP & Nyae Nyae area	8,745
North West (Kunene, Omusati & Erongo)	1,108
Mangetti area	200

The African Elephant Specialist Group is currently producing an updated status report for the African Savanna Elephant (Co-Chairs of IUCN SSC AfESG in litt., 2025). Latest African Elephant Database elephant population estimates for Namibia estimated the population at 25,320 with 21,330 95% lower confidence interval and 29,322 95% confidence intervals (and 348 guesses upper and 342 lower limits (unpublished data (dated 14/08/2025)).

As a part of the KAZA-TFCA, Namibia's suitable habitat and range for elephants is contiguous with neighbouring countries; as such increases in national elephant numbers may be partly explained by immigration across the transboundary area. This may be especially relevant for the elephants of Khaudum-Kavango, Nyae-Nyae Conservancy and Zambezi Region which border Botswana and held approximately 85% of Namibia's elephants according to data from the IUCN African Elephant Status Report 2016 (Thouless et al., 2016; see also Bussiere et al., 2023).

B) Restricted area of distribution

Elephants in Namibia are migratory-nomadic over a very large range with distinct dry season ranges and a much larger wet season dispersal range (estimated at over 100,000 km²). Elephant distribution in Namibia has been expanding as a result of population increases and increasing habitat availability including through community conservancies.

The species is or may be affected by trade

Table 2. Trophy exports from Namibia (Source: SS)

Year	Quantity trophies
2000	43
2001	34
2002	33
2003	48
2004	43
2005	48
2006	32
2007	20
2008	81
2009	25
2010	90

Year	Quantity trophies
2011	48
2012	63
2013	67
2014	52
2015	53

Table 3. Exports reported by Namibia in CITES Trade Database (downloaded 14/07/25)

Year	Trophies	Tusks
2016	94	32
2017	90	
2018	69	15
2019	33	16
2020	24	20
2021		52
2022	63	30
2023	48	8

In addition to trophy exports, small numbers of leather goods and hair are exported originating from trophy hunted animals and recovered from elephants destroyed for management reasons. *52 live wild animals were also reported as exported for re-introduction and zoos between 2014 and 2023 (some to ex-situ destinations) (CITES Trade Database, 2025).*

The proponents state that no elephants have been, or will be, killed specifically to obtain ivory or other products for trade. Ivory is recovered from all recorded natural mortalities, as well as from elephants destroyed as problem animals and some ivory of unknown origin. *It is not clear from the SS whether the intention is to trade in ongoing accumulations of raw ivory (from natural mortality and problem animal control) once current registered stocks are exhausted. One of the strategic objectives of Namibia's Elephant Conservation and Management Plan 2021/2022-2030/2031 is to prevent long-term negative and possibly irreversible impacts by elephants on other components of biodiversity and ecosystem processes and functioning. With a build-up of unsustainable, high elephant densities considered a threat, lethal removal and translocation is considered to be used as means of last resort. Sustainable utilisation of elephants for economic and cultural purposes is seen as essential for addressing as an incentive for co-existence of humans and elephants (Ministry of Environment, Forestry and Tourism, 2020).*

Precautionary Measures

Although the SS states that the proponents seek to trade existing stock of 46,268.30 kg of registered raw ivory (whole tusks and pieces) of Namibian origin, owned by the Government of the Republic of Namibia, for commercial purposes, *which equates to the figures given in section 6 of the SS for Natural and Management and those which no documentation is available (Unknown origin). This quantity is not specified as a quota as an integral part of the amendment listing. Furthermore, although the figure for existing stocks is referred to, the proposal is unclear as to whether it also seeks ongoing trade in ivory as part of its aim to achieve full Appendix II status for its elephants, as provided for in Article IV of the Convention. It seems most appropriate to assess whether the precautionary measures in Annex 4 A 2 a) ii are met in either case.*

The species is likely to be in demand for trade, therefore the CoP would need to be satisfied with:

- A) Implementation by the range States of the requirements of the Convention, in particular Article IV; and
- B) appropriate enforcement controls and compliance with the requirements of the Convention

A) Implementation by the range States of the requirements of the Convention, in particular Article IV; and

Legal acquisition.

Namibia continues to exercise strict control over all ivory stocks. Separate storage is used for confiscated tusks. All ivory specimens are marked and a detailed database is maintained for all ivory specimens.

Non-detriment Findings

Namibia has not exploited elephants directly for commercial trade or domestic consumption, except through sport hunting and photographic tourism *and for the live trade*. Additionally, over the past years, Namibia experienced a severe drought, and Cabinet took a decision to remove a small number of elephants in 2024.

The level of sport hunting is largely determined by the 0.5% of standing population guideline (Martin, 1986). MEFT has established a national annual export quota through CITES of 90 trophy hunted elephants per year (180 tusks per year). *Live elephant exports have averaged 5.2 per year between 2014 and 2023 according to the CITES Trade Database.*

Future management of elephant populations resulting in killing is likely to be on the basis of human-wildlife conflict or *to prevent long-term negative and possibly irreversible impacts by elephants on other components of biodiversity and ecosystem processes and functioning from build-up of unsustainable high elephant densities. Lethal removal and translocation would be used as means of last resort (Ministry of Environment, Forestry and Tourism, 2020). It is not clear whether Namibia intends to trade in any new stocks of ivory once the existing stock of 46,268.30 kg has been exported.*

B) appropriate enforcement controls and compliance with the requirements of the Convention Namibia

The SS states that Namibia has complied with every requirement of CITES concerning the conservation of the African elephant, but remains concerned over the high costs and security implications of holding large ivory stocks. The proponents emphasize that no elephants have been, or will be, killed specifically to obtain ivory or other products for trade. Ivory is recovered from all recorded natural mortalities, as well as from elephants destroyed as problem animals.

Namibia has reported exporting 26 live elephants to United Arab Emirates (UAE) in 2021, an ex-situ location, but as Appendix-I specimens and therefore not under the conditions of Annotation A10.

The most recent Proportion of Illegally Killed Elephants (PIKE) estimates for Southern Africa in CoP20 Doc 76.4 show that PIKE likely increased between 2003 and 2011 and subsequently decreased from 2011 to 2024. Over the last five years (2020-2024) there was no evidence of an increase or decline. The unweighted PIKE estimate for Southern Africa in 2024 is below the average continental PIKE estimate for the same year (See CoP20 Doc 76.4).

The latest Elephant Trade Information System (ETIS) analysis provided in CoP 20 Doc 76.5 shows that globally illegal trade in ivory appears to have reduced in the past 10 years since the peak in 2014 (Figure 1).

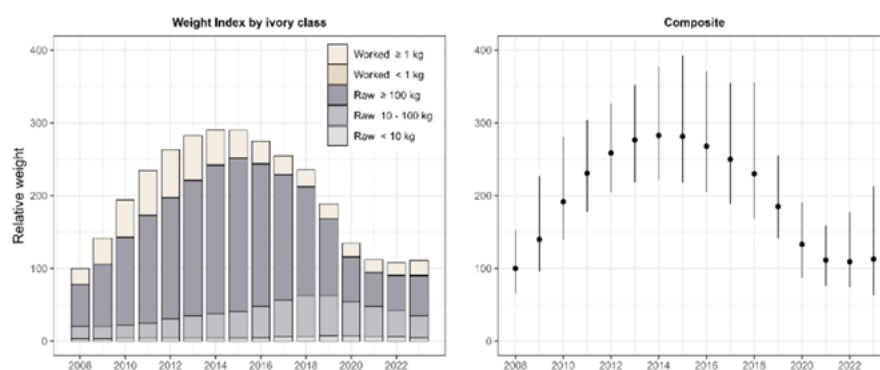


Figure 1. ETIS trend analysis results – Weight index. Weight Index median estimates are presented with their breakdown by ivory type and weight class (left); the bars are stacked in the order seen in the legend (the contribution of the small worked ivory class is barely visible due to its relatively negligible contribution to the overall Weight Index). The composite index across all weight classes (right) shows the median estimates (bold dots) accompanied with 90% quantile-based credible intervals. The Weight Index is derived from the Transaction Index (Figure 3), with the transactions within each weight class assigned a weight sampled from a model¹¹ fitted to the reported weight data. (Source: CoP20 Doc 76.5).

According to the latest ETIS report (CoP20 Doc 76.5) Namibia, had no clear increasing trends nor prominence in the total amounts of seized ivory in recent years. All seizures with a link to Namibia since 2021 were made by Namibia itself in country, with some of these having been large (>100 kg). Namibia has never been included in the National Ivory Action Plan (NIAP) process.

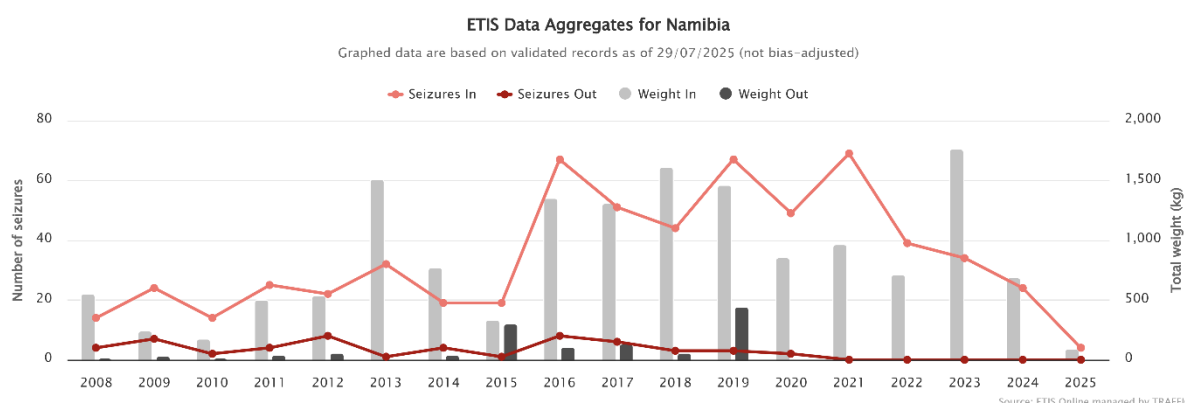


Figure 2. ETIS Data Aggregates for Namibia (Source: ETIS online downloaded, 29 July 2025)

The elephant population of Namibia is mainly found in its north-western and north-eastern region, which is part of the KAZA TFCA where there is movement of elephants between countries. Namibia's neighbour to the north, Angola has been identified in the latest ETIS analysis as a Category B (markedly affected) Party; during the period 2021 – 2023 the largest seizure of raw ivory included a 7-tonne consignment seized by Viet Nam, reportedly shipped from Angola (CoP20 Doc 76.5), as well as being implicated in other large-scale worked ivory seizures. It is conceivable that some of the ivory could have originated in Namibia or from the population of elephants that migrate between the countries. Angola has been in the NIAP process since CoP17.

Management

The Proponent argues that, as with proceeds of previous one-off sales, that the controlled ivory trade will directly benefit the survival of the species as all revenue will be reinvested in elephant conservation in Namibia, including rural community conservation programmes, and monitoring of the impact of trade will be supported.

The SS provides the following detail on international and national controls.

International

Permit control: The MET permit office at Windhoek issues all permits relating to elephants or elephant derivatives. No competencies are delegated to local or regional authorities. (The Directorate of Veterinary Services in Windhoek issues all veterinary permits).

Marking of ivory: All ivory is marked in accordance with Resolution Conf. 10.10, and the marks are included in a database of ivory of known Namibian origin showing the source of each specimen. All specimens of ivory are furthermore marked in a standardized way derived from the domestic permit control system.

Customs and border control: Namibian Customs Officers check CITES, veterinary and transit permits. Where necessary, they refer to the Namibian Police or district veterinary officer. Customs representatives played a key role in supervising the experimental trades of ivory in 1999 and 2008.

Law enforcement: Law enforcement is a joint effort by the Ministry of Environment, Forestry and Tourism, the Protected Resources Sub-Division of the Namibian Police, and the Customs Service. The incidence of ivory confiscations in Namibia points to effective law enforcement, especially by the Protected Resources Sub-Division of the Namibian Police. Law enforcement agencies rely primarily on information, and well-established informer networks exist and are maintained. This approach has been the most effective in a situation of a low human density and government aiming to remain as small as possible.

Trade controls for raw ivory: Only the Namibian population is included in this proposal. All specimens for export will have been individually marked in accordance with Resolution Conf. 10.10. All other ivory will also be individually marked and registered with the CITES Secretariat to ensure that there can be no mixing of unknown or foreign ivory. All revenue from ivory sales will be used exclusively for elephant conservation and community development and conservation programmes. Namibia will cooperate with neighbouring countries in the monitoring of elephant populations and illegal trade, and will assist within its means, credible international organizations involved in such monitoring.

Trade controls for worked ivory: Possession, manufacturing and trade in all elephant specimens are regulated in Namibia. Persons or companies wishing to manufacture or trade worked ivory elephant have to be registered with the Management Authority and are required to maintain comprehensive records of stocks, manufacturing and trade in accordance with Resolution Conf. 10.10.

Trade controls for hides and leather goods: Possession, manufacturing and trade in all elephant specimens are regulated in Namibia. Persons or companies wishing to manufacture or trade elephant hide or leather goods have to be registered with the Management Authority and are required to maintain comprehensive records of origin of stocks, manufacturing and trade.

Population monitoring is carried out in protected areas and across large parts of the elephants' range in communal lands.

Management is further detailed in Namibia's comprehensive National Elephant Conservation and Management plan 2021/2022-2039/2031 (Ministry of Environment, Forestry and Tourism, 2020)

Trading partners

It is stated that the proposed trade in ivory would be with partners that have been verified by the CITES Secretariat to have sufficient national legislation and domestic trade controls to ensure that ivory imported from Namibia will not be re-exported and will be managed according to all requirements of Res. Conf. 10.10 concerning domestic manufacturing and trade. *No potential partners are proposed in the SS.*

Decision 14.77 directed the Standing Committee, assisted by the Secretariat, to propose "a decision-making mechanism for a process of trade in ivory under the auspices of the Conference of the Parties." However, no decision-making mechanism has been adopted. For previous one-off sales the process to verify trading partners involved numerous missions by the Secretariat and other organisations to potential partners to assess their ivory trade controls and domestic markets and to confirm that market controls were in place and working (see SC55 Doc 10.1 (Rev. 1) and SC57 Doc 33.2). Trading partners were endorsed by the Standing Committee. Previous trading partners were Japan and China.

Resolution Conf. 10.10 (Rev CoP19) preambular text considers that "the legal domestic markets for ivory may increase the risk to elephant populations and local communities, due to the opportunity it creates for the laundering of illegal ivory under the guise of legality". This Resolution also recommends "that all Parties and non-Parties in whose jurisdiction there is a legal domestic market for ivory that is contribution to poaching or illegal trade, take all necessary legislative, regulatory and enforcement measures to close their domestic markets for commercial trade in raw and worked ivory". Historically prominent markets either now have trade bans in force or very restrictive markets for sale of ivory legally imported many years ago.

Additional Information

Threats

Loss of habitat and the fragmentation of range through the interruption of access routes and conflicts with people in the absence of effective incentive mechanisms to maintain such habitat.

References

- Co-Chairs AfESG (2022) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- CoP20 Doc 76.4 (2025) Report on Monitoring the Illegal Killing of Elephants (MIKE)
<https://cites.org/sites/default/files/documents/E-CoP20-076-04.pdf>
- CoP20 Doc 76.5 (2025) Report on Elephant Trade Information System (ETIS)
<https://cites.org/sites/default/files/documents/E-CoP20-076-05.pdf>
- Bussi re, E.M.S. and Potgieter, D. (2023) KAZA Elephant Survey 2022, Volume I: Results and Technical Report, KAZA TFCA Secretariat, Kasane, Botswana
- Gobush, K.S., Edwards, C.T.T., Balfour, D., Wittemyer, G., Maisels, F. and Taylor, R.D. (2022) *Loxodonta africana* (amended version of 2021 assessment). The IUCN Red List of Threatened Species 2022: e.T181008073A223031019. <https://dx.doi.org/10.2305/IUCN.UK.2022-2.RLTS.T181008073A223031019.en>. Accessed on 14 July 2025.
- Ministry of Environment, Forestry and Tourism (2020) National Elephant Conservation and Management Plan 2021/2022-2030/2031
- SC55 Doc 10.1 (Rev. 1) (2007) Control of trade in African elephant ivory
<https://cites.org/sites/default/files/eng/com/sc/55/E55-10-1.pdf>
- SC57 Doc 33.2 (2008) Control of trade in elephant ivory <https://cites.org/sites/default/files/eng/com/sc/57/E57-33-2.pdf>
- SC78 Doc 65.1 Annex 1 (2025) Levels of illegal killing of elephants, illegal and legal trade in elephant specimens, the status of elephant populations and the implementation of the African Elephant Action Plan: a report to the CITES Standing Committee.
- Thouless, C.R., Dublin, H.T., Blanc, J.J., Skinner, D.P., Daniel, T.E., Taylor, R.D., Maisels, F., Frederick, H.L. and Bouche, P. (2016) African elephant status report 2016: an update from the African Elephant Database.

Occasional Paper Series of the IUCN species Survival Commission, No. 60 IUCN/SSC Africa Elephant Specialist Group. IUCN, Gland, Switzerland. pp.vi+309.

Amend the existing annotation for the populations of African Elephant *Loxodonta africana* in Botswana, Namibia, South Africa, and Zimbabwe

Proponents: Botswana, Cameroon, Côte d'Ivoire, Namibia, Zimbabwe

Summary: The African Elephant *Loxodonta africana* populations of Botswana, Namibia, and Zimbabwe were transferred from Appendix I to Appendix II in 1997, as was the population of South Africa in 2000. These transfers were subject to detailed conditions that were further modified during subsequent meetings of the Conference of the Parties and are at present expressed in Annotation A10 (previously Annotation 2) to the CITES Appendices. Annotation A10 allows for trade in various non-ivory specimens and products of *L. africana* under a range of conditions, somewhat different for each of the four range States in question. Regarding trade in ivory, the annotation currently only allows for trade in individually marked and certified ekipas incorporated in finished jewellery for non-commercial purposes for Namibia and ivory carvings for non-commercial purposes for Zimbabwe. However, the annotation also includes wording that provided for limited exports of stockpiled raw ivory in the past, under a series of conditions. One of these conditions, added to the annotation at CoP14 in 2007, was that no further proposals to allow trade in elephant ivory from populations already in Appendix II should be submitted until at least nine years after the date of the last permitted sale of ivory that occurred in 2008, during which time a decision-making mechanism for a process of trade in ivory would be developed. No agreed decision-making mechanism for allowing trade in ivory under the auspices of the Conference of the Parties has been adopted.

The proposal is to amend the existing annotation for the Appendix II populations of *Loxodonta africana* in Botswana, Namibia, South Africa, and Zimbabwe in order to remove redundant text and reference to Resolutions and Decisions, harmonize the trade conditions for the four range States and remove reference to trade in raw ivory. The amendments proposed are as follows (deleted text reflected as ~~strike through~~ and text inserted underlined:

For the exclusive purpose of allowing:

- a) trade in hunting trophies for non-commercial purposes;
- b) trade in live animals to appropriate and acceptable destinations, ~~as defined in Resolution Conf. 11.20 (Rev. CoP18), for Botswana and Zimbabwe and for in situ conservation programmes for Namibia and South Africa;~~
- c) trade in hides;
- d) trade in hair;
- e) trade in leather goods ~~for commercial or non-commercial purposes for Botswana, Namibia and South Africa and for non-commercial purposes for Zimbabwe;~~
- f) trade for non-commercial purposes in individually marked and certified ekipas incorporated in finished jewellery ~~for non-commercial purposes for Namibia and ivory carvings for non-commercial purposes for Zimbabwe;~~
- g) trade in registered raw ivory ~~(for Botswana, Namibia, South Africa and Zimbabwe, whole tusks and pieces) subject to the following:~~
 - i) ~~only registered government-owned stocks, originating in the State (excluding seized ivory and ivory of unknown origin);~~
 - ii) ~~only to trading partners that have been verified by the Secretariat, in consultation with the Standing Committee, to have sufficient national legislation and domestic trade controls to ensure that the imported ivory will not be re-exported and will be managed in accordance with all requirements of Resolution Conf. 10.10 (Rev. CoP18) concerning domestic manufacturing and trade;~~
 - iii) ~~not before the Secretariat has verified the prospective importing countries and the registered government-owned stocks;~~
 - iv) ~~raw ivory pursuant to the conditional sale of registered government-owned ivory stocks agreed at CoP12, which are 20,000 kg (Botswana), 10,000 kg (Namibia) and 30,000 kg (South Africa);~~
 - v) ~~in addition to the quantities agreed at CoP12, government-owned ivory from Botswana, Namibia, South Africa and Zimbabwe registered by 31 January 2007 and~~

- ~~verified by the Secretariat may be traded and dispatched, with the ivory in paragraph g) iv) above, in a single sale per destination under strict supervision of the Secretariat;~~
- ~~vi) the proceeds of the trade are used exclusively for elephant conservation and community conservation and development Programmes within or adjacent to the elephant range; and~~
- ~~vii) the additional quantities specified in paragraph g) v) above shall be traded only after the Standing Committee has agreed that the above conditions have been met; and~~
- ~~h) no further proposals to allow trade in elephant ivory from populations already in Appendix II shall be submitted to the Conference of the Parties for the period from CoP14 and ending nine years from the date of the single sale of ivory that is to take place in accordance with provisions in paragraphs g) i), g) ii), g) iii), g) vi) and g) vii). In addition, such further proposals shall be dealt with in accordance with Decisions 16.55 and 14.78 (Rev. CoP16).~~

~~On a proposal from the Secretariat, t~~The Standing Committee can decide to cause this trade to cease partially or completely in the event of non-compliance by exporting or importing countries, or in the case of proven detrimental impacts of the trade on other elephant populations.

All other specimens shall be deemed to be specimens of species included in Appendix I and the trade in them shall be regulated accordingly.

The proposed amendments would;

- In paragraphs b), g ii) and h remove reference to CITES Resolution and Decisions.
- In paragraph b) remove text stipulating that trade in live animals from Namibia and South Africa is restricted to *in situ* conservation programmes, and harmonize the conditions with those for Botswana and Zimbabwe that live trade is to “appropriate and acceptable destinations”.
- In paragraph e) allow for commercial trade in leather goods from Zimbabwe (and thus all four range States), currently limited to non-commercial goods only.
- In paragraph f) streamline and simplify annotation wording with no substantive impact.
- In paragraph g) remove reference to trade in registered raw ivory from the Annotation and the conditions under which that was permitted.
- In paragraph h) remove reference to not submitting further proposals to allow trade in elephant ivory for nine years after the single sale of ivory that occurred in 2008 (a period that lapsed in 2017).
- In the penultimate paragraph to delete the role of the Secretariat in causing trade to cease.

The proponents argue that Resolutions and Decisions referred to in paragraphs b, g (ii) and h of the annotation are non-binding and regard reference to these in an annotation to a CITES listing as problematic because it implies that the relevant Resolution or Decision should be considered legally binding on the Parties as part of the Appendices to the Convention. Furthermore, Resolution Conf. 4.6 (Rev. CoP18) on *Submission of draft resolutions and other documents for meetings of the Conference of the Parties* directs that when a Resolution is revised, the Secretariat is required to update the references to that Resolution in other current Resolutions, and by extension any annotation in which they are referred to, despite this potentially amending the listing without an amendment proposal being considered by the Conference of the Parties.

Namibia, South Africa and Zimbabwe have all reported the export of small numbers of live elephants, with Namibia and Zimbabwe having reported some to *ex-situ* destinations. According to the current annotation to the Appendix-II listing, Namibia’s exports were restricted to *in-situ* destinations. However, their exports to *ex-situ* destinations were carried out under Appendix-I conditions.

Leather items have been reported as exports from all four Parties with populations in Appendix II, although only South Africa and Zimbabwe have exported commercial items. According to A10, Zimbabwe can only export non-commercial leather items. No concerns have been raised over the impact of this trade.

The proposed amendment to the annotation A10 is based on consensus reached at the CITES Dialogue Meeting for African elephant range States that took place from 23 to 26 September 2024 (see document CoP20 Doc. 76.6).

Analysis: The proposal would have two substantive impacts on trade in elephant products. The first being that the conditions for trade in live animals would be harmonised between the four proponents, Botswana, Namibia, South Africa and Zimbabwe, no longer stipulating that Namibia and South Africa may only export for *in-situ* conservation. Any such exports from all four range States would still be guided by the Resolution on *Trade in elephants* (Res. Conf. 10.10 (Rev. CoP19)*) and Res. Conf. 11.20 (Rev. CoP18), on *Definition of acceptable and appropriate destinations*. The amendment to the wording is unlikely to have significant impact on trade.

The second substantive impact would be that Zimbabwe would be able to trade in commercial leather products, which appears to already be taking place in small quantities with no apparent problems being noted.

Although the deletion of paragraph g) appears to be a significant amendment, no trade in raw ivory has been permitted since the conditional sales agreed at CoP12 and therefore the removal of this condition will have no impact on the current trade in elephants. Paragraph h has been redundant for many years and its removal will have no impact.

Overall, **the amendments are very unlikely to have any negative impact on the trade or conservation of African Elephants from these range States**. Any significant changes in trade (including possible issues in compliance arising from implementation of the existing Resolutions addressing trade in elephants), can be addressed by the Standing Committee through implementation of Res. Conf. 14.3 (Rev. CoP19) on *CITES compliance procedures* or through the Review of Significant Trade process. The Standing Committee can decide to cause this trade to cease partially or completely in the event of non-compliance by exporting or importing countries, or in the case of proven detrimental impacts of the trade on other elephant populations.

Summary of Available Information and Discussion

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

Loxodonta africana

*The CITES Animals Committee agreed (at AC33 in 2024) to submit for adoption by CoP20 a proposed nomenclature update concerning *Loxodonta africana*, contained in Annex 3 to CoP20 Doc. 110 on Standard Nomenclature. If adopted, this would split the current listing of African elephant into two species, the African forest elephant *L. cyclotis* and the African savanna elephant *L. africana*. However, this would not affect the populations covered by this proposal, which would remain *L. africana*.*

Range

The proposal relates to the populations of Botswana, Namibia, South Africa, and Zimbabwe

IUCN Global Category

As a single African species *Loxodonta africana* Vulnerable A2a (assessed 2008, ver 3.1)

New IUCN assessments of the split taxon.

- African Savanna Elephant *Loxodonta africana* Endangered A2bd assessed 2020 (Gobush et al., 2022)
- African Forest Elephant *Loxodonta cyclotis* Critically Endangered A2abd assessed 2020 (Gobush et al., 2021)

Discussion

The African Elephant populations of the four proponent countries were estimated to have totalled 255,851 in 2015 and country totals were estimated as follows: Botswana 131,626, Namibia 22,754, South Africa 18,841, and Zimbabwe 85,000.

The African Elephant Specialist Group is currently producing an updated status report for the African Savanna Elephant (Co-Chairs of IUCN SSC AfESG in litt., 2025). The most recent data on Southern African populations are presented in Table 1.

Table 1. Latest African Elephant Database elephant population estimates for Southern African countries, including 95% confidence intervals and guesses upper and lower limits unpublished data (dated 14/08/2025).

COUNTRY	NO. OF ELEPHANTS			GUESSES	
	ESTIMATE	95% lcl	95% ucl	lower limit	upper limit
Angola	6,008	381	12,470	212	213
Botswana	138,469	125,982	144,849	0	0
Malawi	2,168	2,127	2,211	0	0
Mozambique	9,133	6,703	12,208	0	0
Namibia	25,320	21,330	29,322	342	348
South Africa	45,265	43,327	48,171	1,652	1,652
Swaziland	8			83	83
Zambia	17,144	14,008	20,806	214	314
Zimbabwe	87,159	77,624	96,752	1,351	1,421

Botswana, Namibia and Zimbabwe comprise part of the Kavango Zambezi Transfrontier Conservation Area (KAZA TFCA) spanning five countries which also includes Angola and Zambia.

The amendments to annotation A10 were considered, discussed and agreed by African Elephant range States present at the African Elephant Dialogue Meeting in Maun Botswana from 23 to 26 September 2024 and agreed by consensus.

The proposed amendments are summarised below:

- In paragraphs, b), g ii) and h remove reference to CITES Resolution and Decisions.
- In paragraph b) remove the stipulation that trade in live animals from Namibia and South Africa is only for in situ conservation programmes, harmonizing the conditions with those for Botswana and Zimbabwe that live trade is only to “appropriate and acceptable destinations” and remove the reference to Resolution 11.20 (Rev CoP 18)* that defines “appropriate and acceptable destinations”.
- In paragraph e) allow for commercial trade in leather goods from Zimbabwe (and thus all four range States), previously limited to non-commercial goods only.
- In paragraph f) streamline and simplify annotation wording with no substantive impact.
- In paragraph g) remove reference to trade in registered raw ivory from the Annotation and the conditions under which that was permitted.
- In paragraph h) remove reference to not submitting further proposals to allow trade in elephant ivory for nine years after the single sale of ivory that occurred in 2008 (a period that has already passed). Reference to Decisions that are no longer valid would also be removed.
- In the penultimate paragraph remove the role of the Secretariat in causing the trade to cease.

The effects of these amendments are discussed below.

Remove reference to CITES Resolution and Decisions

The proponents argue that Resolutions and Decisions referred to in paragraphs b, g (ii) and h of the annotation are soft law and reference to these in an annotation to a CITES listing is considered problematic because it implies that the relevant Resolution or Decision should be considered legally binding on the Parties as part of the Appendices to the Convention. Their inclusion is not in accordance with Resolution Conf. 11.21 (Rev. CoP18) on *Use of annotations in Appendices I and II*, 1h) which states that references to Decisions or Resolutions of the Conference of the Parties should not be included in annotations. Furthermore, Resolution Conf. 4.6 (Rev. CoP18) directs that when a Resolution is revised, the Secretariat is required to update the references to that Resolution in other current Resolutions, and by extension any Annotation in which they are referred to, despite this potentially amending the listing without an amendment proposal being considered by the Conference of the Parties. *The implications of removal of reference to these is discussed in the relevant paragraphs below.*

Paragraph b)

Proposed amendment of the text in paragraph b would remove the stipulation that trade in live animals from Namibia and South Africa should only be for *in situ* conservation programmes. All Parties would be able to export live elephants to appropriate and acceptable destinations.

The amendment would remove the wording “as defined in Resolution Conf. 11.20 (Rev CoP18)*. If reference to Resolution Conf 11.20 (Rev CoP18)* is removed from the annotation, the Resolution still defines the term “appropriate and acceptable destinations” to mean *in situ* conservation programmes or secure areas in the wild, within the species’ natural and historical range in Africa, except in exceptional circumstances where, in consultation with the Animals Committee, through its Chair with the support of the Secretariat, and in consultation with the IUCN elephant specialist group, it is considered that a transfer to ex-situ locations will provide demonstrable in-situ conservation benefits for African elephants, or in the case of temporary transfers in emergency situations”. However, this definition would not be considered binding as Resolutions are generally intended to provide long-standing guidance on how to interpret the provisions of the Convention.

Namibia, South Africa and Zimbabwe have reported exports of live wild animals variously for educational, reintroduction to the wild, circus or travelling exhibition, and zoos (see Table 2). Namibia and Zimbabwe have both exported live specimens to ex-situ locations. It is not clear whether in each case the Animals Committee has been consulted. Namibia reported exporting 26 live elephants to United Arab Emirates (UAE) in 2021, an ex-situ location but as Appendix-I specimens and therefore not under the conditions of Annotation A10, and a further seven in 2022 to UAE as Appendix-II specimens. There are some discrepancies in the exporter and importer figures over this time period with UAE reporting the import of 26 live, DRC 15, China 131, North Macedonia 1 and Mozambique 124 from the same range States (Namibia, South Africa, Zimbabwe).

Table 2. Direct trade in live wild elephants reported by the Proponents between 2014 and 2023. (Source: CITES trade database downloaded 24 July 2025).

Exporter	Purpose	Importer (Number)				
		UAE	Botswana	DRC	China	Mozambique
Namibia	Reintroduction or introduction into the wild			19		
	Zoo	33				
South Africa	Reintroduction or introduction into the wild					132
	Circus or travelling exhibition		2			
Zimbabwe	Educational				56	
	Reintroduction or introduction into the wild					
	Zoo	4			67	

For similar terms used in annotations the Parties have often included definitions in the interpretation section of the Appendices. However, this could then be considered binding, which the Proponents clearly want to avoid. Furthermore, future changes to a definition might require an amendment proposal to be considered by the Conference of the Parties.

Paragraph e)

Proposed amendments to paragraph e) to allow for commercial trade in leather goods from Zimbabwe, currently limited to non-commercial trade only, thus harmonising the conditions of trade for all four range States. According to CITES trade data the four range States have reported the export of small quantities of small and large leather products. Only South Africa and Zimbabwe have reported exporting items for commercial purposes. No concerns have been raised over the impact of Zimbabwe’s trade in commercial leather items.

Table 3. Direct trade in leather items (small and large) reported by the proponents between 2014 and 2023. (Source CITES trade data downloaded 24 July 2025).

Exporter	Purpose	Total	Unit
Botswana	Hunting /Personal	21	Number of specimens/Blank
Namibia	Hunting /Personal	37	Number of specimens/Blank
South Africa	Hunting /Personal	192	Number of specimens/Blank
	Commercial	65	kg

Exporter	Purpose	Total	Unit
		4576	m2
		2870	Number of specimens/Blank
Zimbabwe	Hunting /Personal	4727	Number of specimens/Blank
	Personal	2	g
	Circus or travelling exhibition	930	Number of specimens
	Commercial	1081	Number of specimens/Blank

Paragraph g)

Paragraph g related to conditions under which a previous one-off sale in registered raw ivory was permitted, *which is no longer relevant and therefore removal would have no impact. Any future trade would likely incorporate similar conditions in an annotation that would need to be adopted by the CoP as a special measure in line with Res Conf 9.24 (rev CoP17) Annex 4 A2 ii.*

Paragraph h)

Changes to paragraph h) remove reference to not submitting further proposals to allow trade in elephant ivory for nine years after the sale of ivory that occurred in 2008 which has already passed and therefore has been redundant for many years. Reference to Decisions that are no longer valid would also be removed. *As pointed out by the Proponents paragraph h appears to be redundant.*

Penultimate paragraph

The role of the Secretariat in causing trade to cease would be removed. Were there to be any concerns due to changes in trade patterns these are likely to be addressed by the Standing Committee through implementation of Res. Conf. 14.3 (Rev. CoP19) on CITES compliance procedures or through the Res. Conf. 12.8 (Rev. COP18) on the Review of Significant Trade process.

Threats

Generally, for elephants in Africa, commercial logging, plantations for biofuels and extractive industries like logging and mining not only destroy habitat but also facilitate accessibility to remote elephant habitat where they are susceptible to poaching. Moreso, poverty, armed conflict and the displacement of people by civil conflict also add to elephant habitat loss and fragmentation. All these factors push elephants into smaller islands of protected areas and hinder elephants' freedom to roam.

Poaching of African Savanna Elephants for ivory has been a major cause of individual death and population decline (Wittemyer et al., 2014; Thouless et al., 2016). After a sustained period of intense poaching between the late 1970s and 1989, many African Savanna Elephant populations (e.g., in Kenya, Tanzania, Zambia, Uganda) experienced two to three decades of recovery. The IUCN Red List for the African Savanna Elephant notes rapid land use change by humans as a factor driving the loss and fragmentation of habitat across their range (Thouless et al., 2016; Mpakairi et al., 2019). Land conversion is a product of the ongoing expansion of the human population in turn leads to increase in human-elephant conflict (e.g., Pozo et al., 2018; Gobush et al., 2021).

The most recent report from the Monitoring Illegal Killing of Elephants (MIKE) programme Proportion of Illegally Killed Elephants (PIKE) estimates for Southern Africa in CoP20 Doc 76.4 show that PIKE likely increased between 2003 and 2011 and subsequently decreased from 2011 to 2024. Over the last five years (2020-2024) there was no evidence of an increase or decline. The unweighted PIKE estimate for Southern Africa in 2024 is below the average continental PIKE estimate for the same year (See CoP20 Doc 76.4).

The most recent ETIS report has not identified any of the range States for the populations that would be affected by this proposal as requiring attention under the National Ivory Action Plan process (see paragraph 27 of CoP20 Doc 76.5 Annex 2. Baruch-Mordo et al., 2025).

Conservation, management and legislation

Elephant populations are managed according to elephant management plans and strategies at national level and spatially-explicit management plans that are responsive to local dynamics. Robust control measures are already in place within the legal framework of the proponents, at national level. The comprehensive commitments under various SADC regional initiatives and agreements ensure accountability and safeguards for compliance. There are some known elephant range expansion initiatives in Botswana, South Africa (Biosphere reserve agreements signed in 2018), Namibia (Community conservancies), Zimbabwe (Hwange Sanyati Biological Corridor, Community conservancies) and six transfrontier conservation areas initiatives, including the KAZA TFCA.

Southern Africa has made significant progress in developing and implementing National Elephant Action Plans (NEAPs), which provide an essential framework for guiding elephant conservation and management. Angola (2018–2028), Botswana (2018–2028), Malawi (2015–2025), Namibia (2021–2031), Zambia (2021–2025), and Zimbabwe (2021–2025) have all adopted NEAPs that are currently under implementation, while South Africa recently launched its National Elephant Heritage Strategy (2024–2034). These strategies collectively demonstrate the region's strong commitment to sustainable elephant conservation, addressing key issues such as habitat protection, human–elephant conflict mitigation, community engagement, law enforcement, and transboundary collaboration. Importantly, several countries, including Malawi, Zambia, and Zimbabwe, are approaching the end of their NEAP cycles, underscoring the need for timely review, renewal, and alignment with evolving conservation priorities. South Africa has also developed the Norms and Standards of Elephants in 2008. A revised version was published for implementation (Government Gazette No. 47984, 3 February 2023) and then withdrawn in Government Gazette No. 48349 on 31 March 2023, so the 2008 version is still in effect. Since 2008 elephants have been managed in accordance with the National Norms and Standards for the Management of Elephants in South Africa (Government Gazette No. 30833, 29 February 2008). The species is listed as protected in terms of section 56 of NEMBA and various provincial ordinances and acts provide further legislative protection. South Africa has a National Red-List Assessment for African Savanna Elephants, published in 2016 that list the species as Least Concern. In 2019 a process was initiated to develop a National Elephant Conservation Strategy, which was revised to a National Elephant Heritage Strategy, which was released for public comment in 2025 (Government Gazette No. 52004, 27 January 2025), and expected to be published for implementation in 2025. This Heritage strategy is intended to serve as the National Elephant Action Plan. The strategy will provide a high-level vision and strategic objectives for the management and long-term conservation of elephants in South Africa, while contributing to the well-being of people. The strategy will also assist in the further revision of the current Elephant Norms and Standards.

Some community-based conservation programmes in which revenue from the trophy hunting of elephants is fed directly to local communities have proved effective in increasing tolerance to elephants, and thus indirectly in reducing levels of human–elephant conflict (Naidoo et al., 2006; Blanc, 2008). Transboundary elephant populations are increasingly being co-managed by relevant range States and large-scale conservation and management efforts are in place at the national and regional levels (Blanc, 2008). Although up to 70% of the species' range is believed to lie in unprotected land, most large populations occur within protected areas (Blanc, 2008). For example, Namibia's elephants occur across the northern region of the country, mostly in national parks and community areas, while South Africa's largest population inhabits Kruger National Park, and the majority of the remaining population exists in relatively small, fenced areas, many of which are privately owned (Thouless et al., 2016).

References

- Bussi re, E.M.S. and Potgieter, D. (2023) KAZA Elephant Survey 2022, Volume I: Results and Technical Report, KAZA TFCA Secretariat, Kasane, Botswana.
- CoP20 Doc 76.4 (2025) *Report on Monitoring the Illegal Killing of Elephants (MIKE)*
<https://cites.org/sites/default/files/documents/E-CoP20-076-04.pdf>
- CoP20 Doc 76.5 (2025) *Report on Elephant Trade Information System (ETIS)*
<https://cites.org/sites/default/files/documents/E-CoP20-076-05.pdf>
- Gobush, K.S., Edwards, C.T.T., Balfour, D., Wittemyer, G., Maisels, F. and Taylor, R.D. (2022) *Loxodonta africana* (amended version of 2021 assessment). *The IUCN Red List of Threatened Species 2022*: e.T181008073A223031019. <https://dx.doi.org/10.2305/IUCN.UK.2022-2.RLTS.T181008073A223031019.en>.
- Gobush, K.S., Edwards, C.T.T., Maisels, F., Wittemyer, G., Balfour, D. and Taylor, R.D. (2021) *Loxodonta cyclotis* (errata version published in 2021). *The IUCN Red List of Threatened Species 2021*: e.T181007989A204404464. <https://dx.doi.org/10.2305/IUCN.UK.2021-1.RLTS.T181007989A204404464.en>. Accessed on 6 August 2025.
- SC78 Doc 65.1 Annex 1 (2025) Levels of illegal killing of elephants, illegal and legal trade in elephant specimens, the status of elephant populations and the implementation of the African Elephant Action Plan: a report to the CITES Standing Committee.

Inclusion of hornbill genera *Ceratogymna* and *Bycanistes* in Appendix II

Proponents: Cameroon, Congo, Gabon, Niger, Nigeria, Senegal, Sierra Leone, Togo

Summary: The genera *Ceratogymna* (two species) and *Bycanistes* (seven species) comprise nine large-bodied arboreal hornbill species found within Africa (a total of four genera occur there). The hornbills (Bucerotidae) are a diverse lineage of birds with species extending across Africa and Asia. Their unique morphology includes a horn-like casque above the beak, typically more pronounced in males. Many of the Asian hornbill species have been exploited for their casques and skulls, with four species currently included in Appendix I and 20 in Appendix II. African hornbills are threatened across their ranges by habitat loss and degradation resulting from deforestation, climate change and conversion for agriculture. More recently, trade trends have prompted concern about increasing pressure from the regional and international curio and pet trade. No African hornbill species are currently included in the Appendices. There is limited national protection, management, and population monitoring.

All the African hornbill species within the *Bycanistes* and *Ceratogymna* genera have large ranges, but information on population sizes is sparse, with population estimates of more than 10,000 for *Ceratogymna atrata* and *Ceratogymna elata* and no information on the *Bycanistes* species. Two species within the genera, *Bycanistes cylindricus* and *C. elata*, are listed as Vulnerable on the IUCN Red List, while *C. atrata* has been assessed as Near Threatened in 2025. All other species are listed as Least Concern. There is evidence of population reductions in almost all species, except for *C. elata* with an unknown population trend, as well as evidence of local extirpations in some range States. Hornbills have a unique mating strategy, are long-lived and have a low reproductive output, making them particularly vulnerable to exploitation. They are monogamous with females sealing themselves inside natural tree cavities while incubating eggs until the chicks have hatched; they receive all their food from the male through a small slit in the cavity opening and are entirely dependent on him at this time, making both females and chicks vulnerable to the poaching of males.

In addition to threats from hunting and deforestation, there are some indications that African hornbills are being traded internationally to supply the demand for hornbill products in Asia. This trend has been attributed to the lower levels of trade regulation currently in place for African hornbill species. Trade monitoring and control is complicated by the similarities between species in their traded parts and derivative forms, both within the African species and between the African and Asian species, including skulls, feathers, casques, skeletons, and skins. These parts and derivatives are difficult to identify to the species level, as are whole or live females and juveniles. Male skulls/casques are generally distinguishable at species level but identifying those of females and immatures may be challenging. International trade has been recorded in the USA and Asia, with regional trade within Africa. Online surveys have noted more than 500 individual posts containing offers for sale of *Bycanistes* and *Ceratogymna* species between 2010 and 2024, with sellers from Cameroon, Canada, Europe, and the USA. Target countries for the adverts include various Asian countries and the USA; however, the source of the individuals in the posts is unclear and hard to verify. Regional trade is also known to occur in Benin, Nigeria, South Africa, and Togo, with estimated numbers of more than 450 annually between Cameroon and Nigeria alone. Without clear population estimates it is difficult to directly assess the impacts of trade on these species in the wild.

Across Africa, national protection for *Bycanistes* and *Ceratogymna* hornbill species is highly variable. Many countries offer no formal protection, some countries provide conditional or species-specific safeguards, and a few countries enforce stronger protections through permit systems or outright bans on hunting and trade. No commercial captive-breeding facilities are known to exist, and breeding in captivity is known to be challenging. However, these species are recorded in low numbers in zoos and other holding institutions across various countries.

Analysis: The two African hornbill genera *Bycanistes* and *Ceratogymna* comprise nine species, including six assessed as Least Concern, one Near Threatened and two Vulnerable on the IUCN Red List. Population estimates are unknown for most species, with numbers believed to be around or

exceeding 10,000 for both *C. atrata* and *C. elata*; populations are reported to be decreasing for most species. The species generally have large geographic ranges and are predominantly threatened by decreasing extent and quality of habitat caused by deforestation and logging. While national and low levels of international trade in some species are reported along with a few seizures, the impact of this trade on wild populations is unclear. Overall, there is **insufficient information on the status and trends of wild populations, and on the impacts of collection for trade for any species of either genus to determine whether they meet the criteria for inclusion in Appendix II** in Res. Conf. 9.24 (Rev CoP17).

Although there are similarities in the traded forms of Asian CITES-listed hornbill species and *Bycanistes* and *Ceratogymna* species, further evidence would be necessary to determine whether the species enter Asian markets and create enforcement problems for control in the listed species. Therefore, at present it is **unclear whether these species meet criterion A in Annex 2b** of Res. Conf. 9.24 (Rev. CoP17).

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

Genus: *Ceratogymna* (Bonaparte, 1854)

- *Ceratogymna atrata* (Temminck, 1835)
- *Ceratogymna elata* (Temminck, 1831)

Genus: *Bycanistes* (Cabanis & Heine, 1860)

- *Bycanistes brevis* (Friedmann, 1929)
- *Bycanistes bucinator* (Temminck, 1824)
- *Bycanistes cylindricus*
 - Subspecies: *B. c. cylindricus* (Temminck, 1831) *B. c. albotibialis* (Cabanis & Reichenow, 1877)
- *Bycanistes fistulator*
 - Subspecies: *B. f. fistulator* (Cassin, 1852) *B. f. sharpii* (Elliot, 1873)*** *B. f. duboisi* (Sclater, 1922)
- *Bycanistes subcylindricus*
 - Subspecies: *B. s. subcylindricus* (Sclater, 1870) *B. s. subquadratus* (Cabanis, 1880)

The proponents suggest following the more recent taxonomic and nomenclatural changes within the genus *Bycanistes*, but these are not yet recognised by CITES. They do not explicitly propose to designate an updated standard nomenclature reference for the two genera *Bycanistes* and *Ceratogymna*, so the adopted 'default' nomenclature standard, being Dickinson (Ed.) 2003, applies until an updated reference is adopted by the Parties (see Nomenclature notes on the proposals submitted by the Secretariat in CoP20 Inf. 6). A comparison of these changes appears in Table 1. For simplicity, the latest (del Hoyo and Collar, 2014) nomenclature will be used in this analysis, as per the Proposal.

Table 1: Taxonomy and nomenclature per Dickinson (2003) and del Hoyo and Collar (2014) used in this analysis for the genus *Bycanistes*.

Dickinson (Ed.) 2003	del Hoyo and Collar (2014)	Change
	<i>Bycanistes albotibialis</i>	Elevated to species level
<i>Bycanistes brevis</i>	<i>Bycanistes brevis</i>	No change
<i>Bycanistes bucinator</i>	<i>Bycanistes bucinator</i>	No change
<i>Bycanistes cylindricus</i>	<i>Bycanistes cylindricus</i>	Subspecies removal
Subspecies: <i>B. c. cylindricus</i>		Elevated to species level
Subspecies: <i>B. c. albotibialis</i>		Elevated to species level
<i>Bycanistes fistulator</i>	<i>Bycanistes fistulator</i>	Subspecies removal
Subspecies: <i>B. f. fistulator</i>		Elevated to species level
	<i>Bycanistes sharpii</i>	Elevated to species level
Subspecies: <i>B. f. sharpii</i>	Subspecies: <i>B. s. sharpii</i>	Removed from <i>B. fistulator</i>

Dickinson (Ed.) 2003	del Hoyo and Collar (2014)	Change
Subspecies: <i>B. f. duboisi</i>	Subspecies: <i>B. s. duboisi</i>	Removed from <i>B. fistulator</i>
<i>Bycanistes subcylindricus</i>	<i>Bycanistes subcylindricus</i>	
Subspecies: <i>B. s. subcylindricus</i>	Subspecies: <i>B. s. subcylindricus</i>	Elevated to species level
Subspecies: <i>B. s. subquadratus</i>	Subspecies: <i>B. s. subquadratus</i>	

Range

All nine species within the genera *Ceratogymna* and *Bycanistes* are endemic to Africa. The genus *Ceratogymna* is found in West and Central Africa; *Bycanistes* occurs in West, Central, East, and Southern Africa.

Table 2: *Bycanistes* and *Ceratogymna* species and geographic range

Species	Geographic Range
<i>Bycanistes albotibialis</i>	Angola; Benin; Cameroon; Central African Republic; Congo; The Democratic Republic of the Congo; Equatorial Guinea; Gabon; Nigeria; Sudan; Uganda
<i>Bycanistes brevis</i>	Ethiopia; Kenya; Malawi; Mozambique; South Sudan; Sudan; Republic of Tanzania; Zambia; Zimbabwe. Extant and vagrant in Eritrea. Unconfirmed in South Africa.
<i>Bycanistes bucinator</i>	Angola; Botswana; Burundi; The Democratic Republic of the Congo; Eswatini; Kenya; Malawi; Mozambique; Namibia; South Africa; Republic of Tanzania; Zambia; Zimbabwe. In South Africa, the species is found along the eastern coast in the following provinces – KwaZulu-Natal, Eastern Cape and Western Cape.
<i>Bycanistes cylindricus</i>	Côte d'Ivoire; Ghana; Guinea; Liberia; Sierra Leone; Togo. Unconfirmed in Togo.
<i>Bycanistes fistulator</i>	Benin; Côte d'Ivoire; Ghana; Guinea; Guinea Bissau; Liberia; Mali; Nigeria; Senegal; Sierra Leone; Togo.
<i>Bycanistes sharpii</i>	Angola; Cameroon; Central African Republic; Congo; The Democratic Republic of the Congo; Equatorial Guinea; Gabon; Nigeria; Uganda.
<i>Bycanistes subcylindricus</i>	Angola; Burundi; Cameroon; Central African Republic; Congo; The Democratic Republic of the Congo; Côte d'Ivoire; Equatorial Guinea; Gabon; Ghana; Guinea-Bissau; Kenya; Liberia; Nigeria; Rwanda; Sierra Leone; South Sudan; Sudan; Republic of Tanzania; Togo; Uganda.
<i>Ceratogymna atrata</i>	Angola; Cameroon; Central African Republic; Congo; The Democratic Republic of the Congo; Côte d'Ivoire; Equatorial Guinea; Gabon; Ghana; Guinea; Liberia; Nigeria; Sierra Leone; South Sudan; Sudan; Togo; Uganda.
<i>Ceratogymna elata</i>	Cameroon; Côte d'Ivoire; Ghana; Guinea; Guinea Bissau; Liberia; Mali; Nigeria; Senegal; Sierra Leone. Possibly present in Benin, but it is disputed. Possibly extinct in Togo.

IUCN Global Category

Ceratogymna atrata Near Threatened (assessed 2025, ver. 3.1)

Ceratogymna elata Vulnerable A2cd+3cd+4cd (assessed 2025, ver. 3.1)

Bycanistes albotibialis Least Concern (assessed 2024, ver. 3.1)

Bycanistes brevis Least Concern (assessed 2018, ver. 3.1)

Bycanistes bucinator Least Concern (assessed 2018, ver. 3.1)

Bycanistes cylindricus Vulnerable A2cd+3cd+4cd (assessed 2018, ver. 3.1)

Bycanistes fistulator Least Concern (assessed 2018, ver. 3.1)

Bycanistes sharpii Least Concern (assessed 2024, ver. 3.1)

Bycanistes subcylindricus Least Concern (assessed 2025, ver. 3.1)

Biological and trade criteria for inclusion in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)

A) Trade regulation needed to prevent future inclusion in Appendix I

B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences

There are limited data on total population sizes or geographic ranges for species within both genera *Bycanistes* and *Ceratogymna*. Most species' population trends are decreasing, except for *B. subcylindricus* with only local decreases documented, and *Ceratogymna elata*, the population trend of which is currently unknown.

C. atrata is estimated to have a population more than 10,000 individuals and has been recorded at densities of 7.2 ind./km² (Santini et al., 2024). However, population size estimates could not be accurately determined from this density figure, due to the impact of hunting and deforestation in other areas of the species' range (Jensen and

Arcilla, 2024; BirdLife International in litt., 2025). *C. atrata* is thought to have been extirpated from parts of Cameroon, Congo, and Nigeria (Kemp and Kirwan, 2020).

C. elata was previously recorded at 0.35 ind./km², with estimates of 12,500 individuals in total (BirdLife International, 2016b). However, the species' range size was reassessed as considerably larger than previous estimates (Jung et al., 2020). Again, hunting and deforestation make population estimates difficult, with the population now considered unknown, but likely small (BirdLife International in litt., 2025). *C. elata* has an Area of Habitat of 308,836 km² (Ridley et al., in prep).

The population of *B. cylindricus* is suspected to have been in rapid decline due to habitat destruction and degradation, as well as unsustainable hunting pressure (BirdLife International, 2018c). Its Area of Habitat is 255,728 km² (Ridley et al., in prep). For both *B. albotibialis* and *B. sharpii*, tree cover decreases in the species' range are thought to be indicative of decreases in species population size (BirdLife International, 2024a; 2024b). The *B. subcylindricus* population size is still unquantified, but noted to be 'substantial', with local extirpation in Ghana (Holbech et al., 2018). Population estimates are unknown for *B. brevis*, *B. bucinator*, *B. cylindricus*, and *B. fistulator*.

Table 3: Summary of known information on population and geographic range sizes and trends for *Bycanistes* and *Ceratogymna* species.

Species	Population Size & Trend	Geographic Range & Trend
<i>Bycanistes albotibialis</i>	Population size unknown Decreasing at 11-14% over 20 years (BirdLife International, 2024a)	EOO: 3,170,000 km ² (BirdLife International, 2024a) Decreasing
<i>Bycanistes brevis</i>	Population size unknown Decreasing (BirdLife International, 2018a)	EOO: 2,420,000 km ² (BirdLife International, 2018a) Range trend unknown
<i>Bycanistes bucinator</i>	Population size unknown Decreasing (BirdLife International, 2018b)	EOO: 6,520,000 km ² (BirdLife International, 2018b)
<i>Bycanistes cylindricus</i>	Population size unknown Decreasing (BirdLife International, 2018c)	EOO: 556,000 km ² (BirdLife International, 2018c) AOH: 255,728 km ² (Ridley et al., in prep. 2025) Range trend unknown
<i>Bycanistes fistulator</i>	Population size unknown Decreasing (BirdLife International, 2018d)	EOO: 1,880,000 km ² (BirdLife International, 2018d) Range trend unknown
<i>Bycanistes sharpii</i>	Population size unknown Decreasing at 11-14% over 20 years (BirdLife International, 2024b)	EOO: 5,200,000 km ² (BirdLife International, 2024b) Decreasing
<i>Bycanistes subcylindricus</i>	Population size unknown but considered 'substantial' (Poonswad et al., 2013), Local decreases (Holbech et al., 2018)	EOO: 7,210,000 km ² (BirdLife International, 2017) Range trend unknown
<i>Ceratogymna atrata</i>	>10,000 individuals (Santini et al., 2024) Decreasing (BirdLife International, 2016a)	EOO: 5,280,000 km ² (BirdLife International, 2016a) Range trend unknown
<i>Ceratogymna elata</i>	12,500 individuals (BirdLife International, 2016b) Decreasing in 2016, current population trend unknown (BirdLife International in litt., 2025)	AOH: 308,836 km ² (Ridley et al., in prep) Range trend unknown

National trade

National trade and utilisation of *Ceratogymna* and *Bycanistes* hornbills are widespread across their range, driven primarily by subsistence hunting and local consumption. In Cameroon's Ebo Forest and Littoral Region, surveys and questionnaires revealed high levels of hornbill hunting, with *B. albotibialis* and *C. atrata* being the most frequently consumed bird species. Similar patterns of wild meat hunting have been documented in Ghana and Congo, where hornbill meat is regularly consumed in households. These practices raise concerns about sustainability, as hornbills' slow reproductive rates and specialised life history traits make them particularly vulnerable to overexploitation.

International trade

The SS indicates that the demand for these species internationally, in conjunction with other species threats, is reported to be increasing their risk of extinction. Berryman in litt. (2025) notes that African and Asian hornbill species share many life-history traits (e.g. long generation length, monogamous breeding, nesting behaviour) and biological characteristics that make them vulnerable to exploitation. Hunting and trade have had a significant impact

on Asian hornbills, making it reasonable to infer that African hornbills are similarly at risk, especially given observations in reduced density of some species (Berryman in litt., 2025).

International trade in African hornbills has been noted to occur since at least the 1980s, with concerns about the scale of trade raised by Trail (2007) and in CoP8 Prop. 29 (CITES Secretariat, 2025). Hunting for international trade, both for the pet market and for skulls, heads, and casques as curios, occurs across the continent. Both *Bycanistes* and *Ceratogymna* species are traded live, dead, as feathers, heads, skulls, skins, skeletons, and casques. Juveniles and females also in trade, with some trade in live juveniles destined for Asian markets (e.g. Su et al., 2024). According to the SS, *C. atrata*, *C. elata* and *Bycanistes* species are also used in traditional medicine, with preference for larger individuals. International trade is inferred through the widespread appearance of species in markets outside of their natural distribution range (e.g. Su et al., 2024).

Markets and household surveys show that *B. albotibialis* and *C. atrata* are the most commonly traded species. Over 1,400 *C. atrata* carcasses were found in market surveys in Equatorial Guinea between 1997 and 2010, and it is estimated that more than 450 carcasses are traded annually between Cameroon and Nigeria. Markets in Benin, Nigeria, South Africa, and Togo all contained non-native *Bycanistes* and *Ceratogymna* species for sale in a study by Tinsman et al. (2025), and *C. atrata* and *C. elata* were reported as being highly prevalent in trade by Donald et al. (2024).

While South Africa reports no significant trade of *B. bucinator*, other regions face increasing pressure, not only from local consumption but also from emerging foreign demand, particularly for hornbill heads from Cameroon for the USA market. This shift suggests that national trade is evolving and may pose growing threats to hornbill populations if not addressed through conservation and regulation (Tinsman et al., 2025). According to Kemp in litt. (2025), trade has shifted from wild meat for subsistence, with heads as a byproduct, to the main traded form now being heads, possibly for their casques, with some live trade in juveniles destined for Asian markets (e.g. Su et al., 2024).

Data for the USA

LEMIS data show that between 2008 and 2022, the USA reported importing 173 whole organism equivalents (i.e. live individuals, skulls, trophies, bodies, skins and skeletons) from *Ceratogymna* species, and 106 from *Bycanistes* species, for either personal use, use as hunting trophies, or commercial purposes (see Table 4). The main commodity from *Ceratogymna* species was skulls and from *Bycanistes* species trophies and live individuals, and most commodities were reported to be wild-sourced. Ghana and the United Republic of Tanzania were amongst the top origin countries for both genera, but Cameroon was the main origin country for *Ceratogymna* species, and Mozambique and Mali for *Bycanistes* species. These figures likely only show a subset of international trade in these species as they are limited to imports into the USA, and the other significant market outside of Africa is believed to be Asia (Kemp in litt., 2025).

Table 4. Imports of whole organism equivalents (WOE) from *Ceratogymna* and *Bycanistes* species between 2008 and 2022 as reported by the USA for personal use, use as hunting trophies, or commercial purposes from all sources. Source: LEMIS.

Genus	Imports of whole organism equivalents	Commodities	Source	Purpose	Top origin countries*
<i>Ceratogymna</i>	173	Skulls (78%), trophies (8%), live (4%), bodies (4%), skins (3%), skeletons (2%)	Wild-sourced (98%), captive-bred (2%)	Commercial (88%), personal (6%), hunting trophy (6%)	Cameroon (77%), Ghana (8%), United Republic of Tanzania (6%)
<i>Bycanistes</i>	106	Trophies (51%), live (26%), skins (16%), skulls (4%), bodies (3%)	Wild-sourced (78%), captive-bred (28%)	Hunting trophy (46%), commercial (46%), personal (8%)	Mozambique (30%), Mali (23%), Ghana (20%), United Republic of Tanzania (18%)

* All top origin countries were also exporters apart from the United Republic of Tanzania, where specimens were re-exported by Canada

The top species in exports were *Ceratogymna atrata* (53% of all imports of *Ceratogymna* species) and *Bycanistes fistulator* (34% of all imports of *Bycanistes* species) (Figures 1 and 2). Out of all species imported to the USA, whole organism equivalents from captive-bred sources were reported in notable proportions only for *Bycanistes fistulator* (67% of 36 WOE imported) and *Bycanistes subcylindricus* (all four WOE imported). The remaining imports of WOE for each species were all from wild sources aside from one hunting trophy and one skull from *Ceratogymna atrata*, as well as one live individual of *Ceratogymna elata*.

Most imports of WOE of *Ceratogymna* species occurred in 2008, with average of eight per year between 2009 and 2022, whilst most imports of WOE of *Bycanistes* species occurred in 2018, with an average of two per year between 2019 and 2021. LEMIS data between the second quarter of 2021 and 2024 were limited to imports for personal use and use as hunting trophies only. There were no imports of WOE for these purposes for either genus between 2023 and 2024 but there may be commercial imports of WOE from each genus between 2021 and 2024 that are not shown in these data.

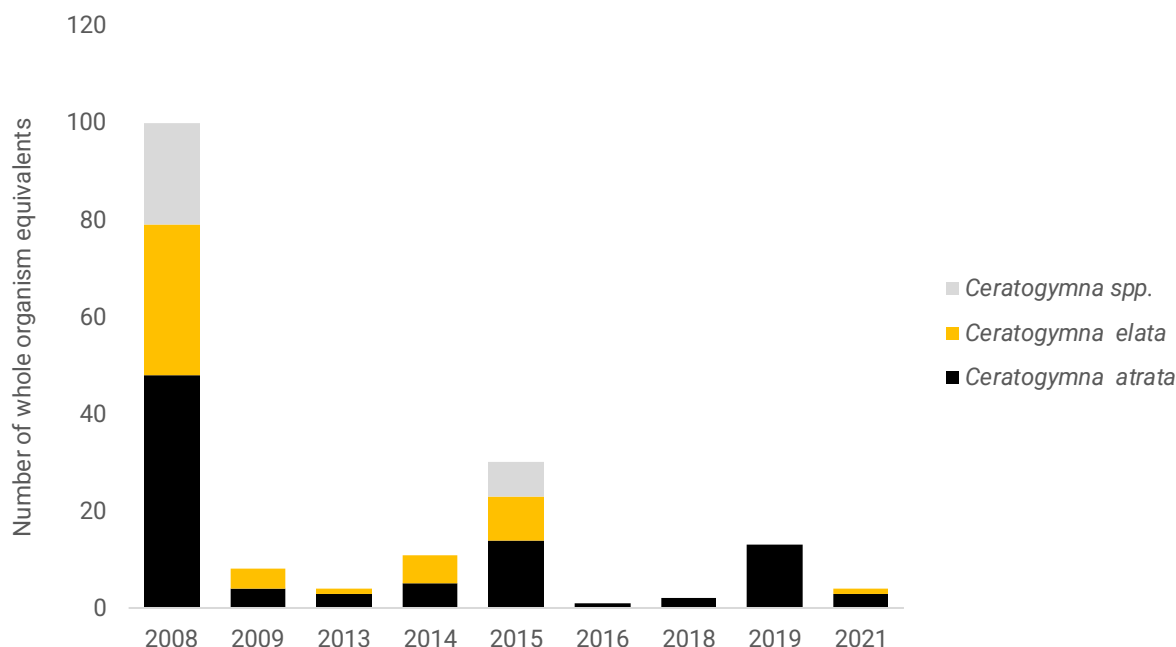


Figure 1. *Ceratogymna* species imported into the USA as whole organism equivalents (skulls, trophies, live, bodies, skins and skeletons) for personal use, use as hunting trophies, or commercial purposes from all sources and as reported by the USA between 2008 and 2022. Data from the second quarter of 2021 onwards was limited to imports for personal use and use as hunting trophies only. Source: LEMIS.

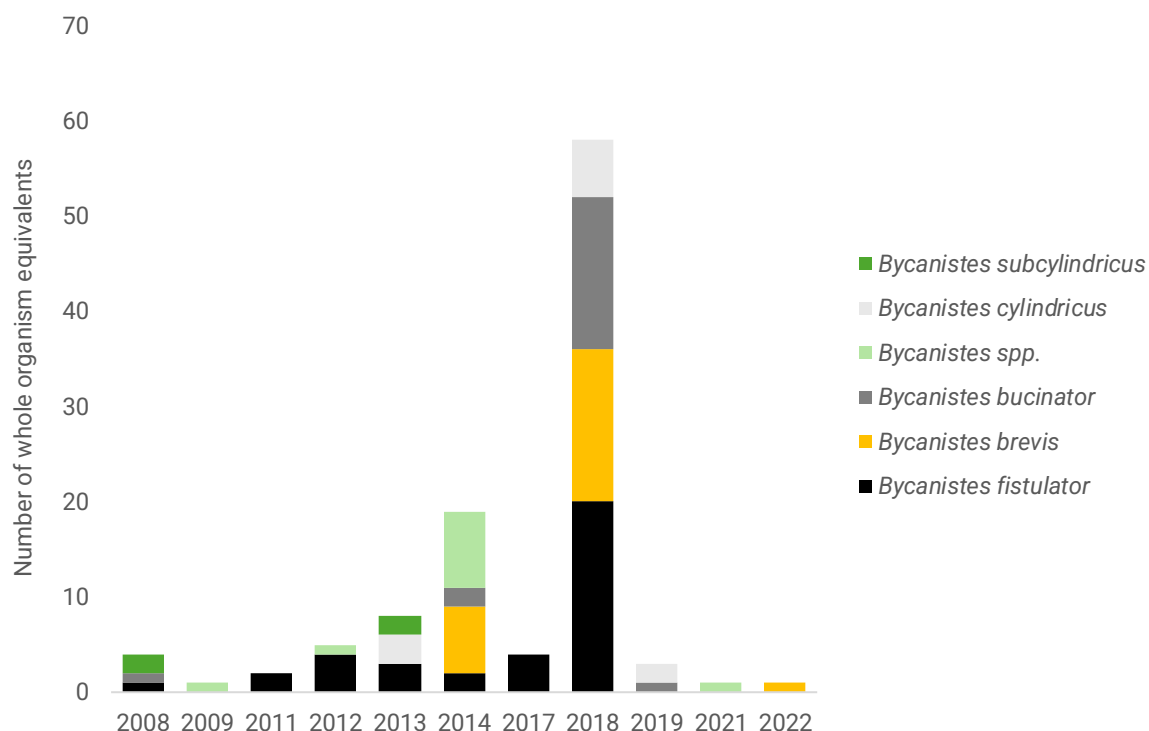


Figure 2. *Bycanistes* species imported into the USA as whole organism equivalents (skulls, trophies, live, bodies, skins and skeletons) for personal use, use as hunting trophies, or commercial purposes from all sources and as

reported by the USA between 2008 and 2022. Data from the second quarter of 2021 onwards was limited to imports for personal use and use as hunting trophies only (Source: LEMIS).

International trade at low levels is documented, but population impacts are relatively unknown. Hunting of both *Bycanistes* and *Ceratogymna* species has been shown to reduce population sizes and, in some cases, result in local extirpations (e.g. Holbech et al., 2018). Drivers of hunting include subsistence (Holbech et al., 2018; Whytock et al., 2016; 2018), but also include international trade (Su et al., 2024).

Illegal trade

TRAFFIC's WiTIS data on illegal trade only indicates one seizure including *Bycanistes* or *Ceratogymna* parts: in 2021, two skulls each of *C. atrata*, *C. elata*, and *B. albotibialis* were found in a seizure of more than 100 wildlife parts in the USA. Online trade is evident on social media and online stores (e.g. eBay, Etsy, Instagram), with sellers from Cameroon, Canada, Europe, and the USA, and more than 500 individual posts between 2010 and 2024 (Su et al., 2024; Tinsman et al., 2025).

Inclusion in Appendix II to improve control of other listed species

A) Specimens in trade resemble those of species listed in Appendix II under Res. Conf. 9.24 (Rev. CoP17) Annex 2 a or listed in Appendix I

Currently, no African hornbill species are listed in the CITES Appendices, but 24 Asian hornbill species are either Appendix I or Appendix II. Kemp in litt. (2025) indicates that trade in live African hornbills has been shown to be destined for Asia, with the juveniles and females difficult to distinguish from Asian species. The SS notes that female and juvenile casques and skulls are also difficult to distinguish from Asian species, as are all feathers, with the high volume of African hornbill skulls and feathers in trade a complication for enforcing protections for Asian hornbills listed on CITES Appendices. According to Kemp in litt. (2025) and Skinner in litt. (2025), the most traded African hornbill species are *C. atrata* and *B. albotibialis* (although this is not reflected in the LEMIS database for trade involving the USA, where *B. fistulator* predominates among *Bycanistes* species), with international trade apparent, and difficulties in distinguishing individual species in their traded forms. The CITES Secretariat (2025) notes that this has been the case since the Australasian hornbill listing at CoP8 in Kyoto 1992 (e.g., CoP8 Prop. 29).

B) Compelling other reasons to ensure that effective control of trade in currently listed species is achieved

Not applicable.

Additional Information

Threats

Ceratogymna and *Bycanistes* species are largely forest-dwelling, with specialised hole-nesting requirements. As a result, they are particularly vulnerable to habitat loss or degradation and fragmentation, including as a result of climate change, and logging and deforestation. Life history characteristics make these genera reliant on old-growth forest and vulnerable to logging as well as the removal of males during the breeding season. Females seal themselves and their eggs in a tree cavity, and are fed by the male until the chicks are ready to fledge. If males, with larger casques and more distinct plumage are taken during breeding season, the knock-on effects are that the female and next generation will also be lost.

Hunting is a threat to these species, largely for wild meat.

Conservation, management and legislation

A summarized table of national protections throughout *Bycanistes* and *Ceratogymna* range States, based on Tinsman et al. (2025), is presented below (Table 5). Partial protection for *Bycanistes* or *Ceratogymna* species is afforded in 18 of 35 range States. For Guinea and Sierra Leone, the status of protection of hornbill species is unclear. Seven countries require permits to hunt these species, with four additional countries providing some additional protection. All hornbill species appear to be protected in Namibia and Senegal. The wide ranges of these species mean that several *Bycanistes* and *Ceratogymna* species are found in protected areas, but local extirpations and population decreases have occurred, including within these protected areas (e.g., Holbech et al., 2018). At least one systematic monitoring scheme is known for *B. brevis* (BirdLife International, 2018a), and conservation sites have been identified throughout the ranges of all species, including at least one protected area (BirdLife International, 2016a, b; 2017; 2018a, b, c, d; 2024a, b).

Table 5. Summary of *Bycanistes* and *Ceratogymna* species protections across range States, per the SS (countries where no protection is provided, have been omitted from the table).

Country	Protection Listed in National Legislation
Burundi	Protected if considered threatened by international conventions
Cameroon	Illegal to kill <i>C. elata</i>
Côte d'Ivoire	Protected, permits needed for hunting

Country	Protection Listed in National Legislation
Eritrea	No species may be hunted or exported without a permit
Eswatini	No species may be hunted without a permit
Ethiopia	No protection, fee for live trade of <i>B. brevis</i>
Gabon	No hunting without a permit
Guinea	Protected if threatened with extinction
Guinea Bissau	Hornbills protected in 1955, current protection unknown
Malawi	<i>B. brevis</i> is protected
Mali	Hunting with a permit
Namibia	All hornbills protected
Republic of the Congo	<i>C. atrata</i> , <i>B. fistulator</i> are partially protected and may be hunted with a permit, no commerce, transport, or trafficking of any hornbill species
Senegal	All species fully protected
Sierra Leone	Possible protection: prohibition of use of birds used in the plumage trade
Togo	<i>C. atrata</i> and <i>C. elata</i> partially protected, permit required for hunting
Uganda	Permit needed for hunting or capture

Artificial Propagation/captive breeding

Their nesting requirements and monogamous mating system make hornbills difficult to successfully breed in captivity (e.g., Fourage *et al.*, 2022). No known commercial captive breeding facilities exist, and so international trade is primarily from wild sources. *Known ex-situ populations, according to ZIMS (2025) are as follows:*

- *Ceratogymna atrata*: 68 individuals in 27 holding institutions across 16 countries, with one population management program implemented by the European Association of Zoos and Aquaria (EAZA).
- *Ceratogymna elata*: Four individuals in two holding institutions across two countries.
- *Bycanistes albotibialis*: Two individuals in one holding institution.
- *Bycanistes brevis*: 142 individuals in 62 holding institutions across 22 countries, with one population management program implemented by the European Association of Zoos and Aquaria (EAZA).
- *Bycanistes bucinator*: 118 individuals in 59 holding institutions across 23 countries, with one population management program implemented by the Association of Zoos and Aquariums (AZA).
- *Bycanistes fistulator*: Ten individuals in three holding institutions across two countries, with one population management program implemented by the Association of Zoos and Aquariums (AZA).
- *Bycanistes subcylindricus*: 27 individuals in 14 holding institutions across nine countries.

Implementation challenges (including similar species)

Sexual dimorphism and age-related variation in casque size, as well as similarities between species, particularly between females and juveniles of different species, make identification of parts, females, and juveniles challenging. In addition, some CITES-listed Asian hornbill species are difficult to differentiate from African hornbills in their traded forms (feathers, skulls, skeletons, females, and juveniles). Challenges also exist with differentiation from other hornbill genera in the family Bucerotidae, both in Africa and in Asia, including large ground hornbills *Bucorvus*, smaller African hornbills *Lophoceros*, *Horizoceros*, and *Tockus*, and Asian *Berenicornis*, *Rhinoplax*, *Buceros*, *Anorrhinus*, *Ocyrceros*, *Anthracoseros*, *Aceros*, *Penelopides*, and *Rhyticeros*.

Potential risk(s) of a listing

Regulation of *Ceratogymna* spp. and *Bycanistes* spp. could impact international trade in other, non-CITES species within the family Bucerotidae, although for the most part these species lack the desired enlarged casques, and tend to be traded live, suggesting different drivers of trade. *Tinsman et al. (2025) suggest that including the Asian hornbills in CITES Appendices in 1992 may have unintentionally shifted demand to unprotected African hornbills, which appear in trade in greater numbers than their Asian counterparts.*

References

- Berryman, A. (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- BirdLife International (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- BirdLife International (2016a) *Ceratogymna atrata*. The IUCN Red List of Threatened Species 2016: e.T22682621A92954110. <https://dx.doi.org/10.2305/IUCN.UK.2016-3.RLTS.T22682621A92954110.en>. Accessed on 28 July 2025.
- BirdLife International (2016b) *Ceratogymna elata*. The IUCN Red List of Threatened Species 2016: e.T22682627A92954374. <https://dx.doi.org/10.2305/IUCN.UK.2016-3.RLTS.T22682627A92954374.en>. Accessed on 28 July 2025.
- BirdLife International (2017) *Bycanistes subcylindricus* (amended version of 2016 assessment). The IUCN Red List of Threatened Species 2017: e.T22682577A111897019. <https://dx.doi.org/10.2305/IUCN.UK.2017-1.RLTS.T22682577A111897019.en>. Accessed on 28 July 2025.

- BirdLife International (2018a) *Bycanistes brevis*. The IUCN Red List of Threatened Species 2018: e.T22682570A130081772. <https://dx.doi.org/10.2305/IUCN.UK.2018-2.RLTS.T22682570A130081772.en>. Accessed on 28 July 2025.
- BirdLife International (2018b) *Bycanistes bucinator*. The IUCN Red List of Threatened Species 2018: e.T22682548A130081383. <https://dx.doi.org/10.2305/IUCN.UK.2018-2.RLTS.T22682548A130081383.en>. Accessed on 28 July 2025.
- BirdLife International (2018c) *Bycanistes cylindricus*. The IUCN Red List of Threatened Species 2018: e.T22682585A129918692. <https://dx.doi.org/10.2305/IUCN.UK.2018-2.RLTS.T22682585A129918692.en>. Accessed on 28 July 2025.
- BirdLife International (2018d) *Bycanistes fistulator*. The IUCN Red List of Threatened Species 2018: e.T22725962A132234122. <https://dx.doi.org/10.2305/IUCN.UK.2018-2.RLTS.T22725962A132234122.en>. Accessed on 28 July 2025.
- BirdLife International (2024a) *Bycanistes albotibialis*. The IUCN Red List of Threatened Species 2024: e.T22682609A264023462. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T22682609A264023462.en>. Accessed on 28 July 2025.
- BirdLife International. (2024b) *Bycanistes sharpii*. The IUCN Red List of Threatened Species 2024: e.T22725974A263962879. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T22725974A263962879.en>. Accessed on 28 July 2025.
- CITES Secretariat (2025) *Notification to the Parties No. 2025/102: Proposals to amend Appendices I and II. Provisional assessments by the Secretariat. Convention on International Trade in Endangered Species of Wild Fauna and Flora.* <https://cites.org/sites/default/files/notifications/E-Notif-2025-102.pdf>
- Cronin, D.T., Woloszynek, S., Morra, W.A., Honarvar, S., Linder, J.M., Gonder, M.K., O'Connor, M.P. and Hearn, G.W. (2015) Long-term urban market dynamics reveal increased bushmeat carcass volume despite economic growth and proactive environmental legislation on Bioko Island, Equatorial Guinea. *PLoS One* 10:e0137470. <https://doi.org/10.1371/journal.pone.0137470>.
- del Hoyo, J., and Collar, N.J. [with Christie, D.A., Elliott, A. and Fishpool, L.D.C]. (2014) *HBW and BirdLife International Illustrated Checklist of the Birds of the World. Volume 1: Non-passerines*. Lynx Edicions BirdLife International, Barcelona, Spain and Cambridge, UK.
- Donald, P. F., Fernando, E., Brown, L., Busana, M., Butchart, S. H. M., Chng, S., de la Colina, A., Ferreira, J. M., Jain, A., Jones, V. R., Lapido, R., Malsch, K., McDougall, A., Muccio, C., Nguyen, D., Outhwaite, W., Petrovan, S. O., Stafford, C., Sutherland, W. J., Tallowin, O. and Safford, R. (2024) Assessing the global prevalence of wild birds in trade. *Conservation Biology* 38(5). <https://doi.org/10.1111/cobi.14350>
- Fa, F. E., Seymour, S., Dupain, J., Amin, R., Albrechtsen, L. and MacDonald (2006) Getting to grips with the magnitude of exploitation: Bushmeat in the Cross-Sanaga Rivers region, Nigeria and Cameroon. *Biological Conservation* 129:497-510.
- Fourage, A., Nijman, V. and Shepherd, C.R. (2022) *Hornbills in zoos in Thailand: species, numbers, and their welfare*. Hornbill Natural History & Conservation. Hornbill Specialist Group. Volume 3. Pp. 1-16.
- Holbech, L.H., Annorabah, N.N.D., Phalan, B. and Arcilla, N. (2018) Uncontrolled hunting and habitat degradation decimate and extirpate forest hornbills in Ghana, West Africa. *Biological Conservation* 223:104-111.
- Jensen, C. and Arcilla, N. (2024) *Investigating the impacts and implications of hunting and wildlife trade on the conservation status of forest hornbills, July, 2024*. Project Report. African Bird Club. Available at: https://www.africanbirdclub.org/wp-content/uploads/2024/09/HornbillProjectReport_JensenArcillaFinal.pdf
- Kemp, L. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK. IUCN Hornbill Specialist Group.
- Kemp, A.C. and Kirwan, G.M. (2020) Black-casqued Hornbill (*Ceratogymna atrata*), version 1.0. In *Birds of the World* (J. del Hoyo, A. Elliott, J. Sargatal, D. A. Christie, and E. de Juana, Editors). Cornell Lab of Ornithology, Ithaca, NY, USA. <https://doi.org/10.2173/bow.blchor1.01>
- Poonswad, P., Kemp, A. C. and Strange, M. (2013) *Hornbills of the world: A photographic guide*. Draco Publishing and Distributions Pte. Ltd., Singapore and Hornbill Research Foundation, Thailand.
- Species360 Zoological Information Management System (ZIMS) (2025). Available zims.Species360.org.
- Su, S., Guetse, F. and Arcilla, N. (2024) A price on their heads? Assessing foreign demand as a driver of hornbill hunting in Cameroon. *Global Ecology and Conservation* 51:e02905. <https://doi.org/10.1016/j.gecco.2024.e02905>.
- Tinsman, J., Woodward, A., Su, S., Skinner, D.H., Kemp, L.V., Abeh, A.D., Afiademanyo, K.M., Arcilla, N., Brouwer, K., Chaffra, A.S., Forzi, F., Guetse, F., Holbech, L.H., Ikome, D., Koutchoro, A.M., Odoukpe, S.G.K., Orenstein, R., Nshom, D., Russo, N.J., Smith, T.B., Terrill, R.S., Trail, P.W. and Terrill, J.A.O. (2025) Intense international exploitation of African hornbills necessitates urgent conservation measures, including CITES listing. *Biological Conservation* 308:111105. <https://doi.org/10.1016/j.biocon.2025.111105>.
- Trail, P.W. (2007) African hornbills: keystone species threatened by habitat loss, hunting and international trade. *Ostrich* 78:609-13. <https://doi.org/10.2989/OSTRICH.2007.78.3.7.318>.

Transfer of White-Backed Vulture *Gyps africanus* and Ruppell's Vulture *Gyps rueppelli* from Appendix II to Appendix I

Proponents: Benin, Burkina Faso, Burundi, Cameroon, Chad, Congo, Gambia, Guinea, Niger, Nigeria, Senegal, Sierra Leone, Togo

Summary: White-Backed Vulture *Gyps africanus* and Ruppell's Vulture *Gyps rueppelli* are both large, old-world vultures with expansive ranges across sub-Saharan Africa, with *G. rueppelli* also occurring in southwestern Europe and the Middle East. Historically reported to be "common", both species have experienced significant declines across their ranges in recent years. The global population of *G. africanus* was inferred to be around 27,000 individuals in the mid-2010s, estimated to have declined by 80% (range: 63-89%) over three generations (40 years). In 2021, the population of *G. rueppelli* was inferred to be fewer than 30,000 individuals with an estimated decline of over 90% (range: 88-98%) over three generations (also ~40 years). Declines of *G. africanus* in Burkina Faso and Mali have been rapid and local extirpations are evident in parts of Ghana, Niger, and Nigeria. Likewise, *G. rueppelli* has experienced 93-99% declines in Burkina Faso, Mali, and Niger, with severe population declines and local extirpations elsewhere in West Africa. Both were globally assessed as Critically Endangered on the IUCN Red List in 2021.

Vultures are scavengers, forming an important part of the ecosystem by breaking down carcasses. They are primarily threatened by reduced food abundance, contamination of their food source, habitat loss, accidental and deliberate poisoning, and hunting for trade. Additional pressures include human disturbance of nesting sites, collision with wind turbines and electrocution from power cables.

Gyps rueppelli and *G. africanus* were included in Appendix II of CITES in 1979 as part of the listing of the Order Falconiformes. Vulture parts are traded nationally and internationally for belief-based use or as wild meat, with body parts commonly used in traditional practices to treat physical and mental ailments and to bring good luck when gambling or competing. The growing profitability of regional belief-based trade is reportedly driving increased exploitation and there are estimates of several thousand individuals traded annually across West and Central Africa, mainly for belief-based use.

Reported international trade in vultures has primarily involved live individuals imported by European countries. Overall figures are low: between 1979 and 2023, around 900 *G. africanus* and just under 700 *G. rueppelli*, were reported in trade, mostly as live, wild-sourced specimens, with the majority originating in Guinea, United Republic of Tanzania and Cameroon and destined for countries in Europe. Some illegal trade has been reported, including seizures of over 883 individuals between 2012 and 2025.

Analysis: Neither *Gyps africanus* nor *G. rueppelli* has a restricted area of distribution. Up-to-date population information is lacking, although according to the most recent estimates, neither species has a small population. However, both are recorded as having undergone steep population declines and local extirpations in recent years, ascribed to a range of factors, including habitat loss, decreasing food availability and deliberate and inadvertent poisoning. As a result, both have been assessed as Critically Endangered on the IUCN Red List both in 2021 with **marked historic declines and meet criterion C** in Annex 1 of Res. Conf. 9.24 (Rev. CoP17).

Reported legal commercial international trade in wild-sourced individuals of both species has been very limited and would be unlikely to have had an adverse impact on wild populations. However, there are reports of illegal cross-border trade, particularly in West and Central Africa, of vulture parts and derivatives used for a variety of belief-based purposes. Levels of such trade are unknown but may be affecting the species in combination with other threats. Considering the Critically Endangered status of both species, **Appendix-I listing seems proportionate**, however, with most of the trade across porous country borders, the benefits of the listing will require significant enforcement efforts.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

Both *G. africanus* and *G. rueppelli* were first listed on Appendix II in 1979 under the Order listing Falconiformes species.

Taxonomy

Gyps africanus (Salvadori, 1865); *Gyps rueppelli* (Brehm, 1852)

Standard nomenclature: Dickinson, E.C. (Ed.) 2003. The Howard and Moore Complete Checklist of the Birds of the World. Revised and enlarged 3rd Edition. Christopher Helm, London. 1039 pp. in combination with Dickinson, E.C. 2005. Corrigenda 4 (02.06.2005) to Howard & Moore Edition 3 2003. (Available on the CITES website).

Range

G. africanus:

Native, Extant (resident): Angola; Benin; Botswana; Burkina Faso; Burundi; Cameroon; Central African Republic; Chad; The Democratic Republic of the Congo; Côte d'Ivoire; Eritrea; Eswatini; Ethiopia; Gambia; Ghana; Guinea; Guinea-Bissau; Kenya; Malawi; Mali; Mauritania; Mozambique; Namibia; Rwanda; Senegal; Sierra Leone; Somalia; South Africa; South Sudan; Sudan; United Republic of Tanzania; Togo; Uganda; Zambia; Zimbabwe

Native, Possibly Extant (resident): Niger; Nigeria

Extant & Vagrant (resident): Liberia

Extant & Vagrant: Lesotho

G. rueppelli:

Native, Extant (resident): Algeria; Benin; Burkina Faso; Burundi; Cameroon; Central African Republic; Chad; Côte d'Ivoire; Djibouti; Eritrea; Ethiopia; Gambia; Ghana; Guinea; Guinea-Bissau; Kenya; Mali; Mauritania; Niger; Nigeria; Rwanda; Senegal; Somalia; South Sudan; Sudan; United Republic of Tanzania; Togo; Uganda

Native, Extant (non-breeding): Portugal; Spain

Native, Extant (passage): Gibraltar; Morocco

Extant & Vagrant (resident): The Democratic Republic of the Congo

Extant & Vagrant (non-breeding): Egypt; Sierra Leone; Zambia

Extant & Vagrant (seasonality uncertain): Botswana; Lesotho; Malawi; Mozambique; Namibia; South Africa; Zimbabwe

Extant & Origin Uncertain (non-breeding): Saudi Arabia

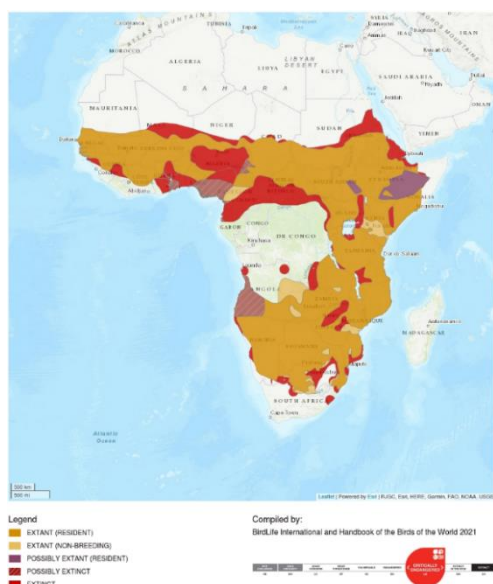


Figure 1. White-backed Vulture (*Gyps africanus*) distribution map. (BirdLife International and Handbook of the Birds of the World 2021. *Gyps africanus*. The IUCN Red List of Threatened Species. Version 2025-1)

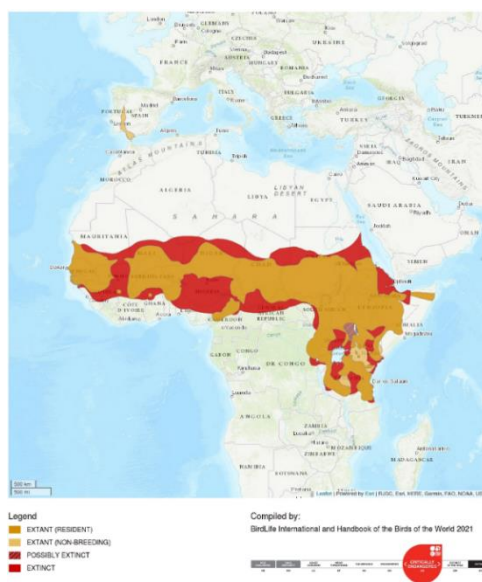


Figure 2. Rüppell's Vulture (*Gyps rueppelli*) distribution map. (BirdLife International and Handbook of the Birds of the World 2021. *Gyps rueppelli*. The IUCN Red List of Threatened Species. Version 2025-1)

IUCN Global Category

Both species: Critically Endangered A2abcd+3bcd+4abcd (assessed 2021, ver. 3.1)

Biological criteria for inclusion in Appendix I**A) Small wild population**

For *G. africanus* the global population of the species was estimated at approximately 270,000 individuals in 1992. However, rapid declines in recent years suggest that the current population may be more than an order of magnitude lower.

For *G. rueppelli* estimates from the early 1990s suggested a population of approximately 11,000 breeding pairs of the species, equating to around 22,000 mature individuals and possibly up to 30,000 individuals in total. These figures included significant populations in the United Republic of Tanzania, Kenya, Ethiopia, Sudan, and West Africa. However, subsequent extremely rapid declines indicate that the current population is likely to be much lower.

B) Restricted area of distribution

G. africanus occurs in 41 countries in Africa (Figure 1), while *G. rueppelli* occurs in 31 countries in Africa and the Middle East, as well as in Europe (Figure 2). Both species occur across a wide variety of habitats. *The current Area of Habitat for G. africanus is 9,305,418 km², and for G. rueppelli, 5,560,410 km² (Ridley et al. in prep).* Both *G. africanus* and *G. rueppelli* have been extirpated from many parts of their range.

C) Decline in number of wild individuals

G. africanus is estimated to have undergone a population decline of 4.1% annually, equating to 81% (63-89%) over three generations (39.6 years). *Some estimates have put this up to 90% (Botha et al., 2017).* *G. rueppelli* has undergone a population decline of 5.8% per year, equating to 92.5% (88-98%) over three generations (43.32 years). *Some estimates have put this up to 99% (Botha et al., 2017).* Population declines are attributed to poisoning, habitat loss, declines in prey species, hunting for trade, persecution, and collisions with power lines.

Both species have historically been viewed as “common” across their ranges but have been subject to significant declines across their ranges in recent years. Abundance of *G. africanus* has declined by 97% in rural areas since 1970, as well as by 50% in protected areas. Local extirpations and declines are evident, including in parts of Botswana, Ghana, Kenya, Niger, Nigeria, South Africa, and Sudan. Likewise, abundance of *G. rueppelli* declined by 97% over 56 years in West Africa, particularly in Burkina Faso, Mali, and Niger, with 30% declines in protected areas. Severe population declines and local extirpations were also evident in parts of Benin, Burkina Faso, Côte d'Ivoire, Ghana, Guinea-Bissau, Kenya, Mali, Niger, Nigeria, Sudan, United Republic of Tanzania, Togo, and Uganda. BirdLife International (2021a, b) indicates declines of Gyps vultures of 52-69% in the Masai Mara and other parts of Kenya, which is inferred to be representative of declines across East Africa.

Trade criteria for inclusion in Appendix I**The species is or may be affected by trade****Legal Trade**

According to CITES Trade Data, exporters reported direct commercial exports of 911 *G. africanus* whole organism equivalents (WOE) items between 1979 and 2023, of which most were live individuals (906, 99%) with two trophies, two bodies, and one skull. Virtually all this trade was in wild-sourced (817, 90%) or unreported (90, 10%) WOE items. This WOE trade was predominantly exported by Guinea (650, 71%), United Republic of Tanzania (88, 10%), and Cameroon (78, 9%), and imported by Portugal (345, 38%), the UK (150, 16%), and Austria (123, 14%).

Exporters reported 692 WOE items of *G. rueppelli* in direct commercial exports between 1979 and 2023. Similar to *G. africanus*, most of these were exported by Guinea, Cameroon and the United Republic of Tanzania, and imported mainly by Portugal, France, and the UK (see Table 1). Around 90% of live individuals of both *G. africanus* and *G. rueppelli* in direct exports were reported to be wild-sourced (816 live individuals of *G. africanus* and 593 of *G. rueppelli*), but 9% did not have a source stated.

According to LEMIS data for *G. africanus* the USA cleared the import of 15 wild-sourced trophies for hunting purposes, with one from the United Republic of Tanzania in 2008 and 14 originating from Mozambique in 2018. During the same period the USA reported refusing entry to one wild-sourced trophy for hunting purposes in 2008, and one live wild-sourced individual for zoological purposes in 2009, both originating from the United Republic of Tanzania.

For *Gyps rueppellii* all trade was cleared and comprised of 13 live, three trophies and one body. Of the live individuals, seven were wild-sourced for zoological (6) and commercial purposes (1) and six were captive-born for zoological purposes. The three wild-sourced trophies for hunting purposes originated from the United Republic of Tanzania in 2015. The one body was captive-bred traded for educational purposes.

International Illegal Trade

The illegal trade in vultures across West Africa is a widespread and growing issue, largely driven by demand for their body parts in traditional medicine and as bushmeat. Despite national protections and international regulations like CITES, vultures are being traded across borders without permits, making much of this activity unlawful. Interviews with market traders and investigations reveal that vultures sold in local markets are often sourced from neighbouring countries, especially as populations decline and vultures become increasingly rare in certain regions.

According to WITIS records for G. africanus, between 2012 and 2025, a total of 883 live and dead individuals were reportedly seized and reported in poaching incidents. Of these seizures and incidents, 302 originated in South Africa (34%), 270 in Zimbabwe (31%), 252 from unknown country (29%), eight from Uganda (1%) and one from the United Republic of Tanzania (<1%). No records were present for G. rueppellii.

In Nigeria, trade is believed to be a major driver of vulture population decline, with estimates suggesting that over 1,500 vultures are traded annually. Nigeria and Benin are central hubs in this trade network, with vultures being brought in from countries such as Niger, Sudan, Cameroon, Chad, Burkina Faso, and even as far as the Central African Republic and Mauritania. In Burkina Faso, vultures have reportedly been deliberately poisoned for export, a practice linked to their scarcity elsewhere and the high prices they fetch—up to USD15 per bird, with poachers earning over USD1,000 from a single event.

The trade is not only regional but international, with smoked vulture meat known to be trafficked across borders. In Benin, vultures are protected by law, yet they are still sold in markets, sourced from at least ten foreign countries. Similarly, in Nigeria, although some vultures originate locally, the majority are imported due to declining local populations. This illegal and unregulated trade poses a serious threat to vulture populations across the continent, exacerbated by poisoning and driven by belief-based uses and economic incentives.

Additional Information

Threats

One of the major threats to both *G. africanus* and *G. rueppellii* is intentional poisoning for belief-based use and the wild meat trade. *Both species are among the most commonly traded vulture species in markets. This is a particularly large threat in Western and Southern Africa, with indications of increasing occurrence in Eastern Africa (Botha et al., 2017). Chandra et al. (2024) indicate evidence of significant cross-border trade in vulture parts, with limited trade through CITES permits.*

Additional *G. africanus* threats include sentinel poisoning (poisoning of a poached carcass to avoid vultures alerting wildlife authorities to its position), unintentional poisoning through baiting, habitat loss and degradation, decline of food availability, persecution, hunting for trade, *electrocution on energy infrastructure, human disturbance including nest harvesting (Botha et al., 2017). Poisoning occurs through exposure to insecticides and pesticides, lead ammunition, and non-steroidal anti-inflammatory drugs (BirdLife International, 2021a). Mass poisoning events to prevent vultures from exposing poached carcasses are also known to occur, and as G. africanus feeds on carcasses in large numbers, up to 468 have been known to have died at a single site. G. africanus is more vulnerable to nest harvesting due to its preferred breeding sites in trees and is vulnerable to collisions with vehicles as well as wind energy plants. The availability of suitable trees for nesting may also impact breeding success (BirdLife International, 2021a).*

Additional threats to G. rueppellii include declines of food availability, human disturbance including nest and egg harvesting, disturbance by rock climbers, and electrocution and collision with wind turbines and power lines with poorly designed poles and pylons (Botha et al., 2017; BirdLife International, 2021b).

Habitat loss for both species occurs as a result of conversion of land to agriculture, loss of habitat quality, and climate change shifting suitable habitat, both for vultures and their prey species.

Conservation, management and legislation

G. rueppellii and *G. africanus* were included in Appendix II of CITES in 1979 as part of the Order listing of Falconiformes spp. Both vulture species are listed in the Convention on Migratory Species (CMS) Appendix I; this requires the Parties that are range States protect them, including prohibiting their capture or killing. At the 2014 Conference of the Parties of the CMS, a set of guidelines to address poisoning was adopted. *G. africanus* and *G. rueppellii* are covered by a Multispecies Action Plan (MsAP) for the conservation of African-Eurasian vultures adopted by Parties to the CMS, with objectives aiming at reducing and halting the trade in vulture parts and sentinel poisoning of vultures. Under the CMS, both species fall under the Memorandum of Understanding on the Conservation of Migratory Birds of Prey in Africa and Eurasia (Raptors MoU) Category 1. In 2024, the West African Vulture Conservation Action Plan (WAVCAP) was launched to build upon the groundwork laid by the CMS MsAP through collaborative efforts involving individuals and institutions dedicated to vulture conservation. The Plan was adopted by the CMS Parties through its Standing Committee in March 2025, and focuses on reducing

the threat posed by belief-based use, with a vision to ensure that by 2043, vulture populations across West Africa have achieved sustainable levels, have become protected by effective legal frameworks, and are thriving in a healthy environment in harmony with people. The Plan targets actions to be carried out across 16 countries: Benin, Burkina Faso, Cameroon, Chad, Côte d'Ivoire, Gambia, Ghana, Guinea, Guinea-Bissau, Liberia, Mali, Niger, Nigeria, Senegal, Sierra Leone, and Togo. *In the African Convention on the Conservation of Nature and Natural Resources (ACNNR), vultures are classified as Class A, with member states committing to protecting the species throughout their national range and preventing illegal offtake. The 2015 African Strategy on Combating Illegal Exploitation and Illegal Trade in Wild Fauna and Flora in Africa sets forth a framework for reducing and preventing illegal offtake and trade of wild species (Chandra et al., 2024).*

Regional and national campaigns and projects aimed at raising awareness of the threats to both species, and to tackle illegal poisoning, have been undertaken by numerous agencies, including BirdLife, the Hawk Conservancy Trust, and the Endangered Wildlife Trust. *At the 2012 Pan-Africa Vulture Summit, recommendations included: 1) Regulate import, manufacture and sale of poisons; 2) Legislate and enforce measures to prosecute those involved in illegal killing and trade in vulture species; 3) Protect and effectively manage breeding sites; 4) Ensure new energy infrastructure is 'vulture-friendly' and modify existing unsafe infrastructure; 5) Support activities to conserve vulture populations, including research and outreach activities (BirdLife International, 2021a).*

There are ongoing monitoring efforts for *G. africanus* in Botswana, Ethiopia, Kenya, Namibia, South Africa, and the United Republic of Tanzania, with plans to extend these to Mozambique, Zambia, and Zimbabwe. No systematic monitoring occurs for *G. rueppellii*. *In South Africa, G. africanus is listed as nationally critically endangered by the Eskom Red Data Book of Birds of South Africa, Lesotho and Swaziland, and endangered in Namibia and Uganda (BirdLife International, 2021a).*

In CoP20 Doc. 74 a suite of revised CITES decisions on West African vultures are directed in part to West African range States specifically on the White-backed Vulture (Gyps africanus) and Rüppell's Vulture (Gyps rueppellii) which are both explicitly prioritized among seven West African vulture species.

Country Protection afforded to Vultures

Most West African range States have prohibited the killing and harvest of vultures and their eggs (Table 1). According to 2021 report on West African vultures, legal protection was confirmed in ten of the 16 range States under review and was considered likely to be in place for a further two range States. Due to their wide ranges, both species occur in a number of protected areas across Africa.

Table 1. West African range States legal protection for vultures including *G. africanus* and *G. rueppellii*

Country	Legal status of species
Benin	All vulture species are fully protected by Law No. 2002-16 dated 18 October 2004, on the wildlife regime in the Republic of Benin. Vultures are in Appendix I of the decree N°2011-394 of 28 May 2011. Appendix I covers species that are fully protected and therefore prohibited from hunting and trade. Benin's nature protection and hunting law (Loi No. 87-014 of 21 September 1987) lists all vultures as totally protected species (Article 2).
Burkina Faso	Vultures have been wholly protected since 1960 through several legislative decrees the most recent being N° 1996-061-PRES/PM/MEE/MATS/MEFP/MCIA/MTT of 11 March 1996 concerning the regulation of the exploitation of Burkina Faso's wildlife. <i>In response to the CITES Notification (No. 2023/049) on West African vultures Burkina Faso provided information relating to their enforcement actions, including the number of investigations, prosecutions and convictions.</i>
Cabo Verde	<i>In response to the CITES Notification (No. 2023/049) on West African vultures Cabo Verde indicated that while legal provisions were in-place for vultures, they are not effectively implemented.</i>
Cameroon	All Vultures are protected under an equivalent of CITES Appendix 2.
Chad	Law N°14/PR/2008 covers forestry, wildlife and fishery resources and organises fauna species into two categories: fully protected (list A) and partially protected species (List B). Decree N°380/PR/PM/MAE/2014 establishes the terms and conditions for the application of the wildlife regime - all vultures are listed as fully protected species (Appendix 1, List A).
Côte d'Ivoire	All migratory raptors are covered by the Raptors MOU and are fully protected by Law n° 94- 442 of 16 August 1994 (that amends Law n° 65-255 of 4 August 1965) on the

Country	Legal status of species
	protection of Fauna and hunting. Hunting by use of poison is prohibited (Article 6). Decree no 003/SEPN/CAB of 20 February 1974 relates to the closure of hunting.
Gambia	Taking of all CMS Appendix I birds is prohibited.
Ghana	Taking of all CMS/CITES birds is prohibited by the Wild Animals Preservation Act, 1961 (Act 43). <i>In response to the CITES Notification (No. 2023/049) on West African vultures Ghana described the processes in place for collaboration between relevant enforcement authorities.</i>
Guinea	Taking of all CMS Appendix I birds is prohibited. The wildlife and hunting regulations code includes articles for the full protection of species on the IUCN Red List in the Critically Endangered and Endangered categories. Vultures fall into this category and are therefore fully protected by law. <i>In response to the CITES Notification (No. 2023/049) on West African vultures Guinea provided information relating to their enforcement actions, including the number of investigations, prosecutions and convictions..</i>
Guinea Bissau	Taking of all CMS Appendix I birds is prohibited. Decree-Law No. 2/2004; Article 16 of 11 this law states that the hunting of protected species is prohibited. Articles 47 and 56 of Guinea's 1997 wildlife protection and hunting law (Loi L/97/038/AN) state that the list of fully and partially protected species is set by Decree.
Liberia	Taking of all CMS Appendix I birds is prohibited by the Environmental Protection and Management Law (2003). All vultures are "fully protected" under Liberia's Wildlife and National Parks Act of 1988 (Forestry Development Authority Regulation No. 25).
Mali	Taking of all CMS Appendix I birds is prohibited. Article 71 of Mali's 2018 law (Law No. 2018-036 of 27 June 2018) states that the list of protected species is set by Decree; this Decree includes <i>G. africanus</i> and <i>G. rueppelli</i> as totally protected species.
Niger	Taking of all Appendix I birds is prohibited by Law 98-07 of 29 April 1998 establishing the regime for hunting and protection of wild fauna. All vulture species are listed as fully protected species (List 1), i.e. those whose hunting, capture, possession, transport or marketing are prohibited in Niger's Hunting and Wildlife Protection Law (Law 98-07 of 29 April 1998) (Republic of Niger, 1998). Vultures are also protected under Law 98-56 of 29 December 1998 on the management of the environment; Law 98-042 of 7 December 1998 of the regime for fishing; and Law 2004-040 of 8 June 2004 on the regime for forestry.
Nigeria	Vultures are protected under the National Wildlife Species Protection Act 2015 of Nigeria. As a signatory to CITES, all vultures listed under the second schedule are legally protected by the Endangered Species Act (ESA) of 2016 of Nigeria. The National Environment Standards and Regulations Enforcement Agency (NESREA) is the main agency tasked with implementing this convention alongside other agencies like the Nigerian customs, police and the judiciary. Taking of all Appendix I birds is prohibited.
Senegal	Taking of all CMS or CITES birds is prohibited.
Sierra Leone	All vulture species are protected nationally by the Wildlife Conservation Act of 1972.
Togo	Taking of all CMS or CITES birds is prohibited.

Artificial Propagation/captive breeding

The CITES Trade Database shows several exports of captive-bred *G. africanus* from South Africa. Captive breeding of the species occurs in the DeWildt Cheetah and Wildlife Trust and from the Pretoria Zoological Gardens. From 2015 to 2021, twenty *G. africanus* have been released to the wild by the VulPro's conservation breeding programme in South Africa. Individuals of captive-bred *G. rueppelli* are kept in zoos, for example, in Italy.

For *G. africanus*, there are 39 institutions holding 142 captive individuals across 14 countries globally, with two population management programs operated by the Association of Zoos and Aquariums (AZA), the European Association of Zoos & Aquaria (EAZA), and the Pan-African Association of Zoos and Aquaria (PAAZA). For *G. rueppelli*, 49 institutions hold 251 captive individuals across 11 countries globally, with two population management programs operated by the Association of Zoos and Aquariums (AZA) and the European Association of Zoos & Aquaria (EAZA) (ZIMS, 2025).

Implementation challenges (including similar species)

Botha and Bowden *in litt.* (2025) indicate that as numbers of *G. africanus* and *G. rueppelli* decline, trade in the Hooded Vulture (*Necrosyrtes monachus*) as a substitute has become more common, reducing species numbers dramatically. Species are traded in parts, reducing identification capabilities, and other raptors have also been observed to be traded as *G. africanus* or *G. rueppelli* once plucked. Parts traded include heads, legs, eggs, feathers, bones, and faeces (Chandra *et al.*, 2024). Additional species thought to be at threat include *G. bengalensis*, *G. coprotheres*, *G. fulvus*, *G. himalayensis*, *G. indicus*, and *G. tenuirostris*, all of which are listed in Appendix II of CITES.

References

- BirdLife International. (2021a) *Gyps africanus*. The IUCN Red List of Threatened Species 2021: e.T22695189A204461164. <https://dx.doi.org/10.2305/IUCN.UK.2021-3.RLTS.T22695189A204461164.en>
- BirdLife International. (2021b) *Gyps rueppelli*. The IUCN Red List of Threatened Species 2021: e.T22695207A204723468. <https://dx.doi.org/10.2305/IUCN.UK.2021-3.RLTS.T22695207A204723468.en>
- Botha, A. J., Andevski, J., Bowden, C. G. R., Gudka, M., Safford, R. J., Tavares, J. and Williams, N. P. (2017) Multi-species Action Plan to Conserve African-Eurasian Vultures. *CMS Raptors MOU Technical Publication No. 5. CMS Technical Series No. 35. Coordinating Unit of the CMS Raptors MOU*, Abu Dhabi, United Arab Emirates.
- Botha, A. and Bowden, C. (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK. IUCN SSC Vulture Specialist Group.
- Buij, R., Nikolaus, G., Whytock, R., Ingram, D.J. and Ogada, D. (2015) Trade of threatened vultures and other raptors for fetish and bushmeat in West and Central Africa. *Oryx* 50(4):606-616. doi:10.1017/S0030605315000514.
- Chandra, S., Citegetse, G., Copsey, J., Deikumah, J.P., Gallo Orsi, U., Henriques, H., Lopes, L. and Onoja, J. (2024) West African Vulture Conservation Action Plan 2023-2043.
- Ogada, D., Shaw, P., Beyers, R.L., Buij, R., Murn, C., Thiollay, J.M., Beale, C.M., Holdo, R.M., Pomeroy, D., Baker, N., Krüger, S.C., Botha, A., Virani, M.Z., Monadjem, A. and Sinclair, A.R.E. (2015) Another continental vulture crisis: Africa's vultures collapsing toward extinction. *Conservation Letters* 9:89-97
- Ridley, F. *et al.* (manuscript in preparation) Advancing understanding of global threat abatement potential for terrestrial vertebrates.
- Species360 Zoological Information Management System (ZIMS) (2025) zims.Species360.org.

Transfer of Peregrine Falcon *Falco peregrinus* from Appendix I to Appendix II

Proponents: Canada and the United States of America

Summary: The Peregrine Falcon *Falco peregrinus* is a falcon in the order Falconiformes with a global distribution encompassing over 200 range States across the Americas, Africa, Asia, Australasia and Europe. It was included in Appendix I in 1977. Currently most Falconiformes are listed in Appendix II, though some are listed in Appendix I and III, and four species are not included in the Appendices.

The global population size of *F. peregrinus* is estimated to be 248,000 to 478,000 mature individuals and increasing, having recovered from severe population declines in the mid-20th century due to the widespread use of pesticides containing DDT, which reduced many bird species' reproductive success. Successful captive breeding and re-introduction programs and regulations on pesticide use have allowed significant species recovery. The extent of occurrence is now estimated to be over 40 million km². Evidence of higher fecundity and juvenile survival rates in highly modified human environments showcase high species adaptivity. Despite decreases reported in parts of North Africa and Ukraine, *Falco peregrinus* was assessed as Least Concern on the global IUCN Red List in 2021 and on the European regional IUCN Red List assessment in 2020. Factors currently impacting *F. peregrinus* populations likely include environmental toxins, habitat loss and modification, illegal killing, and removal from the wild.

Falco peregrinus is currently traded internationally for falconry in Middle Eastern countries. According to the CITES Trade Database almost 13,500 live individuals were reported as exported during the period 2014-2023, the majority (98%) from captive-born sources (C, D and F). A small number of wild specimens (248) were exported during the same period, mainly for re-introduction (80%) and a small number for personal use (12.5%). Trade in captive-produced specimens increased significantly from the previous ten-year period and exports of wild-sourced specimens were less than half the number exported between 2004-2013. Currently the global commercial demand for *F. peregrinus* is largely met by captive-bred birds. Much of the trade is supplied by producers in the UK, Spain and Germany, all of which have CITES-registered captive breeding facilities. There are reports of illegal trade, but it is likely not significant in relation to the increasing wild population.

Parties consulted about their legislative instruments and other tools (including the key trading countries), through a Notification to the Parties in 2024, indicated that they have protection measures in place, deemed effective by the CITES Secretariat. Protection ranges from total prohibition of the capture, killing, owning, transporting, possession and trade of wild specimens and the practice of falconry to countries where falconry is allowed and *F. peregrinus* can be harvested from the wild for national use with certain controls. Most Parties indicated that they would not have concerns with the effectiveness of national/international legislation for ensuring conservation of *F. peregrinus* with an Appendix-II listing. International and regional instruments (particularly in Europe covering the top three exporters) support regional and global efforts aimed at the conservation and sustainable management of *F. peregrinus* and its habitat, and the global population of *F. peregrinus* continues to grow and increase.

A small number of Parties have expressed concern that an Appendix-II listing could increase demand for wild individuals and/or increase illegal trade, which could impact the global population and/or smaller subpopulations. The extensive captive breeding industry and preference for captive-bred individuals for the falconry trade mitigate the risks of increased demand for wild *F. peregrinus*, with offtake for new bloodlines and specific varieties being the main risk. However, falconers often cross falcon species and hybrids to produce offspring that expresses characteristics desired for falconry, wild-sourced individuals in trade are currently minimal, and several breeding operations have confirmed sufficient genetic diversity in captive stock. Avian flu, while impacting some populations, is not considered a serious threat to global populations.

Analysis: *F. peregrinus* has an extremely wide distribution and a large and increasing population in the wild. Populations have recovered since the mid-20th century. The species **does not meet the**

biological criteria for listing in Appendix I.

The trade in live birds for falconry has been increasing although this is mostly supplied by captive-bred birds from countries with registered breeding facilities. Relatively small numbers of wild birds are exported for reintroduction purposes. Illegal trade is limited to low numbers and certain populations, including Pakistan, Iran, the Netherlands, the Russian Federation, and South America, with demand coming primarily from the Middle East.

It is possible that a transfer to Appendix II will stimulate applications for trade in wild individuals, the potential for negative impact on the wild population is likely to be minimal due to the success of the captive breeding industry and the preference within the falconry industry for captive-born individuals, along with evidence of stable and increasing populations of the species.

Given the species is present in over 200 countries, it is difficult to determine whether management and appropriate enforcement and compliance controls in each are such that Parties can be satisfied with implementation by all range States of Article IV. However, the majority of current key trading countries have indicated that national-level controls would not change as a result of a transfer of *F. peregrinus* to Appendix II. The species is unlikely to enter commercial trade from the great majority of range States. It seems likely, therefore, that **precautionary requirements of Annex 4 are satisfied in the greater part of the range and would be proportionate to the anticipated risks to the species.**

There are provisions to monitor trade in wild-caught birds through the *Review of Significant Trade* (Res. Conf. 12.8 (Rev. CoP18)) and captive breeding through the *Review of trade in animal specimens reported as produced in captivity* (Res. Conf. 17.7 (Rev CoP19)), which would allow Parties to address any future concerns.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

Falco peregrinus was first listed on CITES Appendix II in 1975 under the Family listing for Falconidae species. Subspecies *Falco peregrinus anatum*, *F. p. peregrinus*, *F. p. tundrius* were listed on Appendix I in 1975. In 1975, Canada entered a reservation for the Family Falconidae. In 1977, all subspecies of *F. peregrinus* were listed on Appendix I. In 1996, Saudi Arabia entered a reservation for the species, and in 2004 Palau entered a reservation for the species, in both cases at the time of their accession to the Convention.

Taxonomy

First described by Tunstall (1771), del Hoyo and Collar (2014) currently recognize *Falco peregrinus* as a single species. The species was previously split into *F. peregrinus* and *F. peregrinoides* (Sibley and Monroe, 1990, 1993).

Range

Native, Extant (resident): Afghanistan; Algeria; Andorra; Angola; Anguilla; Antigua and Barbuda; Argentina; Armenia; Aruba; Austria; Azerbaijan; Bahamas; Bahrain; Bangladesh; Barbados; Belarus; Belgium; Benin; Bermuda; Bhutan; Bolivia, Plurinational States of; Bonaire, Sint Eustatius and Saba; Botswana; Brazil; Brunei Darussalam; Bulgaria; Burkina Faso; Cabo Verde; Cambodia; Cameroon; Canada; Cayman Islands; Central African Republic; Chad; Chile; Colombia; Comoros; Congo, The Democratic Republic of the; Costa Rica; Croatia; Cuba; Curaçao; Cyprus; Czechia; Côte d'Ivoire; Denmark; Djibouti; Dominica; Dominican Republic; Ecuador; Egypt; El Salvador; Equatorial Guinea; Eritrea; Estonia; Eswatini; Ethiopia; Falkland Islands (Malvinas); Fiji; Finland; France; French Guiana; Gabon; Gambia; Germany; Ghana; Gibraltar; Greece; Greenland; Grenada; Guatemala; Guinea; Guinea-Bissau; Guyana; Haiti; Honduras; Hungary; India; Iran, Islamic Republic of; Iraq; Ireland; Italy; Jordan; Kazakhstan; Kenya; Korea, Republic of; Kuwait; Kyrgyzstan; Lao People's Democratic Republic; Latvia; Lebanon; Lesotho; Liberia; Libya; Lithuania; Madagascar; Malawi; Mali; Martinique; Mauritania; Mayotte; Mexico; Moldova; Mongolia; Montserrat; Morocco; Mozambique; Namibia; Netherlands; Nicaragua; Niger; Nigeria; North Macedonia; Norway; Oman; Palestine, State of; Panama; Paraguay; Peru; Poland; Portugal; Puerto Rico; Qatar; Romania; Rwanda; Saint Kitts and Nevis; Saint Lucia; Saint Vincent and the Grenadines; Saudi Arabia; Senegal; Sierra Leone; Singapore; Sint Maarten (Dutch part); Slovakia; Slovenia; Solomon Islands; Somalia; South Africa; South Georgia and the South Sandwich Islands; South Sudan; Spain; Spain (Canary Is.); Sudan; Suriname; Sweden; Switzerland; Syrian Arab Republic; Taiwan,

Province of China; Tajikistan; Tanzania, United Republic of; Timor-Leste; Togo; Trinidad and Tobago; Tunisia; Turkey; Turkmenistan; Turks and Caicos Islands; Uganda; Ukraine; United Arab Emirates; United Kingdom of Great Britain and Northern Ireland; Venezuela, Bolivarian Republic of; Viet Nam; Western Sahara; Yemen; Zambia; Zimbabwe

Native, Extant (breeding): Albania; Australia; Bosnia and Herzegovina; China; Congo; Georgia; Hong Kong; Indonesia; Israel; Japan; Korea, Democratic People's Republic of; Liechtenstein; Luxembourg; Malaysia; Malta; Montenegro; Myanmar; Nepal; New Caledonia; Pakistan; Papua New Guinea; Philippines; Russian Federation (Central Asian Russia, Eastern Asian Russia, European Russia); Serbia; Sri Lanka; Thailand; United States of America; Uzbekistan; Vanuatu

Native, Extant (non-breeding): Belize; Guadeloupe; Guam; Jamaica; Micronesia, Federated States of; Northern Mariana Islands; Palau; Saint Martin (French part); Uruguay; Virgin Islands, British; Virgin Islands, U.S.

Native, Extant (passage): Saint Pierre and Miquelon; United States Minor Outlying Islands

Native, Extant (seasonality uncertain): Macao; Saint Barthélemy

Extant & Vagrant: Burundi; Christmas Island; Faroe Islands; Iceland; Maldives; Mauritius; Samoa; Seychelles

IUCN Global Category

Least Concern (assessed 2021, ver. 3.1)

Biological criteria for inclusion in Appendix I

A) Small wild population

The global population size is estimated to be 248,000 to 478,000 mature individuals.

B) Restricted area of distribution

F. peregrinus has an extremely large global distribution and habitat range, and occur in 200 countries across North America, Africa, Europe, Asia, Australia, and South America. *The species has extended its distribution range and established new nesting areas (i.e. Cuba), as well as established nesting sites using manmade structures (Regalado and Cables, 2000; Reuter Cortes in litt., 2025).*

C) Decline in number of wild individuals

The most recent IUCN Red List Assessment (2021) indicates that global populations of this species are increasing. *Current regional population estimates are approximately 72,000 for the USA and Canada, 4,500-5,000 for North Africa, and 32,000-62,000 for Europe. The population is estimated to be increasing throughout most of its range, with up to a 127% increase per decade for North America, stable populations in the Arctic and North Africa, with the possibility of localised declines in some countries in North Africa (BirdLife International, 2021), and in the Ukraine (CoP17 Doc. 88.2 Annex 12).*

Trade criteria for inclusion in Appendix I

The species is or may be affected by trade

Trade in live *F. peregrinus* is primarily for the practice of falconry, either for breeding stock or falconry birds. The species is in trade, but international trade is not deemed to be a threat to wild populations. *This species is not currently included in the Review of trade in animal specimens reported as produced in captivity (Res. Conf. 17.7 (Rev. CoP 19)) (UNEP, 2025).*

Between 2014 and 2023, almost 13,500 live individuals were reported as exported (excluding re-exports), the majority (98%) from captive-born sources (C, D and F) predominately reported as for commercial purposes. Of the 248 individuals reported as either wild-sourced or with no source code, 80% (205) were recorded as intended for reintroduction, while 12.5% were reported as for personal use. This contrasts with the previous 10-year period when just over 3,100 live individuals were reported as exports (82% from captive-born sources) and of the wild exports (codes I, O, U, W and blank) totalling 575, almost 60% were for reintroduction (mostly in Kazakhstan and Kyrgyzstan) and almost 40% were as Personal.

Between 2014 and 2023 the main exporters of live birds were Germany, Spain, and the UK, all of which have registered captive breeding facilities, while the main importers were the United Arab Emirates, Qatar, and Bahrain. The UAE exported the majority of wild exports, with Saudi Arabia exporting a smaller number.

Although there is known trade in hybrids of *F. peregrinus*, the CITES trade database only refers to "Falco hybrids" and it is therefore not possible to determine which species are involved.

Illegal trade in falcons is known to occur. There are media reports that wild *F. peregrinus* juveniles and eggs are stolen from nests and laundered into the legal trade, as well as seizures of individuals. Trade in live *F. peregrinus* is primarily for the practice of falconry, either for breeding stock or falconry birds. *TRAFFIC holds seizure records documenting that between 1993 and 2022, 88 individuals were seized in 39 cases. The SS notes illegal trade*

sourced from Pakistan, Iran, the Netherlands, Russian Federation, and South America, with demand coming primarily from the Middle East.

Precautionary Measures

Species not in demand for trade; transfer to Appendix II unlikely to stimulate trade in, or cause enforcement problems for, any other species included in Appendix I

Regarding the precautionary measures outlined in Annex 4 of Res. Conf. 9.24 (Rev. CoP17), it is unlikely that a transfer will stimulate trade in wild *F. peregrinus* that may have a detrimental impact on the status of the species, or be detrimental to its survival in the wild, as there is already a well-established captive-breeding trade which is able to largely satisfy current market demands, and a preference for captive-bred individuals in falconry.

Species likely to be in demand for trade, but its management is such that the CoP will be satisfied with:

A) Implementation by the range States of the requirements of the Convention, in particular Article IV; and *The Review of Significant Trade (Res. Conf. 12.8 (Rev. CoP18)) would monitor wild trade and the Review of trade in animal specimens reported as produced in captivity (Res. Conf. 17.7 (Rev. CoP19)) would monitor captive breeding. These processes would be able to address any concerns that are identified.*

B) appropriate enforcement controls and compliance with the requirements of the Convention

Through responses to a Notification to the Parties submitted at the request of Canada in 2024 (Notification 2024/113), the majority of current key trading countries indicated that national-level controls would not change as a result of a transfer of *F. peregrinus* to Appendix II. *Protection ranges from total prohibition of the capture, killing, owning, transporting, possession and trade of wild specimens to countries where F. peregrinus can be harvested from the wild, details of which can be found in CoP17 Prop 17, and during Committee discussion on the proposal other range States (Argentina, Australia, Brazil, the Democratic Republic of the Congo, Ethiopia, Indonesia, Japan, Kuwait, Mexico, the Russian Federation, South Africa) indicated that they had national measures in place to ensure the conservation of the species in the wild and adequate regulation of trade (CoP17 Com. I Rec. 9). In their analysis, the CITES Secretariat (2025) found that the current legal instruments, trade controls, species management and conservation measures are effective and this would likely remain unchanged with an Appendix II listing.*

Export quota or other special measure

None proposed.

Additional Information

Threats

Historical threats have included widespread use of organochlorine pesticides, including DDT in the mid-20th century, severely impacting *F. peregrinus* and other bird populations. Successful captive breeding and re-introduction, as well as regulations on pesticide use, have resulted in species population recovery, and pesticides are not thought to be currently having a significant impact on populations. Human-mediated threats include other environmental toxins, habitat alteration and destruction of nesting areas, including wind energy developments, rock climbing disturbing nests, and collisions with man-made objects such as power lines and *urban developments (BirdLife International, 2021). Other threats include global warming decreasing food sources (BirdLife International, 2021), the latest (2020–2023) highly pathogenic avian influenza (HPAI) H5 virus, and use in falconry, although the latest IUCN Red List assessment indicates that population-level impacts of falconry offtake are unknown, as the majority of F. peregrinus used in falconry are captive-bred. There is a risk of wild-sourced individuals being used to introduce new bloodlines to captive stock, but this risk is considered low, with low numbers of individuals recorded in CITES Trade Database as taken from the wild, and several breeding operations indicating sufficient genetic diversity in captive stock (e.g., CoP15 Doc. 41.1 Annex 13; CoP15 Doc. 41.1 Annex 14). Sources indicate impacts of avian flu on populations are occurring, especially in coastal populations dependent on seabirds as a food source, with specific impacts in the Netherlands, Peru, and parts of the USA, although the number of individuals affected is a very low proportion of the global population (Caliendo et al., 2024; Castro-Sanguinetti, 2024; Hsueh et al., 2024; Lindquist, 2025; Reuter Cortes in litt., 2025; Varland et al., 2025). Some local illegal persecution and offtake occur, and legal trade numbers are increasing, although the number of permits available for wild-caught specimens is controlled (Millsap and Allen, 2006).*

Conservation, management and legislation

Falconry is a highly organized sport represented globally by the International Association for Falconry and Conservation of Birds of Prey (IAF). The IAF is an accredited member of IUCN and currently has 110 member associations from 87 countries worldwide, totalling 75,000 members. These associations work actively with governments to conserve the species and promote the sport under sustainability and best practices criteria.

Between 2016 and 2024, 33 key trading countries indicated that national protections are in place for *F. peregrinus*, sufficient to ensure the protection of the species if the transfer from Appendix I to Appendix II proceeds. National protection ranges from total prohibition of the capture, killing, owning, transporting, possession and trade of wild specimens to countries where *F. peregrinus* can be harvested from the wild. Many countries indicated that

existing national legal frameworks also specifically consider the protection of habitats for the species, including within protected areas.

International instruments support regional and global efforts aimed at the conservation and sustainable management of *F. peregrinus* and its habitat, including through the Convention on Migratory Species (CMS), under which a Memorandum of Understanding (MOU) concerning the Conservation of Migratory Birds of Prey in Africa and Eurasia is in place and has been signed by 66 countries. Other relevant instruments include EU Wildlife Trade Regulations and the Convention on the Conservation of European Wildlife and Natural Habitats (Bern Convention). *F. peregrinus* is listed as a strictly protected fauna species in Appendix II of the Bern Convention. To implement the Bern Convention in the European Union, the EU Birds Directive was adopted in 1979. The EU Birds Directive protects migratory species and their habitat and generally bans activities which could threaten birds that are naturally occurring in the EU (European Union, 2009). Several of the Parties that provided information are members of the EU and are required to implement both the EU Wildlife Trade Regulations and the EU Birds Directive through national legislation. The European Union range States are not all significant players in the trade of *F. peregrinus*, but it is worth noting that the strict controls in the European Union apply to the majority of *F. peregrinus* trade in the world. The trade restrictions imposed by the EU Wildlife Trade Regulations for *F. peregrinus* are stricter than those required by a CITES Appendix-II listing. According to responding Parties, the trade controls would be unlikely to change in the event of a transfer of *F. peregrinus* to Appendix II.

Management efforts have included close monitoring, collaboration between countries for shared populations, and enforcement of the existing national and international legal frameworks. In many instances, captive breeding and contributions of falconers were key factors in the recovery of *F. peregrinus*. The majority of responding Parties have ongoing efforts for conservation and monitoring of *F. peregrinus* in its territories. Governmental and academic scientists, park wardens and civil society groups monitor nesting sites and wild populations, and generally assure the conservation, protection and population health of the species in the wild.

Artificial Propagation/captive breeding

Extensive captive breeding was used for the successful re-establishment of recovering wild populations following the effects of historical DDT use. Current captive breeding operations supply the falconry industry, with 52 captive breeding operations registered with CITES in Canada, Czechia, Denmark, Germany, Serbia, Spain, the UK, and the USA (CITES, 2025). Captive-bred birds are deemed more desirable for falconry due to being habituated to handling from a young age. CITES trade data indicate that 96% of the legal trade of live *F. peregrinus* between 2015-2023 involved captive-bred falcons, while 83% of all legal trade in *F. peregrinus* (whole and parts) was from captive sources. Captive breeding of *F. peregrinus* is not permitted by all range States.

Berryman in litt. (2025) indicates that captive breeding of F. peregrinus, particularly in North America and Europe, has been successful, with advances in artificial insemination, double-clutching, and hacking techniques utilized to improve captive breeding success (Fleming et al., 2011), and high genetic diversity maintained across global captive breeding programmes (Puchala et al., 2022). Productivity through captive breeding is considered high, and the industry is considered well-regulated. In Canada, wild populations of F. peregrinus do not depend on supplementation from captive sources, and captive breeding programs are well-monitored (Environment and Climate Change Canada, 2017). There are 183 ex-situ F. peregrinus individuals held in 89 institutions across 24 countries (ZIMS, 2025).

Implementation challenges (including similar species)

Trade in all falcons is controlled through CITES under the listing of Falconiformes spp., in either Appendix I or II. While many falcons have similar appearances and juveniles and hybrids can be difficult to identify, good resources exist to aid in the identification of these specimens. In uncertain cases, genetic testing can be used to confirm species identity.

Other considerations

A transfer of *F. peregrinus* to Appendix II could result in increased legal trade of wild birds, with the potential to harm wild populations. However, Parties that have provided information generally indicated that national-level controls were robust and effective at protecting wild falcons from overharvest and unsustainable offtake. National governments control allowable wild take, national use, and issuance of CITES permits for trade (including Non-Detriment Findings for Appendix II species). Most Parties that have provided information indicated that national-level controls would not change as a result of a transfer of *F. peregrinus* to Appendix II.

International instruments are already in place to support regional and global efforts aimed at the conservation and sustainable management and trade of *F. peregrinus* and its habitat. Stakeholders are also engaged in conservation efforts. The International Association for Falconry and Conservation of Birds of Prey and their affiliated national associations work with members and governments towards sustainable use of falcons.

Juvenile birds are the preferred age group associated with take from the wild for falconry, and this age group has high natural mortality and is not yet part of the breeding population. Removal of juvenile birds from *F. peregrinus* populations thus has a low impact on the wild population compared with take of adult birds. Additionally, in healthy populations, the active breeding population can be much smaller than the number of adults, and thus some wild take of adults could occur without harm (and possibly with benefit if prey is scarce) to the wild population. Thus, the structure of healthy *F. peregrinus* populations provides some natural resiliency to either legal or illegal take that would serve to guard against population declines. *In addition, precautionary models to determine sustainable wild take quotas have been developed, with studies showing that up to 5% of the population can be harvested with no negative effects (Millsap and Allen, 2006; Reuter Cortes in litt., 2025).*

It is possible that eased trade restrictions associated with captive-bred birds, resulting from an Appendix-II listing, could stimulate the falconry market. However, it seems unlikely that the change would be large enough to have a negative impact on wild falcon populations. Currently, the market for wild live *F. peregrinus* is small in terms of number of birds and is expected to remain small and specialised for falconry and related purposes, with most of the birds in trade being captive-bred. *F. peregrinus* is not necessarily the most sought-after species for falconry due to its smaller size compared with other falcons, and this is unlikely to change. *A recent study by Panter et al. (2023) shows that between 1975 and 2020, F. peregrinus accounted for only 7.1% of the global trade in raptors.*

The risk of Avian flu spreading with increased trade into wild populations is unlikely, as infected individuals are unlikely to survive long, and many countries impose quarantine procedures for live imports.

References

- Berryman, A. (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- BirdLife International (2021) *Falco peregrinus*. The IUCN Red List of Threatened Species 2021: e.T45354964A206217909. Accessed on 9 July 2025. <https://dx.doi.org/10.2305/IUCN.UK.2021-3.RLTS.T45354964A206217909.en>
- Caliendo, V., Bellido Martin, B., Fouchier, R.A.M., Verdaat, H., Engelsma, M., Beerens, N. and Slaterus, R. (2024) Highly Pathogenic Avian Influenza Contributes to the Population Decline of the Peregrine Falcon (*Falco peregrinus*) in The Netherlands. *Viruses* 17(1):24. doi: 10.3390/v17010024.
- Castro-Sanguinetti, G.R., González-Veliz, R., Callupe-Leyva, A., Apaza-Chiara, A.P., Jara, J., Silva, W., Icochea, E. and More-Bayona, J.A. (2024) Highly pathogenic avian influenza virus H5N1 clade 2.3.4.4b from Peru forms a monophyletic group with Chilean isolates in South America. *Scientific Reports* 14(1):3635. doi: 10.1038/s41598-024-54072-2.
- CITES (2025). List of Species. Available <https://cites.org/eng/common/reg/cb/species.html>. Accessed 16 July 2025.
- Environment and Climate Change Canada (2017) *Management Plan for the Peregrine Falcon anatum/tundrius in Canada*. Species at Risk Act Management Plan Series. Environment and Climate Change Canada, Ottawa. 28pp.
- Fleming, L. V., Douse, A. F. and Williams, W. P. (2011) Captive breeding of peregrine and other falcons in Great Britain and implications for conservation of wild populations. *Endangered Species Research* 14:243-257.
- Hsueh, C.S., Fasina, O., Piñeyro, P., Ruden, R., El-Gazzar, M.M. and Sato, Y. (2024) Histopathologic Features and Viral Antigen Distribution of H5N1 Highly Pathogenic Avian Influenza Virus Clade 2.3.4.4b from the 2022-2023 Outbreak in Iowa Wild Birds. *Avian Diseases* 68(3):272-281. doi: 10.1637/aviandiseases-D-23-00085.
- Lindquist, E. (2025) *Decades after peregrines came back from the brink, a new threat emerges*. The Guardian. Available <https://www.theguardian.com/environment/2025/mar/13/decades-after-peregrines-came-back-from-the-brink-a-new-threat-emerges-aoe>
- Millsap, B. A. and Allen, G. T. (2006) Effects of Falconry Harvest on Wild Raptor Populations in the United States: Theoretical Considerations and Management Recommendations. *Wildlife Society Bulletin* 34(5):1392–1400. <http://www.jstor.org/stable/4134275>.
- Puchala, K. O., Nowak-Życzńska, Z., Sielicki, S. and Olech, W. (2022) Evaluation of the Impact of the Peregrine Falcon (*Falco peregrinus peregrinus*) Reintroduction Process on Captive-Bred Population. *Genes* 13(8):1487.
- Reuter Cortes, A. (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Regalado, P. and Cables, E. (2000) Primer hallazgo de *Falco peregrinus* nidificando en Cuba. *Cotinga* 14:78.
- Species360 Zoological Information Management System (ZIMS) (2025) zims.Species360.org.
- UNEP (2025) The Species+ Website. Nairobi, Kenya. Compiled by UNEP-WCMC, Cambridge, UK. Available at: www.speciesplus.net. Accessed on 18 July 2025.
- Varland, D.E., Buchanan, J.B., Zimmerman, G.S., Bauder, J.M., Fleming, T.L. and Millsap, B.A. (2025) Estimated Annual Abundance of Migratory Peale's Peregrine Falcons in Coastal Washington, USA. *Journal of Raptor Research* 15(3):1-16 <https://doi.org/10.3356/jrr2482>.

Inclusion of Great-billed Seed-finch *Sporophila maximiliani* in Appendix I and *S. angolensis*, *S. atrirostris*, *S. crassirostris*, *S. funerea* and *S. nuttingi* in Appendix II

Proponent: Brazil

Summary: The genus *Sporophila* includes a total of 41 species of small-sized songbirds found in the Neotropics of which six are proposed for inclusion in the Appendices: the Great-billed Seed-finch *Sporophila maximiliani* in Appendix I, owing to concern about over-collection for trade, and the Chestnut-bellied Seed-finch *S. angolensis*, Black-billed Seed-finch *S. atrirostris*, Large-billed Seed-finch *S. crassirostris*, Thick-billed Seed-finch *S. funerea* and Nicaraguan Seed-finch *S. nuttingi* in Appendix II due to their resemblance to *S. maximiliani*. The Great-billed Seed-finch has historically inhabited the Plurinational State of Bolivia (referred to as “Bolivia” hereafter), Brazil, French Guiana, Guyana, Suriname and the Bolivarian Republic of Venezuela (referred to as “Venezuela” hereafter), although its presence throughout this range is no longer certain.

Sporophila species are primarily sought after for their singing capabilities and are entered into singing competitions, which are popular in Brazil, Colombia, Guyana and Suriname.

Capture of wild specimens of Great-billed Seed-finch, habitat loss and degradation have caused rapid population decreases across this species’ range, particularly in Brazil, leading to it being classified as Endangered on the IUCN Red List in 2017. Once widespread, this species is now scarce in the wild with a 2017 global population estimate of below 2500; more recent surveys and information suggest that there may be fewer than 1,000 mature individuals. In Brazil, subpopulations are generally of fewer than 50 mature individuals, possibly occurring in only two areas, and the species is considered by some to be close to extinction in the country. Little information is available on occurrence in other range States. Brazil’s captive population is likely to number over 200,000.

Among the species proposed for inclusion in Appendix II, decreasing population trends for subpopulations of *S. angolensis*, *S. crassirostris*, *S. funerea* and *S. nuttingi* have been reported across their ranges, with the former two having reportedly become extinct in some parts of Brazil (*S. angolensis*), and in Suriname (*S. crassirostris*). Global populations of both *S. angolensis* and *S. crassirostris* were nonetheless reported to be increasing according to their IUCN Red List assessments published in 2018 and 2021 respectively. According to the Red List assessment, *S. atrirostris* was believed to have a stable global population with no evidence of any declines or substantial threats, although BirdLife International believes that this assessment may require review.

Trade is predominantly domestic although there has been export to the USA, Canada and the Netherlands, where Surinamese citizens have established bird singing clubs and tournaments. Online trade of *S. maximiliani* reveals prices in the USA of up to USD800 for females and USD1,000 for males. Most birds in trade appear to be captive-bred; Brazil has a registration system in place for both amateur and commercial breeders of *S. maximiliani*. There are records of wild-sourced specimens of *S. maximiliani*, *S. angolensis* and *S. crassirostris* being exported. Small numbers of wild specimens of *S. maximiliani* have been observed in trade and it is believed that some wild exports are declared as captive-bred. Seizures have been reported mainly within Brazil. The scale of any cross-border trade within the range of *S. maximiliani* remains unknown.

In Brazil, *S. maximiliani* is classified as critically endangered and protected under national legislation, meaning that the capture, transport, storage, custody, handling, processing, and commercialisation of wild individuals are banned. However, the export of captive-bred individuals from registered enterprises is permitted. Legislation is in place in other range States, with a varying degree of protection for wild species.

Males of the six *Sporophila* spp. proposed for inclusion present morphological differences that allow for species-level identification, except for *S. maximiliani* and *S. crassirostris*. However, trade also involves females and juveniles, which are morphologically similar and not easily distinguishable.

Analysis: *Sporophila maximiliani* has a small wild population, possibly of fewer than 1,000 specimens with likely very small subpopulations. Illegal capture of wild specimens has led to the extirpation of subpopulations across the range. The captive population in Brazil held by registered breeders is estimated at over 200,000. Despite recent records mostly involving trade in captive-bred specimens, there is evidence of ongoing wild capture and trade. The species therefore **meets both the biological and trade criteria for inclusion in Appendix I**. Trade in captive-bred specimens could continue from commercial operations breeding the species, which would need to be registered with the CITES Secretariat in accordance with Res. Conf. 12.10 (Rev. CoP15) on *Registration of operations that breed Appendix-I animal species in captivity for commercial purposes*.

Regarding the other five *Sporophila* species proposed as lookalikes, given the resemblance of the females and juveniles of all these species and the evidence that they are found in trade, **if *S. maximiliani* were to be included in Appendix I, the other species would appear to meet the criteria in Annex 2b** of Res. Conf. 9.24 (Rev. CoP17) for inclusion in Appendix II.

Other Considerations: If these species are included in the Appendices, there are other species within the *Sporophila* genus that may merit consideration for inclusion in Appendix II under Annex 2b of Res. Conf. 9.24 (Rev. CoP17).

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

Sporophila maximiliani (Cabanís, 1851)
Sporophila angolensis (Linnaeus, 1766)
Sporophila atrirostris (Sclater & Salvin, 1878)
Sporophila crassirostris (Gmelin, 1789)
Sporophila funerea (Sclater, 1859)
Sporophila nuttingi (Ridgway, 1884)

Range

Sporophila maximiliani – Plurinational States of Bolivia; Brazil; French Guiana; Guyana; Suriname; Bolivarian Republic of Venezuela
Sporophila angolensis – Argentina; Belize; Plurinational States of Bolivia; Brazil; Colombia; Ecuador; French Guiana; Guatemala; Guyana; Honduras; Nicaragua; Panama; Paraguay; Peru; Suriname; Trinidad and Tobago; Bolivarian Republic of Venezuela
Sporophila atrirostris – Plurinational States of Bolivia; Ecuador; Peru
Sporophila crassirostris – Brazil; Colombia; Ecuador; French Guiana; Guyana; Peru; Suriname; Trinidad and Tobago; Bolivarian Republic of Venezuela
Sporophila funerea – Belize; Colombia; Costa Rica; Ecuador; Guatemala; Honduras; Mexico; Nicaragua; Panama
Sporophila nuttingi – Costa Rica; Nicaragua; Panama

IUCN Global Category

Sporophila maximiliani – Endangered (assessed 2017, ver. 3.1)
Sporophila angolensis – Least Concern (assessed 2018, ver. 3.1)
Sporophila atrirostris – Least Concern (assessed 2018, ver. 3.1)
Sporophila crassirostris – Least Concern (assessed 2020, ver. 3.1)
Sporophila funerea – Least Concern (assessed 2020, ver. 3.1)
Sporophila nuttingi – Least Concern (assessed 2020, ver. 3.1)

Biological criteria for inclusion in Appendix I

A) Small wild population

With a global population ranging between 1,000 to 2,500 mature individuals, *Sporophila maximiliani* is documented to be experiencing a decrease in wild specimens across its entire geographical range. Once widespread in Brazil, the country's Ministry of Environment (Ministerio do Meio Ambiente) reported in 2014 that the Brazilian population counted fewer than 250 mature individuals, with each subpopulation reaching fewer than 50 mature specimens. More recent surveys from 2021 indicate a global population of fewer than 1,000 mature individuals and the Brazilian population counting fewer than 100. *This is particularly concerning given the fact that there are no reliable records of the species outside of Brazil from the past 10 years (BirdLife International in litt.,*

2025). According to Marini et al. (2023) and based on recent field searches, it is suspected that the Brazilian population is close to being extirpated in the wild. Ubaid et al. (2021) conducted searches in more than 50 locations in Brazil, accumulating an effort of ~7,000 hours of fieldwork, yet only two small and isolated populations were located.

There is no recent information on the status of the population in Bolivia and Venezuela. Research is ongoing in Bolivia, however previously expected populations of *S. maximiliani* appeared to be *S. crassirostris* (BirdLife International in litt., 2025). Authorities in French Guiana cannot confirm the species' presence in their country whereas authorities in Guyana and Suriname have confirmed its absence. BirdLife International considers *S. maximiliani* to be extirpated in Colombia, however Marini et al. (2023) reports one subpopulation still occurring in the country.

Regarding *S. crassirostris*, according to BirdLife International (2021), its global population was suspected to count 50,000–500,000 mature individuals. The species has been extirpated in countries such as Suriname and large parts of Guyana, and the Brazilian populations are experiencing a decreasing trend. However, the global population is reported to be undergoing a large increase (Partners in Flight, 2019).

All five of the species proposed as lookalikes have been assessed as Least Concern on the IUCN Red List.

- ***S. angolensis***: global population is unknown, and it is suspected to be increasing as ongoing habitat degradation is creating new areas of suitable habitat (BirdLife International, 2018).
- ***S. atriostriis***: global population size has not been quantified, but the species is described as rare and local (BirdLife International, 2018) and its global population trend appears to be stable.
- ***S. crassirostris***: global population is estimated to number between 50,000 and 500,000 mature individuals, and is believed to be increasing significantly.
- ***S. funerea***: the population numbers 500,000–500,000 mature individuals globally but is undergoing a small decline (BirdLife International, 2021).
- ***S. nuttingi***: the population ranges from 50,000 to 500,000 mature specimens, and according to BirdLife International (2021), it is undergoing a large, significant decline.

B) Restricted area of distribution

The SS reports *Sporophila maximiliani* to have a restricted and fragmented distribution, as it appears in a few isolated locations. However, the species' area of habitat (AOH) amounts to 41,760 km² and the species' area of occupancy (AOO) has not been recently estimated. (Ridley et al., in prep.). In Brazil, the species is recorded to be limited to two areas (between the states of Rondônia and Mato Grosso, and between Mato Grosso and Goiás). More studies are needed to assess the species' presence and occurrence in Colombia, French Guiana, Guyana and Suriname. According to de Man (2025), *S. maximiliani* and *S. angolensis* could be found in Trinidad and Tobago before being extirpated or depleted (respectively) due to persecution by cagebird enthusiasts. BirdLife International (2019) confirms that extensive searches by ornithologists have revealed *S. maximiliani* to be extremely rare. It is to be noted here that, as flagged by Valdes in litt. (2025), the difficulty in detection and the absence of verified records are not necessarily equal to the absence of the species in the range States.

S. angolensis, *S. atriostriis*, *S. crassirostris*, *S. funerea* and *S. nuttingi* have all been assessed as having extremely or very large ranges in their IUCN Red List assessments.

C) Decline in number of wild individuals

Sporophila maximiliani is experiencing a rapid decline in the wild, with subpopulations in Brazil having gone extinct in many regions in the north and southeast areas of the country.

Populations of *S. angolensis* have gone extinct in some regions in Brazil, and drastic declines of the species in Trinidad and Tobago have also been recorded. A survey by Silva et al. (2021), which interviewed over 60 Brazilian bird keepers, suggested that the decrease of *S. angolensis* populations in the previous ten years was driven by the increase in demand and the capturing of juveniles, among others.

Decreasing population trends have been recorded in Brazil and at a global scale for *S. funerea* and *S. nuttingi*. Regarding *S. crassirostris*, which is considered a popular songbird in Suriname, Charity and Ferreira (2020) reported the species being extirpated in the country due to decades-long systematic harvesting from the wild and illegal trade, with birds being smuggled from Venezuela via Guyana and Brazil. *S. atriostriis* is suspected to be stable in the absence of evidence for any declines or substantial threats. Nonetheless, it should be noted that stability at a global scale can mask regional collapses in source areas with high trapping pressure (Valdes in litt., 2025).

Trade criteria for inclusion in Appendix I**The species is or may be affected by trade**

Species of the genus *Sporophila* are primarily sought after for their singing capabilities, and are entered into singing competitions, which are popular in Brazil, Colombia, Guyana and Suriname. A 2024 global assessment of the songbird trade conducted by the CITES Secretariat found moderate levels of international trade in *S. maximiliani*, *S. angolensis* and *S. crassirostris*. In the same year, a survey involving 75 songbird experts was published, disclosing extreme concern regarding the domestic and international trade of three species of *Sporophila*.

According to de Man (2025), while trade in *S. angolensis* and *S. crassirostris* is mostly domestic, both species are recorded to be exported to New York (USA) and Toronto (Canada). *S. crassirostris* is also reported to be smuggled to the Netherlands, where Surinamese citizens have established bird singing clubs and tournaments in major cities like Amsterdam and The Hague (Bertrams and Gercama, 2022; de Man, 2025). A systematic review of the most trafficked songbirds in Brazil ranked *S. angolensis* fifth, with a more than 6,500 specimens being seized from 1997 to 2018, and the genus *Sporophila* in 16th position with almost 4,000 individuals (Ferrari et al., 2023). Silva et al. (2021) reiterated that *S. angolensis* is a high-value species in markets in the eastern Amazonia, and therefore is at risk of being moved along distribution chains that can include cross-border flows.

The only trade data available for these species are from the US LEMIS database; imports to the USA are summarised below. Data shows a large proportion of specimens in trade being sourced from one of the many registered captive breeding enterprises.

Table 1. *Sporophila maximiliani* (Great-billed Seed-finch)

Exporter	SS LEMIS 2014-2025 Total No. (percentage wild)	Our analysis 2014-2020 Total No. (percentage wild)
Brazil	239 (3%)	115 (14%)

de Man (2025) reported a number of American websites selling *S. maximiliani* specimens, with prices ranging from USD800 per female specimen to USD1,000 per male (and USD1,600 for a pair). However, according to Valdes in litt. (2025), prices can be as high as USD5,400 per individual in the international market.

Table 2. *Sporophila angolensis* (Lesser Seed-finch)

Exporter	SS LEMIS 2014-2025 Total No. (percentage wild)	Our analysis 2014-2020 Total No. (percentage wild)
Brazil	440 (2%W)	174 (7.5%)
Guyana	930 (69% W)	465 (32%W)
Venezuela	20 (0%)	

In addition, small quantities were reported as scientific purposes.

Table 3. *Sporophila crassirostris* (Large-billed Seed-finch)

Exporter	SS LEMIS 2014-2025 Total No. (percentage wild)	Our analysis 2014-2020 Total No. (percentage wild)
Guyana	756 (52% W)	0

Illegal trade

According to Ortiz-von Halle (2018), the number of birds confiscated in Brazil during the last quarter of the 20th Century remained relatively constant, with a range from 30,000 to 45,000 birds being seized every year. About 80% of these were reported to be songbirds, depleting the populations of several seed-finches (*Sporophila* spp.) across the country.

Between 2019 and July 2024, a total of 611 individuals of *S. maximiliani* were seized by Brazilian authorities, whereas 1,343 specimens of *S. angolensis* and 29 individuals of *S. crassirostris* were admitted into wildlife rehabilitation centres between 2021 to March 2025 after confiscations during inspection operations carried out by law enforcement agencies in the country. Valdes in litt. (2025) highlights a 2023 operation where the majority of 55 birds seized were reported as "bicudos" or *S. maximiliani*, demonstrating active illegal possession and enforcement concern.

The number of the Great-billed Seed-finch in international trade is suspected to be underestimated given that current seizure data are often listed at the genus level due to difficulty in identifying traded specimens to species level (BirdLife International in litt., 2025). Moreover, as suggested by Charity and Ferreira (2020) and Santana Jr. (2020), seizures of *S. maximiliani* are likely becoming uncommon due to the scarcity of the species in the wild. Valdes in litt. (2025) argues that historic depletion of the species has shifted demand to congeners (*S. angolensis*, *S. crassirostris*, and others), which is a common substitution pattern in songbird markets. According to Valdes in litt. (2025), this dynamic sustains trapping pressure even when recent international records appear modest.

WiTIS reports a total of 35 seizures in *Sporophila* specimens in the last ten years, 26 of which occurred in Brazil. Other countries of incident included Colombia ($n = 5$), Guyana ($n = 2$), Trinidad and Tobago ($n = 1$), and the Netherlands ($n = 1$). The search includes a seizure of 60 *S. maximiliani* individuals from 2022 and one incident of a combined total of 330 *S. maximiliani* and *S. angolensis* seized in Brazil the same year. Whilst the vast majority are domestic seizures, the database also reports an incident taking place at Schiphol Airport in the Netherlands from 2020, as well as one 2015 record where a woman from the USA attempted to smuggle 18 live *S. angolensis* specimens from Guyana. The data reported by WiTIS does not indicate whether these were captive-bred or wild-sourced specimens.

No records for any of the six *Sporophila* species were reported by EU-TWIX.

In Guyana, Peru and Suriname, trade and export of wild-caught songbirds is legally permitted, and the borders between Brazil and these countries are extensive and uncontrolled, so that border regions between Brazil, French Guiana, Guyana, Suriname and Venezuela represent illegal transit routes for songbirds, especially for *S. maximiliani* and *S. angolensis* (de Man, 2025). An increasing trend of trapping males of *S. maximiliani* in Bolivia to be illegally sold in rural towns and cities across Brazil has been reported (Hall, 2024; Valdes in litt., 2025). In fact, market studies document organised capture and commercialisation networks that move *Sporophila* species from rural source areas to urban markets, creating conditions conducive to laundering and potential cross-border movement (Valdes in litt., 2025). A study by IFAW that compiled seizure and poaching incidents from 2017 to 2022 in Hispanic America reported that important smuggling routes for seed-finches include Venezuela to Trinidad and Tobago, Guyana, and Suriname. According to the study, *Sporophila* specimens can sell for up to USD5,000 in Trinidad due to local populations being nearly extirpated in these countries (IFAW, 2024).

Valdes in litt. (2025) acknowledged that domestic demand in countries like Brazil, Bolivia, Guyana and Suriname, as well as undocumented border leakage and laundering of captive-bred specimens, still affects wild stocks, especially given very small subpopulations and high prices linked to songbirds like *S. maximiliani*.

Additional Information

Threats

Threats to *Sporophila maximiliani* include illegal capture and trade of wild specimens to supply the illegal trade and captive breeding markets, and habitat loss and degradation. According to BirdLife International (2019) and Marini et al. (2023), 'hunting and trapping' is contributing significantly more to the threat load of the species compared to habitat loss and degradation. However, because seed-finches are highly specialised in the consumption of grass seeds, the conversion of native grasslands into agricultural fields is still affecting most species (de Man, 2025). Additionally, the occurrence of fires in the regions of Brazil where *S. maximiliani* occurs is considered an element of concern impacting the conservation of the species (Machado et al., 2020).

Conservation, management and legislation

Key conservation actions for *Sporophila maximiliani* include advancing scientific research, tightening control over illegal trade, and promoting reintroduction efforts using individuals from captive breeding programmes.

To date, no conservation plans or targeted government-led projects for *S. maximiliani* have been identified in any of the range States. Between 2017 and 2021, the Critical Ecosystem Partnership Fund for the Cerrado (CEPF-Cerrado) and the International Institute of Education of Brazil supported a project aimed at reintroducing Great-billed Seed-finches into the Brazilian Cerrado. The project remains active, and between November 2021 and October 2024, a total of 178 *S. maximiliani* were released into the reserve.

At present, there are no monitoring programmes in place for *S. maximiliani* populations.

Nationally, in Brazil, *S. maximiliani* has been listed as threatened since 1973 and is currently classified as critically endangered. Since 2014, the capture, transport, storage, custody, handling, processing, and commercialisation of wild individuals of species considered threatened with extinction have been banned, except for purposes related to research and conservation. Any wildlife illegally captured and kept is seized by either the Brazilian Institute of Environment and Natural Resources (IBAMA) or the Environmental Military Police (PAm), both law enforcement agencies (Ferrari et al., 2023). The formation of initial captive stocks destined for authorized breeding operations

through the collection of eggs or offspring is permitted under license from the competent authority, and the legal export of captive-bred individuals from these enterprises is permitted.

Bolivia and French Guiana present similar bans on the capture, conditioning, commercialisation and export of wild animals, including S. maximiliani (de Man, 2025). Whereas Guyana and Suriname allow the capture and trade of wild-caught specimens. In Guyana, the Guyana's Wildlife Scientific Committee decides on the capture, keeping and trade for species that are not listed in the CITES Appendices, including the Great-billed Seed-finch (de Man, 2025). The Wildlife Conservation and Management Commission confirmed that S. maximiliani, S. crassirostris and S. angolensis are not currently protected in Guyana and thus may be captured and traded (de Man, 2025).

Captive breeding

There are large captive-bred populations of *Sporophila maximiliani* across Brazil (over 200,000 specimens held by more than 37,000 amateur breeders). Furthermore, at present, 41 commercial breeding enterprises hold a combined total of more than 5,200 individuals of *S. maximiliani* across the country excluding the state of São Paulo (in which an additional 3,000 specimens are held by commercial breeders). Other authorized operations holding *S. maximiliani* include four Wildlife Maintainer (or "Mantenedouro de Vida Silvestre") operations with a total of 45 individuals, four zoos with nine individuals, three conservation breeding programs with 74 individuals, and two wildlife trading operations with six individuals. Valdes in litt. (2025) noted that these very large numbers of captive specimens are indicative of intense extraction pressure on wild populations.

Similarly, *S. angolensis* is bred commercially by 54 enterprises in Brazil, with a total captive population of more than 3,800 individuals. It is to be noted that the SS flagged the laundering of wild-caught songbirds as a concern in Brazil, a practice that might involve falsification of authorisations, false registration of birds, or tampering with identification rings to disguise wild-caught specimens as captive-bred.

Ethnobiological fieldwork and local studies report intentional interspecific breeding and hybrid production among Sporophila congeners in captive contexts to optimise song traits for competition; these high levels of inbreeding may increase incentives to introduce wild genetic material in captive populations (Valdes in litt., 2025). A quantitative genetic analysis conducted by Santana Jr. (2020) found captive individuals of S. maximiliani to exhibit a high level of inbreeding (mean of 8.26%) and an increasing trend (0.57% per year) in the period 2008-2019, which in a few generations could lead to fitness impairment.

Implementation challenges (including similar species)

As noted in the SS, morphological differences among males of the six *Sporophila* spp. (including *S. maximiliani*) are great enough for species-level identification, with the exception of *S. maximiliani* and *S. crassirostris*. However, BirdLife International in litt. (2025), Chng in litt. (2025) and Valdes in litt. (2025) argue that the former is true under normal or ideal observation conditions only: when plumage is in bad condition or viewed under poor lighting conditions, correct identification is likely to be significantly more challenging, especially for officers with little bird knowledge. Finally, Valdes in litt. (2025) notes that the practice of selling multiple *Sporophila* species together, as well as the purposeful hybridisation between congeners, further complicate visual identification and enforcement in markets.

There is evidence of significant trade in females and juveniles too, which are almost impossible to identify (BirdLife International in litt., 2025). De Man (2025) reports females being caught and traded for both captive breeding and singing competition purposes. According to BirdLife International in litt. (2025) and Valdes in litt. (2025), demand in female specimens is also driven by their use in stimulating males in competitions.

Potential benefit(s) of listing for trade regulation

Although export in wild specimens of *Sporophila maximiliani* is already banned in Brazil and French Guiana, this is not the case in all the other range States, where protection is only provided once a species is listed in the CITES Appendices (e.g. Guyana) or considered native (e.g., Suriname). A listing in the CITES Appendices would standardise and render all commercial trade in *S. maximiliani* illegal.

References

- Bertrams, N. and Gercama, I. (2022) De schimmige wereld van de handel in zangvogels [English: the shadow world of the trade in songbirds]. De Groene Amsterdammer Website. Available at: <https://www.groene.nl/artikel/de-schimmige-wereld-van-de-handel-in-zangvogels>. Accessed on 23 July 2025.
- BirdLife International (2025) in litt. to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- BirdLife International (2018) *Sporophila angolensis*. The IUCN Red List of Threatened Species 2018: e.T22723542A132167022. <https://dx.doi.org/10.2305/IUCN.UK.2018-2.RLTS.T22723542A132167022.en>.

- BirdLife International (2018) *Sporophila atirostris*. *The IUCN Red List of Threatened Species* 2018: e.T22723532A132166838. <https://dx.doi.org/10.2305/IUCN.UK.2018-2.RLTS.T22723532A132166838.en>.
- BirdLife International (2019) *Sporophila maximiliani* (amended version of 2017 assessment). *The IUCN Red List of Threatened Species* 2019: e.T22723537A155473409. <https://dx.doi.org/10.2305/IUCN.UK.2019-3.RLTS.T22723537A155473409.en>.
- BirdLife International (2021) *Sporophila crassirostris*. *The IUCN Red List of Threatened Species* 2021: e.T22723527A137077317. <https://dx.doi.org/10.2305/IUCN.UK.2021-3.RLTS.T22723527A137077317.en>.
- BirdLife International (2021) *Sporophila funerea*. *The IUCN Red List of Threatened Species* 2021: e.T22729107A137079233. <https://dx.doi.org/10.2305/IUCN.UK.2021-3.RLTS.T22729107A137079233.en>.
- BirdLife International (2021) *Sporophila nuttingi*. *The IUCN Red List of Threatened Species* 2021: e.T22723522A137075933. <https://dx.doi.org/10.2305/IUCN.UK.2021-3.RLTS.T22723522A137075933.en>.
- Charity, S. and Ferreira, J.M. (2020) Wildlife trafficking in Brazil, TRAFFIC International, Cambridge, UK.
- Chng, S. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- de Man, D. (2025) Management and conservation of songbirds in trade – listing the great-billed seed-finch (*Sporophila maximiliani* Cabanis 1851) in the CITES Appendices. Master's thesis for UNIA Master's degree in Management and Conservation of Species in Trade: the International Framework (15th edition). University of Andalusia, Baeza, Spain.
- Ferrari, G.C.P., Rheingantz, M.L., Rajão, H. and Lorini, M.L. (2023) Wanted: A systematic review of the most trafficked songbirds in a Neotropical hotspot. *Frontiers in Forests and Global Change* 6(930668). doi: 10.3389/ffgc.2023.930668
- Hall, J. (2024) Rewilding efforts throw a lifeline to Brazil's most trafficked endangered bird. Mongabay Website. Available at: <https://news.mongabay.com/2024/05/rewilding-efforts-throw-a-lifeline-to-brazils-most-trafficked-endangered-bird/>. Accessed on 23 July 2025.
- IFAW (2024) Wildlife crime in Hispanic America. An analysis of seizures and poaching incidents in 18 countries (2017-2022).
- Machado, R.B., Silveira, L.F., Soares Gomes da Silva, M.I., Ubaid, F.K., Medolago, C.A., Francisco, M.R. and Dianese, J.C. (2020) Reintroduction of songbirds from captivity: the case of the Great-billed Seed-finch (*Sporophila maximiliani*) in Brazil. *Biodiversity and Conservation* 29. 10.1007/s10531-019-01830-8.
- Marini, A.M., Lima, D.M., de Carvalho, C.B., Ubaid, F.K., da Silva, G.B.M., dos Santos Abreu, T.L., de Oliveira, T.D., Alves, W.N., Dias, F.F. and Alquezar, R.D. (2023) *Sporophila maximiliani*. Sistema de Avaliação do Risco de Extinção da Biodiversidade - SALVE. Instituto Chico Mendes de Conservação da Biodiversidade - ICMBio.
- MMA - Ministério do Meio Ambiente. (2014) Lista Nacional Oficial de Espécies da Fauna Ameaçadas de Extinção. Portaria No 444, de 17 de dezembro de 2014. Diário Oficial da União - Seção 1. N° 245 quinta-feira, 18 de dezembro de 2014.
- Ortiz-von Halle, B. (2018) Bird's-eye view: Lessons from 50 years of bird trade regulation & conservation in Amazon countries. TRAFFIC, Cambridge, UK.
- Partners in Flight. (2019) Avian Conservation Assessment Database, version 2019. Available at: <http://pif.birdconservancy.org/ACAD>.
- Ridley, F. *et al.* (manuscript in preparation) Advancing understanding of global threat abatement potential for terrestrial vertebrates.
- Santana Jr., M.L. (2020) Quantitative genetic analyses provide parameters for selection and conservation of captive Great-billed Seed-finches (*Sporophila maximiliani*). *PLOS ONE* 15(7): e0236647. <https://doi.org/10.1371/journal.pone.0236647>
- Silva, S., Braga, B., Brasil, L., Baia-Junior, P. and Guimaraes, D. (2021) The use of Passeriformes in the eastern Amazonia of Brazil: culture encourages hunting and profit encourages trade. <https://doi.org/10.1017/S0030605320000551>.
- Ubaid, F.K., Malacco, G.B., Medolago, C.A.B. and Silveira, L.F. (2021) Reintroduction of the great-billed seed-finch in the Brazilian Cerrado, Brazil. In: Soorae PS, editor. *Global Conservation Translocation Perspectives*. 7th ed. IUCN SSC Conservation Translocation Specialist Group, Canada, pp. 148-151.
- Valdes, J. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.

Inclusion of Giant Hispaniolan Galliwasp *Caribicus warreni* in Appendix I

Proponent: Dominican Republic

Summary: The Giant Hispaniolan Galliwasp *Caribicus warreni* is a large-sized galliwasp lizard native to the Island of Hispaniola and reported to inhabit the north of the Dominican Republic as well as the northern part of Haiti, including Tortuga Island. The species was globally classified as Vulnerable by IUCN in 2015 primarily due to its restricted and fragmented distribution.

Despite extensive search efforts across the years, there are no global population estimates of *C. warreni*, and the only recorded subpopulation in the Dominican Republic was last recorded in 2021 in the natural monument Loma Isabel de Torres, where it is currently being monitored. The IUCN Red List assessment from 2015 reported that the species had a decreasing population. The extent of occurrence (estimated at 100 km²), area of occupancy (smaller than 10 km²), habitat quality and number of locations are reported to be experiencing a continued decline, primarily due to expanding agricultural and livestock farming activities. Other threats include predation by invasive species, occasional killings by local people and the illegal pet trade.

The Giant Hispaniolan Galliwasp is very rarely utilised nationally, and only traded by reptile enthusiasts who keep them as pets. Dominican authorities have never issued any export permits for *C. warreni*. A moderate level of international trade has been detected in Canada, Czechia, Germany and the USA, as recently as early 2025. Haiti is believed to be the source of most traded specimens, although it is unclear whether specimens in trade continue to be wild-sourced or from captive sources. Giant galliwasp are relatively long-lived (up to 40 years in captivity) and have been bred successfully in captivity producing large litters (up to 34 specimens).

Captive-breeding programmes for the species exist in North America and Europe, with the largest residing at Nashville Zoo in the USA. According to ZIMS, there are currently 264 specimens in 23 different holding institutions across eight countries and territories.

The species is currently listed in the national Red List of Threatened Species of the Dominican Republic as critically endangered, which restricts its use to scientific purposes, research and priority breeding for conservation of the species.

Analysis: Little is known about the population of *Caribicus warreni*, but it is likely small or very small and was last reported as decreasing in a 2015 IUCN Red List global assessment. The species' range is restricted and fragmented, and the extent of occurrence, area of occupancy, habitat quality and number of locations are declining. Together these factors suggest that the species **meets the biological criteria A and B of Annex 1** of Res. Conf 9.24 (Rev. CoP17). Habitat loss and fragmentation are the primary threats to the species. International trade levels are unknown, but specimens have been observed for sale in North American and European pet markets. It is believed that these originated in Haiti, although it is unclear whether there is ongoing export of wild-specimens or such specimens are now originating from captive sources. Despite limited data, due to the likely small and severely fragmented population and its restricted range, as well as the presence of ongoing international trade, the **species appears to meet criteria for inclusion in Appendix I**.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

Caribicus warreni (Schwartz, 1970).

Formerly known as *Celestus warreni*, the species was assigned to *Caribicus* by Schools and Hedges in 2021. The Dominican subpopulation was recognised as a separate species (*Celestus carraui*) before Powell and Henderson (2003) synonymised *C. carraui* as a synonym with *C. w. warreni* and rejected its recognition as a subspecies.

Range

Dominican Republic and Haiti

IUCN Global Category

Vulnerable B1ab(iii) (assessed 2015, ver. 3.1)

Biological criteria for inclusion in Appendix I

A) Small wild population

The global wild population is unknown, and there is only one recorded subpopulation in the Dominican Republic that counts 66 individuals (data collected in 2021). *Herpetological surveys and collections have not been successful across the decades in coming across the species in the field, as encounters have been sporadic and mostly by local people while developing new agricultural areas and cutting down the forest (Inchaustegui in litt., 2025). This is mainly because these animals, although quite big, are very secretive and able to pass mostly undetected (Inchaustegui in litt., 2025).*

The majority of existing reports and collections are from the 1970s in Haiti where, despite search efforts, no sightings or captures have been reported for several years and, therefore, the species is considered to be extremely rare. Although the range in Haiti encompasses part of the mainland and Tortuga Island, the IUCN Red List assessment reported in 2015 that the species occurred in a single locality.

Upon consultation with experts, confirmation was received that no further information on population numbers and distribution exists at present (Inchaustegui in litt., 2025; Powell, in litt., 2025).

B) Restricted area of distribution

Caribicus warreni has been reported to occupy the northern regions of Haiti and Tortuga Island, as well as the northern part of the Dominican Republic. The one recorded subpopulation is found in the latter within the natural monument Loma Isabel de Torres, which was established as a protected area in 1966 (Inchaustegui in litt., 2025) and is included in Management Category III of the IUCN. Within the Dominican Republic, the species' extent of occurrence comprises 100 km², with an area of occupancy of <10 km².

With a total range including Haiti's population, according to the KBA standard, the area of habitat (AOH) amounts to 1,956.42 km² and the species is classified as range restricted (Ridley et al., in prep.). A continued decline in the extent of occurrence, area of occupancy, habitat quality and number of locations is ongoing.

C) Decline in number of wild individuals

There is no data available to quantify whether the species has experienced, or continues to experience, a marked decline in the number of wild individuals.

Trade criteria for inclusion in Appendix I

The species is or may be affected by trade

At a national level, *Caribicus warreni* is not used other than by reptile enthusiasts as pets, although this is not common. In the 1990s, the species was frequently legally exported from Haiti to the USA. Giant galliwasps are long-lived, reaching 25-30 years in the wild according to IUCN (Landestoy et al., 2016), and produce large litters. Individuals may have been legally or possibly illegally introduced into the private sector, where they can occasionally be found for sale in pet shops.

Dominican Republic authorities have never authorised any international trade of the species, nor have they issued any export permits. At present, international trade has been detected primarily in the USA, where people acquire them as pets. *Haiti, which is not a Party to CITES, is likely the principal source of traded specimens (Powell in litt., 2025) or the founding stock for any captive breeding for trade occurring outside the species' range. It is unclear whether specimens in trade continue to be wild or captive sourced.*

Other countries that recorded offers for sale of C. warreni include Canada, Czechia and Germany, with offers recorded as recent as 2023 and 2024 (Inchaustegui in litt., 2025). Noboa in litt. (2025) identified an online advert for a specimen of C. warreni from early 2025 from Czechia. The SS mentions prices abroad to range from USD40 for juveniles to USD225 for adults.

No trade nor seizure records of C. warreni were reported in LEMIS, EU-TWIX nor WiTIS.

Additional Information

Threats

Caribicus warreni faces a number of threats, namely habitat loss and fragmentation, expansion of intensive agriculture and livestock farming practices, presence of invasive species such as ferrets and cats, and illegal trade. According to Inchaustegui in litt. (2025), habitat destruction and fragmentation are likely the primary threat to the species, followed by predation from native species (e.g., Hispaniolan boa) and others (e.g., mongoose, cats, dogs).

The occasional killing of specimens by locals is also mentioned in the SS as a latent threat to *C. warreni*, as people wrongly believe the species to be poisonous. However, according to Inchaustegui in litt. (2025), this might not represent an important threat.

Conservation, management and legislation

The giant galliwasps are listed in the Red List of Threatened Species of the Dominican Republic as critically endangered and considered a priority species for conservation, and therefore it may only be used for scientific purposes, research and priority breeding for conservation of the species.

There are no management measures in place in the Dominican Republic, but a conservation plan for the species is currently being prepared at Nashville Zoo in the USA. The Ministry of Environment and Natural Resources of the Dominican Republic has been monitoring the only recorded population at the natural monument Loma Isabel de Torres since 2018.

Captive breeding

Although the Dominican Republic has no captive-breeding programmes in place for *Caribicus warreni*, Nashville Zoo in the USA has been keeping and breeding this species since 2000. More than 300 juveniles have been produced from the founder population, which comprised nine males and nine females originally captured from the wild in the Dominican Republic. The species' age of first breeding is reported at 3-4 years in captivity, with a litter size of 8 to 34 specimens (Corry et al., 2010). Over the last 20 years, Nashville Zoo has shared juveniles with many zoos all over the USA, and also with a European zoo (i.e., Durrell). The latter obtained 97 juveniles from two females breeding over a period of three years.

At present, the Species360 Zoological Information Management System (ZIMS) database counts a total of 264 captive-bred specimens (66 males, 63 females and 135 unsexed) in 23 ex-situ holding institutions across eight countries and territories (Czechia, Hungary, Jersey (UK), Poland, Slovenia, Spain, the United Kingdom of Great Britain and Northern Ireland and the USA).

Frankfurt Zoo in Germany also reports holding three specimens of *C. warreni* (Noboa in litt., 2025).

Implementation challenges (including similar species)

The genus *Caribicus* includes three species: *Caribicus warreni*, *C. anelpistus* and *C. darlingtoni*, which are all endemic to the Hispaniola region. The three species show marked differences in their colouring and patterns, and in their geographic distribution.

As far as trade through Haiti is concerned, the country has historically represented an escape route for animals with a broad Hispaniolan distribution, as well as Dominican subspecies, which are then sold internationally as "Haitian species" to avoid CITES regulations (Inchaustegui in litt., 2025).

References

- Corry, E., Martin, L., Morton, M.N., Hilton, G.M., Young, R.P. and Garcia, G. (2010) A Species Action Plan for the Montserrat galliwasps *Diploglossus montisserrati*. Department of Environment, Montserrat.
- Inchaustegui, S.J. (2025) in litt. to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Landestoy, M., Inchaustegui, S. and Powell, R. (2016) *Celestus warreni* (errata version published in 2017). *The IUCN Red List of Threatened Species* 2016: e.T39260A115172613. <https://dx.doi.org/10.2305/IUCN.UK.2016-3.RLTS.T39260A71739879.en>.
- Noboa, L. (2025) in litt. to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Powell, R. (2025) in litt. to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Powell, R. and Henderson, R.W. (2003) The taxonomic and conservation status of giant Hispaniolan *Celestus* (Anguidae). *Caribbean Journal of Science* 39(2): 237–240.
- Ridley, F. et al. (manuscript in preparation) Advancing understanding of global threat abatement potential for terrestrial vertebrates.
- Species360 Zoological Information Management System (ZIMS) (2025) *Caribicus warreni*. zims.Species360.org Accessed on 30 July 2025

Inclusion of Mount Elliot Leaf-tailed Gecko *Phyllurus amnicola* in Appendix II

Proponent: Australia

Summary: The Mount Elliot Leaf-tailed Gecko *Phyllurus amnicola* is a moderately sized gecko, one of 11 species in the genus *Phyllurus*, described in 2000 endemic to Queensland, Australia, where it is known from three subpopulations in the Mount Eliot region of Bowling Green Bay National Park in rocky rainforest habitats between 400 and 1000 m above sea level with a fourth subpopulation from nearby Saddle Mountain. Extent of occurrence has been estimated at around 200 km². There are no population estimates but the species is presumed to exist at high densities in areas of suitable habitat, which is extensive within its range.

Leaf-tailed geckos *Phyllurus* species such as *P. amnicola* are relatively slow-growing compared to other gecko species and usually reach reproductive size at 2-3 years of age. They are reported to be in demand in the international pet trade. The commercial export of live, native reptiles from Australia has been prohibited since 1982. Despite this, *P. amnicola* has been identified in international trade, indicating historic or ongoing illegal export. Field studies in 2013 identified a substantial reduction in population density for one of the four subpopulations, believed to be due to a single poaching event.

Phyllurus amnicola was assessed as Near Threatened on the IUCN Red List in 2017 due to an absence of other threats beyond its restricted distribution and an ongoing threat of collection for trade, with three of the four known subpopulations reported to be less accessible to collectors and hence less likely to be impacted by this threat. It is listed as vulnerable under Queensland state legislation. Aside from illegal collection, other potential threats include increasing frequency and intensity of periodic fires impacting habitats and habitat connectivity.

Around 80 live individuals were reported to be in trade based on US trade data between 2012 and 2021. There were also estimates of around 935 individuals offered for sale in over 355 online advertisements between 2014 and 2023, with most of these reported to be offering adult individuals. Twenty individuals directly imported into the USA from Germany and the Netherlands between 2015 and 2018 were reported to be wild-sourced, although this may have been a reporting error. The sources of specimens found for sale online were generally not reported, but where they were, sellers often claimed they were captive-bred. A survey of online platforms in Europe in 2017-2018 indicated that prices sought for specimens of this species varied from USD1470-2385. More recently, two advertisements in 2025 offered individuals for sale for USD220, and USD450, each.

The species was listed in CITES Appendix III in 2022, and since then, 24 specimens have been reported in direct exports for commercial purposes from European countries to Japan. Three of these (exported by Germany in 2022) were reported to be from captive-born sources. Little further information is available on the extent of captive breeding.

Analysis: *Phyllurus amnicola* is characterised by a restricted area of distribution occurring in just four known subpopulations (three on Mount Elliot, and one on Saddle Mountain) in the state of Queensland, Australia, and therefore may already meet the biological criteria B in Annex 1 of Res. Conf. 9.24 (Rev. CoP17). Although no legal commercial export of the species has been permitted from Australia, the species has been identified in international trade in recent years with high prices per individual. Although there is no clear evidence of the impact of trade across all subpopulations, there is evidence that illegal collection has previously led to notable declines in one of the four subpopulations. Although, the reported trade has been relatively low, given the apparent vulnerability of populations of this species to wild harvest, **it may meet the criteria for inclusion in Appendix II.**

All commercial export of wild specimens from Australia is illegal. If the species were to be included in Appendix II, **Australia could also submit a zero quota for wild-sourced commercial exports** to be posted on the CITES website to indicate that trade in wild harvested specimens of the species from Australia is not permitted.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES background

Australia listed 10 *Phyllurus* spp. in CITES Appendix III in 2022.

Taxonomy

Phyllurus amnicola (Hoskin et al., 2000).

Range

Australia

IUCN Global Category

Near Threatened B1a (assessed 2017, ver. 3.1)

Biological and trade criteria for inclusion in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)

A) Trade regulation needed to prevent future inclusion in Appendix I

The Mount Elliot Leaf-tailed Gecko *Phyllurus amnicola* is endemic to Australia and has a very limited natural distribution, being known from only four localities on and around Mount Elliot in the state of Queensland within the protected area of Bowling Green Bay National Park (NP), *with one expert stating the estimated overall extent of occurrence for the species is just 224 km² (Meiri in litt., 2025).* The species is narrowly restricted to rocky rainforest habitats and is found mostly between 400 m and 1,000 m above sea level at each of the known localities.

Three of the four known subpopulations are found on Mount Elliot (Alligator Creek, Western Boulders and Cockatoo Creek), with a more recently discovered subpopulation on the immediately adjacent mountain, Saddle Mountain. The discovery of the Saddle Mountain subpopulation represents the first record of the species outside of Mount Elliot. Although there is a narrow (<2 km) gap in suitable habitat between Mount Elliot and the Saddle Mountain subpopulation, this gap is believed to prohibit gene flow between the subpopulations due to the restricted habitat and dispersal capabilities of the species. The proposal states that despite being a protected area, Mount Elliot is easily accessible and close to a major urban centre, which makes it accessible to poachers, *although the 2017 IUCN Red List Assessment states that aside from the Alligator Creek subpopulation on Mount Elliot, the remainder of the range is much less accessible to collectors (Hoskin et al., 2018).*

This species was considered to be common in suitable habitat in 2017 (Hoskin et al., 2018). The population size for the species is currently unknown but as their preferred rocky rainforest habitat is widespread across Mount Elliot, the species is generally presumed to occur at high densities within areas of suitable habitat. The exception to this is the subpopulation located at Western Boulders on Mount Elliot. At this locality, the species occurs in low densities, likely due to the exposed boulder features with little associated rainforest. Previous surveys recorded 20-30 individuals at the most accessible stretch of habitat on Alligator Creek on Mount Elliot, but field surveys conducted in 2013 detected a substantial reduction in population density, with only a single individual recorded through a targeted search. Poaching by a single collector was determined to be the cause, following the online advertisement of Mount Elliot Leaf-tailed Gecko specimens in the illegal pet trade. It is unknown if specimen numbers have since recovered at this locality, but population recovery *was previously reported* to be likely to be slow and the species' population was estimated to be in decline following the marked impact on the Alligator Creek subpopulation from this single illegal collection event. There is little information on the reproductive characteristics or strategies of this species, but it is likely that it reproduces in a similar manner to other *Phyllurus* spp., which are relatively slow growing compared to other gecko species, and usually reach reproductive size at 2-3 years of age.

Australian leaf-tailed geckos (*Phyllurus* spp.) are highly desirable in international pet markets due to their unique morphological characteristics and rarity. The proposals states that individuals have been detected in international trade in various countries, with evidence as follows:

Data from LEMIS show that:

- From 2008-2018, 72 specimens of Mount Elliot Leaf-tailed Gecko originating from Germany, the Netherlands and the USA were traded internationally with declared values ranging from EUR10 to 70 (USD11-81) per individual (*see below for more details on trade reported in LEMIS*).

Data on 355 online advertisements from different data sources published in a study in 2024 (Chekunov et al., 2024) (additional details from the supplementary files for the study have been included) showed:

- Data from *long term* online trade surveillance: From 2014-2023, there were 227 advertisements for the species from online data sources, including websites, forums and social media pages. These were identified during long term monitoring which started in 2019, on a website which had advertisements dating back as far as 2014. The lead author was not able to comment on clear trends over time, as it is likely that older advertisements would have often been deleted once individuals were sold (Chekunov in litt., 2025).
- Data from *snapshot web* surveillance: From 2019-2022, there were 19 advertisements for the species.
- Data from *snapshot social media* surveillance: From 2016-2023, there were 109 advertisements for the species, with all but one of these from 2017 onwards.⁷

The lead author of this paper stated that based on an analysis of a subset of this data, an average of 2.5 individuals were offered per advertisement (Chekunov in litt., 2025), which would provide an estimate of ~890 individuals offered for sale between 2014 and 2023. They stated that the source is often not stated for Australian reptiles in international trade but that for almost all the advertisements that did provide a source claim, the specimens were reported to be captive-bred, noting that there is no way to verify these claims. The author also stated that whilst unsexed young are occasionally advertised for sale, the species was most often advertised as adult males and females. One expert noted that often, if species such as geckos are captive-bred, they are sold as juveniles (Chng in litt., 2025).

Evidence published in a study in 2019 indicates that:

- From 2017-2018, 45 individuals were detected for sale from traders in Denmark, Czechia, France, the Russian Federation and the USA on European online platforms and social media pages within a six-month period at a substantially higher price of EUR1,250-2,030 (USD1470-2385) each.

Other evidence of international trade

CITES

The species is listed on CITES Appendix III, under a Genus listing for *Phyllurus* spp., which was effective 22 June 2022. According to the CITES Trade Database between 2022 and 2023, 24 live individuals were directly exported for commercial purposes by Czechia (21) and Germany (3) to Japan. Germany reported all three specimens exported in 2022 were captive-born (source code F, meaning produced in a controlled environment from at least one parent taken from the wild) and all individuals exported by Czechia were reported to be captive-bred (source code C).

LEMIS

This data builds upon the evidence provided in the proposal for trade reported in LEMIS from the study published by Chekunov et al. (2024), which stated 72 specimens were traded between 2008-2018. Our analysis found 84 individuals of the species were traded for commercial purposes over a longer time frame between 2008-2021, with 21 in direct exports from the USA to other countries and 63 in direct imports into the USA.

All 21 direct exports of live individuals of *P. amnicola* from the USA were reported to be captive-bred (source code C), most (19) to Germany in 2013, with two exported to Canada in 2021.

Most of the direct imports of the 63 live individuals to the USA were reported as captive-bred (source code C, 43, 68%), but around a third were reported to be wild-sourced (source code W, 20 32%) (Figure 1). Most of the wild-sourced imports were reported to be from the Netherlands (60%; 12), with the remaining from Germany, whereas most of the captive-bred imports were reported to be from Germany (60%; 26), with the remaining from Netherlands (20%; 9) and Canada (20%; 8). Most imports were reported between 2015 and 2020, with all wild-sourced imports reported between 2015 and 2018. Declared values per individual, where they were reported, varied from USD4 to USD21 for imports and USD10 to USD200 for exports, and didn't vary markedly for individuals reported as wild-sourced and captive-bred.

⁷ numbers of individuals for sale per advertisement were not stated for either long term or snapshot web data

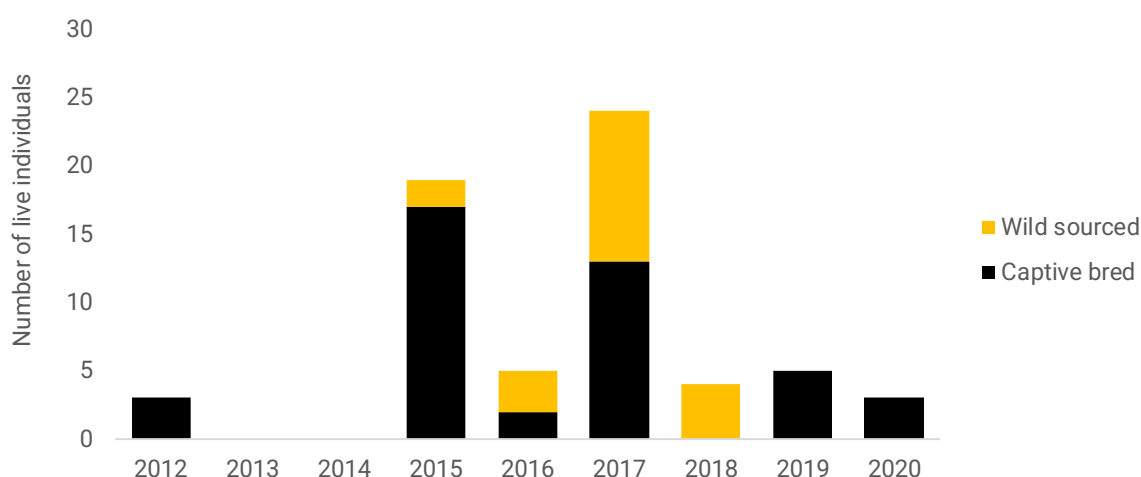


Figure 1. Imports of live individuals of *P. amnicola* for commercial purposes into the USA from 2012-2020 as reported in the LEMIS database.

One direct import of a wild-sourced specimen from Australia for scientific purposes in 2008 was refused entry to the US and re-exported. There was also some trade reported in *Phyllurus* spp for commercial purposes in US trade data. Nine live individuals were reported in direct exports from the USA to countries Germany (4 individuals), Japan (4), and Sweden (1) between 2009 and 2010, with values ranging from USD10 to USD650 per individual. There were also 50 wild-sourced bodies reported in direct imports into the USA from Indonesia in 2011 (value USD3 per individual), and 5 live captive bred individuals from Canada in 2015 (value USD575 per individual).

No seizure records for the species could be found in EU-TWIX or WiTIS based on available data (from 2014-2023 for EU-TWIX and 1993-2025 for WiTIS).

The proposal states that the substantial growth in sale price for specimens indicates that the commercial value of the species is rapidly increasing and that this may, in turn, increase international demand for the species and intensify the threat of illegal collection for individuals in the wild. Based on data available on declared import/export values from LEMIS and online market studies, no clear trend toward increased prices over time could be identified. The highest market prices per individual identified (USD1,470 - 2,385) were for the 45 individuals offered for sale online between 2017 and 2018. One expert identified two online advertisements for the species in 2025 with lower prices than this offered (USD450 from a seller in Canada, and USD220 from a seller in Germany) (Janssen in litt., 2025). The proposal also states that given that the species is only known from four subpopulations, the illegal removal of any individuals from the wild has the potential to significantly reduce population numbers to unviable levels.

The commercial export of live native reptiles from Australia is currently prohibited by Australia's national environmental legislation, so there is no legal commercial international trade in the species. Domestically, there is no lawful market for the species in Australia. The species cannot be taken, held or traded commercially. The export of live Australian reptile species from Australia for commercial purposes has been prohibited since at least 1982 and no permits providing for the legal export of the species from Australia have been issued since the species was first described in 2000. As no legal exports of the species have occurred, any individuals available for sale outside of Australia were almost certainly illegally acquired or are the progeny of illegally exported specimens.

B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences

According to the 2017 IUCN Red List assessment, the harvest of individuals for the international pet trade caused a marked decrease in the number of individuals in one subpopulation. The impact of this threat on the other subpopulations is unknown.

Additional Information

Threats

Phyllurus amnicola was assessed as *Near Threatened* on the IUCN Red List in 2017 due to its' restricted range, being known only in four locations, and the threat from collection, in part after the 'near-extinction' of one subpopulation after a single illegal collection event (Hoskin et al., 2018). The Red List assessment also stated that the extent to which the species is at risk is not clear and, in the absence of other threats aside from illegal collection, it was not thought likely to be at significant risk of extinction (Hoskin et al., 2018).

The proposal states the species is subject to ongoing threats from changed fire regimes, invasive grasses, climate change and illegal collection. As the species' habitat occurs in a protected area it is largely shielded from major threats but changes to fire regimes in the last century have altered the timing, frequency and intensity of fires, thereby increasing the potential threat of burning in rainforest areas of the region. Mount Elliot is frequently burned as part of routine management activities to control invasive grasses that create high fuel loads and reduce the risk of bushfire. The introduction of highly flammable grasses to the area, such as guinea grass *Megathyrsus maximus* and molasses grass *Melinis minutiflora*, has created high fuel loads around the rainforest patches preferred by the species. In combination with the burning of land for cattle grazing, high-frequency hazard reduction burns, late dry-season burning and the increase of severe weather events due to climate changes, these higher fuel loads threaten to destroy the amount of rainforest area available to the species and reduce connectivity in the region.

Although the species primarily occurs in areas of rocky habitat that are buffered from the effects of fire, any loss of local rainforest due to extreme weather events or burning has the potential to reduce the connective pathways between the Mount Elliot and Saddle Mountain populations, thereby increasing habitat fragmentation and degradation. The gap of unsuitable habitat between the populations of Mount Elliot and the one subpopulation on Saddle Mountain has already posed a significant barrier to current and historic gene flow of the species. Adult specimens on Saddle Mountain are ~25% smaller than those at all sites on Mount Elliot and there is ~2.9% genetic divergence between the two areas. With the limited dispersal capabilities of the species, any activities that threaten connective pathways risk creating further population fragmentation and increasing its isolating effects.

Conservation, management and legislation

In Australia, the Mount Elliot Leaf-tailed Gecko is listed as vulnerable under Queensland legislation and recognised as a protected species. The entire known distribution falls within a protected area. Under Queensland legislation, all native reptiles are recognised as a protected species.

The export of live native reptiles from Australia has been regulated from at least 1982 under previous legislation, the Wildlife Protection (Regulation of Exports and Import) Act 1982, when the export of live reptiles was only permitted for specific non-commercial purposes. The current Australian national environmental legislation, the Environment Protection and Biodiversity Conservation Act 1999 (EPBC Act), continues to regulate live animal export. Export permits for live Australian reptiles can only be issued if for specified non-commercial purposes (exhibition, conservation breeding, research and education), as described in the legislation. The species is prohibited from being kept or traded as pets although it is likely that this species is currently held in private collections in Australia. Penalties, ranging from fines to imprisonment, are enforced by the NC Act for offences involving wildlife and protected areas. The proposal states no permits have been granted for the species to be taken from the wild for the purpose of export.

Captive breeding

The IUCN Red List assessment in 2017 reports that captive breeding of the species is underway but does not elaborate on where, or whether this is on a commercial scale. *There are no current records for captive breeding of this species in the Species360 Zoological Information Management System (ZIMS) database for zoos. One online platform in Canada offering the species for sale in 2025 claimed the individuals for sale were currently 'incubating' but this claim could not be verified. One expert stated that based on their biology, the species was likely to have a low captive breeding potential given that they lay approximately two eggs at a time and breed only once or twice a year after reaching sexual maturity (thought to be at around three years of age) (Chekunov in litt., 2025).*

Implementation challenges (including similar species)

The Mount Elliot Leaf-tailed Gecko can be reliably distinguished from other *Phyllurus* species by unique traits such as its flared, leaf-like tail, specific tail banding, scale counts, prominent tail spines, and larger size. These features support accurate identification by CITES export regulators. As live exports of native reptiles for commercial purposes are prohibited under Australian law, and non-commercial international trade requires permits, misidentification at the border is unlikely to pose a regulatory issue.

References

- Chekunov, S., Stringham, O., Toomes, A., Prowse, T. and Cassey, P. (2024) Scale of unregulated international trade in Australian reptiles and amphibians. *Conservation Biology* 38(5): e14355. <https://doi.org/10.1111/cobi.14355>
- Chekunov, S. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Chng, S. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Hoskin, C., Couper, P. and Amey, A. (2018) *Phyllurus amnicola*. *The IUCN Red List of Threatened Species* 2018: e.T102664470A102664473. <https://dx.doi.org/10.2305/IUCN.UK.2018-1.RLTS.T102664470A102664473.en>. Accessed on 18 July 2025.
- Janssen, J. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.

Meiri, S. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.

Inclusion of the Ringed Thin-tail Gecko *Phyllurus caudiannulatus* in Appendix II

Proponent: Australia

Summary: The Ringed Thin-tail Gecko *Phyllurus caudiannulatus* is a moderately sized gecko, one of 11 species in the genus *Phyllurus* endemic to Queensland, Australia where it inhabits subtropical vine forest and adjacent wet sclerophyll forests as well as Hoop Pine *Araucaria cunninghamii* plantations 180-600 m above sea level. It is currently known to occur in just three isolated subpopulations within 15 km of each other, the largest in Bulburin National Park (NP). Overall extent of occurrence has been estimated at just under 350 km², with a possible area of Occupancy (AOO) of 20 km².

There are no population estimates, but the species may occur at high densities, at least in Bulburin NP. The two subpopulations located outside Bulburin NP were only discovered in the last ten years, and there may be other as yet undiscovered subpopulations. A quarter of the modelled potential habitat for the largest subpopulation in Bulburin NP was burnt during bushfires in 2019 and 2020; information on the impact on the species from this is lacking. *Phyllurus caudiannulatus* was assessed as Near Threatened on the IUCN Red List in 2017. It is listed as endangered at both the national and state levels in Australia. It was included in Appendix III (by Australia) in 2022.

Leaf-tailed geckos *Phyllurus* species are said to be in demand in the international pet trade amongst specialist collectors, with the distinctive markings of this species reportedly making it particularly desirable.

Commercial export of live native reptiles is illegal in Australia but the species has appeared in international trade. US LEMIS trade data report 32 live individuals in trade between 2010 and 2018, and there are estimates of around 190 individuals offered for sale in around 80 online advertisements between 2009 and 2022, with most reported as adult individuals. Six individuals in direct exports from the USA in 2017 were reported to be wild-sourced, although these may have been a reporting error; individuals have been offered for sale as captive-bred. Two live captive-bred specimens have been reported in CITES trade data (from Czechia to Japan for commercial purposes).

Sources differ on the likely impact of poaching for the international pet trade on populations of *P. caudiannulatus*, with the proposal stating that although poaching intensity is relatively low, it is likely that demand for the species is increasing, with the species reported to occur in accessible areas making it vulnerable to poaching. Information from the US does not demonstrate a clear trend for increased trade over time, with most offers for sale in 2017 and fairly consistent quantities in other years between 2010 and 2018, averaging two individuals per year. Supplementary data files from a study of online trade show around half of the 76 advertisements were posted in recent years (between 2019 and 2022), but as older advertisements are often deleted once individuals are sold, it is difficult to infer trends based on online trade.

There are no captive breeding populations for the species in Australia and captive breeding potential is likely to be low.

Analysis: *Phyllurus caudiannulatus* is a localised species that occurs in three known subpopulations in the state of Queensland, Australia. Its habitat has been impacted by recent wildfires.

Although no legal commercial export of the species has been permitted from Australia, the species has been identified in international trade in recent years in very small numbers. There is however no evidence of the impact of trade on wild populations, which can be at least locally abundant. Given the small quantities of individuals reported in trade, and a lack of clear evidence that a significant proportion of this trade is from wild-sourced individuals and that this collection is ongoing, **it is unclear that the species meets the criteria in 2a of Res. Conf. 9.24 (Rev. CoP17).**

As all commercial export of wild specimens from Australia is illegal. If the species were to be included in Appendix II, **Australia could also submit a zero quota for wild-sourced commercial exports** to be

posted on the CITES website to indicate that trade in wild harvested specimens of the species from Australia is not permitted.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES background

Australia listed 10 *Phyllurus* spp. in CITES Appendix III in 2022.

Taxonomy

Phyllurus caudiannulatus (Covacevich, 1975)

Range

Australia

IUCN Global Category

Near Threatened B1a (assessed 2017, ver 3.1)

Biological and trade criteria for inclusion in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)

A) Trade regulation needed to prevent future inclusion in Appendix I

The Ringed Thin-tail Gecko *Phyllurus caudiannulatus* is a moderately sized prickly gecko endemic to the south-east region of Queensland, Australia, occurring primarily in the Dawes Range, the Many Peaks Range and surrounding forestry reserves. The population is largely restricted to protected areas and found in only three discrete locations within 15 km of one another. The main subpopulation, located in Bulburin National Park (NP), is geographically isolated from the other two known subpopulations (which were only recently discovered) and is believed to be genetically distinct. Leaf-tailed geckos, including the Ringed Thin-tail Gecko, have limited dispersal ability and tend to remain within their established territories. Dispersal between subpopulations is unlikely due to unsuitable intervening habitats. The proposal states that the two recently discovered populations are small, highly localized, and likely unsustainable due to restricted connectivity, habitat limitations, and varying land uses, *although an expert noted that there are no clear data to show these two populations are below the minimum viable population size, particularly given the species potential to live in high densities in areas of suitable habitat (Janssen in litt., 2025).*

Ringed Thin-tail Geckos can be found in discrete patches of subtropical vine forest and adjacent wet sclerophyll forests and Hoop Pine *Araucaria cunninghamii* plantations 180-600 m above sea level. The species shelters in buttress cavities of trees (especially *Ficus* spp.), under bark and beneath rock piles. Unlike most *Phyllurus* species, Ringed Thin-tail Geckos appear to prefer sheltering in trees and their hollows over layered rock formations and can be observed moving across the forest floor in areas away from rocky substrates. Many individuals of this species have been observed occupying a single tree hollow, indicating that Ringed Thin-tail Geckos may cohabit suitable refuges in large numbers.

Phyllurus caudiannulatus lay two eggs per clutch, typically between October to November, sometimes communally in crevices. There is no published information on the species' reproductive ecology in the wild, but observation of captive individuals suggests sexual maturity (age at first reproduction) may not be reached until three to four years of age, depending on environmental conditions. There are no known records for longevity of Ringed Thin-tail Geckos in the wild. Their lifespan has been estimated at ten years and generation length at six years.

The species is only known from three discrete subpopulations, the largest of which occurs within Bulburin NP. The two subpopulations located outside of Bulburin NP were recorded in the last ten years, so it may be possible that additional subpopulations have not yet been detected. Mapping records from 1975 to 2020 indicate that the estimated Area of Occupancy (AOO) may be as low as 20 km² *and one expert stated the estimated extent of occurrence for this species is 348 km² (Meiri in litt., 2025).* An estimated 28% of the modelled potential habitat of Ringed Thin-tail Geckos in Bulburin NP was burnt during bushfires between 2019-2020. The Ringed Thin-tail Gecko can occur at high densities in areas of Bulburin NP but there is insufficient data to adequately estimate population numbers, as there is no population monitoring data currently in place for the species. The Ringed Thin-tail Gecko population is considered to be naturally fragmented across their distribution and *while its population status was reported as stable in 2017 (Couper et al., 2017), the proponents state it is in a current state of population decline.* The species is listed as endangered at both the national (on the Environment Protection and Biodiversity Conservation Act) and state level (Nature Conservation Act, Queensland) in Australia.

Australian leaf-tailed geckos (*Phyllurus* spp.) are prized by collectors for their uniqueness and rarity, being only found in Australia and nationally protected. The unique pale to white bands on the original tail (not found on regenerated tails) of this species distinguish it from any other leaf-tailed gecko and increase their desirability to collectors, and the species occurs in accessible areas. Their high desirability has led to an increase in the number of *Phyllurus* spp. specimens found in the international pet trade market. The proposal states that individuals have been detected in international trade in various countries, with evidence as follows:

Data from LEMIS show that:

- From 2008-2018, 27 specimens of Ringed Thin-tail Gecko were traded internationally from Germany and the USA to seven countries with declared values ranging from EUR15-500 (USD17-580) per specimen (see below for more details on trade reported in LEMIS).

Data on 355 online advertisements from different data sources published in a study in 2024 (Chekunov et al., 2024) (additional details from the supplementary files for the study have been included) showed:

- Data on 76 online advertisements from different data sources, including:
- Long term online trade surveillance: From 2009-2022, there were 47 advertisements for the species. These were identified during long term monitoring which started in 2019, on a website which had advertisements dating back as far as 2009. The lead author was not able to comment on clear trends over time, as it is likely older advertisements would have often been deleted once individuals were sold (Chekunov in litt., 2025).
- Snapshot web surveillance: From 2020-2022, there were 13 advertisements for the species.
- Snapshot social media surveillance: From 2011-2022, there were 15 advertisements for the species, with all but four of these from 2019 onwards.
- Summary of data retrieved from a literature review: one site found selling the species in Japan.

The lead author of this paper stated that based on an analysis of a subset of this data, an average of 2.5 individuals were offered per advertisement (Chekunov in litt., 2025), which would provide an estimate of 190 individuals offered for sale between 2009 and 2022. They stated that the source is often not indicated for Australian reptiles in international trade but that for almost all the advertisements that did provide a source claim, the specimens are captive-bred, noting that there is no way to verify these claims. The author also stated that whilst unsexed juveniles are occasionally advertised for sale, the species is most often advertised as adult males and females. One expert noted that often, if species such as geckos are captive-bred, they are sold as juveniles (Chng in litt., 2025).

Evidence published in a study in 2019 showed that.

- From 2017-2018, two specimens of Ringed Thin-tail Gecko were offered for sale at EUR500-800 each (USD585-940⁸) by sellers in Germany.

Other evidence of international trade

CITES

The species is listed on CITES Appendix III, under a genus listing for *Phyllurus* spp., which was effective 22 June 2022. Since then, according to the CITES Trade Database, the only trade reported by CITES Parties was in 2023 when two live captive-bred specimens (source code C) were reported in direct exports from Czechia to Japan for commercial purposes (CITES, 2025).

LEMIS

This analysis builds upon the evidence provided in the proposal for trade reported in LEMIS from the study published by Chekunov et al., 2024, which stated 27 specimens were traded between 2008-2018. Our analysis of LEMIS data found 32 individuals of the species were traded for commercial purposes between 2008-2020, with 26 in direct exports from the USA to other countries, and five in direct imports into the USA.

Most (20) of the live individuals of *P. caudiannulatus* reported in direct commercial exports from the USA were captive-bred, with six wild-sourced specimens. As the USA is not a range State for this species, these trade records could be a reporting error but may also be evidence of individuals having been imported from Australia. All exports occurred between 2010 and 2018, with over half of all direct exports (including all those reported as wild-sourced) in 2017 and an average of two individuals a year in other years. Most direct exports of individuals reported as captive-bred were to Germany (6 live individuals) and Hong Kong SAR of China (6), with the remaining to Canada (3), Thailand (2), Taiwan Province of China (2) and Japan (1), whilst the exports of six individuals reported as wild-sourced were to the Republic of Korea. Declared values per individual, where reported, varied markedly from USD5 to

⁸ Based on exchange rates sourced from oanda.com July 2025

USD800 (average USD220). The highest values were reported for captive-bred individuals exported in 2013 (USD800), 2016 (USD500), and 2017 (USD425). The direct imports of five live individuals were all from Germany between 2016 and 2017, with all reported to be captive-bred and values ranging from USD5 to USD13 per individual.

As well as commercial trade, LEMIS data included three live captive-bred individuals of *P. caudiannulatus* reported in direct exports from the USA to Germany for breeding in captivity in 2011, and one direct import into the USA of a wild-sourced specimen (scientific or museum) from Australia in 2016 for scientific purposes (another in 2008 was refused entry by the USA and re-exported). There was also some trade reported in *Phyllurus* spp (e.g. those not identified to the species level) for commercial purposes. Nine live individuals were reported in direct exports from the USA to countries Germany (4 individuals), Japan, (4) and Sweden (1) between 2009 and 2010, with declared values ranging from USD10 to USD650 per individual, 50 wild sourced bodies were reported in direct imports from Indonesia in 2011 (value USD3 per individual), and five live captive-bred individuals from reported imported from Canada in 2015 (value USD575 per individual).

No seizure records for the species could be found in EU-TWIX or WiTIS based on available data (from 2014-2023 for EU-TWIX and 1993-2025 for WiTIS).

Domestically, there is no lawful market for this species in Australia. The species cannot be taken, held or traded commercially. The export of live Australian reptile species from Australia for commercial purposes has been prohibited since at least 1982 and no permits have been issued for the legal export of live Ringed Thin-tail Gecko specimens. The individuals available for sale outside of Australia were potentially illegally acquired or are the progeny of illegally exported specimens. The proposal states that with the high commercial value for Ringed Thin-tail Gecko specimens, international demand will continue to grow, increasing the threat to the species in the wild. It also states that recent indications suggest that trade of this species is growing likely due to increased demand. With growing popularity for rare Australian endemic reptile species in the international pet market, Australian wildlife enforcement authorities are concerned about the associated increase in poaching of specimens from the wild to satisfy overseas demand. *Based on data available on declared values from LEMIS and prices from online market studies, no clear trend toward increased quantities of individuals over time could be identified, although one expert identified an online advertisement for the species in 2025 with a relatively high price of USD600 per individual (Janssen in litt., 2025).*

According to the SS poaching intensity is believed to be relatively low for the Ringed Thin-tail Gecko and the scale of threat from trade impacts is unclear but that as the species occurs in accessible areas, is highly sought-after, commands high prices in the pet market and is subject to incidental collection from visitors to the locality, potential trade impacts could be great, threatening the long-term survivability of the species. The species is restricted to only a small number of fragmented subpopulations, and any depletion through direct take or habitat damage from harvesting and poaching, alongside increasing threats from habitat degradation, climate change, and bushfires, risks detrimentally impacting population dynamics and reducing the wild subpopulations to unviable numbers.

The IUCN Red List assessment in 2017 stated that the export of this species is thought to be historical (e.g., the species entered the international pet trade a long time ago) and that the animal still occurs in high densities within its restricted range (Couper et al., 2018). The assessment acknowledges that the species commands high prices on the market and occurs in an accessible area where it is likely that there may be some incidental collection, but that the scale of this threat is unclear, noting however, that a future impact from collection remains plausible.

B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences

According to the 2017 IUCN Red List assessment, harvest of individuals for the international pet trade had taken place in the past, but that there was no evidence the species' population is continuing to decrease due to overcollection.

Additional Information

Threats

The IUCN Red List lists the species as Near Threatened based on criteria that the species has a restricted range and occurs in a single location defined by a threat from illegal collection (Couper et al., 2018). The Red List assessment also stated that there were no threats to the habitat and no current evidence (as of the assessment in 2017) of continuing decline as a result of collection, but that future impact from collection was plausible based on the high prices for the species in the international pet trade (Couper et al., 2018). The 2017 IUCN Red List assessment is (as of 2025) eight years old, and the proposal outlines some current threats to the habitat. The proposal states the Ringed Thin-tail Gecko population is considered to be naturally fragmented across their distribution and in a current state of decline.

Bushfires have already significantly impacted the species with 28% of the species predicted range in Bulburin NP burnt in 2019-2020. The direct impacts of bushfire and loss of critical habitat structures were likely responsible for high levels of species mortality across the entire area of burnt rainforest in Bulburin NP. These effects have been further compounded by the slow post-fire regeneration of important habitat features and reduction in habitat quality due to increasing drought conditions. Post-fire surveys demonstrated lower detection rates of the species within fire-impacted habitats, indicating a decline in population abundance. A survey of previously recorded Ringed Thin-tail Gecko habitat sites in 2020 revealed that at burnt sites, geckos were only detected on unburnt portions of habitat features, such as fig trees with minimal scorching, and previously occupied sites (including large logs and dead stags) that were burnt in the 2019-2020 bushfires had been vacated. The limited distribution of the species is inferred to be contracting because of increased fire frequency, severe weather events driven by climate change and degradation of habitat quality by invasive weeds.

Despite the species' distribution occurring predominantly in protected areas, there are still many key threats to Ringed Thin-tail Gecko populations. Modelling for the narrow distribution area of the Ringed Thin-tail Gecko predicts an average increase of 0.9°C in mean annual temperatures by 2030. This increase will be associated with a higher frequency of hotter days and heatwaves, as well as increased probability of drought and bushfires. Increased fire frequency has also been associated with invasive weed infestation in the habitat of other *Phyllurus* species. Invasive grasses are well-adapted to exploit post-fire conditions, providing highly flammable fuel loads around rainforest edges and filling canopy gaps. In addition to rendering Ringed Thin-tail Gecko habitats even more susceptible to bushfires, weed invasion can degrade habitat quality and compromise subpopulation connectivity by encroaching on rainforest edges.

Fragmentation of subpopulations poses another threat. Historical clearing of rainforest for timber harvesting within Bulburin State Forest before it was declared a National Park in 2006 removed a considerable amount of habitat for the species. The species is now restricted to three discrete subpopulations across a single location, sometimes seeking shelter in Hoop Pine plantations. The plantations are subject to regular timber harvesting and replanting, resulting in disturbance or removal of important habitat features. Harvesting also decreases connectivity between the subpopulations, increasing the risk of additional population fragmentation.

Conservation, management and legislation

The commercial export of live native reptiles from Australia is currently prohibited by Australia's national environmental legislation and has been regulated since at least 1982 under the Wildlife Protection (Regulation of Exports and Imports) Act 1982, when exports of live reptiles were only permitted for specific non-commercial purposes. Export permits for live Australian reptiles can only be issued for specified non-commercial purposes (exhibition, conservation breeding, research and education), as described in the legislation, and this must be demonstrated prior to the issuing of any export permit. No such permits have been issued for the export of *P. caudiannulatus* from Australia and the illegal export of an Australian native reptile specimen is punishable by up to ten years' imprisonment under national legislation.

The Ringed Thin-tail Gecko is also listed as endangered at both the national and state levels on the EPBC Act and NC Act (Queensland), respectively. Under Queensland legislation, all native reptiles are recognised as a protected species. The NC Act 1992 requires the issuing of permits for scientific research or education, for moving wildlife within, into or out of Queensland, and for taking from and/or releasing specimens in the wild. The species is not permitted to be kept as pets although it is likely that this species is currently held in private collections in Australia. Conservation Advice for *P. caudiannulatus* came into effect from 16 July 2024 when the species was listed as endangered. The Conservation Advice provides a foundation for conservation activities and further planning, identifying priority actions to manage the risks imposed by key threats to the species.

The largest subpopulation of the species occurs within the protected area of Bulburin NP, and this habitat is protected under the Nature Conservation Regulation 2024 and Nature Conservation Regulation 1994 of the NC Act through which all activities must be consistent with the management principles and management plans for the area. Penalties, ranging from fines to imprisonment, are enforced by the NC Act for offences involving wildlife and protected areas. Management of the protected areas in which this species is located is guided by a Management Statement developed by the Queensland Government. Although Ringed Thin-tail Geckos have not been specifically identified for protective measures, rainforest habitats are recognised as a key value and have associated strategic management directions to guide park activities. The two smaller, isolated subpopulations occur in surrounding forestry reserves and in the Many Peaks range. Part of the species' geographic range occurs in State Forests that are subject to logging and replanting of Hoop Pine. As the geographic range of the species spreads across National Park, State Forest and adjacent land, management actions require the maintenance of cooperative relationships between Traditional Custodians, landowners and conservation groups.

Captive breeding

The proposal states that there are no commercial captive-breeding operations for the species and captive breeding of the species is not permitted under Queensland state legislation. It states that there are no captive

populations held within Australia and that no permits have been granted for the species to be taken from the wild for the purpose of export. *The IUCN Red List assessment states that, based on anecdotal evidence, it is likely that the species is bred in captivity (Couper et al., 2018), and one online platform in Canada offering the species for sale in 2025 claimed the individuals for sale were currently 'incubating' but this claim could not be verified.*

An expert also found a social media account of a person who appeared to be breeding the species for personal use (Janssen in litt., 2025). This conflicts with statements from another expert who stated that based on their biology, the species was likely have a low captive breeding potential given that they lay approximately two eggs at a time and breed only once or twice a year after reaching sexual maturity (thought to be at around three years of age) (Chekunov in litt., 2025). There are no current records for captive breeding of this species in the Species360 Zoological Information Management System (ZIMS) database for zoos.

Implementation challenges (including similar species)

The Ringed Thin-tail Gecko is distinguishable from other *Phyllurus* species by its long, cylindrical tail with 5–6 pale bands, which are absent in regenerated tails. It also has a depressed head, conical tubercles, and unique scale patterns on its hindlimbs. Despite morphological similarities with other species, identification by Australian border officials is unlikely to be problematic, as commercial export of live native reptiles is prohibited, and non-commercial trade requires an export permit under national environmental legislation.

References

- Chekunov, S., Stringham, O., Toomes, A., Prowse, T. and Cassey, P. (2024) Scale of unregulated international trade in Australian reptiles and amphibians. *Conservation Biology* 38:e14355. <https://doi.org/10.1111/cobi.14355>
- Chekunov, S. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Chng, S. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Couper, P., Hoskin, C. and Amey, A. (2018) *Phyllurus caudiannulatus*. *The IUCN Red List of Threatened Species* 2018: e.T102664500A102664504. <https://dx.doi.org/10.2305/IUCN.UK.2018-1.RLTS.T102664500A102664504.en>. Accessed on 21 July 2025.
- Janssen, J. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Meiri, S. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.

Transfer of the genus *Amblyrhynchus* spp. represented by the Galápagos Marine Iguana *Amblyrhynchus cristatus* from Appendix II to Appendix I

Proponent: Ecuador

Summary: The Marine Iguana *Amblyrhynchus cristatus*, the sole species in the genus *Amblyrhynchus*, is endemic to the Galápagos Islands, Ecuador. It is known to occur throughout the archipelago inhabiting a range of large to satellite islands. Its area of occupancy has been estimated at 4,368 km² with area of habitat of around 3,250 km².

The Marine Iguana was assessed on the IUCN Red List in 2019 as Vulnerable on the basis that its population had decreased by at least 30% over the last three generations (18-24 years). The species' global population was estimated to be between 19,800-210,000 mature individuals in 2019. Its population undergoes significant fluctuations, with mortality rates ranging from 10% to 90% during El Niño climate events. While climate events are the primary threat to the species, it is also impacted by alien invasive species (e.g., cats, rats, dogs) affecting around 30% of the total population, as well as anthropogenic disturbance, pollution, habitat degradation and disease.

The species *Amblyrhynchus cristatus* has been listed in CITES Appendix II since 1975; Ecuador became a Party to the Convention in the same year. No live wild-collected individuals of the species have been reported as legally exported from Ecuador since the listing, and the export of this species from Ecuador has been prohibited since 1998. Very small quantities of live *A. cristatus* are recorded in the CITES Trade Database, all captive-bred. Between 2012 and 2019, exporters reported eight individuals: four from Uganda to Japan (for captive breeding), two from Switzerland to Uganda (commercial), and two from Austria to Japan (commercial; origin Spain). In contrast, importers reported 11 captive-bred individuals over the same period.

As no legal exports of this species have been authorised by Ecuador, it is questionable whether the "captive-bred" individuals in trade were legally acquired and that trade in *ex situ* captive-bred individuals was carried out in compliance with CITES provisions. There is some evidence of illegal trade in juveniles being smuggled off the islands by sea, with seizures of live individuals of this species being made in 2010, 2014 and 2015, indicating some level of demand, likely from specialist breeders and collectors. However, the impact of this trade on the wild populations of these species is unknown.

Analysis: The Marine Iguana *Amblyrhynchus cristatus* is endemic to the Galápagos Islands, Ecuador. The species is classified as Vulnerable on the IUCN Red List with an estimated population reduction of at least 30% over the past three generations. The species has a somewhat restricted range with an area of habitat around 3,250 km² and is subject to extreme population fluctuations due to climatic events. These fluctuations, combined with ongoing threats from invasive species, habitat degradation, pollution, and disease make the population intrinsically vulnerable. There is some evidence of illegal international trade, likely in wild-caught individuals, although the quantities appear to be limited and impact on the wild population is unlikely to be detrimental. The species therefore **does not appear to meet the criteria for inclusion in Appendix I**. If the Parties decide against the listing, Ecuador could submit a zero quota for wild-sourced commercial exports of the Marine Iguana to be posted on the CITES website to indicate that trade in wild harvested specimens is not permitted.

Other considerations: The proposal is to transfer the genus *Amblyrhynchus* spp., represented by the Galápagos marine iguana, *Amblyrhynchus cristatus*, from Appendix II to Appendix I. The current listing is at the species level. Were this proposal to be accepted the effect of transferring the species or the genus would be the same.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

Amblyrhynchus cristatus has been listed in Appendix II since 1975. Although *Amblyrhynchus* is a monospecific genus, the current listing is at the species level.

Taxonomy

Amblyrhynchus cristatus (Bell, 1825).

The CITES Standard reference is Hollingsworth, B.D. 2004. The evolution of iguanas. Iguanas - biology and conservation. Alberts, A. C. et al., (eds.) University of California Press. London.

Current taxonomy recognises one species (*A. cristatus*) and eleven subspecies (*A. c. cristatus*, *A. c. godzilla*, *A. c. hassi*, *A. c. hayampi*, *A. c. jeffreysi*, *A. c. mertensi*, *A. c. nanus*, *A. c. sielmanni*, *A. c. trillmichi*, *A. c. venustissimus* and *A. c. wikelskii*) (Buckley et al., 2022).

Range

Ecuador

IUCN Global Category

Vulnerable A2abce+4abce (assessed 2019, ver. 2025-1)

Biological criteria for inclusion in Appendix I

A) Small wild population

The 2019 Red List assessment estimated the global population at between 19,800 and 210,000 (Macleod et al. 2020). However, Macleod in litt. (2025) noted the current population is likely to be on the lower side of this estimate, with fewer individuals on Isabella Island than previously estimated.

The species' population has been observed to fluctuate markedly due to the El Niño-Southern Oscillation and La Niña climatic events. Population estimates following these events range from 33,000 individuals after a severe El Niño period, to 350,000 after a prolonged La Niña episode. However, in reference to these population figures, in the SS this information "has been acknowledged by researchers to be inaccurate". During the El Niño that took place in 1997 to 1998, approximately 90% of the Marine Iguanas on North Seymour Island disappeared, reducing the population to less than 150 individuals. A similar situation occurred on Genovesa Island in 1991-1994, which reduced the total population of about 15,000 to approximately 900 individuals.

B) Restricted area of distribution

The Marine Iguana inhabits a range of Galápagos islands, including Española, Fernandina, Floreana, Genovesa, Isabela, Marchena, Pinta, San Cristóbal, Santa Cruz, Santa Fé, Santiago, Baltra, Bartolomé, Pinzón, Plaza Norte, Plaza Sur, Rábida, Seymour Norte, Darwin, Roca Redonda and Wolf, as well as many satellite islets. The distribution of each subspecies is restricted to a single island, except on San Cristóbal which has two subspecies.

Amblyrhynchus cristatus has an estimate area of occupancy of 4,368 km² (Macleod et al., 2020) and an area of habitat of 3,242 km² (Ridley et al., in prep.). Restricted to the Galápagos Islands, the species' distribution is limited to suitable coastal habitats, resulting in a geographically constrained range.

C) Decline in number of wild individuals

Marine Iguanas experience extreme fluctuations in population size caused by El Niño and La Niña climatic events. The El Niño event is associated with mass mortality of Marine Iguanas, with average declines between 30-50%, and extreme declines of up to 90% reported. Climate change predictions indicate El Niño events may increase in frequency, which may cause the extirpation of some subpopulations. While population declines of 70-90% occur, these appear to be part of natural fluctuation cycles, and the 2019 IUCN Red List assessment estimates an overall population reduction of at least 30% over the last three generations.

Furthermore, approximately 20–30% of the total population, equating to between 10,000 and 73,000 individuals, were estimated to be declining due to the continual impacts of invasive alien predators, such as cats (*Felis catus*), dogs (*Canis familiaris*), rats (*Rattus rattus*, *R. norvegicus*) and pigs (*Sus domesticus*). Sharp declines in Marine Iguana subpopulations relating to these threats have been reported since the early 1980s and a recent study conducted in 2021 found an estimated population size reduction of over half at two colonies, at La Lobería and Playa Blanca on San Cristóbal island.

Recent observations suggest that many island populations and subspecies of the Marine Iguana may be smaller than previously estimated, potentially due to recent declines or the imprecise nature of earlier census methods (MacLeod in litt., 2025). Notably, population numbers on Isabela Island were unexpectedly low, indicating that this location may no longer serve as a stronghold for the species (MacLeod in litt., 2025).

Trade criteria for inclusion in Appendix I

Reports from the CITES Trade Database show a total of 15 live individuals exported to Ecuador from the USA in 1989 having originated in Ecuador. The source and purpose of this trade was not reported. *It is not clear whether this was a repatriation of live individuals, and no corresponding import was reported by Ecuador.*

Between 2013 and 2019, a total of eight live captive-bred individuals were reported as exported for commercial (T, 4) and captive breeding (B, 4) purposes. These exports originated from Uganda (4), Switzerland (2), and Austria (2), with Japan importing six individuals and Uganda two.

Auliya et al. (2025) reported that the private zoo, CTC Conservation Center in Uganda, is the only facility in the world that has Marine Iguanas. Uganda has only reported the import of two live captive-bred specimens from Switzerland, and Auliya et al. (2025) and Noboa in litt. (2025) raised concerns about the legality and legitimacy of the acquisition of these specimens (and other breeding stock) unverified outside its native range.

Under CITES Res. Conf. 10.16 (Rev. CoP19) on Specimens of animal species bred in captivity, for a specimen to qualify as "captive-bred" (source code C), the competent government authorities must be satisfied that the breeding stock was legally acquired in accordance with both CITES and national legislation, and that acquisition was not detrimental to wild populations. There must also be evidence of ongoing breeding success and limit additions of wild specimens. Due to the lack of evidence that the "captive-bred" individuals in trade were legally acquired, and that trade was carried out in compliance with CITES requirements, the definition of these Marine Iguanas as "captive-bred" is questionable.

Illegal trade

Auliya et al. (2025) report two smuggling events of Marine Iguanas. One event occurring in 2010 involving two individuals of the species reported at Galápagos airport. In 2015, nine juvenile Marine Iguanas were seized in Puerto Ayora on Santa Cruz Island. A Mexican national was arrested and subsequently convicted of attempting to smuggle these individuals to Uganda.

There is one EU-TWIX seizure record of one Amblyrhynchus cristatus derivative of unknown source, seized in the Netherlands in 2014 and reported to have originated in the Dutch Antilles.

The SS reports that individual iguanas can fetch prices as high as USD25,000, indicating strong demand and/ or limited availability. Juvenile specimens are reportedly preferred in trade due to their ease of concealment. *Although there is currently no clear evidence of a significant increase in demand for Marine Iguanas in legal or illegal trade, this apparent stability should be interpreted with caution (Noboa in litt., 2025). Seizure incidents remain rare, but experts suggest this may underrepresent the true scale of trade pressure (Noboa in litt., 2025). Enforcement efforts, particularly maritime inspections, are reportedly insufficient, allowing traffickers to exploit gaps in surveillance and avoid detection (Noboa in litt., 2025).*

Marine Iguanas are primarily smuggled via maritime routes, making it difficult for authorities to intercept illegal shipments or assess the full extent of the trade (Noboa in litt., 2025). Trafficking networks in the Galápagos reportedly involve individuals with detailed local knowledge and sophisticated concealment methods, further complicating enforcement and obscuring the true scale of demand (Noboa in litt., 2025).

Additional Information

Threats

Marine Iguanas face multiple threats, primarily from climate driven El Niño events, which cause significant population declines of up to 90% due to starvation. Climate change is expected to increase the frequency and severity of El Niño events, potentially leading to the local extinction of some subpopulations. Alien invasive predators pose a major risk to hatchlings and juveniles, with survival rates as low as 1% in predator-affected colonies. Additional threats to the species include human-induced impacts such as tourism and pollution, particularly oil spills, which affect Marine Iguanas both directly and indirectly through habitat loss and degradation due to urban development on inhabited islands such as Santa Cruz.

Conservation, management and legislation

Ecuadorian legislation strictly prohibits the export of endemic species from the Galápagos Islands to both the mainland and abroad. *This was enacted in 1998 with the Special Law for the Conservation and Sustainable Use of the Galápagos Province (Law No. 278), which explicitly banned the transport of native flora and fauna from the Galápagos islands to the mainland or abroad and introduced strict controls on tourism, fishing, and the introduction*

of non-native species. This law was strengthened by the Organic Law of the Special Regime of the Galápagos Province in 2015, which explicitly bans the transport of native flora and fauna by any means outside the islands. Additionally, the Regulations to the Organic Code of the Environment prohibit trade in native, endemic, endangered, or migratory species of wild fauna taken directly from their natural habitat. The Organic Integral Penal Code further reinforces these protections by criminalizing the hunting, possession, transport, and trade of protected wild species, with penalties of up to three years' imprisonment, especially when offences involve endemic or endangered species or occur within protected areas.

Ecuador submitted an information document to the 74th CITES Standing Committee ([CITES 2022 SC74 Inf. 26](#)) flagging the illegal trade in both marine and land iguanas from the Galápagos islands and providing CITES Parties with recommendations relating to the trade in these species.

Several invasive species eradication programmes have been implemented across the Galápagos Islands, targeting feral goats, cats, and black rats on various islands. These efforts aim to support Marine Iguana subpopulations, though managing invasive species remains a significant challenge. While some campaigns have been successful, the full impact of certain species, such as rats, is not fully understood. Notably, a near-total eradication of black rats on Pinzón Island in 1988 was followed by a marked increase in offspring survival.

Artificial Propagation/captive breeding

There is no known record of successful long-term captive breeding of the Marine Iguana (MacLeod in litt., 2025). Historical attempts include brief periods of captivity at Bellevue Zoo (Manchester, UK) and Brookfield Zoo near Chicago (USA), but these did not result in sustained breeding success (MacLeod in litt., 2025). Although Marine Iguana hatchlings are reportedly being traded internationally as captive-bred, there is uncertainty about the legitimacy of these claims, raising concerns that individuals may be illegally sourced from the wild. While it is possible that individuals have been kept or bred illegally, such efforts appear to be largely unsuccessful and remain undocumented, suggesting that the species is poorly suited to captivity (MacLeod in litt., 2025).

There are no current records for this species in the Species360 ZIMS database for zoos.

Implementation challenges (including similar species)

The only similar species are the three Galápagos land iguana species (*Conolophus marthae*, *C. pallidus* and *C. subcristatus*), however, trade in these species from the Galápagos is also prohibited and a proposal to transfer these to Appendix I will also be considered by the CoP (Prop. 23).

Potential benefit(s) of listing for trade regulation

Were the proposal to be accepted, operations breeding *Amblyrhynchus cristatus* for commercial purposes would need to be registered with the Secretariat in accordance with Res. Conf. 12.10 (Rev. CoP15) on Registration of operations that breed Appendix-I animal species in captivity for commercial purposes. *However, given no legal trade has been permitted by Ecuador, it is unlikely that any operation would be able to prove legal acquisition of founding stock.*

Other Considerations:

The proponent states that listing the species in Appendix I is essential to prevent wild collection and to regulate trade in specimens claimed to be captive-bred. If *Amblyrhynchus cristatus* is included in Appendix I, operations breeding the species for commercial purposes would need to be registered with the Secretariat in accordance with Res. Conf. 12.10 (Rev. CoP15) on *Registration of operations that breed Appendix-I animal species in captivity for commercial purposes*. However, Res. Conf. 17.7 (Rev. CoP19) on *Review of trade in animal specimens reported as produced in captivity* provides another mechanism to address concerns related to such trade.

Document CoP20 Doc. 73 on *Trade in threatened endemic species*, submitted by Brazil and Ecuador, raises issues relevant to this proposal. Parties are encouraged to consider the concerns expressed by the proponent regarding the legality of specimens of this species reported in trade as captive-bred, as well as the request from Ecuador outlined in Notification 2025/063.

References

- Auliya, M., Nijman, V., Altherr, S., Aguilera, W.T., Ariano-Sánchez, D., Cantu, J.C., Colosimo, G., Gentile, G., Gerber, G., Grant, T. and Henningheim, E. (2025) Trafficking of Galápagos iguanas as an example of a global problem: CITES permits, laundering and the role of transit countries in Europe and Africa. *Biological Conservation* 305:111104.
- Buckley, L.J., de Queiroz, K., Grant, T.D., Hollingsworth, B.D., Malone, C.L., Pasachnik, S.A., Reynolds, R.G. and Zarza, E. (2022) A checklist of the iguanas of the world (Iguanidae; Iguaninae). 2022 Supplement to: 2016 Herpetological Conservation and Biology 11(Monograph 6):4–46. and 2019 Supplement Iguana Taxonomy Working Group (ITWG).

MacLeod, A., Nelson, K.N. and Grant, T.D. (2020) *Amblyrhynchus cristatus* (errata version published in 2020). *The IUCN Red List of Threatened Species* 2020: e.T1086A177552193.
<https://dx.doi.org/10.2305/IUCN.UK.2020-2.RLTS.T1086A177552193.en>.
MacLeod, A. (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
Noboa, K. (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
Species360 Zoological Information Management System (ZIMS) (2025) zims.Species360.org.

Transfer of Galápagos land iguanas *Conolophus* spp. from Appendix II to Appendix I

Proponent: Ecuador

Summary: The Galápagos land iguanas *Conolophus* spp. endemic to the Galápagos Islands, Ecuador, comprise three species: *Conolophus marthae*, *C. pallidus*, and *C. subcristatus*. The Galápagos Pink Land Iguana *C. marthae* is endemic to Isabela Island, Barrington Land Iguana *C. pallidus* endemic to Santa Fe (Barrington) Island, and the Galápagos Land Iguana *C. subcristatus* occurs on several islands. All are currently included in Appendix II and all three are proposed for transfer to Appendix I.

- *Conolophus marthae* was assessed as Critically Endangered in 2012, based on a very small population estimate of 192 mature individuals confined to a single subpopulation occupying an area of 25 km². The population trend was unknown but subsequently reported to be stable.
- *Conolophus pallidus* was assessed as Vulnerable in 2019, with 3,500–4,000 mature individuals in a single subpopulation and an area of occupancy of 40 km². Its population trend was unknown.
- *Conolophus subcristatus* was assessed as Vulnerable in 2020 based on a best estimate of 10,216 mature individuals in 13 subpopulations and an area of occupancy of 540 km².

Galápagos land iguanas face an array of threats that vary in intensity across the archipelago. One of the most significant natural threats is the El Niño climate phenomenon, which can lead to prolonged droughts and severe food shortages, resulting in high mortality rates. In addition, the species are highly vulnerable to invasive alien species, including feral cats, dogs, pigs, goats, and rats, which impact iguana populations through predation of eggs, juveniles, and adults, and indirectly through habitat degradation. The iguanas' intrinsic biological traits, such as slow reproductive rates, long generation length, and small, isolated populations, further exacerbate their vulnerability. Low genetic diversity and stochastic events such as volcanic eruptions also threaten these species.

Limited trade has been reported in recent years mostly declared as captive-bred, however the legitimacy of this has been questioned as since the species was listed in Appendix II in 1975, no legal exports of live specimens have reportedly been authorized by Ecuador. Some seizures of live individuals demonstrate there is international demand for these species. However, the impact of this trade on the wild populations of these species is unknown.

Analysis: The **Galápagos Pink Land Iguana (*C. marthae*)** has a very small population and the population of **Barrington Land Iguana (*C. pallidus*)** is small. Both of these species have restricted areas of distribution, occur in one location and have a high vulnerability to ongoing extrinsic factors such as climate events resulting in large fluctuations in the number of individuals and quality of habitat. As such, **both appear to meet the biological criteria for inclusion in Appendix I.**

The **Galápagos Land Iguana (*C. subcristatus*)** has a larger population, although estimates vary considerably. It has a somewhat restricted range and is projected to incur ongoing subpopulation decreases due to extrinsic factors including alien invasive species and climate events causing large subpopulation and habitat quality fluctuations. As such, this species **may meet the biological criteria in Annex 1** of Res. Conf. 9.24 (Rev. CoP17).

No legal exports have been permitted by Ecuador; trade in live specimens of uncertain origin has been recorded and there is some evidence of illegal harvesting. It is difficult to ascertain the impact that collection of these species for trade has on their wild populations, though it would likely be more detrimental to *C. marthae* and *C. pallidus* than *C. subcristatus*. Given the uncertainties about trade impact and high level of conservation risk, Appendix I-listing could be justifiable for all three species. If the Parties consider that certain species meet the criteria for inclusion in Appendix I while others do not, Ecuador could submit a zero-export quota for wild-sourced commercial exports of those species to be posted on the CITES website to indicate that trade in wild harvested specimens is not permitted.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

The three species have been listed in Appendix II since 1975.

Taxonomy

Conolophus marthae (Gentile & Snell, 2009), *Conolophus pallidus* (Heller, 1903), *Conolophus subcristatus* (Gray, 1831).

The CITES Standard reference for C. marthae is Gentile, G. and Snell, H. 2009. Conolophus marthae sp. nov. (Squamata, Iguanidae), a new species of land iguana from the Galápagos archipelago. Zootaxa: 2201: 1-10.

The CITES Standard reference for C. pallidus and C. subcristatus is Hollingsworth, B.D. 2004. The evolution of iguanas. Iguanas - biology and conservation. Alberts, A. C. et al., (eds.) University of California Press. London.

Range

Ecuador

IUCN Global Category

Conolophus marthae Galápagos Pink Land Iguana: Critically Endangered B1ab(iii,v)+2ab(iii,v); C2a(ii) (assessed 2012, ver. 2025-1)

Conolophus pallidus Barrington Land Iguana: Vulnerable A1ce; D2 (assessed 2019, ver. 2025-1)

Conolophus subcristatus Galápagos Land Iguana: Vulnerable A2abce; B1ab(i,ii,iii,iv,v)+2ab(i,ii,iii,iv,v);C1 (assessed 2020, ver. 2025-1)

Biological criteria for inclusion in Appendix I

A) Small wild population

Galápagos Pink Land Iguana (*C. marthae*) was estimated at 192 mature individuals in 2012, and between 150-270 individuals between 2009 and 2019 (Garizio et al., 2024).

Barrington Land Iguana (*C. pallidus*) was estimated at 3,500-4,000 mature individuals in 2005, with a total population of between 4,642-8,651 individuals. The SS also reports a total population of 5,016 individuals (ranging from 4,500 to 5,800) and 3,500-4,000 mature individuals.

Galápagos Land Iguanas (*C. subcristatus*) estimated at 11,435 individuals (an estimate made based on population estimate from four locations from 2002–2008). Kumar et al. (2020) gives a best estimate of 10,216 mature individuals (ranging from 8,618 to 18,117) in the 2020 IUCN Red List Assessment. A 2023 study of *Conolophus subcristatus* on Fernandina Island estimated the adult population at 45,600 individuals, based on 12 individuals per ha occurring throughout 3,800 ha of suitable habitat on the island (Ortiz-Catedral et al., 2023). However, concerns have been raised that this estimate may substantially overestimate the population size, as it assumes uniform density across the island based on data collected during a mating congregation period, which may not reflect typical density distribution patterns (Gentile in litt., 2025).

B) Restricted area of distribution

Galápagos Pink Land Iguana (*C. marthae*) is endemic to the Wolf Volcano on Isabela Island, occurring between 600-1700 m above sea level. The species' extent of occurrence was estimated at 25 km² and a recent area of occupancy at 42 km² (Gentile in litt., 2025).

The Barrington Land Iguana (*C. pallidus*) is endemic to Santa Fe (Barrington) Island, occurring between 0-259 m above sea level. The species' extent of occurrence was estimated at 27 km² and its area of occupancy at 40 km².

The Galápagos Land Iguana (*C. subcristatus*) is found on several islands in the Galápagos Archipelago (resident: Isabela, Fernandina, South Plaza, Santa Cruz and some smaller islets; reintroduced: Baltra, North Seymour and Santiago islands; and extirpated: Rabida) and occurs between 0-1707 m above sea level. The extent of occurrence for the species was estimated at 9,524 km², and the area of occupancy at 540 km² based on areas from Isabela (280 km²), Fernandina (184 km²) and the islands of Santa Cruz, Venecia, Plaza Sur, Seymour Norte, Baltra (76 km²).

C) Decline in number of wild individuals

Galápagos Pink Land Iguana (*C. marthae*) population trends in 2012 were noted to be unknown, with genetic evidence indicating a past reduction. *A recent study found no evidence of any significant change in the population size of this species during the sampling period from 2006 to 2021* (Garizio et al., 2024). However, independent data suggest competition with the cooccurring subpopulation of *C. subcristatus* that may have led to a Galápagos Pink Land Iguana population demographic contraction in the past (Gargano et al., 2022; Paradiso et al., 2025).

Barrington Land Iguana (*C. pallidus*) population trend was unknown but noted to be potentially stable in 2019 (Gentile and Grant, 2020). However, prior to 1972, *C. pallidus* was reported to have experienced a population decrease of 50%, evidenced by extremely low genetic diversity.

Galápagos Land Iguana (*C. subcristatus*) population trend was reported to be decreasing in 2020, with an estimated reduction of at least 30% over the past three generations (equivalent to 69 years). The decrease was due to habitat degradation and the impact of invasive species. The species has been nearly extirpated from much of southern Isabela and Santa Cruz, and in northern Isabela, numbers have diminished primarily due to the destructive presence of feral goats and predatory cats. On Fernandina and South Plaza, the species subpopulations were reported to be relatively stable and on Baltra and North Seymour population increases have been observed and attributed to targeted conservation interventions. Future projections predict a 10-15% decline in the species' population over three generations (2020-2089) based on the impact of invasive alien species on subpopulations.

Trade criteria for inclusion in Appendix I

Some caution is merited with interpretation of records in the CITES Trade Database for these species, as some transactions for leather goods appear likely to have been misreported. Some older records for imports of live animals for which no exports were reported may also be inaccurate.

A total of 442 live individuals were reported in trade by importing countries during the period 1974-2022, though more than half of this total (280 animals with no source code noted) was reported in one year as imported to the USA from Mexico, which may have been an erroneous report. Similarly, USA reported imports of 98 wild-collected live specimens from Indonesia and the Solomon Islands in 1995 and 1998 are questionable.

Otherwise, most reported trade in Conolophus species has been small numbers of live specimens each year, predominantly reported as captive-bred. Since 2017, most exports have been reported from Uganda, which itself reported exports of 62 live individuals declared as captive-bred in the years up to 2023. Most of these transactions were reportedly for the purpose of breeding in captivity, with small numbers also reported for zoological and commercial purposes.

Data from Zootierliste, an online database of vertebrates in zoos, indicates that live Galápagos Land Iguana (Conolophus subcristatus) are currently held at the private zoo, CTC Conservation Center in Uganda, as well as in zoos in Japan and South Korea. According to Auliya et al. (2025), the founder of the CTC Conservation Centre was convicted of reptile trafficking in New Zealand in 2010. There are concerns about the legality and legitimacy of the acquisition of these specimens and highlights ongoing international possession of the species outside its native range (Noboa in litt., 2025).

Under CITES Res. Conf. 10.16 (Rev. CoP19) on Specimens of animal species bred in captivity, to qualify for certification as "captive-bred" (source code C), the competent government authorities must be satisfied that the breeding stock was legally acquired in accordance with both CITES and national legislation, and that acquisition was not detrimental to wild populations. There must also be evidence of ongoing breeding success and limited additions of wild specimens. The lack of evidence that the "captive-bred" individuals in trade were legally acquired, and trade was carried out in compliance with CITES requirements, the classification of these Galápagos land iguanas as "captive bred" is questionable.

Illegal trade

Auliya et al. (2025) report four documented smuggling events of the three Conolophus iguanas. These include: four juvenile Galápagos land iguanas smuggled in 2012; two juveniles in 2015; and in 2022, iguanas were offered for sale on social media, and five Galápagos land iguanas were seized. Three of these events resulted in convictions.

The seizure events reported by Auliya et al. (2025) align with records held in TRAFFIC's WiTIS database. No records for these species were reported in the EU-TWIX seizure database.

Currently, there is no clear or definitive evidence indicating a significant increase in demand for Galápagos land iguanas (Noboa in litt., 2025). However, this apparent stability in reported trade volumes and seizures must be interpreted with caution (Noboa in litt., 2025). Although seizure data for Galápagos land iguanas remains low, this potentially underrepresents the true scale of demand and trade pressure (Noboa in litt., 2025). There is a perception

that enforcement efforts, particularly maritime inspections, may be insufficient due to limited logistical capacity, potentially allowing traffickers to exploit gaps in surveillance and avoid detection (Noboa in litt., 2025).

Maritime transport is possibly the primary method used to smuggle Galápagos land iguanas, making it difficult for authorities to detect and intercept illegal shipments or accurately assess the scale of the illegal trade (Noboa in litt., 2025). Noboa in litt. (2025) reported that trafficking networks in the Galápagos islands, facilitated by corruption, involve local individuals who possess detailed knowledge of the islands and discreet methods to conceal specimens during transit. This significantly hampers law enforcement's ability to detect and prevent illegal trade, further obscuring the true scale of demand (Noboa in litt., 2025).

Additional Information

Threats

The Galápagos land iguanas (*Conolophus* spp.) collectively face a range of persistent and ongoing threats, primarily driven by invasive species, human activity, and natural hazards. The most significant threat to the Galápagos land iguanas has been the introduction of invasive alien species, specifically dogs (*Canis familiaris*), cats (*Felis catus*), pigs (*Sus domesticus*), goats (*Capra hircus*), rats (*Rattus rattus*, *R. norvegicus*) and donkeys (*Equus asinus*). Dogs have been especially destructive, with feral dogs responsible for near-extirpation events in Santa Cruz and Isabela during the 1970s and domestic dogs causing major losses (>130 iguanas) in San Cristobal in the early 2000s. Goats and donkeys degrade iguana habitats by consuming vegetation, trampling nests, and competing for resources. Pigs pose a dual threat by both damaging habitat and actively digging up and eating iguana eggs. These combined pressures have led to local extinctions on islands such as Baltra, Rábida, and Santiago. Feral cats remain a widespread and ongoing threat, especially in Isabela and Santa Cruz, where they prey heavily on juveniles, severely limiting recruitment and skewing population structures. In some subpopulations, juveniles make up less than 1% of individuals, indicating intense predation. However, a recent study on *C. marthae* reported that the impact of feral cat predation on hatchlings was not significant enough to cause a demographic decline, although the observed recruitment was not sufficient to augment the population size (Garizio et al., 2024).

Stochastic risks from climatic events, such as El Niño, and volcanic activity presents a major threat to the Galápagos land iguanas (*Conolophus* spp.). Volcanic events on Fernandina and Isabela impacting iguanas that nest in or near active craters. Past eruptions, such as Volcán Chico in 1979, were documented to have wiped out entire subpopulations.

Conservation, management and legislation

Ecuadorian legislation strictly prohibits the export of endemic species from the Galápagos Islands to both the mainland and abroad. This was enacted in 1998 with the Special Law for the Conservation and Sustainable Use of the Galápagos Province (Law No. 278), which explicitly banned the transport of native flora and fauna from the Galápagos islands to the mainland or abroad and introduced strict controls on tourism, fishing, and the introduction of non-native species. This law was strengthened by the Organic Law of the Special Regime of the Galápagos Province in 2015, which explicitly bans the transport of native flora and fauna by any means outside the islands. Additionally, the Regulations to the Organic Code of the Environment prohibit trade in native, endemic, endangered, or migratory species of wild fauna taken directly from their natural habitat. The Organic Integral Penal Code further reinforces these protections by criminalizing the hunting, possession, transport, and trade of protected wild species, with penalties of up to three years' imprisonment, especially when offences involve endemic or endangered species or occur within protected areas.

Ecuador submitted an information document to the 74th CITES Standing Committee (CITES 2022 SC74 Inf. 26) flagging the illegal trade in both marine and land iguanas from the Galápagos islands and providing CITES Parties with recommendations relating to the trade in these species.

The European Union Scientific Review Group (SRG) has maintained a Negative Opinion for trade in *Conolophus subcristatus* since 2020 (Auliya et al., 2025).

Eradication efforts have successfully removed many invasive mammals from several islands and *in situ* captive-breeding programmes established subpopulations on islands unaffected by invasive alien species.

Artificial Propagation/captive breeding

A captive breeding program was initiated by the Galápagos National Park and the Charles Darwin Research Station (CDRS) in the 1970s for *C. subcristatus*. Captive bred individuals were introduced to Venice islet in 1977, and hatchlings subsequently transferred to Santa Cruz Island and reintroduced to Isabela Island (Cayot 2008). The captive breeding programme was discontinued in 2008. According to the Species360 ZIMS database for zoos, only one *Conolophus subcristatus* individual occurs in an ex-situ holding institution in Ecuador. There are no current records for *C. marthae* or *C. pallidus*. A head start program is envisioned in the IUCN Conservation and Management Plan for *C. marthae* (Rueda et al., 2023).

As noted above, the lack of publicly available evidence of ex situ breeding of these species means that the classification of these Galápagos land iguanas as 'captive-bred' remains questionable.

Implementation challenges (including similar species)

The only similar species is the Marine Iguana (*A. cristatus*), however, trade in this species from the Galápagos is also prohibited.

Potential benefit(s) of listing for trade regulation

The proponent states that listing the species in Appendix I is essential to dissuade wild collection and to regulate trade in specimens claimed to be captive-bred. If these species are included in Appendix I, operations breeding them for commercial purposes would need to be registered with the Secretariat in accordance with Res. Conf. 12.10 (Rev. CoP15) on *Registration of operations that breed Appendix-I animal species in captivity for commercial purposes*. However, given no legal trade has been permitted by Ecuador, it is unlikely that any operation would be able to prove legal acquisition of founding stock.

Other Considerations:

The proponent states that listing the species in Appendix I is essential to prevent wild collection and to regulate trade in specimens claimed to be captive-bred. If these species are included in Appendix I, operations breeding them for commercial purposes would need to be registered with the Secretariat in accordance with Res. Conf. 12.10 (Rev. CoP15) on *Registration of operations that breed Appendix-I animal species in captivity for commercial purposes*. However, Res. Conf. 17.7 (Rev. CoP19) on *Review of trade in animal specimens reported as produced in captivity* provides another mechanism to address concerns related to such trade.

Document CoP20 Doc. 73 on *Trade in threatened endemic species*, submitted by Brazil and Ecuador, raises issues relevant to this proposal. Parties are encouraged to consider the concerns expressed by the proponent regarding the legality of *Conolophus* spp. specimens reported in trade as captive-bred, as well as the request from Ecuador outlined in Notification 2025/063.

References

- Auliya, M., Nijman, V., Altherr, S., Aguilera, W.T., Ariano-Sánchez, D., Cantu, J.C., Colosimo, G., Gentile, G., Gerber, G., Grant, T. and Henningheim, E. (2025) Trafficking of Galápagos iguanas as an example of a global problem: CITES permits, laundering and the role of transit countries in Europe and Africa. *Biological Conservation* 305:111104.
- Gargano, M., Colosimo, G., Gratton, P., Marta, S., Brilli, M., Giustini, F., Sevilla, C. and Gentile, G. (2022) Nitrogen and carbon stable isotope analysis sheds light on trophic competition between two syntopic land iguana species from Galápagos. *Scientific Reports* 12(1):16897.
- Garizio, L., Gargano, M., Colosimo, G., Gratton, P., Gerber, G., Lewbart, G., Sevilla, C. and Gentile, G. (2024) First evidence of recruitment in critically endangered Galápagos pink land iguanas (*Conolophus marthae*). *Conservation Science and Practice* 6. 10.1111/csp2.13108.
- Gentile, G. and Grant, T.D. (2020) *Conolophus pallidus*. The IUCN Red List of Threatened Species 2020: e.T5239A3014028. <https://dx.doi.org/10.2305/IUCN.UK.2020-2.RLTS.T5239A3014028.en>. Accessed on 23 July 2025
- Noboa, K. (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Paradiso, C., Gratton, P., Trucchi, E., López-Delgado, J., Gargano, M., Garizio, L., Carr, I.M., Colosimo, G., Sevilla, C., Welch, M.E. and Firdaus-Raih, M. (2025) Genomic insights into the biogeography and evolution of Galápagos iguanas. *Molecular Phylogenetics and Evolution* 204:108294.
- Rueda, D., Castaño, P. A., Campbell, K. J., Colosimo, G., Gerber, G. P., León, P., Tapia, W. and Gentile, G. (Eds.). (2023) Galápagos pink land iguana (*Conolophus marthae*): Conservation and management plan 2022–2027. Gland, Switzerland: IUCN. <https://doi.org/10.2305/TNOP5832>
- Species360 Zoological Information Management System (ZIMS) (2025) zims.Species360.org.

Inclusion of Bale Mountain Adder *Bitis harenni* and Ethiopian Mountains Adder *Bitis parviocula* in Appendix I

Proponent: Ethiopia

Summary: The Bale Mountain Adder *Bitis harenni* and Ethiopian Mountains Adder *Bitis parviocula* are large, morphologically similar viper species endemic to Ethiopia, where they occur in montane forests and grassland. *Bitis parviocula* was assessed in 2014 on the IUCN Red List as Endangered due to its limited range and ongoing decrease in extent and quality of habitat. At that time *Bitis harenni* was considered a subpopulation of *B. parviocula*. It was elevated to species level in 2016 and has not yet been assessed on the IUCN Red List.

Both species occur on either side of the Ethiopian Rift Valley, *B. harenni* having a restricted range and known from two localities, with *B. parviocula* having an estimated area of habitat of over 20,000 km². The current known ranges of both species likely represent an underestimate of their occurrence. There is no detailed population information for either species, both of which are rarely encountered due to their cryptic habits. Their natural habitats are reported to be decreasing in extent and quality due to pressures from deforestation, agricultural expansion including coffee cultivation, and human encroachment.

The only recent trade data available for these species are from the USA; between 2008 and 2020, 141 live captive-bred and wild-sourced *B. parviocula* were recorded as imported. The trade in wild-sourced individuals comprised 31 individuals exported directly from Ethiopia, and 14 originating in Ethiopia and re-exported by Switzerland, both in 2009. Captive-bred individuals were exported from the USA to Germany, the Netherlands, Slovenia, and Ecuador. Individuals appear to fetch a high price based on market observations in the USA and Europe. While the legitimacy of the captive-breeding *B. parviocula* stocks observed in USA and European markets is unclear, experts report that *B. parviocula* has been bred successfully on multiple occasions by both private and public institutions and individuals. No verified records of *B. harenni* in trade exist, and experts found no evidence of illegal trade involving this species. Although *B. parviocula* continues to appear sporadically in trade, including online platforms, the volume appears limited, and evidence of illegal trade is unclear.

The species are protected under national legislation with the collection or export of endemic animals including *B. parviocula* and *B. harenni*, prohibited since 2004, and prohibited without a permit since 1957. As such, the legitimacy of the wild-sourced *B. parviocula* individuals reported as exported from Ethiopia is unclear.

Analysis: *Bitis harenni* and *B. parviocula* are rarely encountered species, with *B. harenni* occupying a restricted range. Current population estimates and trends are unknown for either species. Despite protection under Ethiopian law since 2004 prohibiting trade in both species, *B. parviocula* from Ethiopia in wild-sourced individuals were last recorded in trade in 2009. *Bitis parviocula* has been observed in pet trade markets in Europe and the USA, although infrequently and only in low numbers, and experts report it has been successfully bred in captivity on multiple occasions. There is no evidence that *B. harenni* is affected by trade and it is unclear if the wild population of *B. parviocula* is affected by ongoing harvest and trade, or whether captive breeding is now meeting demand.

While *B. harenni* appears to meet the biological criteria for inclusion in Appendix I based on Criterion B in Annex 1 of Res. Conf. 9.24 (Rev CoP17), the species is not affected by trade and therefore **does not meet the trade criterion**.

As *B. parviocula* does not have a restricted range and only low levels of international trade in predominantly captive-bred individuals have been reported recently, it also **is unlikely to meet either criterion for inclusion in Appendix I**.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

Bitis parviocula (Böhme, 1976)

Bitis harena (Gower et al., 2016)

No standard reference is recommended by the proponent. As noted in CoP20 Inf. 6 an Addendum to document CoP20 Doc. 110 notes that if the Parties adopt Proposal 24, the Parties may wish to consider adopting a nomenclature standard reference for the two species; the following recommendation not only taxonomically and nomenclaturally defines both species, but also provides a functional key to identify and differentiate the species:

Gower, D.J., Wade, E.O.Z., Spawls, S., Böhme, W., Buechley, E.R., Sykes, D., & Colston, T.J. (2016): A new large species of *Bitis* Gray, 1842 (Serpentes: Viperidae) from the Bale Mountains of Ethiopia. *Zootaxa* 4093 (1): 041–063.

<http://doi.org/10.11646/zootaxa.4093.1.3>

Range

Ethiopia

IUCN Global Category

Bitis parviocula: Endangered B2ab(iii) (assessed 2014, ver 3.1) (assessment included *Bitis harena* as a subpopulation; it was further described in 2016)

Bitis harena: Not Evaluated

Biological criteria for inclusion in Appendix I

A) Small wild population

No population estimate was available for either *Bitis parviocula* or *B. harena*. While the proponent states both species were reported to be rare in the wild, Gower in litt. (2025) notes that they are “rarely encountered in the wild”. Colston in litt. (2025) notes that both species depending on the season are regularly encountered by local coffee farmers, and while they are considered rare species, this is likely due to their highly subterranean ecology, particularly *Bitis harena*.

B) Restricted area of distribution

Bitis parviocula and *B. harena* are both endemic to Ethiopia.

Bitis parviocula was reported to be restricted to locations on the southwestern and southern side of the Ethiopian Rift Valley. In 2014, *B. parviocula* (including *B. harena*) was estimated to have an Extent of Occurrence (EOO) of 70,000 km² and Area of Occupancy (AOO) of 20 km² and occur at elevations between 1,700–2,800 m above sea level. However, experts now report this AOO estimate significantly underestimated the species' range (Colston in litt., 2025). An Area of Habitat map, representing both species' range combined, was estimated at 20,413 km² (Ridley et al. in prep.). *Bitis parviocula* is currently known from five locations and *B. harena* from two (Gower et al., 2016), however, Colston in litt. (2025) reported that these localities span vast areas and the species is likely to occur in additional unsurveyed locations. Spawls in litt. (2025) also noted that *B. parviocula* currently has eight observations for the species in iNaturalist (iNaturalist community, 2025) and *B. harena* has been sighted three to four times along a 52 km stretch of road from Rira to Dolo Mena.

Bitis harena is only known from the Bale Mountains National Park (encompassing around 2,150 km²), and adjacent protected area (Colston in litt., 2025).

Bitis parviocula and *B. harena* inhabit dense broadleaf forests (Gower et al., 2016; Spawls et al., 2021; Spawls et al., 2023). *Bitis parviocula* has also been observed in disturbed areas such as coffee plantations and montane grassland, however, the suitability of this habitat for either species is unknown (Gower et al., 2016).

Bitis harena was reported to occur in two locations on the eastern side of the Rift Valley, from the town of Dodola and 70 km away in the Bale Mountains National Park at approximately 2,140 m above sea level. Colston in litt. (2025) noted that while *B. harena* was only known from the Harena Forest of the Bale Mountains, this area is poorly surveyed for herpetofauna, and the species may occur outside this area.

C) Decline in number of wild individuals

Bitis parviocula (including *B. harena*) was reported to have a decreasing population trend in 2014. However, Colston in litt. (2025) reported that no thorough surveys for either species had been carried out and that there was no evidence to suggest either species population is decreasing.

Colston in litt. (2025) reported that for *Bitis parviocula* collection of specimens in 2018, for a 2022 study (Gebrewold et al., 2022), indicated the species had a healthy population, with nine individuals collected in wooded savanna land forest and seven in forested bushy grassland.

Trade criteria for inclusion in Appendix I

The species is or may be affected by trade

The Proponent states that Ethiopia has never authorised exports of *Bitis parviocula* or *Bitis harensa* for any purpose.

A total of 20 individuals were reportedly imported by the USA from Ethiopia in 2007. Furthermore, Spawls stated in 2014 that “a single shipment of 30 individuals was exported from Ethiopia to the USA around 2005, and the species has since bred in captivity and been sold widely on the internet. It is thought that all animals currently in the international pet trade, including those in Europe, are descended from this original import”.

According to the LEMIS database, between 2008-2020, the USA imported 127 live individuals of *B. parviocula* for commercial purposes and re-exported 14 wild-sourced live individuals originating in Ethiopia to Switzerland. This trade comprised captive-bred (94, 74%) and wild-sourced individuals (33, 26%). Trade in wild-sourced individuals included a direct export of 31 individuals from Ethiopia and 14 individuals originating in Ethiopia re-exported by the USA to Switzerland, both in 2009. A direct export of two wild-sourced individuals from the non-range State Czechia to the USA in 2017 is likely to represent an error. Countries exporting captive bred *B. parviocula* individuals to the USA included Germany (37), Ecuador re-exporting from the Netherlands (30), the Netherlands (14) and Slovenia (13).

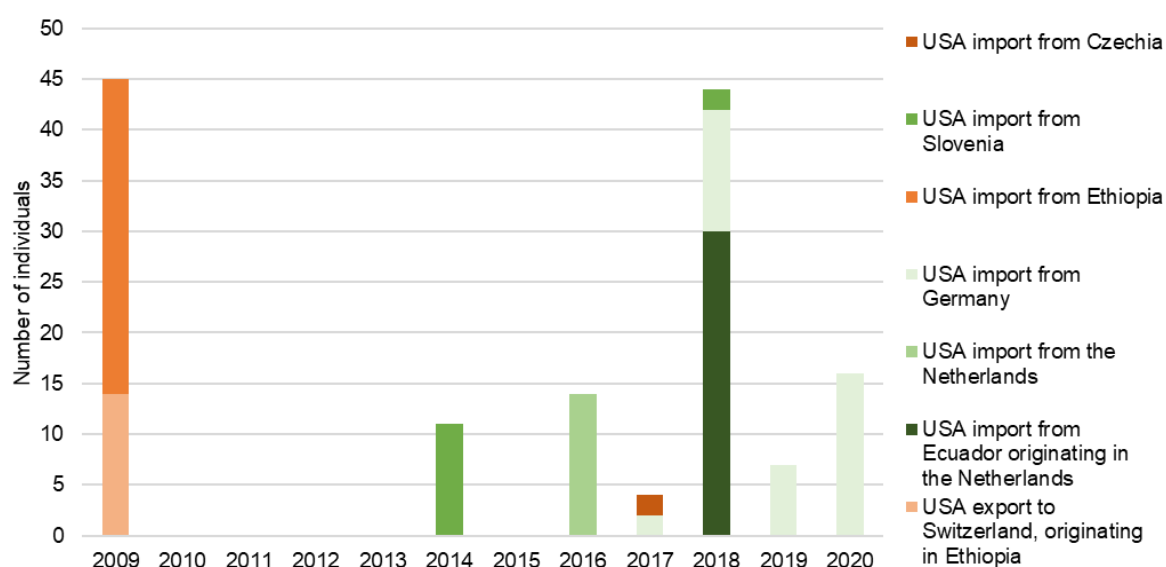


Figure 1. Imports to the USA and exports from the USA of live *B. parviocula* from wild-sourced (orange) and captive-bred (green) individuals for commercial purposes, 2008–2020, as reported by the USA. Source: LEMIS Database.

Additionally, LEMIS reports that 21 wild-sourced individuals of the *Bitis* spp. were imported to the USA from Ethiopia for commercial purposes in 2008, though the specific species is not specified (and the record could relate to other species such as *B. arietans* which is also native to Ethiopia).

According to the reported trade value of *B. parviocula* provided in the LEMIS database, the average price for an individual was USD1237 (ranging from USD10-2600). In the SS, it was stated that trade in *B. parviocula* in Europe and the USA reached USD4000, and in one case in Germany USD7500. Strugariu in litt. (2025) noted that captive-born *Bitis parviocula* individuals are rarely sold, but when they are, they reach prices of USD1500-3000.

A 2019 study exploring the international trade in African snakes not listed in the CITES Appendices, using data from one social media platform, Facebook and five online forums, found one 2016 record of “*Bitis parviocula*” (*B. parviocula*) listed for sale (Juul Jensen et al., 2019). Furthermore, a recent study by Heim (in prep.) which recorded *Bitis* species mentioned in online platforms, noted that while *B. parviocula* was mentioned, *B. harensa* was not. A recent search of online market platforms for the species showed an advert posted by a Swedish individual offering *Bitis parviocula* for sale, reported as “cb2025”, assumed to mean captive-bred individuals born in 2025 (Terraristik.com community, 2025). Further mentions of *B. parviocula* for sale, trade or desired were mentioned in 2014, 2017, 2021, 2023 and 2025 (FaunaClassifieds community, 2025).

Experts note that there is no evidence of illegal smuggling or trade of *B. harena*, nor of individuals of this species being offered for sale (Colston in litt., 2025). Furthermore, the individual exporter responsible for the large number of exports in the early 2000s reported to Colston in litt. (2025) that no shipments of *B. parviocula* had occurred in years.

No records for either *Bitis harena* or *B. parviocula* were reported in the EU-TWIX seizure database or TRAFFIC's WiTIS database.

Additional Information

Threats

These two *Bitis* species were reported to be threatened by habitat loss and degradation due to human activities, such as deforestation and agriculture. These species are also likely to be threatened by human persecution (Spawls et al., 2021). The region where both species occur is under great pressure from agriculture and increasing urbanization, exacerbated by high human population growth (Gower et al., 2016).

Bitis parviocula was reported to be predominantly threatened by decreasing habitat extent and quality.

Conservation, management and legislation

The exploitation of Ethiopia's natural resources, including wild animals, without a permit has been prohibited since 1957 (Penal Code of Ethiopia 1957). In 2004, this law was replaced by the Criminal Code of Ethiopia (2004) specifically prohibiting the possession, collection, transport, and export of endemic animals and their products, which includes *B. parviocula* and *B. harena*. Furthermore, Article 9 of Proclamation No. 541/2007, states that permits may be authorized for the collection of wildlife and their products for scientific or academic purposes. The proponent states "no such permits have ever been issued for either species".

Bitis parviocula occurs in the Sheka Forest Biosphere Reserve (Spawls et al., 2023). The entire range of *Bitis harena* occurs in a protected area in the Harena escarpment within the Bale Mountains National Park (Gower et al., 2013). However, Gower et al. (2013) noted this area was under environmental pressure, with habitat loss and degradation driven by human modification.

Artificial Propagation/captive breeding

According to the Species360 ZIMS database for zoos, a total of 28 *B. parviocula* individuals are held in 12 ex-situ holding institutions, 17 individuals in eight institutions in the USA, six individuals in two institutions in the United Kingdom of Great Britain and Northern Ireland (UK), three individuals in one institution in Sweden and two individuals in one institution in Croatia. Data from an online database of vertebrates in zoos, also indicates that *Bitis parviocula* individuals are currently held at 16 holding institutions, including 12 in the USA, two in the EU, and two in the UK. *Bitis harena* individuals were not recorded in either database.

Bitis parviocula was reported to have been successfully bred in captivity in Europe in 2011-2012, producing 11-16 offspring, and in London Zoo in 2021. Furthermore, experts report *B. parviocula* has been bred numerous times in captivity (Colston in litt., 2025; Spear in litt., 2025), by both professional and private individuals and institutions, including at USA zoological institutions such as St. Louis Zoo where successful breeding of *B. parviocula* has occurred on at least two occasions according to Colston's in litt. (2025) personal recollection. Sundman in litt. (2025) noted *B. parviocula* was rarely observed for sale in Europe.

Doubts have been raised regarding the legality of trade in captive-bred individuals of *B. parviocula*, with concerns that some specimens in trade may originate from wild-caught individuals or their immediate progeny. Observations from social media discussions among traders and keepers have also contributed to scepticism about the claimed captive bred origins of these specimens. As the supporting statement also indicates, there are discrepancies between the proponent's statement that Ethiopia has never authorised trade in these species and reports by experts that legal exports of *B. parviocula* from Ethiopia have taken place (Colston in litt., 2025).

No evidence of *B. harena* in trade has been documented. The proponent states that an image of a snake being sold as *B. parviocula*, was *B. harena*, indicating that illegal trade had taken place in this species. However, experts who described *B. harena* in 2016 report the individual in the image to be a dark specimen of *B. parviocula* (Colston, and Spawls in litt. 2025).

Implementation challenges (including similar species)

B. parviocula and *B. harena* differ from all other *Bitis* species in the subgenus *Macrocerastes* in lacking horns between the nostrils and thus are readily distinguishable from other similar species in trade.

Potential benefit(s) of listing for trade regulation

The inclusion of the species in Appendix I would appear to be in line with existing national legislation. However, the benefits of an Appendix-I listing are unlikely to be realised unless enforcement efforts are increased.

References

- Colston, T. (2025) *in litt.* to the IUCN/TRAFFIC Analysis Team, Cambridge, UK.
- FaunaClassifieds community (2025) Classified listings for *Bitis parviocula* (Serpentes: Viperidae) in the exotic pet trade. Accessed on 18 August 2025.
From: <https://www.faunaclassifieds.com/forums/search/1546876/?q=bitis+parviocula>
- Gower, D.J. (2025) *in litt.* to the IUCN/TRAFFIC Analysis Team, Cambridge, UK.
- Gower, D.J., Abera, R.K., Schwaller, S., Largen, M.J., Collen, B., Spawls, S., Menegon, M., Zimkus, B.M., de Sá, R., Mengistu, A.A., Gebresenbet, F., Moore, R.D., Saber, S.A. and Loader, S.P. (2013) Long-term data for endemic frog genera reveal potential conservation crisis in the Bale Mountains, Ethiopia. *Oryx* 47:59–69.
- Gower, D.J., Wade, E.O., Spawls, S., Böhme, Wolfgang, Buechley, E.R., Sykes, D. and Colston, T.J. (2016) A new large species of *Bitis* Gray, 1842 (Serpentes: Viperidae) from the Bale Mountains of Ethiopia. *Zootaxa* 4093(1):41–63. <http://zoobank.org/urn:lsid:zoobank.org:pub:14DE413C-CCAE-4F43-87FE-BF359FD1BC23>
- Gebrewold, G., Colston, T.J., Abebe, A., Kourouma, K., Najjemba, R., Mulugeta, D., Lumma, H., Abdella, S., Haile, T., Zolfo, M. and Vanlerberghe, V. (2022) Distribution of snake species and snakebites in hotspots of Ethiopia. *The Journal of Infection in Developing Countries* 16(8.1):45S–51S.
- Heim, J. (in prep.) Not my Snake, not my Circus: Customary Illegality and Moral Economies in the *Bitis* Trade. University of Maastricht & University of Cape Town
- iNaturalist community (2025) Observations of *Bitis* species from North-east Africa. Exported from https://www.inaturalist.org/observations?taxon_id=30856 on 18 August 2025.
- Jensen, T.J., Auliya, M., Burgess, N.D., Aust, P.W., Pertoldi, C. and Strand, J. (2019) Exploring the international trade in African snakes not listed on CITES: highlighting the role of the internet and social media. *Biodiversity and conservation* 28(1):1–19. <https://doi.org/10.1007/s10531-018-1632-9>
- Ridley, F. *et al.* (manuscript in preparation) Advancing understanding of global threat abatement potential for terrestrial vertebrates.
- Spawls, S. (2021) *Bitis parviocula*. The IUCN Red List of Threatened Species 2021: e.T22473680A22473691. <https://dx.doi.org/10.2305/IUCN.UK.2021-3.RLTS.T22473680A22473691.en>. Accessed on 1 August 2025.
- Spawls, S. (2025) *in litt.* to the IUCN/TRAFFIC Analysis Team, Cambridge, UK.
- Spawls, S., Mohammad, A. and Mazuch, T. (2023) Handbook of Amphibians and Reptiles of North-east Africa. Bloomsbury Wildlife. 640 pp. ISBN 978-1472991447.
- Spear, S. (2025) *in litt.* to the IUCN/TRAFFIC Analysis Team, Cambridge, UK.
- Species360 Zoological Information Management System (ZIMS) (2025), zims.Species360.org.
- Strugariu, A. (2025) *in litt.* to the IUCN/TRAFFIC Analysis Team, Cambridge, UK.
- Sundman, E. (2025) *in litt.* to the IUCN/TRAFFIC Analysis Team, Cambridge, UK.
- Terraristik.com community (2025) Classified listings for *Bitis* species (Serpentes: Viperidae) in the reptile trade. Accessed on 18 August 2025 from <https://www.terrarium.com/tb/kaufen-und-verkaufen/reptilien/01.01/?split=0&ftsearch=Bitis>.

Inclusion of the Rock Rattlesnake *Crotalus lepidus* and Central Plateau Pygmy Rattlesnake *Crotalus ravus* in Appendix II and the genera *Crotalus* and *Sistrurus* in Appendix II as lookalikes

Proponents: Mexico and Plurinational State of Bolivia

Summary: The Rock Rattlesnake *Crotalus lepidus* and the Central Plateau Pygmy Rattlesnake *Crotalus ravus* and other species belonging to the genera *Crotalus* and *Sistrurus*, are terrestrial, ectothermic reptiles that occupy a range of habitats including arid rocky slopes, montane forests, and scrublands throughout North, Central and South America. *Crotalus ravus* is endemic to the central highlands of Mexico and *Crotalus lepidus* occurs only in the southwestern United States of America and Mexico. *Crotalus durissus unicolor*, included in Appendix III by Honduras in 1987, is the only taxon within the two genera currently included in the Appendices.

The proponents seek to include the genera *Crotalus* and *Sistrurus* in CITES Appendix II in accordance with Res. Conf. 9.24 (Rev. CoP17), with *C. lepidus* and *C. ravus* under Criterion A of Annex 2a and the remainder of the genera under Criterion A of Annex 2b, stating that trade is mostly in parts and derivatives not identifiable to the species level.

Of the 54 recognised *Crotalus* and three *Sistrurus* species, 36 have been assessed on the IUCN Red List, all but three of these completed in 2007 and considered in need of updating. *Crotalus lepidus* and *C. ravus* were both assessed in 2007 as Least Concern with stable populations. In 2006 the US population of *C. lepidus* was assessed as substantial (between 10,000 and one million individuals), with a range of between 200,000 and 2.5 million km². There are no more recent population data for either species. Three other species among those proposed for listing were considered globally threatened, with *Crotalus catalinensis* assessed as Critically Endangered due to over-collection for the exotic pet trade in 2007. A further two species have been assessed as nationally threatened in countries to which they are endemic with one, *Crotalus adamanteus*, assessed as vulnerable in the USA in 2020, partly due to harvesting for the skin trade.

Crotalus and *Sistrurus* species, including *C. lepidus* and *C. ravus*, are primarily affected by habitat loss, climate change, human persecution, road mortality, and illegal collection, with range-restricted species or those with specialised habitats being especially vulnerable. Of the 47 *Crotalus* and *Sistrurus* species that occur in Mexico, 13, including *C. lepidus*, are listed nationally under special protection (which includes species that are not considered threatened), eight (including *C. ravus*) are listed as threatened, and one (*C. transversus*) is listed as endangered. In the USA, 11 species are listed nationally as secure (including *C. lepidus*), two species are listed as apparently secure, and two as vulnerable.

Between 2008 and 2020, there were very few reports of international trade in *C. lepidus* and *C. ravus* in available trade data, with most reported to be from captive-bred individuals. Between 2022 and 2024 there were exports of wild-sourced medical specimens (e.g. capsules containing dried meat) of *C. lepidus* by both the USA and Mexico, equated to an estimated 740 live individuals a year. There is evidence of illegal trade in *Crotalus* and *Sistrurus* species but in low quantities and considered unlikely to be at levels impacting wild populations. The only species currently classified as globally threatened (*C. catalinensis*, *C. pusillus* and *C. stejnegeri*) were subject to very low levels of wild-sourced export for scientific purposes. There was some notable trade in *Crotalus adamanteus*, assessed as vulnerable within the USA, with direct commercial exports of ~60,000 skin pieces from wild sources in 2018 reported in LEMIS. Between 2019 and the first quarter of 2021, however, very little international trade in this species has been reported in US trade data. This species is distributed across seven states in the USA with an extensive range (200,000-2.5 million km²) and large estimated population size (100,000 to 1 million).

Conservation and management of *Crotalus* and *Sistrurus* species are guided by a mix of national and international legislation, including habitat protection laws, wildlife trade regulations, and species-specific assessments. Two taxa are protected under the US Endangered Species Act (ESA), *Crotalus*

willardi obscurus and *Sistrurus catenatus*, and 22 species afforded legal protection under Mexico's environmental regulation (NOM-059-SEMARNAT). However, other species lack protection, or in the case of the USA likely have protection that varies between states.

Mexico allows export of wild specimens of species including *Crotalus* and *Sistrurus* species under certain conditions and has also authorised export permits for *Crotalus* species from captive-bred sources since 2022; in the USA regulations for harvest and export of species (other than those protected under the ESA) is likely to vary by state. Exports of captive-bred individuals reported in US trade data indicate that commercial captive breeding of *C. lepidus* and *C. ravus* is successfully underway, including in non-range State countries. *Crotalus* species are relatively commonly kept and bred in captivity by keepers of exotic pets, with captive-bred individuals of *C. lepidus* selling for USD150-300; there appears to be little motivation to catch this or other species in the wild for export.

Only *Crotalus durissus* is listed in CITES Appendix III (since 1987), and very little trade has been reported in recent years, with none of this wild-sourced. Identification of *Crotalus* and *Sistrurus* species in international trade when traded as parts and derivatives, such as skins and skin pieces, makes species identification difficult given that a complete specimen is generally required to correctly identify specimens at the species level.

Analysis: Of the lead species proposed for inclusion in Appendix II, *Crotalus lepidus* and *Crotalus ravus* were both assessed as Least Concern on the IUCN Red List in 2007, with stable population trends and no major threats identified. Reported international trade in these species is minimal, with most reported to be in captive-bred specimens and negligible illegal trade. Although recent population information is lacking there are no indications of collection for export having impact on wild populations, and it seems **very unlikely that either *C. lepidus* or *C. ravus* meet the criteria for inclusion in Appendix II under Annex 2aB** of Res. Conf. 9.24 (Rev CoP17). Based on levels of international trade and threat from trade for other species in both genera it is unlikely that any of these meet the criteria for inclusion in Appendix II under Annex 2a of the Resolution. Even though the limited trade in wild-sourced *Crotalus* and *Sistrurus* species is largely in parts and derivatives, which are difficult to assign to species without complete specimens, given that no species in either genus appears to meet the criteria set out in Annex 2a, **neither genus merits listing for 'lookalike' reasons under criteria in Annex 2b.**

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES listing history

Crotalus durissus was listed in CITES Appendix III by Honduras in 1987.

Taxonomy

The Reptile Database (Uetz et al., 2025) is proposed as the standard nomenclature reference for the taxa considered in the proposal (*Crotalus* spp. and *Sistrurus* spp.). Res. Conf. 12.11 (Rev. CoP19) on *Standard nomenclature* currently recognizes McDiarmid et al. (1999) for non-colubrid snakes; this reference still recognises *Sistrurus ravus* and does not reflect the current status of taxonomic changes in *Crotalus* and *Sistrurus*. The proponents state that they consulted the Nomenclature specialist of the CITES Animals Committee on the proposed taxonomic reference. *Currently, 54 species of the genus Crotalus and three species of the genus Sistrurus are recognised by the suggested Standard Reference, the Reptile Database (Uetz et al., 2025).*

Range

Crotalus ravus Mexico

Crotalus lepidus United States of America and Mexico.

Crotalus and Sistrurus species range throughout North, Central and South America, from Canada south to Argentina.

IUCN Global Category

Of the 54 species currently recognised in the genus *Crotalus*, and three in the genus *Sistrurus* (Uetz et al., 2025), 34 *Crotalus* species, and two *Sistrurus* species have been assessed on the IUCN Red List (Table 1). Some species have also been assessed nationally in Mexico and the USA.

Table 1. Species, IUCN Red List global threat category, range State(s) and known national threat status of the 57 species included in the listing proposal. Species in bold indicate the species is being proposed under Criterion A of Annex 2a, with the remaining being proposed under Criterion A of Annex 2b. National threat/ protection status was retrieved from: Mexico: NOM-059-SEMARNAT-2010, last updated 2019 and provided in the proposal, and NatureServe (USA). Range States were derived from those stated in the proposal but cross-referenced with IUCN Red List data, and expert input, where possible, noting where the sources differed. IUCN Red List population trends are indicated using the following symbols: – stable, ▼ declining and ? unknown.

Species name	IUCN Red List global category (ver. 3.1), year assessed, population trend	Range State(s)	Threat status in Mexico based on NOM-059 ⁹ (last updated in 2019)	US NatureServe assessment and review date
<i>Crotalus adamanteus</i>	Least Concern 2007, ▼	USA	N/A	Vulnerable (2020)
<i>Crotalus angelensis</i>	Least Concern 2007, –	Mexico	Not assessed	N/A
<i>Crotalus aquilus</i>	Least Concern 2007, ▼	Mexico	Special Protection	N/A
<i>Crotalus armstrongi</i>	Not Evaluated	Mexico	Not assessed	N/A
<i>Crotalus atrox</i>	Least Concern 2007, –	Mexico, USA	Special Protection	Secure (2006)
<i>Crotalus basiliscus</i>	Least Concern 2007, –	Mexico	Special Protection	N/A
<i>Crotalus brunneus</i>	Not Evaluated	Mexico	Not assessed	N/A
<i>Crotalus campbelli</i>	Not Evaluated	Mexico	Not assessed	N/A
<i>Crotalus catalinensis</i>	Critically Endangered 2007, ▼	Mexico	Threatened	N/A
<i>Crotalus cerastes</i>	Least Concern 2007, –	Mexico, USA	Special Protection	Secure (2023)
<i>Crotalus cerberus</i>	Not Evaluated	USA	-	
<i>Crotalus concolor</i>	Not Evaluated	USA	-	
<i>Crotalus culminatus</i>	Not Evaluated	Mexico	Not assessed	N/A
<i>Crotalus durissus</i>	Least Concern 2014, ?	Argentina, Brazil, Bolivia, Colombia, Guyana, French Guiana, Paraguay, Suriname, Uruguay, Venezuela. Peru*	N/A	N/A
<i>Crotalus enyo</i>	Least Concern 2007, –	Mexico	Threatened	
<i>Crotalus ericsmithi</i>	Not Evaluated	Mexico	Not assessed	
<i>Crotalus estebanensis</i>	Least Concern 2007, –	Mexico	Not assessed	
<i>Crotalus exiguus</i>	Not Evaluated	Mexico	Not assessed	
<i>Crotalus helleri</i>	Not Evaluated	Mexico, USA	Not assessed	

⁹ The proposal refers to some updated recommendations for most Mexican species' threat status resulting from a recent workshop, but the status included here from the NOM-059 (also referred to in the proposal) is the official status of the species in Mexico and refers to published national red list assessments (Mosig in litt., 2025)

Species name	IUCN Red List global category (ver. 3.1), year assessed, population trend	Range State(s)	Threat status in Mexico based on NOM-059 ⁹ (last updated in 2019)	US NatureServe assessment and review date
<i>Crotalus horridus</i>	Least Concern 2007, ▼	USA**	N/A	Apparently secure (2014)
<i>Crotalus intermedius</i>	Least Concern 2007, –	Mexico	Threatened	
<i>Crotalus lannomi</i>	Data Deficient 2007, ?	Mexico	Threatened	
<i>Crotalus lepidus</i>	Least Concern 2007, –	Mexico, USA	Special Protection	Secure (2006)
<i>Crotalus lorenzoensis</i>	Least Concern 2007, –	Mexico	Not assessed	N/A
<i>Crotalus lutosus</i>	Not Evaluated	USA	N/A	Not assessed
<i>Crotalus mictlantecuhtli</i>	Not Evaluated	Mexico	Not assessed	N/A
<i>Crotalus mitchellii</i>	Least Concern 2007, –	Mexico	Special Protection	N/A
<i>Crotalus molossus</i>	Least Concern 2007, –	Mexico, USA	Special Protection	Secure (2006)
<i>Crotalus morulus</i>	Not Evaluated	Mexico	Not assessed	N/A
<i>Crotalus oreganus</i>	Least Concern 2007, –	Canada, Mexico, USA***	Not assessed	Secure (2016)
<i>Crotalus ornatus</i>	Not Evaluated	Mexico, USA	Not assessed	Not assessed
<i>Crotalus polisi</i>	Not Evaluated	Mexico	Not assessed	N/A
<i>Crotalus polystictus</i>	Least Concern 2007, ▼	Mexico	Special Protection	N/A
<i>Crotalus pricei</i>	Least Concern 2007, –	Mexico, USA	Special Protection	Secure (2006)
<i>Crotalus pusillus</i>	Endangered 2007, ?	Mexico	Threatened	N/A
<i>Crotalus pyrrhus</i>	Not Evaluated	Mexico, USA	Not assessed	Not assessed
<i>Crotalus ravus</i>	Least Concern 2007, –	Mexico	Threatened	N/A
<i>Crotalus ruber</i>	Least Concern 2007, ▼	Mexico, USA	Special Protection	Apparently secure (2012)
<i>Crotalus scutulatus</i>	Least Concern 2007, –	Mexico, USA	Special Protection	Secure (2006)
<i>Crotalus simus</i>	Least Concern 2012, ?	Belize, Costa Rica, El Salvador, Guatemala, Honduras, Mexico, Nicaragua	Threatened	N/A
<i>Crotalus stejnegeri</i>	Vulnerable 2007, ▼	Mexico	Threatened	N/A
<i>Crotalus stephensi</i>	Not Evaluated	USA	N/A	Not assessed
<i>Crotalus tancitarensis</i>	Data Deficient 2007, ?	Mexico	Not assessed	N/A
<i>Crotalus thalassoporus</i>	Not Evaluated	Mexico	Not assessed	N/A
<i>Crotalus tigris</i>	Least Concern 2007, –	Mexico, USA	Special Protection	Secure (2006)
<i>Crotalus tlaloci</i>	Not Evaluated	Mexico	Not assessed	N/A
<i>Crotalus totonacus</i>	Least Concern 2007, –	Mexico	Not assessed	N/A

Species name	IUCN Red List global category (ver. 3.1), year assessed, population trend	Range State(s)	Threat status in Mexico based on NOM-059 ⁹ (last updated in 2019)	US NatureServe assessment and review date
<i>Crotalus transversus</i>	Least Concern 2007, –	Mexico	Endangered	N/A
<i>Crotalus triseriatus</i>	Least Concern 2007, –	Mexico	Not assessed	N/A
<i>Crotalus tzabcan</i>	Least Concern 2012, ?	Belize, Guatemala, Mexico	Not assessed	N/A
<i>Crotalus unicolor</i>	Not Evaluated	Venezuela ****	N/A	N/A
<i>Crotalus vegrandis</i>	Not Evaluated	Venezuela	N/A	N/A
<i>Crotalus viridis</i>	Least Concern 2007, –	Canada, Mexico, USA	Not assessed	Secure (2016)
<i>Crotalus willardi</i>	Least Concern 2007, –	USA, Mexico	Special Protection	Secure (2006)
<i>Sistrurus catenatus</i> **	Least Concern 2007, ?	Canada, Mexico, USA*****	Not assessed	Vulnerable (2015)
<i>Sistrurus miliarius</i>	Least Concern 2007, –	USA	N/A	Secure (2006)
<i>Sistrurus tergeminus</i>	Not Evaluated	USA, Mexico	Not assessed	Not assessed

* Proposal stated potentially Peru, but IUCN RL states Peru as a confirmed range State and this is also confirmed by Rodrigues et al. (2022).

** Proposal listed Canada as a range state but IUCN RL states extinct in Canada, which was also noted in expert feedback (Spear in litt., 2025).

*** Expert feedback stated that *Crotalus oreganus* as currently represented in the reptile database only occurs in the USA and Canada, with *Crotalus helleri* considered a subspecies of *C. oreganus* by some and is the only member of the complex to occur in Mexico (Spear in litt., 2025).

**** Expert feedback stated that this species is also present in Aruba (Spear in litt., 2025).

***** Expert feedback stated *Sistrurus catenatus* is only found in the USA and Canada, and that previously *S. tergeminus* used to be considered part of *S. catenatus*, hence the distribution into Mexico in some accounts, such as the red list.

Biological and trade criteria for inclusion in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)

A) Trade regulation needed to prevent future inclusion in Appendix I

B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences

This analysis focuses on the two species proposed for inclusion in Appendix II according to Criterion A of Annex 2a (*Crotalus ravus* and *Crotalus lepidus*) but also notes where other species in the genera *Crotalus* and *Sistrurus* are found in international trade and assesses whether they could also meet this criterion.

Population size and trends

Of the 54 species currently recognised in the genus *Crotalus*, and three in the genus *Sistrurus* (Uetz et al., 2025), 34 *Crotalus* species, and two *Sistrurus* species have been assessed on the IUCN Red List. Most (~85%) of these species occur in Mexico, and around half of these have been assessed and assigned a threat status nationally (see Table 1). Mexico assigns threat statuses of either under special protection (said to be equivalent to an IUCN Red List assessment of Least Concern/Data Deficient), threatened (equivalent to Vulnerable or Near Threatened), or endangered (equivalent to Endangered or Critically Endangered) (MacipRíos et al., 2015). Twelve of these species are endemic to Mexico, and only one (*Crotalus transversus*) has been assessed as endangered nationally.

Around half of the 57 species in the two genera (23) occur in the USA, and most of these (15) have national assessments published in NatureServe. Most of these assessments consider them either 'secure' or 'apparently secure' with only two species assessed as vulnerable (*Crotalus adamanteus* in 2020 and *Sistrurus catenatus* in 2015). Out of these two species, only *Crotalus adamanteus* is endemic to the USA. NatureServe defines vulnerable as 'At moderate risk of extirpation in the jurisdiction due to a fairly restricted range, relatively few populations or occurrences, recent and widespread declines, threats, or other factors',¹⁰

¹⁰ https://help.natureserve.org/biotics/content/record_management/Element_Files/Element_Tracking/ETRAK_Definitions_of_Heritage_Conservation_Status_Ranks.htm

According to the IUCN Red List assessments, many of which were published in 2007 and require updating, the population trends were declining for seven of the *Crotalus* species (*C. adamanteus*, *C. aquilus*, *C. catalinensis*, *C. horridus*, *C. polystictus*, *C. ruber* and *C. stejnegeri*). The remainder were assessed as either stable (22), unknown (7) or not assessed (18). Of the two lead species, both *Crotalus lepidus* and *Crotalus ravus* had a stable population based on IUCN Red List assessments in 2007. No updated global population estimates are available for *C. ravus* and *C. lepidus*.

Crotalus ravus

The species was considered abundant in the 1970s and until 2007, it could be occasionally found in the vicinity of Mexico City in the 1980s and 1990s, but the supporting statement states that it is currently rarely observed based on anecdotal evidence. One expert noted that on iNaturalist *Crotalus ravus* is the 5th most observed Mexican rattlesnake on the site with 693 observations, which did not appear to be consistent with being rarely observed (Spear in litt., 2025). *C. ravus* reported to have a high intrinsic vulnerability due to the species' restricted distribution, presence in few physiographic units and high human persecution due to being a venomous snake. In Mexico, *C. ravus* was considered threatened, with recent assessments (arising from a workshop and not yet published in NOM-059, the official national list of species at risk) indicating that the species was endangered.

Crotalus lepidus

The proposal states that this species has only been rarely recorded due to its preference for inaccessible terrain. One expert noted that on iNaturalist the species has 2,029 observations, with most of these in the USA (Spear in litt., 2025). In the USA, the species was assessed as "secure" nationally in 2006 but has not been assessed since. *Crotalus lepidus* was considered common in the USA in the 1990s, and the national assessment in 2006 stated the extent of occurrence, area of occupancy, number of subpopulations, and population size were probably relatively stable, with national population estimates between 10,000 and 1 million individuals, and an estimated range in the USA of 200,000-2,500,000 km² (NatureServe, 2025a). In Mexico, *C. lepidus* is "Subject to Special Protection" in the national list of species at risk (NOM-059-SEMARNAT-2010), which is said to be equivalent to an IUCN Red List assessment of Least Concern/Data Deficient (MacipRios et al., 2015). In a study carried out in the Mapimí Biosphere Reserve, Chihuahua, Mexico, *C. lepidus* was noted to be rare and had the lowest abundance of recorded snake species.

Other species

Of the species in the two genera, only three species have been assessed as globally threatened:

- *Crotalus catalinensis* was assessed as Critically Endangered in 2007, being known from only one locality on the Santa Catalina Island in the Gulf of California in Mexico, having a decreasing population trend and primarily threatened by illegal collection for the international pet trade. In a rapid online survey, no evidence of recent international trade was found (August 2025).
- *Crotalus pusillus* was assessed as Endangered in 2007 with an unknown population trend and being known from less than five locations in an area with declining habitat extent and quality.
- *Crotalus stejnegeri* was assessed as Vulnerable in 2007 due to its extent of occurrence of less than 20,000 km², fragmented range and decreasing extent and quality of habitat.

Two species have also been assessed as nationally threatened in the countries where they are endemic:

- *Crotalus transversus* is endemic to Mexico and assessed as nationally endangered (last national update 2019), but no data on population size or trends could be found either in the national assessment database or in other literature.
- *Crotalus adamanteus* is endemic to the USA and was assessed as vulnerable by the USA in 2020 (NatureServe, 2025b). The assessment said the species has a 'limited' range across seven states in the Southeast (although with a wide ranging and potentially large of estimate of 200,000-2,500,000 km²). The national assessment also stated *Crotalus adamanteus* had experienced long term declines in the area of occupancy, number of subpopulations, and population size of 30-70%, and short-term declines of 10-30%, due to threats including loss of habitat, heavy exploitation for the skin trade, and general persecution, as well as fragmented populations. The population size was estimated to be at least 100,000 individuals but potentially up to 1 million. The species was reported to be locally common in suitable habitat in areas of Georgia, South Carolina, and Florida, but rare in North Carolina and Louisiana.

Domestic trade

Rattlesnakes are used domestically in Mexico for human food, medicine, esoteric uses, manufacturing of products with their skin, soaps, oils, jewellery, and taxidermy. In an ethnobiological study in the mountains of the state of Aguascalientes, 83% of the people interviewed said they consumed rattlesnakes, including *C. lepidus*, *C. aquilus*, *C. molossus*, *C. polystictus*, *C. pricei* and *C. scutulatus*. In the state of Chihuahua, the Rarámuri indigenous people use native rattlesnakes, *C. lepidus*, *C. atrox* and *C. willardi*, crushed and diluted in alcohol as a remedy for rheumatism. *Crotalus ravus* is sold in Mexico City markets primarily for medicinal uses against diabetes and

cancer, along with other *Crotalus* species; the skin is sold whole or in sachets as amulets to ward off the evil eye, provide luck in business, and attract good fortune; the rattle, head, and fangs are also used as amulets to make necklaces. The retail price of *C. ravus* for the pet market in Mexico is reported to be around USD112.

International trade

The proposal states that both species, *C. lepidus* and *C. ravus* are present in legal international trade, mainly from captive breeding in small numbers. International trade in *C. ravus* is reported to be primarily in live individuals, and for *C. lepidus* mostly in medical derivatives and extracts (dried meat pills, powders, etc.), as well as live individuals and skins, with its products reported to be sold online and shipped to countries such as the USA, Russian Federation, India, Kuwait, Bahrain, Argentina and Guatemala.

In Mexico, Article 84 of the General Wildlife Law (2000¹¹) states that any harvest of terrestrial and native wildlife species is permissible if it can be shown that harvest rates are lower than the natural renewal of the populations subject to use and that harvest will not have a detrimental effect on populations. Based on the trade data provided by Mexico, it appears that export permits for wild-sourced specimens/samples from *Crotalus lepidus* and *Crotalus ravus* have been authorised for scientific purposes between 2010 and 2024, but that no commercial exports of wild-sourced specimens from either species have been authorised. The SS states that captive breeding of the species in Mexico began in 2022 and that export permits for captive-bred specimens of *Crotalus* species have been issued since then. The SS provides a summary of data on the legal international trade of the two species from Mexico and the USA, based on Mexican trade databases. This is summarised in Table 2 below.

Table 2. A summary of legal trade reported in national trade databases in Mexico outlined in the SS

Trade data source	Time frame	<i>Crotalus lepidus</i>		<i>Crotalus ravus</i>	
		Wild-sourced	Captive-bred	Wild-sourced	Captive-bred
Export authorisations from Mexico provided by the CITES Management Authority of Mexico ¹²	2010-2024	Six scientific records totalling 17 live specimens and an unstated quantity of scientific samples	Eight live specimens for commercial purposes	15 scientific samples	13 live individuals (purpose not stated)
Import authorisations from the USA provided by the CITES Management Authority of Mexico	2010-2024	30 live individuals imported for commercial purposes (source not stated)			
Exports and re-exports (aggregated) from Mexico provided by PROFEPA (Federal Attorney for Environmental Protection)	2010-2024	74 heads (specimens) of <i>Crotalus culminatus</i> , <i>C. lepidus</i> , <i>C. molossus</i> , <i>C. polystictus</i> , <i>C. ravus</i> , <i>C. tzabcan</i> and <i>C. willardi obscurus</i> (source/ purpose/ individual quantities not stated)			

Other evidence of international trade for commercial purposes

*A summary of available international trade data for *C. lepidus*, *C. ravus* and other species identified as either globally threatened (*Crotalus catalinensis* and *Crotalus stejnegeri*) or endemic and nationally threatened (*Crotalus transversus* and *Crotalus adamanteus*) are provided below. The main data source consulted was LEMIS (2008 to the first quarter of 2021). The proposal analysed LEMIS data for a different period between 2012 and 2024, and where applicable, additional data based on this dataset is referenced. Where possible, data were also sourced from available literature on the species, including a 2024 study of online trade in *Crotalus* species (Cantú and Sánchez, 2024).*

¹¹ Last reformed in 2021

¹² information provided by the General Directorate of Wildlife within the Mexican Secretariat of Environment and Natural Resources (DGV-S-SEMARNAT)

Crotalus lepidus (occurs in the USA and Mexico)

- 49 live captive-bred individuals were reported in direct commercial imports into the USA; from Germany (41) and Slovenia (8) between 2008 and 2016 for commercial purposes in LEMIS, with none reported between 2017 and 2021.
- 169 live captive-bred and 29 wild-sourced individuals were reported in direct commercial exports from the USA in LEMIS between 2008 and 2018, with none reported between 2019 and 2021.
- Authors of a study on legal and illegal trade of rattlesnakes in Mexico carried out a survey of online platforms in the USA and the EU in 2024 and identified five advertisements for live individuals of *C. lepidus* in the USA (average USD415 each) and three in the EU (average EUR290 each equivalent to USD335) (Cantú and Sánchez, 2024). It is not clear if the advertisements in the USA were for domestic or international sale.

Crotalus ravus (endemic to Mexico)

- 30 live captive bred individuals were reported in direct commercial imports into the USA; from Germany (21) and Costa Rica (9) between 2008 and 2018 for commercial purposes in LEMIS, with none reported between 2019 and 2021.
- One live captive-bred individual was reported in a direct commercial export from the USA in 2014 in LEMIS.
- The proposal states four live captive-bred individuals were reported exported from Mexico to the USA (assumedly between the latter half of 2021 and 2024, as these data were not present in LEMIS data between 2008 and the first quarter of 2021).
- Authors of a study on legal and illegal trade of rattlesnakes in Mexico carried out a survey of online platforms in the USA and the EU in 2024, and found three advertisements for live individuals of *C. ravus* in the USA (~USD300 each) and four in the EU (~USD360 each) (Cantú and Sánchez, 2024).
- The proposal also states that in 2018, a survey of European online pet retailers found *C. ravus* for sale online but does not refer to the source of this information. Cantú and Sánchez (2024) also state that *C. ravus* was found in a survey of European online platforms in 2018 (assumedly the same study), but the source cited is not published so it is not possible to know how many were found offered for sale.

Crotalus catalinensis (endemic to Mexico)

- 2 live captive-bred individuals reported in commercial re-exports from the USA to Germany (origin Germany) in LEMIS in 2014.
- A study published in 2019 stated that this species attracts the interest of persons who illegally collect wildlife for the pet trade because it occurs only on the island of Santa Catalina, but that the actual impact of such exploitation on the population of *C. catalinensis* is unknown (Arnaud and Martins, 2019).

Crotalus stejnegeri (endemic to Mexico)

- No evidence of international trade found.

Crotalus transversus (endemic to Mexico)

- No evidence of international trade found.

Crotalus adamanteus (endemic to the USA)

- Direct commercial exports from the USA of wild-sourced commodities including ~60,550 skin pieces, 1,750 skins and 160 shoes reported between 2008 and 2020 in LEMIS. Most skin pieces (~60,000) were exported in 2018 with none reported exported since, and most skins (1,000 out of the 1,750) in 2015. Since 2019, only eight shoes and ~260 skins have been reported in commercial exports.
- The national assessment in 2020 for this species in the USA reports that the impact of the skin trade on this species may be more damaging than that of rattlesnake roundups (which likely contributed to localised declines), stating that harvest for the skin trade probably accounts for at least ten times the number of rattlesnake deaths caused by roundups across the species range (NatureServe, 2025b).
- In a rapid online survey, no international trade was observed in any product (August 2025).

Only one species - *Crotalus durissus* – is listed in the CITES Appendices. This species was listed in Appendix III by Honduras in 1987 and in the last ten years, around 300 medicinal specimens (units unstated) from captive-bred sources and 49 live individuals (30 wild-sourced and the remaining captive-bred) have been directly exported for commercial purposes. All medicinal specimens were reported exported in 2020 (by France) and most live individuals between 2015 and 2016 (by Mexico and Guyana), with an average of 2 per year since by Guyana only.

Other notable trade

Two other species were found in international trade in notable quantities in recent years, based on the USA trade data in LEMIS¹³ (Table 3). Neither species has been assessed as globally or nationally threatened.

Table 3. A summary of evidence of notable recent international trade for other *Crotalus* and *Sistrurus* species.

Source: LEMIS

Species name	Evidence of notable recent international trade
<i>Crotalus atrox</i> (Least Concern see Table 1)	Direct commercial exports from the USA of 105 live captive-bred individuals, 1,070 wild-sourced skins and ~ 297,000 wild-sourced skin pieces reported between 2016 and 2020.
<i>Crotalus horridus</i> (Least Concern see Table 1)	Direct commercial imports from France into the USA of 1,700 wild-sourced specimens (scientific or museum) reported between 2016 and 2020

Illegal trade

The SS states that although there is legal use primarily in captivity for the sale of live specimens, the use of rattlesnakes nationally is mostly wild-caught and unlicensed. There have been seizures of *Crotalus ravus* (66 live specimens and 73 parts and derivatives) and *Crotalus lepidus* (eight live specimens) reported nationally in Mexico between 2000 and 2023, as well as 216 live specimens and 2,471 parts and derivatives of *Crotalus* spp.

Other evidence of illegal international trade in the SS is based mostly on inferred illegal trade from data reported in LEMIS. The proponents state that using LEMIS data generated by Cantú and Sánchez pers. comm (2025), it was estimated that at least 727 *C. lepidus* individuals (illegal, wild-sourced) were reported in international trade between Mexico and the USA based on conversions for around 220,000 capsules exported in the years 2022-2024 (e.g. years when no authorised wild harvest had been permitted from Mexico).¹⁴ The SS also states that in 2012, 2013, and 2014, the USA recorded 17 live specimens of *C. ravus* from Germany and Costa Rica, even though Mexico did not begin legal commercial export of the species until 2022, indicating illegal trade may have occurred (LEMIS, 2024). Our analysis of LEMIS data between 2008 and 2020 confirmed this and identified that the USA also reported importing 13 live captive-bred individuals of *C. ravus* from Germany in 2008 (2), 2009 (7), 2016 (2) and 2018 (2).

No seizure records for any of the globally threatened, or nationally threatened and endemic, *Crotalus* and *Sistrurus* species (outlined in the section on international trade above) could be found from 2014-2023 in EU-TWIX and from 1993-2025 in WITIS.

Expert consultations

One expert on trade in venomous snakes for use as pets stated that most *Crotalus* species can be caught and exported legally from the USA, but they are also very commonly kept and bred in captivity throughout the world (with price estimates of less than USD100-400 for most). The expert noted that captive-born individuals of *Crotalus lepidus* sold for between USD150-300, with little reason to catch wild-sourced individuals for sale (Spear in litt., 2025). The expert also stated that in the last ten years, the only wild-sourced rattlesnakes they had observed offered for sale in Europe were a few individuals of *Crotalus tigris* that were thought to be legally imported from the USA. *Crotalus tigris* was assessed as Least Concern on the IUCN Red List in 2007 and has not been assessed as nationally threatened in either Mexico or the USA.

Inclusion in Appendix II to improve control of other listed species

A) Specimens in trade resemble those of species listed in Appendix II under Res. Conf. 9.24 (Rev. CoP17) Annex 2 a or listed in Appendix I

The proposal states that more than 95% of international trade involving *Crotalus* and *Sistrurus* species involves parts and derivatives, making species level identification difficult, given that a complete specimen is generally required to correctly identify the species. Both genera possess a rattle at the tail tip, but differ in head scalation. *Sistrurus* has large, regularly arranged crown scales, while *Crotalus* has many small, irregularly arranged scales. *Crotalus ravus* may be an exception to this, as most international trade identified by the proponents and in our research was in live individuals.

¹³ Notable recent trade was defined as 100+ whole organism equivalents or 1,000+ parts and derivatives in the last five years of legal trade data available in US LEMIS (2016-2020)

¹⁴ This is based on trade described earlier of medical specimens reported in LEMIS data for *C. lepidus* totalling ~68,000 specimens (when including commercial exports from the USA to Mexico, and from Mexico to the USA). The proposal states that based on information on the number and presentation of LEMIS specimens provided by the USA, a higher number of ~218,250 capsules were instead estimated. Cantú and Sánchez pers. comm. (2025) reported that during a visit to the "Mercado de Sonora" in Mexico City (known for its illegal sale of specimens), vendors claimed that ~2 capsules can be produced per linear centimetre of rattlesnake (~500 mg per capsule), so an average 50 cm *C. lepidus* specimen could produce up to ~100 capsules. This would equate to at least ~2,183 snakes between 2022 and 2024 (or ~727 annually).

Additional Information

Threats

Rattlesnakes are largely threatened by being perceived as harmful (poisonous) organisms that serve no ecological, economic, or cultural function. According to a study, mountain species (including *C. ravus* and *C. lepidus*) are affected by habitat loss and fragmentation, collection for remedies, fur trade, crafts, esotericism, pets, climate change, and death by aversion. Likewise, rattlesnakes are commonly killed upon encounter.

The ecosystems in Mexico where *C. ravus* and *C. lepidus* live are changing rapidly: temperate forest habitat lost between 2,600 and 3,480 km² annually between 1990 and 2005, while the loss of 8,160 km² was reported in 2023 (conversion to agricultural lands, pastures and urbanisation). The ecosystems of the Chihuahuan Desert (where *C. lepidus* is also found) are subject to conversion due to cattle ranching and mining. Climate change is also estimated to directly affect the mountain ecosystems in which *C. ravus* and *C. lepidus* live. A study analysed the vulnerability of 14 North American *Crotalus* species to climate change, and although *C. ravus* had the greatest niche breadth, it was the fourth most vulnerable species to climate change. *C. ravus* is also the fourth most trafficked rattlesnake species and seized by PROFEPA in Mexico.

Conservation, management and legislation

Mexico

In Mexico, as indicated above, the species is regulated by Article 84 of the General Wildlife Law (2000). Article 420 of the Mexican Federal Penal Code imposes penalties on anyone who fails to comply with the provisions of the General Wildlife Law, including irregularities regarding the legal origin and traceability of specimens, and the use of supporting documentation.

The management of the *Crotalus* genus is carried out through UMAs (Multicultural UMAs) and PIMVS (National Parks and Wildlife Conservation Institutes). The Species Conservation Action Program (PACE) for rattlesnakes (*Crotalus* spp.) includes objectives, goals, conservation strategies, and success indicators. There is no program for monitoring wild populations of *Crotalus* spp. A study found that *C. ravus* is found in 10 Natural Protected Areas (NPAs) in seven states in the centre of the country (~11% of its total distribution), and *C. lepidus* is found in 35 NPAs, with ~8% of its distribution in said areas.

USA

*The proposal does not state the regulations related to the USA for wild harvest and trade of *Crotalus* and *Sistrurus* species in the USA, but under the US ESA, the only rattlesnakes currently listed as threatened, and therefore legally protected, include the Eastern massasauga rattlesnake (*Sistrurus catenatus*) species and New Mexico ridge-nosed rattlesnake (*Crotalus willardi obscurus*) subspecies. A proposal submitted to CoP19 by the USA for inclusion of *Crotalus horridus* in Appendix II (CoP19 Prop 21) stated that the species was not listed or afforded direct national protection, but that protections vary across its range, with 18 of the extant 31 states where it distributes in the USA directly prohibiting harvest. It is likely that regulations relating to harvest of other *Crotalus* and *Sistrurus* species in the USA also differ according to state. An expert on trade in venomous snakes for use as pets stated that most *Crotalus* species can be caught and exported legally from the USA (Spear in litt., 2025).*

Artificial Propagation/captive breeding

The proposal states that in Mexico, there are several management units (i.e., UMAs and PIMVs) which reproduce *C. ravus* and *C. lepidus* in captivity. Captive breeding of the species in Mexico began in 2022 and export permits for captive-bred specimens of *Crotalus* species have been issued since then. *The USA has reported direct commercial exports and imports of live captive-bred individuals of *C. lepidus* and *C. ravus*, with some imports from non-range States for the two species, which indicates commercial captive breeding of these species occurs.*

References

- Arnaud, G. and Martins., M. (2019) "Living Without a Rattle: The Biology and Conservation of the Rattlesnake, *Crotalus catalinensis*, from Santa Catalina Island, Mexico." *In Islands and Snakes: Isolation and Adaptive Evolution*, pp. 241–257. Oxford University Press.
- Cantú, J.C. and Sánchez, M.E. (2024) *The Legal and Illegal Trade of Rattlesnakes in Mexico*. Teyeliz A.C. 52 pp.
- Macip-Rios, R., Ontiveros, R., López-Alcaide, S. and Casas-Andreu, G. (2015) The conservation status of the freshwater and terrestrial turtles of Mexico: A critical review of biodiversity conservation strategies. *Revista Mexicana de Biodiversidad* 86(4):1048–1057. <https://doi.org/10.1016/j.rmb.2015.09.013>
- Mosig, P. (2025) in litt. to the IUCN/TRAFFIC Analyses Team, Cambridge, UK
- NatureServe (2025a) *Crotalus lepidus*. NatureServe Explorer. Accessed on 25 June 2025. <https://explorer.natureserve.org/>
- NatureServe (2025b) *Crotalus adamanteus*. NatureServe Explorer. Accessed on 25 June 2025. <https://explorer.natureserve.org/>
- Spear, S. (2025) in litt. to the IUCN/TRAFFIC Analyses Team, Cambridge, UK

- Rodrigues, C.R., Molina, D.A.M., de Souza, D.L.N., Cardenas, J., Costal-Oliveira, F., Guerra-Duarte, C. and Chávez-Olórtegui, C. (2022). Biological and proteomic characterization of the venom from Peruvian Andes rattlesnake *Crotalus durissus*. *Toxicon* 207:31–42.
- Uetz P., Freed P., Reyes F., Aguilar R., Kudera J. and Hošek J. (2025) The Reptile Database. <http://www.reptile-database.org>. Accessed on 20 May 2025.

Transfer of Home's Hinge-back Tortoise *Kinixys homeana* from Appendix II to Appendix I

Proponents: Cameroon, Guinea, Nigeria, Togo

Summary: *Kinixys homeana*, commonly known as Home's Hinge-back Tortoise, is a medium-sized terrestrial tortoise native to the humid tropical forests of West and Central Africa. Its indigenous range includes Liberia, Côte d'Ivoire, Ghana, Togo, Benin, Nigeria and Cameroon, with unconfirmed reports from Central African Republic and the Democratic Republic of the Congo likely reflecting misidentifications or traded specimens. The species occurs in lowland evergreen and semi-deciduous forests, often in association with swampy habitats, streams and mangroves. It is a habitat specialist that prefers closed-canopy forests providing cool, moist, and shaded microhabitats, though in regions such as the Niger Delta in Nigeria, it now persists primarily in mature secondary forests, as primary forest has been lost. The species has a low reproductive rate (clutches of 2-4 eggs once or twice a year), is long-lived (~60 years) and relatively slow to mature (estimated at ~10 years), with an estimated generation time of 15 years. It has been included in Appendix II since 1975 when the whole genus *Kinixys* was listed.

Kinixys homeana was assessed on the IUCN Red List as Critically Endangered in 2019 based on inferred and projected population declines of more than 80% over three generations (~45 years). Populations of *K. homeana* are increasingly fragmented and isolated within forest patches, with evidence of significant population declines in parts of its range. In Nigeria alone, populations of the species are estimated to have dropped by approximately 90% over the past three decades. This decline is attributed to widespread deforestation, agricultural encroachment, infrastructure development, and intensive harvesting for local consumption (for food, traditional medicine and belief-based purposes), as well as for international trade, particularly for the exotic pet market. The IUCN Red List assessment concluded that the surviving population was likely to be large, possibly very large, though with a sharply declining trend.

International trade volumes have been relatively limited in recent years, likely linked to the inclusion of the species in the Review of Significant Trade process. This has resulted in several range States establishing zero export quotas, in some cases to avoid formal trade suspensions being advised under CITES. Benin is currently subject to a CITES recommendation to suspend trade in *K. homeana*. In the recent past, specimens exported from range States were often declared as ranches with a small proportion declared as captive-bred. All are presumed to have originated from the wild, as no facilities have successfully reproduced the species in captivity in significant numbers.

Analysis: *Kinixys homeana* does not have a restricted range nor, based on currently available information, does it have a small population. However, its distribution is fragmented and its population is believed to have undergone a marked recent decline (over 80%) over the past three generations (45 years), which is projected to continue. This decline is attributed to high levels of habitat loss and overexploitation for both domestic use across its range, and to a lesser extent international trade (including regional). However, it does appear to adapt to secondary or modified habitats. The species has a low reproductive rate. Combined with delayed maturity this limits its capacity to withstand overharvesting and, while not large in volume, the extent and impact of international trade, consisting mainly or entirely of wild-sourced specimens has contributed to pressure on already vulnerable populations. Recent reductions in reported trade reflect export restrictions resulting from the Review of Significant Trade process, but there is potential for trade to resume and there may also be unreported regional trade. On this basis, *Kinixys homeana* **may meet both the biological and trade criteria for inclusion in Appendix I** of Res. Conf. 9.24 (Rev. CoP17).

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

Since 1975, it has been listed in CITES Appendix II at the genus level (*Kinixys* spp.), and since 1977 it has been included in the family listing in Testudinidae spp.

Taxonomy

Kinixys homeana (Bell, 1827)

CITES Standard Reference: Fritz, C. and Havaš, P. 2007. Checklist of chelonians of the world. Vertebrate Zoology: 57: 149-368.

Range

Extant: Benin, Cameroon, Côte d'Ivoire, Ghana, Liberia, Nigeria, Togo

Possibly Extant: Equatorial Guinea

IUCN Global Category

Critically Endangered A2bcd+4bcd (assessed 2019, ver. 3.1)

Biological criteria for inclusion in Appendix I

A) Small wild population

In suitable forest habitats, the average density of *K. homeana* varied between 0.40 individuals per ha and 1.40 individuals per ha. However, the density ranged from 0.15 to 0.90 tortoises per ha in harvested areas, and from 1.65 to 2.85 tortoises per ha in areas where these animals are traditionally venerated and hence not hunted. Lawson (2006) extrapolated densities as low as 0.03 tortoises/ha for *K. homeana* in Cameroon, and annual exploitation rates as high as 0.007 tortoises/ha. Population sizes of *K. homeana* are strongly depressed in areas where these tortoises are actively hunted by human populations.

Regarding Luiselli et al. (2021a), it is estimated that in 1960 there were at least 5 million individuals of Kinixys homeana in Nigeria compared to an estimate of about 500,000 today. This estimate incorporates various assumptions: 1) the density of tortoises has remained constant in remaining suitable habitats (which is unlikely given hunting pressures), 2) local hunting pressures have remained constant over the years (which is unlikely considering human population growth from about 35 million in 1959 to approximately 120 million in 2001), and 3) suitable habitat for K. homeana has declined by at least 90% over the last 45 years, based on known habitat loss. The IUCN Red List assessment was based on observations in Nigeria and extrapolated range-wide, however, according to Luiselli et al. (2021a), there is no indication that Nigeria represents an exceptional case of decline (apart from its high human population size), rather it is a best-studied example indicative of trends across much of the region.

B) Restricted area of distribution

Kinixys homeana is found in Liberia, Côte d'Ivoire, Ghana, Togo, Benin, Nigeria, and Cameroon. Specimens have also been reported from the Central African Republic and the Democratic Republic of the Congo (DRC), but these are most likely misidentified individuals or trade animals. Nigeria contains the largest proportion of the species' indigenous range, followed by Cameroon.

According to the 2021 IUCN Red List assessment (Luiselli et al., 2021a), this forest tortoise occupies a formerly extensive indigenous range along the coastal region of the Gulf of Guinea, within the continuous Guinea-Congo rainforest belt. Kinixys homeana prefers moist forests characterized by a closed canopy and shady microhabitats. It has also been reported in association with streams and swampy areas.

Research by Luiselli et al. (2000) in the River Niger Delta in southern Nigeria has shown populations of K. homeana to be habitat specialists associated with mature secondary forests around mangroves and mature secondary swamp forests. With the complete loss of primary forests in this area, these tortoises now inhabit the nearest and most mature secondary forests available. Many populations of this species are severely fragmented and isolated inside remnant patches of forest (Luiselli et al., 2021a).

C) Decline in number of wild individuals

The Home's Hinge-back Tortoise's population is experiencing a serious decline across much of its range, mainly due to extensive habitat loss, intensive harvesting for subsistence and traditional medicine, and exploitation for international trade. There is clear evidence that the wild populations of *K. homeana* are heavily declining and collapsing throughout the species' range, with cases of extirpation even inside protected areas.

Kinixys homeana shows a consistent pattern of decline across much of its range in West and Central Africa. Reports suggest local extirpations, increasing rarity, and severe population declines, based on field surveys, interviews, and market data. The species appears to be under significant pressure from hunting, habitat loss, and collection for subsistence and trade, primarily for domestic markets, but also for international trade. In some areas, the species may already be commercially extinct in the wild.

Benin: According to recent interviews and field data collected during 2018 surveys in the Oueme Region along the southeastern coastal border with Nigeria, the species is almost nationally extirpated in Benin.

Cameroon: In 2000, very strong hunting pressure on the species was reported in the Western and South-West Provinces.

Côte d'Ivoire: Surveys in 2020 revealed that the species is (almost certainly) extirpated from coastal forest sites where in 2010–2012 it was noted to be relatively common. A survey of people's perceptions regarding the species indicates a less steep decline compared to Togo and Nigeria.

Ghana: Collectors report tortoises are found with increasing difficulty. Intensive surveys in 2016 and 2017 failed to locate any specimens in the Muni-Pomadze site, indicating this population may no longer exist. The species has become increasingly rare in Ghana.

Liberia: In January–February 2018, no individuals were found during two weeks of field surveys in the mature forests of the Nimba mount region. The species has recently become increasingly rare throughout Liberia.

Nigeria: The population of *K. homeana* in Nigeria has experienced a severe decline, from an estimated 5 million individuals in 1960 to about 500,000 today. Most surveyed localities show the species as rare, declining, or presumably extirpated; particularly in areas affected by deforestation and subsistence hunting. Although it remains locally common in a few culturally protected sites (e.g., parts of Bayelsa and Rivers), even these populations are reportedly in decline. By 2022–2023, the species had nearly disappeared from the wildmeat trade in nine southern markets, suggesting potential commercial extinction of wild populations at the local level.

Togo: The species is apparently rare in the mature and altered forests of the country. It is reported to be in decline by turtle hunters and traders.

Suitable habitat for *K. homeana* has declined by at least 90% in the last 45 years (*i.e., three generations*). Habitat loss is particularly strong in southern Nigeria, where heavy deforestation occurs due to oil and natural gas industry development. Ghana and Togo have among of the highest deforestation rates in Africa, with the former, for instance, at 2% (135,000 ha) loss per year. Thus, approximately 1,931,000 ha (26%) of *K. homeana* habitat was lost between 1990 and 2005.

Trade criteria for inclusion in Appendix I

The species is or may be affected by trade

Legal trade

According to currently available data from the CITES Trade Database, a total of 114,240 live specimens were exported mainly from Benin, Ghana and Togo between 1975 and 2018: 64,876 from Togo, 30,111 from Benin, 16,076 from Ghana, and 3,177 from all other countries combined.

The work of Luiselli *et al.* (2021a) indicates that trade volumes have remained stable or increased over the years (around 86,000 individuals of *K. homeana* were recorded in legal international trade between 1990 and 2020). The main importers are the USA, Japan, Mexico, the United Arab Emirates, Spain, Germany and France. Most of the trade consists of individuals of wild origin and the species is known to be difficult to maintain and breed in captivity. The species is likely to have disappeared from areas where it is easily caught.

Trade in *K. homeana* appears to be significant and potentially problematic. Since around 1995, there has been a reported shift in CITES source codes from wild to ranched specimens, though discrepancies in trade volumes and source codes between exporting and importing countries suggest potential misreporting. Between 2000 and 2023, over 10,000 individuals were exported from the wild alongside tens of thousands reportedly from ranching operations. However, the validity of the “ranching” designation is questioned, as there is little evidence of legitimate ranching practices in West and Central Africa. Observations in Togo and Ghana indicate that some traded specimens do not originate from farms, and the few tortoise farms visited appear to focus on other species, casting further doubt on the legal origin of many exported individuals.

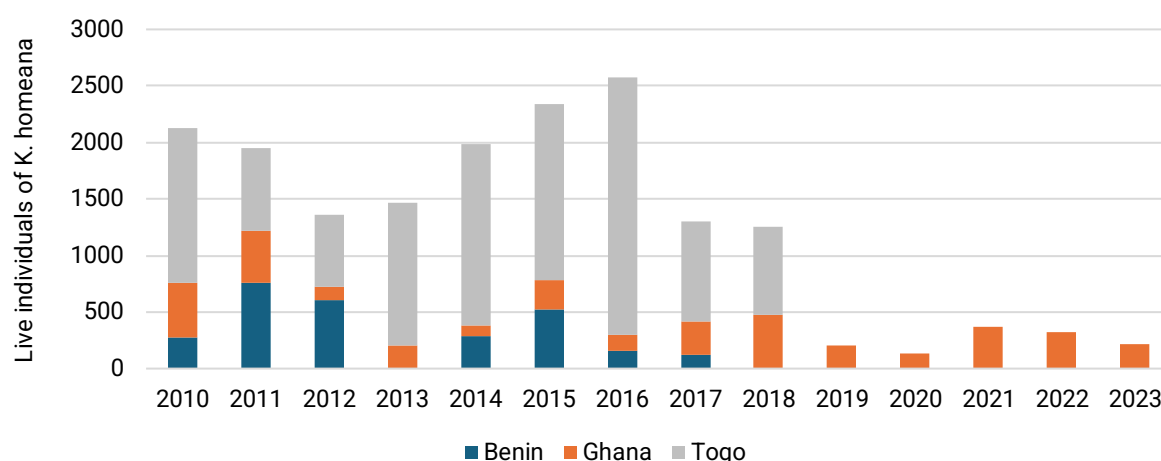


Figure 1. Direct trade in live specimens of *K. homeana* from Benin, Ghana and Togo, 2010-2023, for commercial purposes.

Benin, Ghana and Togo are currently in the Review of Significant Trade (RST) process for *Kinixys homeana*. As a result of this process, Benin is subject to a suspension of trade in all specimens of the species. Liberia is also subject to a suspension of trade since 2016, but this includes all CITES-listed species, and it is unrelated to trade in *K. homeana*. Following recommendations from the Animals Committee and the Standing Committee, Togo has established a zero-export quota for live, ranched and wild specimens, and Ghana has established a zero-export quota for trade in all specimens of *K. homeana*.

The RST for *K. homeana* was initiated in 2011, when the 25th meeting of the Animals Committee (AC25) agreed to include the species along with all its range States. In 2014, the AC27 categorised *K. homeana* as a species of possible concern for Benin and Togo, and of least concern for Côte d'Ivoire, the Democratic Republic of the Congo, Equatorial Guinea and Gabon, which were subsequently excluded from the process. Benin and Togo remained under review.

For Benin, the Standing Committee recommended a suspension of all trade in *K. homeana* at its 66th meeting (SC66) in 2016. Despite this recommendation, Species+ records indicate that Benin set export quotas in 2017 for 50 wild-taken, 800 ranched, and 30 captive-bred specimens. However, since 2019, Benin has maintained a zero-export quota for all sources of specimens.

Togo, historically authorized substantial exports, up to 2,500 ranched and 500 wild-taken individuals annually, remained under the RST. At its 67th meeting, the Standing Committee (SC67) requested Togo to set an export quota of 400 live ranched specimens (under 10 cm in size) and zero for wild specimens for 2017. In 2018, Togo switched the zero-export quota to ranched specimens and set an export quota of 500 for wild-taken specimens. Since 2020, Togo has maintained a zero-export quota for all specimens of the species, in line with the Animals and Standing Committees' recommendations.

Regarding Ghana, the country maintained an annual export quota of 340 wild specimens for over two decades (since 1997). However, in 2023, the AC32 agreed to include Ghana in the RST for *K. homeana*. Following this inclusion, Ghana established zero-export quotas for all specimens of the species for 2024 and 2025.

Suspected Illegal Trade

According to LEMIS data analysed from 2008 to 2023, 11 shipments of *K. homeana*, totalling 221 individuals, were refused entry to the USA at the point of import, potentially indicating illegal trade; 209 of these were live specimens, while the rest were carapaces. Twenty-three of these were recorded as having been taken from the wild, while the rest were reported as ranched. The average value of live specimens was ~USD400, while that of carapaces was ~USD50. In terms of the rejected entries, Ghana was recorded as the exporting country for 128 live specimens (98 of which originated in Benin and 30 in Togo), while Togo was recorded as the exporting country for 81. Nigeria and Cameroon exported 10 and two carapaces, respectively.

The CITES Annual Illegal Trade Database (AITR) has recorded seven seizures since it was established in 2016 (CITES Secretariat, 2025). These were reported by Spain, the United Kingdom of Great Britain and Northern Ireland, and the USA (CITES Secretariat, 2025). A total of 114 specimens were reported to have been seized, including 105 live specimens and nine carapaces (CITES Secretariat, 2025). The low number of seizures of this species may reflect the low prices they fetch; online searches indicate that a medium-sized adult can be purchased for less than USD100 (CITES Secretariat, 2025).

According to data from the EU-TWIX Database, only two live individuals of *Kinixys homeana* were seized in the period from 2014-2023. These were seized by the UK in 2017 and recorded as having been exported by Ghana. Additionally, two carapaces were seized: one by Belgium in 2015, the country of origin of which is unknown; and another by the UK in 2019, originating in and exported from Nigeria.

TRAFFIC's WITIS database contains details of 126 individuals (125 live and one carapace) seized during four events since 2010. Two of these seizures were made by the UK, involving three specimens exported by Ghana and Nigeria (in 2017 and 2019). One seizure was made by France, involving 25 individuals from Benin (in 2012). The final seizure was made by Belgium, involving 95 individuals from Togo (in 2013).

Additional Information

Threats

The main threats to *K. homeana* include: 1) subsistence hunting by local people; 2) habitat loss through agricultural and industrial expansion with massive deforestation; and 3) trade for the international pet industry.

Exploitation for local egg and meat consumption and exploitation for domestic trade (pets, food, medicine), represent 50% of all threats for *K. homeana*. The exploitation for international trade accounts for 10% of the extinction risk in the species. The threat of international pet trade is potentially relevant in Ghana, Togo, and Benin, the main export countries for this species.

For *Kinixys* spp., Luiselli et al. (2021b) recommend prioritising projects focusing on bushmeat consumption, habitat degradation, and potential shifts in habitat use, primarily in the rainforest belt of West Africa, as well as the effects of livestock overgrazing and climate change, specifically in relation to desertification impacting the Sahel region. For *K. homeana* specifically, reinforcing international policy is also recommended.

Conservation, management and legislation

Kinixys homeana occurs in various protected areas in Nigeria, Cameroon, Côte d'Ivoire and Ghana.

It is legally protected in Togo under the law N° 2008-005, which regulates the environment protection and wildlife conservation. No exports of wildlife for commercial purposes are permitted from Nigeria.

As of November 2023, the legislative progress of some of the range States of the species in implementing CITES is still ongoing. Ghana and Côte d'Ivoire are rated in category 3, which means their legislation is believed generally to not meet any of the four requirements for effective implementation of CITES. Togo, Benin and Liberia are in category 2, with legislation that is believed generally to meet one to three of the four requirements for effective implementation of CITES. Nigeria and Cameroon are in Category 3 for which legislation is believed generally to meet all four requirements for effective implementation of CITES.

Currently, none of the three traditional main exporters of this species are exporting *K. homeana* individuals. This is because Benin is subject to a recommendation to suspend trade in this species, while Togo and Ghana have established a zero-export quota as a result of a recommendation by the Animals and Standing Committees.

Artificial Propagation/captive breeding

Kinixys homeana is known to be difficult to keep and breed in captivity. Wild individuals usually arrive in suboptimal health conditions (i.e., heavily parasitized) in the western countries.

The exported animals have been variously declared as originating from the wild, from ranching operations, or captive breeding facilities. However, there is no documentation available to show that these facilities operate effectively (Luiselli et al., 2021a).

Ex-situ conservation breeding programs on various *Kinixys* species is ongoing in the USA to establish assurance colonies and to better understand behaviour and husbandry. Assurance colonies have been established since 2013, and a studbook has been developed for this species through the Association of Zoos and Aquariums. By 2021, over 150 *K. homeana* had been produced and greater understanding of the captive management and husbandry learned.

According to the Zoological Information Management System (ZIMS), a total of 115 Home's Hinge-back Tortoise individuals are currently kept in 25 institutions in six countries worldwide; most of the facilities are in the USA, a few in Europe and Asia, but none in Africa.

Implementation challenges (including similar species)

Since the species is traded across porous borders, regulating and halting the regional component of this transboundary trade will be challenging.

References

- CITES Secretariat (2025) Notification to the Parties No. 2025/102: Proposals to amend Appendices I and II. Provisional assessments by the Secretariat. Convention on International Trade in Endangered Species of Wild Fauna and Flora. Available at: <https://cites.org/sites/default/files/notifications/E-Notif-2025-102.pdf>
- Lawson, D.P. (2006) Habitat use, home range, and activity patterns of hingeback tortoises, *Kinixys erosa* and *K. homeana*, in southwestern Cameroon. *Chelonian Conservation and Biology* 5(1):48–56. [https://doi.org/10.2744/1071-8443\(2006\)5\[48:HUHRAA\]2.0.CO;2](https://doi.org/10.2744/1071-8443(2006)5[48:HUHRAA]2.0.CO;2)
- Luiselli, L., Agyekumhene, A., Akani, G.C., Allman, P., Diagne, T., Eniang, E.A., Mifsud, D.A., Petrozzi, F. and Segniagbeto, G.H. (2021a) *Kinixys homeana*. *The IUCN Red List of Threatened Species* 2021: e.T11003A18341580. <https://dx.doi.org/10.2305/IUCN.UK.2021-2.RLTS.T11003A18341580>
- Luiselli, L., Diagne, T. and McGovern, P. (2021b) Prioritizing the next decade of freshwater turtle and tortoise conservation in West Africa. *Journal for Nature Conservation* 60(125977) <https://www.sciencedirect.com/science/article/abs/pii/S1617138121000248?via%3Dihub>
- Luiselli, L., Angelici, F.M. and Akani, G.C. (2000) Habitat use, thermal ecology, and activity patterns of sympatric hinge-back tortoises (*Kinixys belliana* and *Kinixys homeana*) in the Niger Delta, Nigeria. *Chelonian Conservation and Biology* 3(4):735–742.

Inclusion of four species of water frogs in Appendix II: *Pelophylax epeiroticus*, *P. lessonae*, *P. ridibundus* and *P. shqipericus*

Proponents: European Union, Israel, North Macedonia

Summary: *Pelophylax epeiroticus*, *P. lessonae*, *P. ridibundus* and *P. shqipericus* belong to the taxonomically complex water frog genus *Pelophylax*, which currently contains 14 recognized species, with numerous clades awaiting taxonomic resolution. The four species are medium-sized Western Palearctic water frogs distributed across Europe, western Asia and northern Africa extensively harvested and traded for food, chiefly as frogs' legs for the EU market. Three of the species (*P. epeiroticus*, *P. ridibundus* and *P. shqipericus*) are proposed for inclusion in Appendix II under Criterion B in Annex 2a of Res. Conf. 9.24 (Rev. CoP17), with *P. lessonae* proposed as a 'lookalike' species under Criterion A in Annex 2b. No *Pelophylax* species are currently included in the Appendices. It is proposed that entry into force of the listing would be delayed by 18 months, i.e. until 5 June 2027.

The total EU imports of frogs' legs was estimated at 40,700 tonnes between 2010 and 2019, purported to equate to between 800 million and 2 billion frogs. The largest importers, accounting for the vast majority of demand, are Belgium, France, and the Netherlands, with most frogs believed to be sourced from wild populations in Indonesia, Viet Nam, Türkiye, and Albania. International trade involves a large number of species, including other *Pelophylax* species and frogs of other genera, and many frog species are also eaten locally in parts of their native range. A lack of species-specific data prevents accurate assessment of the impact of trade on the four focal species.

Pelophylax epeiroticus is restricted to wetland habitats along the Ionian coast from western Greece to southwestern Albania. The species was globally classified as Near Threatened on the IUCN Red List in 2023 due to a suspected population decline linked to habitat degradation. No population estimates are available, but it has been described as "quite abundant" and "common" across much of its range, with some subpopulations reported to be decreasing, particularly in Greece. The species is legally protected in Albania but is known to be harvested in both countries in its range. International trade in *P. epeiroticus* is poorly documented due to a lack of species-specific trade codes. Commercial trade in *P. epeiroticus* is presumed to be wild-sourced and primarily for human consumption.

Pelophylax ridibundus is widely distributed across Europe and parts of western Asia and was globally classified as Least Concern on the IUCN Red List in 2021 with a stable global population trend. It is reportedly the most commonly harvested *Pelophylax* species for international trade, particularly for frogs' legs. The trade in *P. ridibundus* is again poorly documented but has been reported from Türkiye, Bosnia and Herzegovina, Croatia, Montenegro, North Macedonia, Serbia, Slovenia, and likely Romania. Trade volumes in Türkiye, where the species is reportedly collected almost exclusively from the wild, suggest substantial extraction pressure. Mislabelling in trade has been reported at the species and genus level, but no seizures are recorded in EU-TWIX or WiTIS databases. Between 2015 and 2024, EU imports of frogs' legs from Türkiye, likely primarily *P. ridibundus*, totalled 1,221 tonnes, equivalent to an estimated 24–61 million individuals. Trade declined by 31% over the period, though it is unclear whether this reflected reduced demand, reporting inconsistencies, or population declines.

P. ridibundus is listed in Annex V of the EU Habitats Directive, allowing regulated use with monitoring. In Türkiye, the species is subject to national harvest regulations, whereby wild harvest is permitted but this is reportedly poorly monitored. Illegal harvest in this species has been reported in Montenegro.

Pelophylax shqipericus is endemic to the western Balkans, with a range limited to Albania and Montenegro. There are introduced populations in Italy and possibly Croatia. No population size or trend estimates are available. In 2023 it was globally classified as Vulnerable in the IUCN Red List because of its restricted distribution and ongoing threats from habitat loss and pollution. It was also reported to be threatened by over-collection in northern parts of its range, specifically at the transboundary Lake Skadar. Trade data specific to this species is lacking. Commercial trade in *P. shqipericus* for human consumption is presumed to be in wild-sourced individuals. In Montenegro, sourcing for export of frogs, including this species, during the 20th century are reported to have shifted toward illegal harvesting, although the extent of this illicit trade remains uncertain. No

seizures were recorded in EU-TWIX or WiTIS databases. Although included in Annex D of the EU Wildlife Trade Regulations, no EU trade has been recorded.

Pelophylax lessonae has a very wide range spanning much of Europe, from France in the west, east to the Russian Federation, and Scandinavia in the north, south to Bulgaria. It is currently globally assessed as Least Concern on the IUCN Red List (2022), although regional declines have been documented, particularly in mixed populations where hybridisation with *P. ridibundus* and the hybrid *P. kl. esculentus* has led to reduced frequency of *P. lessonae*. International trade is expected to be negligible, with no significant commercial trade reported although identification without molecular tools is unreliable.

The four *Pelophylax* species are morphologically similar, but live individuals can be generally differentiated based on a combination of morphological and morphometric traits.

Pelophylax shqipericus and *P. epeiroticus* show only slight morphological differences, despite clear genetic and bioacoustic distinctions. *P. ridibundus* tends to be larger and exhibits more pronounced sexual dimorphism, while *P. lessonae* is generally smaller with distinctive dorsal striping and males displaying white vocal sacs during the breeding season. However, most trade in *Pelophylax* species for human consumption is of skinned and processed legs, either fresh, chilled or frozen, or as live individuals intended for consumption as frogs' legs. Trade in legs would be difficult to differentiate by species.

Analysis: Of the four *Pelophylax* species considered, *P. epeiroticus* and *P. shqipericus* have relatively restricted distributions and are classified as Near Threatened and Vulnerable on the IUCN Red List, respectively. Their conservation status reflects concerns about collection pressure for international trade. *P. ridibundus* has a much wider distribution and is listed as Least Concern, although some populations are reported to be under significant harvest pressure. *P. lessonae* is also listed as Least Concern, but concerns have been raised primarily in relation to trade in hybrids involving this species and *P. ridibundus*, which can also hybridise with *P. epeiroticus* and *P. shqipericus*.

All four species are believed to contribute to the high-volume frogs' legs trade, particularly for EU markets, with exports from range States such as Albania (where all of the proposed species except *P. lessonae* are native) and Türkiye (where of the four proposed species only *P. ridibundus* occurs) being especially notable. However, available trade data do not distinguish volumes at the species level, further obscured by misidentification in national reporting. While *P. ridibundus* is thought to be the most heavily traded species, this cannot be confirmed due to the lack of species-specific data. Other species, including other *Pelophylax* species also occurring in the main exporting countries also contribute to the frogs' legs trade. Overall, **the uncertainty surrounding species-specific trade volumes and their impact on individual species, particularly for *P. lessonae*, hampers assessment of whether any of them meet the criteria for inclusion in Appendix II under Res. Conf. 9.24 (Rev. CoP17).**

In the case of *P. lessonae* the main concern appears to be trade in hybrids with *P. ridibundus*. Should *P. ridibundus* be listed, trade in hybrids occurring across Eastern Europe would be subject to the same controls, in line with guidance under Res. Conf. 10.17 (Rev. CoP14) on *Animal Hybrids*.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

As noted in Proposal 27 to include four species of Eurasian frogs in Appendix II, the phylogeny, taxonomy and nomenclature of these frogs has been exceedingly complicated, due to extensive hybridization between taxa and extensive geographic and individual variability in morphology. Annex 1 to the Proposal is a Checklist extracted from the Amphibians of the World database, from which extracts for other amphibian groups are already adopted as CITES standard references for amphibian nomenclature. The proposal notes that this Annex be adopted as the standard nomenclature reference for these taxa if the Proposal is adopted; it would then be included in the Annex to

Res. Conf. 12.11 (Rev.) (CoP20 Inf. 6). *Pelophylax ridibundus* now comprises the formerly recognised species of *P. bedriagae*, *P. caralitanus*, *P. cerigensis*, *P. cypriensis*, *P. kurtmuelleri*, *P. persicus*, and *P. terentievi*.

Range

Table 1. Species and geographic range included in this proposal

Species	Distribution
<i>Pelophylax epeiroticus</i>	Albania, Greece
<i>Pelophylax lessonae</i>	Austria, Belarus, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Czech Republic, Denmark, Estonia, Finland (introduced), France, Germany, Hungary, Italy, Latvia, Liechtenstein, Lithuania, Luxembourg, Netherlands, Norway, Poland, Republic of Moldova, Romania, Russian Federation, Serbia, Slovakia, Slovenia, Spain (introduced), Sweden, Switzerland, Ukraine, United Kingdom of Great Britain and Northern Ireland (UK) (introduced).
<i>Pelophylax ridibundus</i>	Afghanistan, Albania, Armenia, Austria, Azerbaijan, Bahrain, Belarus, Belgium (introduced), Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Egypt, Estonia (introduced), Finland (introduced), France, Georgia, Germany, Greece, Hungary, Iraq, Israel, Italy (introduced), Jordan, Kazakhstan, Kyrgyzstan, Latvia, Lebanon, Liechtenstein, Lithuania, Luxembourg, Malta (introduced), Montenegro, Netherlands, North Macedonia, Pakistan (uncertain), Poland, Republic of Moldova, Romania, Russian Federation, Saudi Arabia, Serbia, Slovakia, Slovenia, Spain (introduced), Switzerland (introduced), Syrian Arab Republic, Tajikistan, Türkiye, Turkmenistan, Ukraine, UK (introduced), Uzbekistan.
<i>Pelophylax shqipericus</i>	Albania, Croatia (introduced), Italy (introduced), Montenegro

IUCN Global Category

Table 2. Species included in this proposal, the associated IUCN Red List (version 2025-1) global category (all using IUCN Red List Categories and Criteria version 3.1). Since the 2021 assessment of *Pelophylax ridibundus* several formerly recognised species have been synonymised with *P. ridibundus*, including *P. bedriagae*, *P. caralitanus*, *P. cerigensis*, *P. cypriensis*, *P. kurtmuelleri*, *P. persicus*, and *P. terentievi*.

Species	IUCN Global Category	Year assessed
<i>Pelophylax epeiroticus</i>	Near Threatened B1ab(iii)	2023
<i>Pelophylax lessonae</i>	Least Concern	2022
<i>Pelophylax ridibundus</i>	Least Concern	2021
<i>Pelophylax shqipericus</i>	Vulnerable B1ab(iii,v)	2023

Biological and trade criteria for inclusion in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)

A) Trade regulation needed to prevent future inclusion in Appendix I

Pelophylax epeiroticus Epirus Water Frog

Distribution

The Epirus Water Frog *Pelophylax epeiroticus* is restricted to western Greece along the Ionian coast and from the district Sarandë in south-western Albania, from sea level up to around 500 metres above sea level. Currently, known only from wetland localities west of the Pindos Mountain range, potential observations to the east of the Pindos range outside its current geographic range require validation (Mizsei et al., 2024). The species' estimated extent of occurrence is 25,660 km², with an area of occupancy of 332–400 km², and its area of habitat estimated at 4,541 km² (Ridley et al., in prep).

Biology

Pelophylax epeiroticus reach sexual maturity on average after one year. Breeding taking place from February to May. Clutch size for the species was noted at 2,575 eggs but varies with female age and size. No information on the species' metamorphosis age, hibernation pattern or survival rate was identified.

Population trends

Populations of the species were historically described as "quite abundant" and "common" throughout much of its range, however, some subpopulations are believed to be decreasing due to a reduction in the quality and extent of suitable habitat. The species' population trend in Greece was reported to be decreasing (Mizsei et al., 2024b).

Pelophylax ridibundus Marsh Frog

Distribution

The species has a widespread distribution in western, central and eastern Europe, western and central Asia, and northern Africa (Egypt). The species is found at elevations up to 2,500 metres above sea level.

Biology

Pelophylax ridibundus reaches sexual maturity at two years in males, and three years in females in the wild, one year in captivity. Breeding takes place from January to June, with indications of late breeding periods in some subpopulations (e.g., May to late July for a population along the floodplains of Dniro River in Ukraine, and May/June to mid-September depending on altitude). *Pelophylax ridibundus* was reported to be capable of spawning multiple clutches within a breeding season, with clutch size noted to range between 3,853 to 16,000 eggs. Metamorphosis in *P. ridibundus* is dependent on environmental factors, but typically occurs between April and November, lasting approximately 45 to 65 days. Survival rates of *P. ridibundus* juveniles in their first year is approximately 50%, while the survival rates of adults are approximately 70%. Longevity in *P. ridibundus* was eight to nine years in males and nine to ten years in females.

Population trends

The species is currently generally considered to be common to abundant throughout its range. Population densities of the species vary significantly across regions. In the Republic of Moldova, densities range from 63 individuals/hectare (ha) in lakes to 150–175 individuals/ha in riverine ponds. The species was reported to be relatively sparse on Karpathos Island but more abundant on the island of Rhodes. Subpopulations are considered common in Egypt and Lebanon, though earlier reports in 2013 described the species as rarely observed in Egypt. In Israel, subpopulations declined significantly between the 1950s and 1970s but have since stabilized in 2020 due to the creation of suitable artificial habitats.

Trade

In Türkiye, in 2021, *P. ridibundus* was extensively harvested for the frog-leg trade, which may be contributing to population declines in some areas.

***Pelophylax shqipericus* Albanian Water Frog**Distribution

This species is native to western coastal Albania and southern Montenegro at elevations below 500 m above sea level. There is also evidence of limited introduction to mainland Italy and Croatia.

Biology

P. shqipericus was reported to have a clutch size of 3,000 eggs with breeding taking place from March to May. No information on the species' sexual maturity, metamorphosis age, hibernation pattern or survival rate was identified.

Population trends

The species is considered to be "quite rare" but locally abundant. Populations are severely fragmented, and the species extent of occurrence is estimated at 10,387 km². The population was suspected to be decreasing due to a reduction in the extent and quality of habitats and over collection for use as fishing bait (Borzee in litt., 2025). However, no rate of population decrease was available.

Trade

Harvesting for human consumption is a major threat to *P. shqipericus*, especially in Lake Skadar and surrounding areas, where collection occurs even during the breeding season, as reported in 2010, 2015 and 2020. The species is also collected for the international pet trade. Previously, commercial collection for human consumption was reported to be a significant threat to *P. epeiroticus*, with reports of large-scale harvesting in Albania. However, information from a 2019 Europe Red List Assessment Workshop indicated that there was no evidence of excessive collection in Albania at the time, and while it is no longer considered a major threat in 2024, concerns remain regarding ongoing collection and misreporting.

***Pelophylax lessonae* Pool Frog**Distribution

The Pool Frog *Pelophylax lessonae* has a broad distribution across much of Europe, native to over 30 countries, and has been introduced to the UK and Spain. The species typically inhabits lowland areas with stagnant or slow-flowing water bodies such as ponds, swamps, and ditches, often surrounded by dense herbaceous vegetation.

Biology

Pelophylax lessonae reaches sexual maturity at 2–3 years of age, with a maximum lifespan of 6–12 years depending on population. Breeding occurs from late March to June, with males forming choruses and engaging in axillary amplexus. Clutch sizes range from 440 to 4,400 eggs, with metamorphosis typically occurring between July and August, though some tadpoles may overwinter. Hibernation takes place from late autumn to early spring, primarily in aquatic environments, though terrestrial hibernation in burrows has been recorded.

Population trends

Pelophylax lessonae is currently listed as Least Concern on the IUCN Red List (2022) with an unknown population trend. Regional declines have been documented due to hybridisation with *P. ridibundus* and *P. kl. esculentus*, however, it remains widespread and common in Poland, the Baltic Republics, Republic of Moldova, Russia and Ukraine.

B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences

National use

All four species, *P. epeiroticus*, *P. ridibundus*, *P. shqipericus*, and *P. lessonae* were reported to be exploited for the frog-leg trade for human consumption at the local or national level in some range States, and to a lesser extent collected live for the pet trade or ornamental purposes.

In Albania, where all species except *P. lessonae* occur, frogs' legs from wild-caught *Pelophylax* species are consumed locally, particularly in central and southern regions, and live frogs are also collected for the pet trade. *P. epeiroticus* was once harvested in "enormous numbers" in the country. In Türkiye, wild harvest of *Pelophylax* species has occurred at large volumes for over 40 years, with individual collectors reportedly catching 40–60 kg of frogs per day. Despite this, domestic consumption is minimal and largely limited to tourist-focused restaurants, with most harvest destined for international trade.

In Montenegro, harvesting of *Pelophylax* species, including the protected *P. shqipericus*, continues in areas such as the Bojana/Buna River delta, driven by tourist demand despite being nationally protected. In northern Italy, harvests of *P. ridibundus*, or misidentified *Pelophylax*, or hybrids of *P. ridibundus*, for local consumption are complemented by individuals imported from Albania and Türkiye, among others (Borzee in litt., 2025).

International trade

Based on the available information in the SS, the only species of the *Pelophylax* genus being traded internationally for commercial purposes are *P. epeiroticus*, *P. ridibundus*, and *P. shqipericus*, and the hybrid *P. kl. esculentus* (*P. ridibundus* x *P. lessonae*). Trade volumes for *Pelophylax* species are uncertain, and there are no species-specific Harmonised System tariff codes (HS Codes) for trade in any commercial forms. *Pelophylax shqipericus* is included in Annex D of the EU Wildlife Trade Regulations which monitors EU trade in non-CITES listed species. No EU trade in *P. shqipericus* has been recorded. In the SS, this was reported to be due to a lack of reporting rather than an absence of trade.

The total EU trade in frogs' legs has been estimated at 40,700 tonnes between 2010 and 2019, equating to between 800 million and 2 billion frogs (Auliya et al., 2023). According to EUROSTAT data, imports of frogs' legs under HS code 02089070 (frogs legs; fresh, chilled or frozen) by the EU from the four *Pelophylax* species' range States outside the EU, over the most complete recent ten years, 2015–2024, were imported predominately from Türkiye and Albania (EUROSTAT, 2025). However, it is important to note that EUROSTAT data are not species-specific, and no information could be found to confirm whether Albania or Türkiye export frog genera other than *Pelophylax* for the frogs' legs trade. Additionally, EUROSTAT does not indicate whether the exporting country is also the country of origin. In the absence of evidence for captive-breeding facilities in Albania, it is assumed that the majority of exports involve wild-sourced specimens.

International trade exports in *P. ridibundus* is reported to take place from Türkiye, Bosnia and Herzegovina, Croatia, Montenegro, North Macedonia, Serbia, Slovenia, and likely Romania (Borzee in litt., 2025).

Trade from Türkiye

Pelophylax species native to Türkiye: *P. ridibundus*, *P. bedriagae*, *P. caralitanus* (according to the IUCN Red List). Imports to the EU from Türkiye, 2015–2024, totalled 1,221 tonnes and averaged 122 tonnes per year (EUROSTAT, 2025). Applying a conversion factor presented in Veith et al. (2000) estimating that one kilogram of frogs' legs represented between 20 to 50 individuals, total trade from Türkiye equated to 24.4 – 61.0 million individuals, and averaged 2.4–6.1 million individuals per year. This trade decreased by 31% over the ten-year period. It was reported that in Türkiye frogs are harvested almost exclusively from the wild, and that it was unclear whether the decrease in exports was due to reporting issues, a decrease in demand, or reduced availability due to population reductions. No domestic consumption of frogs was reported in Türkiye and all harvest was therefore assumed to be for international trade.

It was reported that the only *Pelophylax* species subject to harvest and wild-sourced export in Türkiye was *P. ridibundus* [previously referred to as *P. ridibunda*], noting that captive-bred individuals were also exported and that "*P. esculenta*" (*P. kl. esculentus*) was imported from Azerbaijan for processing and re-export.

P. bedriagae is considered to contribute to the frog's legs trade (IUCN SSC Amphibian Specialist Group, 2022), and *P. caraltanus* is the largest edible frog in Türkiye and is commercially collected for trade (IUCN SSC Amphibian Specialist Group, 2023).

The hybrid, *P. kl. esculentus* [as *P. esculenta*], is considered by Frost (2025) to be a synonym of both *P. lessonae* and *P. ridibundus*. The only *Pelophylax* species native to Azerbaijan is *P. ridibundus*. A 2021 study reported that frog-exporting companies in Türkiye often mislabel all exports as *P. esculentus*. Turkish exporters were also reported to import frogs from the Syrian Arab Republic, a country that is a range State only for *P. ridibundus* [previously referred to as *P. bedriagae*].

Trade from Albania

Pelophylax species native to Türkiye: ***P. epeiroticus*, *P. shqipericus*, *P. kurmuelleri*** (according to the IUCN Red List).

Albania was reported to be the EU's fourth largest supplier of frogs' legs. It was reportedly presumed that these frogs were sourced from wild subpopulations, based on the lack of evidence of established commercial breeding farms for the three *Pelophylax* species. No information was provided in the SS on whether Albania frog legs' exports included other *Pelophylax* species, although according to the IUCN Red List *P. kurmuelleri* is (Andreone et al., 2024). It was noted that Albania is also a range State of *Rana dalmatina* identified as being a species of economic value in Türkiye. However, morphological differences between *Rana* and *Pelophylax* species enable differentiation, even in the frog leg trade form (Borzee in litt., 2025). The SS notes that as no trade in *P. shqipericus* was reported in the EU Wildlife Trade Regulations, this implied that either all wild-sourced trade in *Pelophylax* frogs' legs from Albania were only *P. epeiroticus* or *P. ridibundus*, or that *P. shqipericus* was included in exports but mislabelled.

Imports to the EU from Albania, 2015-2024, totalled 377 tonnes and averaged 38 tonnes per year (EUROSTAT, 2025). Applying the same Veith et al. (2000) conversion factor, total trade from Albania equated to 7.5 – 18.8 million individuals, and averaged 0.8-1.9 million individuals per year. This trade decreased by 89% over the ten-year period, however, no data was provided in 2021 and no explanation for this was provided.

The SS noted that the trade data derived from various sources was unreliable due to the discrepancies in reporting. These discrepancies were reportedly caused by factors including incomplete reporting, differences in calculating trade volumes in each dataset, or a lack of reporting of live frog trade, such as in the Eurostat Comext database.

Data from LEMIS on trade into and from the USA from 2008 to 2020 only include one species of the genus, *Pelophylax fukiensis*, with 18 export events involving 124 wild-sourced specimens traded for commercial purposes to various countries between 2018 and 2020.

Illegal trade

The illegal harvest and collection of *Pelophylax shqipericus* for commercial purposes has been identified as a threat to the species, as evidenced by a 2020 study in the Bojana/Buna River delta area of southern. In Montenegro, a shift has been observed from legal exports during the 20th century to the illegal collection of green frogs, including *P. ridibundus* and *P. shqipericus*, although the scale of this activity appears to be smaller than in the past.

A 2019 study reported uncontrolled and illegal harvest of frogs has been reported in Edirne Province, Türkiye, and a 2017 newspaper report in Türkiye highlighted a case of large-scale illegal harvesting and sale of frogs to frog farms.

No records for any *Pelophylax* species were reported in the EU-TWIX seizure database or TRAFFIC's WiTIS database.

Inclusion in Appendix II to improve control of other listed species

A) Specimens in trade resemble those of species listed in Appendix II under Res. Conf. 9.24 (Rev. CoP17) Annex 2 a or listed in Appendix I

The four *Pelophylax* species are morphologically similar, but differentiation among the species was possible based on a combination of morphological and morphometric traits. However, comprehensive differentiation among hybrids is not possible (Borzee in litt., 2025).

Pelophylax lessonae and *P. ridibundus* were noted to be significantly morphologically different, but that their hybrids possess a combination of traits from their putative parent species. *Pelophylax lessonae* does not occur in the main exporting countries of Albania and Türkiye.

In accordance with the treatment of animal hybrids in CITES (Res. Conf. 10.17 (Rev. CoP14)), if at least one of the animals in the recent lineage is of a species included in Appendix II, and there are no specimens of an Appendix-I species in such lineage, the hybrids shall be treated as specimens of species included in Appendix II.

Pelophylax species are primarily traded as frogs' legs (skinned and processed, either fresh, chilled or frozen), or as live individuals intended for human consumption as frogs' legs, with trade as pets or ornamental purposes occurring to a much lesser extent. *It is assumed that in the form of processes legs it would be difficult to differentiate any species.*

Additional Information

Threats

All four species, *Pelophylax epeiroticus*, *P. shqipericus*, *P. lessonae* and *P. ridibundus*, face varying degrees of threat, primarily linked to habitat degradation, pollution, disease, and harvest pressure, though the severity and impact differ across species and regions. *Some of the species' subpopulations, recognised previously as valid species, and likely to be resurrected in the future, are experiencing higher levels of threat due to habitat loss, climate change and exploitation (Borzee in litt., 2025).*

Pelophylax epeiroticus and *P. shqipericus* were reported in 2024 to be particularly vulnerable to wetland loss and fragmentation due to urbanisation, tourism, agriculture, and infrastructure development. Recent reports, 2023-2025, note that in Albania and Montenegro, aquatic pollution from agrochemicals, industrial waste, and mining is widespread, with additional impacts from illegal construction, sand excavation, and solid waste disposal.

P. ridibundus is generally adaptable and resistant to pollution, but in 2023 reported to be experiencing localised declines due to habitat loss and prolonged droughts. In eastern Europe and Türkiye, frog-leg trade and industrial pollution have been linked to significant population declines, *with modelled Pelophylax subpopulations under current harvest levels in Türkiye suggesting a high probability of local extirpation within 50 years (Çiçek et al., 2021).*

P. ridibundus reportedly lacks specific legal protection in both Albania and Greece (Jablonski in litt., 2025).

Additional threats include climate change, particularly droughts affecting breeding sites, and genetic erosion through hybridisation with sympatric species. For *P. shqipericus*, competition from the introduced *Aquarana catesbeiana* may also pose a risk.

Conservation, management and legislation

All four species in this proposal are included in Appendix III of the EU Bern Convention (Convention on the Conservation of European Wildlife and Natural Habitats) as protected fauna species. *Pelophylax ridibundus* and *P. lessonae* have been protected since 1992 by the EU Habitats Directive (Council Directive 92/43/EEC). Wherein *P. ridibundus* (as "*Rana ridibunda*") is listed in Annex V of the Directive which requests strict regulations against over-exploitation, and *P. lessonae* (as "*Rana lessonae*") is listed in Annex IV includes animal and plant species of community interest in need of strict protection. *Pelophylax shqipericus* is included in Annex D of the EU Wildlife Trade Regulations which monitors trade volumes of non-CITES listed species into the EU.

Pelophylax epeiroticus (listed as *Rana epeirotica*) is listed as vulnerable in Albania's Red List of Wild Flora and Fauna (Order No. 1280, 20/11/2013). The Albanian law on wild fauna (No.10006 23.10.2008) contains provisions for the protection of species based on species' national Red List status, including the prohibition of capturing or killing individuals of wild fauna. A moratorium on hunting of wild fauna in Albania was declared by law in 2014 (Order No. 7/2014) for a period of two years. The moratorium was extended for five years in 2016 (Order No. 61/2016) and for an additional three years in 2022 (Order No. 60/2022), with exceptions for hunting "overpopulated" species, and for scientific research. The implementation of the hunting ban across the country was reported to have varied, with a lack of controls and regular monitoring of populations reported in some areas (Ruppert, 2018). It is not clear whether the capture of frogs would be included within the scope of this legislation. In Greece, *P. epeiroticus* is protected under Presidential Decree 67/1981, which lists protected native species of fauna and provides legal protection against collection, killing, or disturbance of listed species.

Pelophylax shqipericus was noted to be found within the Lake Skadar protected area in Albania and Montenegro (IUCN SSC Amphibian Specialist Group 2020b; 2022b), however, it is unclear whether *P. epeiroticus* occurs in any protected areas in Albania.

*In Türkiye, frog harvesting was reported to be regulated by the Communiqué on the Regulation of Commercial Fisheries No. 5/1, with closed harvest seasons for particular provinces during which take is prohibited (CITES MA of Türkiye in litt. to the European Commission, 2023 in UNEP-WCMC, 2023). Frog harvesting was only permitted with a licence, and individuals could not be processed or traded without a "Special Product Certificate of Origin" (CITES MA of Türkiye in litt. to the European Commission, 2023 in UNEP-WCMC, 2023). For *P. ridibundus* [as *P. ridibunda*] a minimum catch weight of 30 g was reported to have been adopted and in 2023, monitoring of *P. ridibundus* [P.*

ridibunda] was reported to take place across a range of harvest locations in ten provinces (CITES MA of Türkiye in litt. to the European Commission, 2023 in UNEP-WCMC, 2023).

Artificial Propagation/captive breeding

According to the Species360 ZIMS database for zoos, a total of 19 *Pelophylax ridibundus* individuals are held in nine ex-situ holding institutions around the world. There are no current records for *P. epeiroticus*, *P. shqipericus* or *P. lessonae*.

Approximately 12 frog farms were reported to have been established in Türkiye by studies in 2015 and 2021, and these farms contributed an average of 45 t annually to frog exports and totalled 180 t during 2014-2017. This appears to comprise only a small proportion of exports described above. Several farms were noted to breed *P. ridibundus*, including a farm in Aydıncık, Adana, İstanbul, and Mersin.

Frog farming of *Pelophylax* spp. on a commercial level with the aim to replace large-scale imports of wild-caught animals was reported to have had variable success in European countries by studies in 2009, 2019 and 2022, but failed in Egypt as reported in 2013. The captive breeding of *P. lessonae* and *P. kl. esculentus* was reported to be challenging.

Implementation challenges (including similar species)

The continued collection of *Pelophylax* species in Albania presents an ongoing implementation challenge, particularly due to persistent misidentification and misreporting in trade records involving closely related congeners. This undermines efforts to monitor and regulate harvest and trade effectively.

Potential risk(s) of a listing

Çiçek et al. (2021) mentioned that frog harvesters in Türkiye traditionally have low incomes, no permanent jobs and no social benefits, choosing to harvest frogs due to the lack of capital investment required. As the impact of the Appendix-II listing of these *Pelophylax* species on livelihoods is unclear, potential negative impacts on livelihoods should be considered in line with Res. Conf. 16.6 (Rev. CoP18) on CITES and livelihoods.

Experts report that in the event the species were listed, it would prove challenging to undertake Non-Detriment Findings for the species due to the lack of population size estimates (Cogalniceanu in litt., 2025).

Other comments

Studies have identified multiple invasive alien frog lineages, primarily *Pelophylax ridibundus*, across approximately 20 invasion sites, mainly in Western Europe (Bellati et al., 2019; Di Nicola and Ribo Ferrier, 2022; Dufresnes et al., 2024). Genetic analyses traced these lineages predominantly to the Balkans and Anatolia, consistent with export patterns linked to the frog leg trade and the pet industry (Dufresnes et al., 2024). These findings underscore the importance of considering the risks associated with international trade in live *Pelophylax* species, as the subsequent release of potentially invasive alien frogs may pose significant threats to native amphibian populations through competition, hybridization, and disease transmission (Altherr; and Cogalniceanu in litt., 2025).

References

- AC Nomenclature Specialist (2025) in litt. to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Altherr, S. (2025) in litt. to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Andreone, F., Haxhiu, I., Vogrin, M., Lymberakis, P., Uzzell, T., Litvinchuk, S. and Jablonski, D. (2024) *Pelophylax kurtmuelleri*. The IUCN Red List of Threatened Species 2024: e.T58637A229003815. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T58637A229003815.en>. Accessed on 7 September 2025.
- Auliya, M., Altherr, S., Nithart, C., Hughes, A. and Bickford, D. (2023) Numerous uncertainties in the multifaceted global trade in frogs' legs with the EU as the major consumer. *Nature Conservation* 51:71–135. <https://doi.org/10.3897/natureconservation.51.938682025>
- Bellati, A., Bassu, L., Nulchis, V. and Corti, C. (2019) Detection of alien *Pelophylax* species in Sardinia (western Mediterranean, Italy). *BioInvasions Records* 8(1):8–25 <https://doi.org/10.3391/bir.2019.8.1.02>
- Borzee, A. (2025) in litt. to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Çiçek, K., Ayaz, D., Afsar, M., Bayrakçı, Y., Pekşen, Ç.A., Cumhuriyet, O., İsmail, İ.B., Yenmiş, M., Üstündağ, E., Tok, C.V. and Bilgin, C.C. (2021) Unsustainable harvest of water frogs in southern Turkey for the European market. *Oryx* 55(3):364–372.
- Cogalniceanu, D. (2025) in litt. to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Di Nicola, M. and Ribo Ferrier, J. (2022) First case of predation of the Tyrrhenian endemic Sardinian treefrog *Hyla sarda* by an alien water frog *Pelophylax* sp. in northern Sardinia, Italy. *The Herpetological Bulletin* 161:40–41.
- Dufresnes, C., Monod-Broca, B., Bellati, A., Canestrelli, D., Ambu, J., Wielstra, B., Dubey, S., Crochet, P.-A., Denoël, M. and Jablonski, D. (2024) Piecing the barcoding puzzle of Palearctic water frogs (*Pelophylax*) sheds light on amphibian biogeography and global invasions. *Global Change Biology* 30(3): e17180.

- EUROSTAT (2023) EU trade since 1988 by HS2-4-6 and CN8 [DS-045409__custom_7109450]. Import data 2014–2024 for the commodity groups 02082000 and 02089070, frog legs fresh, refrigerated, or frozen. Available at: https://ec.europa.eu/eurostat/databrowser/view/ds-045409__custom_17617140/default/table. Accessed on 29 July 2025.
- IUCN SSC Amphibian Specialist Group (2022) *Pelophylax bedriagae*. *The IUCN Red List of Threatened Species* 2022: e.T58893474A173829632. <https://dx.doi.org/10.2305/IUCN.UK.2022-1.RLTS.T58893474A173829632.en>. Accessed on 7 September 2025.
- IUCN SSC Amphibian Specialist Group (2023) *Pelophylax caralitanus*. *The IUCN Red List of Threatened Species* 2023: e.T135806A176124496. <https://dx.doi.org/10.2305/IUCN.UK.2023-1.RLTS.T135806A176124496.en>. Accessed on 7 September 2025.
- Mizsei, E., Haxhiu, I., Sotiropoulos, K., Lymberakis, P., Jablonski, D. and Uzzell, T. (2024a) *Pelophylax epeiroticus*. *IUCN Red List of Threatened Species* 2024: E.T58592A207981785. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T58592A207981785.en>. Accessed 28 July 2025.
- Mizsei, E., Crnobrnja-Isailović, J., Jablonski, D. and Uzzell, T. (2024b) *Pelophylax shqipericus*. *The IUCN Red List of Threatened Species* 2024: e.T58715A207982996. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T58715A207982996.en>. Accessed on 28 July 2025.
- Species360 Zoological Information Management System (ZIMS) (2025) Available at: zims.Species360.org.
- UNEP-WCMC (2023) Review of EU trade in *Pelophylax* species from Albania and Türkiye. UNEPWCMC, Cambridge.

Transfer of Oceanic Whitetip Shark *Carcharhinus longimanus* from Appendix II to Appendix I

Proponents: Argentina, Bahamas, Brazil, Comoros, Dominican Republic, Ecuador, European Union, Fiji, Gabon, Honduras, Lebanon, Oman, Panama, Samoa, Senegal, Seychelles, Sri Lanka, Sudan, Togo, United Kingdom of Great Britain and Northern Ireland

Summary: The Oceanic Whitetip Shark *Carcharhinus longimanus* is a globally distributed, large (3 m), pelagic shark species that is highly migratory, has low-moderate productivity, and relatively low reproductive rates. It can be found in epipelagic tropical and subtropical waters between 30° North latitude and 35° South latitude.

The main threat to *Carcharhinus longimanus* is overexploitation, primarily as a result of incidental secondary catch in commercial fisheries. Although it is not typically targeted, its habit of staying near the surface in tropical waters—where fishing activity is intense—makes it especially vulnerable. Most of the catch is taken as secondary catch of commercial pelagic fleets in offshore and high-seas waters but it is also captured in coastal longlines, gillnets, trammel nets and sometimes trawls. Its fins are large and of high value within the trade.

In 2013, *Carcharhinus longimanus* was included in CITES Appendix II under the family listing for Carcharhinidae. Since 2014, 45,109 kg of fins have been reported as exported for commercial purposes from India, Kenya, Oman, Papua New Guinea, Senegal, Seychelles, Sri Lanka and Yemen, along with 509 individual fins.

Having once been one of the most abundant pelagic shark species in tropical seas worldwide, *Carcharhinus longimanus* was globally assessed in 2018 as Critically Endangered on the IUCN Red List, with populations severely reduced in all oceans. Based on modelling of Catch Per Unit Effort (CPUE) and stock assessment data, the global population was estimated as to have undergone a reduction possibly more than 98%, with the highest probability of more than an 80% reduction over three generation lengths (61.2 years). The South Atlantic, which comprises a small proportion of the range, was the only region where the species was assessed as Least Concern, based in part on conflicting CPUE data. *Carcharhinus longimanus* has been the subject of the CITES Review of Significant Trade process in 2023. The process resulted in Senegal establishing zero export quotas in 2024 for wild-sourced trade and for all specimens taken in “the marine environment not under the jurisdiction of any State”; the same quotas were established by Kenya and Yemen in 2025. Oman also established a zero-export quota for commercial trade in 2024.

All tuna Regional Fisheries Management Organizations (tRFMOs) established retention bans between 2010 and 2013, and over 31 countries have fully protected the species. *Carcharhinus longimanus* appears to have relatively high at-vessel survivorship compared to other sharks, suggesting individuals caught and released alive may be able to survive. However, there is currently insufficient evidence to determine the impact of these retention bans and more recent zero-export quotas and whether global population abundance continues to decline, has reached a plateau, or is showing signs of increase.

Analysis: *Carcharhinus longimanus* does not have a restricted area of distribution and it is unlikely to have a small population. An assessment in 2018 found that the global population was estimated to have undergone a reduction potentially over 98%, with the highest probability of more than an 80% reduction over three generations. This meets the criteria for inclusion in Appendix I for a historical extent of decline for a commercially exploited aquatic marine species with low-moderate productivity as set out in the footnote to Annex 5 of Res. Conf. 9.24 (Rev. CoP17). Despite retention bans, there is evidence that incidentally caught *Carcharhinus longimanus* are still retained for sale. Substantial trade levels have been reported by CITES Parties within the past five years and there is direct and indirect

evidence of ongoing and illegal trade in fins, likely placing pressure on any recovery of the species. The species therefore **appears to meet the criteria for inclusion in Appendix I**.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

The Oceanic Whitetip Shark was included in CITES Appendix II at CoP13 with delayed entry coming into effect until 14 September 2014.

A family listing for all other Carcharhinidae spp. was adopted at CoP19 in November 2022 with entry into effect delayed by 12 months (UNEP, 2025).

Taxonomy

Carcharhinus longimanus (Poey, 1861)

Range

Carcharhinus longimanus occurs worldwide in tropical and temperate waters.

American Samoa; Angola; Anguilla; Antigua and Barbuda; Argentina; Aruba; Australia; Bahamas; Bangladesh; Barbados; Belize; Benin; Bermuda; Bonaire, Sint Eustatius and Saba; Brazil; British Indian Ocean Territory (Chagos Archipelago); Brunei Darussalam; Cabo Verde; Cambodia; Cameroon; Cayman Islands; China; Christmas Island; Cocos (Keeling) Islands; Colombia (Colombia (mainland), Colombian Caribbean Is.); Comoros; The Democratic Republic of the Congo; Cook Islands; Costa Rica; Cuba; Curaçao; Côte d'Ivoire; Djibouti; Dominica; Dominican Republic; Ecuador (Galápagos, Ecuador (mainland)); Egypt; El Salvador; Equatorial Guinea; Eritrea; Fiji; France (Clipperton I., France (mainland)); French Guiana; French Polynesia; Gabon; Gambia; Ghana; Grenada; Guadeloupe; Guam; Guatemala; Guinea; Guinea-Bissau; Guyana; Haiti; Honduras; Hong Kong SAR of China; India; Indonesia; Israel; Jamaica; Japan; Kenya; Liberia; Macao SAR of China; Madagascar; Malaysia; Maldives; Marshall Islands; Martinique; Mauritania; Mauritius; Mexico; Montserrat; Morocco; Myanmar; Nauru; New Caledonia; New Zealand; Nicaragua; Niue; Northern Mariana Islands; Oman; Pakistan; Palau; Panama; Papua New Guinea; Peru; Philippines; Pitcairn; Portugal (Madeira, Azores); Puerto Rico; Réunion; Saint Barthélemy; Saint Helena, Ascension and Tristan da Cunha; Saint Kitts and Nevis; Saint Lucia; Saint Martin (French part); Saint Vincent and the Grenadines; Samoa; Sao Tome and Principe; Saudi Arabia; Senegal; Seychelles; Sierra Leone; Singapore; Sint Maarten (Dutch part); Slovenia; Solomon Islands; Somalia; South Africa; Spain (Canary Is., Spanish North African Territories, Spain (mainland)); Sri Lanka; the Sudan; Suriname; Taiwan Province of China; United Republic of Tanzania; Thailand; Togo; Tokelau; Tonga; Trinidad and Tobago; Turks and Caicos Islands; Tuvalu; United States (Hawaiian Is.); United States Minor Outlying Islands (Johnston I., Wake Is.); Uruguay; Vanuatu; Venezuela (Bolivarian Republic of); Viet Nam; Virgin Islands, British; Virgin Islands, U.S.; Yemen (Rigby et al., 2019)



Figure 1. Map of the Oceanic Whitetip Shark's distribution (Rigby et al., 2019).

FAO Fishing Areas: 21 (Atlantic – northwest), 27 (Atlantic – northeast), 31 (Atlantic - western central), 34 (Atlantic - eastern central), 41 (Atlantic – southwest), 47 (Atlantic – southeast), 51 (Indian Ocean – western), 57 (Indian Ocean – eastern), 61 (Pacific – northwest), 71 (Pacific - western central), 77 (Pacific - eastern central), 81 (Pacific – southwest) and 87 (Pacific – southeast) (Rigby et al., 2019)

IUCN Global Category

Critically Endangered A2bd (assessed 2018, ver 3.1)

Biological criteria for inclusion in Appendix I**A) Small wild population**

See C) below.

B) Restricted area of distribution*Carcharhinus longimanus* is globally distributed (Figure 1).**C) Decline in number of wild individuals**

There are no data available on the global population size of *Carcharhinus longimanus* and therefore abundance indices are used to determine population trends (Table 1).

In 2018, the IUCN Red List global assessment categorised the species as Critically Endangered A2bd ver 3.1. The global population was estimated to have undergone a reduction of >98%, with the highest probability of a >80% reduction over three generation lengths (61.2 years). *The IUCN Red List assessment examined population declines over the past three generation lengths which is required when assessing population reduction under Criterion A2. The JARA analysis provides a projection which gives similar results to extrapolating backwards to attain a three-generation length period, as recommended when estimating population reduction from annual rates of change (IUCN Standards and Petitions Subcommittee 2019) (Sherley et al., 2020). As such, the JARA declines in the Carcharhinus longimanus Red List assessment Supplementary Information are historical extents of decline (Rigby in litt., 2025).*

Table 1. Abundance indices available to assess population trends of *Carcharhinus longimanus*. *Denotes data used in the latest IUCN Red List global assessment.

Ocean	Indicator	Extent of trend	Time period	Source
Eastern Pacific Ocean	Nominal catches	80-95% decline	1994-2009	Hall and Roman, 2013
Western Central Pacific Ocean	CPUE	90% decline	1995-2010	Brodziak et al., 2013*
	Spawning stock biomass	95% decline	1996-2016	Tremblay-Boyer et al., 2019
	Spawning biomass	86% decline	1995-2009	Rice and Harley, 2012*
	CPUE	+80% decline	1996-2014	Rice et al., 2015*
	Catch rates	90% decline	1995-2010	Clarke et al., 2012
	CPUE	76% decline (deep sets) 53% decline (shallow sets)	1995-2000 and 2001-2006	Walsh et al., 2009
Southwest Atlantic	CPUE	111% increase	2004-2010	Tolotti et al., 2013*
	CPUE	85% decline	1980-2011	Barreto et al., 2015
Northwest Atlantic	CPUE	55% decline	1992-2015	Young et al., 2017*
	Catch rates	50% decline	1992-2005	Baum and Blanchard, 2010
	CPUE	9% decline (observer) 57% decline (logbook)	1992-2005	Cortes et al., 2007
	CPUE	70% decline	1992-2000	Baum et al., 2003
	CPUE	88% decline	1957-1999	FAO, 2013; Baum and Meyers, 2004
	CPUE	88% decline	1957-1999	FAO, 2013; Baum and Meyers, 2004
Indian Ocean	CPUE	40% decline	2000-2010	Yokawa and Semba, 2012
	CPUE	25-30% decline	1998-2011	Ramos-Cardelle et al., 2012*
	CPUE	90%	1987-1988 and 2000-2004	FAO, 2012

Population trend data were analysed over three generation lengths using a Bayesian state-space framework (Table 2). The analysis yielded a median change over three generation lengths and the probability of the most likely IUCN Red List category percent change over three generations (Rigby et al., 2019).

Table 2. Median change over three generation lengths and the most likely IUCN Red List category.

Region	Data Length (years)	Proportional Area Weighting	Median Change	Likely Status
N. Atlantic	24	0.15	-93.1	CR
S. Atlantic	7	0.08	+150.9	LC
N. Pacific	16	-	-100	CR
Pacific 1	15	0.57	-100	CR
Pacific 2	19	0.57	-98.6	CR
Indian	14	0.20	-92.9	CR

Note: Much of the population trend data used for the IUCN Red List assessment was fisheries dependent data Catch per unit effort (CPUE). However, nearly all of the time-series were prior to the 2014 listing of *Carcharhinus longimanus* in CITES Appendix II, but some may have just extended a few years past the various RFMO bans on retention. Therefore, any impact of the prohibition on retention is unlikely to have impacted the CPUE time-series as the trends are all significantly declining and would not impact the overall trend. The exception is for the South Atlantic where the time-series is short and susceptible to changes in fishing strategy during those years and may not be representative of the region. There is observer coverage on the tuna longline and purse seine vessels which is mandated at high levels for purse seine vessels but not longline vessels and although fishers' behaviour may change when observers are onboard, this is unlikely to be the case with high levels of coverage. For the longline vessels, there may be some lower level of catch reporting following prohibition but even the limited observer coverage will provide some accurate estimates of *Carcharhinus longimanus* catches that could then be checked against the broader logbook reporting. However, as noted in the SS, there is illegal take of *Carcharhinus longimanus* and this may be from the RFMOs where catch is not reported.

Eastern Pacific Ocean

Hall and Roman (2013) found that in the tropical purse seine fishery, from 1993 to 2009, *Carcharhinus longimanus* made up approximately 9% of the estimated annual shark catch. However, after 1994, catch rates were significantly reduced. This reduction represented an estimated 80–95% population reduction for the species.

Western Central Pacific Ocean

Brodziak et al. (2013) utilised data from the Hawaii-based pelagic longline fishery. The standardized CPUE for *Carcharhinus longimanus* in the Hawaii-based pelagic longline fishery experienced a 90% decline between 1995 and 2010. This decline was considered particularly concerning because the species was already subject to fishing pressure even before 1995.

Tremblay-Boyer et al. (2019) conducted a significant regional stock assessment for the *Carcharhinus longimanus* in the Western Central Pacific Ocean (WCPO) and concluded that the adult biomass of *Carcharhinus longimanus* in the WCPO is predicted to be below 5% of unfished levels. Furthermore, they determined that the population is at risk of extinction in the long-term if current fishing mortality is not reduced.

Rice and Harley (2012) conducted a stock assessment based on observer longline data for 1995–2009 which indicated an 86% reduction in spawning stock biomass, with the stock overfished and overfishing occurring. The trend analysis of the same time-series for 1995–2009 (15 years) revealed annual rates of reduction of 14.1%, consistent with an estimated median reduction of 100.0% over three generation lengths (61.2 years), with the highest probability of >80% reduction over three generation lengths (Rigby et al., 2019).

Rice et al. (2015) provided updated standardised CPUE data from the WCPO observer longline data from 1996–2014 which indicated continuing declines with the stock likely maintaining an overfished status. The trend analysis of the same time series for 1996–2014 (19 years) revealed annual rates of reduction of 5.3%, consistent with an estimated median reduction of 98.6% over three generation lengths (61.2 years), with the highest probability of >80% reduction over three generation lengths (Rigby et al., 2019).

Clarke et al. (2012) found that longline catch rates for *Carcharhinus longimanus* were reduced consistently: annual values decreased by 90% from 1996 to 2009 and uncertainty in the estimates was low.

Walsh et al. (2009) calculated a recent rate of decline of 76% (deep sets) or 53% (shallow sets) over an approximately ten-year period (1995–2000 vs 2004–2006).

Southwest Atlantic

Tolotti et al. (2013) provided information on *Carcharhinus longimanus* population trends, specifically focusing on the Southwest Atlantic Ocean. They showed a slight increase in CPUE over a short period from 2004 to 2010 in this region. However, it was noted that this slight increase may reflect changes in fishing practices rather than a true population.

Barreto *et al.* (2016) focussed on the Southwest Atlantic Ocean. Fisheries data from 1980 to 2011 indicated a significant reduction in *Carcharhinus longimanus* populations of 85%.

Northwest Atlantic

Young *et al.* (2017) utilized observer data from the U.S. Northwest Atlantic Pelagic Longline Fishery, covering the period from 1992 to 2015. Their analysis indicated a large decline in *Carcharhinus longimanus* abundance from the 1990s to the 2000s in this region. Following this reduction, the data suggested a relatively stable trend or an approximate 4% decline per year since then. Between 1992 and 2015, standardised CPUE decreased by 55%.

Baum and Blanchard (2010) provided information on the population trend of Carcharhinus longimanus in the Northwest Atlantic Ocean. The population of Carcharhinus longimanus was reduced by an estimated 50% between 1992 and 2005.

Cortes et al. (2007) found less severe declines, but in a shorter period (1992-2003/5). Reductions of 57% in logbook CPUE from the commercial longline fishery, and a 9% decrease in observer CPUE from the same fleet. The observer CPUE is considered to be more reliable than logbook CPUE.

Baum et al. (2003) estimated an extent of decline of 70% in CPUE based on logbook records in the Northwest Atlantic Pelagic longline fishery between 1992 and 2000.

Baum and Meyers (2004) estimated a 99.9% decrease in abundance in *Carcharhinus longimanus* in the USA Gulf of Mexico from the mid-1950s to late 1990s which was estimated using logbook (1950s) and observer (1990s) data from pelagic tuna longline fisheries. The magnitude of this decline has been questioned because of differences in fishing gear and depth between these fishing operations (Burgess *et al.*, 2005). Reanalysis of these data accounting for some of these biases suggest a decline of ~ 88% over this period (FAO, 2013).

Indian Ocean

Yokawa and Semba (2012) focused on an update of the standardized CPUE of *Carcharhinus longimanus* caught by the Japanese longline fishery in the Indian Ocean from 2000 to 2010. The authors interpreted the decline in CPUE of about 40% in the period as an indication of decrease in abundance of the population.

Ramos-Cartelle *et al.* (2012) used set records of the Spanish longline fishery targeting swordfish in the period 1998-2011 to calculate standardised CPUE. The historical trend showed large fluctuations and the fit to the data indicated that the CPUE showed a general decreasing trend in 1998-2007, followed by an increase thereafter. Overall, the magnitude of the decrease in CPUE was estimated to be about 25-30% over this period. More specifically, over the 14-year period assessed (1998-2011), an annual reduction rate of 5% was found, which is consistent with a 92.9% reduction over three generation lengths.

Trade criteria for inclusion in Appendix I

The species is or may be affected by trade

The species has low-moderate productivity and relatively low reproductions rates.

The species is used for its meat, fins, liver oil, and skin. Carcharhinus longimanus accounted for 1.8% in 1991–2001 and 0.6% in 2014, of the fin imported in Hong Kong SAR of China. In some regions, the meat of juvenile Carcharhinus longimanus is sold fresh for human consumption at local markets (Rigby et al., 2019).

Legal trade

According to exporter reported data in the CITES Trade Database, 45,109 kg of fins were exported between 2014 and 2023 from India, Kenya, Oman, Papua New Guinea, Senegal, Seychelles, Sri Lanka and Yemen for commercial purposes, that had been sourced from the wild (Figure 2).

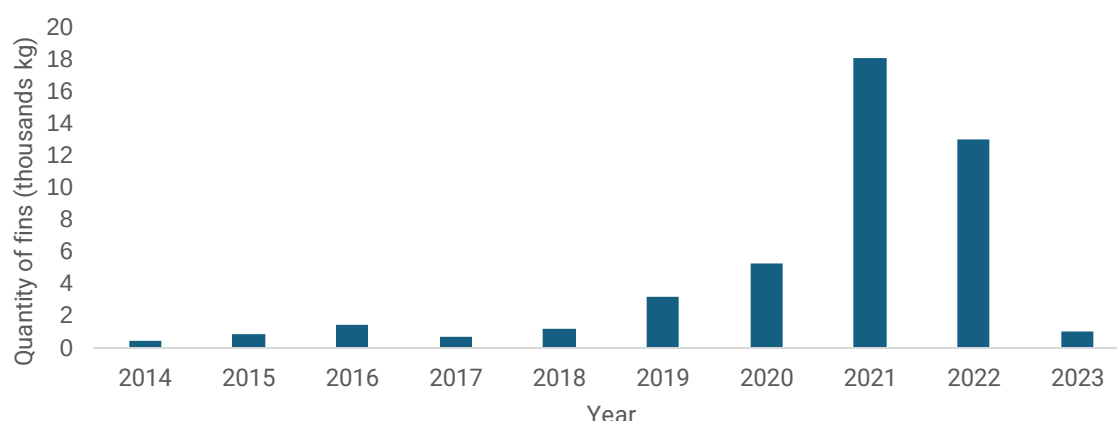


Figure 2. Quantity of *Carcharhinus longimanus* fins exported between 2014 and 2023 (CITES Trade Database, 2025).

Additionally, 2,371 individual fins were exported for commercial purposes, of which 1,862 fins were pre-convention specimens, and 509 fins were wild sourced. One pre-convention “bone” was exported as well (CITES Trade Database, 2025).

Carcharhinus longimanus has been included in the CITES Review of Significant Trade (RST) process in 2023. The process resulted in Senegal establishing zero-export quotas for all specimens taken in “the marine environment not under the jurisdiction of any State” and wild-sourced trade in the species in 2024, and Oman establishing an equivalent zero-export quota for commercial trade in 2024. Furthermore, a zero-export quota for all specimens taken in “the marine environment not under the jurisdiction of any State” and wild-sourced commercial trade in the species was established for Kenya and Yemen in 2025 (UNEP, 2025).

Illegal trade

The SS states that between 2014-2021, 31.5 metric tonnes of *Carcharhinus longimanus* fins were imported by China (97% Hong Kong SAR of China, 3% Taiwan Province of China) from only five exporters (Colombia, Oman, Seychelles, Sri Lanka and Yemen). However, during the same time period Hong Kong SAR of China imported 22,483 metric tonnes of dried shark fins, which means that legally imported *Carcharhinus longimanus* were only ~ 0.014% of the total imports (i.e. 30.6 t/ 22,483 t). Over this same period, random monthly sampling and DNA-based identification of fin-trimmings in the Hong Kong SAR of China retail market, which are a byproduct of fin processing thought to represent recent (< 1 year) imports, found that *Carcharhinus longimanus* were relatively common. The fin trimmings provide an index of species presence over time, with the species contribution to trimmings considered to be proportional to the number and volume of fins of that species that are imported within the last year. *Carcharhinus longimanus* averaged 0.71% (range 0.22-1.37%) of all trimmings sampled per year and were detected in from 2.5-10% of monthly sampling events, with a stable trend over time. This is incongruent with the very small legal volumes that were reported (i.e., ~ 0.014% of total fin imports by weight) and is potentially indicative of illegal trade into Hong Kong SAR of China.

In order to determine the species composition of the shark fin trade, Clarke et al. (2006) analysed 1999–2001 Hong Kong SAR of China trade auction data in conjunction with species-specific fin weights and genetic information to estimate the annual number of globally traded shark fins. Using this approach, the authors discovered that *Carcharhinus longimanus* are sold under their own category “Liu Qiu” and represented approximately 1.8% of the Hong Kong SAR of China shark fin market from 1999 to 2001. This equates to approximately 700,000 *Carcharhinus longimanus* (range: 200,000–1,200,000 individuals), with an equivalent median biomass of around 21,000 mt (range 9,000–48,000 mt), traded annually. Since initial studies of Clarke et al. (2006), genetic tests of fins conducted in various fish markets in Taiwan Province of China, United Arab Emirates, and Indonesia confirm the ongoing utilization of *Carcharhinus longimanus* fins in markets around the world. Although *Carcharhinus longimanus* has been prohibited from retention by all the relevant major tuna RFMOs (between 2010 and 2013), *Carcharhinus longimanus* fins were still commonly found in the Hong Kong SAR of China fin market in a survey conducted from 2014 to 2016. *Carcharhinus longimanus* is commonly utilized in the international shark fin trade because its fins are highly valuable, selling for USD45 to USD85 per kg and comprises part of the “first choice” category in the Hong Kong SAR of China fin market (Young and Carlson, 2020). Therefore, trade of *Carcharhinus longimanus* fins is ongoing even subsequent to implementation of tuna RFMO no-retention measures.

Young and Carlson (2020) reported that the *Carcharhinus longimanus* is caught in numerous commercial fisheries globally and has been heavily impacted by fishing pressure in all oceans and regions where the species occurs. This is likely due to the species’ distribution in tropical and subtropical waters where fishing effort is often most

concentrated, combined with its tendency to remain in shallow surface waters where chances of interactions with fishing gear are prevalent. Consequently, incidental secondary catch and opportunistic utilization of the species' fins for international trade have resulted in large population declines globally, and these threats are ongoing. Although *Carcharhinus longimanus* may be moderately productive relative to other shark species, it is unlikely able to withstand such heavy harvest pressure.

Additional Information

Threats

Young and Carlson (2020) reported that overexploitation due to incidental secondary catch in commercial fisheries is the single most important threat to *Carcharhinus longimanus*. *Carcharhinus longimanus* is generally not a targeted species, but its tendency to remain in surface waters (<152 m depth) and in tropical latitudes where fishing pressure is often most concentrated for target species such as tuna, results in frequent interactions in numerous fisheries throughout its global range. This species is commonly caught with pelagic longlines, purse seines, handlines, troll and occasionally pelagic and even bottom trawls; but undoubtedly, the largest source of mortality of *Carcharhinus longimanus* is secondary catch-related mortality in commercial longline fisheries. Additionally, *Carcharhinus longimanus* is a preferred species for opportunistic retention because its large fins obtain a high price per kg in the Asian fin market. This high value for *Carcharhinus longimanus* fins incentivizes the opportunistic retention and subsequent finning of *Carcharhinus longimanus* when caught, and thus represents the main economic driver of mortality of this species in commercial fisheries throughout its global range.

Productivity

Reproduction is placental viviparous with litter sizes of 1–15 pups and litter size increases with female size; gestation period is 10–12 months with most likely a biennial reproductive cycle and size at birth of 57–77 cm TL. Consequently, the rate of population increase is very low and has been estimated at 0.039–0.067, or 0.110, although these may be overestimates as they are based on younger age-at-maturity and maximum age than has since been reported. There is regional variation in age estimates: female age-at-maturity is 4.5–8.8, 6.5, and 15.8 years, and maximum age is 11, 17, and 24.9 years in Northwest Pacific, Southwest Atlantic, and Western Central Pacific, respectively. Studies have verified annual periodicity of band formation, but none have yet validated the age estimates. Using the precautionary approach, the older age-at-maturity of 15.8 years and maximum age of 24.9 years are used to calculate a generation length of 20.4 years across all regions (Rigby et al., 2019).

Conservation, management and legislation

The success of actions agreed through international wildlife and fisheries treaties depend on implementation at the domestic level; for sharks, such follow up actions have to date been lacking (Rigby et al., 2019).

Carcharhinus longimanus was the first (and is still currently the only) shark species to be subject to prohibitions on retention, transshipment, storage, and landing by all four major Regional Fishery Management Organizations (RFMOs) focused on tuna fisheries: the International Commission for the Conservation of Atlantic Tunas (2010), the Inter-American Tropical Tuna Commission (2011), the Western and Central Pacific Fisheries Commission (2012), and the Indian Ocean Tuna Commission (2013) (Rigby et al., 2019).

In 2018, *Carcharhinus longimanus* was added to Annex 1 of the Convention on Migratory Species (CMS) Memorandum of Understanding (MoU) for Migratory Sharks, even though the species is not (yet) included on the CMS Appendices. The MoU is aimed at facilitating regional conservation of listed shark and ray species (Rigby et al., 2019).

To allow recovery, it is recommended that all *Carcharhinus longimanus* retention and landings be prohibited, in line with tuna RFMO obligations, at least as long as the global population is classified as Critically Endangered or Endangered. Initiatives to prevent capture, minimize secondary catch mortality, promote safe release, and improve catch (including discard) reporting are also urgently needed, as is full implementation of additional commitments agreed through international treaties (Rigby et al., 2019).

Young and Carlson (2020) reported that *Carcharhinus longimanus* appears to have relatively higher at-vessel survivorship compared to some other sharks therefore, a large proportion of individuals caught and released alive may be able to survive. However, estimates of post-release survival rates are unknown for the *Carcharhinus longimanus*, and no-retention measures do not mitigate for potential post-release mortality that may occur because of the interaction. The SS states that Mucientes et al. (2022) reported that 56% of *Carcharhinus longimanus* were alive at the vessel in the Eastern Tropical Atlantic, and notes that Musyl et al. (2011), Hutchinson and Bigelow, (2019) and Sabarros et al. (2023) reported that they survive when released. Musyl et al. (2011) found that 95% had at vessel survival at a longline in Hawaii, Hutchinson and Bigelow (2019), found that high survival (92%) was observed when they were caught alive and in good condition and when the line is cut and gear removed, when the condition was not known, 67% survived in a longline in WCPO, and Sabarros et al. (2023) found there was a 0% post-release mortality for individuals in a good condition and 20% for individuals injured. However, it should be noted, that even though survival is high, it's still not 100%.

It is unknown whether global population abundance has continued to decline, has stabilised or has recently increased (NMFS, 2024)

Implementation challenges (including similar species)

The SS states that *Carcharhinus longimanus* are the only truly oceanic shark in the Family Carcharhinidae and there are no species with a similar morphology and ecological niche. *Carcharhinus longimanus* fins, because of their shape and colouring, are easily visually identifiable from other species of sharks – with a range of visual ID tools available to assist Governments in visually identifying traded fins of the species:

<https://citessharks.org/visual-identification-tools>.

Rapid DNA testing to distinguish between *Carcharhinus longimanus* meat from other species already exists and is in use by Governments around the world. Governments have used these tools in their implementation of the Appendix II listing, therefore transferring the species to Appendix I would require no new tools to be developed. *However, it's not clear whether the tool is routinely used by Governments and therefore identifying Carcharhinus longimanus is still an implementation challenge (Burgener in litt., 2025).*

References

- Barreto, R., Ferretti, F., Flemming, J.M., Amorim, A., Andrade, H., Worm, B. and Lessa, R. (2016) Trends in the exploitation of South Atlantic shark populations. *Conservation Biology* 30(4):792-804.
- Baum J.K. and Blanchard, W. (2010) Inferring shark population trends from generalized linear mixed models of pelagic longline catch and effort data. *Fisheries Research* 102(3):229–239.
- Baum, J.K., Myers, R.A., Kehler, R.A., Worm, B., Harley, S.J. and Doherty, P.A. (2003) Collapse and conservation of shark populations in the Northwest Atlantic. *Science*, 299:389–392.
- Baum, J.K. and Myers, R.A. (2004) Shifting baselines and the decline of pelagic sharks in the Gulf of Mexico. *Ecology Letters* 7:135–145.
- Brodziak, J. and Walsh, W.A. (2013) Model selection and multimodel inference for standardizing catch rates of bycatch species: a case study of oceanic whitetip shark in the Hawaii-based longline fishery. *Canadian Journal of Fisheries and Aquatic Sciences* 70:1723-1740.
- Burgener, M. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Clarke, S.C., Magnussen, J.E., Abercrombie, D.L., Mcallister, M.K. and Shivji, M.S. (2006) Identification of shark species composition and proportion in the Hong Kong shark fin market based on molecular genetics and trade records. *Conservation Biology* 20:201–211.
- Clarke, S.C., Harley, S.J., Hoyle, S.D. and Rice, J.S. (2012) Population trends in Pacific oceanic sharks and the utility of regulations on shark finning. *Conservation Biology* 27:197–209.
- Cortes, E., Brown, C.A. and Beerkircher, L.R. (2007) Relative abundance of pelagic sharks in the western North Atlantic Ocean, including the Gulf of Mexico and Caribbean Sea. *Gulf and Caribbean Research* 19:37–52.
- FAO (2012) Report of the fourth FAO expert advisory panel for the assessment of proposals to amend Appendices I and II of CITES concerning commercially-exploited aquatic species. In: *FAO Fisheries and Aquaculture Report No. 1032*, Rome.
- Hall, M. and Roman, M. (2013) Bycatch and Non-Tuna Catch in the Tropical Tuna Purse Seine Fisheries of the World. FAO Fisheries and Aquaculture Technical Paper.
- Mucientes, G., Vedor, M., Sims, D.W. and Queiroz, N. (2022) Unreported discards of internationally protected pelagic sharks in a global fishing hotspot are potentially large. *Biological Conservation*, 269:109534.
- Musyl, M.K., Brill, R.W., Curran, D.S., Fragoso, N.M., McNaughton, L.M., Nielsen, A., Kikkawa, B.S. and Moyes, C.D. (2011) Postrelease survival, vertical and horizontal movements, and thermal habitats of five species of pelagic sharks in the central Pacific Ocean. *Fishery Bulletin* 109:341–368.
- National Marine Fisheries Service (2024) Recovery Plan for the Oceanic Whitetip Shark (*Carcharhinus longimanus*). NMFS, Office of Protected Resources. Silver Spring, Md. 69 pages.
- Ramos-Cardelle, A., García-Cortés, B., Ortiz de Urbina, J., Fernández-Costa, J., González-González, I. and Mejuto, J. (2012) Standardized catch rates of the oceanic whitetip shark (*Carcharhinus longimanus*) from observations of the Spanish longline fishery targeting swordfish in the Indian Ocean during the 1998-2011 period. *IOTC-2012-WEPB08-27*. 15 pp
- Rice, J. and Harley, S. (2012) Stock assessment of oceanic whitetip sharks in the western and central Pacific Ocean. WCPFC-SC8-2012/SA_WP-06 Rev 1. Scientific Committee Eight Regular Session 7-15 August Busan, Republic of Korea. Western and Central Pacific Fisheries Commission.
- Rice, J., Tremblay-Boyer, L., Scott, R., Hare, S. and Tidd, A. (2015) Analysis of stock status and related indicators for key shark species of the Western Central Pacific Fisheries Commission. In: *Western Central Pacific Fisheries Commission* (ed.), Scientific Committee Eleventh Regular Session, 5-13 August 2015, Pohnpei, Federated States of Micronesia.
- Rigby, C.L., Barreto, R., Carlson, J., Fernando, D., Fordham, S., Francis, M.P., Herman, K., Jabado, R.W., Liu, K.M., Marshall, A., Pacoureau, N., Romanov, E., Sherley, R.B. and Winker, H. (2019) *Carcharhinus longimanus*. *The IUCN Red List of Threatened Species* 2019: e.T39374A2911619. <https://dx.doi.org/10.2305/IUCN.UK.2019-3.RLTS.T39374A2911619.en>. Accessed on 01 August 2025.

- Rigby, C.L. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Sabarros, P.S., Mollier, E., Tolotti, M., Romanov, E.V., Krug, I. and Bach, P. (2023) Post-release mortality of oceanic whitetip sharks caught by purse seiners–POREMO project (2023) IOTC-2023-WPEB19-18_Rev1 Presented at the 19th Session of the IOTC Working Party on Ecosystems and Bycatch, La Réunion, France.
- Sherley, R.B., Winker, H., Rigby, C.L., Kyne, P.M., Pollom, R., Pacoureaux, N., Herman, K., Carlson, J.K., Yin, J.S., Kindsvater, H.K. and Dulvy, N.K. (2020) Estimating IUCN Red List population reduction: JARA – A decision-support tool applied to pelagic sharks. *Conservation Letters* 13(2): p.e12688.
- Tolotti, M.T., Travassos, P., Frédou, F.L., Wor, C., Andrade, H.A. and Hazin, F. (2013) Size, distribution and catch rates of the oceanic whitetip shark caught by the Brazilian tuna longline fleet. *Fisheries Research* 143:136-142.
- Tremblay-Boyer, L., Carvalho, F., Neubauer, P. and Pilling, G. (2019) Stock assessment for oceanic whitetip shark in the Western and Central Pacific Ocean. WCPFC-SC15-2019/SA-WP-06. Report to the WCPFC Scientific Committee. Fifteenth Regular Session, 12–20 August 2018, Pohnpei, Federated States of Micronesia
- UNEP (2025) *Carcharhinus longimanus*. Species+ Website. Nairobi, Kenya. Compiled by UNEP-WCMC, Cambridge, UK. Available at: https://www.speciesplus.net/species#/taxon_concepts/3712/legal Accessed on 18 August 2025
- Walsh, W.A., Bigelow, K.A. and Sender, K.L. (2009) Decreases in shark catches and mortality in the Hawaii-based longline fishery as documented by fishery observers. *Marine and Coastal Fisheries* 1:270–282.
- Yokawa, K. and Semba, Y. (2012) Update of the standardized CPUE of oceanic whitetip shark (*Carcharhinus longimanus*) caught by Japanese longline fishery in the Indian Ocean. IOTC-2012-WPEB08-26. 5 pp.
- Young, C.N., Carlson, J., Hutchinson, M., Hutt, C., Kobayashi, D., McCandless, C.T. and Wraith, J. (2017) Status review report: oceanic whitetip shark (*Carcharhinus longimanus*). Final Report to the National Marine Fisheries Service, Office of Protected Resources.
- Young, C. N. and Carlson, J. K. (2020) The biology and conservation status of the oceanic whitetip shark (*Carcharhinus longimanus*) and future directions for recovery. *Reviews in Fish Biology and Fisheries* 3:293–312. doi: 10.1007/s11160-020-09601-3

Inclusion of Tope (*Galeorhinus galeus*) and Smoothhounds (genus *Mustelus*) in Appendix II

Proponents: Brazil, Ecuador, European Union, Panama, Senegal

Summary: Tope (also known as School Shark, Soupfin Shark, and Oil Shark) *Galeorhinus galeus*, and smoothhounds (*Mustelus* spp.) are members of the family Triakidae, small to medium-sized sharks generally found on continental shelves in warm and temperate waters worldwide comprising around 40 species in nine genera. Three species (Tope, Nownose Smoothhound *Mustelus schmitti*, and Common Smoothhound *M. mustelus* – referred to here as the lead species) are proposed for inclusion in Appendix II in accordance with Article II paragraph 2(a) of the Convention. All other smoothhound species in the genus *Mustelus* (28 species total) are proposed for inclusion as 'lookalike' species under Article II paragraph 2(b). According to the proponents, other genera in this family are not being considered in the proposal as other species are not traded in significant volumes and do not present 'lookalike' implementation challenges.

Tope are found most frequently at depths of 0 to 200 m. Nownose Smoothhound and Common Smoothhound live and feed near the bottom in sandy, muddy, and rocky habitats; Nownose Smoothhound are generally found in waters from 2 to 195 m and Common Smoothhound usually occur in waters less than 50 m deep. Tope and smoothhounds are known, or inferred, to have low productivity based on estimates of natural mortality (less than 0.2) or generation length (greater than five years). Frequency of reproduction ranges from annual (Nownose Smoothhound) to triennial (Tope), with an average litter size ranging from 4 to 5 pups for Nownose Smoothhound, 1 to 25 pups (with an average of 11) for Common Smoothhound, and 20 to 35 pups for Tope. The three-generation period (which can be used to determine the recent rate of decline) is estimated to be 79 years for Tope, 28 years for Nownose Smoothhound, and 53 years for Common Smoothhound.

Tope and all smoothhound species are subject to targeted or incidental catch by commercial fisheries, and some are caught by artisanal or recreational fisheries. Many smoothhound species are captured in trawl, gillnet or longline fisheries; a few are primarily taken by demersal trawl fisheries or may be caught with other gear types, including line, seine, set net, or trammel gear fisheries. Although Tope was historically targeted for its liver oil, this species and smoothhounds are now primarily traded internationally for their meat. A study estimating unreported shark landings from 2012 to 2019 found that Tope and Common Smoothhound were among the top ten species in the global shark meat trade, and Nownose Smoothhound was among the top 20 species. Fins of these species are traded internationally in various forms (typically dried or frozen) and fins of multiple species are typically combined within similar size and grade categories. An analysis of the trade in small shark fins through Hong Kong SAR of China found that one-third of species present in samples collected between 2014 and 2021 were members of the family Triakidae, with Tope the 14th most highly traded species in the fin trade; over 20% were species of Smoothhound sharks.

Among the lead species, *Galeorhinus galeus* is the most widespread, with populations occurring in all major oceans and the Mediterranean. It was globally assessed in 2020 as Critically Endangered on the IUCN Red List (with estimated population reductions greater than 80% over the past three generations) due to levels of exploitation. Reductions in Tope populations were estimated based on fishery independent surveys, stock assessments, and catch per unit effort (CPUE) data from demersal trawl, longline, and gillnet fisheries. In some regions, notably the north-east Pacific and the Celtic Sea ecoregion in the north-east Atlantic, regulatory measures and a decline in commercial landings has led to some stock recovery, but elsewhere stocks that have been assessed remain highly depleted or continue to decline. *Mustelus schmitti* has a much more restricted distribution than Tope, being confined to waters of the south-east Atlantic off Brazil, Uruguay and Argentina. It was also globally assessed by IUCN as Critically Endangered in 2019, due to fishing pressure, lack of refuge at depth, and documented decreases in both research trawl surveys and fishery landings data. Catches in Uruguay and Argentina have stabilised since 2017 at around 3000 mt per year. *Mustelus mustelus* occurs in the Mediterranean and along the west coast of Africa. It was globally assessed by IUCN as Endangered (with suspected population reductions of 50-79% over the past three generations) based on decreasing trends in catch, fishing intensity, absence of management, limited areas of refuge at

depth, and declines in other shark species in 2020.

Of the remaining *Mustelus* species proposed for listing under Article II 2(b), three have been globally assessed on the IUCN Red List as Critically Endangered (*Mustelus fasciatus*, *M. mento*, and *M. whitneyi*), five as Endangered (*M. griseus*, *M. higmani*, *M. manazo*, *M. minicanis*, and *M. mosis*), four as Near Threatened (*M. asterias*, *M. canis*, *M. lunulatus*, *M. norrisi*), and four as Vulnerable (*M. andamanensis*, *M. dorsalis*, *M. punctulatus*, and *M. widodoi*). Nine were Least Concern (*M. albipinnis*, *M. antarcticus*, *M. californicus*, *M. henlei*, *M. lenticulatus*, *M. palumbes*, *M. ravidus*, *M. sinusmexicanus* and *M. stevensi*), and one was Data Deficient (*M. walkeri*).

Legislation and regulations are in place in some countries for Tope. The species is included in Appendix II of the Convention on Migratory Species (CMS), Annex I of the CMS Memorandum of Understanding on the Conservation of Migratory Sharks (Shark MOU), and Annex II of the Protocol concerning Specially Protected Areas and Biological Diversity in the Mediterranean (SPA/BD Protocol) under the Convention for the Protection of the Marine Environment and Coastal Region of the Mediterranean (Barcelona Convention). Limited national measures are in place for Narnownose Smoothhound and Common Smoothhound.

There is currently no method to visually distinguish meat, the primary product in trade, of Tope and smoothhound species. The remaining smoothhounds are also proposed for inclusion based on the similarity in appearance of products in trade and overlapping habitat with Tope, Narnownose Smoothhound, and Common Smoothhound. Since multiple species are often caught in the same fisheries and it is challenging to distinguish between them and their products, data on catch, landings, and trade are often aggregated. Apart from two species (*M. californicus* and *M. stevensi*, which are not currently known to be exploited), their meat is generally consumed domestically, although the meat of *M. lenticulatus* may be traded internationally and the fins of some smoothhound species are in international trade.

Analysis: Where regulations impacting the fishing pressure on Tope and Narnownose Smoothhound have been enacted and effectively implemented, they appear to have resulted in some levels of population recovery. However, based on population reductions in other regions, **Tope *Galeorhinus galeus* appears to meet the criteria for inclusion in Appendix II** in Annex 2aA of Res. Conf. 9.24 (Rev. CoP17) as declines appear consistent with the indicative guidelines for inclusion in Appendix II of commercially exploited low-productivity aquatic species suggested in the footnote to Annex 5 of the Resolution. Despite indications of stabilising catches in Argentina and Uruguay, **historic declines in Narnownose Smoothhound *M. schmitti* are also consistent with the guidelines for an Appendix-II listing.**

Common Smoothhound populations have also reduced, but there is insufficient information on historic population levels to determine if these are in line with those suggested in the footnote to Annex 5 to constitute a marked decline. In a large part of its range (coastal Africa), fishing pressure is generally increasing, and many countries have not implemented effective management regulations. Evidence of ongoing decreases in some regions, contraction of the range in one region, and demand for the species in international trade, imply that regulation is needed to ensure that the harvest is not reducing the wild populations to a level at which their survival may be threatened by continued harvesting or other influences, indicating that ***M. mustelus* meets the criteria for inclusion in Appendix II in Annex 2aB** of the Resolution.

The remaining *Mustelus* species are primarily killed for their meat, which is largely consumed domestically. The fins of some of these species enter international trade. Based on the difficulty distinguishing between the meat and fins of different smoothhound species, if Tope, Narnownose Smoothhound or Common Smoothhound were to be included in Appendix II, products of other Smoothhound species that enter international trade would resemble specimens of species included in Appendix II; enforcement officers who encounter these products would be unlikely to be able to distinguish them. Therefore, these species **appear to qualify for inclusion in Appendix II in accordance with the criteria in Annex 2bA.**

Summary of Available Information

Text in non-italics is based on information in the Supporting Statement (SS); text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

In addition to the three lead species in the proposal – Galeorhinus galeus, Mustelus mustelus, and M. schmitti – another 26 Mustelus species are included in the Proposal. The CITES standard reference is the Taxonomic Checklist of Fish species listed in the CITES Appendices and the Annexes of EC Regulation 338/97 (Elasmobranchii, Actinopteri, Coelacanthi, and Dipneusti, except the genus Hippocampus). Information extracted from Eschmeyer, W.N., and Fricke, R. (eds): Catalogue of Fishes, an online reference, version update from 3 February 2015. See Annex 6 of CoP17 Doc. 81.1 at <https://cites.org/sites/default/files/eng/cop/17/WorkingDocs/E-CoP17-81-01-A6.pdf>. No checklist concerning these species has been adopted or proposed for CITES. If the proposal is adopted, such a checklist would need preparation at the earliest reasonable opportunity, ideally for the 34th meeting of the Animals Committee. As of 30 July 2025, Eschmeyer's Catalog of Fishes and FishBase list 27 valid species of Mustelus (Mustelus walkeri White & Last 2006, in the proposal, is synonymized with M. antarcticus by White et al 2021), indicating the need for evaluation and review (AC nomenclature specialist in litt., 2025).

Range

Tope has a wide but patchy distribution across temperate waters in most oceans (Figure 1). This species occurs in the same geographic region as other Smoothhound species, including Nownose Smoothhound (Figure 2) and Common Smoothhound (Figure 3) in the Atlantic Ocean.

Tope *Galeorhinus galeus*

Range: Extant (resident)

Albania; Algeria; Angola; Argentina; Australia; Belgium; Bosnia and Herzegovina; Brazil; Canada; Chile; Croatia; Cyprus; Denmark; Ecuador; Egypt; Faroe Islands; France; Germany; Gibraltar; Greece; Iceland; Ireland; Israel; Italy; Lebanon; Libya; Malta; Mauritania; Mexico; Monaco; Montenegro; Morocco; Namibia; Netherlands; New Zealand; Norway; Palestine, State of; Peru; Portugal; Senegal; Slovenia; South Africa; Spain (Canary Is.); Sweden; Syrian Arab Republic; Tunisia; Turkey; United Kingdom; USA (Hawaiian Is.); Uruguay; Western Sahara

Possibly Extant (resident)

Benin; Cameroon; Congo; The Democratic Republic of the Congo; Côte d'Ivoire; Equatorial Guinea; Gabon; Gambia; Ghana; Guinea; Guinea-Bissau; Liberia; Nigeria; Sierra Leone; Togo

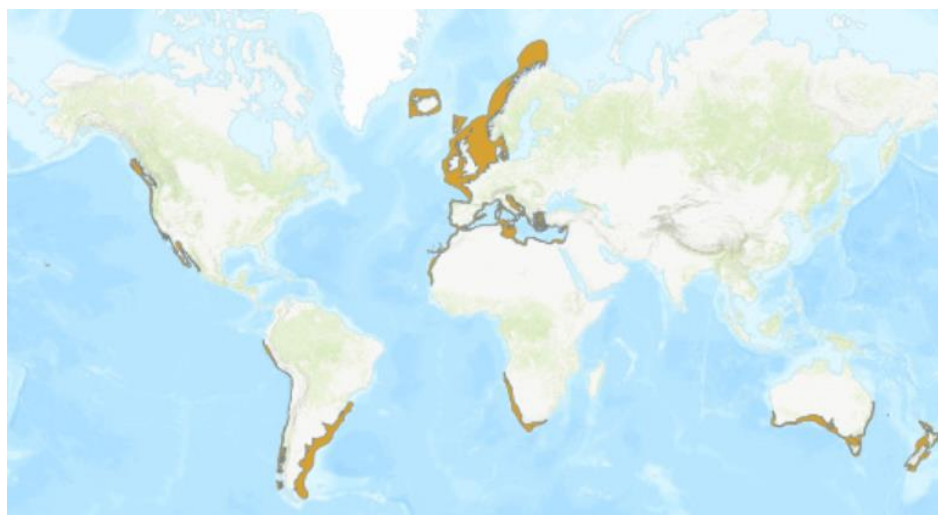


Figure 1. Tope Distribution Map (IUCN SSC Shark Specialist Group 2019. *Galeorhinus galeus*. The IUCN Red List of Threatened Species. Version 2025-1).

Nownose Smoothhound *Mustelus schmitti*

Range: Argentina; Brazil; Uruguay



Figure 2. Narrownose Smoothhound Distribution Map. (IUCN Red List Assessment 2018).

Common Smoothhound *M. mustelus*

Range: Extant (resident)

Albania; Algeria; Angola; Benin; Bosnia and Herzegovina; Cabo Verde; Cameroon; Congo; The Democratic Republic of the Congo; Croatia; Cyprus; Côte d'Ivoire; Egypt; Equatorial Guinea; France; Gabon; Gambia; Ghana; Gibraltar; Greece; Guinea; Guinea-Bissau; Israel; Italy; Lebanon; Liberia; Libya; Malta; Mauritania; Monaco; Montenegro; Morocco; Namibia; Nigeria; Palestine, State of; Portugal (Madeira); Sao Tome and Principe; Senegal; Sierra Leone; Slovenia; South Africa; Spain (Canary Is.); Syrian Arab Republic; Togo; Tunisia; Türkiye; Western Sahara

Presence Uncertain

Ireland; UK



Figure 3. Common Smoothhound Distribution Map (IUCN SSC Shark Specialist Group 2018. *Mustelus schmitti*. The IUCN Red List of Threatened Species. Version 2025-1).

IUCN Global Category and population trend information**Table 1.** Tope, Narnose Smoothhound, and Common Smoothhound, IUCN Red List global category (version 3.1), and key population trend information used in this analysis. Population trend information taken from the IUCN Red List or Supplemental Information for each species, unless otherwise noted.

Species, IUCN Red List Category and generation length	Population Trend Information (also see section on population trends below)
Tope (<i>Galeorhinus galeus</i>)	Estimated population reduction over the past three generations (global): >80%
Critically Endangered A2bd ver. 3.1 (Walker et al., 2020, assessed 2020)	Average annual rates of reduction (Southwest Atlantic/Argentina): 5.9%; (South Africa): 3.1%; (Australia): 2.8%; (Northeast Atlantic and Mediterranean): 1.7%; (New Zealand): 0.05%
Generation length = 26.3 years	Biomass (South Africa): Decrease to 10-14% of historic (pre-exploitation) levels based on data from 1952-2016 (Winker et al., 2019)
Narnose Smoothhound (<i>Mustelus schmitti</i>)	Suspected population reduction over the past three generations (global): >80%
Critically Endangered A2bd ver. 3.1 (Pollom et al., 2020, assessed 2019)	CPUE estimates (Brazil): Decreased 90% for adult catches between 1972 and 2002 (discussed in Pollom et al., 2020c).
Generation length = 9.3 years	Decrease in landings: (Uruguay): Peak of 3,212 tons in 1999 with steady decrease until 2012 (Casselberry and Carlson, 2015)
	Biomass (Argentina): Decrease of 50% between 1994 and 2003 (discussed in Cortés, 2007)
Common Smoothhound (<i>Mustelus mustelus</i>)	Suspected population reduction over the past three generations (global): 50-70%
Endangered A2bd ver. 3.1 (Jabado et al., 2021, assessed 2020)	Estimated population reduction from standardised CPUE data** (1920s-1990s) (Western Mediterranean/Adriatic Sea): 88%
Generation length = 17.7 years	Changes in abundance index from scientific bottom trawl survey data (1996-2019) (North-Central Adriatic Sea): Decreases between 2002-2011, followed by increases until 2019
	Average annual rates of reduction: (Mauritania): 4.8%; (S. Africa) 1.7%
	Change in CPUE: (Gabon) 10.5 kg/day to 4 kg/day

* According to ICES (2019), this trend is largely driven by higher catch rates at the start of the time-series, with data from the latter part of the time-series indicating more stable trends.

** Estimate for Common Smoothhound and Blackspotted Smoothhound (*Mustelus punctulatus*) combined.

Biological and trade criteria for inclusion in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)**A) Trade regulation needed to prevent future inclusion in Appendix I****Biology**

All species included in this proposal are members of the family Triakidae. Tope is the only member of the genus *Galeorhinus*. The other 28 species in the proposal are members of the genus *Mustelus* (Smoothhounds). For these species, reproduction is aplacental viviparity (embryos develop in eggs retained by the mother). Pups are born after gestation in nursery grounds, where they spend their early years. All three coastal shark species are highly dependent on shallow, protected bays and estuaries, which serve as critical pupping grounds and nursery habitats for juveniles up to two years of age.

An estimator of natural mortality (M) for sharks and rays is the reciprocal of average lifespan ($M=1/w$) where average lifespan 'w' is the generation length of the species (Pardo et al., 2016). These CITES mortality thresholds can be back converted to calculate the corresponding generation length productivity thresholds ($GL = M-1$) and these are as follows: high productivity (natural mortality > 0.5 or a generation length of < 2 years), medium productivity (natural mortality 0.2–0.5 or generation lengths 2–5 years), and low productivity (natural mortality < 0.2 or generation length >5 years). Tope, Narnose Smoothhound, and Common Smoothhound are characterized by low productivity (Table 2).

Table 2. Life history characteristics of the lead species.

Species	Maximum Size (cm)	Average size at maturity (cm) Total Length	Average litter size (pups)	Frequency of reproduction/ gestation period	Estimated generation length (years)	Productivity
Tope (<i>Galeorhinus galeus</i>)	Nearly 200	107 to 170 (males) 118 to 185 (females)	20 to 35	Triennial /12 months	26.3	Low
Narrownose Smoothhound (<i>Mustelus schmitti</i>)	110	54-70 (males); 56-79 (females)	4 to 5	11 to 12 months	9.3	Low
Common Smoothhound (<i>Mustelus mustelus</i>)	200	70 to 112 (males); 107 to 124 (females)	1 to 25 (average 11)	7 to 12 months	17.6	Low

Population trends

Tope and all the Smoothhound sharks were recently globally assessed on the IUCN Red List from 2015-2025; most species were assessed between 2019 and 2025.

Tope *Galeorhinus galeus*

This species has a widespread distribution with multiple populations and is considered data-rich regarding population trends. Genetic and tagging data support up to six separate subpopulations of Tope. In the early 1990s, Tope accounted for 6% of shark landings and was the third most important shark species in the Northeast Atlantic. This species experienced a steady decline in landings from approximately 1,100 tonnes in 1982 to 225 tonnes in 1992.

The International Council for the Exploration of the Sea (ICES) conducted an exploratory trend analysis using three standardized fishery-independent CPUE datasets: trawl surveys from the northern (2005–2018) and southern (1997–2016) areas of the Celtic Seas Ecoregion, and longline data from the Azores (1990–2015). Trawl survey data from the North Sea collected between 1992 and 2016 was removed from analysis due to concerns about possible taxonomic confusion between Tope and Smoothhounds. The combined data showed a decline in Tope CPUE until the early 2000s, followed by a gradual increase that coincided with decreasing commercial landings. Analysis of the data (1990–2018) indicated an average annual decline of 1.7%, with the highest probability outcome of a reduction exceeding 80% over the last three generations (79 years). ICES advised a reduction in annual landings to a precautionary level of no more than 241 tonnes for Tope between 2024 to 2027.

In the Mediterranean, Tope is now rarely recorded following severe historic depletions.

In South Africa, Tope was historically targeted for its liver oil, and today, *Tope remains one of the most commercially valuable and most heavily depleted sharks in South Africa, with a status of overfished (DFFE, 2025).* Based on scientific survey data and commercial catches from 1952 to 2016, a JABBA (Just Another Bayesian Biomass Assessment) fisheries stock assessment identified a decline of Tope stocks to 10–14% of pre-exploitation biomass.

In the Southwest Atlantic, artisanal catches in Uruguay averaged 28.7 metric tonnes (t) annually between 2004 and 2012. Nominal CPUE data from Argentine demersal trawl fisheries from 1992 to 2015 revealed an annual reduction rate of 5.9%, with declines greater than 80% although catches may have become more stable between 2003 and 2015 (Sharkipedia 2025a; 2025b; 2025c). Scientific fisheries surveys indicate that the biomass of the Tope population in southern Brazil declined by 90% between 1972 and 2000; the species has been classified as critically endangered in Brazil.

In the Northeast Pacific, the discovery of the high vitamin A content in Tope liver oil in 1937 led to a significant expansion of the Californian fishery during the 1940s. Between 1937 and 1949, over 800,000 individuals were killed for their livers. The 1994 gillnet ban in the Southern California Bight, which was enacted to protect seabass species, also reduced shark bycatch. The enactment of measures to discontinue targeted fishing and minimize bycatch after 60 years resulted in the recovery of the Tope population to at least 10% of its historical level, or roughly 1,500 tonnes. In Canada, Tope was designated as a species of Special Concern by the Committee on the Status of Endangered Wildlife in Canada (COSEWIC) in 2007, with this status reaffirmed in 2021. In 2022, a

petition was made under the USA's Endangered Species Act (ESA) to list Tope as an endangered species and to designate critical habitat concurrent with the listing.

Australia reports the world's third-largest landings of Tope, and it is estimated that the population has decreased to less than 10% of its historical level over three generations. *Biomass was estimated to remain below 20% of unexploited levels (Australia) based on fishing mortality rate in 2017-2018 (Patterson et al., 2019). A close-kin analysis by Thomson et al. (2020) provided evidence of multiple stocks that are somewhat reproductively isolated with overlapping spatial distribution; some stocks were likely severely depleted, while stocks remaining in sufficient numbers are small but likely to be increasing. Thomson et al. (2020) found that while the median current trend of the stock(s) sampled showed a positive trend, the overall trend in abundance could be downwards given the wide confidence intervals in their findings. The most recent stock status report (Woodhams et al., 2023) states there is no reliable modelling to indicate that the stock would recover at the rate of fishing mortality in 2022–2023, and if recovery did occur, it would be very slow and unlikely to be consistent with the applicable domestic fisheries law. The conclusion reached is that the Southern Australia biological stock is classified as a depleted stock.*

In New Zealand, the stock – which is part of a subpopulation shared with Australia – is estimated to have declined by 30 to 49% over three generations based on standardised CPUE data (1990-2016) in nine different areas.

The 2020 IUCN Red List global assessment concluded that *Galeorhinus galeus* experienced a global population decline between 70 to 99% over the past three generations (79 years), with a median estimate of 88% and the highest probability exceeding 80%. Consequently, Tope is classified as Critically Endangered globally with population reductions greater than 80% in several regions.

Narrownose Smoothhound *Mustelus schmitti*

This species has a more restricted range, but there are latitudinal differences in its biology. *In Uruguay, Narrownose Smoothhound is the species caught in the highest volumes among total catch of chondrichthyans (Silveira et al., 2018), and has been one of the predominant species caught in the Argentine Sea (Perez et al., 2011). In the Common Fishing Zone of Argentina-Uruguay, the species has shown signs of overexploitation due to historic landings (CTMFM, 2022). On average, 167 t were reported annually in artisanal fisheries in Uruguay between 2004 and 2012. Catch data for Uruguay and Argentina from 1960 to 2023 show a peak in catches in 1988, followed by a decline until 2017, after which catches have stabilized at around 3,000 tonnes annually. In Argentina, an analysis of estimated annual standardised abundance indices using Argentine commercial fishing fleet data from 1992 to 2008 showed reductions with an increasing trend since 2002 (Perez et al., 2011).*

In Brazil, total biomass declined by 855 tonnes between 1975 and 1995; the species was classified as critically endangered in Brazil in 2004 (Casselberry and Carlson, 2015).

Narrownose Smoothhound was globally assessed as Critically Endangered in 2020 due to intense unmanaged fishing pressure, the absence of deepwater refuges, and documented population declines. Its population was estimated to have decreased by over 80% in the past 28 years (three generations).

Common Smoothhound *Mustelus mustelus*

Populations of Common Smoothhound are fragmented and genetically isolated. Fisheries operate in many regions within its range. Analysis of several data sets from the species' range based on both catch trends and scientific observations, which generally show declines, leading to its classification as Endangered, with an estimated global population reduction of 50–79% over the past 53 years. Species-specific genetic analyses suggest that overexploitation has significantly reduced genetic diversity of Common Smoothhound in the Mediterranean and West African waters. These analyses also showed contraction of its range in the Mediterranean. Reported Mediterranean landings of the three species of Smoothhounds in this region combined declined from a peak of 14,000 tonnes during the 1980s, to less than 3,000 tonnes since the late 1990s.

Estimated population reduction from standardised CPUE data for both Common Smoothhound and Blackspotted Smoothhound in the western Mediterranean and Adriatic Sea from the 1920s until the 1980s and 1990s revealed an 88% historical decline, consistent with the indicative guidelines in the footnote for the application of decline for commercially exploited aquatic species characterized by low productivity. Another study estimating indexes of abundance in the North-Central Adriatic Sea using data from a scientific bottom trawl survey (1996-2019) showed fluctuations in abundance of these two species, with decreases between 2002 and 2011, followed by increases until 2019 (Maioli et al., 2023).

In the Eastern Atlantic, in waters off Mauritania, annual standardised abundance index estimates from trawl surveys between 1982 and 2015 showed annual reduction rates of 4.8%, with an estimated decrease greater than 80% over the past three generations.

In South Africa, a stock assessment by da Silva et al. (2019) concluded that current catch levels of Common Smoothhound are unsustainable, projecting continued stock decline unless annual mortality is reduced from 100 to 200 t to below 75 t.

B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences

Meat

A recent study estimating the likely composition of shark and ray species in the global meat trade from 2012 to 2019 concluded that small coastal shark species, including Tope, Smoothhounds, and Dogfish (Family Squalidae) – were the second most abundant group in trade after pelagic sharks, with 78% of their meat exported, primarily to Europe and Australia. Tope and Common Smoothhound were among the top ten species within the global meat trade, and Narrownose Smoothhound is among the top 20 species (MacNeil et al., 2025).

Estimates indicate that average annual global landings of Tope from 2012 to 2019 were around 5,067 tonnes, ranging between 4,163 tonnes and 6,480 tonnes. Estimates of average annual exports of Tope meat again place New Zealand first with 1,969 tons, followed by South Africa with 185 tons and France with 97 tons. These three countries export more than 80% of the global annual *G. galeus* catch (MacNeil et al., 2025). For Narrownose Smoothhound, meat exports were estimated to be 1,411 tonnes per year, on average, with some years reaching 3,273 tonnes. In the case of Common Smoothhound, global export volumes for this species are estimated at around 8,859 tonnes per year, with maximum values that exceed 17,600 tonnes. *The study also found that the estimates of Tope, Smoothhounds, and Dogfish in trade is likely 5% higher than reported (MacNeil et al., 2025).*

Fins and Other Products

Although Tope and Smoothhound shark species are primarily traded for their meat, the fins of these species are traded internationally. One analysis of the trade in small shark fins through Hong Kong SAR of China from 2014 to 2021 found that one-third of species present in samples were members of the family Triakidae. The same study found that Tope was the 14th most highly traded species in the fin trade, and over 20% were species of Smoothhound sharks.

Tope liver oil has not been an important product or traded internationally in significant quantities for more than 80 years. Yet, it is possible that small quantities of liver oil, cartilage and skin from Tope and Smoothhound species enter international trade as byproducts of these meat fisheries. *A review of U.S. trade data showed imports of over 600,000 units (referred to as number of specimens) of Tope extract between 2019 and 2020 from New Zealand. Approximately 56% of these imports were reported as sourced from animals bred in captivity, which seems unlikely and is probably the result of a reporting error, as well as parts and derivatives thereof (LEMIS). Trade was only reported for these years in a LEMIS data set comprised of records between 2008 and 2021.*

Inclusion in Appendix II to improve control of other listed species

A) Specimens in trade resemble those of species listed in Appendix II under Res. Conf. 9.24 (Rev. CoP17) Annex 2 a or listed in Appendix I

Tope occurs in the same geographic region as other Smoothhound species. Due to skin pigmentation and similarities in appearance, range States for Tope and Smoothhound sharks may not report species-specific landings data. Instead, these sharks are typically grouped under broad categories such as ‘dogfish,’ ‘smoothhounds,’ or ‘houndsharks.’ They may also be categorized using local terms, such as ‘cazón,’ ‘cação,’ or ‘gatuço’.

Of the Smoothhound species proposed as lookalikes, 26 out of 28 species are harvested for their meat. With the exception of two species (*M. californicus* and *M. stevensi*, which are not currently known to be utilized), their meat is consumed domestically. However, the meat of *M. lenticulatus* may be traded internationally, and the fins of some Smoothhound species (*M. albiginnis*, *M. widodoi*, *M. albiginnis*, *M. lunulatus*, *M. palumbes*, and *M. sinusmexicanus*) may be in international trade. Although the meat of most species is consumed or sold domestically, the meat of Tope and other Smoothhound sharks cannot be distinguished from the meat of these species. *Carcasses entering the shark meat market have usually had their fins and head removed, which poses a challenge to reliable species identification based on the morphology of captured individuals (Almerón-Souza et al., 2018).*

B) Compelling other reasons to ensure that effective control of trade in currently listed species is achieved

In several areas, Tope and Smoothhound sharks have proven responsive to effective fisheries management following intense fishing pressure and depletion. Following the enactment of management measures, stocks in the Northeast Atlantic, Northeast Pacific, and Australia are currently stable, and there is a possibility that the Northeast Pacific stocks are on a trajectory to recovery. *According to Dulvy in litt., (2025), the regulation of international trade could complement national management efforts, avoiding the risk that international demand will pose considerable additional pressures upon national fisheries.*

Additional Information

Threats

The main threat to Tope and Smoothhound sharks is overexploitation in some regions. These low productivity species are targeted catch in some fisheries and secondary retained catch in other fisheries. Although populations have stabilized in some areas because of management, other populations are at risk of further reductions and possible collapse.

Another threat is coastal habitat loss and degradation due to these species' reliance on shallow-water nursery grounds. Climate change may put these areas at risk, reducing the availability of suitable habitat. *An analysis of the vulnerability, resilience risk, and exposure of chondrichthyan species within Australian waters to habitat reduction due to climate change stressors between 2000 and 2006 and in 2018 revealed that Tope is among the species at relatively high risk of being negatively impacted by climate change due to its dependence on inshore nursery habitat (Walker et al., 2021).*

*Illegal trade is another potential threat to these species. A seizure of 3,300 kg of shark fins in Brazil during 2008 included fins of Narnose Smoothhound (WiTIS); this species is designated as an endangered species and is prohibited from harvest. A study assessing seafood samples purchased between May and June 2022 from randomly selected outlets in seven states and territories in Australia found that Tope was mislabelled in eight incidents as 'flake' (which is only approved for *M. antarcticus* and *M. lenticulatus*) (Cundy et al., 2023). In 2024, there were several seizures in Colombia of meat from Smoothhound species in violation of domestic law.*

Conservation, management and legislation

Some management measures have been enacted for Tope, Narnose Smoothhound, and Common Smoothhound at the national and regional level (Table 3).

National/Regional

Table 3: National and Regional Measures with the range of the species denoted.

Region	Tope (<i>Galeorhinus galeus</i>)	Narnose Smoothhound (<i>Mustelus schmitti</i>)	Common Smoothhound (<i>Mustelus mustelus</i>)
Northeast Pacific North America. Canada has designated this species as a species of Special Concern under its Species at Risk Act of 2009, which requires development of a management plan. Targeted catch and retention of this species have also been prohibited in British Columbia. In the USA, a petition to list this species as threatened or endangered under its ESA and designate critical habitat is under review. In Mexico, gear restrictions and designation of seasonal and permanent closed areas provide some protection for Tope.	x		
Northwest Atlantic and Mediterranean Tope is prohibited from catch by longline fishing under European Union (EU) regulations. In the United Kingdom (UK), Tope may only be caught with rod and line gear, and recreational anglers are required to release their catch. A bycatch limit of 45 kg per day has also been established in commercial fisheries (ICES, 2019). States in the Mediterranean are required to ensure the maximum protection possible and recovery of Tope since the species is listed in Annex II of the SPA/BD Protocol to the Barcelona Convention. With respect to Smoothhound species, the EU requires fixed nets with mesh sizes exceeding 220 mm to target these species. Although several Smoothhound species are included in Annex III of the SPA/BD Protocol, which requires appropriate measures to ensure the conservation of the species, limited national measures are in place for Narnose Smoothhound and Common Smoothhound.	x		x

Region	Tope (<i>Galeorhinus galeus</i>)	Narrownose Smoothhound (<i>Mustelus schmitti</i>)	Common Smoothhound (<i>Mustelus mustelus</i>)
<u>Coastal Africa</u> Tope and Common Smoothhound are protected in Cabo Verde. Management measures have been implemented in South Africa for both species, including a minimum mesh size to reduce the catch of pregnant females and a slot size of 70 to 130 cm TL to protect juveniles and females. Recreational fishers may only catch one shark per day.	x		x (significant part of the range may have no regulations)
<u>Southwest Atlantic</u> Argentina has established fishery regulations for Tope, including a ban on bottom trawling within parts of its Exclusive Economic Zone. Use of this gear is also prohibited in an area of shared fishing and management between Argentina and Uruguay. <i>To restore the population biomass at levels that will produce maximum sustainable yield, the Joint Technical Commission of the Maritime Front (CTMFM) launched a recovery plan for Narrownose Smoothhound (CTMFM Resolution No. 13/20) (CTMFM, 2025).</i> Brazil has classified Tope and Narrownose Smoothhound as critically endangered in their national regulations, so these species are prohibited from catch, transport, storage, and commercialization. <i>However, according to Begossi et al. (2017), implementation of the prohibition on catch of Narrownose Smoothhound was suspended in 2015 and the first half of 2016 and faced a legal challenge in 2019 (as cited in Pollom et al. (2020c)).</i>	x	x	
<u>Oceania</u> Australia banned targeted commercial fishing for Tope in 2001, and it has established management measures to rebuild the species, including gear restrictions, mesh-size limits, length limits, seasonal closures, and a bycatch quota. Within the same region, New Zealand manages Tope under its Quota Management System, and vessels are issued Individual Transferable Quotas. There is also a recreational bag limit of 20 to 30 fish per day.	x		

International

Limited protections for Tope and Smoothhounds have been adopted in international fora. Tope is included in Appendix II of CMS and Annex 1 of the CMS Memorandum of Understanding on the Conservation of Migratory Sharks (Shark MOU). *These Annexes list migratory species that have an unfavourable conservation status and require international agreements for their conservation and management or would significantly benefit from the international cooperation that could be achieved by an international agreement. Range States are encouraged to conclude global or regional Agreements for the conservation and management of these species (CMS, 1979; CMS Shark MOU, 2023).*

Artificial Propagation/captive breeding

There is no information or literature available demonstrating that Tope and Smoothhound sharks are captive-bred. However, analysis of U.S. trade data showed exports of Tope extract by New Zealand to the USA that were reported as sourced from animals bred in captivity, as well as parts and derivatives thereof (LEMIS). This may, however, be a reporting error by the exporter.

Implementation challenges (including similar species)

A recent study that quantifies the international trade in shark and ray meat from 2012 to 2019 found that blue shark and 156 other species make up a majority of the shark and ray meat in trade (MacNeil et al., 2025). Given the difficulty distinguishing shark meat in trade and the variety of species that are traded internationally (Almerón-Souza et al., 2018; Bornatowski et al., 2014), implementation of the CITES listing for Tope and Smoothhounds could be a challenge for customs officers and wildlife inspectors. However, it may be possible to distinguish the meat of Tope

and Smoothhounds from the meat of other shark and ray species due to quality and price differentiation (Sant in litt., 2025).

References

- AC Nomenclature Specialist (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Almerón-Souza, F., Sperb, C., Castillho, C.L., Figueiredo, P.I.C.C., Gonçalves, L.T., Machado, R., Oliveira, L.R., Valiati, V.H. and Fagundes, N.J.R. (2018) Molecular identification of shark meat from local markets in Southern Brazil based on DNA barcoding: Evidence for mislabeling and trade of endangered species. *Frontiers in Genetics* 9. <https://doi.org/10.3389/fgene.2018.00138>
- Begossi, A., Salivonchik, S., Hallwass, G., Hanazaki, N., Lopes, P.F. and Silvano, R.A. (2017) Threatened fish and fishers along the Brazilian Atlantic Forest Coast. *Ambio* 46(8):907–914.
- Bornatowski, H., Braga, R.R. and Vitule, J.R.S. (2014) Threats to sharks in a developing country: The need for effective and simple conservation measures. *Natureza & Conservação* 12(1):11–18.
- Carlson, J., Charvet, P., Blanco-Parra, M.P., Briones Bell-Iloch, A., Cardenosa, D., Derrick, D., Espinoza, E., Morales-Saldaña, J.M., Naranjo-Elizondo, B., Pacoureaux, N., Pérez Jiménez, J.C., Schneider, E.V.C., Simpson, N.J., Talwar, B.S. and Pollom, R. (2021) *Mustelus canis*. *The IUCN Red List of Threatened Species* 2021: e.T39359A2908200. <https://dx.doi.org/10.2305/IUCN.UK.2021-1.RLTS.T39359A2908200.en>
- Carlson, J., Charvet, P., Blanco-Parra, M.P., Briones Bell-Iloch, A., Cardenosa, D., Derrick, D., Espinoza, E., Morales-Saldaña, J.M., Naranjo-Elizondo, B., Pacoureaux, N., Pérez Jiménez, J.C., Schneider, E.V.C., Simpson, N.J., Talwar, B.S. and Pollom, R. (2021) *Mustelus norrisi*. *The IUCN Red List of Threatened Species* 2021: e.T161518A890179. <https://dx.doi.org/10.2305/IUCN.UK.2021-1.RLTS.T161518A890179.en>
- Carlson, J., Pollom, R., Derrick, D., Pacoureaux, N. and Pérez-Jiménez, J.C. (2021) *Mustelus sinuatus*. *The IUCN Red List of Threatened Species* 2021: e.T60206A3092676. <https://dx.doi.org/10.2305/IUCN.UK.2021-1.RLTS.T60206A3092676.en>
- Cassellberry, G.A. and Carlson, J.K. (2015) Endangered Species Act Review of the Narnose Smoothhound (*Mustelus schmitti*). Report to the National Marine Fisheries Service, Office of Protected Resources. SFD Contribution PCB-15-07.
- CMS (1979) Convention on the Conservation of Migratory Species of Wild Animals. Article IV Migratory Species to be the Subject of the Agreements: Appendix II. Available at: <https://www.cms.int/sites/default/files/instrument/CMS-text.en.PDF>
- CMS Shark MOU (2023) Memorandum of Understanding on the Conservation of Migratory Sharks. Section 3 Fundamental Principles. Available at: https://www.cms.int/sharks/sites/default/files/basic_page_documents/Sharks_MOU_Text_annexes_2023_e.pdf.
- Cortés, F. (2007) Sustentabilidad de la explotación del gatuzo, *Mustelus schmitti*, en el ecosistema costero bonaerense (34 - 42°S). Licenciatura en Ciencias Biológicas, Tesis de Grado. Universidad Nacional de Mar del Plata. Instituto Nacional de Investigación y Desarrollo Pesquero.
- CTMFM (2025) Aprovechamiento Integral y Sustentable del Tiburón Gatuzo (*Mustelus schmitti*) Bajo el Modelo de Economía Circular. In: 19° Simposio Científico Libro de Resúmenes Volumen 27. Comisión Técnica Mixta del Frente Marítimo (CTMFM). Available at: <https://ctmfm.org/publicaciones/volumen-27/>
- De Silva, C., Kerwath, S.E., Parker, D. and Winker, H. (2019) Assessment of smoothhound shark *Mustelus mustelus* in South Africa. Technical Report. DEFF/FISHERIES/LSWG/AUG 2019 (assessment only). DOI:10.13140/RG.2.2.31225.83048.
- Department of Forestry, Fisheries and the Environment (DFFE) (2025). Status of the South African marine fishery resources 2025. Cape Town, South Africa. 136pp.
- Dulvy, N.K. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Dulvy, N.K., Acuña, E., Bustamante, C., Cevallos, A., Herman, K. and Velez-Zuazo, X. (2020) *Mustelus whitneyi*. *The IUCN Red List of Threatened Species* 2020: e.T63129A124461599. <https://dx.doi.org/10.2305/IUCN.UK.2020-3.RLTS.T63129A124461599.en>
- Dulvy, N.K., Acuña, E., Bustamante, C., Concha, F., Herman, K. and Velez-Zuazo, X. (2021) *Mustelus mento* (amended version of 2020 assessment). *The IUCN Red List of Threatened Species* 2021: e.T63128A203720849. <https://dx.doi.org/10.2305/IUCN.UK.2021-2.RLTS.T63128A203720849.en>
- Dulvy, N.K., Bin Ali, A., Derrick, D., Dharmadi, Fahmi, Sherman, C.S. and Sianipar, A. (2021) *Mustelus widodoi*. *The IUCN Red List of Threatened Species* 2021: e.T161571A124509147. <https://dx.doi.org/10.2305/IUCN.UK.2021-2.RLTS.T161571A124509147.en>
- Finucci, B. and Kyne, P.M. (2018) *Mustelus lenticulatus*. *The IUCN Red List of Threatened Species* 2018: e.T39356A116736541. <http://dx.doi.org/10.2305/IUCN.UK.2018-2.RLTS.T39356A116736541.en>
- Fisheries New Zealand (2025) Fisheries Assessment Plenary, May 2025: stock assessments and stock status. Compiled by the Fisheries Science Team, Fisheries New Zealand, Wellington, New Zealand. 1955 p.
- ICES (2019) Working Group on Elasmobranch Fishes (WGEF). *ICES Scientific Reports* 1(25). 988pp. International Council for the Exploration of the Seas, Copenhagen, Denmark.
- Jabado, R.W., Chartrain, E., Cliff, G., Da Silva, C., De Bruyne, G., Derrick, D., Dia, M., Diop, M., Doherty, P., El Vally, Y., Leurs, G.H.L., Meissa, B., Metcalfe, K., Pacoureaux, N., Pires, J.D., Seidu, I., Serena, F., Soares, A.-L., Tamo,

- A., VanderWright, W.J., Williams, A.B. and Winker, H. (2021) *Mustelus*. *The IUCN Red List of Threatened Species* 2021: e.T39358A124405881. <https://dx.doi.org/10.2305/IUCN.UK.2021-2.RLTS.T39358A124405881.en>
- Jabado, R.W., Dulvy, N.K., Farrell, E.D., Buscher, E. and Derrick, D. (2021) *Mustelus punctulatus*. *The IUCN Red List of Threatened Species* 2021: e.T161485A124493853. <https://dx.doi.org/10.2305/IUCN.UK.2021-1.RLTS.T161485A124493853.en>
- Jabado, R.W., Ellis, J.R., McCully, S., Dulvy, N.K., Farrell, E.D., Mancusi, C. and Derrick, D. (2021) *Mustelus asterias*. *The IUCN Red List of Threatened Species* 2021: e.T39357A124405496. <https://dx.doi.org/10.2305/IUCN.UK.2021-1.RLTS.T39357A124405496.en>
- Kyne, P.M., Charvet, P., Areano, E.M., Cevallos, A., Espinoza, M., González, A., Herman, K., Mejía-Falla, P.A., Morales-Saldaña, J.M. and Navia, A.F. (2020) *Mustelus dorsalis*. *The IUCN Red List of Threatened Species* 2020: e.T44580A124433729. <https://dx.doi.org/10.2305/IUCN.UK.2020-3.RLTS.T44580A124433729.en>
- Kyne, P.M. and Rigby, C.L. (2016) *Mustelus walkeri*. *The IUCN Red List of Threatened Species* 2016: e.T195426A68634512. <http://dx.doi.org/10.2305/IUCN.UK.2016-1.RLTS.T195426A68634512.en>
- MacNeil, M.A., Mull, C.G., Barbosa Martins, A., Babcock, E.A., Tyabji, Z., Andorra, A., Clarke, S., Jabado, R.W., Sant, G., Cinner, J.E., Gephart, J.A., N Dulvy, N.K., Oakley-Cogan, A., Kasana, D., Warwick, L., Simpfendorfer, C.A., Fowler, S., Freire, M.D.A., Bariche, M., Beaufort, O., Bizzarro, J.J., Braccini, M., Sales, J.B.D.L., Bustamante, C., Carlson, J., Charvet, P., Cuevas, J.M., Fernandes, C.A.F., Fernando, D., Finucci, B., Rodriguez, E.G., Gonzalez-Pestana, A., Cardoso, L.G., Hauser-Davis, R.A., Muttaqin, E., Blanco-Parra, M.D.P., Polo-Silva, C.J., Ready, J.S., Ruiz-García, D., Saldaña-Ruiz, L.E., Seidu, I., Sosa-Nishizaki, O., Tanna, A., Vögler, R., Werner, L., Wosnick, N. and Chapman, D. (2025) Hidden Shark and Ray Diversity in the Global Meat Trade. *bioRxiv*. <https://doi.org/10.1101/2025.04.24.650194>
- Maioli, F., Weigel, B., Chiarabelli, E., Manfredi, C., Anibaldi, A., Isailović, I. and Casini, M. (2023) Influence of ecological traits on spatio-temporal dynamics of an elasmobranch community in a heavily exploited basin. *Scientific Reports* 13(1):9596.
- McAuley, R.B. and Kyne, P.M. (2019) *Mustelus ravidus*. *The IUCN Red List of Threatened Species* 2019: e.T63165A68634279. <http://dx.doi.org/10.2305/IUCN.UK.2019-1.RLTS.T63165A68634279.en>
- McAuley, R.B. and Kyne, P.M. (2019) *Mustelus stevensi*. *The IUCN Red List of Threatened Species* 2019: e.T42716A68634394. <http://dx.doi.org/10.2305/IUCN.UK.2019-1.RLTS.T42716A68634394.en>
- Molina, J.M. and Cazorla, A.L. (2011) Trophic ecology of *Mustelus schmitti* (Springer, 1939) in a nursery area of northern Patagonia. *Journal of Sea Research* 65:381–389.
- Pardo, S., Kindsvater, H., Cuevas-Zimbrón, E., Sosa-Nishizaki, O., Pérez-Jiménez, J.C. and Dulvy, N.K. (2016) Growth, productivity and relative extinction risk of a data-sparse devil ray. *Scientific Reports* 6:33745. <https://doi.org/10.1038/srep33745>
- Pérez, M., Massa, A. and Cordo, H.D. (2011) Estimación de un Índice de Abundancia Anual de Gatuza (*Mustelus schmitti*) a Partir de Datos de la Flota Comercial Argentina. Período 1992-2008. Instituto Nacional de Investigación y Desarrollo Pesquero (INIDEP) Informe Técnico 79. Mar del Plata, Argentina. 52pp.
- Pollom, R., Avalos, C., Bizzarro, J., Burgos-Vázquez, M.I., Cevallos, A., Espinoza, M., González, A., Herman, K., Mejía-Falla, P.A., Morales-Saldaña, J.M., Navia, A.F., Pérez Jiménez, J.C. and Sosa-Nishizaki, O. (2020a) *Mustelus albiguttatus*. *The IUCN Red List of Threatened Species* 2020: e.T161451A124487621. <https://dx.doi.org/10.2305/IUCN.UK.2020-3.RLTS.T161451A124487621.en>
- Pollom, R., Barreto, R., Charvet, P., Chiaramonte, G.E., Cuevas, J.M., Herman, K., Montealegre-Quijano, S., Motta, F., Paesch, L. and Rincon, G. (2020b) *Mustelus fasciatus*. *The IUCN Red List of Threatened Species* 2020: e.T44581A2995765. <https://dx.doi.org/10.2305/IUCN.UK.2020-3.RLTS.T44581A2995765.en>
- Pollom, R., Barreto, R., Charvet, P., Chiaramonte, G.E., Cuevas, J.M., Herman, K., Montealegre-Quijano, S., Motta, F., Paesch, L. and Rincon, G. (2020c) *Mustelus schmitti*. *The IUCN Red List of Threatened Species* 2020: e.T60203A3092243. <https://dx.doi.org/10.2305/IUCN.UK.2020-3.RLTS.T60203A3092243.en>
- Pollom, R., Charvet, P., Avalos, C., Blanco-Parra, M.P., Briones Bell-Iloch, A., Derrick, D., Espinoza, E., Faria, V., Herman, K., Lasso-Alcalá, O.M., Mejía-Falla, P.A., Morales-Saldaña, J.M., Naranjo-Elizondo, B., Navia, A.F. and Pérez Jiménez, J.C. (2020d) *Mustelus higmani*. *The IUCN Red List of Threatened Species* 2020: e.T60204A3092518. <https://dx.doi.org/10.2305/IUCN.UK.2020-3.RLTS.T60204A3092518.en>
- Pollom, R., Da Silva, C., Fernando, S., Leslie, R., McCord, M.E. and Winker, H. (2020e) *Mustelus palumbes*. *The IUCN Red List of Threatened Species* 2020: e.T60247A124455981. <https://dx.doi.org/10.2305/IUCN.UK.2020-2.RLTS.T60247A124455981.en>
- Pollom, R., Jabado, R.W., Bennett, R., Elhassan, I., Fernando, S., Kuguru, B., Leslie, R., Moore, A., Simpfendorfer, C. and Spaet, J. (2019) *Mustelus mosis*. *The IUCN Red List of Threatened Species* 2019: e.T161480A139617441. <http://dx.doi.org/10.2305/IUCN.UK.2019-3.RLTS.T161480A139617441.en>
- Pollom, R., Lasso-Alcalá, O., Mejía-Falla, P.A., Navia, A.F. and Herman, K. (2020f) *Mustelus minicanis*. *The IUCN Red List of Threatened Species* 2020: e.T60205A124452001. <https://dx.doi.org/10.2305/IUCN.UK.2020-2.RLTS.T60205A124452001.en>
- Rigby, C.L., Bin Ali, A., Bineesh, K.K., Chen, X., Derrick, D., Dharmadi, Ebert, D.A., Fahmi, Fernando, D., Gautama, D.A., Haque, A.B., Ho, H., Hsu, H., Krajangdara, T., Maung, A., Vo, V.Q., Sianipar, A., Tanay, D., Utzurrum, J.A.T., Yuneni, R.R. and Zhang, J. (2020a) *Mustelus manazo*. *The IUCN Red List of Threatened Species*

- 2020: e.T161633A124518703. <https://dx.doi.org/10.2305/IUCN.UK.2020-3.RLTS.T161633A124518703.en>
- Rigby, C.L., Chen, X., Ebert, D.A., Ho, H., Hsu, H., Vo, V.Q., Zhang, J. and Derrick, D. (2020b) *Mustelus griseus*. *The IUCN Red List of Threatened Species* 2020: e.T161715A124532728. <https://dx.doi.org/10.2305/IUCN.UK.2020-3.RLTS.T161715A124532728.en>
- Rigby, C.L., Ebert, D.A. and Charles, R. (2025 in press) *Mustelus californicus*. *The IUCN Red List of Threatened Species* 2025: e.T161334A124467241
- Rigby, C.L., Pérez-Jiménez, J.C., Ebert, D.A. and Charles, R. (2025 in press) *Mustelus henlei*. *The IUCN Red List of Threatened Species* 2025: e.T161648A124521269
- Rigby, C.L., Saldaña Ruiz, L. and Rohner, C.A. (2025 in press) *Mustelus lunulatus*. *The IUCN Red List of Threatened Species* 2025: e.T161640A124519657
- Sant, G. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Sharkipedia (2025a) *Galeorhinus galeus*. Available at: <https://www.sharkipedia.org/species/925>
- Sharkipedia (2025b) *Galeorhinus galeus*. Available at: <https://www.sharkipedia.org/trends/3790>
- Sharkipedia (2025c) *Galeorhinus galeus*. Available at: <https://www.sharkipedia.org/trends/3791>
- Sherman, C.S., Bineesh, K.K. and Charles, R. (2022) *Mustelus andamanensis*. *The IUCN Red List of Threatened Species* 2022: e.T211716547A211716832. <https://dx.doi.org/10.2305/IUCN.UK.2022-2.RLTS.T211716547A211716832.en>
- Silveira, S., Laporta, M., Pereyra, I., Mas, F., Doño, F., Santana, O. and Fabiano, G. (2018) Análisis de la captura de condriktios en la pesca artesanal oceánica de Uruguay, Atlántico Sudoccidental. *Frente Marítimo* 25:303–324.
- Walker, T.I. (2016) *Mustelus antarcticus*. *The IUCN Red List of Threatened Species* 2016: e.T39355A68634159. <http://dx.doi.org/10.2305/IUCN.UK.2016-1.RLTS.T39355A68634159.en>
- Walker, T.I., Day, R.W., Awruch, C.A., Bell, J.D. Braccini, J.M., Dapp, D.R., Finotto, L., Frick, L.H., Garcés-García, K.C., Guida, L., Huveneers, C., Martins, C.L., Rochowski, B.E.A., Tovar-Ávila, J., Trinnie, F.I. and Reina, R.D. (2021) Ecological vulnerability of the chondrichthyan fauna of southern Australia to the stressors of climate change, fishing and other anthropogenic hazards. *Fish and Fisheries* 22(5):1105–1135. <https://doi.org/10.1111/faf.12571>
- Walker, T.I., Rigby, C.L., Pacoureau, N., Ellis, J., Kulka, D.W., Chiaramonte, G.E. and Herman, K. (2020) *Galeorhinus galeus*. *The IUCN Red List of Threatened Species* 2020: e.T39352A2907336. <https://dx.doi.org/10.2305/IUCN.UK.2020-2.RLTS.T39352A2907336.en>
- Winker, H., Parker, D., da Silva, C. and Kerwath, S. (2019) First comprehensive assessment of soupfin shark *Galeorhinus galeus* in South Africa. *Fisheries/LSWG/#05/2019*. Department of Agriculture, Forestry and Fisheries, South Africa. 30pp.
- Woodhams, J., Braccini, M., Peddemors, V., Bell, J., Bradshaw, S. and Drew, M. (2023) Status of Australian Fish Stocks Report School Shark *Galeorhinus galeus*. Fisheries Research and Development Corporation. Available at: <https://www.fish.gov.au/report/302-School-Shark-2023>

Transfer of manta and devil rays Family Mobulidae (Mobulids) from Appendix II to Appendix I

Proponents: Bahamas, Belize, Brazil, Comoros, Dominican Republic, Ecuador, Fiji, Gabon, Jamaica, Maldives, Panama, Samoa, Senegal, Seychelles, Sudan, Togo

Summary: Nine species of manta and devil rays, in the genus *Mobula* (family Mobulidae) are proposed for transfer from Appendix II to Appendix I in accordance with Article II, paragraph 1 of the Convention and under Criterion C (i & ii) in Annex 1 of Res. Conf. 9.24 (Rev. CoP17).

Mobula species are highly migratory cartilaginous fish found in a diversity of marine habitats in tropical and subtropical waters worldwide. Bentfin Devil Ray (*M. thurstoni*), Oceanic Manta Ray (*M. birostris*), Sicklefins Devil Ray (*M. tarapacana*) and Spinetail Devil Ray (*M. mobular*) are generally found in oceanic habitats along coastlines at oceanic islands, offshore pinnacles, and seamounts, but seasonally can be found along productive coastlines with regular upwelling. The Reef Manta Ray (*M. alfredi*) is generally found in shallow, productive coral reef habitats of island groups, atoll archipelagos, and continental coastlines. Longhorned Pygmy Devil Ray (*M. eregoodoo*) and Shorthorned Pygmy Devil Ray (*M. kuhlii*) occur in coastal and continental shelf waters in the Indian Ocean and western Pacific Ocean, while Atlantic Pygmy Devil Ray (*M. hypostoma*) is found in similar habitats within the Atlantic Ocean. Munk's Pygmy Devil Ray (*M. munkiana*) is found in warm, shallow waters along the coast, often in bays, estuaries, and along continental shelves in the Eastern Pacific Ocean.

Manta rays (*Mobula alfredi* and *M. birostris*) were included as "*Manta alfredi*" and "*Manta birostris*" in Appendix II at CoP16 (entering into effect in September 2014), but taxonomic changes adopted at CoP19 have subsumed *Manta* into *Mobula*. Remaining members of the genus were included in Appendix II in 2017 following CoP17.

All *Mobula* species are characterized by low productivity due to their slow growth, low fecundity, and long generation times. Most are relatively large with a wingspan – referred to as disc width – of one to five metres. Manta rays are typically larger than devil rays, and the Oceanic Manta Ray can reach up to seven metres disc width. Age at maturity varies with species but is generally 5-10 years. *Mobula* spp. reproduce by aplacental viviparity, and are thought to produce one pup per pregnancy, generally every one to three years. The lifespan is estimated at 15 to 20 years for devil rays, and around 40 years in manta rays.

Mobula species are impacted by overfishing, the gill plate trade (legal and illegal), degradation of coastal and coral reef habitats, boat strikes, increases in surface temperature associated with climate change, and disturbances from unregulated tourism activities. Their aggregating behaviour may increase access to food, courtship and mating, and avoidance of predators, but may also facilitate increased exploitation.

There are no historical baseline population data and global population numbers are unknown for all species, but population trends have been estimated based on available information from monitored populations, catch and landings data (including landings by vessel), and evidence of depletion. All *Mobula* spp. were assessed for the IUCN Red List between 2018 and 2020. Two species (Munk's Pygmy Devil Ray and Reef Manta Ray) were classified as Vulnerable, the remaining seven as Endangered because of estimated population reductions of 50-79% over the past three generations based on several factors, including levels of exploitation and slow life histories. More recently, Bentfin Devil Ray, Sicklefins Devil Ray, and Spinetail Devil Ray have been assessed as Critically Endangered on the IUCN Red List 2025-2 update, due to suspected global population reductions of greater than 80% over the past three generations.

All Mobulids are expected to face further population reduction in the future due to the demand for their high-value gill plates in international trade, which are prized for use in traditional medicine in Asia. There is also commercial trade in fins, skins, and live animals. *Mobula* spp. were included in the

CITES Review of Significant Trade process, in combination with Sri Lanka, at the 32nd meeting of the CITES Animals Committee (AC32) in 2023, resulting in Sri Lanka establishing a zero-export quota for all species. Current levels of international trade in gill plates may exceed those prior to their inclusion in Appendix II, although records of international trade have improved since the listing.

Mobula species are protected under several international instruments. Measures prohibiting the retention, transshipment, and landing of Mobulids have been adopted by four tuna Regional Fishery Management Organizations (tRFMOs). At least 44 countries have developed policies or legislation providing full or partial protection for these species.

Analysis: Mobulids are characterized by very low productivity and low fecundity and are vulnerable to overexploitation due to their interactions with a range of fisheries and the high value of their gill plates in international trade. Global population sizes of all *Mobula* species are unknown, but population estimates for the manta rays suggest that their populations are not small. They do not have a restricted distribution.

Overall, the available data for **Bentfin Devil Ray *M. thurstoni*, Oceanic Manta Ray *M. birostris*, Sicklefins Devil Ray *M. tarapacana*, and Spinetail Devil Ray *M. mobular* indicate declines consistent with the indicative guidelines** for inclusion in Appendix I of commercially exploited low-productivity aquatic species suggested in the footnote to Annex 5 of Res. Conf. 9.24 (Rev. CoP17). **These species are affected by trade and meet the biological and trade criteria for inclusion in Appendix I.**

Decreases of **Atlantic Pygmy Devil Ray *M. hypostoma*, Longhorned Pygmy Devil Ray *M. eregoodoo*, and Shorthorned Pygmy Devil Ray *M. kuhlii* may demonstrate declines consistent with the guidelines** for inclusion in Appendix I if genus-level data for *Mobula* species are considered in assessing population trends. All of these species do appear to be affected by trade.

Based on the available data, **Munk's Pygmy Devil Ray *M. munkiana* and Reef Manta Ray *M. alfredi***, similarly affected by trade, **do not appear to meet the decline guidelines** for inclusion in Appendix I.

A split-listing for the family Mobulidae, with some species listed in Appendix I and some in Appendix II, could lead to regulatory challenges as products in trade would be hard to distinguish. Article II of the treaty does not provide for listing 'lookalike' species in Appendix I, so if this challenge arises, the **Parties could consider annotating the listing of species remaining in Appendix II with a zero quota for commercial trade in wild-caught specimens.**

Summary of Available Information

Text in non-italics is based on information in the Supporting Statement (SS); text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

All *Mobula* species were included in Appendix II in April 2014.

Taxonomy

Mobula alfredi and *Mobula birostris* were originally listed as the genus *Manta* (2014), these species were moved into the genus *Mobula* in 2023, following taxonomic changes adopted at CoP19 (Checklist of CITES Species: <https://checklist.cites.org/>). There are two standard taxonomic references for Mobulids. The references for *Mobula hypostoma*, *Mobula birostris*, and *Mobula alfredi* can be found in White and Last (2016). References for all the other Mobulids can be found in Eschmeyer et al. (2017) (CITES Res. Conf. 12.11 (Rev. CoP19) *Standard Nomenclature*). According to Bucair et al. (2025), *Manta yarae* is recognized as a new *Manta* species and is likely a split from *M. birostris*.

Range and IUCN Global Category (in alphabetical order by scientific name)

Reef Manta Ray *Mobula alfredi*

Vulnerable A2bcd+3d (assessed 2018, ver. 3.1) (Marshall et al., 2022e)

Distribution in tropical and subtropical coastal and pelagic waters throughout the Indian and Pacific Oceans from the surface to 432 m deep. The Reef Manta Ray is found in the following FAO Fishing Areas: 51, 57, 61, 71, 77, 81.

Range: Extant (resident)

Australia; British Indian Ocean Territory (Chagos Archipelago); Cocos (Keeling) Islands; Cook Islands; Djibouti; Egypt; Eritrea; Fiji; French Polynesia; Guam; India (Andaman Is.); Indonesia; Islamic Republic of Iran; Japan; Madagascar; Maldives; Marshall Islands; Mayotte; Federated States of Micronesia; Mozambique; Myanmar; New Caledonia; Northern Mariana Islands; Oman; Pakistan; Palau; Papua New Guinea; Philippines; Saudi Arabia; Seychelles; Solomon Islands; Somalia; South Africa; Sudan; Taiwan Province of China; United Republic of Tanzania; Thailand; Timor-Leste; United Arab Emirates; USA (Hawaiian Is.); U.S. Minor Outlying Islands (Howland-Baker Is.); Vanuatu; Yemen

Possibly Extant (resident)

American Samoa; Bahrain; Bangladesh; Brunei Darussalam; Cambodia; China; Christmas Island; Disputed Territory (Paracel Is., Spratly Is.); Hong Kong SAR of China; Iraq; Kenya; Kiribati; Republic of Korea; Kuwait; Macao; Malaysia; Mauritius; Nauru; Qatar; Réunion; Samoa; Singapore; Sri Lanka; Tokelau; U.S. Minor Outlying Islands (Johnston I., Wake Is.); Viet Nam; Wallis and Futuna

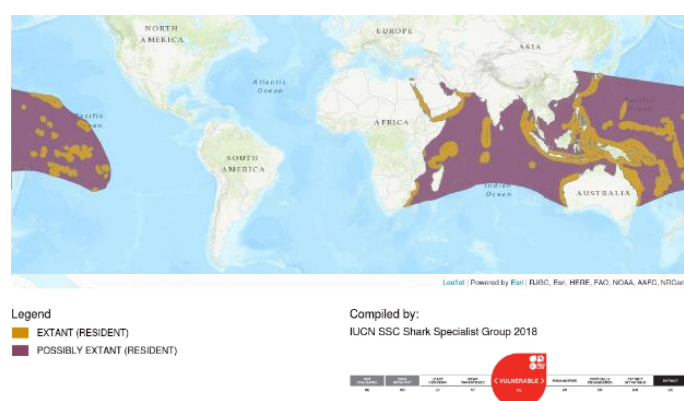


Figure 1. Reef Manta Ray (*Mobula alfredi*) Distribution Map (IUCN SSC Shark Specialist Group 2018. *Mobula alfredi*. The IUCN Red List of Threatened Species. Version 2025-1).

Oceanic Manta Ray *Mobula birostris*

Endangered A2bcd+3d (assessed 2019, ver. 3.1)

Circumglobal distribution in tropical and temperate waters throughout the Pacific, Indian, and Atlantic Oceans in coastal and pelagic waters from the surface to a depth of 1,000 m. The Oceanic Manta Ray is found in the following FAO Fishing Areas: 27, 31, 34, 41, 47, 51, 57, 61, 71, 77, 87.

Range: Extant (resident)

Angola; Anguilla; Antigua and Barbuda; Aruba; Australia; Bahamas; Barbados; Belize; Benin; Bermuda; Bonaire, Sint Eustatius and Saba (Bonaire, Saba, Sint Eustatius); Brazil; Cabo Verde; Cameroon; Cayman Islands; China; Christmas Island; Cocos (Keeling) Islands; Colombia (Colombia (mainland), Colombian Caribbean Is.); Comoros; Congo; Costa Rica (Costa Rica (mainland)); Cuba; Curacao; Djibouti; Dominica; Dominican Republic; Ecuador (Ecuador (mainland), Galapagos); Egypt; El Salvador; Equatorial Guinea (Annobon, Equatorial Guinea (mainland)); Eritrea; French Guiana; Gabon; Gambia; Grenada; Guadeloupe; Guam; Guatemala; Guyana; Haiti; Honduras; India (Andaman Is.); Indonesia; Islamic Republic of Iran; Jamaica; Japan; Kenya; Madagascar; Malaysia; Maldives; Martinique; Mauritania; Mauritius; Mexico; Montserrat; Mozambique; Myanmar; New Caledonia; New Zealand; Nicaragua; Nigeria; Northern Mariana Islands; Oman; Pakistan; Panama; Papua New Guinea; Peru; Philippines; Portugal (Azores, Madeira, Portugal (mainland), Selvagens); Puerto Rico (Navassa I., Puerto Rico (main island)); Reunion; Saint Barthelemy; Saint Helena, Ascension and Tristan da Cunha Ascension, Saint Helena (main island)); Saint Kitts and Nevis; Saint Lucia; Saint Martin (French part); Saint Vincent and the Grenadines; Sao Tome and Principe; Saudi Arabia; Senegal; Seychelles; Sint Maarten (Dutch part); Solomon Islands; Somalia; South Africa; Spain (Canary Is., Spain (mainland)); Sri Lanka; Sudan; Suriname; Taiwan Province of China; United Republic of Tanzania; Thailand; Togo; Tokelau; Trinidad and Tobago; Turks and Caicos Islands; United Arab Emirates; USA (Hawaiian Is.); Uruguay; Venezuela, Bolivarian Republic of; Virgin Islands, British; Virgin Islands, U.S.; Western Sahara; Yemen

Possibly Extant (resident)

American Samoa; Argentina; Bahrain; Cook Islands; Costa Rica (Cocos I.); Côte d'Ivoire; Fiji; French Polynesia; Ghana; Guinea; Guinea-Bissau; Hong Kong SAR of China; Iraq; Kiribati; Democratic People's Republic of Korea; Republic of Korea; Kuwait; Liberia; Macao; Marshall Islands; Federated States of Micronesia; Namibia; Nauru; Niue; Norfolk Island; Palau; Qatar; Russian Federation; Samoa; Sierra Leone; Tonga; Tuvalu; U.S. Minor Outlying Islands (Howland-Baker Is., Johnston I., Midway Is., U.S. Line Is., Wake Is.); Vanuatu; Wallis and Futuna

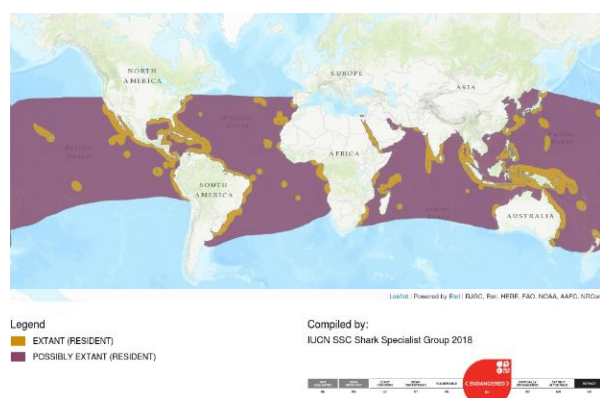


Figure 2. Oceanic Manta Ray (*Mobula birostris*) Distribution Map (IUCN SSC Shark Specialist Group 2018. *Mobula birostris*. The IUCN Red List of Threatened Species. Version 2025-1).

Longhorned Pygmy Devil Ray *Mobula eregoodoo*

Endangered A2bd+3d (assessed 2020, ver. 3.1)

Found in temperate, subtropical, and tropical waters. It is an inshore and offshore pelagic species that occurs in continental shelf areas from the surface to depths of 50 m. The Longhorned Pygmy Devil Ray is found in the following FAO Fishing Areas: 51, 57, 61, 71, 81.

Range: Extant (resident)

Australia; Bangladesh; British Indian Ocean Territory; Cambodia; Egypt; Eritrea; Ethiopia; India; Indonesia; Israel; Jordan; Malaysia (Peninsular Malaysia, Sarawak); Oman; Pakistan; Papua New Guinea; Qatar; Saudi Arabia; South Africa; Taiwan Province of China; United Arab Emirates

Possibly Extant (resident)

Bahrain; British Indian Ocean Territory (Chagos Archipelago); Brunei Darussalam; China; Christmas Island; Cocos (Keeling) Islands; Comoros; Djibouti; Guam; Islamic Republic of Iran; Iraq; Kenya; Kuwait; Madagascar; Maldives; Mauritius; Mayotte; Mozambique; Myanmar; Philippines; Réunion; Seychelles; Singapore; Somalia; Sri Lanka; Sudan; United Republic of Tanzania; Thailand; Timor-Leste; Viet Nam; Yemen

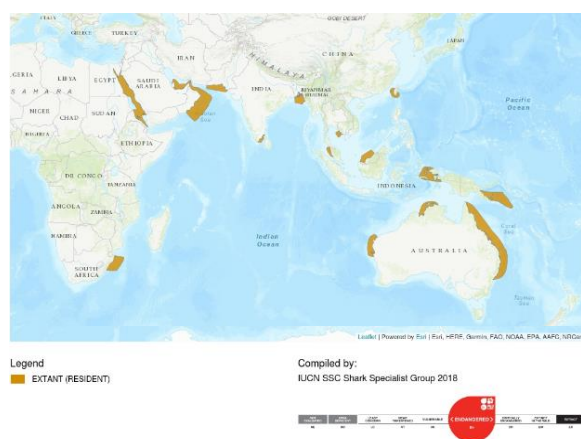


Figure 3. Longhorned Pygmy Devil Ray (*Mobula eregoodoo*) Distribution Map (IUCN SSC Shark Specialist Group 2018. *Mobula eregoodoo*. The IUCN Red List of Threatened Species. Version 2025-1).

Atlantic Pygmy Devil Ray *Mobula hypostoma*

Endangered A2cd+3d (assessed 2018, ver. 3.1)

Endemic to the Atlantic Ocean, where it has patchy distribution, to depths of 100 m. Atlantic Pygmy Devil Ray is found in the following FAO Fishing Areas: 21, 31, 34, 41.

Range: Extant (resident)

Angola; Antigua and Barbuda; Argentina; Bahamas; Barbados; Belize; Brazil; Cabo Verde; Cayman Islands; Cuba; Dominica; Dominican Republic; Gambia; Grenada; Guadeloupe; Guinea; Guinea-Bissau; Haiti; Honduras; Jamaica; Martinique; Mauritania; Mexico; Montserrat; Puerto Rico (Navassa I., Puerto Rico (main island)); Saint Lucia; Saint

Vincent and the Grenadines; Sao Tome and Principe; Senegal; Sierra Leone; Turks and Caicos Islands; USA; Uruguay; Western Sahara

Possibly Extant (resident)

Anguilla; Aruba; Benin; Bonaire, Sint Eustatius and Saba (Bonaire, Saba, Sint Eustatius); Cameroon; Colombia (Colombia (mainland), Colombian Caribbean Is.); The Democratic Republic of the Congo; Curaçao; Côte d'Ivoire; Equatorial Guinea; Gabon; Ghana; Guyana; Liberia; Nicaragua; Nigeria; Panama; Saint Barthélemy; Saint Kitts and Nevis; Saint Martin (French part); Sint Maarten (Dutch part); Suriname; Togo; Trinidad and Tobago; Venezuela, Bolivarian Republic of; Virgin Islands, British; Virgin Islands, U.S.

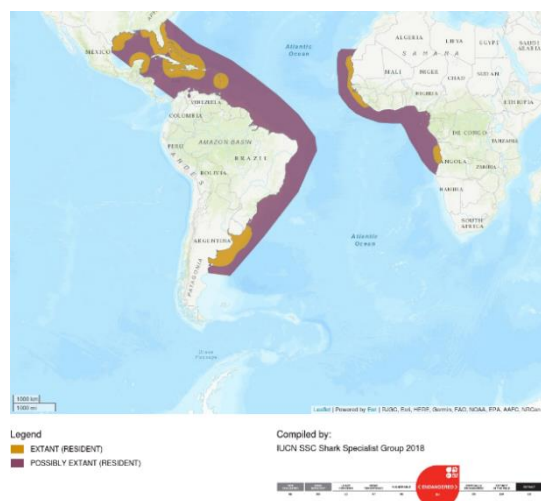


Figure 4. Atlantic Pygmy Devil Ray (*Mobula hypostoma*) Distribution Map (IUCN SSC Shark Specialist Group 2018. *Mobula hypostoma*. The IUCN Red List of Threatened Species. Version 2025-1).

Shorthorned Pygmy Devil Ray *Mobula kuhlii*

Endangered A2bd+3d (assessed 2020, ver. 3.1)

Indo-Pacific distribution in temperate, subtropical, and tropical waters. It is a mainly inshore continental shelf species that occurs in coastal areas from the surface to depths of 50 m. The Shorthorned Pygmy Devil Ray is found in the following FAO Fishing Areas: 51, 57, 61, 71.

Range: Extant (resident)

Australia; India; Indonesia; Islamic Republic of Iran; Kenya; Malaysia; Maldives; Mozambique; Myanmar; Oman; Pakistan; Papua New Guinea; Philippines; Qatar; Seychelles; Singapore; Solomon Islands; Somalia; South Africa; Sri Lanka; United Republic of Tanzania; Thailand; Timor-Leste; United Arab Emirates; Yemen

Possibly Extant (resident)

Bangladesh; British Indian Ocean Territory (Chagos Archipelago); Brunei Darussalam; Cambodia; China; Christmas Island; Cocos (Keeling) Islands; Comoros; Disputed Territory (Paracel Is., Spratly Is.); Guam; Hong Kong SAR of China; Macao; Madagascar; Mauritius; Mayotte; Réunion; Taiwan Province of China; Viet Nam

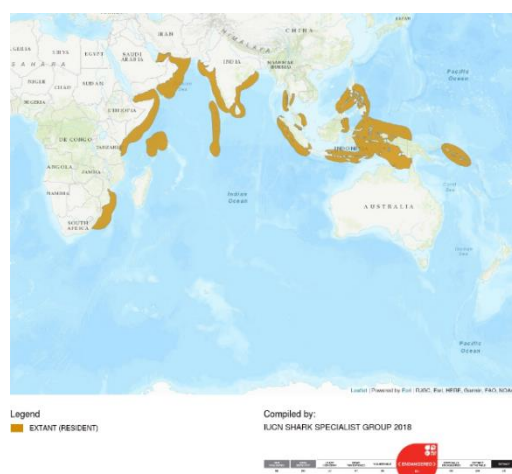


Figure 5. Shorthorned Pygmy Devil Ray (*Mobula kuhlii*) Distribution Map (IUCN Shark Specialist Group 2018. *Mobula kuhlii*. The IUCN Red List of Threatened Species. Version 2025-1).

Spinetail Devil Ray (*Mobula mobular*)

Critically Endangered A2bd+3d (assessed 2025, ver. 3.1)

Circumglobal pelagic species found in tropical and subtropical waters from the surface to 1,112 m deep. The Spinetail Devil Ray is found in the following FAO Fishing Areas: 21, 27, 31, 34, 37, 41, 47, 51, 57, 61, 67, 71, 77, 81, 87

Range: Extant (resident)

Albania; Algeria; Angola; Australia; Bangladesh; Bosnia and Herzegovina; Brazil; Cabo Verde; Cambodia; Central African Republic; Chile; China; Colombia; Comoros; Congo; The Democratic Republic of the Congo; Costa Rica; Croatia; Côte d'Ivoire; Cyprus; Ecuador; Egypt; El Salvador; Eritrea; Fiji; France; French Polynesia; Gabon; Ghana; Greece; Guatemala; Honduras; Hong Kong SAR of China; India; Indonesia; Israel; Italy; Japan; Kenya; Democratic People's Republic of Korea; Republic of Korea; Lebanon; Libya; Macao; Madagascar; Malaysia; Maldives; Malta; Marshall Islands; Mauritania; Mayotte; Mexico; Monaco; Montenegro; Morocco; Mozambique; Myanmar; New Zealand; Nicaragua; Nigeria; Oman; Pakistan; State of Palestine; Panama; Papua New Guinea; Peru; Philippines; Portugal (Azores, Madeira); Saudi Arabia; Senegal; Seychelles; Solomon Islands; Somalia; South Africa; Spain; Sri Lanka; Syrian Arab Republic; Sudan; Taiwan Province of China; United Republic of Tanzania; Thailand; Togo; Tunisia; Tuvalu; Türkiye; United Arab Emirates; U.S. Minor Outlying Islands (Wake Is.); USA (California, Florida, Georgia, Hawaiian Is., Maine, North Carolina, South Carolina, Texas); Uruguay; Bolivarian Republic of Venezuela; Viet Nam; Western Sahara; Yemen

Possibly Extant (resident)

American Samoa; Anguilla; Antigua and Barbuda; Aruba; Bahamas; Bahrain; Barbados; Belize; Benin; Bermuda; Bonaire, Sint Eustatius and Saba; British Indian Ocean Territory (Chagos Archipelago); Brunei Darussalam; Cameroon; Cayman Islands; Chile, Easter Island; Christmas Island; Cocos (Keeling) Islands; Cook Islands; Cuba; Curacao; Djibouti; Dominica; Dominican Republic; Equatorial Guinea; French Guiana; Gambia; Gibraltar; Grenada; Guadeloupe; Guam; Guinea; Guinea-Bissau; Guyana; Haiti; Islamic Republic of Iran; Iraq; Jamaica; Jordan; Kiribati; Kuwait; Liberia; Malawi; Martinique; Mauritius; Federated States of Micronesia; Montserrat; Nauru; New Caledonia; Niue; Norfolk Island; Northern Mariana Islands; Palau; Pitcairn; Puerto Rico; Qatar; Réunion; Saint Barthelemy; Saint Helena, Ascension and Tristan da Cunha; Saint Kitts and Nevis; Saint Lucia; Saint Martin (French part); Saint Vincent and the Grenadines; Samoa; Sao Tome and Principe; Sierra Leone; Singapore; Sint Maarten (Dutch part); Slovenia; Suriname; Timor-Leste; Tokelau; Tonga; Trinidad and Tobago; Turks and Caicos Islands; U.S. Minor Outlying Islands (Howland-Baker Is., Johnson Is.); Vanuatu; Virgin Islands, British; Virgin Islands, U.S.; Wallis and Futuna

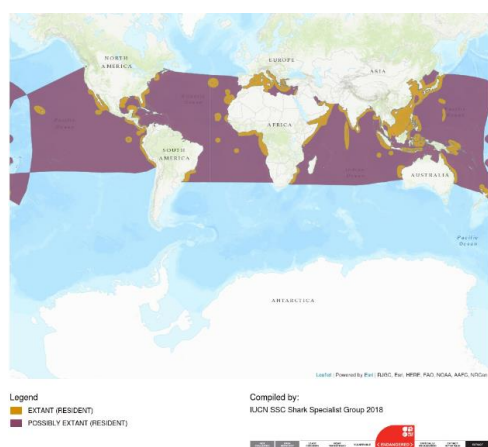


Figure 6. Spinetail Devil Ray (*Mobula mobular*) Distribution Map (IUCN SSC Shark Specialist Group 2018. *Mobula mobular*. The IUCN Red List of Threatened Species. Version 2025-1).

Munk's Pygmy Devil Ray (*Mobula munkiana*)

Vulnerable A2d+3d (assessed 2018, ver. 3.1)

Occurs in shallow, tropical Eastern Pacific waters from the Gulf of California to Peru down to depths of 30 m. Munk's Pygmy Devil Ray is found in the following FAO Fishing Areas: 77, 87.

Range: Extant (resident)

Colombia (Malpelo I.); Costa Rica (Cocos I.); Ecuador (Ecuador (mainland), Galápagos); El Salvador; Guatemala; Honduras; Mexico; Nicaragua; Panama; Peru

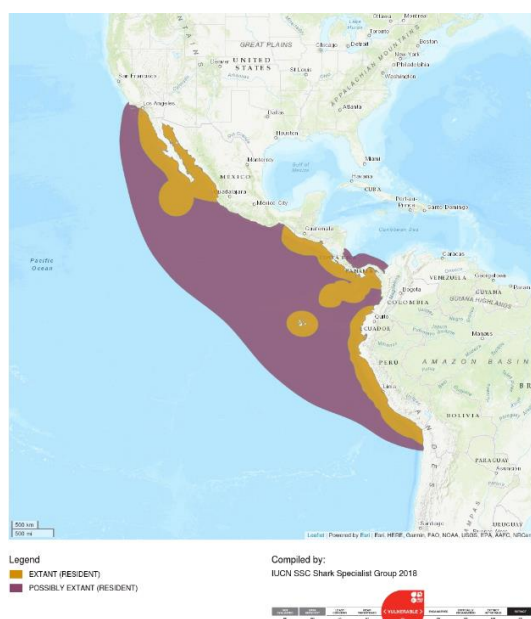


Figure 7. Munk's Pygmy Devil Ray (*Mobula munkiana*) Distribution Map (IUCN SSC Shark Specialist Group 2018. *Mobula munkiana*. The IUCN Red List of Threatened Species. Version 2025-1).

Sicklefin Devil Ray (*Mobula tarapacana*)

Critically Endangered A2bd+3d (assessed 2025, ver. 3.1)

Circumglobal but patchy distribution in temperate, subtropical, and tropical waters of the Indian, Pacific, and Atlantic Oceans, with populations mostly occurring in areas of high upwelling-related productivity. The species is primarily oceanic, but it is also found in coastal waters. The Sicklefin Devil Ray is found in the following FAO Fishing Areas: 21, 27, 31, 34, 41, 47, 51, 57, 61, 71, 81, 87.

Range: Extant (resident)

Angola; Aruba; Australia; Bahamas; Bangladesh; Barbados; Belize; Bonaire, Sint Eustatius and Saba (Bonaire); Brazil (Fernando de Noronha); Cabo Verde; Cameroon; Cayman Islands; Chile; China; Colombia (mainland, Caribbean

Is.); Comoros; Congo; Costa Rica (mainland, Cocos I.); Cote d'Ivoire; Curacao; Disputed Territory (Paracel Is., Spratly Is.); Ecuador (mainland, Galapagos); Egypt; Fiji; France (Clipperton Is.); French Polynesia; Ghana; Guadeloupe; Haiti; Honduras; Hong Kong SAR of China; India; Indonesia; Japan; Macao; Malaysia; Maldives; Martinique; Mauritania; Mayotte; Mexico; Myanmar; Nauru; Nicaragua; Pakistan; Palau; Panama; Peru; Philippines; Puerto Rico; Saint Helena, Ascension and Tristan da Cunha; Saint Helena, Ascension and Tristan da Cunha (Ascension, Saint Helena (main island)); Saint Lucia; Sao Tome and Principe; Senegal; South Africa; Sri Lanka; Suriname; Taiwan Province of China; United Republic of Tanzania; Trinidad and Tobago; Turks and Caicos Islands; United Arab Emirates; United States (Hawaiian Is.); Uruguay; Bolivarian Republic of Venezuela; Virgin Islands, British; Virgin Islands, U.S.

Possibly Extant (resident)

American Samoa; Anguilla; Antigua and Barbuda; Bahrain; Benin; Bermuda; Bonaire, Sint Eustatius and Saba (Saba, Sint Eustatius); Brazil (Trindade); British Indian Ocean Territory (Chagos Archipelago); Brunei Darussalam; Cambodia; Chile (Easter Is.); Christmas Island; Cocos (Keeling) Islands; Colombia (Colombia (mainland), Colombian Caribbean Is.); The Democratic Republic of Congo; Cook Islands; Cuba; Djibouti; Dominica; Dominican Republic; El Salvador; Equatorial Guinea (Annobon, mainland); Eritrea; French Guiana; Gabon; Gambia; Grenada; Guam; Guatemala; Guinea; Guinea-Bissau; Guyana; India (Andaman Is., Nicobar Is.); Islamic Republic of Iran; Israel; Jamaica; Jordan; Kenya; Kiribati; Democratic People's Republic of Korea; Republic of Korea; Liberia; Madagascar; Marshall Islands; Mauritius; Federated States of Micronesia; Montserrat; Morocco; Mozambique; New Caledonia; Nigeria; Niue; Oman; Papua New Guinea; Pitcairn; Portugal (Azores, Madeira, mainland); Qatar; Réunion; Saint Vincent and the Grenadines; Saint Barthelemy; Saint Kitts and Nevis; Samoa; Saudi Arabia; Seychelles; Sierra Leone; Singapore; Sint Maarten (Dutch part); Solomon Islands; Somalia; Spain (Canary Is., North African Territories, mainland); Sudan; Thailand; Timor-Leste; Togo; Tokelau; Tonga; Tuvalu; U.S. Minor Outlying Islands (Howland-Baker Is., Johnston I., Midway Is., U.S. Line Is., Wake Is.); Vanuatu; Viet Nam; Wallis and Futuna; Western Sahara; Yemen

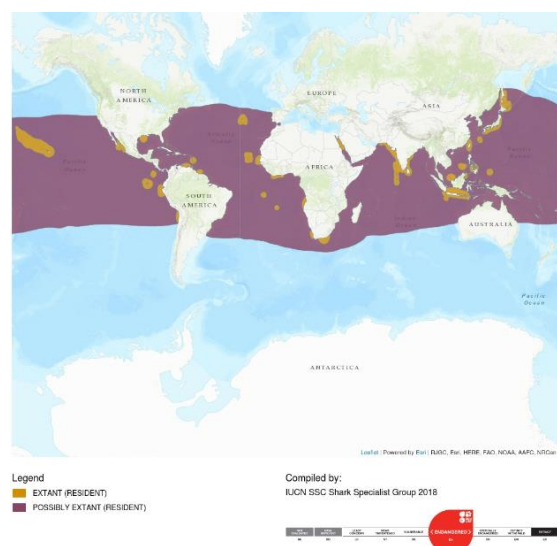


Figure 8. Sicklefin Devil Ray (*Mobula tarapacana*) Distribution Map (IUCN SSC Shark Specialist Group 2018. *Mobula tarapacana*. The IUCN Red List of Threatened Species. Version 2025-1).

Bentfin Devil Ray *Mobula thurstoni*

Critically Endangered A2bd+3d (assessed 2025, ver. 3.1)

Circumglobal in temperate, subtropical, and tropical waters worldwide. It occurs in both shallow, neritic waters and in pelagic waters from the surface down to depths of 100 m. Bentfin Devil Ray is found in the following FAO Fishing Areas: 31, 34, 41, 47, 51, 57, 61, 71, 77, 81, 87.

Range: Extant (resident)

Angola; Australia; Bangladesh; Belize; Brazil; British Indian Ocean Territory (Chagos Archipelago); Brunei Darussalam; Cabo Verde; Chile; Christmas Island; Cocos (Keeling) Islands; Colombia (mainland, Colombian Caribbean Is.); Comoros; Congo; The Democratic Republic of Congo; Costa Rica; Cote d'Ivoire; Ecuador (mainland, Galapagos); Egypt; El Salvador; Fiji; French Polynesia; Gambia; Ghana; Grenada; Guatemala; Guinea; Honduras; Hong Kong SAR of China; India (Andaman Is., Nicobar Is.); Indonesia; Japan; Kenya; Republic of Korea; Liberia; Malaysia (Peninsular Malaysia, Sabah, Sarawak); Maldives; Mauritania; Mauritius; Mayotte; Mexico; Myanmar; New Caledonia; Nicaragua; Oman; Pakistan; Panama; Peru; Philippines; Portugal (Madeira, Selvagens); Reunion; Russian Federation (Central Asian Russia, Kuril Is.); Sao Tome and Principe; Saudi Arabia; Senegal; Singapore; Solomon Islands; Somalia; South Africa; Sri Lanka; Taiwan Province of China; United Republic of Tanzania; Thailand; Timor-Leste; Tonga; United Arab Emirates; USA (Hawaiian Is.); Uruguay; Vanuatu; Bolivarian Republic of Venezuela

Possibly Extant (resident)

American Samoa; Anguilla; Antigua and Barbuda; Aruba; Bahamas; Bahrain; Barbados; Benin; Bermuda; Bonaire, Sint Eustatius and Saba (Bonaire, Saba, Sint Eustatius); Brazil (Trindade); Cambodia; Cameroon; Cayman Islands; China; Cuba; Curacao; Disputed Territory (Paracel Is., Spratly Is.); Djibouti; Dominica; Dominican Republic; Equatorial Guinea (Annobon, mainland); Eritrea; France (Clipperton Is., mainland); French Guiana; Gabon; Guadeloupe; Guam; Guinea-Bissau; Guyana; Haiti; India (Andaman Is.); Islamic Republic of Iran; Iraq; Jamaica; Jordan; Kiribati (Gilbert Is., Kiribati Line Is., Phoenix Is.); Democratic People's Republic of Korea; Kuwait; Macao; Madagascar; Marshall Islands; Martinique; Federated States of Micronesia; Montserrat; Morocco; Mozambique; Namibia; Nauru; New Zealand; Nigeria; Niue; Norfolk Island; Northern Mariana Islands; Palau; Papua New Guinea; Pitcairn; Puerto Rico (Navassa I., main island); Qatar; Saint Barthelemy; Saint Helena, Ascension and Tristan da Cunha (Ascension, Saint Helena (main island)); Saint Kitts and Nevis; Saint Lucia; Saint Martin (French part); Saint Vincent and the Grenadines; Samoa; Seychelles; Sierra Leone; Sint Maarten (Dutch part); Spain (Canary Is.); Sudan; Suriname; Togo; Tokelau; Trinidad and Tobago; Turks and Caicos Islands; Tuvalu; USA (Hawaiian Is.); U.S. Minor Outlying Islands (Howland-Baker Is., Johnston Is., Midway Is., U.S. Line Is., Wake Is.); Viet Nam; Virgin Islands, British; Virgin Islands, U.S.; Wallis and Futuna; Western Sahara; Yemen

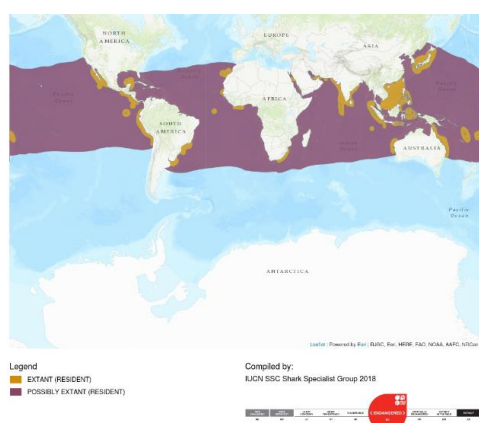


Figure 9. Bentfin Devil Ray (*Mobula thurstoni*) Distribution Map (IUCN SSC Shark Specialist Group 2018. *Mobula thurstoni*. The IUCN Red List of Threatened Species. Version 2025-1).

Biological criteria for inclusion in Appendix I**A) Small wild population**

Historical baseline population data is not available for Mobulids. Global population numbers are unknown for Devil Rays, but local estimates have been reported, such as a population of up to 12,700 Spinetail Devil Ray *Mobular mobular* in the north-west Mediterranean.

Some population estimates are available for the Manta Rays. Photo-identification studies at specified aggregation sites found minimum estimates of 42 to 500 individuals over a 10-year period in Brazil, Indonesia, Japan, Mexico, Mozambique, Myanmar, and Thailand.

Based on several data sets, Oceanic Manta Ray *Mobular birostris* population sizes appear to be less than 1,000 individuals at some locations. The SS includes a reference to Stevens *et al.* (in prep.), which suggests global estimates of *M. birostris* may be approximately 150,000 individuals.

The population size of Reef Manta Ray *M. alfredi* in some regions is also estimated to also be fewer than 1,000 individuals. The SS refers to a global population of approximately 80,000, according to Stevens *et al.* (in prep.), with the exceptions of Ecuador and the Maldives. In Ecuador, the population of Reef Manta Ray is estimated to be more than 22,000; in the Maldives, studies across the country have catalogued almost 5,000 individuals, and its population is extrapolated to be nearly 10,000.

B) Restricted area of distribution

Mobulids are distributed globally, primarily in tropical and subtropical waters. Within these broad ranges, populations of Mobulids are sparsely distributed and highly fragmented, likely due to their resource and habitat needs, in addition to over-exploitation. These species are highly migratory characterised by unique filter-feeding adaptations.

C) Decline in number of wild individuals

They are vulnerable to overexploitation due to their conservative life history, which is defined by late maturity, long lives, low fecundity, and long gestation periods. They are extremely biologically vulnerable due to their slow growth and late maturation, and *Mobula* species are among the least fecund of all elasmobranchs. After reaching

maturity at an average of 10 years, they typically give birth to only one pup every two to three years, although the Manta Rays may take longer (Table 1).

A simple but robust estimator of natural mortality (M) for sharks and rays is the reciprocal of average lifespan ($M=1/w$) where average lifespan 'w' is the generation length of the species (Pardo et al., 2016). Hence, these CITES mortality thresholds can be back converted to calculate the corresponding generation length productivity thresholds ($GL = M-1$) and these are as follows: high productivity (natural mortality > 0.5 or a generation length of < 2 years), medium productivity (natural mortality $0.2-0.5$ or generation lengths $2-5$ years), and low productivity (natural mortality < 0.2 or generation length > 5 years).

Table 1. Life history characteristics of *Mobula* species (in alphabetical order by scientific name).

Species	Maximum Size (cm) disc width (DW)	Average size at maturity (cm) DW	Average litter size (pups)	Frequency of reproduction/gestation period	Estimated generation length (years)	Productivity
Reef Manta Ray (<i>Mobula alfredi</i>)	500	270 to 300 (males); 300 to 350 (females)	1	1 to 7 years	29	Low
Oceanic Manta Ray (<i>Mobula birostris</i>)	700	350 to 400 (males); 380 to 400 (females)	1	4 to 5 years	29	Low
Longhorned Pygmy Devil Ray (<i>Mobula eregoodoo</i>)	130	99 (males); 92.5 (females)	1	1 to 3 years	12.8	Low
Atlantic Pygmy Devil Ray (<i>Mobula hypostoma</i>)	120	114 (males); 111 (females)	1	1 to 3 years	12.8	Low
Shorthorned Pygmy Devil Ray (<i>Mobula kuhlii</i>)	135	115 (males); 116 cm (females)	1	1 to 3 years	12.8	Low
Spinetail Devil Ray (<i>Mobula mobular</i>)	350	203 to 220 (males); 207 to 240 (females)	1	1 to 3 years	16	Low
Munk's Pygmy Devil Ray (<i>Mobula munkiana</i>)	110	87 (males); 97 (females)	1	1 to 3 years	12.8	Low
Sicklefin Devil Ray (<i>Mobula tarapacana</i>)	370	234 to 252 (males); 265 to 280 (females)	1	1 to 3 years	16	Low
Bentfin Devil Ray (<i>Mobula thurstoni</i>)	197	148 to 158 (males); 150 to 164 (females)	1	1 to 3 years	12	Low

Population trend information

All Mobulids were assessed as Endangered or Vulnerable on the IUCN Red List between 2018 and 2020 (Table 2). Bentfin Devil Ray, Sicklefin Devil Ray, and Spinetail Devil Ray are classified as Critically Endangered on the IUCN Red List (2025 in press).

Four *Mobula* species were assessed in 2019 for the United Arab Emirates (UAE) National Red List of Marine Species. Bentfin Devil Ray and Spinetail Devil Ray are assessed as vulnerable, and Longhorned Pygmy Devil Ray and Shorthorned Pygmy Devil Ray are assessed as near threatened (Ralph et al., 2021).

Table 2. Family Mobulidae, IUCN Red List global category (all using IUCN Red List version 3.1), and key population trend information used in this analysis. Population trend information taken from the IUCN Red List assessment or Supplemental Information for each species, unless otherwise noted.

Species, IUCN Red List Category and generation length	Population Trend Information
Reef Manta Ray (<i>Mobula alfredi</i>) Vulnerable A2bcd+3d ver. 3.1 (Marshall, 2022e) Generation length = 29 years	• Suspected population reduction over three generations (global): approximately 30 to 49%. • Catch data (East Flores and Lambata, Indonesia): Decrease to zero since 1996; historical annual catches of Manta Rays from 1959 until 1970 recorded high of 360 individuals. • Standardised sightings-per-unit-effort (Inhambane, Mozambique): Decrease of 94% between 2003 and 2016; similar study at same site found 99% decrease from 2003 to 2023 (Venables et al., 2024).
Oceanic Manta Ray (<i>Mobula birostris</i>) Endangered A2bcd+3d ver. 3.1 (Marshall, 2022a) Generation length = 29 years	• Suspected population reduction over three generation lengths (global): 50 to 79%. • Catch data from observer program on tuna purse seine fleet (IATTC Convention Area): Decrease of 98% from 1,169 individuals in 2010 to 16 in 2014; although data set began in 1993, observed catches do not appear to distinguish this species until approximately 1998; catches ranged from 99 individuals in 1998 and decreased to a low of 13 in 2002 before reaching peak in 2010 followed by decreases (Lezama-Ochao et al., 2019). • Catch data (East Flores and Lambata, Indonesia): Decrease to zero since 1996; historical annual catches of Manta Rays from 1959 until 1970 recorded high of 360 individuals. • Landings (Cilacap, Lamakera, and Tanjung Luar, Indonesia): Decreases of 71%, 75%, and 95% respectively (from 2001-2005 to 2013-2014) with possible local extinction of Manta species at Lamakera. • Landings (Karachi Fish Harbour, Pakistan): Decrease of 98% at major fish landing sites based on two study intervals of two years (2013-2015 and 2016-2018) (decreases since 2014 with no observations during 2018) (Moazzam, 2018); continued monitoring shows that landings have further declined recently from zero to two individuals per month despite increased effort by the gillnetting fleet (Moazzam Khan in litt., 2025).* • Landings per active vessel (Threspuram, India): Decrease of 82% between 2013 and 2023 (corrected for fishing effort and season) (Chopra et al., 2025 in review). • Standardised time series data of shark and ray observations systematically collected by divers between 1993 and 2013 in a Marine Protected Area in Eastern Tropical Pacific (Cocos Island, Costa Rica): 89% decrease in Manta Rays. • Standardised sightings-per-unit-effort (Inhambane, Mozambique): Decrease of 94% between 2003 and 2016.
Longhorned Pygmy Devil Ray (<i>Mobula eregoodoo</i>) Endangered A2bd+3d ver. 3.1 (Rigby et al., 2022a) Generation length = 12.8 years	• Suspected population reduction over three generations (global): 50 to 79%. • Catch in trial gillnets (New Wales, Australia): Decrease of 78% over two years (2016 to 2018). • Landings (Karachi Fish Harbour, Pakistan): Monthly catches peaked at 54 individuals in October 2013, then decreased rapidly from 2014 to an absence of any individuals from 2017 onwards, despite regular fisheries monitoring. • Landings (India): 51% decrease over 10 years (6.3 t between 1993 and 1995 to 3.1 t between 2002 and 2004) (may include Longhorned Pygmy Devil Ray). • Landings (Cilacap, Lamakera, and Tanjung Luar, Indonesia): Decreases of 64%, 75%, and 94% respectively (from 2001-2005 to 2013-2014) (Lewis et al. 2015) (may include Longhorned Pygmy Devil Ray).
Atlantic Pygmy Devil Ray (<i>Mobula hypostoma</i>) Endangered A2cd+3d ver. 3.1 (Marshall et al., 2022f)	• Suspected population reduction over past three generations (global): 50 to 79%. • Catch-per-unit-effort (CPUE) (Uruguay): 201 individuals caught between 1998 to 2013 -- equates to CPUE of 0.028 individuals per 1,000 hooks for 0.05% of the total longline catch; peak catch between 40 and 50

Species, IUCN Red List Category and generation length	Population Trend Information
Generation length = 12.8 years	from 2009 to 2011 decreased to less than 5 annually in 2013 (90% reduction). - Localized depletions (Brazil): Species is absent from many areas believed to be a part of their former geographic range and there have been few sightings in some countries, particularly in Brazil. - Sightings per hour (Senegal): Common in Senegal in the 1960s, observations are now rare across the northernmost part of its range, pointing to a severe depletion (de Boer et al., 2024). - Landings (Republic of Guinea): Decreasing Mobula species, including the Atlantic Pygmy Devil Ray, and decrease in maximum observed sizes observed between 2001 and 2010. - Trawl survey data (east coast of the USA): 113 individuals were caught in 15 out of 17 survey years (from 1989 to 2005), suggesting possible increase
Shorthorned Pygmy Devil Ray (<i>Mobula kuhlii</i>) Endangered A2bd+3d ver. 3.1 (Rigby et al., 2022b) Generation length = 12.8 years	- Suspected population reduction over three generation lengths (global): 50 to 79%. - Landings (Pakistan): Decrease of 98% based on two study intervals of two years (2013-2015 and 2016-2018) (decreases after 2014 – from peak of 54 in 2013 – with less than five observations annually in 2015-2017 and no observations during 2018) (Moazzam, 2018); continued monitoring shows that landings have further declined recently from zero to two individuals per month despite increased effort by the gillnetting fleet (Moazzam Khan in litt., 2025).* - Landings (Cilacap, Indonesia): Decreased to zero from 2013 to 2014 after comprising approximately 15% of catch in first data set from 2001 to 2005 (Lewis et al., 2015). - Landings (Tanjung Luar, Indonesia) Decrease of 88% (from 2001-2005 to 2013-2014) (Lewis et al., 2015). - Standardised sightings-per-unit-effort (Inhambane, Mozambique): Decrease of 81% between 2003 and 2016.
Spinetail Devil Ray (<i>Mobula mobular</i>) Critically Endangered A2bd+3d ver. 3.1 (Jabado et al., 2025a in press) Generation length = 16 years	- Suspected population reduction over three generations (global): >80% (Jabado et al., 2025a in press). - Landings (Karachi Fish Harbour, Pakistan): Decrease of 98% based on two study intervals of two years (2013-2015 and 2016-2018) (regularly observed in 2013-2014 with peak of 47 followed by decreases of one or two specimens annually after April 2015) (Moazzam, 2018); continued monitoring shows that landings have further declined recently from zero to two individuals per month despite increased effort by the gillnetting fleet (Moazzam Khan in litt., 2025).* - Landings of registered vessels (Peru): Decrease of 81% over eight years (from 2015 to 2023) (Rojas-Perea et al., 2025). - Landings per registered vessel (East Java, Indonesia): Decrease of 93% between 2015 and 2023 (Laglbauer et al., 2025 in press). - Landings per active vessel (Threspuram, India): Decrease of 86% between 2013 and 2023 (corrected for fishing effort and season) (Chopra et al., 2025 in review). - Landings per registered vessel (CPUE proxy) (Sri Lanka): Estimated catches decreased by an order of magnitude between 2011 and 2019 (Fernando and Stewart, 2021). - Landings (Sri Lanka): Decrease of 89.3% estimated between 2010-2024 based on more than 1,400 surveys across 38 landing sites nationwide (estimated to be >99% population reduction over the last three generations) (discussed in Jabado et al., 2005a in press).
Munk's Pygmy Devil Ray (<i>Mobula munkiana</i>) Vulnerable A2d+3d ver. 3.1 (Marshall et al., 2022d) Generation length = 12.8 years	- Suspected population reduction over three generations (global): approximately 30 to 49%. - Catch data from fishery observers on tuna purse seine fleet (IATTC Convention Area): Decrease of 85% from 171 in 2008 to 25 in 2014; although the data set began in 1993, this species does not appear to be distinguished in trade until after 2001; observed catches show overall increases until peak in 2008 followed by decreases (Lezama-Ochao et al., 2019).

Species, IUCN Red List Category and generation length	Population Trend Information
Sicklefin Devil Ray (<i>Mobula tarapacana</i>) Critically Endangered A2bd+3d ver. 3.1 (Jabado et al., 2025b in press) Generation length = 16 years	- Standardised time series data of shark and ray observations systematically collected by divers between 1993 and 2013 in a Marine Protected Area in Eastern Tropical Pacific (Cocos Island, Costa Rica): Decrease of 78% (within range of Munk's Pygmy Devil Ray). - Suspected population reduction over three generations (global): >80% (Jabado et al., 2025b in press). - Catch data from fishery observers on tuna purse seine fleet (IATTC Convention Area): Decrease of 90% from 283 in 2008 to 33 in 2014; although the data set began in 1993, this species does not appear to be distinguished in trade until after 2001; observed catches fluctuated between 2001-2004 until peak in 2008 followed by decreases (Lezama-Ochao et al., 2019). - Catch (Pakistan): Decrease of 98% based on two study intervals of two years (2013-2015 and 2016-2018) (rarely seen with no observations after September 2016) (Moazzam, 2018); landings have further declined recently from zero to two individuals per month despite increased effort by the gillnetting fleet (Moazzam Khan in litt., 2025).* - Landings (Cilacap and Tanjung Luar, Indonesia): Decreases of 77% and 99% respectively (from 2001-2005 to 2013-2014) (Lewis et al., 2015). - Landings per active vessel (Threspuram, India): Decrease of 91% between 2012 and 2023 (corrected for fishing effort and season) (Chopra et al., 2025 in review). - Standardised time series data of shark and ray observations systematically collected by divers between 1993 and 2013 in a Marine Protected Area in Eastern Tropical Pacific (Cocos Island, Costa Rica): Decrease of 78% (with reports of Sicklefin Devil Ray). - Landings per registered vessel (CPUE proxy) (Sri Lanka): Estimated catches decreased by an order of magnitude between 2011 and 2019 (Fernando and Stewart, 2021).
Bentfin Devil Ray (<i>Mobula thurstoni</i>) Critically Endangered A2bd+3d ver. 3.1 (Jabado et al., 2025c in press) Generation length = 12 years	- Suspected population reduction over three generations (global): >80% (Jabado et al., 2025c in press). - Catch data from fishery observers on tuna purse seine fleet (IATTC Convention Area): Decrease of 94% from 2143 in 2002 to 127 in 2014; although the data set began in 1993, this species does not appear to be distinguished in trade until 1994 at the earliest; observed catches ranged from zero to eight individuals from 1994-1999 followed by increases until peak in 2002 followed by decreases (Lezama-Ochao et al., 2019). - Localized depletions (Bohol Sea, Philippines): inferred localized depletions in 2002 after fishery expansion (50% of population comprised of Bentfin Devil Ray). - Landings per registered vessels (Peru): Decrease of 98% over 10 years (between 2013 and 2023) (Rojas-Perea et al., 2025). - Landings (Cilacap, Indonesia) Decreased to zero from 2013 to 2014 after comprising approximately 15% of catch in first data set from 2001 to 2005 (Lewis et al., 2015). - Landings (Tanjung Luar, Indonesia) Decrease of 75% from 2001-2005 to 2013-2014 (Lewis et al., 2015). - Landings per active vessel (Tharuvaikulam, India): Decrease of 66% between 2013 and 2023 (corrected for fishing effort and season) (Chopra et al., 2025 in review). - Landings per registered vessel (CPUE proxy) (Sri Lanka): Estimated catches decreased by an order of magnitude between 2011 and 2019 (Fernando and Stewart, 2021).
<i>Mobula</i> spp.	- Bycatch estimates (Eastern Tropical Pacific Ocean): Decrease of approximately 93% (150 t per year by 2006 reduced to 10 per year by 2009). - Catch reports (Indian Ocean): Decrease of 86% in reported catch by Pakistan from 14 individual <i>Mobula</i> spp. in 2018 to 2 in 2023 to the Indian Ocean Tuna Commission.

Species, IUCN Red List Category and generation length	Population Trend Information
	• Landings of registered vessels (Peru): Decrease of 84% over eight years (from 2015 to 2023) (Rojas-Perea et al., 2025). • Landings (India): Decreased 51% over 10 years (6.3 t between 1993 and 1995 to 3.1 t between 2002 and 2004). • Landings (Cilacap, Lamakera, and Tanjung Luar, Indonesia): Decreases of 64%, 75%, and 94% respectively (from 2001-2005 to 2013-2014) (Lewis et al., 2015). • Standardised time series data of shark and ray observations systematically collected by divers between 1993 and 2013 in a Marine Protected Area in Eastern Tropical Pacific (Cocos Island, Costa Rica): 89% decrease in Manta Rays. • Landings (Republic of Guinea): Decreasing <i>Mobula</i> species, including the Atlantic Pygmy Devil Ray, and decrease in maximum observed sizes observed between 2001 and 2010. • Survey data (Sri Lanka): Based on data and analyses, the five species found in Sri Lanka are being unsustainably harvested; <i>M. mobular</i> (the species likely to have the largest population among all the mobulids), is being overfished (Fernando in litt., 2025). • NDF data (India): Shows landings of all <i>Mobula</i> spp. peaked in 2016 with 5,875 t landed, declining to 1,878 t in 2020, an estimated overall decline of 68% (uncorrected for effort) (discussed in Jabado et al., 2025a in press)

* The second survey was conducted in 2016 after a prohibition was put into place on harvest of mobulids (Moazzam, 2018). According to Moazzam Khan in litt. (2025), implementation and enforcement are non-existent.

Trade criteria for inclusion in Appendix I

The species is or may be affected by trade

Mobulids were listed in Appendix II due to concerns regarding the international trade in gill plates (see CoP16 Prop 46 (entry into effect September 2014) and CoP17 Prop 44 (entry into effect April 2017)). *Mobulid* gill plates are sold in the Chinese medicinal market and are highly valued in international trade. The average price per kg paid by gill raker dealers for dried gill rakers of an Oceanic Manta ray less than 300 cm DW in Sri Lanka during 2011 was USD141.02 (15,500 Sri Lankan Rupee) (Fernando and Stevens, 2011). Based on studies in 2015 and 2022, average prices for gill plates were significantly higher in 2022 relative to 2015 in the dried seafood market in Hong Kong SAR of China; prices were above USD300 per kg, and gill plates of manta rays could be sold for approximately USD450 per kg.

After their inclusion in CITES Appendix II, there has not been a significant reduction in the availability of gill plates (Hau and Shea, 2023). Evidence suggests that the trade is growing and migrating to online platforms. Online research in 2024 identified five Chinese platforms selling dried *Mobulid* gill plates with prices starting at USD249 per kg and reaching up to USD1,260 per kg. Despite the listing of *Mobula* species in CITES Appendix II, the demand for gill plates continues to drive the decline of these species in international trade.

Legal Trade

Products in international trade are primarily gill plates, but there is also commercial trade in wild-sourced fins, skins, and live animals for aquaria. Between 2021 and 2023, the CITES Trade Database had records of more than 80,000 kg of gill plates, approximately 2,500 kg of fins (including dried fins), 250 kg of skins, and 800 kg of live *Mobulids* in international trade. Most gill plates were exported from Sri Lanka to Hong Kong SAR of China, and most fins were exported from Yemen to Hong Kong SAR of China.

According to LEMIS, 25 live *Mobulids*, 8 fins, 101 skins (whole or piece), 66 small leather products, and 193 shoes of *Mobula* species were imported into the USA between 2008 and 2021. With the exception of four pre-Convention skins and one live *Mobulid* (recorded as captive-bred), all exports were wild-sourced and for commercial purposes. During this same timeframe, the USA authorized export of 81 live animals.

Illegal Trade

Illegal trade in *Mobula* species is known to occur based on seizures and an analysis of trade statistics. Between 2008 and 2021, 655.59 kg (and 50 quantities) of *Mobulid* meat were seized in the USA from Mexico; there were also seizures of 12 fins in 2015 and four shoes in 2018 from Mexico. Eight quantities of *Mobulid* meat were seized in the USA from Vietnam in 2008, and there were two seizures of *Mobulid* medicinal products from China in 2018 (each comprised of 94 quantities).

WiTIS did not show any records of seizures of Mobulids.

Only five Parties (less than 5% of range States) have reported trade in Mobulids to CITES, but analysis of the CITES Trade Database between 2017 and 2021 highlighted an increase and high volume in the reported trade of gill plates globally. Consequently, *Mobula* species were included in the Review of Significant Trade in combination with Sri Lanka at AC32 in 2023. An assessment of the consumption and trade in Mobulid gill plates and meat from November 2023 to February 2024 revealed there is significantly more international trade in *Mobula* species than records in the CITES Trade Database show. More than 100 expert interviews covering 75 countries revealed that Mobulid gill plates are exported from at least from 14 Parties (Bangladesh, Hong Kong SAR and Mainland China, Ghana, India, Indonesia, Myanmar, Philippines, Senegal, Somalia, South Africa, Sri Lanka, Thailand, United Arab Emirates and Yemen); extracted in the Chagos archipelago, Viet Nam, Mozambique, Benin, Gabon, and Republic of Congo; and exported internationally across Asia and Africa, with five major destinations (Hong Kong SAR and Mainland China, Singapore, South Korea, and Thailand). This contradicts records found in the CITES Trade Database showing exports from only four Parties (Sri Lanka,¹⁵ India, Oman, and Yemen) and two importers (Hong Kong SAR of China and Singapore). Gill plates are being transported and concealed with other products and may be taking similar routes as shark fins.

Additional Information

Threats

Mobulid populations are impacted by overfishing by industrial and artisanal fleets in the Atlantic, Indian, and Pacific Oceans. They are caught by a variety of gear, including driftnets, gillnets, longlines, purse seines, traps, and trawls, as directed catch or bycatch. The seasonal aggregating behaviour of Mobulid species may increase their risk of overexploitation.

The species are currently subject of a CITES Review of Significant Trade Process in combination with Sri Lanka (CITES, 2023; 2025).

Mobula species are targeted and retained when caught incidentally for their high value gill plates (which are used to filter feed plankton, small fish, and crustaceans). Meat is also used and sold; the most desirable meat from the pectoral fins is separated from the gill plates and the rest of the carcass.

Degradation of coastal and coral reef habitats, boat strikes, increases in surface temperature associated with climate change, and disturbances from unregulated tourism activities may also impact these species (Osgood et al., 2020).

Conservation, management and legislation

Several international, regional, and national protections and legislation for Mobulids have been enacted in recent years. These species were included in Appendix I and II under CMS in 2015, so they are prohibited from being taken from the wild by Parties to CMS. The Mediterranean population of Spinetail Devil Ray is listed in Annex II of the SPA/BD Protocol of the Barcelona Convention, and Oceanic Manta Ray is listed in Annex III of the SPAW Protocol.

In 2015, the Inter-American Tropical Tuna Commission (IATTC) adopted a resolution aimed at conserving Mobulids caught in association with fisheries within the IATTC Convention Area (IATTC 2019). Both the Indian Ocean Tuna Commission (IOTC) and the Western and Central Pacific Fisheries Commission (WCPFC) adopted similar measures in 2019. The International Commission for the Conservation of Atlantic Tunas (ICCAT) adopted a conservation and management measure for mobulids in 2023. However, ICCAT's measure did not enter into force until its Standing Committee on Research and Statistics provided favourable scientific advice at its 2024 meeting. These measures prohibit the retention, transshipment, or landing of mobulids caught in the Convention area. The measures also require prompt release, encourage best practices for handling and release, require fishery observers to record Mobulid catch data, and call for research to identify important reproduction areas that could inform future management for *Mobula* species.

At the national level, laws and regulations have been promulgated in the last two decades for the conservation of Mobulids. Currently, at least 44 countries have developed policies or legislation providing full or partial protection for these species. The establishment of Marine Protected Areas in some locations, such as the Galápagos National Reserve in Ecuador, has also provided refugia for these species.

¹⁵ According to Fernando in litt. (2025), export records of Mobulid fins and skins found in the CITES Trade Database between 2017 and 2023 were likely the result of a coding error. Throughout years of surveying (over 4,300 surveys in Sri Lanka and from available knowledge in neighbouring countries like India), there are no records or evidence to suggest that "finning" or "skinning" of *Mobula* occurs.

Artificial Propagation/captive breeding

There is no literature or information available regarding captive breeding of *Mobulids*.

Implementation challenges (including similar species)

Distinguishing between some *Mobulids* can be challenging due to their visual similarities and highly variable patterning and colouration. *Mobulid* gill plates, once extracted and dried for trade, are also difficult to distinguish and identify by species. The identification of other products is a challenge; meat as is usually dried and upon death, these species experience significant changes in their skin patterns and colours.

A new Field Guide to the Manta and Devil Rays of the World published this year may provide a tool to assist CITES Parties. The guide includes updated information on species, key species features, and post-mortem colour change illustrations, among other information. An updated *Mobulid* gill plate identification key is also available in that publication.

References

- Bucair, N., Hinojosa-Alvarez, S., Marshall, A.D., Pate, J., Bezerra Francini, C.L., Guedes Garrido, A., Cruz Capel, K.C., Silva Loboda, T., Sirino Monteiro, J., Malavasi Bruno, C.E., Vaga, C.F., Dove, A.D.M., Hoopes, L.A., Perry, C. and Kitahara, M.V. (2005) An integrative taxonomy investigation unravels a cryptic species of *Mobula Rafinesque*, 1810 (*Mobulidae*, *Myliobatiformes*), from the Atlantic Ocean. *Environmental Biology of Fishes*. <https://doi.org/10.1007/s10641-025-01727-2>.
- Chopra, M., Rowlands, G., Stevens, G.M.W., Fernando, D., Mohanraj, T., Laglbauer, B.J.L., Karnad, D. and Davis, K.J. (2025) Fewer devil rays in the sea: Evidence of declining *mobulid* populations off India's southeastern coast (in review).
- CITES (2023) AC32 Summary Record (Annex 1). Available at: <https://cites.org/sites/default/files/eng/com/ac/32/E-AC32-SR.pdf>
- CITES (2025) SC78 Summary Record. Available at: <https://cites.org/sites/default/files/eng/com/sc/78/E-SC78-SR.pdf>
- de Boer, M.N., Wieczorek, A. and Notarbartolo di Sciara, G. (2024) First records of *Mobula hypostoma* (Bancroft, 1831), Atlantic Pygmy Devil Ray, (*Myliobatiformes*, *Mobulidae*) and other devil rays in shallow waters of the Republic of the Congo, Gulf of Guinea (West Africa). *Check List* 20(3):646–652. <https://doi.org/10.15560/20.3.646>
- Eschmeyer, W.N., Fricke, R. and Van der Lann, R. (editors) (2017) Species information extracted from Catalog of Fishes: Genera, Species, References. Online version of 28 April 2017. AC29 Doc. 35 (Annex 1). Available at: <https://cites.org/sites/default/files/eng/com/ac/29/E-AC29-35-A4.pdf>
- Fernando, D. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Fernando, D. and Stevens, G. (2011) A Study of Sri Lanka's Manta & *Mobula* Ray Fishery. Dorchester, UK: The Manta Trust.
- Fernando, D. and Stewart, J.D. (2021) High bycatch rates of manta and devil rays in the "small-scale" artisanal fisheries of Sri Lanka. *PeerJ* 9:e11994 <http://doi.org/10.7717/peerj.11994>
- Hau, L. and Shea, S. (2023) Re-visit of *Mobula* gill plates in the Hong Kong dried seafood market after CITES Appendix II listing. Bloom Association, Hong Kong.
- Jabado, R.W., Marshall, A., Stevens, G., Laglbauer, B., D'Costa, N., Barros, N., Carter, R., Doherty, P., Fernando, D., García, E., Metcalfe, K., Morey, G. and Mvomo Minko, Y. (2025a in press) *Mobula mobular*. *The IUCN Red List of Threatened Species* 2025: e.T110847130A279075986
- Jabado, R.W., Marshall, A., Stevens, G., Laglbauer, B., Barros, N., D'Costa, N., Carter, R., De Bruyne, G., Doherty, P., Fernando, D., Metcalfe, K., Mvomo Minko, Y. and Rohner, C.A. (2025b in press) *Mobula tarapacana*. *The IUCN Red List of Threatened Species* 2025: e.T60199A279077133
- Jabado, R.W., Marshall, A., Stevens, G., Laglbauer, B., Barros, N., D'Costa, N., Carter, R., De Bruyne, G., Doherty, P., Fernando, D., Metcalfe, K., Mvomo Minko, Y., Rambahinirison, J. and Rohner, C.A. (2025c in press) *Mobula thurstoni*. *The IUCN Red List of Threatened Species* 2025: e.T60200A279078290
- Laglbauer, B.J.L., Salim, M.G., Fahmi, Oktaviyani, S., Gozali, I.C., Tawang, F., Rizal, H.S., Rosady, V.P., Rudianto, D., Ender, I., Fontes, J.M.R., Afonso, P., Bennett, M.B. and Stevens, G.M.W. (2025) High take of *mobulid* rays in East Java, Indonesia: landing trends and socio-economic context (in press)
- Lawson, J.M., Fordham, S.V., O'Malley, M.P., Davidson, L.N.K., Walls, R.H.L., Heupel, M.R., Stevens, G., Fernando, D., Budziak, A., Simpfendorfer, C.A., Ender, I., Francis, M.P., Notarbartolo di Sciara, G. and Dulvy, N.K. (2017) Sympathy for the devil: a conservation strategy for devil and manta rays. *PeerJ* 5:e3017. <http://dx.doi.org/10.7717/peerj.3027>
- Lewis, S.A., Setiasih, N., Dharmadi, Fahmi, O'Malley, M.P., Campbell, S.J., Yusuf, M. and Sianipar, A. (2015) Assessing Indonesian Manta and Devil Ray Populations Through Historical Landings and Fishing Community Interviews. *PeerJ Preprints*. Available at <https://peerj.com/preprints/1334/>
- Lezama-Ochoa, N., Hall, M. A., Roman-Verdesoto, M. H. and Vogel, N. (2019) Spatial and temporal distribution of *mobulid* ray species in the eastern Pacific Ocean ascertained from observer data from the tropical tuna purse-seine fishery. *Environmental Biology of Fishes* 102:1–17. <https://doi.org/10.1007/s10641-018-0832-1>

- Marshall, A., Barreto, R., Carlson, J., Fernando, D., Fordham, S., Francis, M.P., Derrick, D., Herman, K., Jabado, R.W., Liu, K.M., Rigby, C.L. and Romanov, E. (2022a) *Mobula birostris* (amended version of 2020 assessment). *The IUCN Red List of Threatened Species* 2022: e.T198921A214397182. <https://dx.doi.org/10.2305/IUCN.UK.2022-1.RLTS.T198921A214397182.en>
- Marshall, A., Barreto, R., Carlson, J., Fernando, D., Fordham, S., Francis, M.P., Herman, K., Jabado, R.W., Liu, K.M., Rigby, C.L. and Romanov, E. (2022d) *Mobula munkiana* (amended version of 2019 assessment). *The IUCN Red List of Threatened Species* 2022: e.T60198A214407880. <https://dx.doi.org/10.2305/IUCN.UK.2022-1.RLTS.T60198A214407880.en>
- Marshall, A., Barreto, R., Carlson, J., Fernando, D., Fordham, S., Francis, M.P., Herman, K., Jabado, R.W., Liu, K.M., Pacoureau, N., Rigby, C.L., Romanov, E. and Sherley, R.B. (2022e) *Mobula alfredi* (amended version of 2019 assessment). *The IUCN Red List of Threatened Species* 2022: e.T195459A214395983. <https://dx.doi.org/10.2305/IUCN.UK.2022-1.RLTS.T195459A214395983.en>
- Marshall, A., Barreto, R., Carlson, J., Fernando, D., Fordham, S., Francis, M.P., Herman, K., Jabado, R.W., Liu, K.M., Rigby, C.L. and Romanov, E. (2022f) *Mobula hypostoma* (amended version of 2019 assessment). *The IUCN Red List of Threatened Species* 2022: e.T126710128A214399766. <https://dx.doi.org/10.2305/IUCN.UK.2022-1.RLTS.T126710128A214399766.en>
- Moazzam, M. (2018). Unprecedented decline in the catches of mobulids: an important component of tuna gillnet fisheries of the Northern Arabian Sea. IOTC-2018-WPEB14-30. Cape Town, South Africa.
- Moazzam Khan, M. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Osgood, G.J., White, E.R. and Baum, J.K. (2020) Effects of climate-change drive gradual and acute temperature changes on shark and ray species. *Journal of Animal Ecology* 90:2547-2559.
- Pardo, S., Kindsvater, H., Cuevas-Zimbrón, E., Sosa-Nishizaki, O., Pérez-Jiménez, J.C. and Dulvy, N.K. (2016) Growth, productivity and relative extinction risk of a data-sparse devil ray. *Scientific Reports*: 6(33745). <https://doi.org/10.1038/srep33745>
- Ralph, G.M., Stump, E., Linardich, C., Bullock, R.W., Carpenter, K.E., Allen D.J., Hilton-Taylor, C., Al Mheiri, R., and Alshamsi, O. (2021) UAE National Red List of Marine Species: Reef-building corals, cartilaginous fishes and select bony fishes. Ministry of Climate Change and Environment, Dubai, United Arab Emirates.
- Rojas-Perea, S., D'Costa, N.G., Kanagusuaku, K., Eschobedo, R., Rodríguez, F., Mendoza, A., Maguiño, R., Flores, R., Laglbauer, B.J.L., Stevens, G.M.W. and Kelez, S. (2025) Fisheries, trade, and conservation of manta and devil rays in Peru. *Environmental Biology of Fishes* 108:725–748.
- Rigby, C.L. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Rigby, C.L., Barreto, R., Carlson, J., Fernando, D., Fordham, S., Francis, M.P., Jabado, R.W., Liu, K.M., Marshall, A. and Romanov, E. (2022a) *Mobula eregoodoo* (amended version of 2020 assessment). *The IUCN Red List of Threatened Species* 2022: e.T41832A214376402. <https://dx.doi.org/10.2305/IUCN.UK.2022-1.RLTS.T41832A214376402.en>
- Rigby, C.L., Barreto, R., Carlson, J., Fernando, D., Fordham, S., Francis, M.P., Jabado, R.W., Liu, K.M., Marshall, A. and Romanov, E. (2022b) *Mobula kuhlii* (amended version of 2020 assessment). *The IUCN Red List of Threatened Species* 2022: e.T161439A214405747. <https://dx.doi.org/10.2305/IUCN.UK.2022-1.RLTS.T161439A214405747.en>
- Sant, G. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Venables, S.K., Rohner, C.A., Flam, A.L., Pierce, S.J. and Marshall, A.D. (2024) Persistent declines in sightings of manta and devil rays (Mobulidae) at a global hotspot in southern Mozambique. *Environmental Biology of Fishes* 108:749–765. <https://doi.org/10.1007/s10641-024-01576-5>
- White, W.T. and Last, P.R. (2016). Devilrays, Family Mobulidae. pp. 741-749 in Last, P.R., White, W.T., de Carvalho, M.R, Séret, B., Stehmann, M.F.W. and Naylor, G.J.P. (editors). *Rays of the World*. CSIRO Publishing, Comstock Publishing Associates. i-ix + 1-790.

Transfer of Whale Shark *Rhincodon typus* from Appendix II to Appendix I

Proponents: Argentina, Bahamas, Bangladesh, Belize, Comoros, Dominican Republic, Ecuador, Fiji, Gabon, Maldives, Panama, Philippines, Samoa, Senegal, Seychelles, Sri Lanka, Togo

Summary: The Whale Shark *Rhincodon typus* is the largest living fish, found in all tropical and warm-temperate waters of the Atlantic, Pacific, and Indian Oceans, generally between 30 degrees North and 35 degrees South of the Equator. It is pelagic and highly migratory. The species was included in Appendix II in 2003 following CoP12 and is proposed for transfer to Appendix I in accordance with Criterion C (i & ii) in Annex 1 of Res. Conf. 9.24 (Rev. CoP17).

The Whale Shark is characterized by very low productivity due to its slow growth, late maturity (~25 years in males, probably longer in females), and long generation time. It is believed to have high fecundity, with a single female able to produce up to 300 embryos on a biennial or longer basis. Whale Shark pups are approximately between 50 and 60 cm in length at birth. Based on its size and birth in open ocean environments, the Whale Shark is expected to have high early mortality, although it can live more than 50 years.

The Whale Shark is threatened by a range of factors including overfishing, illegal trade, vessel strikes, pollution, and habitat loss due to climate change. There are no stock assessments for Whale Shark, but population size and trends can be estimated based on genetic studies and data from standardized surveys, aerial surveys, observations, logbooks, and landings. Population estimates have ranged from ~100,000 adults (estimated in 2009) to 120,000-240,000 females (estimated in 2007). Population decreases equal to or greater than 50% have been inferred based on count data, modelled population estimates and habitat availability. The species was assessed by IUCN as Endangered in 2016 and was recently reassessed as Endangered for the forthcoming 2025 IUCN Red List (generation length having been reevaluated since the 2016 IUCN Red List assessment). This is based on an inferred population reduction of more than 30% over the last three generations (120 years) for the Atlantic where around 25% of the population is inferred to occur. In the Indo-Pacific, where the remainder of the population occurs, a reduction of more than 65% is inferred over the last three generations. Combining data from these regions a global population reduction of 50–79% over the last three generations was inferred, but the basis for the higher end of this estimate is unclear.

The species was listed in Appendix II at CoP12, effective from 2003, based on concerns over international trade, particularly in fins and meat. Since the listing, reported trade to CITES has been minimal. Fins were found in market surveys after their inclusion in Appendix II of CITES, but legal trade does not appear to be impacting the species, likely due to the numerous protections that have been put in place over the last 20 years. Illegal trade is reported in meat, fins, and gill plates, and there are records of eight seizures involving live Whale Sharks or fins between 2014 and 2023.

Protective measures have been put into place for Whale Shark at the national, regional, and international level, including under several tuna Regional Fishery Management Organizations (RFMOs). The Convention on Migratory Species of Wild Animals (CMS) prohibits take of the species by those nations that are Parties to CMS and the Whale Shark is also included in Annex I of the CMS Memorandum of Understanding on the Conservation of Migratory Sharks (Shark MOU).

Analysis: Whale Shark populations are not small, nor do they have a restricted area of distribution at present. Available data show that the population size of Whale Sharks has decreased significantly in some locations and is projected to continue to decrease, due to fishing, ship strikes and climate change. According to the footnote in Annex 5 of Res. Conf. 9.24 (Rev CoP17) on commercially exploited aquatic species, the likely decrease of 50%, perhaps somewhat higher, over 120 years for a low productivity species would not be considered a marked historic decline consistent with the indicative guidelines for inclusion in Appendix I (which suggests a reduction to 15-20% of historic

baseline). Levels of reported legal trade are minimal, likely due to national and international protection measures. Available evidence of illegal trade levels suggests they are unlikely to be affecting the population overall.

On this basis and taking into account the wide range of conservation measures adopted in recent years, **this species does not appear to meet the biological or trade criteria** for inclusion in Appendix I.

Summary of Available Information

Text in non-italics is based on information in the Supporting Statement (SS); text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

Whale Sharks were included in Appendix II of CITES in 2003.

Taxonomy

Rhincodon typus (Smith, 1828).

Range

Range: Extant (resident)

American Samoa; Angola; Anguilla; Antigua and Barbuda; Argentina; Aruba; Australia; Bahamas; Bahrain; Bangladesh; Barbados; Belize; Benin; Brazil; Brunei Darussalam; Cabo Verde; Cambodia; Cameroon; Cayman Islands; Chile; China; Colombia; Congo; The Democratic Republic of the Congo; Cook Islands; Costa Rica; Cuba; Curaçao; Côte d'Ivoire; Djibouti; Dominica; Ecuador; Egypt; El Salvador; Equatorial Guinea; Eritrea; Ethiopia; Fiji; French Guiana; French Polynesia; Gabon; Gambia; Ghana; Grenada; Guadeloupe; Guatemala; Guinea; Guinea-Bissau; Guyana; Haiti; Honduras; India; Indonesia; Islamic Republic of Iran; Iraq; Israel; Jamaica; Japan; Jordan; Kenya; Kiribati; Liberia; Madagascar; Malaysia; Maldives; Marshall Islands; Martinique; Mauritania; Mexico; Micronesia, Federated States of; Montserrat; Morocco; Mozambique; Myanmar; Namibia; Nauru; New Caledonia; New Zealand; Nicaragua; Nigeria; Niue; Oman; Pakistan; Panama; Papua New Guinea; Peru; Philippines; Pitcairn; Portugal; Puerto Rico; Qatar; Saint Helena, Ascension and Tristan da Cunha; Saint Kitts and Nevis; Saint Lucia; Saint Martin (French part); Saint Vincent and the Grenadines; Samoa; Sao Tome and Principe; Saudi Arabia; Senegal; Sierra Leone; Sint Maarten (Dutch part); Solomon Islands; Somalia; South Africa; Sudan; Suriname; Taiwan Province of China; United Republic of Tanzania; Thailand; Togo; Tokelau; Tonga; Turks and Caicos Islands; Tuvalu; United Arab Emirates; USA; Uruguay; Vanuatu; Bolivarian Republic of Venezuela; Viet Nam; Virgin Islands, British; Virgin Islands, U.S.; Wallis and Futuna; Western Sahara; Yemen

Extant & Vagrant (resident)

Canada

Whale Sharks are circumglobal, found in the waters of the Atlantic, Pacific, and Indian Oceans (Figure 1). They are often found between 30 degrees North and 35 degrees South, and are rarely found where the surface temperatures are less than 21 degrees Celsius.



Figure 1. Map of the Whale Shark's distribution (IUCN Shark Specialist Group 2016. *Rhincodon typus*. The IUCN Red List of Threatened Species. Version 2025-1).

IUCN Global Category and population trend information

Endangered A2bd+4bd (assessed 2025, ver. 3.1)

Biological criteria for inclusion in Appendix I

A) Small wild population

As of January 2021, there were 12,355 individual Whale Sharks listed in the Wildbook for Sharks (<https://www.sharkbook.ai>), which provides a visual database of sharks that have been encountered or individuals that have been catalogued. Genetic studies have sought to estimate the number of breeding adults. Using mitochondrial DNA, Castro et al. (2007) estimated that there are between 119,000 and 238,000 female Whale Sharks, and Schmidt et al. (2009) estimated ~100,000 ($\geq 76,000$) adult sharks based on microsatellite analysis. According to Schmidt (2021), these figures should be interpreted as rough estimates and may not reflect recent decreases in population. More recent genetic studies have not attempted to estimate population sizes due to the unvalidated assumptions (Pierce in litt., 2025).

Whale Sharks often aggregate to feed on seasonally available patches of plankton, which is their primary food source (Rohner and Prebble, 2022). As a result, in some locations, 500 or more individual Whale Sharks have been documented (Pierce et al., 2016). The largest aggregations are related to extremely productive events (such as mass fish spawning) that draw in Whale Sharks from hundreds to thousands of km (Pierce in litt., 2025). According to Kindsvater et al. (2016), although the populations of some species can reach very high densities in productive environments, they are not able to withstand high levels of fishing pressure. While large aggregations continue to be reported at some sites, aggregate-wide trends show declines in size structure and encounter rates at multiple sites (Jabado in litt., 2025). Overall, there are (i) declines in abundance at the largest aggregation(s), (ii) very steep declines in the abundance of largest individuals, and (iii) evidence of declines at multiple sites consistent with a global population reduction of 50–79% over the past three generations (Dulvy in litt., 2025).

B) Restricted area of distribution

The species is circumglobal and highly migratory. Mitochondrial and microsatellite DNA analyses indicate that Whale Shark populations in the Atlantic and Indo-Pacific are functionally separate. Based on counts, modelled population estimates, and habitat availability, it is likely that approximately 75% of the global Whale Shark population occurs in the Indo-Pacific, and 25% in the Atlantic. A recent study showed an increasing threat of core habitat loss for Whale Sharks due to warming sea surface temperatures, with estimated losses of greater than 50% in some national waters predicted by 2100 due to geographic shifts of over 1,000 km² per year.

C) Decline in number of wild individuals

As in 2016, the Whale Shark was assessed as Endangered A2bd+4bd on the IUCN Red List in 2025. It has also been assessed as endangered on the national red list of the United Arab Emirates (UAE) (Ralph et al., 2021). Based on the species' distribution, separate analyses were conducted for the Whale Shark subpopulations in the Indo-Pacific and in the Atlantic.

Different data sets present a range of declines in the Indo-Pacific from 40 to 92%, inferring an overall decline of 63% over the last three generations (75 years), resulting in a subpopulation assessment of Endangered A2bd+4bd. In the Atlantic, the overall population decline was estimated to be $\geq 30\%$ with a subpopulation assessment of Vulnerable A2b+4b. The overall global decline is inferred to be $>50\%$ over the last three generations (75 years) (Pierce et al., 2016). The data sets with estimates of population trend information are provided below (Table 1). Since the 2016 IUCN Red List Assessment, Pierce et al. (2021b) created a demographic model for the species resulting in a revised generation length which is estimated at 40 years. The 2025 assessment using this generation length also assessed the species as Endangered. Based on count data, modelled population estimates and habitat availability, 75% of the global Whale Shark population is inferred to occur in the Indo-Pacific, and 25% in the Atlantic. An overall decline of $>65\%$ is inferred in the Indo-Pacific over the last 120 years (three generations). In the Atlantic Ocean, the overall population decline is considered to be lower at $\geq 30\%$. Combining data from both regions, based on abundance data and actual levels of exploitation, a population reduction of 50–79% was inferred over the last three generations (120 years) (Pierce et al., 2025).

Table 1. Whale Shark key population trend information used in this analysis. Population trend information was taken from the IUCN Red List assessment (2016) and Supplemental Information unless otherwise noted.

Region/Ocean Basin	Population Trend Information*
Indo-Pacific	Sightings data (Andaman Sea, Thailand): Decreased by approximately 92% from 1.58 per trip between 1992 and 1993 to 0.13 per trip between 2000 and 2001. Standardised sightings data (Inhambane, Mozambique): Decrease from 2005 to 2011 of 79% (Rohner et al., 2013); updated data shows further decline of 87% between 2005 and 2019, with a 99% decline of sharks $>7m$ total length (Auditore et al., 2025 in revision). Catch per unit effort (CPUE) data (Philippines): Local decreases of approximately 60%; declining from 4.44 to 1.7 in Pamilacan and 10 to 3.8 in Guiwanon between two surveys conducted in 1993 and 1997.

Region/Ocean Basin	Population Trend Information*
	CPUE data (Maldives): Decreased from 30/year at one fishing location (B. Dhonfanu) in the 1980s to less than 20/year nationwide by 1993 Abundance (Maldives): Decrease of the resident population documented from 2014-2019 (Harvey-Carroll et al., 2021). CPUE data (Hongchun Harbour, Taiwan Province of China): Decrease of approximately 80% (between 50-60 individuals to less than 10 between the mid-1980s and 1995). CPUE data (Taiwan Province of China): Decreased by approximately 58% from 272 sharks in 1997 (over 12 months) to 113 sharks between 2001 and 2002 (over 15 months). Mean size (total length) of landed sharks (Taiwan Province of China and Mainland China): Decreased from 8.27 m before 2004 to 6.3 m between 2008-2011 in southern Chinese waters and decreased from approximately 4.6 m in 2002 to approximately 4.4 m in 2007 in waters off Taiwan Province of China. Sightings data (Western and Central Indian Ocean): Decreased approximately 66% from 600 in the 1990s to approximately 200 across 2000 to 2007. Sightings data (Western Central Pacific Ocean): Decreased approximately 50% from purse-seine vessels fishing for tuna in the from 2003 to 2012. Increase in occurrence based on sightings from 0.003 in 2000 to 0.012 in 2010 in the Western Central Pacific Ocean; model performance for dataset was poor (Sequiera et al., 2014). Interactions with tuna fisheries in the Pacific decreased from between 13 and 17 interactions per 1,000 sets from 2006 to 2008 to approximately eight interactions per 1,000 sets between 2009 and 2016 (Neubauer et al., 2018). Tourist sightings data (Ningaloo Reef, Australia): 40% decrease in the rate of sightings and a decline in mean shark length of 1.6 m between 1995 and 2004; the data was considered controversial and shown to be due to recruitment of smaller individuals by follow-up analyses conducted from 1995 to 2008. Genetic diversity study (Ningaloo Reef, Australia): Decrease in genetic diversity over five years for mtDNA (from 2007 to 2012).
Atlantic*	Sightings (Belize): Decreased from a mean of 4 to 6 sharks per day between 1998-2001 to less than 2 per day in 2003 in Belize with reports from diving guides indicating that numbers remained low until at least 2016, to current cessation of the aggregation and rare visitation of Whale Sharks to the site as of 2025 (R. Graham unpubl. data cited in Pierce and Norman, 2016). Sightings data recorded in logbooks of tuna purse-seine vessels (Atlantic): decreased by approximately 70% from 1980 to 2010; according to Pierce in litt. (2025), however, updated results from the region were reinterpreted and there was too much variation to provide a clear trend. Relative abundance (Northern Mexican Caribbean): Observation of 17 Whale Sharks per 100 km² from 2016-2021 with significant decrease in density by 2021, with 8 Whale Sharks per 100 km² (Cárdenas-Palomo et al., 2025) Sightings (Azores): Increase from 4 in 1998 – during and after 2008 – to approximately 90 in 2013 (96% increase); this was strongly correlated with the location of the 22°C isotherm, indicating that the sighting trend was due to environmental conditions.

* According to Pierce in litt. (2025), there have not been many publications from the Atlantic due to the Whale Sharks largely disappearing.

Trade criteria for inclusion in Appendix I

The species is or may be affected by trade

The species was listed in the Appendices due to concerns over the trade in its fins and meat (see CoP12 Prop 35). According to the CITES Trade Database, there has been little commercial trade since its inclusion. Most reported trade of Whale Shark is in specimens for scientific purposes. The species has not been included in the Review of Significant Trade process.

Legal trade

The CITES Trade Database records show low levels of trade in Whale Sharks and their products; most trade records are composed of small (a few grams) tissue samples for scientific research or exports of a small number of live animals for display in public aquaria.

Six live specimens for zoological purposes were reportedly imported by the USA from Taiwan Province of China, as reported by the USA. The export of 395 live "specimens" for scientific purposes to Australia in 2021 is likely to be an error (Pierce in litt., 2025).

Practically no commercial trade has been reported in any parts or derivatives of this species including fins or meat.

Market Data

The SS documented fishing and national use of Whale Sharks (some of which may have been caught in areas beyond national jurisdiction). Whale Shark fins were observed in market surveys after the species was included in Appendix II. *Genetic studies revealed that Whale Shark fins were found in markets in Canada, and their gill plates were found in markets in China and Sri Lanka during 2011 and 2012 (Steinke et al., 2017). A small amount of Whale Shark gills was observed in three years of trader surveys in China and Southeast Asia in 2011, 2013, and 2015 (O'Malley et al., 2017).*

Illegal trade

Although there are few records of international trade in Whale Shark according to the CITES Trade Database, there is known illegal trade in live animals, meat, and fins of Whale Sharks. *A survey and genetic analysis of tissue samples from three landings and nine shark processing centres between 2016 and 2017 in Bangladesh revealed CITES-listed sharks and rays, including Whale Sharks (Haque et al., 2019).*

WiTIS includes four seizures of live Whale Sharks that occurred between 2014 and 2023 (totalling five individuals) two illegal captures of Whale Sharks between 2019 and 2020, and four seizures of shark fins between 2014 and 2018 (comprising a total of more than 2,000 kg of fins from mixed species) that contained fins from Whale Sharks. In 2017, Whale Shark fins were among thousands that were seized from a Chinese-flagged fishing vessel crossing through the Galápagos Marine Reserve without authorization (Bonaccorso et al., 2021). Whale shark fins were also discovered in a shipment destined for Hong Kong SAR of China in 2018. Indian officials successfully stopped the poaching of a Whale Shark in 2023. According to LEMIS, the USA seized Whale Shark medicinal parts and products exported from China in 2010, and there were seizures of Whale Shark meat originating from Mexico (2017) and Haiti (2018).

Additional Information**Threats**

The meat, fins, and liver oil of Whale Sharks have a high commercial value, encouraging retention of bycatch or targeting of the species for international trade. Consequently, although there are protections in place at the national, regional, and international level, there is international demand for Whale Sharks and their products, and seizure data indicate that illegal exports are occurring.

Due to rising ocean surface temperatures, Whale Sharks are also threatened by projected decreases in their available suitable habitat. Pollution also adversely affects these species. Beyond these threats, Whale Sharks face the risk of mortality due to ship strikes. A recent study predicts increasing co-occurrence of Whale Sharks and shipping traffic based on predictions of habitat shifts with rising ocean surface temperatures.

Conservation, management and legislation

Measures have been adopted for Whale Sharks in many countries and territories that include bans on shark fishing or provide protection specifically for this species. Protections are in place in American Samoa, Australia, Bahamas, Belize, Cambodia, the UK (overseas territories), China, Republic of the Congo (Congo-Brazzaville), Cook Islands, Costa Rica, Djibouti, Dominican Republic, Ecuador, Egypt, El Salvador, French Polynesia, Guatemala, Guadeloupe, Guyana, Honduras, Hong Kong SAR of China, Indonesia, India, Kuwait, Maldives, Malaysia, Marshall Islands, Mexico, Mozambique, Myanmar, New Caledonia, New Zealand, Nicaragua, Palau, Panama, Philippines, Reunion, Samoa, Saudi Arabia, Seychelles, South Africa, Taiwan Province of China, Israel, Thailand, Tokelau, United Arab Emirates, and the USA. Marine protected areas have also been established where shark fishing is prohibited. These areas include Cocos Island in Costa Rica (Pacific Ocean), Malpelo Island in Colombia (Pacific Ocean), the Galapagos Islands in Ecuador, the Banc d'Arguin National Park in Mauritania, and the Marine Protected Areas in Guinea-Bissau.

In some locations where Whale Sharks provide ecotourism opportunities, regulations have been enacted to ensure these species are not harassed. Such regulations have been established in Australia, Belize, Ecuador

(Galápagos Islands), Mexico, and Saint Helena Island (UK). In the Philippines, local ordinances are in place to regulate tourism activities.

As discussed in Pierce et al. (2021a), the Whale Shark is listed as a Highly Migratory Species on Annex I of the United Nations Convention on the Law of the Sea (UNCLOS). Under UNCLOS, the Agreement relating to the Conservation and Management of Straddling Fish Stocks and Highly Migratory Fish Stocks outlines principles for the conservation and management of fisheries resources within national Exclusive Economic Zones and in areas beyond national jurisdiction. The Agreement mandates that management must take a precautionary approach and be based on the best available scientific information.

Whale Shark is included in Appendices I and II of CMS, which requires CMS Parties to prohibit take of this species. The Whale Shark is also included in Annex I of the CMS Shark MOU. Between 2013 and 2023, several tuna RFMOs adopted conservation and management measures for the Whale Shark. The Western and Central Pacific Fisheries Commission (WCPFC) (2022), the Indian Ocean Tuna Commission (2013), and the Inter-American Tropical Tuna Commission (2019) prohibit vessels from setting a purse seine on a school of tuna associated with a Whale Shark. *The International Commission for the Conservation of Atlantic Tunas adopted the same requirement (2023), but the Commission also prohibits vessels from retaining, transshipping, or landing any specimen of Whale Shark (ICCAT 2023). According to Jabado in litt. (2025) records from RFMOs indicate that this species is still being captured. Whale Sharks are more likely to survive fisheries interactions with purse seine vessels, especially if they are releasing the animals according to safe release protocols. However, these prohibitions have not had a direct impact on population trends yet because of the species' generation length. According to experts, trade flows have yet to be effectively regulated and enforced, and closing international trade through an Appendix-I listing could remove the incentive for such fisheries to catch and retain Whale Shark (Dulvy in litt., 2025).*

Implementation challenges (including similar species)

Whale Sharks have a distinct morphology that differs from all other shark species. They also have a characteristic pattern of yellow or white lines and spots on dorsal and lateral surfaces. Due to their large size, unique shape and pattern, whale sharks can be differentiated. Whale shark fins, dressed carcasses, meat with skin attached, and gill rakers can be identified visually. The meat is also distinctive with a unique soft spongy texture with very large, segmented blocks of muscle tissue.

Potential risk(s) of a transfer from Appendix II to I

The transport of biological samples of sharks and rays for research and data collection purposes can be difficult due to the timeframes and requirements for researchers to acquire CITES permits. The CITES Animals Committee and Standing Committee have been considering how to address the challenges associated with the transport of biological samples of sharks and rays, and several draft decisions will be considered at CoP20 that may lead to actions to improve this situation (CoP20 Doc. 88.1).

References

- Auditore, L.M., Pierce, S.J., Prebble, C.E.M., Flam, A., Marshall, A.D., Venables, S.K., Watts, A., Reeve-Arnold, K.E., Domenico, G., Keeping, J., Méndez-Jiménez, A., Riensema, A., Rosero, M., Cumbana, L., Tilley, A., Westling, A., Williams, C. and Rohner, C.A. (2025) A 15-year time-series shows major declines in whale sharks in southern Mozambique. *Aquatic Conservation: Marine and Freshwater Ecosystems* (in revision).
- Bonaccorso, E., Ordóñez-Garza, N., Pazmiño, D.A., Hearn, A., Páez-Rosas, D., Cruz, S., Muñoz-Pérez, J.P., Espinoza, E., Suárez, J., Muñoz-Rosado, L.D., Vizuete, A., Chaves, J.A., Torres, M.L., Bustos, W., Rueda, D., Hirschfield, M. and Guayasamin, J.M. (2021) International fisheries threaten globally endangered sharks in the Eastern Tropical Pacific Ocean: the case of the Fu Yuan Yu Leng 999 reefer vessel seized within the Galápagos Marine Reserve. *Scientific Reports* 11(14959). <https://doi.org/10.1038/s41598-021-94126-3>
- Cárdenas-Palomo, N., Mimila-Herrera, E., Trujillo-Córdova, J.A., Velázquez Abunader, I. and Herrera-Silveira, J. (2025) Whale shark abundance, spatial distribution and effective tourism areas in the northern Mexican Caribbean. *Journal of Ecotourism* 24(1):58-74.
- Castro, A.L.F., Stewart, B.S., Wilson, S.G., Hueter, R.E., Meekan, M.G., Motta, P.J., Bowen, B.W. and Karl, S.A. (2007) Population genetic structure of Earth's largest fish, the Whale Shark (*Rhincodon typus*). *Molecular Ecology* 16:5183-5192.
- Dove, A.D.M., Meekan, M.G. and McClain, C.R. (2021) How and Why is the Whale Shark the World's Largest Shark? pp 1-12 in Dove, A.D.M. and Pierce, S.J. (editors) *Whale Sharks Biology, Ecology, and Conservation*. Boca Raton: CRC Press.
- Dulvy, N.K. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Haque, A.B., Das, S.A. and Biswas, A.R. (2019) DNA analysis of elasmobranch products originating from Bangladesh reveals unregulated elasmobranch fishery and trade on species of global conservation concern. *PLoS ONE* 14(9):e0222273.

- Harvey-Carroll, J., Stewart, J.D., Carroll, D., Mohamed, B., Shameel, I., Zareer, I.H., Araujo, G. and Rees, R. (2021) The impact of injury on apparent survival of Whale Sharks (*Rhincodon typus*) in South Ari Atoll Marine Protected Area, Maldives. *Scientific Reports* 11:1–15.
- ICCAT (2023) Recommendation by ICCAT for the Conservation of Whale Sharks (*Rhincodon typus*) Caught in Association with ICCAT Fisheries. Available at: <https://www.iccat.int/Documents/Recs/compendiopdf-e/2023-12-e.pdf>
- Jabado, R. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Kindsater, H.K., Mangel, M., Reynolds, J.D. and Dulvy, N.K. (2016) Ten principles from evolutionary ecology essential for effective marine conservation. *Ecology and Evolution* 6:2125–2138.
- Neubauer, P., Richard, Y. and Clarke, S. (2018) Risk to the Indo-Pacific Ocean whale shark population from interactions with Pacific Ocean purse-seine fisheries, 52 pp. Report prepared for the Western and Central Pacific Fisheries Commission.
- O'Malley, M.P., Townsend, K.A., Hilton, P., Heinrichs, S. and Stewart, J.D. (2017) Characterization of the trade in manta and devil ray gill plates in China and South-East Asia through trader surveys. *Aquatic Conservation: Marine and Freshwater Ecosystems* 27:394–413.
- Pierce, S.J. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Pierce, S.J. and Norman, B. (2016) *Rhincodon typus*. *The IUCN Red List of Threatened Species* 2016: e.T19488A2365291. <http://dx.doi.org/10.2305/IUCN.UK.2016-1.RLTS.T19488A2365291.en>
- Pierce, S.J., Grace, M.K. and Araujo, G. (2021a) Conservation of Whale Sharks. pp. 267–300 in Dove, A.D.M. and Pierce, S.J. (editors) *Whale Sharks Biology, Ecology, and Conservation*. Boca Raton: CRC Press.
- Pierce, S.J., Pardo, S.A., Rohner, C.A., Matsumoto, R., Murakumo, K., Nozu, R., Dove, A.D.M., Perry, C. and Meekan, M.G. (2021b) Whale Shark Reproduction, Growth, and Demography. pp. 13–45 in Dove, A.D.M. and Pierce, S. J. (editors) *Whale Sharks: Biology, Ecology, and Conservation*. Boca Raton: CRC Press.
- Pierce, S.J., Rohner, C.A., Perry, C.T., Jabado, R.W., Norman, B., Reynolds, S., Womersley, F., Robinson, D., Graham, R. and Araujo, G. (2025 in press) *Rhincodon typus*. *The IUCN Red List of Threatened Species* 2025: e.T19488A126673248.
- Ralph, G.M., Stump, E., Linardich, C., Bullock, R.W., Carpenter, K.E., Allen D.J., Hilton-Taylor, C., Al Mheiri, R., and Alshamsi, O. (2021) UAE National Red List of Marine Species: Reef-building corals, cartilaginous fishes and select bony fishes. Ministry of Climate Change and Environment, Dubai, United Arab Emirates.
- Rohner, C.A. and Prebble, C.E.M. (2021) Whale Shark Foraging, Feeding, and Diet. pp. 153–180 in Dove, A.D.M. and Pierce, S.J. (editors) *Whale Sharks Biology, Ecology, and Conservation*. Boca Raton: CRC Press.
- Schmidt, J.V. (2021) Genetic Population Structure of Whale Sharks. pp. 83–103 in Dove, A.D.M. and Pierce, S.J. (editors) *Whale Sharks Biology, Ecology, and Conservation*. Boca Raton: CRC Press.
- Schmidt, J.V., Schmidt, C.L., Ozer, F., Ernst, R.E., Feldheim, K.A., Ashley, M.V. and Levine, M. (2009) Low genetic differentiation across three major ocean populations of the Whale Shark, *Rhincodon typus*. *PloS ONE* 4(4):e4988.
- Sequeira, A.M.M., Mellin, C., Fordham, D.A., Meekan, M.G. and Bradshaw, C.J.A. (2014) Predicting current and future global distributions of whale sharks. *Global Change Biology* 20:778–789.
- Steinke, D., Bernard, A.M., Horn, R.L., Hilton, P., Hanner, R. and Shivji, M.S. (2017) DNA analysis of traded shark fins and mobulid gill plates reveals a high proportion of species of conservation concern. *Scientific Reports* 7(9505).

Retain giant guitarfishes *Glaucostegus* spp. in Appendix II with a zero-export quota for wild specimens traded for commercial purposes

Proponents: Bangladesh, Benin, Brazil, Burkina Faso, Burundi, Cabo Verde, Central African Republic, Comoros, Congo, Gabon, Gambia, Guinea, Guinea-Bissau, Maldives, Mali, Niger, Nigeria, Panama, Sierra Leone, Sudan, Togo

Summary: Giant guitarfish species in the genus *Glaucostegus*, the only genus in the family Glaucostegidae, are shark-like batoid species occurring in the coastal waters of the Mediterranean and Black Sea, Atlantic, Pacific, and Indian Oceans. The genus *Glaucostegus* was included in Appendix II of CITES at CoP18 in 2019 due to concerns that the international trade in their fins could be causing significant population declines. The intention of the listing was to cover all species within the genus. While currently adopted CITES nomenclature recognises nine species within the genus, the proposal and this analysis consider seven species within the genus in line with recent taxonomic revisions and recommendations that will be considered by CoP20 (in CoP20 Doc 110). This does not change the scope of the listing, which is to regulate trade in all *Glaucostegus* species. The current proposal seeks to add an annotation to this listing specifying: “a zero annual export quota for wild-taken specimens traded for commercial purposes”. According to the supporting statement, this quota is proposed to ensure international trade is paused until population levels are able to support sustainable trade levels, prevent an Appendix-I listing in the future, and prevent the extinction of Glaucostegidae.

Giant guitarfish can reach more than 2 m in total length. Species in the family Glaucostegidae grow slowly, mature late, have a generation length between 10 and 15 years, and exhibit low productivity.

Similar to wedgefish, Rhinidae (the subject of proposal CoP20 Prop. 33), and sawfishes (Family Pristidae, also listed in Appendix I), giant guitarfish are generally found in shallow coastal and continental shelf waters up to depths of 120 m. Most species are strongly associated with soft-bottom habitats in shallow warm-temperate to tropical coastal waters less than 50 m deep.

Giant Guitarfish are subject to intense fishing pressure, and they are primarily impacted by unregulated fisheries. The fins of *Glaucostegus* species are highly valued on the international market, so they are targeted and commonly retained as bycatch in coastal fisheries. Given their dependence on shallow coastal environments, other factors affecting these species are habitat degradation, coastal development, and pollution.

Despite the extent of their capture in fisheries, there are no stock assessments for these data-limited species, and catches of giant guitarfish have historically only been reported at the genus or family level. Members of the Glaucostegidae family are similar in appearance and can be difficult to distinguish from each other, consequently, there are limited species-specific data, making it difficult to determine decreases in population for individual *Glaucostegus* species.

Due to the lack of species-specific data, population trends for *Glaucostegus* species are estimated or inferred using catch, landings, effort, and size data that often include data reported at the family-level, as well as information on potential extinctions of local populations. Decreases in other batoid species with similar life history characteristics and habitats - such as wedgefishes or sawfishes - are also used to infer population reductions in Giant Guitarfish species. All seven *Glaucostegus* species are globally assessed as Critically Endangered on the IUCN Red List between 2018 and 2022, due to severe population decreases. Based on an analysis of variable data sets over a timeframe from 1948 until 2015, population decreases of greater than 80% are inferred based on trends in landings and modelled population estimates.

Analysis of the fin market in Hong Kong SAR of China from 2014 to 2021 indicated continued international trade in their fins. Between 2021 and 2023, the CITES Trade Database recorded commercial exports of about 150,000 kg of fins, 60,000 kg of skins, and over 3,000 kg of meat from Giant Guitarfishes in international trade, mostly from Indonesia to Hong Kong SAR of China. Around

50% of the reported trade volume is related to *G. typus*, 20% to *G. cemiculus*, and the remainder was reported at a genus level, with limited records for other species. Evidence suggests there is illegal trade in the fins of Glaucostegidae based on discrepancies in trade data and recent seizures between 2019 and 2024.

There are few conservation measures in place for giant guitarfishes. They are not protected under any other international Convention. At the regional level, Blackchin Guitarfish (*G. cemiculus*) is listed in Annex II of the Protocol concerning Specially Protected Areas and Biological Diversity in the Mediterranean (SPA/BD Protocol) under the Convention for the Protection of the Marine Environment and Coastal Region of the Mediterranean (Barcelona Convention). However, these species are not directly managed by Regional Fishery Management Organizations (RFMOs), and few national management and conservation measures are in place for them.

Analysis: There is currently no guidance on how to assess a proposal to add an annotation to the listing of a species already in Appendix II. This is equivalent to the inclusion of an export quota or other special measure for a species being transferred from Appendix I to II, as provided for as precautionary measures under Annex 4, paragraph A.2.a) iii).

Giant Guitarfishes are characterized by slow growth, late maturity, and low productivity. Their reliance on shallow, coastal water habitats where fisheries occur combined with the high value and demand for their fins in international trade, make them highly vulnerable to overexploitation.

In the absence of species-specific data for *Glaucostegus* species, information on related species (Rhinobatidae (guitarfishes), unspecified Glaucostegidae (giant guitarfishes), or Rhinidae (Wedgefishes)) have been used to help infer population trends. Given the fishing pressure on these coastal, low productivity species and on-going exploitation pressure, it is likely that population declines, estimated at over 80% in the respective IUCN Red List assessments, have continued and would already meet the criteria in Annex 1 paragraph C of Res. Conf 9.24 (Rev CoP17).

In this light, the **proposed annotation of a zero-quota for wild specimens traded for commercial purposes, does appear to be an appropriate precautionary measure** pending improved circumstances.

Summary of Available Information

Text in non-italics is based on information in the Supporting Statement (SS); text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

All giant guitarfish species in the family Glaucostegidae were included in Appendix II in 2019.

Taxonomy

The following nine Glaucostegus spp. species are currently listed in Appendix II:

Glaucostegus cemiculus Geoffroy Saint-Hilaire, 1817

Glaucostegus granulatus Cuvier, 1829

Glaucostegus halavi Forsskal, 1775

Glaucostegus microphthalmus (Teng 1959)

Glaucostegus obtusus Muller & Henle, 1841

Glaucostegus petiti (Chabanaud 1929)

Glaucostegus spinosus (Günther 1870)

Glaucostegus thouin Anonymous, 1798

Glaucostegus typus Bennet, 1830

According to the Animals Committee (AC) Nomenclature Specialist in litt. (2025), Proposal 32 lists seven species as being validly recognized in the genus *Glaucostegus*; this contrasts with the currently adopted CITES nomenclature recognizing nine species. *Glaucostegus microphthalmus* is at present considered a synonym of *G. typus*; *Glaucostegus petiti* is a synonym of *G. cemiculus*; *Glaucostegus spinosus* is based on a single specimen from

1870, most likely a juvenile of a previously described species, and is best considered invalid for the time being; and the recently described *Glaucostegus younholeei* is included in the Proposal but not yet recognized by CITES.

In all these updates, the Proposal is consistent with the recommendations from the AC for updated nomenclature of CITES-listed fishes and is fully consistent with the Fish Checklist proposed for adoption at CoP20 in Doc 110 (AC Nomenclature Specialist in litt., 2025).

IUCN recognises seven species in line with the SS and the Standard Reference proposed for *Glaucostegus* in CoP20 Doc 110.

Range and IUCN Global Category (in alphabetical order by scientific name)

Blackchin Guitarfish *Glaucostegus cemiculus*

Critically Endangered A2bd (assessed 2018, ver. 3.1)

Eastern Atlantic species of giant guitarfish, whose range extends from Portugal, throughout the Mediterranean and as far south as Angola. Blackchin Guitarfish is found in the following FAO Fishing Areas: 27, 34, 37, 47

Range: Extant (resident)

Albania; Algeria; Angola; Benin; Bosnia and Herzegovina; Cabo Verde; Cameroon; Congo; The Democratic Republic of the Congo; Côte d'Ivoire; Croatia; Cyprus; Egypt; Equatorial Guinea; France; Gabon; Gambia; Ghana; Greece; Guinea; Guinea-Bissau; Israel; Italy; Lebanon; Liberia; Libya; Malta; Mauritania; Monaco; Montenegro; Morocco; Nigeria; Portugal; Senegal; Sierra Leone; Slovenia; Spain; Syrian Arab Republic; Togo; Tunisia; Turkey; Western Sahara

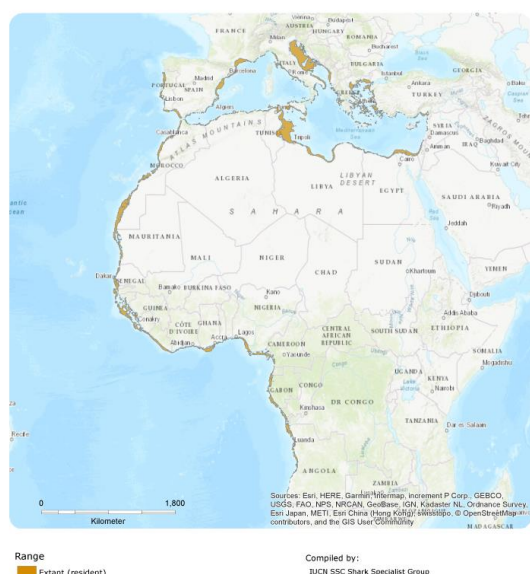


Figure 1. Blackchin Guitarfish (*Glaucostegus cemiculus*) Distribution Map. (IUCN SSC Shark Specialist Group 2018. *Glaucostegus cemiculus*. The IUCN Red List of Threatened Species. Version 2025-1).

Sharpnose Guitarfish *Glaucostegus granulatus*

Critically Endangered A2bd (assessed 2022, ver. 3.1)

Found in the northwestern Indian Ocean, from the United Arab Emirates (including the Arabian/Persian Gulf) to Myanmar. Sharpnose Guitarfish is found in the following FAO Fishing Areas: 51, 57

Range: Extant (resident)

Bangladesh; India; Islamic Republic of Iran; Iraq; Kuwait; Myanmar; Oman; Pakistan; Saudi Arabia; Sri Lanka; United Arab Emirates

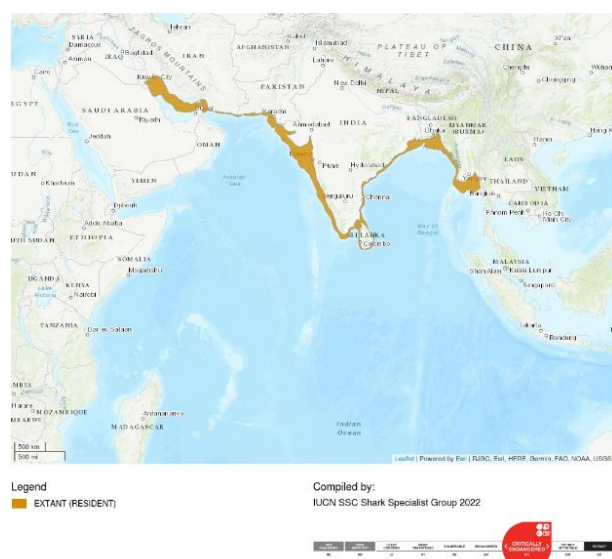


Figure 2. Sharpnose Guitarfish (*Glaucostegus granulatus*) Distribution Map. (IUCN SSC Shark Specialist Group 2022. *Glaucostegus granulatus*. The IUCN Red List of Threatened Species. Version 2025-1).

Halavi Guitarfish *Glaucostegus halavi*

Critically Endangered A2bd (assessed 2018, ver. 3.1)

Occurs in the northern Indian Ocean, Persian Gulf and Arabian and Red Seas. Halavi Guitarfish is found in the following FAO Fishing Area: 51

Range: Extant (resident)

Bahrain; Djibouti; Egypt; Eritrea; India (Gujarat); Islamic Republic of Iran; Israel; Kenya; Oman; Pakistan; Qatar; Saudi Arabia; Sudan; United Arab Emirates; Yemen



Figure 3. Halavi Guitarfish (*Glaucostegus halavi*) Distribution Map. (IUCN SSC Shark Specialist Group 2018. *Glaucostegus halavi*. The IUCN Red List of Threatened Species. Version 2025-1).

Widenose Guitarfish *Glaucostegus obtusus*

Critically Endangered A2bd (assessed 2018, ver. 3.1) (Kyne and Jabado, 2019b)

Found in the Northern Indian Ocean, from Pakistan to Thailand. Widenose Guitarfish is found in the following FAO Fishing Areas: 51, 57, 71

Range: Extant (resident)

Bangladesh; India; Indonesia; Malaysia; Myanmar; Pakistan; Singapore; Sri Lanka; Thailand



Figure 4. Widenose Guitarfish (*Glaucostegus obtusus*) Distribution Map. (IUCN SSC Shark Specialist Group 2018. *Glaucostegus obtusus*. The IUCN Red List of Threatened Species. Version 2025-1).

Clubnose Guitarfish *Glaucostegus thouin*

Critically Endangered A2bd (assessed 2018, ver. 3.1)

Found in the Indo-west Pacific, from India to Indonesia. There are reliable records of this species in the Red Sea. Clubnose Guitarfish is found in the following FAO Fishing Areas: 51, 57, 71

Range: Extant (resident)

Bangladesh; Brunei Darussalam; Cambodia; India; Indonesia; Malaysia; Myanmar; Singapore; Sri Lanka; Thailand; Viet Nam

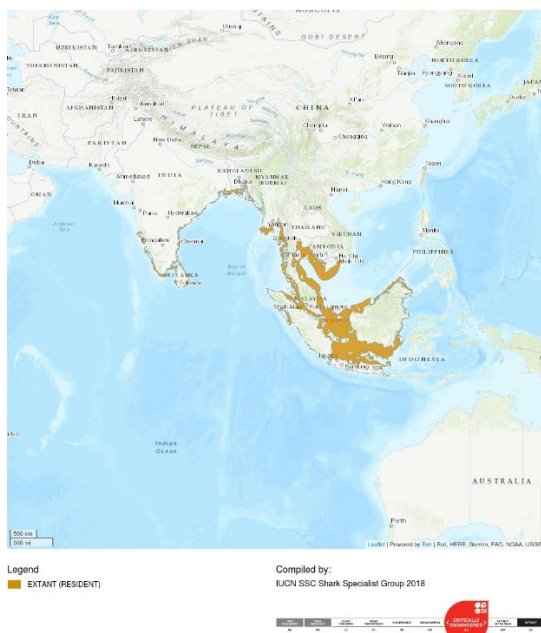


Figure 5. Clubnose Guitarfish (*Glaucostegus thouin*) Distribution Map. (IUCN SSC Shark Specialist Group 2018. *Glaucostegus thouin*. The IUCN Red List of Threatened Species. Version 2025-1).

Giant Guitarfish *Glaucostegus typus*

Critically Endangered A2bd (assessed 2018, ver. 3.1)

Occurs in the Indo-West Pacific from India to the East China Sea and Australia. Giant Guitarfish is found in the following FAO Fishing Areas: 51, 57, 61, 71, 81

Range: Extant (resident)

Australia (New South Wales, Northern Territory, Queensland, Western Australia); Bangladesh; Brunei Darussalam; Cambodia; China; India; Indonesia; Malaysia; Myanmar; Papua New Guinea; Singapore; Solomon Islands; Sri Lanka; Taiwan Province of China; Thailand; Timor-Leste; Viet Nam



Figure 6. Giant Guitarfish *Glaucostegus typus* Distribution Map. (IUCN SSC Shark Specialist Group 2018. *Glaucostegus typus*. The IUCN Red List of Threatened Species. Version 2025-1).

Bangladeshi Guitarfish *Glaucostegus younholeei*

Critically Endangered A2bd (assessed 2022, ver. 3.1)

Found in the northeast Bay of Bengal, Bangladesh. It may be more wide-ranging and its distribution will be better defined as more information becomes available. Bangladeshi Guitarfish is found in the following FAO Fishing Area: 57

Range: Extant (resident)

Bangladesh

A map of the distribution of Bangladeshi Guitarfish (*Glaucostegus younholeei*) is not available.

Biological information

A) Wild population

The global population size is unknown for any *Glaucostegus* species. Based on the limited catch and landings data available, the wild population does not appear to be small as defined in Annex 5.

B) Area of distribution

Giant Guitarfishes are primarily coastal species found in the Mediterranean and Black Sea, and Atlantic, Indian, and Pacific Oceans, typically habiting waters less than 100 m. According to Moore (2017), several species of Guitarfishes in the Order Rhinopristiformes (which includes Giant Guitarfishes) are restricted to a particular area, or few specimens have been found, and the distribution of some species may render them susceptible to extinction. Although this publication refers to species in the Order Rhinopristiformes, it suggests that some populations of Giant Guitarfishes may face the same risks of a restricted area of distribution.

C) Population trends

All the species in this proposal are in the Glaucostegidae family. These shark-like batoid species are highly susceptible to overexploitation due to their residence in coastal and continental shelf waters in areas with intense fishing pressure and large size (Dulvy et al., 2014). Giant Guitarfishes grow slowly to sometimes more than 2 m, mature late, and exhibit low productivity (or are inferred to exhibit low productivity based on their similarity to other species) (Table 1).

A simple but robust estimator of natural mortality (M) for sharks and rays is the reciprocal of average lifespan ($M=1/w$) where average lifespan 'w' is the generation length of the species (Pardo et al., 2016). Hence, these CITES mortality thresholds can be back converted to calculate the corresponding generation length productivity thresholds ($GL = M^{-1}$) and these are as follows: high productivity (natural mortality > 0.5 or a generation length of < 2 years), medium productivity (natural mortality $0.2-0.5$ or generation lengths $2-5$ years), and low productivity (natural mortality < 0.2 or generation length > 5 years).

Table 1. Life history characteristics of giant guitarfishes (*Glaucostegus* species) (in alphabetical order by scientific name).

Species	Maximum Size (cm) total length (TL)	Average size at maturity (cm) TL	Average litter size (pups)	Frequency of reproduction / gestation period	Estimated generation length (years)	Productivity
Blackchin Guitarfish (<i>Glaucostegus cemiculus</i>)	265	155 (males); 163 (females)	5 to 12 (Tunisia); 16 to 24 (Senegal)	1 year	15	Low
Sharpnose Guitarfish (<i>Glaucostegus granulatus</i>)	229	N/A	6 to 18	N/A	15	Low
Halavi Guitarfish (<i>Glaucostegus halavi</i>)	187	N/A	up to 10	N/A	10	Low
Widenose Guitarfish (<i>Glaucostegus obtusus</i>)	93	48 (males)	4 to 10	N/A	10	Low
Clubnose Guitarfish (<i>Glaucostegus thouin</i>)	300	unknown	small	unknown	15	Low
Giant Guitarfish (<i>Glaucostegus typus</i>)	270	150 to 180 (both sexes)	small	N/A	15	Low
Bangladeshi Guitarfish (<i>Glaucostegus younholei</i>)	93	unknown	unknown	unknown	10	Low

All giant guitarfishes are considered to have low productivity based on estimates of natural mortality (less than 0.2) or generation length (greater than five years). Horswill et al. (2025) also provides evidence that these are low productivity species.

Population trend information

All *Glaucostegus* species were globally assessed as Critically Endangered on the IUCN Red List between 2018 and 2022 (Table 2). *Halavi Guitarfish* (*Glaucostegus halavi*) and *Sharpnose Guitarfish* (*Glaucostegus granulatus*) were assessed in 2019 as vulnerable and endangered, respectively, for the United Arab Emirates (UAE) National Red List of Marine Species (Ralph et al., 2021).

In the absence of species-specific time-series data to assess population reductions, the IUCN Red List assessments have relied on relevant historical accounts and contemporary datasets for landings and catch rates to infer population reduction where landings have decreased while fishing efforts remained stable or increased (Dulvy in litt., 2025).

Table 2. Family Glaucostegidae, IUCN Red List global category (all using IUCN Red List version 3.1), and key population trend information used in this analysis. Population trend information taken from the IUCN Red List assessment for each species and Supplemental Information, unless otherwise noted (in alphabetical order by scientific name).

Species, IUCN Red List Category and generation length	Population Trend Information
Blackchin Guitarfish <i>(Glaucostegus cemiculus)</i> <i>Critically Endangered A2d</i> <i>(Kyne and Jabado, 2019)</i> Generation length = 15 years	• <i>Inferred population reduction over last three generations (global): >80%.</i> • <i>Catch estimates (Ghana): 71% of fishers estimate that catches decreased by 80-90% (Seidu et al., 2022).</i> • <i>Local Ecological Knowledge (Ghana): Fishers noted severe depletion in the early 2000s with declines continuing to 2020 (based on 33 interviews with fishers in five coastal communities considered hotspots of elasmobranch fisheries in western Ghana between June 2019 and July 2020) (Seidu et al., 2021).</i> • <i>Reduction in catch size (Banc D'Arguin National Park, Mauritania): 95% of Blackchin Guitarfish caught are smaller than their size-at-maturity; analysis of the change in the average size of landed individuals from 1998 to 2007 highlights a significant size decrease (Diop and Dossa, 2011).</i> • <i>Landings (Senegal): Decrease by 80% in 7 years from 4,050 t in 1998 to 821 t in 2005 (unpublished data).</i> • <i>Trawl surveys (Mediterranean): Blackchin Guitarfish no longer recorded in surveys conducted between 1948 and 2005 or landings in several places across the Mediterranean.</i> • <i>Trawl surveys (Mediterranean): Blackchin Guitarfish no longer recorded in surveys conducted in the Mediterranean (MEDITS) from 1994 to 2015</i>
Sharpnose Guitarfish <i>(Glaucostegus granulatus)</i> <i>Critically Endangered A2bd</i> <i>(Kyne et al., 2022)</i> Generation length = 15 years	• <i>Inferred population reduction over last three generations (global): >80%.</i> • <i>Catch rate (Gulf of Thailand): 93% decrease in 'Rhinobathidae' (likely to include Guitarfishes and Wedgefishes) from peak catches in 1968 to 1972.</i> • <i>Landings (Indonesia): Decreases in 'whitespotted wedgefishes' (group may include Giant Guitarfishes) of 88% from 28,492 t/year to 3,540 t/year in 2015 (est. >99% reduction over last three generations).</i> • <i>Research trawl survey (Java Sea, Indonesia): Decrease in 'rays' from 1976 to 1997 estimated to be at least 90%.</i> • <i>Landings (Islamic Republic of Iran): Decrease of 'giant guitarfish' (likely to include all locally occurring Giant Guitarfishes and Wedgefishes) from 1997 to 2016 by 66% (est. 91% decrease over last three generations).</i> • <i>Catch data for various demersal rays (Maharashtra, India): Decrease in catch rate from 0.65 kg/hr in 1990 to 0.24 kg/hr in 2004 (63% over 15 years; est. 95% reduction over last three generations).</i> • <i>Landings (Balochistan, Pakistan): Decrease of 81% based on 'rhinobatids' (likely to include all locally occurring Guitarfishes and Wedgefishes) from 921 t in 1994 to 178 t in 2011 (est. >98% reduction over last three generations).</i> • <i>Landings (Sindh, Pakistan): Decrease of 72% based on reductions in 'rhinobatid' (likely to include all locally occurring Guitarfishes and Wedgefishes) from 902 in 1999 to 252 in 2011 (est. >98% reduction over last three generations).</i> • <i>Research trawl surveys (United Arab Emirates): Sharpnose Guitarfish was the most common guitarfish in 2002, but was not captured during similar surveys in 2016</i>
Halavi Guitarfish <i>(Glaucostegus halavi)</i> <i>Critically Endangered A2bd</i> <i>(Kyne and Jabado, 2019)</i> Generation length = 10 years	• <i>Inferred population reduction over last three generations (global): >80%.</i> • <i>Catch rate (Gulf of Thailand): 93% decrease in 'Rhinobathidae' (likely to include Guitarfishes and Wedgefishes) from peak catches in 1968 to 1972.</i> • <i>Landings data (Indonesia): Decreases in 'whitespotted wedgefishes' (group may include Giant Guitarfishes) of 88% from 28,492 t/year to 3,540 t/year in 2015 (est. >99% reduction over last three generations).</i> • <i>Research trawl survey (Java Sea, Indonesia): Decrease in 'rays' from 1976 to 1997 estimated to be at least 90%.</i> • <i>Landings (Islamic Republic of Iran): Decrease of 'giant guitarfish' (likely to include all locally occurring Giant Guitarfishes and Wedgefishes) from 1997 to 2016 by 66% (est. 91% decrease over last three generations).</i> • <i>Catch data for various demersal rays (Maharashtra, India): Decrease in catch rate from 0.65 kg/hr in 1990 to 0.24 kg/hr in 2004 (63% over 15 years; est. 95% reduction over last three generations).</i>

Species, IUCN Red List Category and generation length	Population Trend Information
	- Landings (Balochistan, Pakistan): Decrease of 81% based on 'rhinobatids' (likely to include all locally occurring Guitarfishes and Wedgefishes) from 921 t in 1994 to 178 t in 2011 (est. >98% reduction over last three generations). - Landings (Sindh, Pakistan): Decrease of 72% based on reductions in 'rhinobatid' (likely to include all locally occurring Guitarfishes and Wedgefishes) from 902 in 1999 to 252 in 2011 (est. >98% reduction over last three last generations).
Widenose Guitarfish (<i>Glaucostegus obtusus</i>) Critically Endangered A2bd (Kyne and Jabado, 2021) Generation length = 10 years	- Inferred population reduction over last three generations (global): >80%. - Catch rate (Gulf of Thailand): 93% decrease in 'Rhinobathidae' (likely to include Guitarfishes and Wedgefishes) from peak catches in 1968 to 1972. - Landings data (Indonesia): Decreases in 'whitespotted wedgefishes' (group may include Giant Guitarfishes) of 88% from 28,492 t/year to 3,540 t/year in 2015 (est. >99% reduction over last three generations). - Research trawl survey (Java Sea, Indonesia): Decrease in 'rays' from 1976 to 1997 estimated to be at least 90%. - Landings (Islamic Republic of Iran): Decrease of 'giant guitarfish' (likely to include all locally occurring Giant Guitarfishes and Wedgefishes) from 1997 to 2016 by 66% (est. 91% decrease over last three generations). - Catch data for various demersal rays (Maharashtra, India): Decrease in catch rate from 0.65 kg/hr in 1990 to 0.24 kg/hr in 2004 (63% over 15 years; est. 95% reduction over last three generations). - Landings (Balochistan, Pakistan): Decrease of 81% based on 'rhinobatids' (likely to include all locally occurring Guitarfishes and Wedgefishes) from 921 t in 1994 to 178 t in 2011 (est. >98% reduction over last three generations). - Landings (Sindh, Pakistan): Decrease of 72% based on reductions in 'rhinobatid' (likely to include all locally occurring Guitarfishes and Wedgefishes) from 902 in 1999 to 252 in 2011 (est. >98% reduction over last three generations).
Clubnose Guitarfish (<i>Glaucostegus thouin</i>) Critically Endangered A2bd (Kyne and Jabado, 2021) Generation length = 15 years	- Inferred population reduction over last three generations (global): >80%. - Catch rate (Gulf of Thailand): 93% decrease in 'Rhinobathidae' (likely to include Guitarfishes and Wedgefishes) from peak catches in 1968 to 1972. - Catch data for various demersal rays (Maharashtra, India): Decrease in catch rate from 0.65 kg/hr in 1990 to 0.24 kg/hr in 2004 (63% over 15 years; est. 95% reduction over last three generations). - Landings data (Indonesia): Decreases in 'whitespotted wedgefishes' (group may include Giant Guitarfishes) of 88% from 28,492 t/year to 3,540 t/year in 2015 (est. >99% reduction over last three generations). - Research trawl survey (Java Sea, Indonesia): Decrease in 'rays' from 1976 to 1997 estimated to be at least 90%.
Giant Guitarfish (<i>Glaucostegus typus</i>) Critically Endangered A2bd (Kyne et al., 2019) Generation length = 15 years	- Inferred population reduction over last three generations (global): >80%. - Catch rate (Gulf of Thailand): 93% decrease in 'Rhinobathidae' (likely to include Guitarfishes and Wedgefishes) from peak catches in 1968 to 1972. - Landings data (Indonesia): Decreases in 'whitespotted wedgefishes' (group may include Giant Guitarfishes) of 88% from 28,492 t/year to 3,540 t/year in 2015 (est. >99% reduction over last three generations). - Research trawl survey (Java Sea, Indonesia): Decrease in 'rays' from 1976 to 1997 estimated to be at least 90%. - Catch data for various demersal rays (Maharashtra, India): Decrease in catch rate from 0.65 kg/hr in 1990 to 0.24 kg/hr in 2004 (63% over 15 years; est. 95% reduction over last three generations).
Bangladeshi Guitarfish (<i>Glaucostegus younholeei</i>) Critically Endangered A2bd (Kyne et al., 2022) Generation length = 10 years	- Inferred population reduction over last three generations (global): >80%. - Catch data for various demersal rays (Maharashtra, India): Decrease in catch rate from 0.65 kg/hr in 1990 to 0.24 kg/hr in 2004 (63% over 15 years; est. 95% reduction over last three generations).

Species, IUCN Red List Category and generation length	Population Trend Information
<i>Glaucostegus</i> spp.	- Landings (Islamic Republic of Iran): Decrease of 'giant guitarfish' (likely to include all locally occurring Giant Guitarfishes and Wedgefishes) from 1997 to 2016 by 66% (est. 91% decrease over last three generations). - Catch rate (Gulf of Thailand): 93% decrease in 'Rhinobathidae' (likely to include Guitarfishes and Wedgefishes) from peak catches in 1968 to 1972. - Landings data (Indonesia): Decreases in 'whitespotted wedgefishes' (group may include Giant Guitarfishes) of 88% from 28,492 t/year to 3,540 t/year in 2015 (est. >99% reduction over last three generations). - Research trawl survey (Java Sea, Indonesia): Decrease in 'rays' from 1976 to 1997 estimated to be at least 90%. - Trawl surveys (Java Sea and North Natuna Sea, Indonesia): Only three <i>Rhynchobatus</i> recorded; inferred localized depletion. - Catch data for various demersal rays (Maharashtra, India): Decrease in catch rate from 0.65 kg/hr in 1990 to 0.24 kg/hr in 2004 (63% over 15 years; est. 95% reduction over last three generations). - Landings (Balochistan, Pakistan): Decrease of 81% based on 'rhinobatids' (likely to include all locally occurring Guitarfishes and Wedgefishes) from 921 t in 1994 to 178 t in 2011 (est. >90% reduction over last three generations). - Landings (Sindh, Pakistan): Decrease of 72% based on reductions in 'rhinobatid' (likely to include all locally occurring Guitarfishes and Wedgefishes) from 902 in 1999 to 252 in 2011 (est. >90% reduction over last three generations).

Trade assessment

The species is or may be affected by trade

Giant guitarfishes were listed in Appendix II due to concerns that the levels of international trade in the fins of these species may be driving their declines (see CoP18 Prop. 43). Analyses of the Hong Kong SAR of China fin market between 2014 and 2021 found that Giant Guitarfish fins comprised approximately 0.16% of the overall shark fin trade. Fins of these species were found both before and after their inclusion in Appendix II. Market studies conducted in 2018 in Guangzhou and Hong Kong SAR of China revealed that the fins from shark-like batoids – including giant guitarfishes and wedgefishes – are categorized in the highest valued categories of fins, "Qun chi" (recognized as the "King of shark fins"), according to Hau et al. (2018). The average price for these fins were USD276 per kg and USD185 per kg in Guangzhou and Hong Kong SAR of China, respectively.

Typically, the meat of *Glaucostegus* species is consumed domestically as either fresh or dried. In Bangladesh, for example, the meat of Sharpnose Guitarfish (*Glaucostegus granulatus*) and Widenose Guitarfish (*Glaucostegus obtusus*) is dried for consumption and the fins are exported to Asian and European markets. The skin may be dried and traded internationally as a luxury leather product.

Legal Trade

Between 2021 through 2023, the CITES Trade Database had records of around 150,000 kg of Giant Guitarfish fins and more than 60,000 kg of skins in international trade. Most were commercial exports from Indonesia to Hong Kong SAR of China. More than 3,000 kg of meat were commercially exported, mostly from Indonesia to Hong Kong SAR of China. There were also records of commercial exports of powder and bones.

From 2021 to 2023, approximately 50% of the total trade volume (in kg) reported by exporters (excluding re-exports) in the CITES Trade Database was for *Glaucostegus typus*, the most frequently reported species in trade. This was followed by *G. cemiculus*, which accounted for about 20% of the volume reported by exporters, while the remaining 30% was recorded at the genus level. A small number of records were also reported for *Glaucostegus granulatus*, *G. halavi*, *G. obtusus*, and *G. thouin*, although these were few in comparison.

According to LEMIS, two specimens of Giant Guitarfish (*Glaucostegus typus*) were exported from the USA to Australia in 2016 for scientific purposes.

Illegal Trade

Due to the high prices and demand for giant guitarfish fins, recent seizures suggest that illegal trade is also adversely affecting these species. Since 2021, Hong Kong SAR of China has intercepted seven illegal shipments of *Glaucostegus* species transiting through its ports, which total more than 5,118 kg of fins. According to the provisional assessments by the Secretariat (No. 2025/102) there are two records of *Glaucostegus* spp. in the CITES Illegal Trade Database (accessed July 2025), which included one body of *G. cemiculus* and 0.45 kg of meat reported at the genus level.

WiTIS reports 11 incidents of guitarfish seizures between 2019 and 2024. Of these seizures, two involved illegal transport of fins from Giant Guitarfish (*Glaucostegus typus*) (1,200 kg) and Blackchin Guitarfish (*Glaucostegus cemiculus*) (0.29 kg). Eight were reports of illegal harvest and poaching of Blackchin Guitarfish (*Glaucostegus cemiculus*), totalling 129 individuals. One incident involved illegal bottom trawling, and one Halavi Guitarfish (*Glaucostegus halavi*) was found dead onboard.

Additional Information

Threats

The greatest threat to giant guitarfishes is unmanaged fisheries that either target or incidentally catch these species. On a global scale, *Glaucostegus* species are subject to high levels of fishing pressure as they are captured in industrial, artisanal, and subsistence fisheries. These species are generally retained for their fins and meat.

Conservation, management and legislation

Beyond their inclusion in Appendix II of CITES, there are few measures for giant guitarfishes at the regional or national level. *Blackchin Guitarfish* (*Glaucostegus cemiculus*) is listed on Annex II of the SPA/BD Protocol of the Barcelona Convention since 2012. The SPA/BD Protocol prohibits the landing of these species in the Mediterranean and requires that they be released unharmed and alive to the highest extent possible (UNEP/MAP-SPA/RAC 2018). None of the RFMOs have adopted conservation and management measures for these species.

Nationally, all giant guitarfish are protected in Bangladesh under Schedule I of the Bangladesh Wildlife (Conservation and Security) Act of 2012, but the legislation has yet to be implemented. Israel has protections for all its sharks and rays. Two provinces in Pakistan have protections for Guitarfish species. Both Saudi Arabia and Sudan ban shark fishing, and the United Arab Emirates bans the export of shark fins.

Artificial Propagation/captive breeding

There is no literature or information available regarding captive breeding of *Glaucostegus* species.

Implementation challenges (including similar species)

All *Glaucostegus* species resemble one another, but tools are available to assist CITES Parties with differentiation between the family Glaucostegidae and other shark and ray species. *Identification guides are available to assist CITES Parties with visual identification of Giant Guitarfish carcasses and distinguish these species from other shark and ray species included in the CITES Appendices* (Jabado, 2019; 2024).

Visual identification and differentiation of giant guitarfish fins and meat may be a challenge. Giant guitarfish fins are morphologically similar to the fins of other CITES-listed species, such as the large hammerhead shark species listed in Appendix II and Sawfishes that are listed in Appendix I.

References

- AC Nomenclature Specialist (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Diop, M. and Dossa, J. (2011) *30 Years of Shark Fishing in West Africa: development of fisheries, catch trends, and their conservation status in Sub-Regional Fishing Commission member countries.* www.iucnssg.org/uploads/5/4/1/2/54120303/30years_eng.pdf
- Dulvy, N.K. (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Dulvy, N.K., Fowler, S.L., Musick, J.A., Cavanagh, R.D., Kyne, P.M., Carlson, J.K., Davidson, L.N.K., Fordham, S.V., Francis, M.P., Pollock, C.M., Simpfendorfer, C.A., Burgess, G.H., Carpenter, K.E., Compagno, J.V., Ebert, D.A., Gibson, C., Heupel, M.R., Linvingson, S.R., Sanciangoco, J.C., Stevens, J.D., Valenti, S. and White, W.T. (2014) Extinction risk and conservation of the world's sharks and rays. *eLife* 3:3:e00590. DOI: 10.7554/eLife.00590
- Hau, C.Y., Abercrombie, D.L., Ho, K.Y.K. and Stan, S.K.H. (2018) *"King of Shark Fins" not quite sharks...so what is in my shark fin soup? A rapid survey on the availability of shark-like batoid fins in Hong Kong SAR and Guangzhou, China retail markets.* BLOOM Hong Kong Shark Foundation. 30 pp.
- Horswill, C., Kindsvater, H. K., Dulvy, N. K., Mull, C. G., Judah, A., D'Aberto, B. M., Matthiopoulos, J. and Mangel, M. (2025) Imputation of fisheries reference points for endangered data-poor fishes, with application to rhino rays. *Fish and Fisheries* 26(5):848–865. doi:<https://doi.org/10.1111/faf.70003>
- Jabado, R.W. (2019) *Wedgefishes and Giant Guitarfishes: A Guide to Species Identification.* Wildlife Conservation Society, New York, United States. 30 pp.
- Jabado, R.W. (2024) *Sharks and Rays Listed on the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES): Whole Carcass.* Wildlife Conservation Society, New York. 194 pp.
- Kyne, P.M., Haque, A.B., Charles, R. and Jabado, R.W. (2022a) *Glaucostegus younholeei.* The IUCN Red List of Threatened Species 2022: e.T214418407A214418413. <https://dx.doi.org/10.2305/IUCN.UK.2022-2.RLTS.T214418407A214418413.en>

- Kyne, P.M., Haque, A.B., Charles, R. and Jabado, R.W. (2022b) *Glaucostegus granulatus*. *The IUCN Red List of Threatened Species* 2022: e.T60166A215829219. <https://dx.doi.org/10.2305/IUCN.UK.2022-2.RLTS.T60166A215829219.en>
- Kyne, P.M. and Jabado, R.W. (2019a) *Glaucostegus cemiculus*. *The IUCN Red List of Threatened Species* 2019: e.T104050689A104057239. <http://dx.doi.org/10.2305/IUCN.UK.2019-2.RLTS.T104050689A104057239.en>
- Kyne, P.M. and Jabado, R.W. (2019) *Glaucostegus halavi*. *The IUCN Red List of Threatened Species* 2019: e.T161408A124479984. <http://dx.doi.org/10.2305/IUCN.UK.2019-2.RLTS.T161408A124479984.en>
- Kyne, P.M. and Jabado, R.W. (2021a) *Glaucostegus thouin* (amended version of 2019 assessment). *The IUCN Red List of Threatened Species* 2021: e.T60175A207731709. <https://dx.doi.org/10.2305/IUCN.UK.2021-3.RLTS.T60175A207731709.en>
- Kyne, P.M. & Jabado, R.W. (2021b) *Glaucostegus obtusus* (amended version of 2019 assessment). *The IUCN Red List of Threatened Species* 2021: e.T60170A207283191. <https://dx.doi.org/10.2305/IUCN.UK.2021-3.RLTS.T60170A207283191.en>
- Kyne, P.M., Rigby, C.L., Dharmadi, Gutteridge, A.N. and Jabado, R.W. (2019) *Glaucostegus typus*. *The IUCN Red List of Threatened Species* 2019: e.T104061138A68623995. <http://dx.doi.org/10.2305/IUCN.UK.2019-2.RLTS.T104061138A68623995.en>
- Last, P. R., White, W.T., de Carvalho, M. R., Séret, B., Stehmann, M.F. and Naylor, G.J.P. (2016) *Rays of the world*. CSIRO Publishing, Clayton South.
- Moore, A.B.M. (2017) Are guitarfishes the next sawfishes? Extinction risk and an urgent call for conservation action. *Endangered Species Research* 34:75-88. <https://doi.org/10.3354/esr00830>
- Seidu, I., Cabada-Blanco, F., Brobbey, L.K., Asiedu, B., Barnes, P., Seidu, M. and Dulvy, N.K. (2022) "Every fish in the sea is meat and so are guitarfishes." Socio-economic drivers of a guitarfish fishery in Ghana. *Marine Policy* 143. <https://doi.org/10.1016/j.marpol.2022.105159>
- Pardo, S., Kindsvater, H., Cuevas-Zimbrón, E., Sosa-Nishizaki, O., Pérez-Jiménez, J.C. and Dulvy, N.K. (2016) Growth, productivity and relative extinction risk of a data-sparse devil ray. *Scientific Reports* 6:33745. <https://doi.org/10.1038/srep33745>
- Ralph, G.M., Stump, E., Linardich, C., Bullock, R.W., Carpenter, K.E., Allen D.J., Hilton-Taylor, C., Al Mheiri, R. and Alshamsi, O. (2021) *UAE National Red List of Marine Species: Reef-building corals, cartilaginous fishes and select bony fishes*. 2021. Ministry of Climate Change and Environment, Dubai, United Arab Emirates.
- Seidu, I., Brobbey, L.K., Danquah, E., Oppong, S.K., van Beuningen, D. and Dulvy, N.K. (2021) Local ecological knowledge, catch characteristics and evidence of elasmobranch depletions in Western Ghana. *bioRxiv*. doi: <https://doi.org/10.1101/2021.01.14.426682>
- UNEP/MAP-SPA/RAC (2018) SAP/RAC: SPA-BD Protocol - Annex II: List of endangered or threatened species. https://www.rac-spa.org/sites/default/files/annex/annex_2_en_20182.pdf

Retain wedgefishes Family Rhinidae in Appendix II with a zero-export quota for wild specimens traded for commercial purposes

Proponents: Bangladesh, Benin, Brazil, Burkina Faso, Burundi, Central African Republic, Comoros, Congo, Gabon, Gambia, Guinea, Guinea-Bissau, Maldives, Mali, Niger, Nigeria, Senegal, Sierra Leone, Sudan, Togo

Summary: Wedgefishes in the family Rhinidae comprise 10 or 11 recognised species of medium to large rays that generally inhabit warm temperate to tropical inshore waters on the continental shelf ranging from the Eastern Atlantic, across the Western and Eastern Indian Ocean and extending far into the Pacific. The family Rhinidae was included in Appendix II at CoP18 in 2019 due to concerns that the high value of their fins in international markets was driving declines. The proposal is to retain the family in Appendix II with the following annotation: “a zero annual export quota for wild-taken specimens traded for commercial purposes.” According to the supporting statement, this quota is proposed to provide Wedgefishes an opportunity to recover to a point where trade will not threaten them with extinction.

Although there is limited information on their life history characteristics and general biology, wedgefishes appear to be characterized by low productivity due to their relatively low fecundity (2-19 pups), slow growth rates, and estimated late age at maturity (10-15 years).

Like giant guitarfishes *Glaucostegus* spp. (the subject of proposal CoP20 Prop. 32) and sawfishes (Family Pristidae, also listed in Appendix I), wedgefishes are bottom dwelling fish that are often found in shallow coastal waters and are rarely found at depths below 400 m. They inhabit enclosed bays and estuaries and are found on coral reefs. Based on their size and use of coastal habitat, wedgefishes are susceptible to exploitation by demersal trawl, net, and line fisheries as either target or incidental catch. Meat of these species is usually consumed locally, but wedgefish fins are known as being some of the most valuable fins traded internationally. Snouts, skin, and other products are also traded.

As some species in this family are morphologically similar, share common names, and have overlapping ranges, there has been confusion and possible misidentification in reporting and catch data. Information is not available to determine the precise population size of any species in the family and there is a lack of species-specific data to calculate population trends.

All members of the Rhinidae were globally assessed as Critically Endangered on the IUCN Red List between 2018 and 2023 except for the Eyebrow Wedgefish (*Rhynchobatus palpebratus*), which was assessed as Near Threatened in 2018. Some species appear to have restricted areas of distribution and are still being heavily exploited in coastal fisheries. Because of the paucity of species-specific status information, population decreases have been inferred from fisheries data. Declines over three generations have been estimated at greater than or equal to 80% for the Critically Endangered species and 20-30% for *Rhynchobatus palpebratus*. Observations of reductions of over 90% have been observed in some fishing areas, and one species, *R. cooki*, has only had one confirmed record in the last 23 years. The distributions of species such as the False Shark Ray (*Rhynchorhina mauritaniensis*), Japanese Wedgefish (*Rhynchobatus mononoke*), and Taiwanese Wedgefish (*Rhynchobatus immaculatus*) are notably limited.

Analysis of the fin market in Hong Kong SAR of China between 2015-2017 and 2020-2021 revealed that wedgefish fins continued to be traded internationally. According to the CITES Trade Database, approximately 127,000 kg of dried fins, 136,000 kg of fins, 35,000 kg of skins, and 4,600 kg of meat from wild-sourced Rhinidae species were commercially exported between 2021 and 2023. Across different products exported for commercial purposes, Bottlenose Wedgefish (*Rhynchobatus australiae*) comprised about 40% of the reported trade volume during this period.

There is little management of wedgefishes across their range, but some countries have enacted legislation or regulations for these species. Bottlenose Wedgefish (*Rhynchobatus australiae*) is listed

in Appendix II of the Convention on the Conservation of Migratory Species of Wild Animals (CMS), and three species are listed in Annex 1 the CMS Memorandum of Understanding on the Conservation of Migratory Sharks (Shark MOU).

Analysis: There is currently no guidance on how to assess a proposal to add an annotation to the listing of a species already in Appendix II. This is equivalent to the inclusion of an export quota or other special measure for a species being transferred from Appendix I to II, as provided for as precautionary measures under Annex 4, paragraph A.2.a) iii).

Rhinidae species are mostly found in inshore continental areas, and their areas of habitat overlap with areas with high levels of fishing pressure. There is international demand for their fins, which are extremely valuable, as well as local use for meat, both of which provide an incentive to retain catch even when incidental.

There are no estimates of the global populations of wedgefishes, but data that are available on catch and landings suggest that the populations of most species are not small, possibly with the exception of *R. cooki* which has only been recorded once in the past two decades. In the absence of species-specific data for Rhinidae species, fisheries information for related species (Rhinobatidae (guitarfishes), Glaucostegidae (Giant Guitarfishes), or unspecified Rhinidae (Wedgefishes)) have been used to help infer population trends of over 80% declines for all species except *Rhynchobatus palpebratus*. Given the continued fishing pressure on these low productivity species, it is likely that the population declines identified in the respective IUCN Red List assessments have continued, meaning that all species except the Eyebrow Wedgefish (*Rhynchobatus palpebratus*), may already meet the criteria in Annex 1 of Res. Conf. 9.24 (Rev. CoP17).

In this light, the proposed **annotation of a zero-quota for wild specimens traded for commercial purposes, does appear to be an appropriate precautionary measure for all species of the family** pending improved circumstances.

Summary of Available Information

Text in non-italics is based on information in the Supporting Statement (SS); text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

All Wedgefishes in the family Rhinidae were included in Appendix II in 2019.

Taxonomy

The SS names 11 species as well as any other putative species of Family Rhinidae; however, the currently accepted Standard Reference (Eschmeyer *et al.*, 2019) recognises the following 10 species.

Rhina ancylostomus (Bloch and Schneider, 1801)
Rhynchobatus australiae (Whitley, 1939)
Rhynchobatus cooki (Last, Kyne and Compagno, 2016)
Rhynchobatus djiddensis (Forsskal, 1775)
Rhynchobatus immaculatus (Last, Ho and Chen, 2013)
Rhynchobatus laevis (Bloch and Schneider, 1801)
Rhynchobatus luebberti (Ehrenbaum, 1915)
Rhynchobatus palpebratus (Compagno and Last, 2008)
Rhynchobatus springeri (Compagno and Last, 2010)
Rhynchobatus mauritaniensis (Séret and Naylor, 2016)

Nomenclature changes proposed at CoP20 will include a new species *Rhynchobatus mononoke*. CITES Standard Reference: ESCHMEYER, W.N., R. FRICKE, & R. VAN DER LAAN (eds.) (2019) Taxonomic Checklist of Fish taxa included in the Appendices at the 18th Meeting of the Conference of the Parties. Species information extracted from Catalog of Fishes: Genera, Species, References, an online reference, version of 4 May 2020, accessed 5 May 2020. See Annex 3 of AC31 Doc. 37 <https://cites.org/sites/default/files/eng/com/ac/31/Docs/E-AC31-37-A3.pdf>

Range and IUCN Global Category (in alphabetical order by scientific name)**Bowmouth Guitarfish *Rhina ancylostomus***

Critically Endangered A2bd (assessed 2018, ver. 3.1)

Occurs in Indo-West Pacific. Bowmouth Guitarfish is found in the following FAO Fishing Areas: 51, 57, 61, 71, 81

Range: Extant (resident)

Australia (New South Wales, Northern Territory, Queensland, Western Australia); Bahrain; Bangladesh; Brunei Darussalam; Cambodia; China; Djibouti; Egypt; Eritrea; India; Indonesia; Islamic Republic of Iran; Iraq; Israel; Japan; Kenya; Democratic People's Republic of Korea; Republic of Korea; Kuwait; Madagascar; Malaysia; Maldives; Mauritius; Mozambique; Myanmar; New Caledonia; Oman; Pakistan; Papua New Guinea; Philippines; Qatar; Réunion; Saudi Arabia; Seychelles; Singapore; Somalia; South Africa (KwaZulu-Natal); Sri Lanka; Sudan; Taiwan Province of China; United Republic of Tanzania; Thailand; Timor-Leste; United Arab Emirates; Viet Nam; Yemen



Figure 1. Bowmouth Guitarfish (*Rhina ancylostomus*) Distribution Map. (IUCN SSC Shark Specialist Group 2018. *Rhina ancylostomus*. The IUCN Red List of Threatened Species. Version 2025-1).

Bottlenose Wedgefish *Rhynchobatus australiae*

Critically Endangered A2bd (assessed 2018, ver. 3.1)

Occurs in the Indo-West Pacific. Bottlenose Wedgefish is found in the following FAO Fishing Areas: 51, 57, 61, 71, 81

Range: Extant (resident)

Australia (New South Wales, Northern Territory, Queensland, Western Australia); Bahrain; Bangladesh; Brunei Darussalam; Cambodia; China; Djibouti; Egypt; Eritrea; India; Indonesia; Islamic Republic of Iran; Iraq; Israel; Kenya; Kuwait; Malaysia; Maldives; Mozambique; Myanmar; Oman; Pakistan; Papua New Guinea; Philippines; Qatar; Saudi Arabia; Seychelles; Singapore; Solomon Islands; Somalia; Sri Lanka; Sudan; Taiwan Province of China; United Republic of Tanzania; Thailand; Timor-Leste; Viet Nam; Yemen

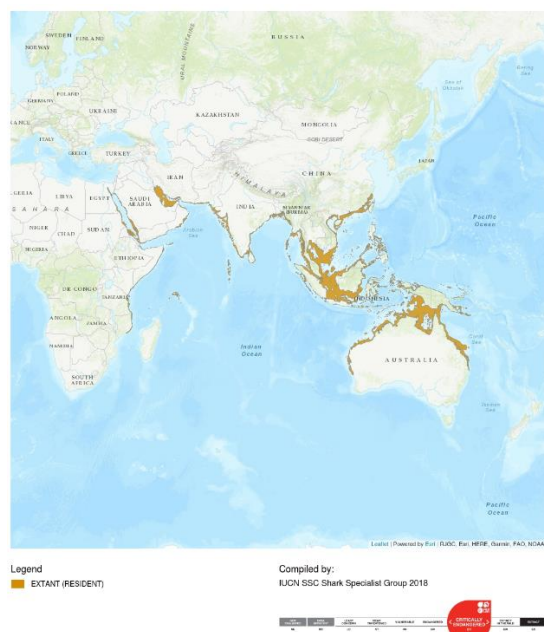


Figure 2. Bottlenose Wedgefish (*Rhynchobatus australiae*) Distribution Map. (IUCN SSC Shark Specialist Group 2018. *Rhynchobatus australiae*. The IUCN Red List of Threatened Species. Version 2025-1).

Clown Wedgefish *Rhynchobatus cooki*

Critically Endangered A2bd (assessed 2019, ver. 3.1)

Known from specimens collected from fish markets in Singapore and Jakarta (Indonesia) in the Western Central Pacific; origin of these landings is unknown, but several areas are suggested within the Indo-Malay Archipelago. Clown Wedgefish found in the following FAO Fishing Area: 71

Range: Extant (resident)

Indonesia (Jawa); Singapore



Figure 3. Clown Wedgefish (*Rhynchobatus cooki*) Distribution Map. (IUCN SSC Shark Specialist Group 2019. *Rhynchobatus cooki*. The IUCN Red List of Threatened Species. Version 2025-1).

Whitespotted Wedgefish *Rhynchobatus djiddensis*

Critically Endangered A2bd (assessed 2018, ver. 3.1)

Found in Western Indian Ocean from South Africa to Oman and the Arabian/Persian Gulf, but it may not be present further east (Last et al., 2016). The species' distribution may not be fully defined due to confusion with other members of the *Rhynchobatus djiddensis* species complex. Whitespotted Wedgefish is found in the following FAO Fishing Areas: 47, 51

Range: Extant (resident)

Bahrain; Djibouti; Egypt; Eritrea; Islamic Republic of Iran; Iraq; Israel; Kenya; Kuwait; Mozambique; Oman; Qatar; Saudi Arabia; Somalia; South Africa; Sudan; United Republic of Tanzania; United Arab Emirates; Yemen

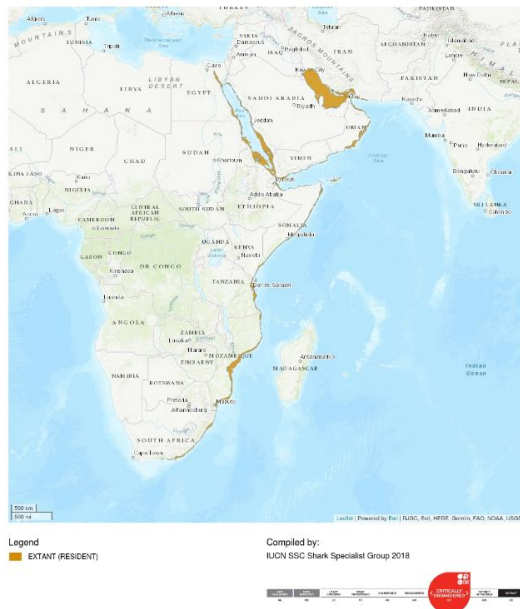


Figure 4. Whitespotted Wedgefish (*Rhynchobatus djiddensis*) Distribution Map. (IUCN SSC Shark Specialist Group 2018. *Rhynchobatus djiddensis*. The IUCN Red List of Threatened Species. Version 2025-1).

Taiwanese Wedgefish *Rhynchobatus immaculatus*

Critically Endangered A2bd (assessed 2018, ver. 3.1)

Known only from Taiwan Province of China in the Northwest Pacific. Taiwanese Wedgefish is found in the following FAO Fishing Area: 61

Range: Extant (resident)

Taiwan Province of China

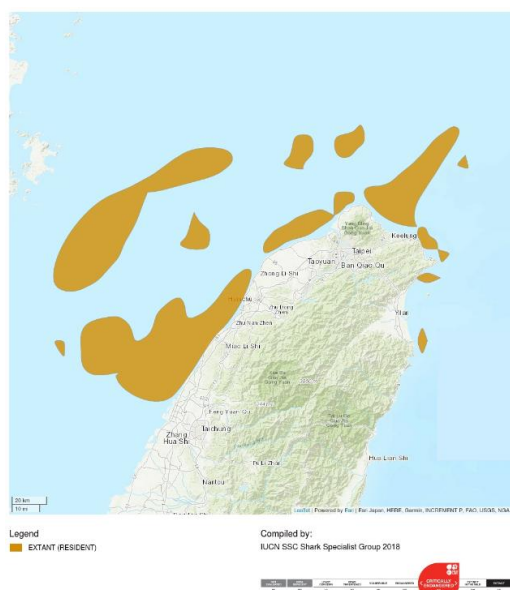


Figure 5. Taiwanese Wedgefish (*Rhynchobatus immaculatus*) Distribution Map. IUCN SSC Shark Specialist Group 2018. *Rhynchobatus immaculatus*. The IUCN Red List of Threatened Species. Version 2025-1).

Smoothnose Wedgefish *Rhynchobatus laevis*

Critically Endangered A2bd (assessed 2024, ver. 3.1)

Occurs from close inshore to depths of at least 60 m on the continental shelf and prefers shallow bays and river mouths. Widespread but patchy distribution in Indo-West Pacific. Smoothnose Wedgefish is found in the following FAO Fishing Areas: 51, 57, 61, 71

Range: Extant (resident)

Bahrain; Bangladesh; China; India; Indonesia; Islamic Republic of Iran; Iraq; Kuwait; Oman; Pakistan; Qatar; Saudi Arabia; Sri Lanka; Taiwan Province of China; United Arab Emirates

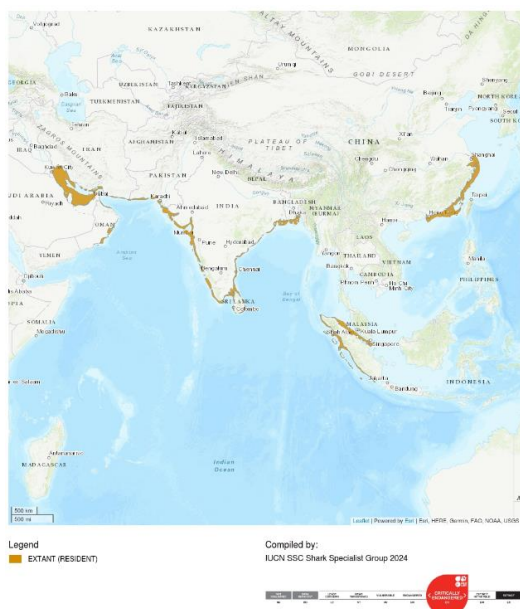


Figure 6. Smoothnose Wedgefish (*Rhynchobatus laevis*) Distribution Map. (IUCN SSC Shark Specialist Group 2024. *Rhynchobatus laevis*. The IUCN Red List of Threatened Species. Version 2025-1).

African Wedgefish *Rhynchobatus luebberti*

Critically Endangered A2d (assessed 2018, ver. 3.1)

Occurs in the eastern Atlantic from Mauritania to the Democratic Republic of the Congo and Angola. African Wedgefish is found in the following FAO Fishing Areas: 34, 47

Range: Extant (resident)

Angola; Benin; Cameroon; Congo; The Democratic Republic of the Congo; Côte d'Ivoire; Equatorial Guinea; Gabon; Gambia; Ghana; Guinea; Guinea-Bissau; Liberia; Mauritania; Nigeria; Sao Tome and Principe; Senegal; Sierra Leone; Togo

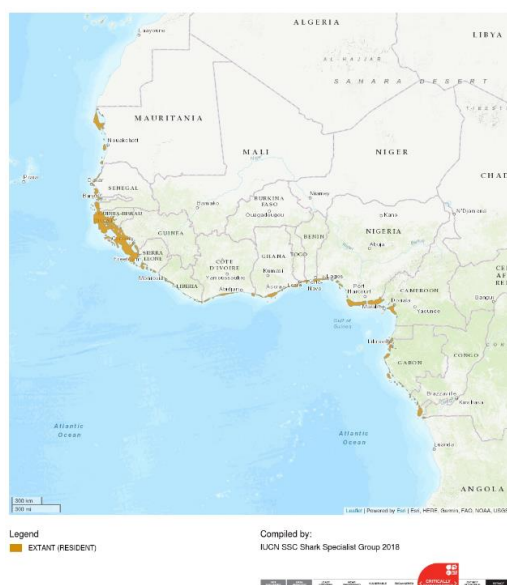


Figure 7. African Wedgefish (*Rhynchobatus luebberti*) Distribution Map. (IUCN SSC Shark Specialist Group 2018. *Rhynchobatus luebberti*. The IUCN Red List of Threatened Species. Version 2025-1).

False Shark Ray (*Rhynchorhina mauritaniensis*)

Critically Endangered A2d (assessed 2018, ver. 3.1)

Currently only known from a single location, the Banc d'Arguin of Mauritania in the Eastern Central Atlantic. False Shark Ray is found in the following FAO Fishing Area: 34

Range: Extant (resident)

Mauritania

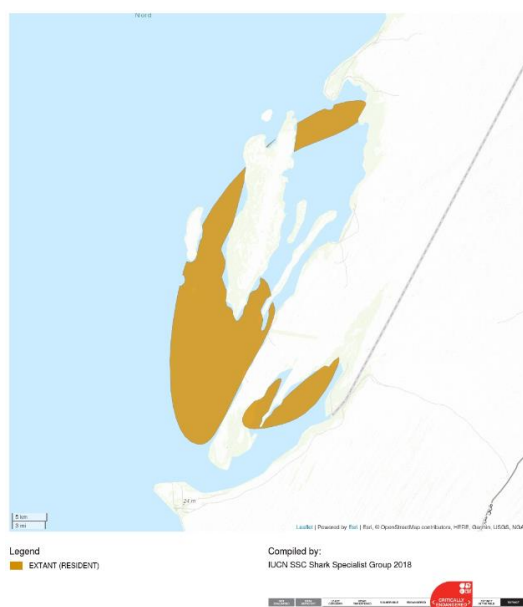


Figure 8. False Shark Ray (*Rhynchorhina mauritaniensis*) Distribution Map. (IUCN SSC Shark Specialist Group 2018. *Rhynchorhina mauritaniensis*. The IUCN Red List of Threatened Species. Version 2025-1).

Japanese Wedgefish (*Rhynchobatus mononoke*)

Critically Endangered A2bd (assessed 2023, ver. 3.1)

Endemic to southern Japan. Japanese Wedgefish is found in the following FAO Fishing Area: 61

Range: Extant (resident)

Japan

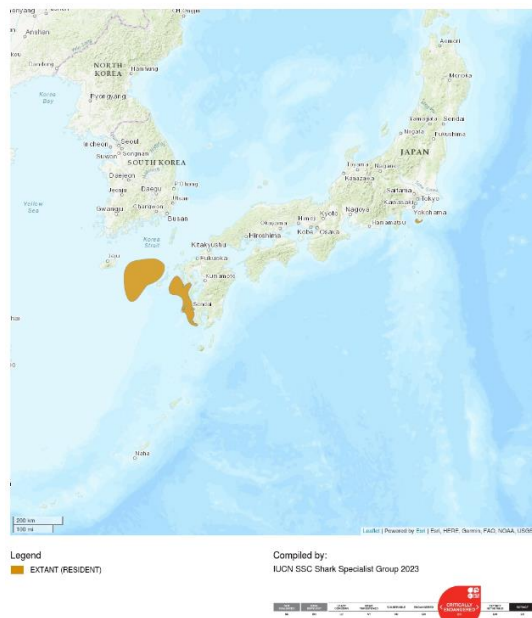


Figure 9. Japanese Wedgefish (*Rhynchobatus mononoke*) Distribution Map. (IUCN SSC Shark Specialist Group 2023. *Rhynchobatus mononoke*. The IUCN Red List of Threatened Species. Version 2025-1).

Eyebrow Wedgefish *Rhynchobatus palpebratus*

Near Threatened A2bd (assessed 2018, ver. 3.1)

Distribution is poorly defined due to confusion with *Rhynchobatus djiddensis*. According to Last et al. (2016) the species only occurs across northern Australia and southern Papua New Guinea, but a single record from the Andaman Sea off Thailand and two specimens from Taiwan Province of China suggest its occurrence in the Eastern Indian-Western Pacific region. Eyebrow Wedgefish is found in the following FAO Fishing Areas: 57, 61, 71

Range: Extant (resident)

Australia (Northern Territory, Queensland, Western Australia); Indonesia; Papua New Guinea; Taiwan Province of China; Thailand

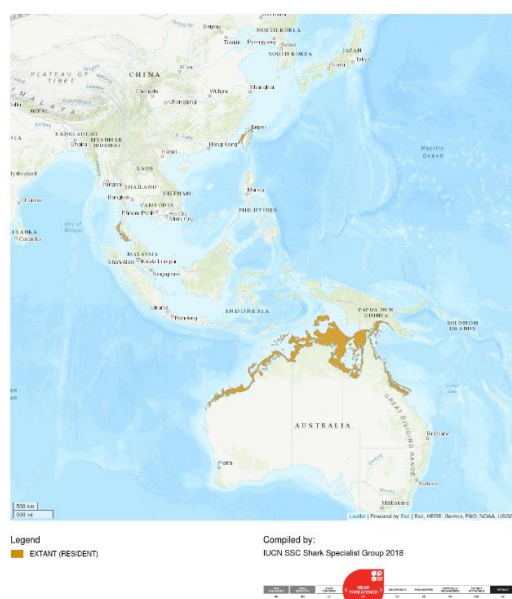


Figure 10. Eyebrow Wedgefish (*Rhynchobatus palpebratus*) Distribution Map. (IUCN SSC Shark Specialist Group 2018. *Rhynchobatus palpebratus*. The IUCN Red List of Threatened Species. Version 2025-1).

Broadnose Wedgefish *Rhynchobatus springeri*

Critically Endangered A2bd (assessed 2018, ver. 3.1)

Found in the Indo-Malay Archipelago. Broadnose Wedgefish is found in the following FAO Fishing Areas: 57, 71

Range: Extant (resident)

Cambodia; Indonesia (Jawa, Kalimantan, Sumatera); Malaysia; Philippines; Singapore; Thailand; Viet Nam



Figure 11. Broadnose Wedgefish (*Rhynchobatus springeri*) Distribution Map. (IUCN SSC Shark Specialist Group 2018. *Rhynchobatus springeri*. The IUCN Red List of Threatened Species. Version 2025-1).

Biological information

A) Wild population

The global population size is unknown for any Wedgefish species. Based on the limited catch and landings data available, the wild population does not appear to be small as defined in Annex 5.

B) Area of distribution

Wedgefishes are coastal species that occur in shallow, inshore, continental areas stretching from the eastern Atlantic, across the Western and Eastern Indian Ocean and extending far into the Pacific. Several species have a

patchy or sparse distribution that appears to be restricted, such as the False Shark Ray (*Rhynchorhina mauritaniensis*; Figure 7), Japanese Wedgefish (*Rhynchobatus mononoke*; Figure 8), and Taiwanese Wedgefish (*Rhynchobatus immaculatus*; Figure 9). This suggests that some populations of Wedgefishes may have a restricted area of distribution.

C) Population trends

The family Rhinidae is comprised of bottom-dwelling rays that are vulnerable to overexploitation due to their habitat overlapping with coastal fisheries, susceptibility to various types of fishing gear, and a lack of management to ensure their harvest is sustainable. They also have low reproductive potential, and their fins have a high value in international trade, which encourages harvest and retention when they are caught as bycatch.

A simple but robust estimator of natural mortality (M) for sharks and rays is the reciprocal of average lifespan ($M=1/w$) where average lifespan 'w' is the generation length of the species (Pardo et al., 2016). Hence, these CITES mortality thresholds can be back converted to calculate the corresponding generation length productivity thresholds ($GL = M^{-1}$) and these are as follows: high productivity (natural mortality >0.5 or a generation length of <2 years), medium productivity (natural mortality $0.2-0.5$ or generation lengths $2-5$ years), and low productivity (natural mortality <0.2 or generation length >5 years).

Table 1. Life history characteristics of Wedgefishes (Family Rhinidae) (in alphabetical order by scientific name).

Species	Maximum Size (cm) total length (TL)	Average size at maturity (cm) TL	Average litter size (pups)	Frequency of reproduction/ gestation period	Estimated generation length (years)	Productivity
Bowmouth Guitarfish (<i>Rhina ancylostomus</i>)	270	150 to 175 (males); 180 (females)	2 to 11	N/A	15	Low
Bottlenose Wedgefish (<i>Rhynchobatus australiae</i>)	300	110 to 130 (males); 155 (females)	7 to 19	N/A	15	Low
Clown Wedgefish (<i>Rhynchobatus cooki</i>)	81	>70.5 (males)	Unknown	Unknown	10	Low
Whitespotted Wedgefish (<i>Rhynchobatus djiddensis</i>)	310	150 (males)	4	Unknown	15	Low
Taiwanese Wedgefish (<i>Rhynchobatus immaculatus</i>)	99 recorded (likely 150)	Unknown	small	Unknown	10	Low
Smoothnose Wedgefish (<i>Rhynchobatus laevis</i>)	>200	130 (males)	small	Unknown	15	Low
African Wedgefish (<i>Rhynchobatus luebberti</i>)	300	N/A	2 to 5	N/A	15	Low
False Shark Ray (<i>Rhynchorhina mauritaniensis</i>)	275	Unknown	Unknown	Unknown	15	Low
Japanese Wedgefish (<i>Rhynchobatus mononoke</i>)	300	116 to 129 (males)	Unknown	Unknown	15	Low
Eyebrow Wedgefish (<i>Rhynchobatus palpebratus</i>)	262	103 (males)	small	N/A	15	Low
Broadnose Wedgefish (<i>Rhynchobatus springeri</i>)	213	115 (males)	small	N/A	15	Low

There is limited information on life history characteristics of Wedgefishes, but they are considered to have low productivity based on estimates of natural mortality (less than 0.2) or generation length (greater than five years). Horswill et al. (2025) also provides evidence that these are low productivity species.

Population trend information

All wedgefishes in the Rhinidae family were globally assessed as Critically Endangered on the IUCN Red List between 2019 and 2024 except for Eyebrow Wedgefish (*Rhynchobatus palpebratus*) which was assessed as Near Threatened in 2018 (Table 2).

In the absence of species-specific time-series data to assess population reductions, the IUCN Red List Assessments have relied on relevant historical accounts and contemporary datasets for landings and catch rates to infer population reduction where landings have decreased while fishing efforts remained stable or increased (Dulvy in litt., 2025).

Table 2. Family Rhinidae, IUCN Red List global category (all using IUCN Red List version 3.1), and key population trend information used in this analysis (in alphabetical order by scientific name). Population trend information taken from the IUCN Red List Assessment and Supplemental Information for each species, unless otherwise noted.

Species, IUCN Red List Category and generation length	Population Trend Information
Bowmouth Guitarfish <i>(Rhina ancylostomus)</i> <i>Critically Endangered A2bd</i> (Kyne et al., 2019) Generation length = 15 years	• <i>Inferred population reduction over last three generations (global): >80%.</i> • <i>Catch rate (Gulf of Thailand): 93% decrease in 'Rhinobathidae' (likely to include Guitarfishes and Wedgefishes) from peak catches in 1968 to 1972.</i> • <i>Landings (Indonesia): Decrease of 88% for 'whitespotted wedgefishes' (likely includes all locally occurring Wedgefishes, including Bowmouth Guitarfish) from 28,492 t in 2005 to 3,540 in 2015 (est. >99% reduction over last three generations).</i> • <i>Landings (Islamic Republic of Iran): Decrease of 66% from 880 t in 1997 to 295 t in 2016 (est. 91% reduction over last three generations) (likely to include all locally occurring Giant Guitarfish and Wedgefish species, including Bowmouth Wedgefish).</i> • <i>Catch data for various demersal rays (Maharashtra, India): Decrease in catch rate from 0.65 kg/hr in 1990 to 0.24 kg/hr in 2004 (63% over 15 years; est. 95% reduction over last three generations).</i> • <i>Landings (Tamil Nadu, India): Decrease of 86% from 278.7 t in 2002 to 38.3 t in 2006 (includes Bowmouth Wedgefish).</i> • <i>Anecdotal reports (Mozambique): Artisanal longline fishing led to decreases in Bowmouth Guitarfish.</i> • <i>Landings (Balochistan, Pakistan): Decrease of 81% based on 'rhinobatids' (likely to include locally occurring Wedgefishes, including Bowmouth Wedgefish) from 921 t in 1994 to 178 t in 2011 (est. >98% reduction over last three generations).</i> • <i>Landings (Sindh, Pakistan): Decrease of 72% based on reductions in 'rhinobatid' (likely to include locally occurring Wedgefishes, including Bowmouth Wedgefish) from 902 in 1999 to 252 in 2011 (est. >98% reduction over three generations).</i> • <i>Research trawl survey (Java Sea, Indonesia): Decrease in 'rays' from 1976 to 1997 estimated to 90%.</i>
Bottlenose Wedgefish <i>(Rhynchobatus australiae)</i> <i>Critically Endangered A2bd</i> (Kyne et al., 2019) Generation length = 15 years	• <i>Inferred population reduction over the last three generations (global): >80%.</i> • <i>Catch rate (Gulf of Thailand): 93% decrease in 'Rhinobathidae' (likely to include Guitarfishes and Wedgefishes) from peak catches in 1968 to 1972.</i> • <i>Landings (Indonesia): Decrease of 88% for 'whitespotted wedgefishes' (likely includes all locally occurring Wedgefishes, including Bottlenose Wedgefish) from 28,492 t in 2005 to 3,540 in 2015 (est. >99% reduction over last three generations).</i> • <i>Research trawl survey (Java Sea, Indonesia): Decrease in 'rays' from 1976 to 1997 estimated to 90%.</i> • <i>Landings (Islamic Republic of Iran): Decrease of 66% from 880 t in 1997 to 295 t in 2016 (est. 91% reduction over last three generations) (likely to include locally occurring Giant Guitarfish and Wedgefish species, including Bottlenose Wedgefish).</i> • <i>Catch data for various demersal rays (Maharashtra, India): Decrease in catch rate from 0.65 kg/hr in 1990 to 0.24 kg/hr in 2004 (63% over 15 years; est. 95% reduction over last three generations).</i> • <i>Landings (Tamil Nadu, India): Decrease of 87% from 114.9 t in 2002 to 14.9 t in 2006 (includes Bottlenose Wedgefish).</i> • <i>Landings (Balochistan, Pakistan): Decrease of 81% based on 'rhinobatids' (likely to include locally occurring Wedgefishes, including Bottlenose Wedgefish) from 921 t in 1994 to 178 t in 2011 (est. >98% reduction over last three generations).</i> • <i>Landings (Sindh, Pakistan): Decrease of 72% based on reductions in 'rhinobatid' (likely to include locally occurring Wedgefishes, including Bottlenose Wedgefish) from 902 in 1999 to 252 in 2011 (est. >98% reduction over last three generations).</i>

Species, IUCN Red List Category and generation length	Population Trend Information
Clown Wedgefish (<i>Rhynchobatus cooki</i>) <i>Critically Endangered A2bd</i> (Kyne et al., 2019) Generation length = 10 years	• Inferred population reduction over last three generations (global): >80%. • There has only been one confirmed record of the species in the last 23 years with six records identified on social media posts. • Landings (Indonesia): Decrease of 88% for 'whitespotted wedgefishes' (likely includes all locally occurring Wedgefishes, including Clown Wedgefish) from 28,492 t in 2005 to 3,540 in 2015 (est. >99% reduction over last three generations). • Research trawl survey (Java Sea, Indonesia): Decrease in 'rays' from 1976 to 1997 estimated to 90%.
Whitespotted Wedgefish (<i>Rhynchobatus djiddensis</i>) <i>Critically Endangered A2bd</i> (Kyne et al., 2019) Generation length = 15 years	• Inferred population reductions over the last three generations (global): >80%. • Risk assessment (South Africa): Decrease in abundance of 65.1% for Whitespotted Wedgefish over a period of three generations, along with a significant decline in CPUE over a 37-year monitoring period (7,703 were caught between 1977 and 2017 and 2,856 were caught between 1981 and 2017) (Daly et al., 2020). • Anecdotal reports (Mozambique): Artisanal longline fishing led to decreases in Whitespotted Wedgefish; this was one of the main target species of the fishery due to its abundance on reef before longline fisheries began in the early 2000s and are only seen in low numbers now. • Landings (Islamic Republic of Iran): Decrease of 66% from 880 t in 1997 to 295 t in 2016 (est. 91% population reduction over last three generations) (likely to include all locally occurring Wedgefish species, including Whitespotted Wedgefish).
Taiwanese Wedgefish (<i>Rhynchobatus immaculatus</i>) <i>Critically Endangered A2bd</i> (Kyne and Ebert, 2019) Generation length = 10 years	• Inferred population reduction over last three generations (global): >80%. • Levels of exploitation are high in Taiwan Province of China; species vulnerable to trawl fisheries in the north.
Smoothnose Wedgefish (<i>Rhynchobatus laevis</i>) <i>Critically Endangered A2bd</i> (Kyne et al., 2024) Generation length = 15 years	• Inferred population reduction over last three generations (global): >80%. • Catch rate (Gulf of Thailand): 93% decrease in 'Rhinobathidae' (likely to include Giant Guitarfishes and Wedgefishes) from peak catches in 1968 to 1972. • Landings (Indonesia): Decrease of 88% for 'whitespotted wedgefishes' (likely includes all locally occurring Wedgefishes) from 28,492 t in 2005 to 3,540 in 2015 (est. >99% population reduction over last three generations). • Research trawl survey (Java Sea, Indonesia): Decrease in 'rays' from 1976 to 1997 estimated to 90%. • Landings (Islamic Republic of Iran): Decrease of 66% from 880 t in 1997 to 295 t in 2016 ((likely to include locally occurring Giant Guitarfish and Wedgefish species, including Smoothnose Wedgefish) (est. 91% population reduction over last three generations). • Catch data for various demersal rays (Maharashtra, India): Decrease in catch rate from 0.65 kg/hr in 1990 to 0.24 kg/hr in 2004 (63% over 15 years; est. 95% reduction over last three generations). • Landings (Tamil Nadu, India): Decrease of 87% in 'guitarfishes' (includes locally occurring Wedgefishes) from 114.9 t in 2002 to 14.9 t in 2006. • Landings (Balochistan, Pakistan): Decrease of 81% based on 'rhinobatids' (likely to include locally occurring Wedgefishes, including Smoothnose Wedgefish) from 921 t in 1994 to 178 t in 2011 (est. >98% reduction over last three generations). • Landings (Sindh, Pakistan): Decrease of 72% based on reductions in 'rhinobatid' (likely to include locally occurring Wedgefishes, including Smoothnose Wedgefish) from 902 in 1999 to 252 in 2011 (est. >98% reduction over last three generations).

Species, IUCN Red List Category and generation length	Population Trend Information
African Wedgefish (<i>Rhynchobatus luebberti</i>) <i>Critically Endangered A2d</i> (Kyne and Jabado, 2019) Generation length = 15 years	• Inferred population reduction over last three generations (global): >80%. • Annual surveys of fish landing and processing sites (The Gambia): No records of African Wedgefish between 2010 and 2016, suggesting a possible local extirpation. • Trawl surveys (Guinea): Catch rates of 30-34 kg/hr in the 1960s; with no records found in recent fish market surveys. • Artisanal gillnet fishery (Mayumba, Gabon): Individuals identified in surveys decreased from 40 between February 2013 and October 2015 to 5 between May and October 2018. • Fishery observers on national trawlers (Gabon): No sightings between 2015 and 2019 despite many species of rays recorded.
False Shark Ray (<i>Rhynchorhina mauritaniensis</i>) <i>Critically Endangered A2d</i> (Kyne et al., 2019) Generation length = 15 years	• Inferred population reduction over last three generations (global): >80%. • Species moderately abundant across its range in 1960s; subsequently declined and is now rarely encountered.
Japanese Wedgefish (<i>Rhynchobatus mononoke</i>) <i>Critically Endangered A2bd</i> (Jabado et al., 2024) Generation length = 15 years	• Inferred population reduction over last three generations (global): >80%. • Annual catch (Japan): Decreased catch in rays of 71% from 18,069 t in 1956 to 5,235 t in 2006; reconstructed catch data showed 98% decrease in landings over 65 year period from 1,300 t annually in the 1950s to around 14 t annually from 2010 to 2014 (noting some decrease may be due to reduced fishing effort). • Considered extremely rare across its range beginning in approximately 2002; occasionally recorded at fish markets in Kagoshima Prefecture, there were few records of the species in the year 2022
Eye-brow Wedgefish (<i>Rhynchobatus palpebratus</i>) <i>Near Threatened A2bd</i> (Kyne and Rigby, 2019) Generation length = 15 years	• Inferred population reduction over last three generations (global): 20-30%. • Inferred population reduction over last three generations (Asian part of its range): >80% . • Landing site surveys (Andaman Coast, Thailand): No records recently (no dates provided); although most of the species' currently recognized distribution is in areas with low fishing pressure and management measures are in place, high levels of exploitation cited just outside these areas. • Catch rate (Gulf of Thailand): 93% decrease in 'Rhinobathidae' (likely to include Giant Guitarfishes and Wedgefishes) from peak catches in 1968 to 1972. • Landings (Indonesia): Decrease of 88% for 'whitespotted wedgefishes' (likely includes all locally occurring Wedgefishes, including Eye-brow Wedgefish) from 28,492 t in 2005 to 3,540 in 2015 (est. >99% population reduction over last three generations). • Research trawl survey (Java Sea, Indonesia): Decrease in 'rays' from 1976 to 1997 estimated to 90%.
Broadnose Wedgefish (<i>Rhynchobatus springeri</i>) <i>Critically Endangered A2bd</i> (Kyne, 2019) Generation length = 15 years	• Inferred population reduction over last three generations (global): >80%. • Catch rate (Gulf of Thailand): 93% decrease in 'Rhinobathidae' (likely to include Giant Guitarfishes and Wedgefishes) from peak catches in 1968 to 1972. • Landings (Indonesia): Decrease of 88% for 'whitespotted wedgefishes' (likely includes all locally occurring Wedgefishes, including Broadnose Guitarfish) from 28,492 t in 2005 to 3,540 in 2015 (est. >99% reduction over last three generations). • Research trawl survey (Java Sea, Indonesia): Decrease in 'rays' from 1976 to 1997 estimated to 90%.

Species, IUCN Red List Category and generation length	Population Trend Information
Rhinidae spp.	- Catch rate (Gulf of Thailand): 93% decrease in 'Rhinobathidae' (likely to include Giant Guitarfishes and Wedgefishes) from peak catches in 1968 to 1972. - Landings (Indonesia): Decrease of 88% for 'whitespotted wedgefishes' (likely includes all locally occurring Wedgefishes) from 28,492 t in 2005 to 3,540 in 2015 (est. >99% reduction over last three generations). - Research trawl survey (Java Sea, Indonesia): Decrease in 'rays' from 1976 to 1997 estimated to 90%. - Landings (Islamic Republic of Iran): Decrease of 66% from 880 t in 1997 to 295 t in 2016 (est. 81% reduction over last three generations) (likely to include all locally occurring Giant Guitarfish and Wedgefish species). - Catch data for various demersal rays (Maharashtra, India): Decrease in catch rate from 0.65 kg/hr in 1990 to 0.24 kg/hr in 2004 (63% over 15 years; est. 86% reduction over last three generations). - Catch data for various demersal rays (Maharashtra, India): Decrease in catch rate from 0.65 kg/hr in 1990 to 0.24 kg/hr in 2004 (63% over 15 years; est. 95% reduction over three generations). - Landings (Tamil Nadu, India): Decrease of 87% in 'guitarfishes' (includes locally occurring Wedgefishes) from 114.9 t in 2002 to 14.9 t in 2006. - Landings (Balochistan, Pakistan): Decrease of 81% based on 'rhinobatids' (includes locally occurring Wedgefishes) from 921 t in 1994 to 178 t in 2011 (est. 94-95% reduction over last three generations). - Landings (Sindh, Pakistan): Decrease of 72% based on reductions in 'rhinobatid' (includes locally occurring Wedgefishes) from 902 in 1999 to 252 in 2011 (est. 94-95% reduction over last three generations).

Trade assessment

The species is or may be affected by trade

The family Rhinidae was listed in Appendix II in 2019 to address concerns that wedgefish species may become threatened with extinction unless trade is regulated due to the high value of wedgefish fins in international markets, restricted habitat use, and overharvest in coastal fisheries (see CoP18 Prop. 44). Wedgefish fins are of high value and sell for as much as USD680 at first point of sale. A genetic study of processed fin trimmings (which are the by-products after preparing imported fins for consumption) purchased regularly from the Hong Kong SAR of China shark fin market over two years (2015-2017) prior to the inclusion of these species in Appendix II showed the presence of at least three species in the family Rhinidae: Bottlenose Wedgefish (*Rhynchobatus australiae*), Smoothnose Wedgefish (*Rhynchobatus laevis*), and Whitespotted Wedgefish (*Rhynchobatus djiddensis*) (Cardena et al., 2020). These species together made up at least 0.1% of the trimmings collected, placing them in the top 20 most common among 86 species or species groups recorded in the fin trade.

Genetic analyses of processed fin trimmings using the same methodology after the CITES listing in 2020-2021 showed the presence of the same species in the shark fin market, as well as two other species: Broadnose Wedgefish (*Rhynchobatus springeri*) and Bowmouth Guitarfish (*Rhina ancylostomus*) (Chapman et al., 2025). In addition to fins, snouts, meat, and skins of Wedgefishes are also traded internationally.

According to the SS, a study in Singapore revealed that out of 106 products labelled as shark, 17% were of Bottlenose Wedgefish (*R. australiae*).

Legal Trade

In 2021 through 2023, the CITES Trade Database records show significant commercial exports of wedgefish fins (Table 3 and Table 4). Indonesia was the largest exporter of fins to Hong Kong SAR of China; records show four species of Rhinidae fins being exported. There was also commercial trade in bodies, bones, gill plates, meat, skins, and specimens.

Table 3. Exporter reported commercial trade of wild-sourced Wedgefish products between 2021 and 2023 (in alphabetical order by scientific name).

Species	Dried Fins (kg)	Fins (kg)	Total Commercial Exports (kg)	Percent of Total (rounded to the nearest value)
Bowmouth Guitarfish (<i>Rhina ancylostomus</i>)	3515	12509	17,454	5%
Bottlenose Wedgefish (<i>Rhynchobatus australiae</i>)	36158	47126	142,075	41%
Whitespotted Wedgefish (<i>Rhynchobatus djiddensis</i>)	366	17590	18,156	5%
Smoothnose Wedgefish (<i>Rhynchobatus laevis</i>)	6796	4416	11,5613	3%
African Wedgefish (<i>Rhynchobatus luebberti</i>)	63196	14201	77,397	22%
Eyeblink Wedgefish (<i>Rhynchobatus palpebratus</i>)	0	1447	1447	0.4%
Broadnose Wedgefish (<i>Rhynchobatus springeri</i>)	16552	20779	60,044	17%
<i>Rhynchobatus</i> spp.	72	17668	20919	6%

Table 4. Other Exporter Reported Commercial Exports Reported as *Rhynchobatus* spp.

Species	Bones, Meat, Skins	Bodies (kg)	Gill plates (kg)	Specimens (kg)
<i>Rhynchobatus</i> spp.	0	159	100	2920

Illegal Trade

According to LEMIS, the USA had one seizure of an unspecified amount of Rhinidae fins in 2020 from Cambodia.

There are no reports of seizures of wedgefishes in WiTIS.

Additional Information

Threats

Wedgefishes are highly vulnerable to various fishing gear types, such as demersal trawl and gillnets, and as a result, unregulated coastal fisheries are a primary threat to these species. Their risk increases due to the high value for their fins in international trade, which motivates fishers to target wedgefishes and retain them when caught as bycatch. The demand and high prices for Rhinidae fins also provides an incentive to engage in illegal trade in these species.

Other threats to the species are significant habitat modification, such as due to fishing gear and installation of dams (Moore, 2017).

Conservation, management and legislation

There is little management of wedgefishes across their range, but some countries, such as Bangladesh, India, Pakistan, and Australia, have adopted measures. Whitespotted Wedgefish (*Rhynchobatus djiddensis*) is listed on Schedule 1 of the Bangladesh Wildlife Protection Act of 2012, and this species is protected from exploitation under India's Wildlife Protection Act. Pakistan promulgated regulations prohibiting the catch of wedgefish and guitarfish species, and Australia manages a complex of wedgefish that includes Bottlenose Wedgefish (*Rhynchobatus australiae*).

Since wedgefishes are coastal species and most RFMOs have competency over waters beyond national jurisdiction, these species fall outside the scope of regional fishery management bodies. *Consequently, there is a lack of management, monitoring, and evaluation of Rhinidae species.*

At the international level, some actions have been taken to promote the conservation of wedgefishes. In addition to the inclusion of the family Rhinidae in Appendix II of CITES, Bottlenose Wedgefish (*Rhynchobatus australiae*) was listed in Appendix II of the Convention on the Conservation of Migratory Species of Wild Animals (CMS) in

2017. This Appendix covers migratory species with an unfavourable conservation status that require international agreements for their conservation and management; the Convention encourages range States to conclude global or regional Agreements for the conservation and management of species listed in Appendix II. Bottlenose Wedgefish (*Rhynchobatus australiae*), Whitespotted Wedgefish (*Rhynchobatus djiddensis*), and Smoothnose Wedgefish (*Rhynchobatus laevis*) were listed in Annex 1 the CMS Memorandum of Understanding on the Conservation of Migratory Sharks (Shark MOU) in 2018, which calls for international efforts under the Agreement in favour of the conservation of these species.

Implementation challenges (including similar species)

Species in the family Rhinidae can be challenging to distinguish, but identification guides are available to assist CITES Parties with differentiation between the wedgefishes and other CITES-listed sharks and rays.

References

- Cardeñosa, D., Fields, A.T., Babcock, E.A., Shea, S.K., Feldheim, K.A. and Chapman, D.D. (2020) Species composition of the largest shark fin retail-market in mainland China. *Scientific Reports* 10:12914 <https://doi.org/10.1038/s41598-020-69555-1>
- Chapman, D.D., Cardeñosa, D., Shea, K.H., Babcock, E.A., Jabado, R.W., Zhang, H., Gale, S.W. and Feldheim, K.A. (2025) Ongoing trade of fins from critically endangered rays (Wedgefish and giant guitarfish) despite international regulations and a novel PCR test to detect rays among 'shark' fins. *Conservation Genetics* <https://doi.org/10.1007/s10592-025-01716-z>
- Daly, R., Parker, D., Cliff, G., Jordaan, G.L., Nomfundo, N., Bennett, R.H. and Mann, B.Q. (2020) Long-term catch trends and risk assessment of the Critically Endangered white-spotted Wedgefish (*Rhynchobatus djiddensis*) from South Africa. *Aquatic Conservation of Marine and Freshwater Ecosystems* 31:777-788.
- Dulvy, N.K. (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Horswill, C., Kindsvater, H.K., Dulvy, N.K., Mull, C.G., Judah, A., D'Aberto, B.M., Matthiopoulos, J., and Mangel, M. (2025) Imputation of fisheries reference points for endangered data-poor fishes, with application to rhino rays. *Fish and Fisheries* 26(5):848–865. <https://doi.org/10.1111/faf.70003>
- Jabado, R.W. (2019) *Wedgefishes and Giant Guitarfishes: A Guide to Species Identification*. Wildlife Conservation Society, New York, United States. 30 pp.
- Jabado, R.W. (2024) *Sharks and Rays Listed on the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES): Whole Carcass*. Wildlife Conservation Society, New York. 194 pp.
- Jabado, R.W., Charles, R., Wosnick, N. and Kyne, P.M. (2024) *Rhynchobatus mononoke*. *The IUCN Red List of Threatened Species* 2024: e.T214417427A214417808. <https://dx.doi.org/10.2305/IUCN.UK.2024-1.RLTS.T214417427A214417808.en>
- Jabado, R.W., Kyne, P.M., Pollom, R.A., Ebert, D.A., Simpfendorfer, C.A., Ralph, G.M. and Dulvy, N.K. (2017) *The Conservation Status of Sharks, Rays, and Chimeras in the Arabian Sea and Adjacent Waters*. Environment Agency – Abu Dhabi, UAE and IUCN Species Survival Commission Shark Specialist Group, Vancouver, Canada. 236 pp.
- Kyne, P.M. (2019) *Rhynchobatus springeri*. *The IUCN Red List of Threatened Species* 2019:e.T60182A124448942. <http://dx.doi.org/10.2305/IUCN.UK.2019-2.RLTS.T60182A124448942.en>
- Kyne, P.M., Charles, R., Simeon, B. and Jabado, R.W. (2024) *Rhynchobatus laevis*. *The IUCN Red List of Threatened Species* 2024: e.T253117160A253117341. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T253117160A253117341.en>
- Kyne, P.M. and Ebert, D.A. (2019) *Rhynchobatus immaculatus*. *The IUCN Red List of Threatened Species* 2019: e.T104019954A104020006. <http://dx.doi.org/10.2305/IUCN.UK.2019-2.RLTS.T104019954A104020006.en>
- Kyne, P.M., Gledhill, K. and Jabado, R.W. (2019) *Rhynchobatus djiddensis*. *The IUCN Red List of Threatened Species* 2019: e.T39394A121035795. <http://dx.doi.org/10.2305/IUCN.UK.2019-2.RLTS.T39394A121035795.en>
- Kyne, P.M. and Jabado, R.W. (2019a) *Rhynchobatus luebberti*. *The IUCN Red List of Threatened Species* 2019: e.T60180A124448712. <http://dx.doi.org/10.2305/IUCN.UK.2019-2.RLTS.T60180A124448712.en>
- Kyne, P.M. and Jabado, R.W. (2019) *Rhynchorhina mauritaniensis*. *The IUCN Red List of Threatened Species* 2019: e.T130391495A130392523. <http://dx.doi.org/10.2305/IUCN.UK.2019-2.RLTS.T130391495A130392523.en>
- Kyne, P.M. and Rigby, C.L. (2019) *Rhynchobatus palpebratus*. *The IUCN Red List of Threatened Species* 2019: e.T195475A2382420. <http://dx.doi.org/10.2305/IUCN.UK.2019-2.RLTS.T195475A2382420.en>
- Kyne, P.M., Rigby, C.L., Dharmadi and Jabado, R.W. (2019) *Rhynchobatus australiae*. *The IUCN Red List of Threatened Species* 2019: e.T41853A68643043. <http://dx.doi.org/10.2305/IUCN.UK.2019-2.RLTS.T41853A68643043.en>
- Kyne, P.M., Rigby, C.L., Dharmadi and Jabado, R.W. (2019) *Rhina ancylostoma*. *The IUCN Red List of Threatened Species* 2019: e.T41848A124421912. <http://dx.doi.org/10.2305/IUCN.UK.2019-2.RLTS.T41848A124421912.en>
- Kyne, P.M., Rigby, C.L. and Last, P.R. (2019) *Rhynchobatus cooki*. *The IUCN Red List of Threatened Species* 2019: e.T60181A151858712. <http://dx.doi.org/10.2305/IUCN.UK.2019-2.RLTS.T60181A151858712.en>
- Moore, A.B.M. (2017) Are guitarfishes the next sawfishes? Extinction risk and an urgent call for conservation action. *Endangered Species Research* 34:75-88. <https://doi.org/10.3354/esr00830>

- Pardo, S., Kindsvater, H., Cuevas-Zimbrón, E., Sosa-Nishizaki, O., Pérez-Jiménez, J.C. and Dulvy, N.K. (2016) Growth, productivity and relative extinction risk of a data-sparse devil ray. *Scientific Reports* 6:33745. <https://doi.org/10.1038/srep33745>
- Stobutzki, I.C., Miller, M.J, Heales, D.S. and Brewer, D.T. (2002) Sustainability of elasmobranchs caught as bycatch in a tropical prawn (shrimp) trawl fishery. *Fishery Bulletin* 100(4):800-821.

Inclusion of Gulper Sharks of the Family Centrophoridae in Appendix II

Proponents: Brazil, Comoros, Dominican Republic, Ecuador, European Union, Lebanon, Nigeria, Panama, Senegal, Syrian Arab Republic, United Kingdom of Great Britain and Northern Ireland

Summary: The Dwarf Gulper Shark *Centrophorus atromarginatus* and the Gulper Shark *Centrophorus granulosus* (lead species) are proposed to be included in Appendix II in accordance with Article II paragraph 2(a) of the Convention and satisfying Criterion A and B in Annex 2a of Res. Conf. 9.24 (Rev. CoP17). All 14 other gulper shark species in the Family Centrophoridae¹⁶ are proposed for inclusion in Appendix II in accordance with Article II paragraph 2(b) of the Convention and satisfying Criterion A in Annex 2b (lookalike species) of the Resolution.

The Centrophoridae comprises two genera (*Centrophorus* and *Deania*) of deepwater sharks, typically found at depths between 200 and 1,500 m, occurring in cold-temperate to tropical waters. They are recorded throughout the world with few exceptions (Northeast Pacific and at high latitudes). The *Centrophorus atromarginatus* occurs in association with the seafloor on the upper continental slope at depths between 100 and 540 m, and *C. granulosus* is found on or near continental and insular shelves and slopes below 50 m, but primarily at depths between 300 and 1,100 m.

Gulper sharks are characterized by low productivity, with slow growth rates, long gestation periods, low fecundity, and late maturity. Their maximum size depends on the species, reaching up to 176 cm in *C. granulosus*. Breeding is generally every two years with small litter sizes: 1-2 in *C. atromarginatus* and 4-11 (usually 4-6) in *C. granulosus*. Age parameters for *C. atromarginatus* are unknown, but are inferred to be similar to *C. granulosus* with a generation length between 27 and 28 years.

All gulper shark species have IUCN Red List global assessments with *C. atromarginatus* classified as Critically Endangered and *C. granulosus* as Endangered. Among the other species, seven are assessed as Endangered (*C. harrissoni*, *C. isodon*, *C. lesliei*, *C. longipinnis*, *C. squamosus*, *C. tessellatus*, and *C. uyato*), and three as Vulnerable (*C. lusitanicus*¹⁶, *C. moluccensis*, and *Deania quadrispinosa*) due to levels of exploitation. Of the remaining species, two are classified as Near Threatened (*D. calceus* and *D. profundorum*) and one (*C. seychellorum*) as Least Concern, with one (*C. westraliensis*) listed as Data Deficient.

The main threats to sharks in the Centrophoridae are commercial and artisanal fisheries. They are targeted and retained when caught as bycatch because they have some of the highest value shark liver oil on the international market. Unlike coastal sharks, deepwater sharks have very large livers that can account for up to 30% of their body weight. They are also used for meat, fins, and possibly fish meal. The Dwarf Gulper Shark and Gulper Shark (*C. granulosus*) have a squalene content above 70%, squalene being the traded product extracted from shark liver oil, used in cosmetics, pharmaceuticals, food and dietary supplements. In addition to fishing, gulper sharks also face threats from deep-sea mining, marine pollution, and climate change.

Although there is a lack of species-specific data for gulper sharks, decreases in population have been estimated from catch per unit effort (CPUE) data, catch and landings data, fishing effort, and reported exports of shark liver oil for deepwater sharks. *C. atromarginatus* was assessed globally as Critically Endangered due to an over 80% reduction over the past three generations, caused by limited biological productivity and high levels of exploitation across most of its range. The Gulper Shark (*C. granulosus*) was assessed globally as Endangered due to a 50-79% population reduction over the past three generations, based on abundance data and levels of exploitation. These estimates were derived based on analysis of limited available data sets collected between 1975 and 2022.

Gulper shark fisheries are mostly unregulated, except for in parts of the Northeast Atlantic and

¹⁶ A recent taxonomic revision reduced the number of species within the genus *Centrophorus*. Evidence suggests that *Centrophorus lusitanicus* refers to the true *C. granulosus* from off Portugal and must be considered a junior synonym of the latter species (Fishbase 2025).

Southwest Pacific where concerns about their status have led to some fisheries management measures. Where management is lacking, these deepwater fisheries can be considered boom-and-bust fisheries.

All species of gulper sharks, except one, Seychelles Gulper Shark (*C. seychellorum*), have been identified in the shark liver oil trade or inferred to be traded based on their range and interactions with fisheries that have reported trade of *Centrophorus* species. Taxonomy of the Centrophoridae family is complex, and while it is possible to distinguish Gulper sharks of the genus *Deania* from those in the genus *Centrophorus*, it is difficult to visually distinguish between species within the genus *Centrophorus* as morphological changes between a juvenile and adult may be greater than the differences between species. Consequently, species-specific identification is challenging, and gulper shark catches and landings are often nationally reported under a generic category, such as by genus. The main commodity in trade is the highly valued liver oil, which cannot be differentiated by species visually or easily with other techniques.

Analysis: Species in the family Centrophoridae are among the most valuable shark species in international trade due to the use of liver oil in cosmetics, pharmaceuticals, food and dietary supplements. However, information on their conservation status – particularly species-specific information – is patchy in geographic coverage and timeframes.

Limited fishing and trade data sets for deepwater shark species are used as a basis for estimation of population trends of **Dwarf Gulper Shark *C. atromarginatus* and Gulper Shark *C. granulosus*, declines for both species would appear consistent with the indicative guidelines for inclusion in Appendix II** of commercially exploited low-productivity aquatic species suggested in the footnote to Annex 5 of Res. Conf. 9.24 (Rev. CoP17) (**thereby meeting Criterion 2aA of the Resolution**). Fishing for trade in liver oil does appear to be a significant factor affecting these species.

The taxonomic uncertainty and difficulty in identifying gulper sharks have caused some confusion with respect to the distribution of the various species. Different species are often reported under a general category, and some studies only refer to the genus *Centrophorus*. It is likely that most gulper sharks are retained when caught as bycatch due to the high value of shark liver oil on the international market. Only one species has not been identified in trade (*C. seychellorum*). It is not possible to differentiate liver oil by species visually or easily with other techniques. Therefore, the **regulation of all species in the family Centrophoridae may be necessary to ensure that international trade in *C. atromarginatus* and *C. granulosus* is effectively controlled consistent with criterion 2bA.**

Summary of Available Information

Text in non-italics is based on information in the Supporting Statement (SS); text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

A recent taxonomic revision reduced the number of species within the genus *Centrophorus*. Evidence suggests that *Centrophorus lusitanicus* refers to the true *C. granulosus* from off Portugal and must be considered a junior synonym of the latter species (Fishbase, 2025).

Range

Generally, Gulper sharks are deepwater species whose distribution has been recorded worldwide, except in the Northeast Pacific and at very high latitudes, occurring in cold-temperate to tropical waters at depths of mainly 200–1,500 m. Among the lead species, the Dwarf Gulper Shark has a sporadic distribution in parts of the Pacific and Indian Oceans (Figure 1). The Gulper Shark (*C. granulosus*) is widely distributed across the globe, but its distribution is patchy (Figure 2). This species may also be found in Areas Beyond National Jurisdiction.

Range: Extant (resident)

India; Indonesia; Japan; Oman; Papua New Guinea; Somalia; Sri Lanka; Taiwan Province of China



Figure 1. Map of the Dwarf Gulper Shark Distribution (IUCN SSC Shark Specialist Group 2024. *Centrophorus atromarginatus*. The IUCN Red List of Threatened Species. Version 2025-1).

Range: Extant (resident)

Angola; Anguilla; Antigua and Barbuda; Aruba; Australia; Bahamas; Barbados; Belize; Benin; Brazil; Cameroon; Cayman Islands; China; Colombia; Costa Rica; Cuba; Curaçao; Côte d'Ivoire; Dominica; Dominican Republic; Equatorial Guinea; France; French Guiana; Gabon; Gambia; Ghana; Grenada; Guadeloupe; Guatemala; Guyana; Haiti; Honduras; India (Andaman Is.); Indonesia; Jamaica; Japan; Madagascar; Maldives; Martinique; Mauritania; Mauritius; Mayotte; Mexico; Montserrat; Morocco; Mozambique; Myanmar; Namibia; Nicaragua; Nigeria; Panama; Papua New Guinea; Philippines; Portugal (Madeira); Puerto Rico; Saint Kitts and Nevis; Saint Lucia; Sao Tome and Principe; Senegal; Seychelles; Somalia; South Africa; Spain (Canary Is.); Sri Lanka; Suriname; Taiwan Province of China; Thailand; Togo; Trinidad and Tobago; UK; USA; Venezuela, Bolivarian Republic of; Virgin Islands, British; Virgin Islands, U.S.; Western Sahara



Figure 2. Map of the Gulper Shark Distribution IUCN SSC Shark Specialist Group 2024. *Centrophorus granulosus*. The IUCN Red List of Threatened Species. Version 2025-1).

IUCN Global Category and population trend information**Table 1.** Gulper sharks, IUCN Red List global category (version 3.1), and key population trend information used in this analysis. Population trend information taken from the IUCN Red List for each species and Supplemental Information, unless otherwise noted.

Species	IUCN Red List Category	Population Trend Information
Lead Species		
Dwarf Gulper Shark (<i>Centrophorus atromarginatus</i>)	Critically Endangered A2bd (Rigby et al., 2024)	- Inferred population reduction over past three generations (global): greater than 80%. - CPUE data (Lombok, Indonesia): Decrease by 88% from 2016 to 2018; data stayed at 2018 level in 2019 (Rigby et al., 2020). - Landings (all <i>Centrophorus</i> species are included; Dwarf Gulper Shark was reported, but landings may include Gulper Shark (<i>C. granulosus</i>)) (Cochin, India): Decreased from 114 t in 2008 to 39 t between 2009 and 2011 (66% decrease over 3 years with est. >99% reduction over three generation lengths). - Survey observations (Taiwan Province of China): Common species during 1988 survey, but was rarely found by the 2000s and absent since 2009 (Ebert et al., 2013) (est. >99% reduction over three generation lengths).
Gulper Shark (<i>Centrophorus granulosus</i>)	Endangered A2bd (Finucci et al. 2024)	- Estimated population reduction over the last three generations (global): 50 to 79%. - CPUE data (Gulf of Mexico): Annual rate of change in catch of Gulper Shark (<i>C. granulosus</i>) was 10% from 2011 to 2018 (>80% decrease over three generation lengths). - CPUE data (East Nusa Tenggara, Indonesia): Decrease from 146 kg per trip in 2011 to 77 kg per trip in 2014 (Gulper Shark (<i>C. granulosus</i>) comprised 15-20% of catch) (47% reduction in three years; est. >99% reduction over three generation lengths). - CPUE data (Lombok, Indonesia): Decrease by 88% from 2016 to 2018; data stayed at 2018 level in 2019 (Rigby et al., 2020). - CPUE data (South Andaman Islands, India): Small-scale fishery CPUE targeting deepwater sharks (Gulper Shark (<i>C. granulosus</i>) reported with <i>Squalus megalops</i>) for liver oil increased to peak of 112 kg per trip during 1988 and was 49 kg per trip in early 1992 (Soundararajan and Roy 2004) (56% reduction with est. >99% decline over three generation lengths). - Catch data (Southern Indian Ocean): Gulper Shark (<i>C. granulosus</i>) reported to the Southern Indian Ocean Fisheries Agreement (SIOFA) decreased from 133.7 t in 2015 to 9.2 t in 2022 (93% decrease with >99% decrease over three generation lengths). - Landings (Portugal): Decrease from 1056 t in 1990 to 17 t in 2009 (ICES WGEF, 2022) (98% decrease with est. >99% decrease over three generation lengths). - Landings (Mauritania): Reported squalid sharks (only livers recorded) decreased from 158 t in 1992 to 22 t in 2001 (86% decrease with est. >99% decrease over three generation lengths). - Landings (all shark species) (Taiwan Province of China): Decrease from 32,400 t in 1975 to 4,023 t in 2015 (88% reduction). - Annual exports of squalene shark liver oil (Philippines): Decreased from peak of 336 t in 1980 to 45 t (87% decrease with est. >99% decrease over three generations). - Catch estimates from high-value shark liver oil of Gulper sharks (<i>C. granulosus</i>) (Maldives): Estimated catch of 330 t in 1982 decreased to 70 t (Anderson and Ahmed, 1993) until 1989 (Ali, 2015) and appeared to collapse by the 2000s based on very low export figures after 1996 (Kyne and Simpfendorfer, 2007) (est. 97% decrease with >99% decrease over three generations).

Biological and trade criteria for inclusion in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)**A) Trade regulation needed to prevent future inclusion in Appendix I****Biology**

All the species covered by this proposal are members of the Centrophoridae family. These species are deepwater sharks that spend most of their lifecycle at depths below 200 m. They are vulnerable to overexploitation due to their conservative life history, which is defined by late maturity, long lives, low fecundity, and long gestation periods. *Gulper sharks have some of the slowest life histories of deepwater species (Simpfendorfer and Kyne, 2009), and the slowest life histories of any other animals targeted for their oil, such as whales, sirenians, and pinnepedes (Finucci et al., 2024d).* The lead species in the proposal have particularly conservative life histories; reproduction is viviparous with low litter sizes (Table 2).

A simple but robust estimator of natural mortality (M) for sharks and rays is the reciprocal of average lifespan ($M=1/w$) where average lifespan 'w' is the generation length of the species (Pardo et al., 2016). Hence, these CITES mortality thresholds can be back converted to calculate the corresponding generation length productivity thresholds ($GL = M-1$) and these are as follows: high productivity (natural mortality > 0.5 or a generation length of < 2 years), medium productivity (natural mortality $0.2-0.5$ or generation lengths $2-5$ years), and low productivity (natural mortality < 0.2 or generation length > 5 years).

Table 2: Life history characteristics of the lead species. Estimated generation length is from the IUCN Red List assessment for each species.

Species	Maximum Size (total length, TL, cm)	Average size at maturity (TL, cm)	Average litter size (pups)	Frequency of reproduction/gestation period	Estimated generation length (years)	Productivity
Dwarf Gulper Shark (<i>Centrophorus atomarginatus</i>)	99	56 (males); 75 (females)	1 to 2	Unknown; possibly 2 to 3 years	27.8	Low
Gulper Shark (<i>Centrophorus granulosus</i>)	176	111 (males); 143 (females)	4 to 11	2 years	27.75	Low

Population size

There is no information on population size for any of the Gulper sharks.

Population trends

All gulper sharks were recently assessed on the IUCN Red List between 2019 and 2024. At its 33rd meeting (2024), the CITES Animals Committee took note that gulper sharks have been previously identified as species of concern.

Dwarf Gulper Shark is targeted in nearly all its range, leading to a global status of Critically Endangered with an inferred decline greater than 80%. Gulper Shark (*C. granulosus*) has also been targeted and retained as bycatch in much of its range, which has resulted in significant and rapid declines in some locations. Gulper Shark was assessed as globally Endangered with a global decline between 50 and 80%. Due to taxonomic uncertainty and identification issues, there has been some confusion over the species-specific distribution of Gulper Sharks, and the species are often reported under a generic category in catch and landings data. Consequently, the available data and studies on populations used to infer decreases of the lead species often aggregate *Centrophorus* species.

*According to Simpfendorfer in litt. (2025) this is a common occurrence in assessment of sharks and rays that stems from the lack of species-specific data. The consequence of this data aggregation is that you often are confronted with information that includes multiple species; such grouped data is used to make inferences about decline in sharks and rays because it remains the best data available. The danger with this type of use is that the grouped data will provide an "average" across the mix of species included, and so some species will decline more than others. Those that are most likely to decline are those with the slowest life histories (i.e., lowest productivity) (Simpfendorfer in litt., 2025). This would include the Dwarf Gulper Shark, Gulper Shark (*C. granulosus*), and Leafscale gulper shark (*C. squamosus*).*

Indo-Pacific Ocean

Within the Indo-Pacific Ocean, Gulper sharks are targeted in demersal deepwater longline fisheries in Indonesia. Data from one of these targeted demersal longline fisheries landed in Tenau, East Nusa Tenggara, documented a decline in CPUE from 146 kg per trip to 77 kg per trip between 2011 and 2014, which calculates to a 47% decline over just three years, equating to an estimated population decline greater than 99% over three generation lengths (83 years). Gulper Shark (*C. granulosus*) was found to account for between 15 and 20% of the catch, so the decrease is inferred for this species. In Lombok, Indonesia, the estimated CPUE for *Centrophorus* species declined by 88% between 2016 and 2018 and stayed at this level in 2019. Based on the species' ranges, both Dwarf Gulper Shark and Gulper Shark (*C. granulosus*) could have been included in the catch of these sharks; the reduction in gulper sharks could be inferred for both species.

Pacific Ocean

In the Pacific Ocean, a population decline of greater than 80% is inferred for Dwarf Gulper Shark in Taiwan Province of China. This species was relatively common in landings in the late 1980s but was rare by the 2000s and has not been seen since 2009; decreases occurred as fishing effort increased. Although landings data in Taiwan Province of China included records of Dwarf Gulper Shark, data for other shark species (shelf and deepwater species) are combined and are not species-specific. As a result, fisheries data of all shark species were examined to assess the population trends of Gulper Shark (*C. granulosus*). Decreases from 32,400 t in 1975 to 4,023 t in 2015 were found, even though both industrial and artisanal fishing efforts have increased substantially since 1975, inferring an 88% decline for Dwarf Shark.

In the Philippines, *Centrophorus* species were first targeted in the 1960s for their liver oil, which was exported to Japan, Hong Kong SAR of China, and the USA. In many areas of the Philippines, this target fishery was highly profitable but usually collapsed after 10 years. Annual exports of shark liver oil from the Philippines peaked at 336 t in 1980 and declined to 45 t in 1993, a reduction of 87%. This decrease is inferred for Gulper Shark (*C. granulosus*) based on the species' range.

Indian Ocean

In the Maldives, Gulper sharks were targeted for their liver oil; the fishery reached its peak between 1982 and 1984. Subsequently, there was a drastic decline, and stocks collapsed in the early 2000s, after approximately 20 years of exploitation. A 97% population reduction in *Centrophorus* species was estimated from the fishery targeting Gulper sharks for their liver oil for export. Since Gulper Shark (*C. granulosus*) was the most common species taken in the fishery, this population decrease is inferred for this species.

At Cochin, in southwestern India, *Centrophorus* species are suspected to have collapsed due to the rapid development of targeted fishing for their liver oil. A population decline greater than 99% in these species over three generation lengths (83 years) is estimated based on significant decreases in landings of *Centrophorus* species for their liver oil. Landings data – that included reports of Dwarf Gulper Shark – decreased from 114 t to 39 t between 2008 and 2011, a 66% reduction in landings over four years. This decrease equates to a decline of more than 99% over three generation lengths (83 years) for Dwarf Gulper Shark. Off the South Andaman Islands, India, a 56% reduction in mean CPUE of Gulper Shark (*C. granulosus*) over four years from 1988 to 1992 was estimated in a small-scale fishery targeting these species for its liver oil. The CPUE peaked at 112 kg per trip in mid-1988 and decreased to 49 kg per trip by early 1992. This reduction in CPUE, which was reported for Gulper Shark (*C. granulosus*) and *Squalus megalops*, infers a population decline greater than 99% for Gulper Shark (*C. granulosus*) over three generation lengths (83 years).

According to data reported to SIOFA, Gulper Shark (*C. granulosus*) catch in the Southern Indian Ocean by the European Union was 106.1 t in 2008, peaked at 133.7 t in 2015, and rapidly declined to 2.3 t by 2020. A further 9.2 t was reported in 2022, and no catch was reported in 2021 or 2023, resulting in an estimated decline of 93% that is inferred for Gulper Shark (*C. granulosus*).

Atlantic Ocean

In the Gulf of Mexico, an estimated population decline of greater than 80% is inferred for Gulper Shark (*C. granulosus*) over three generation lengths (83 years) based on CPUE data. This decrease was extrapolated from CPUE data between 2011 and 2018 that showed an annual rate of decline of 10%. Across the Atlantic, landings of squalid sharks – including Gulper Shark (*C. granulosus*) – off the coast Mauritania decreased from 158 t in 1992 to 22 t in 2001. This calculates to an 86% decrease over 10 years, and an estimated population reduction greater than 99% over three generation lengths for Gulper Shark (*C. granulosus*). In Portugal, Gulper Shark (*C. granulosus*) landings from the Portuguese longline fishery declined from a peak of 400 t in 1996 to 17 t in 2009, a 96% decline.

B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences

The liver oil of Gulper sharks, including Dwarf Gulper Shark and Gulper Shark (*C. granulosus*), is the primary product used from these species and, due to its high value in international trade, is the main driver of their exploitation and retention when caught as bycatch in fisheries. *Shark liver oil is among the most widely used shark products (after fins and meat) and is twice as valuable as shark-meat exports in some countries, such as India (Dent and Clarke, 2015).* Unlike coastal sharks, deepwater sharks have very large livers that can account for up to 30% of their body weight. Their squalene content is also higher, making them preferred for shark liver oil over coastal species. The liver oil of gulper sharks contains the highest percentage of squalene of any shark family and is considered the most valuable of shark liver oils. *According to Finucci et al. (2024d), liver-oil fisheries are not sustainable and result in steep population reductions and elevated extinction risk. These boom-and-bust fisheries peak within a few years with collapse soon after.*

Liver oil, which provides a natural source of squalene, is used in a variety of products, including cosmetics, pharmaceuticals, and other industrial applications. Worldwide squalene production is mostly used by the cosmetics industry (65%), followed by the pharmaceutical industry (20%), food (10%), and other applications (5%). *Marine fish are the main commercial source of squalene for use in dietary supplements and nutraceuticals (Patel et al., 2020). In response to the demand for sustainable substitutes, advances in biotechnology have developed synthetic techniques to produce squalene from plants. However, these yields are insufficient to meet the global demand (Lozano-Grande et al., 2018).*

Inclusion in Appendix II to improve control of other listed species

A) Specimens in trade resemble those of species listed in Appendix II under Res. Conf. 9.24 (Rev. CoP17) Annex 2 a or listed in Appendix I

All species of gulper sharks, except one (*C. seychellorum*), have been identified in the shark liver oil trade or inferred to be traded based on their range and interactions with fisheries that have reported trade of *Centrophorus* species. *Most (93.3%) Gulper shark species have been identified in trade or inferred to be traded on the basis of information from regional fisheries that report catch under generic categories (e.g., Centrophorus spp.) (Finucci et al., 2024d).* Yet, the Centrophoridae family is one of the most taxonomically complex shark families, and it is difficult to visually distinguish between species within the genus *Centrophorus*. Traders use hand-held refractometers to check squalene content of shark liver oil, and squalene content can also be determined using Raman Spectroscopy. Gulper sharks could potentially be identified in liver oil due to their high level of squalene, but refractometers may not be precise enough to separate species of gulper sharks. *DNA work is underway that would allow identification by species, even for oil that has been processed (Sant in litt., 2025). To assist with species and product identification, an identification guide has also been developed for gulper sharks that provides a tool for visual identification and guidance for conducting genetic sampling (Finucci et al., 2025; Sant in litt., 2025).*

B) Compelling other reasons to ensure that effective control of trade in currently listed species is achieved

Other lookalike species in the family Centrophoridae may qualify for listing under Annex 2a, Criterion A and/or B. To illustrate, the Leafscale Gulper Shark (*C. squamosus*), which is found in the Northeast Atlantic, is considered to have a poor status by the Convention for the Protection of the Marine Environment of the North-East Atlantic (OSPAR) and zero catches have been recommended by ICES for 2024-2027. Due to levels of exploitation, seven species (*C. harrissoni*, *C. isodon*, *C. lesliei*, *C. longipinnis*, *C. squamosus*, *C. tessellatus*, and *C. uyato*) are assessed as Endangered, and three species (*C. lusitanicus*¹⁶, *C. moluccensis*, and *Deania quadrispinosa*) are globally listed as Vulnerable. The rest of the species are classified as either Near Threatened (*Deania calceus* and *D. profundorum*), Least Concern (*C. seychellorum*), or Data Deficient (*C. westraliensis*).

Additional Information

Threats

Although gulper sharks are primarily caught and used for their liver oil, they may also be used for their meat, fins, and possibly fish meal. The meat is sold fresh, dried, and salted and generally sold in local domestic markets. In some countries it is popular for the boneless flakes and is commonly used domestically for fish and chips. In some regions, the meat is highly valued and may be traded internationally. Individuals that are too small for human consumption are likely used locally in fish meal production. Byproducts from gulper shark liver oil production may also be used in fish meal and in poultry feed. Gulper shark fins are of low value, but they have been reported in the international fin trade in low quantities. *One study showed that they accounted for approximately 0.4% of fin samples in Hong Kong SAR of China (Cardeñosa et al., 2020).*

Since the international trade in gulper sharks is not currently regulated, there is limited information on illegal trade in the species. There are no reports of seizures in Gulper sharks in WiTIS, and there were no reports of trade of gulper sharks in LEMIS between 2008 and 2021. *Notably, a recent study suggests that the use of squalene from sharks may not be accurately represented in products. Despite the availability of synthetic squalene, there is*

evidence that 26% of cosmetics contain shark-derived squalene, including samples of cosmetics claiming to use vegan or plant-based squalene (Nicholson, 2022).

Beyond the threats of fishing and international trade, gulper sharks are also threatened by deep sea mining, pollution, and habitat loss due to climate change.

Conservation, management and legislation

National

At the national level, there are no species-specific measures for Dwarf Gulper Shark or Gulper Shark (*C. granulosus*). These fisheries are mostly unregulated, except for in parts of the Northeast Atlantic and Southwest Pacific where concerns about their status have led to some fisheries management measures. In Australia, *C. harrissoni* and *C. uyato* are listed as Conservation Dependent under the Environment Protection and Biodiversity Conservation Act 1999. To halt their decline and support recovery, the species are strictly managed and monitored, including through a network of spatial closures, zero retention, move-on provisions, and handling practices.

Regional and International

Since 2013, the North East Atlantic Fisheries Commission (NEAFC) has required that Contracting Parties prohibit vessels flying their flag in the Regulatory Area from directed fishing for specified deepwater sharks. These include Gulper Shark (*C. granulosus*), Leafscale Gulper Shark, and *Deania calceus*. Gulper Shark (*C. granulosus*) (and Leafscale Gulper Shark) were included in 2008 on the OSPAR List of threatened and/or declining species and habitats, which is intended to guide the OSPAR Commission in priority setting for further conservation work. In northeast Atlantic waters, Gulper Shark (*C. granulosus*) (and Leafscale Gulper Shark) were included in 2013 in the 'deep-sea shark' category; a NEAFC Recommendation in 2013 banned directed fishing for these 'deep-sea shark' species in European Community and international waters with no allowances for bycatch. A temporary restrictive bycatch allowance (10 t) was enacted in 2017 and 2018, but it was discontinued after 2020 with a return to a ban on directed fishing and bycatch retention for 'deep-sea shark' species in those fisheries, in European Union and international waters, and in Northwest African waters of the Fishery Committee for the Eastern Central Atlantic (CEFAC). This legislation is also applicable in the UK's waters and to UK vessels and is addressed as 'Retained European Union Law' (<https://www.sharktrust.org/pages/category/fisheries-advisories>).

Since 2005, gillnets are banned at depths greater than 200 m in Azores, Madeira, and Canary Islands, and since 2007, the use of gillnets by European Community vessels are banned at depths below 600 m depth, with an exceptional provision that sets maximum bycatch limits of 5% deepwater shark for those fisheries targeting European Hake (*Merluccius merluccius*) and monkfish (*Lophius spp.*) at depths between 200 and 600 m. In European waters, demersal trawl fishing has been prohibited at depths greater than 800 m since 2016.

In the Southern Ocean, SIOFA banned gillnets in 2017 (CMM 2016/05), and in 2019, it adopted a conservation and management measure that banned directed fishing for all deepwater shark species previously targeted and listed as 'high risk and of concern', which includes Gulper Shark (*C. granulosus*).

The South East Atlantic Fisheries Organisation (SEAFO) has recommended a ban on targeting of deepwater sharks (Recommendation 1/2008), but it has not yet been adopted. The South Pacific Regional Fisheries Management Organisation (SPRFMO) banned gillnets in 2010 (CMM 08-2023). Neither Dwarf Gulper Shark nor Gulper Shark (*C. granulosus*) are present in the SPRFMO Convention Area, but other gulper sharks occur there.

Two other Regional Fisheries Management Organizations (RFMOs) and one Regional Fishery Body include the range of one or both lead *Centrophorus* species. Gulper Shark (*C. granulosus*) is found in the area of competence of the Northwest Atlantic Fisheries Organization (NAFO) and the Western Central Atlantic Fishery Commission (WECAFC). The Dwarf Gulper Shark and the Gulper Shark (*C. granulosus*) are both found in the Convention Area of the North Pacific Fisheries Commission (NPFC). However, no measures have been adopted by these organizations for the management of these species.

In the Mediterranean, one species of Gulper shark (*C. uyato*) is listed on Annex III of the Convention for the Protection of the Marine Environment and Coastal Region of the Mediterranean (Barcelona Convention) Protocol concerning Specially Protected Areas and Biological Diversity in the Mediterranean (SPA/BD Protocol), which includes species whose exploitation is regulated.

Given their ranges, Gulper sharks may be caught in areas beyond national jurisdiction and could potentially fall under the jurisdiction of RFMOs that ban the use of deepwater gillnets. However, the distinction between gulper sharks and other deepwater shark species is that gulper sharks are primarily caught within national Exclusive Economic Zones (national waters). Therefore, the responsibility falls to the flag State to manage the harvest of these species by its vessels (Sant in litt., 2025).

Artificial Propagation/captive breeding

There is no literature or information available regarding captive breeding of gulper sharks.

Implementation challenges (including similar species)

Nearly two-thirds of deepwater sharks (approximately 58%) have been used for their liver oil. The family of Centrophoridae represents just over one-quarter (26%) of all deepwater shark species known to be used for the liver oil. Based on one study, other species identified in the trade besides *Centrophorus* species include dogfishes (Squalidae), sleeper sharks (Somniosidae), and cowsharks (Hexanchidae) (Finucci et al., 2024b). Identification of different species in shark liver oil would be a challenge for enforcement officers and Customs or wildlife inspectors. According to Sant in litt. (2025) quality and price differentiation could apply; since the oil from gulper sharks is much higher than the oil from other shark species, the oil from gulper sharks is used in different products (e.g., cosmetics) whereas lower quality oil from other sharks may be used in other [less valuable] products (e.g., industrial products). Traders are aware of the different quality, and it is unlikely that they would mix gulper shark oil with the oil of lower quality species since this would diminish the quality and value (price). However, monitoring the international trade of shark liver oil that is comprised of multiple species would make it difficult to identify and quantify the species composition of shark liver oil (Finucci et al., 2024d).

Other comments

The European Union and the UK submitted Document CoP20 88.2, which is relevant to this proposal. The document provides some information on deepwater elasmobranchs and proposes a set of draft decisions that would call on the Secretariat, subject to external funding, to contract technical and scientific experts to prepare an overview of the conservation status, population status and trends of these species. This report would highlight specific gaps in knowledge by species at global and regional scales, focusing on any deep-water sharks that are included in the CITES Appendices at CoP20 and, an overview of the international trade and management of species of deepwater elasmobranchs and their products.

Products of CITES-listed species, particularly meat, processed fins, and highly processed parts and derivatives, should be appropriately marked to make them identifiable and help ensure traceability. In the case of high value liver oil, supply chain traceability is particularly important to exclude products of illegal, unreported, and unregulated fishing (Sant in litt., 2025).

References

- Akhilesh, K.V. and Ganga, U. (2013) A note on the targeted fishery for deep-sea Oil Sharks at Cochin Fisheries Harbour. *Marine Fisheries Information Service; Technical & Extension Series* 218: 22-23.
- Ali, K. (2015) *Status of the Shark Fishery Ban in the Maldives and the Implementation of the National Plan of Action on Sharks – An Update with Notes on Turtles and Seabirds*. IOTC-2015-WCEB11-12 Rev_1.
- Anderson, R.C. and Ahmed, H. (1993) *The Shark Fisheries of the Maldives. A Review*. Ministry of Fisheries and Agriculture, Republic of the Maldives and the Food and Agriculture Organization of the United Nations. <https://www.fao.org/fishery/docs/CDrom/bobp/cd1/Bobp/Publns/MIS/0007.pdf>
- Cardeñosa, D., Fields, A., Babcock, E.A., Shea, S.K.H., Feldheim, K.A. and Chapman, D. (2020) Species composition of the largest shark fin retail-market in mainland China. *Scientific Reports* 10:12914. <https://doi.org/10.1038/s41598-020-69555-1>
- Clarke, M., White, W. and Compagno, L.J.V. (2009) *Centrophorus lusitanicus*. *The IUCN Red List of Threatened Species* 2009: e.T161442A5424877. <http://dx.doi.org/10.2305/IUCN.UK.20092.RLTS.T161442A5424877.en>
- Dent, F. and Clarke, S. (2015) *State of the global market for shark products*. FAO Fisheries and Aquaculture Technical Paper No. 590. Rome, FAO. 187 pp.
- Ebert, D.A., White, W.T., Ho, H.-C., Last, P.R., Nakaya, K., Séret, B., Straube, N., Naylor, G.J.P. and de Carvalho, M.R. (2013) An annotated checklist of the chondrichthyans of Taiwan. *Zootaxa* 3752(1):279-386.
- Finucci, B., Bineesh, K.K., Cheok, J., Cotton, C.F., Dharmadi, Kulka, D.W., Neat, F.C., Pacoureaux, N., Rigby, C.L., Tanaka, S. and Walker, T.I. (2024a) *Centrophorus moluccensis* (amended version of 2024 assessment). *The IUCN Red List of Threatened Species* 2024: e.T42838A267277845. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T42838A267277845.en>.
- Finucci, B., Cheok, J., Cotton, C.F., Kulka, D.W., Neat, F.C., Rigby, C.L., Tanaka, S. and Walker, T.I. (2020) *Centrophorus harrissoni*. *The IUCN Red List of Threatened Species* 2020: e.T41740A68614118. <https://dx.doi.org/10.2305/IUCN.UK.2020-3.RLTS.T41740A68614118.en>
- Finucci, B., Cheok, J., Cotton, C.F., Kulka, D.W., Neat, F.C., Pacoureaux, N., Rigby, C.L., Tanaka, S. and Walker, T.I. (2024b) *Deania calceus* (amended version of 2024 assessment). *The IUCN Red List of Threatened Species* 2024: e.T41798A267283710. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T41798A267283710.en>.
- Finucci, B., Cheok, J., Cotton, C.F., Kulka, D.W., Neat, F.C., Rigby, C.L., Tanaka, S. and Walker, T.I. (2024c) *Deania quadrispinosa* (amended version of 2020 assessment). *The IUCN Red List of Threatened Species* 2024: e.T161635A267298077. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T161635A267298077.en>.

- Finucci, B., Green, M.E., Rigby, C.L. and Simpfendorfer, C.A. (2025) *Gulper Shark Identification Guide*. University of Tasmania, Hobart, Australia. 37 pp.
- Finucci, B., Rigby, C. and Daley, R. (2024d) Global Conservation Status of Gulper Sharks (Genus *Centrophorus*). *Science* 383:1135-1141.
- Finucci, B., Rigby, C.L., Bineesh, K.K., Charles, R., Cheok, J., Cotton, C.F., Kulka, D.W., Neat, F.C., Pacoureau, N., Tanaka, S. and Walker, T.I. (2024e) *Centrophorus granulosus*. *The IUCN Red List of Threatened Species* 2024: e.T162293947A254964078. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T162293947A254964078.en>
- Finucci, B., Rigby, C.L., Bineesh, K.K., Cheok, J., Cotton, C.F., Kulka, D.W., Neat, F.C., Pacoureau, N., Rohner, C.A., Tanaka, S. and Walker, T.I. (2024f) *Centrophorus squamosus*. *The IUCN Red List of Threatened Species* 2024: e.T41871A254975849. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T41871A254975849.en>.
- Finucci, B., Rigby, C.L., Bineesh, K.K., Charles, R., Cotton, C.F., Kulka, D.W., Neat, F.C., Pacoureau, N., Tanaka, S. and Walker, T.I. (2024g) *Centrophorus uyato*. *The IUCN Red List of Threatened Species* 2024: e.T41745A254988584. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T41745A254988584.en>.
- Finucci, B., Rigby, C.L., Charles, R., Cotton, C.F., Kulka, D.W., Neat, F.C., Tanaka, S. and Walker, T.I. (2024h) *Centrophorus leslei*. *The IUCN Red List of Threatened Species* 2024: e.T162292832A254971919. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T162292832A254971919.en>.
- Finucci, B., Rigby, C.L., Cheok, J., Cotton, C.F., Kulka, D.W., Neat, F.C., Rohner, C.A., Tanaka, S. and Walker, T.I. (2024i) *Deania profundorum*. *The IUCN Red List of Threatened Species* 2024: e.T161551A255043036. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T161551A255043036.en>.
- Fishbase. (2025) *Centrophorus lusitanicus* Barbosa du Bocage & de Brito Capello, 1864 Lowfin gulper shark. *Fishbase*. <https://www.fishbase.se/summary/Centrophorus-lusitanicus>. Viewed 30 July 2025.
- ICES (2022) Working Group on Elasmobranch Fisheries (WGEF). ICES Scientific Reports. 4:74. 848 pp. <http://doi.org/10.17895/ices.pub.21089833>
- Kyne, P.M. and Simpfendorfer, C.A. (2007) *A Collation and Summarization of Available Data on Deepwater Chondrichthians: Biodiversity, Life History and Fisheries*. <https://www.researchgate.net/publication/228826523>
- Lozano-Grande, M.A., Gorinstein, S., Espitia-Rangel, E., Davila-Ortiz, G. and Martinez-Ayala, A., L. (2018) Plant Sources, Extraction Methods, and Uses of Squalene. *International Journal of Agronomy*. 1829160, 13pp. <https://doi.org/10.1155/2018/1829160>
- Nicholson, S.J. (2022) *Sharkless Makeup> Sharkless Oceans*. Masters thesis. Environmental Applied Science and Management, Toronto, Ontario, Canada. https://rshare.library.torontomu.ca/articles/thesis/Sharkless_Makeup_Sharkless_Oceans/26053096?file=4710750
- Pardo, S., Kindsvater, H., Cuevas-Zimbrón, E., Sosa-Nishizaki, O., Pérez-Jiménez, J.C. and Dulvy, N.K. (2016) Growth, productivity and relative extinction risk of a data-sparse devil ray. *Scientific Reports* 6:33745. <https://doi.org/10.1038/srep33745>
- Patel, A., Mu, L., Shi, Y., Rova, U., Christakopoulos, P. and Matsakas, L. (2020) Novel Biorefinery Approach Aimed at Vegetarians Reduces the Dependency on Marine Fish Stocks for Obtaining Squalene and Docosahexaenoic Acid. *ACS Sustainable Chemistry & Engineering* 8(23):8803-8813. <https://dx.doi.org/10.1021/acssuschemeng.0c02752>
- Pollom, R. (2019) *Centrophorus seychellorum*. *The IUCN Red List of Threatened Species* 2019: e.T161417A124481302. <https://dx.doi.org/10.2305/IUCN.UK.2019-2.RLTS.T161417A124481302.en>.
- Rigby, C.L., Bin Ali, A., Bineesh, K.K., Derrick, D., Dharmadi, Ebert, D.A., Fahmi, Fernando, D., Gautama, D.A., Maung, A., Tanay, D., Utzurrum, J.A.T. and Yueni, R.R. (2020) *Centrophorus longipinnis*. *The IUCN Red List of Threatened Species* 2020: e.T162292301A162292797. <https://dx.doi.org/10.2305/IUCN.UK.2020-3.RLTS.T162292301A162292797.en>
- Rigby, C.L., Braccini, J., Charles, R., Daley, R.K., Ebert, D.A. and Simpfendorfer, C. (2024) *Centrophorus westraliensis*. *The IUCN Red List of Threatened Species* 2024: e.T195485A124547736. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T195485A124547736.en>.
- Rigby, C.L., Ebert, D.A. and Herman, K. (2024) *Centrophorus atromarginatus* (amended version of 2024 assessment). *The IUCN Red List of Threatened Species* 2024: e.T161384A267274043. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T161384A267274043.en>
- Rigby, C.L., Ebert, D.A. and Herman, K. (2024) *Centrophorus isodon* (amended version of 2020 assessment). *The IUCN Red List of Threatened Species* 2024: e.T44563A267274929. <https://dx.doi.org/10.2305/IUCN.UK.2024-2.RLTS.T44563A267274929.en>.
- Rigby, C.L., Finucci, B., Walls, R.H.L., Derrick, D., Herman, K. and Tanaka, S. (2021) *Centrophorus tessellatus*. *The IUCN Red List of Threatened Species* 2021: e.T60208A124452245. <https://dx.doi.org/10.2305/IUCN.UK.2021-1.RLTS.T60208A124452245.en>.
- Sant, G. (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Simpfendorfer, C. A. and Kyne, P. M. (2009) Limited potential to recover from overfishing raises concerns for deep-sea sharks, rays and chimaeras. *Environmental Conservation* 36(2):97-103.

- Simpfendorfer, C. (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Soundararajan, R. and Roy, S.D. (2004) Distributional record and biological notes on two deep-sea sharks, *Centrophorus acus* Garman and *Squalus megalops* (Macleay), from Andaman waters. *Journal of Marine Biological Association of India* 46(2):178-184.

Inclusion of eels *Anguilla* species in Appendix II

Proponents: European Union, Honduras

Summary: Eels, *Anguilla* spp., are a very distinctive group of elongated fish distributed throughout temperate and tropical waters across the Atlantic, Indian, and Pacific Oceans. They migrate from freshwater or estuarine habitats to marine spawning grounds, completing a complex life cycle across oceanic and continental ranges. They pass through various life stages (marine leptocephali, glass eels, elvers, yellow eels and 'silver' eels) and breed only once (i.e. are semelparous), with each female potentially producing millions of eggs. The European Eel *Anguilla anguilla*, was included in Appendix II at CoP14 in 2007. The Japanese Eel *Anguilla japonica* and American Eel *A. rostrata* are now proposed for inclusion in Appendix II in accordance with Article II, paragraph 2(a) of the Convention and Annex 2a Criterion B of Res. Conf. 9.24 (Rev. CoP17); the remaining 14 non-listed *Anguilla* species are proposed for inclusion as lookalike species in accordance with Annex 2b.

Eels are highly sought after as food species, particularly in East Asia, notably China and Japan. Nearly 90% of global eel production for human consumption relies on input of live wild-caught glass eels, elvers or yellow eels (seedstock) that are raised in aquaculture systems, because commercial-scale captive breeding remains unfeasible for all species. The vast majority of eel aquaculture production is in East Asia. With increasing difficulty in obtaining adequate seedstock of *A. japonica* there has been growing reliance on international trade of other species to meet demand.

The life history of eels makes stock assessment extremely difficult and no global population sizes are known. However sometimes drastically decreasing harvests, both of seedstock for aquaculture and of larger (mainly 'silver') eels for consumption are persuasive indicators of decreasing populations. *Anguilla japonica* and *A. rostrata* were categorised as Endangered in 2018 and 2020 on the IUCN Red List respectively, based on an approximate 50% decrease in abundance over three generations across their respective ranges. The main identified threats include overfishing, habitat degradation, pollution, migration barriers such as hydroelectric dams and climate change impacts. The significance of any one threat or the synergy it may have with other threats is still poorly understood. Notably it is difficult to assess the effects of harvest on stock levels and recruitment rates.

The trade includes seedstock destined for aquaculture as well as yellow and silver eels for direct human consumption (in both cases, mostly *A. japonica*, *A. rostrata*, and *A. anguilla*, to a lesser extent *A. bicolor*, but also some exports of farmed *A. marmorata*, *A. australis*, and *A. dieffenbachii*). According to the UN Comtrade database, global exports of anguillid products between 2014 and 2023 amounted to 920,000 tonnes, or ~80,000 to ~110,000 tonnes annually, just under half comprising processed or preserved eels and one third live eels, with the remainder largely frozen. However, the global Harmonised System (HS) codes used that relate to *Anguilla* spp. do not distinguish between species or life stages. Trade in *Anguilla* species is dynamic and constantly evolving, with historic intensive harvesting of glass eels for aquaculture initially concentrated on *A. japonica* for East Asian farms.

During the 1980s and 1990s European glass eel demand surged for aquaculture and consumption in East Asia, largely due to shortages of *A. japonica*. Following a decline of over 90% in *A. anguilla* recruitment and increased regulatory controls for the species (including its inclusion in Appendix II in 2007), commercial demand shifted toward other species, notably *A. rostrata* and, to a lesser extent, tropical eels such as Shortfin Eel *A. bicolor*. Since 2014, reported glass eel inputs into aquaculture ponds in China show that inputs of *A. rostrata* have equalled or surpassed *A. japonica* in volume, highlighting a significant shift in sourcing for eel farming operations. Additionally, East Asian imports of *A. rostrata* in 2022 were more than five times the annual average recorded between 2004 and 2021. This corresponded with sharp price increase for *A. rostrata* seedstock landed in Canada, from USD314/kg in 2009 to USD3,492/kg in 2023. A study investigating the species composition of eel products in Japan found that approximately 60% of imported eel products were *A. rostrata*.

Enforcement is complicated by the absence of species-specific HS codes and the fact that processed products are visually indistinguishable between species. DNA-based market studies have revealed frequent mislabelling, with substitution of species detected in products sold on domestic markets

within producing countries and in products exported to international markets. Such identification challenges hinder monitoring of trade volumes by species, complicate the detection of illegal trade, and reduce the effectiveness of management measures. However, large variations in market price, for example between *A. japonica* and *A. marmorata*, may serve as incentives for traders to separate the two in some cases.

Several national and subnational measures regulate eel harvest and trade across range States. For *A. japonica*, management is subject to an informal agreement among China, Japan, the Republic of Korea, and Taiwan Province of China, which includes (voluntary) limits on seed (glass eel) input to aquaculture. The reported quantity of eel seed input into aquaculture ponds was expected to approach 125 tonnes for the 2024-2025 fishing season, substantially exceeding the agreed threshold (78.8 tonnes), which has not been modified since the agreement was established in 2014. Other species are managed nationally through seasonal restrictions, minimum size limits, gear limitations, and licensing systems; however, implementation and enforcement vary widely. The USA has implemented domestic restrictions for *A. rostrata*. Canada also recently developed new legislation for harvest of *A. rostrata*.

Analysis: Eels *Anguilla* spp. are a highly valuable fisheries resource that in the case of temperate species, notably the CITES-listed *A. anguilla*, and the non-listed *A. japonica* and *A. rostrata*, has been harvested in large quantities, particularly as seedstock for aquaculture, mainly in China and Japan. Resulting eel products are largely consumed nationally but are also re-exported to a range of other markets, including south-east Asia, Europe and the USA.

Eel stocks of some species, have decreased in several range States, including Japan, European countries, Canada and the USA. Eels are affected by a wide range of factors including habitat loss and fragmentation, pollution, and barriers to migration (notably dams and weirs) as well as harvest and it is difficult to separate the effects of these when assessing changes to populations. *A. anguilla* recruitment levels have largely stabilised, albeit at very low levels compared with historic baselines.

In some jurisdictions, notably Canada and the USA for *A. rostrata* and China, Japan and Republic of Korea for *A. japonica*, harvest is regulated through a variety of mechanisms, including quotas, closed seasons, licenses and Total Allowable Catches (TACs). Stocks have decreased or appear to have decreased in some or possibly all of these range States. However, the distinctive nature of eel biology makes it difficult to ascribe a single cause to such decreases. As such, **it is not possible to say with certainty whether either *A. rostrata* or *A. japonica* meets the criteria for inclusion in Appendix II set out in Annex 2a** of Res. Conf. 9.24 (Rev CoP17). The complexity of eel fisheries and management, whereby wild seedstock is harvested to replenish reduced stocks, compounds this difficulty.

The most heavily traded species of Anguillid are *A. japonica* and *A. rostrata* and the already Appendix-II listed *A. anguilla* and products of other eel species are also known to enter trade. Processed eel products and glass eels or elvers, some of the main products in trade, may closely resemble each other. These would be difficult for non-specialists to distinguish by species, particularly in consignments of mixed species and almost impossible without expensive and impractical genetic testing in products destined for the retail market. Therefore, **all currently unlisted *Anguilla* species meet criteria for inclusion in Appendix II in Annex 2b** of Res. Conf. 9.24 (Rev. CoP17) as lookalikes of *A. anguilla*.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES background

The European Eel *Anguilla anguilla* was included in CITES Appendix II at CoP14 in 2007, its listing entering into effect 13 March 2009. A zero-export quota for *A. anguilla* by EU Member States was introduced in 2010 as a positive Non-Detriment Finding (NDF) could not be made by the EU Scientific Review Group (European Commission, 2010), alongside an import ban by the European Union.

Taxonomy

The genus *Anguilla* comprises 17 species, of which this proposal includes 16 species:

Anguilla australis Richardson 1841 – Australian shortfinned eel
Anguilla bengalensis Gray 1831 – Bengal eel, Indian mottled eel
Anguilla bicolor McClelland 1844 – Shortfin eel, Indian bicolor eel
Anguilla borneensis Popta 1924 – Borneo eel, Indonesian longfinned eel
Anguilla celebesensis Kaup 1857 – Celebes longfin eel
Anguilla dieffenbachii Gray 1842 – New Zealand longfin eel
Anguilla interioris Whitley 1938 – New Guinea eel, Highlands long-finned eel
Anguilla japonica Temminck & Schlegel 1846 – Japanese eel, Freshwater eel
Anguilla luzonensis Watanabe, Aoyama & Tsukamoto 2009 – Luzon eel, Philippine mottled eel
Anguilla marmorata Quoy & Gaimard 1824 – Giant Mottled eel, Giant long-finned eel
Anguilla megastoma Kaup 1857 – Polynesian longfin eel, Pacific long-finned eel
Anguilla mossambica (Peters 1852) – Mozambique eel, African longfin eel
Anguilla nebulosa McClelland 1844 – African Bengal eel, Indian mottled eel
Anguilla obscura Günther 1872 – Polynesian shortfin eel, Brown eel
Anguilla reinhardtii Steindachner 1867 – Australian longfin eel, Congor eel
Anguilla rostrata Lesueur 1817 – American eel, Common eel

The Proposal utilises CITES Standard Reference for fishes: Taxonomic Checklist of Fish species listed in the CITES Appendices and the Annexes of EC Regulation 338/97 (Elasmobranchii, Actinopteri, Coelacanthi, and Dipneusti, except the genus *Hippocampus*). Information extracted from Eschmeyer, W.N. & Fricke, R. (Eds.): Catalog of Fishes, an online reference, version update from 8 July 2025.

Range

Anguilla spp. are globally distributed, except for the South Atlantic and eastern Pacific Oceans, and generally categorised as either temperate (*A. australis*, *A. dieffenbachii*, *A. japonica*, *A. rostrata*) or tropical (all other unregulated species in the genus).

Table 1. Scientific, common names and range States and dependent territories of the two lead and 14 other *Anguilla* species.

Species	Common name	Range States and dependent territories
Lead species		
<i>Anguilla japonica</i>	Japanese eel, Freshwater eel	China, Japan, Philippines, Republic of Korea, Thailand, Viet Nam, Guam (USA), Hong Kong SAR of China, Taiwan, Province of China
<i>Anguilla rostrata</i>	American eel, Common eel	Antigua and Barbuda, Bahamas, Barbados, Belize, Brazil, Canada, Colombia, Costa Rica, Cuba, Dominica, Dominican Republic, Grenada, Guatemala, Haiti, Honduras, Italy (introduced), Jamaica, Mexico, Nicaragua, Panama, Saint Kitts and Nevis, Saint Lucia, Saint Vincent and the Grenadines, Trinidad and Tobago, USA, Venezuela (Bolivarian Republic of), <u>France</u> : Guadeloupe, Martinique, Saint Barthélemy, Saint Martin, Saint Pierre and Miquelon <u>Denmark</u> : Greenland <u>Netherlands</u> : Aruba, Bonaire, Saint Eustatius and Saba, Curaçao, Sint Maarten <u>United Kingdom</u> : Anguilla, Bermuda, British Virgin Islands, Cayman Islands, Montserrat <u>USA</u> : Puerto Rico, Virgin Islands of the United States
Other species		
<i>Anguilla australis</i>	Australian shortfinned eel	Australia, New Zealand, New Caledonia (France)
<i>Anguilla bengalensis</i>	Bengal eel, Indian mottled eel	Bangladesh, Eswatini, Ethiopia, India, Indonesia, Kenya, Madagascar, Malawi, Malaysia, Mozambique, Myanmar, Nepal, South Africa, Sri Lanka, United Republic of Tanzania, Yemen, Zimbabwe
<i>Anguilla bicolor</i>	Shortfin eel, Amalona	Australia, Bangladesh, Cambodia, China, India, Indonesia, Japan, Kenya, Madagascar, Malawi, Malaysia, Maldives, Mauritius, Mozambique, Myanmar, Oman, Papua New Guinea, Philippines, Republic of Korea, Seychelles, Solomon Islands, Somalia, South Africa, Sri Lanka, Thailand, United Republic of Tanzania, Viet Nam, Yemen, Commonwealth of the Northern Mariana Islands (USA), Mayotte (France), Réunion (France), Taiwan Province of China
<i>Anguilla borneensis</i>	Borneo eel, Indonesian longfinned eel	Indonesia, Malaysia, Philippines
<i>Anguilla celebesensis</i>	Celebes longfin eel	Indonesia, Malaysia, Philippines

Species	Common name	Range States and dependent territories
<i>Anguilla dieffenbachii</i>	New Zealand longfin eel	New Zealand
<i>Anguilla interioris</i>	New Guinea eel, Highlands long-finned eel	Indonesia, Papua New Guinea, Philippines
<i>Anguilla luzonensis</i>	Luzon eel, Philippine mottled eel	Philippines (Luzon Island, Mindanao Island), Taiwan Province of China
<i>Anguilla marmorata</i>	Giant Mottled eel, Giant long-finned eel	Australia, Cambodia, China, Comoros, Eswatini, Fiji, Hong Kong SAR of China, India, Indonesia, Japan, Kenya, Lao PDR, Madagascar, Malaysia, Mauritius, Mozambique, Myanmar, Palau, Papua New Guinea, Papua New Guinea, Philippines, Republic of Korea, Samoa, Solomon Islands, South Africa, Sri Lanka, Tanzania, Thailand, Tonga, USA, Vanuatu, Viet Nam, Zimbabwe, Commonwealth of the Northern Mariana Islands (USA), Guam (USA), Mayotte (France), French Polynesia (France), New Caledonia (France), Réunion (France), Taiwan Province of China, Wallis and Futuna Islands (France)
<i>Anguilla megastoma</i>	Polynesian longfin eel, Pacific long-finned eel	Fiji, Indonesia, Papua New Guinea, Samoa, Solomon Islands, Tonga, Vanuatu, French Polynesia (France); New Caledonia (France); Pitcairn Islands (UK)*; Wallis and Futuna Islands (France)
<i>Anguilla mossambica</i>	Mozambique eel, African longfin eel	Comoros, Eswatini, Kenya, Madagascar, Mauritius, Mozambique, Seychelles, South Africa, United Republic of Tanzania, Zimbabwe, Mayotte (France), Réunion (France)
<i>Anguilla nebulosa</i>	African Bengal eel, Indian mottled eel	Bangladesh, Indonesia, Myanmar Little information found on distribution; FishBase reports 'East Africa to Sumatra, Indonesia' (FishBase, 2024).
<i>Anguilla obscura</i>	Polynesian shortfin eel, Brown eel	Australia, Fiji, Indonesia, Papua New Guinea, Samoa, Solomon Islands, Tonga, Vanuatu, French Polynesia (France); New Caledonia (France); Wallis and Futuna Islands (France)
<i>Anguilla reinhardtii</i>	Australian longfin eel, Congor eel	Australia, Papua New Guinea, New Zealand, New Caledonia (France)

IUCN Global Category

Table 2. Genus *Anguilla*, IUCN Red List global category (all using IUCN Red List Categories and Criteria version 3.1) and population trend information for the two lead and 14 other species.

Species	IUCN Red List Global Category	Population Trend Information
Lead species		
<i>Anguilla japonica</i>	Endangered A2bd (assessed 2018)	Decreasing
<i>Anguilla rostrata</i>	Endangered A2bd (assessed 2020)	Stable
Other species		
<i>Anguilla australis</i>	Near Threatened (assessed 2018)	Unknown
<i>Anguilla bengalensis</i>	Near Threatened A2cd (assessed 2019)	Unknown
<i>Anguilla bicolor</i>	Near Threatened A2bcde (assessed 2019)	Unknown <i>Anguilla bicolor</i> was suspected to experience a population size reduction close to 30% over the next three generations (21 years), from 2019-2040 (Pike et al., 2020).
<i>Anguilla borneensis</i>	Vulnerable A2c+4c (assessed 2018)	Unknown
<i>Anguilla celebesensis</i>	Data Deficient (assessed 2018)	Unknown
<i>Anguilla dieffenbachii</i>	Endangered A2bd (assessed 2018)	Unknown
<i>Anguilla interioris</i>	Data Deficient (assessed 2018)	Unknown
<i>Anguilla luzonensis</i>	Vulnerable D2 (assessed 2018)	Unknown
<i>Anguilla marmorata</i>	Least Concern (assessed 2018)	Unknown
<i>Anguilla megastoma</i>	Data Deficient (assessed 2018)	Unknown
<i>Anguilla mossambica</i>	Near Threatened A2bcde (assessed 2018)	Unknown
<i>Anguilla nebulosa</i>	Considered a synonym of <i>A. bengalensis</i> which was Near Threatened (assessed 2019))	Unknown
<i>Anguilla obscura</i>	Data Deficient (assessed 2018)	Unknown
<i>Anguilla reinhardtii</i>	Least Concern (assessed 2018)	Unknown

Biological and trade criteria for inclusion in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)**A) Trade regulation needed to prevent future inclusion in Appendix I****Biology**

All species in this proposal belong to the *Anguilla* genus (17 species, with *Anguilla anguilla* already in Appendix II). The *Anguilla* species life cycle includes oceanic spawning, larval drift as leptocephali, transformation into glass eels and elvers, growth as yellow eels, and eventual migration to sea as silver eels to spawn and die. Spawning has never been directly observed, but all species are believed to breed in oceans, with the prevailing currents driving larval dispersal. Anguillids are semelparous, breeding only once in their lifetime at a relatively advanced age. Individual studies found that estimated age of female silver eels ranged from 11–29 years for *A. rostrata* and 4–22 years for *A. japonica*. The IUCN assessments assumed a generation length of 8.4 years for *A. japonica* and 12 years for *A. rostrata*. *Fecundity is high but variable across geographic locations. Females are estimated to produce millions of eggs (MacGregor et al., 2018), with A. rostrata showing relatively higher fecundity (millions of eggs per kg) compared to other species like the Australian short-finned eel A. australis and the New Zealand long-finned eel A. dieffenbachii (MacNamara and McCarthy, 2012). A. japonica was estimated as "medium to high productivity" based on a natural mortality rate (M) during continental life stages ranging from 0.31 to 0.78/year (Lin and Sun, 2013). However, this figure does not account for losses as eggs/larvae or migrating silver eels. Tagging studies (albeit with small sample sizes) suggest heavy predation on silver eels. In the Gulf of St. Lawrence, six of seven (86%) transmitting tagged A. rostrata eels showed predation signs (Béguet-Ponté et al., 2012). Whilst 58% predation was recorded for A. dieffenbachii silver eels in the South Pacific (Jellyman and Tsukamoto, 2010; Schabetsberger et al., 2013). For A. rostrata M values at age five ranged from 0.04-0.44 (females) and 0.05-0.39 (males), indicating "low or medium productivity. Temperate species spawn once seasonally, but some tropical species spawn year-round. Early larval stages are especially vulnerable to climate-driven changes in temperatures and ocean currents, impacting their recruitment to inland freshwater habitats. Aquaculture data also illustrate survival challenges. The Giant Mottled Eel A. marmorata in Viet Nam showed a 56% survival rate through glass eel and elver stages in the Van Xuan hatchery in Nha Trang hatchery between 2017 to early 2020, with lower rates experienced by other farms (Nguyen and Arai, 2021).*

Most anguillids, including the lead species, form single panmictic populations, meaning that global stocks interbreed regardless of geography. Consequently, localised mortality or mismanagement in one area can have global consequences for species status.

Population size

Global population sizes are unknown.

Population trends

Species of the Anguilla genus were assessed on the IUCN Red List from 2018 to 2020.

Anguilla japonica

The 2018 IUCN Red List assessment reported at least a 50% decrease over the past three generations (24 years) across *A. japonica*'s range. This assessment was based primarily on declining yellow and silver eel catch-per-unit-effort (CPUE) trends in Japan. Excluding re-stocked populations (*i.e. supplemented*), Kaifu and Yokouchi (2019) found significant CPUE decreases for yellow and silver Japanese Eels in four of six Japanese fisheries cooperatives in coastal and estuarine waters from 2003-2018. The same study found glass eel CPUE in nine prefectures to have significantly decreased between 1977-2018. Chen et al. (2014) estimates that, on average, 76.8% of *A. japonica*'s effective habitat in 16 major rivers across Japan, Republic of Korea, Taiwan Province of China, and mainland China, was lost between the 1970s–2010s. Despite fluctuating catches, the Fisheries Agency of Japan and the Japan Fisheries Research and Education Agency (FRA) reported that available data "indicate that the population has been experiencing a long-term decrease and now remains low". Tanaka (2014; 2025) concluded that stock abundances had increased since 1990 based on standardised CPUE data from twelve Japanese lakes inferred from observed data. However, the methodology behind this assessment has been questioned with one reanalysis finding decreasing trends (Kaifu in litt., 2025). A recent analysis of yellow and silver eel CPUE showed statistically significant decreases in seven of eight areas, with projected three generation decreases (24 years) of 79.2 %- 99.9% (Kaifu et al., 2025).

In other parts of its range, the species was categorised as 'possibly Extinct in the Wild' in Viet Nam's Red Data Book in 2007 and is anecdotally extremely rare in China's Yangtze and Pearl, though range-wide stock assessments are lacking.

Anguilla rostrata

The 2020 IUCN Red List assessment reported a stable population trend for *A. rostrata*, but when considering the population trend over the past three generations (36 years), trends varied by region, showing a mixture of declining, stable, and increasing trends. An average downward trend in recruitment and continental population size corresponded to an estimated 50% reduction in abundance over the past three generations. The Atlantic States Marine Fisheries Commission (2023) classified the Atlantic coast stock (from Maine to Florida) as

depleted and at or near historically low levels, with decreases in stock for multiple decades. In Canada, abundance indices suggest populations were stable or decreasing, with only two of twelve fishery-independent datasets indicating increasing abundance. Benchetrit and McCleave's 2016 study highlighted that while there is less research on the southern populations in the wider Caribbean region, they are facing similar major threats, including habitat loss and degradation, pollution, and commercial fishing pressure, particularly in the past decade. *Research has also shown that the widespread declines are most severe at the extremities of the species' range, which includes the Caribbean (Casselman and Cairns, 2009).* Localised increased abundance, such as in the St Lawrence Estuary, Canada, are likely due to the presence of stocked eels (deliberately introduced by humans), rather than recovery of the wild populations (Cornic and Cairns, 2021).

B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences

Anguillid eels are in demand for both domestic markets within producing countries and in products exported to international markets, targeting all continental life stages (glass eels, elvers, yellow and silver eels) for direct consumption and aquaculture stocking. *Commercial eel aquaculture began in Asia in 1931, initially using A. japonica, with A. anguilla increasingly used by the 1990s due to A. japonica glass eel shortages (Ringue et al., 2002). Anguillid resources remain among the most economically important inland fishery resources in Southeast Asia (SEAFDEC/IFRDMD, 2022), with 2.78 million tonnes of live eels produced via aquaculture from 2014- 2023 (FAO data).*

The primary species targeted for consumption and aquaculture are *A. anguilla*, *A. japonica* and *A. rostrata*, with *A. japonica* historically considered the most commercially important. However, decreases in the availability of *A. japonica* and *A. anguilla* have intensified demand for *A. rostrata*. *This shift is increasingly evident at the consumer level as well, with A. rostrata making up a substantial proportion of retail eel products. A 2023 survey of 327 anguillid products across 86 retailers in Singapore found A. rostrata accounted for 70% of identifications (Choo et al., 2024). This shift mirrors historic cycles of overexploitation in eel harvest and trade, where one species' decline drives demand for another to supply international demand.* Shifts in demand between the key anguillid species in trade are also reflected in price data: in Canada, *A. rostrata* elvers values rose from USD314/kg in 2009 to USD3,492/kg in 2023, and increased *A. rostrata* glass eel harvesting has been observed in Caribbean countries such as Haiti and the Dominican Republic.

Anguilla japonica

According to the FAO production database, 98% of anguillid aquaculture production between 2014–2023 was recorded as *A. japonica*. *However, this figure does not necessarily reflect actual production but rather reflects default reporting of A. japonica for any anguillid originally from East Asia due to the species' geographic origin (Gollock et al., 2018; Crook et al., 2014). This highlights mismatches in reported data and poor traceability for anguillid species.* National production statistics published by Japan's Fisheries Agency estimated that domestic glass eel catches peaked at over 200 tonnes/year in the late 1950s, declining steadily over the next decades; and have since fluctuated between ~5- 28 tonnes/year (1990–2024). Japan's eel imports (reflect live adult eels and processed eels converted into live weight equivalent) surged in the mid-1980s due to decreasing native stocks and farming of *A. anguilla* in China, peaking at 133,211 tonnes imported in 2000, and stabilising at ~30,000-40,000 tonnes/year over the past decade (Japan Fisheries Agency, 2024). *A. japonica glass eel input into aquaculture ponds reported by China, Japan, Republic of Korea, and Taiwan Province of China from 2011-2024 highlight this species still dominates in farms in Japan and the Republic of Korea, increasingly reliant on imports of A. japonica, especially for the Republic of Korea where imports often account for most of the seed stock.*

Although China, Japan, the Republic of Korea, and Taiwan Province of China agreed in 2014 to a voluntary upper limit of 78.8 tonnes for *A. japonica* eel seed input under the Joint Statement adopted at the Seventh Meeting of the Informal Consultation on International Cooperation for the Conservation and Management of Japanese Eel Stock and Other Relevant Eel Species, recent figures suggest this measure has not been effective. For the 2024–2025 fishing season, a Japanese aquaculture newspaper reported that total input by these four participants was approaching 125 tonnes (*Nihon Yōshoku Shimbun*, 2025) substantially exceeding the agreed threshold 78.8 tonnes. The fact that a non-binding target has been exceeded by such a margin demonstrates the limitations of voluntary arrangements and underscores the need for enforceable international trade regulation.

Anguilla rostrata

Since 2014, *A. rostrata* inputs to aquaculture ponds in mainland China have equalled or surpassed *A. japonica*, with 30 metric tonnes of *A. rostrata* recorded in 2023-2024 fishing season, compared to 9 tonnes of *A. japonica*. Customs data analysed in 2024 showed a marked increase in *A. rostrata* glass eel/elver imports into East Asia from 2 tonnes in 2004 to 157 tonnes in 2022, comprising 89% of total live eel fry imports that year. *This sudden surge was largely due to reported shipments from Haiti reported by Hong Kong SAR of China (Shiraishi and Kaifu, 2024).* While exports from Haiti dropped sharply to just 1 tonne in 2024, reported import volumes from *A. rostrata* range States to East Asia remain significantly higher than those observed prior to 2021 (96.9 tonnes in 2023, and to 73.2 tonnes in 2024).

A. rostrata glass eel commercial landings in Maine's (USA) fishery averaged at 4-4.4 tonnes annually from 2014–2022. Coastal yellow eel fishery catches decreased from 481 tonnes in 2014 to 120–150 tonnes in 2020–2022. As a result, the coastal wide quota was lowered to 235 tonnes. While in Canada, adult harvest decreased to ~300 tonnes annually, compared to 700 – 1000 tonnes in the 1970-1990s. Canadian elver harvests averaged 7095 kg in 2019–2022, worth ~USD24 million. Mexico, Cuba, and the Dominican Republic were the only countries in the Central and South American and Caribbean region to report *A. rostrata* landings to the FAO. According to national Customs data, the Dominican Republic exported approximately 18 tonnes of glass eels between 2015 and 2022, averaging 2 tonnes annually, with exports sometimes having exceeded the permitted quota (Diario Libre, 2020). The value of glass eel exports to the Dominican Republic has grown significantly, making up ~20% of all fishery exports in 2021 (Crook in litt., 2025; Diario Libre, 2022). Haiti reported a total of 36.83 tonnes of elver landings from 2013-2016, and Jamaica reported 363 kg of landings over the same period.

As pressure mounts on temperate species, traders have also turned to tropical anguillids – trade in these other species for aquaculture purposes is ongoing (Crook in litt., 2025). In particular, demand has risen for *A. bicolor*, *A. marmorata* and *A. mossambica*. UN Comtrade data between 2014-2023 shows considerable trade in *Anguilla* (live, frozen or products) from various range States of tropical species (Thailand, Myanmar, Philippines, Indonesia, Malaysia). China (526,498 tonnes, 57%), and Indonesia (83,685 tonnes, 9%) accounted for the most exports of eel products (across the four HS codes specific to *Anguilla* spp. in the UN Comtrade Database). The Republic of Korea also reported importing ~26.9 tonnes of *A. bicolor* and ~6.7 tonnes of *A. marmorata* eel inputs into aquaculture ponds between 2011-2024. The Philippines supplied ~30% of East Asia's live glass eel imports in 2012-2013 and is home to at least five *Anguilla* species: *A. bicolor*, *A. marmorata*, *A. japonica*, *A. celebesensis* and *A. luzonensis* (Crook et al., 2014).

According to the UN Comtrade Database, over 920,000 tonnes of eel products were exported between 2014–2023; prepared/preserved products increased over the ten years (60% of global exports in 2023), while live eel exports have halved from 41,599 tonnes in 2014 to 23,609 tonnes in 2023. Yet, species-specific reporting remains limited, and records are often inferred by geographic origin rather than the species. The limitations of this approach include the country/territory reported as the 'origin' of the trade may in fact represent a re-exporter, as eels are farmed, processed and traded across the world, several anguillid species can occur in one country and non-native *Anguilla* species are farmed around the world.

Illegal trade

There are numerous documented cases involving the proposed species that violate national legislation. In 2024, two Dominican nationals were charged with attempting to illegally export 30 kg (110,000 individual juvenile *A. rostrata*) from Puerto Rico to the Dominican Republic, in violation of the US Lacey Act and Endangered Species Act (Agencia EFE, 2024). In December 2024, ~60,000 live *A. rostrata* were seized in Sint Maarten, originating from the Dominican Republic, with fraudulent documents. Moreover, there is emerging evidence of significant eel trafficking operating from Haiti, with a recent UN Security Council Briefing reporting that eel trade was being used to launder money from drug operations. Canada has also reported high levels of unauthorised elver harvesting of *A. rostrata*, which had become increasingly difficult to control, prompting a two-year fishery closure. For *A. japonica*, seizures have involved large volumes of smuggled fry, including 1.5 tonnes confiscated at Quanzhou port and 1.8 tonnes at Shenzhen port in mainland China, while in Hong Kong SAR of China, authorities intercepted 100 kg of smuggled fry illegally exported from Taiwan Province of China and an additional 70 kg transiting from Japan to the Republic of Korea. Other anguillid species have also featured in illegal trade: in 2022, 3.9 tonnes of live *A. bicolor* elvers were seized at Chennai Airport, India; in 2023, Indonesia intercepted 200 kg of *A. marmorata* elvers; in New Zealand, both *A. australis* and *A. dieffenbachii* were subject to illegal harvest and trafficking cases; and in Mozambique, seizures of 150 kg of live *A. mossambica* elvers intended for illicit export were reported. Collectively, these cases demonstrate that illegal, unreported, and unregulated exploitation of anguillid eels is widespread, transboundary, and increasingly linked to organised criminal networks.

WiTIS recorded a total of 133 seizure incidents involving anguillid eels between 2009 and 2024, *A. anguilla* (121), *A. rostrata* (five seizure incidents in the USA – with a total known mass of 332 kg for three of the incidents), two incidents in Canada involving 26 kg and one incident involving 100,000 live eels in Puerto Rico), *A. japonica* (320,000 live eel in Taiwan Province of China and 60 kg of eel fry in Japan) and *A. marmorata* (two [volumes not specified], both seized in China).

Inclusion in Appendix II to improve control of other listed species

A) Specimens in trade resemble those of species listed in Appendix II under Res. Conf. 9.24 (Rev. CoP17) Annex 2 a or listed in Appendix I

There is significant overlap in morphological characteristics among *Anguilla* species, especially in juvenile and processed forms. In 2024, Watanabe and collaborators grouped *A. anguilla*, *A. japonica*, and *A. rostrata* with *A. borneensis*, *A. dieffenbachii*, and *A. mossambica* based on their lack of variegated markings and long dorsal fins. Species in this group are difficult to distinguish, facilitating misdeclaration. According to UN Comtrade data, over 60% of the global eel trade (by import volume) consisted of processed products in 2023. These products (e.g.

fillets, grilled eel [unagi]) are visually indistinguishable by species (Schroeder et al., 2022). Molecular authentication studies indicate that substantial proportions - often around half or more - of eel samples were mislabelled, containing species other than that stated on the label (Nijman and Stein, 2022). Additionally, the glass eel fishing season for *A. anguilla*, *A. rostrata*, and *A. japonica* partially overlap, so trade in glass eels of these species can occur during the same months, increasing the risk of substitution or mislabelling (Shiraishi in litt., 2025). The overlapping geographical range of multiple eel species, particularly tropical ones, also increases the likelihood of inadvertent mixing or substitution during harvest, trade, and processing.

There is evidence of *A. anguilla* being frequently mis-declared as non-CITES listed species such as *A. rostrata*, *A. japonica* and *A. bicolor* to circumvent trade restrictions. This practice of misdeclaration is widely documented across the EU by enforcement authorities (Crook, 2010), with enforcement efforts to ensure the legality of shipments hampered by the use of mixed *Anguilla* species stocking in Chinese eel farms. There is also a lack of clarity over precise methods used in farming operations in the region (Shiraishi and Crook, 2015). This is further exacerbated by the lack of traceability along the eel supply chain. Several seizures illustrate the scale of mislabelling: in 2009, two 25 tonne shipments of frozen eels from mainland China to Poland declared as *A. japonica* were DNA tested and found to contain mixtures of *A. anguilla*, *A. rostrata*, and *A. japonica* (Crook, 2010). Outside the EU, mislabelling/declaration is also widespread: between 2017 and 2018, a supermarket survey in Hong Kong SAR of China found 45% of eel products tested (22/49) were genetically identified as *A. anguilla*, often labelled generically as “eel” or “product of China” (Richards et al., 2020). Illegal fishing and trade also affect other range States: In the Dominican Republic, illegal fishing and trade, including attempts to launder eels caught illegally in Puerto Rico and smuggled across the border with Haiti, have been reported (Crook in litt., 2025, US Department of Justice, 2024).

The illegal trafficking of European glass eels (*A. anguilla*) continues to thrive, generating substantial illicit revenue and involving millions of individual eels (Europol, 2023). This illegal trade often uses transit hubs (e.g., Hong Kong SAR of China, Malaysia) and complicates accurate trade monitoring and enforcement, as it obscures the true volume and origin of traded eels, including potentially tropical species.

Additional seizures and trafficking patterns provide further evidence of ongoing illegal trade:

- **Hong Kong SAR of China:** In March 2024, the Agriculture, Fisheries and Conservation Department (AFCD) seized live European eel fry in a shipment that contained both *A. mossambica* and *A. anguilla* mixed together (Hong Kong Government Information Services, 2024).
- **Senegal:** *A. anguilla* glass eels exported from France to Dakar were subsequently re-exported to East Asia falsely labelled as *A. mossambica* (Robin des Bois, 2025).
- **Netherlands to Malaysia:** In early 2023, 170,000 *A. anguilla* glass eels (54 kg) were found smuggled in suitcases at Schiphol Airport, potentially re-exported via Southeast Asia to Hong Kong SAR of China. It is unclear whether these glass eels were deliberately misidentified as non-CITES species (NL Times, 2023).
- **North America:** In the USA, a major seafood dealer and eight individuals were indicted for international wildlife trafficking involving approximately 138 shipping containers containing all or mostly *A. anguilla* but were falsely declared as *A. rostrata* (U.S. Department of Justice, 2022). In Canada, over USD3 million worth of eel meat (90 tonnes declared as *A. rostrata* but mixed with *A. anguilla*) was seized in Ontario in 2023, highlighting the scale of illegal processed eel trade (North Shore Daily Post, 2023).
- **Puerto Rico:** In 2023, a significant illegal wildlife smuggling operation was shut down involving trafficking of eels linked to regional networks, including those operating in the Dominican Republic, highlighting ongoing illicit trade routes in the Caribbean region (Dominican Today, 2023).
- According to the EU-TWIX database¹⁷, a total of 144 seizure records involving significant quantities of *A. anguilla* were reported between 2014 and 2023. Noting that misdeclaration of *A. anguilla* as non-CITES listed eel species is an issue, in 2020 one seizure of 110 kg of *A. anguilla* (valued at EUR660 000) at Lisbon Airport was intercepted on export to Malaysia involving four Chinese nationals and reported the concealment method as ‘mixed species’.

Additional Information

Threats

Anguillids face multiple, interacting threats due to their complex catadromous life history. These include changes in oceanic currents and other climate-driven factors affecting larval transport and recruitment; barriers to migration (e.g. dams) making upstream areas inaccessible and impeding the downstream migration of silver eels; loss and degradation of freshwater and estuarine habitats; disease and parasites; contamination and other

¹⁷ Note: Data for use in the Analyses were downloaded with prior authorisation from EU Member States (EU MS). However, Spanish data was not authorised for use due to the Spanish authority’s interpretation of Article 47 of Regulation (EU) No 952/2013 of the European Parliament and of the Council of 9 October 2013 laying down the Union Customs Code - which only provides for the exchange of information between the competent authorities and the Commission. Given the prominence of Spain as a reporter of European eel seizures, the TWIX data presented here should be interpreted as illustrative and not exhaustive of eel volumes reported by EU MS.

forms of pollution impacting gonad development, reproductive success, and the development of embryos and larvae; and fisheries and trade. *It is not possible to easily separate the effects of these drivers, and cumulative mortality across life stages needs to be accounted for when assessing sustainability.*

Conservation, management and legislation

Various regional and national management measures exist to regulate eel fisheries. *Coordination across jurisdictions remains limited for most species, despite this being identified as fundamental to effectively managing the species (CITES Notification No. 2023/062 – seeking information on the status, management and trade in eels *Anguilla* spp.). Most measures focus primarily on fisheries, compared to other factors that affect populations and catch like pollution, migration barriers and habitat degradation.*

A. japonica

In East Asia, *A. japonica* management efforts focused on limiting wild-sourced eel seed for aquaculture. China, Japan, Republic of Korea, and Taiwan Province of China maintain limits - *at no more than 80% of the 2013-2014 input (which equates to 78.8 tonnes) (Joint Press Release, 2025).* Japan codified this limit into domestic quotas, allocated to individual aquaculture facilities. *Although there are reports of potential underreporting, national-level compliance is at least nominally enforced. For other participating countries, it remains unclear whether they have similarly translated these voluntary thresholds into enforceable national law (Shiraishi in litt., 2025).*

Some countries, such as Japan and Republic of Korea, have introduced seasonal closures, size limits and restocking programmes to support wild populations, *though evidence for the benefits of restocking for the species as a whole is lacking (ICES, 2023; 2024b).* Japan prohibits silver eel catch in most prefectures, and China banned commercial fishing in the Yangtze River and adjoining lakes for ten years starting in 2020 to address fish population declines.

A. rostrata

Canada has implemented quotas/seasonal closures for *A. rostrata* fisheries including a TAC of 9,960 kg/year for elvers since 2005. The USA also has a coastwide commercial quota for *A. rostrata* yellow eel of 235 tonnes for the 2025-2027 fishing period. An annual glass eel quota of 4.4 tonnes per year for the Maine fishery has been in place since 2015.

*The Dominican Republic, Canada and the USA discussed a tri-partite agreement to tackle illegal eel trade (Dominican Today, 2023). However, the current status of this regulatory framework is unknown. The Dominican Republic also proposed listing *A. rostrata* in CITES Appendix III in 2022, in recognition of declines (Choo et al., 2024). The Dominican Republic reduced its glass eel export quota from 2,500 kg to 1,500 kg in 2021 as a precautionary measure, with some leeway permitted for the sale of a small additional quantities derived from seizures (Diario Libre, 2023).*

Other *Anguilla* species

In Southeast Asia, some countries (the Philippines and Indonesia) have size-based export restrictions and ASEAN Member States established a regional initiative for tropical anguillids (CITES, 2018). Regional coordination for tropical species is limited, with no binding fisheries management agreements or stock recovery frameworks. *Recent shifts in demand between eel species, particularly toward *A. rostrata* and *A. bicolor*, suggest that management gaps are leading to serial depletion and redirection of exploitation (Choo et al., 2024).*

Artificial Propagation/captive breeding

Eel farms depend entirely on wild juveniles (glass eels, elvers), as commercial captive breeding is not yet viable, therefore fishing pressure for glass eels remains strong. *Eels only mature and breed after encountering certain environmental conditions - such as temperature change, fasting, exercise, and sensory signals - during migration (Churcher et al., 2014; Sudo and Yada, 2022). Captive environments cannot adequately replicate these cues, which prevents natural gonadal development and breeding (Tsukamoto and Miller, 2017). The leptocephali, feed on marine snow - biological debris originating at the top of the ocean, such as the faeces or decaying carcasses of other fish - which is difficult to replicate in aquaculture. Therefore, leptocephali larvae often die in captivity (Tsukamoto and Miller, 2021). Studies to assess the culture potential of *A. bicolor pacifica* and *A. marmorata* consistently demonstrate the biological or technical hurdles that prevent a shift from reliance on wild-caught juveniles for global production (Aya, 2023). Japan aims to commercialise captive-bred *A. japonica* by 2028 (Japanese FRA, n.d).*

Implementation challenges (including similar species)

*Glass eels are small and difficult to detect, therefore prone to smuggling. Some of the issues that occurred for *A. anguilla* upon listing could arise again. In particular, the making of an NDF could be challenging (and this is what resulted in the zero-export quota from the EU), as well as ensuring traceability along the whole supply chain (given the lack of marking techniques, or accurate record keeping by fisheries organisations/authorities), and the issue of overlapping jurisdictions (Musing et al., 2018). Data gaps also remain for *A. rostrata* fisheries and trade in the Caribbean, and the situation in Southeast Asia is similarly unclear (SEAFDEC/IFRDMD, 2022), where eight anguillid*

species/sub-species occur. The presence of multiple *Anguilla* species in Southeast Asia makes management and enforcement more complex. Including all *Anguilla* species in Appendix II would require capacity building efforts especially in Southeast Asia.

Potential risk(s) of a listing

A genus listing of *Anguilla* could reduce the incentive for enforcement officers to conduct DNA testing, as shipments of live glass eels might only be reported to genus level during controls, resulting in less species-specific data on *A. anguilla*. Customs would likely focus on verifying CITES documentation rather than species identification. DNA analysis would still likely be required for processed products, which remain morphologically difficult to distinguish from non-anguilla eel species products e.g. filets. Species identification to detect *A. anguilla* may still be necessary, particularly in the EU where a zero-export quota and import ban for *A. anguilla* apply, or if seized eels are reintroduced to watercourses. Domestic permitting systems will also need to ensure that legal trade can continue without causing mortality of live glass eels due to delays, as reported by the Atlantic States Marine Fisheries Commission (ASMFC).

Potential benefit(s) of listing for trade regulation

A genus listing could help close loopholes in *A. anguilla* trade by simplifying enforcement. Shipments of *Anguilla* without CITES permits could be intercepted more efficiently, and reporting by Parties in CITES Annual Legal and Illegal Trade Reports would improve global trade monitoring, data availability, and traceability through permitting. Even with a genus-level listing, re-exports could still pose challenges. This is because shipments may contain pre-Convention or mixed-species eels, making it difficult to verify the presence of *A. anguilla* and fully ensure traceability and legality without additional checks.

Inclusion of *Anguilla* spp., and in particular *A. japonica* and *A. rostrata*, in Appendix II would reinforce domestic management efforts and establish a mechanism to ensure that harvest for export was sustainable and not reducing the wild population, in combination with other pressures, to a level at which its survival might be threatened.

References

- Agencia EFE (2024) Acusan en Puerto Rico a dos dominicanos por contrabando de 110.000 anguilas americanas. Listín Diario Website. Available at: https://listindiario.com/las-mundiales/ee-uu/20240301/acusan-puerto-rico-dos-dominicanos-contrabando-110-000-anguilas-americanas_797967.html
- Arai, T. (2015) Save our eels: Protection or extinction? *Fish for the People* 13(3):12-18
<http://hdl.handle.net/20.500.12066/973>
- Aya, F.A. (2023) Promoting tropical eel culture in the Philippines: Comparative performance of *Anguilla bicolor pacifica* and *A. marmorata* in captivity. *Fish for the People* 20(3):35–38.
<http://hdl.handle.net/20.500.12066/7176>
- Arai, K., Itakura, H., Yoneta, A., Kaifu, K., Shirai, K., Miyake, Y. and Kimura, S. (2019) Anthropogenic impacts on the distribution of wild and cultured Japanese eels in the Tone River watershed, Japan, from otolith oxygen and carbon stable isotopic composition. *Environmental Biology of Fishes* 102(11):1405–1420.
- Béguer-Ponté, S., Castonguay, M. and Dodson, J.J. (2012) Predation of American eels during their oceanic migration. *ICES Journal of Marine Science* 69(3):533–543. <https://doi.org/10.1093/icesjms/fsr198>
- Benchetrit, J. and McCleave, J. (2016) Current and historical distribution of the American eel *Anguilla rostrata* in the countries and territories of the Wider Caribbean. *ICES Journal of Marine Science* 73(1):122–132.
<https://academic.oup.com/icesjms/article/73/1/122/2457886>
- Casselman, J.M. and Cairns, D.K. (editors) (2009) Eels at the edge: science, status, and conservation concerns. American Fisheries Society, Symposium 58, Bethesda, Maryland.
- Choo, J.S.Y., Rabbani, G., Lim, E.X.Y. and Wainwright, B.J. (2024) A shift in the trade? An investigation of the eel trade reveals a likely species switch. *Conservation Science and Practice* 7(4):e70013.
<https://doi.org/10.1111/csp2.70013>
- Churcher, A.M., Pujolar, J.M., Milan, M., Hubbard, P.C., Martins, R.S., Saraiva, J.L., Huertas, M., Bargelloni, L., Patarnello, T., Marino, I.A., Zane, L. and Canário, A.V. (2014) Changes in the gene expression profiles of the brains of male European eels (*Anguilla anguilla*) during sexual maturation. *BMC Genomics* 15(799).
<https://doi.org/10.1186/1471-2164-15-799>
- CITES (2018) Status and Resources Management of Tropical Anguillid Eels in Southeast Asia. AC30 Inf. 11. Available at: <https://cites.org/sites/default/files/eng/com/AC/30/Inf/E-AC30-Inf-11.pdf>
- Crook, V. (2025) *in litt.* to IUCN/SSC Anguillid Eel Specialist Group
- Crook, V. (2014) Slipping away: International *Anguilla* eel trade and the role of the Philippines. TRAFFIC and ZSL, UK.
- Dekker, W. (2004) Status of the European eel stock and its fishery. *Marine Ecology Progress Series* 269:219–234.

- Del Cid, M. (2023) Autoridades de RD luchan contra el contrabando de anguila desde Haití. Diario Libre Website. Available at: <https://www.diariolibre.com/planeta/medioambiente/2023/05/02/codopesca-lucha-contras-el-contrabando-de-anguila-desde-haiti/2303132>
- Dominican Today (2023) Dominican Republic talks with the US and Canada to combat illegal trafficking of rostrata eels. Dominican Today Website. Available at: <https://dominantoday.com/dr/world/2023/06/12/dominican-republic-talks-with-the-us-and-canada-to-combat-illegal-trafficking-of-rostrata-eels/>
- European Commission (2010) Summary of Conclusions from the 53rd Meeting of the Committee on Trade in Wild Fauna and Flora. Brussels, 6 December 2010. Available at: https://circabc.europa.eu/sd/a/4f201784-1bf1-443a-ae27-84aab31d2861/53_summary_com.pdf
- Fisheries Research Agency of Japan (n.d.). Japanese eel (*Anguilla japonica*) information. Available at: https://www.fra.go.jp/home/kenkyushokai/unagi/unag_info.html. Accessed on 1 September 2025.
- Hong Kong Government Information Services (2024) AFCD seized endangered European eel fry (with photo). Available at: <https://www.info.gov.hk/gia/general/202403/15/P2024031500633.htm>
- ICES (2024a) Joint EIFAAC/ICES/GFCM Working Group on Eels (WGEEL). ICES Scientific Reports. Report. <https://doi.org/10.17895/ices.pub.27233457.v4>
- ICES (2024b) European eel (*Anguilla anguilla*) throughout its natural range. ICES Advice on fishing opportunities and conservation. Ecoregions in the Northeast Atlantic. Available at: https://www.fishsec.org/app/uploads/2024/11/ele.2737.nea_.pdf
- ICES (2023) European eel (*Anguilla anguilla*) throughout its natural range. ICES Advice on fishing opportunities and conservation. Ecoregions in the Northeast Atlantic. Available at: <https://www.fishsec.org/app/uploads/2023/07/221103-ICES-Advice-on-fishing-opportunities-for-eel-2023.pdf>
- Itakura, H., Arai, K., Kaifu, K., Shirai, K., Yoneta, A., Miyake, Y., Secor, D.H. and Kimura, S. (2018) Distribution of wild and stocked Japanese eels in the tone river watershed revealed by otolith stable isotopic ratios. *Journal of Fish Biology* 93(5):805–813. <https://doi.org/10.1111/jfb.13782>
- Jellyman, D. and Tsukamoto, K. (2010) Vertical migrations may control maturation in migrating female New Zealand longfin eels (*Anguilla dieffenbachii*). *Marine Ecology Progress Series* 404:233–243. <https://doi.org/10.3354/meps08488>
- Joint Press Release (2025) Eighteenth Meeting of the Informal Consultation on International Cooperation for Conservation and Management of Japanese Eel Stock and Other Relevant Eel Species (Informal Consultation). Available at: https://www.maff.go.jp/j/pr/event/attach/pdf/kaigi_release_2506-12.pdf
- Kaifu, K. (2025) *in litt.* to IUCN/SSC Anguillid Eel Specialist Group.
- Kaifu, K., Itakura, H., Kotani, T., Shinoda, A., Han, Y., Yoshinaga, T. and Wakiya, R. (2025) A decade after being listed as Endangered: Japanese eel stock inferred from fishery-dependent and independent monitoring records. *Regional Studies in Marine Science* 104456. <https://doi.org/10.1016/j.rsma.2025.101847>
- Kaifu, K., Yokouchi, K., Higuchi, T., Itakura, H. and Shirai, K. (2018) Depletion of naturally recruited wild Japanese eels in Okayama, Japan, revealed by otolith stable isotope ratios and abundance indices. *Fisheries Science* 84(5):757–763.
- MacGregor, R.B. and Jessop, B.M. (2018) American Eel Fecundity and Ovary Maturation in Relation to Body Size and Geographic Distribution. *Marine and Coastal Fisheries* 10(2):169–183. <https://academic.oup.com/mcf/article/10/2/169/7827089>
- MacNamara, R. and McCarthy, K.T. (2012) Size-related variation in fecundity of European eel (*Anguilla anguilla*). *ICES Journal of Marine Science* 69(8):1333–1337. <https://doi.org/10.1093/icesjms/fss123>
- Miller, M.J. and Tsukamoto, K. (2017) The ecology of oceanic dispersal and survival of anguillid leptocephali. *Dispersal during Early Life History of Fish* 1(1):958–971. <https://doi.org/10.1139/cjfas-2016-0281@cjfas-ddel/issue01>
- Musing, L., Shiraishi, H., Crook, V., Gollock, M., Levy, E. and Kecse-Nagy, K. (2018) Implementation of the CITES Appendix II listing of European Eel *Anguilla anguilla*. ZSL and TRAFFIC. AC30 Doc. 18.1 (Annex 1). Available at: <https://cites.org/sites/default/files/eng/com/ac/30/E-AC30-18-01-A1.pdf>
- Nijman, V. and Stein, F.M. (2022) Meta-analyses of molecular seafood studies identify the global distribution of legal and illegal trade in CITES-regulated European eels. *Current Research in Food Science* 5:191–195. <https://doi.org/10.1016/j.crf.2022.01.009>
- Nihon, Y.S. (2025) 日本養殖新聞 [Japan Aquaculture Newspaper]. Tokyo: Nihon Yōshoku Shimbunsha.
- NL Times (2023) Some 170,000 endangered glass eels found in suitcases at Schiphol Airport. NL Times Website. Available at: <https://nltimes.nl/2023/03/01/170000-endangered-glass-eels-found-suitcases-schiphol-airport>. Accessed on 31 July 2025.

- North Shore Daily Post (2023) Eel Meat Worth More Than \$4.3 Million Seized in Ontario. North Shore Daily Post Website. Available at: https://www.northshoredailypost.com/eel-meat-worth-more-than-4-3-million-seized-in-ontario/?utm_source=chatgpt.com Accessed on 31 July 2025.
- Nguyen, T.T.D. and Arai, T. (2021) An overview of the Anguillid eel culture in Vietnam. *Journal of Aquaculture & Marine Biology* 10(2):64–67. <https://medcraveonline.com/JAMB/an-overview-of-the-anguillid-eel-culture-in-vietnam.html>
- Pike, C., Crook, V., Jacoby, D. and Gollock, M. (2020) *Anguilla bicolor* (amended version of 2019 assessment). *The IUCN Red List of Threatened Species* 2020: e.T166894A176494582. <https://dx.doi.org/10.2305/IUCN.UK.2020-3.RLTS.T166894A176494582.en>
- Richards, J.L., Sheng, V., Yi, C.W., Ying, C.L., Ting, N.S., Sadovy, Y. and Baker, D. (2020) Prevalence of critically endangered European eel (*Anguilla anguilla*) in Hong Kong supermarkets. *Science Advances* 6(10):eaay0317. Available at: <https://doi.org/10.1126/sciadv.aay0317>
- Ringuet, S., Muto, F. and Raymakers, C. (2002) Eels: their harvest and trade in Europe and Asia. *TRAFFIC Bulletin* 19:2–27.
- Robin des Bois (2025) Prison ferme pour des trafiquants de civelles. Available at: <https://robindesbois.org/en/prison-ferme-pour-des-trafiquants-de-civelles/>. Accessed on 31 July 2025.
- Schabetsberger, R., Økland, F., Aarestrup, K., Kalfatak, D., Sichrowsky, U., Tambets, M., Dall'Olmo, G., Kaiser, R. and Miller, P.I. (2013) Oceanic migration behaviour of tropical Pacific eels from Vanuatu. *Marine Ecology Progress Series* 475:177–190. <https://doi.org/10.3354/meps10254>
- Schroeder, B., Mariani, S., Rico, A. and McDevitt, A.D. (2022) Molecular identification of seafood mislabelling in eel products reveals the presence of threatened species. *Food Control* 132:108534. <https://doi.org/10.1016/j.foodcont.2021.108534>
- SEAFDEC/IFRDMD (2022) Overview and Perspective of Anguillid Eels Fishery in Southeast Asia. Southeast Asian Fisheries Development Center / Inland Fishery Resources Development & Management Department.
- Seah, Y.G. and Wainwright, B.J. (2025) An Assessment of the Fish Maw Trade in Singapore and Malaysia Reveals Threatened Species and Highlights the Need for a More Complete Assessment of the Conservation Status of the World's Fishes. *Conservation Letters* 18:e13115. <https://doi.org/10.1111/conl.13115>
- Shiraishi, H. (2025) *in litt.* to IUCN/SSC Anguillid Eel Specialist Group
- Shiraishi, H. and Crook, V. (2015) Eel market dynamics: an analysis of *Anguilla* production, trade and consumption in East Asia. Tokyo, Japan. TRAFFIC. Available at: <https://www.traffic.org/publications/reports/eel-market-dynamics-an-analysis-of-anguilla-production-trade-and-consumption-in-east-asia/>
- Shiraishi, H. and Kaifu, K. (2024) Early warning of an upsurge in international trade in the American Eel. *Marine Policy* 159:105938. <https://doi.org/10.1016/j.marpol.2023.105938>
- Sudo R, Yada T. (2022) Anguillid Eels as a Model Species for Understanding Endocrinological Influences on the Onset of Spawning Migration of Fishes. *Biology* 11(6):934. <https://doi.org/10.3390/biology11060934>
- Tanaka, E. (2014) Stock assessment of Japanese eels using Japanese abundance indices. *Fisheries Science* 80:1129–1144. <https://doi.org/10.1007/s12562-014-0807-x>
- Tanaka, E. (2025) Updated stock assessment of Japanese eels using Japanese abundance indices. *Fisheries Science* <https://doi.org/10.1007/s12562-025-01912-3>
- Tejero Puntos, S. (2020) Exportaciones de angulas se disparan en 2019 en medio de la fiebre por su pesca. Diario Libre Website. Available at: <https://www.diariolibre.com/actualidad/medioambiente/exportaciones-de-angulas-se-disparan-en-2019-en-medio-de-la-fiebre-por-su-pesca-CA16707261>
- Tsukamoto, K. and Miller, M.J. (2021) The mysterious feeding ecology of leptocephali: a unique strategy of consuming marine snow materials. *Fisheries Science* 87:11–29. <https://doi.org/10.1007/s12562-020-01477-3>
- U.S. Department of Justice (2024) Two men arrested and indicted for illegally exporting juvenile American eels from Puerto Rico. U.S. Attorney's Office, District of Puerto Rico. Available at: <https://www.justice.gov/usao-pr/pr/two-men-arrested-and-indicted-illegally-exporting-juvenile-american-eels-puerto-rico>
- U.S. Department of Justice (2022) Major Seafood Dealer and Eight Individuals Indicted for International Wildlife Trafficking. U.S. Department of Justice Website. Available at: https://www.justice.gov/archives/opa/pr/major-seafood-dealer-and-eight-individuals-indicted-international-wildlife-trafficking?utm_source=chatgpt.com Accessed on 31 July 2025.
- Wisky, P. (2022) Piden más control a la pesca de angula en el país. Diario Libre Website. Available at: <https://www.diariolibre.com/planeta/medioambiente/2022/10/11/piden-mas-control-a-la-pesca-de-angula-en-el-pais/2106636>

Yoneta, A., Itakura, H., Arai, K., Kaifu, K., Yoshinaga, T., Miyake, Y., Shirai, K. and Kimura, S. (2019) The natural distribution range of the Japanese eel in Japan based on otolith stable isotope analysis and a literature survey. *Nippon Suisan Gakkaishi (Journal of the Japanese Society of Fisheries Science)* 85(2):150–161.

Inclusion of Sea Cucumbers *Actinopyga echinites*, *A. lecanora*, *A. mauritiana*, *A. miliaris*, *A. palauensis* and *A. varians* in Appendix II

Proponent: European Union

Summary: Sea cucumbers (class Holothuroidea) are marine echinoderms with a worldwide distribution, comprising around 1,800 species. In dried form, known as trepang or bêche-de-mer, sea cucumbers are traded for food and use in traditional medicine, particularly in Asia. The genus *Actinopyga* contains 18 species, including several commercially exploited species of medium value.

Actinopyga echinites (Deep-water Redfish), *A. mauritiana* (Surf Brownfish), *A. miliaris* (Blackfish) and *A. varians* (Surf Redfish, previously considered part of *A. mauritiana*) are proposed for inclusion in Appendix II in accordance with Article II, Paragraph 2a of the Convention, and *A. lecanora* and *A. palauensis* in accordance with Paragraph 2b due to their similarity in appearance.

The six *Actinopyga* species are found across the Indo-Pacific region, with some range overlap and typically inhabit coral reefs, lagoons, and seagrass beds at depths of less than 30 m. Their sedentary nature, accessibility in shallow waters, assumed longevity (12 years or more), slow recovery rates, and density-dependent reproduction make them vulnerable to overexploitation.

No global population estimates exist, but localised declines have been widely documented. Examples include commercial extinction of *A. echinites* and *A. miliaris* in parts of Fiji; limited recovery of *Actinopyga* populations was noted in the Egyptian Red Sea despite fishery closures. Three of the lead species (*A. echinites*, *A. mauritiana* and *A. miliaris*) were assessed as Vulnerable on the IUCN Red List in 2010 (now in need of updating), with decreasing population trends, and estimated 30-40% global decreases over three generations based on estimates of depletion and over-exploitation across their ranges. *Actinopyga varians* was assessed as Data Deficient in 2012 after its recognition as a valid species separate from *A. mauritiana*.

Sea cucumber fisheries have developed globally, and many species have been subject to overfishing. Although aquaculture of sea cucumbers has expanded in some countries (e.g., China), for *Actinopyga* spp., captive breeding remains largely undeveloped. Current and future trade in *Actinopyga* is expected to continue relying on harvest from wild populations.

The majority of sea cucumber harvest is intended for export: a 2013 review found that out of 77 fisheries in major sea cucumber fishing regions, only four (in the Cook Islands, Samoa, Guam and Nauru) were solely for subsistence use. In Papua New Guinea, *A. mauritiana*, *A. miliaris* and *A. echinites* are among the top-targeted species in the bêche-de-mer fishery, and prices for dried *Actinopyga* species are among the highest paid to local fishers. In the Solomon Islands, *A. miliaris* and *A. echinites* are also key harvested species of economic importance, often as secondary targets after higher-value taxa.

Assessing trade volumes by individual species is challenging due to sea cucumbers often being traded without specific taxonomic identification and there are no reliable estimates of the volume of *Actinopyga* species in international trade. However, several sources indicate that *A. echinites*, *A. mauritiana*, and *A. miliaris* represent a notable proportion of global holothurian exports. According to LEMIS, between 2009 and 2024, the USA imported notable volumes of *A. echinites* (4,597 kg), *A. miliaris* (2,946 kg) and *A. mauritiana* (648 kg), mostly from Fiji, Taiwan Province of China and Indonesia. *A. varians* was not reported in LEMIS. From 2009 to 2020, the USA also imported 180 kg of *A. spinea* and 687 kg of unspecified *Actinopyga* spp. along with other species from the genus not found in the same area (i.e. *A. agassizii*). *Actinopyga spinea* occurs in the Western Central Pacific, overlapping with the ranges of some of the species proposed for inclusion in the Appendices. Identification guides note that *A. spinea* is frequently grouped with *A. miliaris* by fishers and processors and is probably misidentified as that species. Regarding *A. lecanora* and *A. palauensis*, a study of 38 retail shops in Hong Kong SAR of China in 2022 found that they were not present despite previous records, possibly reflecting reduced availability or mislabelling.

While illegal trade in sea cucumbers is recognised, there is little direct evidence specific to *Actinopyga*. There is a range of domestic management measures including moratoria, area closures, licensing, size limits, export restrictions and quotas, but enforcement remains inconsistent and large areas are not effectively regulated. Species-level identification in trade remains a major challenge, as morphological features overlap considerably, especially in processed forms, resulting in frequent mislabelling and mixed consignments.

Analysis: The six species of *Actinopyga* sea cucumber proposed for inclusion in Appendix II are shallow-water Indo-Pacific species that are widely harvested for international trade where they are known as trepang or bêche-de-mer. Their life-histories make them intrinsically vulnerable to overharvest and there is evidence of significant depletion of some of the species, including *A. echinites* and *A. miliaris*, in at least some parts of their range. While species-level trade volumes remain unquantified, shipment records indicate that *Actinopyga* species make a significant contribution to the overall bêche-de-mer trade. Distinguishing species in processed form is complicated by the fact they are often traded in mixed consignments. However, there is a lack of recent information on the global status of and trends in populations of the species proposed and little is known about the current overall impact of wild harvest on any of them. Where information is available, it generally indicates population decreases. Overall, there is **insufficient information at present to determine whether any of the species proposed meets the criteria in Annex 2a** of Res. Conf. 9.24 (Rev. CoP17). Therefore, it is also **unclear whether regulation of trade in *A. lecanora* or *A. palauensis* is required to effectively control the trade in other *Actinopyga* species and should be included in Appendix II** as lookalikes under the criteria in Annex 2b.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

The proposed Standard Reference is a supplement to the Taxonomic Checklist of selected Sea Cucumber taxa relevant to CITES following CoP19 extracted from the World Register of Marine Species (WoRMS) accessed 5 June 2025, with edits from the Nomenclature Specialist of the CITES Animals Committee.

The genus *Actinopyga* comprises 18 recognised species, of which this proposal includes six species:

- *Actinopyga echinites* Jaeger, 1833 – Deep-water Redfish
- *Actinopyga mauritiana* Quoy & Gaimard, 1834 – Surf Brownfish
- *Actinopyga miliaris* Quoy & Gaimard, 1834 – Hairy Blackfish
- *Actinopyga lecanora* Jaeger, 1833 – Stonefish
- *Actinopyga palauensis* Panning, 1944 – Panning's Blackfish
- *Actinopyga varians* Selenka, 1867 – Surf Redfish

Actinopyga varians was formerly considered as part of *A. mauritiana*. The *A. mauritiana* species complex, once considered a single taxon spanning the Indian and Pacific Oceans, has since been split into the Pacific form (*A. varians*) and the Indian Ocean form (*A. mauritiana*). Accordingly, references in the literature to *A. mauritiana* in the Pacific are assumed to refer also to *A. varians*.

Range

Actinopyga echinites

Widely distributed across the Western Central Pacific, Asia, Africa, and the Indian Ocean regions. It has been specifically documented in: American Samoa (USA); Australia; Brunei Darussalam; Cambodia; China; Colombia; Comoros; Egypt; Eritrea; Fiji; Guam (USA); India; Indonesia; Israel; Japan; Jordan; Kenya; Kiribati; Madagascar; Malaysia; Maldives; Marshall Islands; Mauritius; Mayotte (France); Federated States of Micronesia; Mozambique; Myanmar; Nauru; New Caledonia; Niue; Commonwealth of the Northern Mariana Islands (USA); Palau; Papua New Guinea; Philippines; Réunion; Samoa; Saudi Arabia; Seychelles; Singapore; Solomon Islands; Somalia; Sri Lanka; Sudan; United Republic of Tanzania; Thailand; Tokelau; Tonga; Tuvalu; Vanuatu; Viet Nam; Wallis and Futuna Islands (France); Yemen.



Figure 1. Occurrence of *A. echinites*, extracted from World Register of Marine Species. Available from <https://www.marinespecies.org> at VLIZ. Accessed 2025-08-21. doi:10.14284/170

Actinopyga mauritiana

Distributed across the islands of the western Indian Ocean; while formerly thought to also occur in the Pacific Ocean, this is now believed to represent *A. varians*. It is reported to occur in Comoros; Djibouti; Egypt; Eritrea; India; Kenya; Madagascar; Mayotte (France); Mozambique; Réunion (France); Seychelles; Somalia; Sri Lanka; Sudan; United Republic of Tanzania.

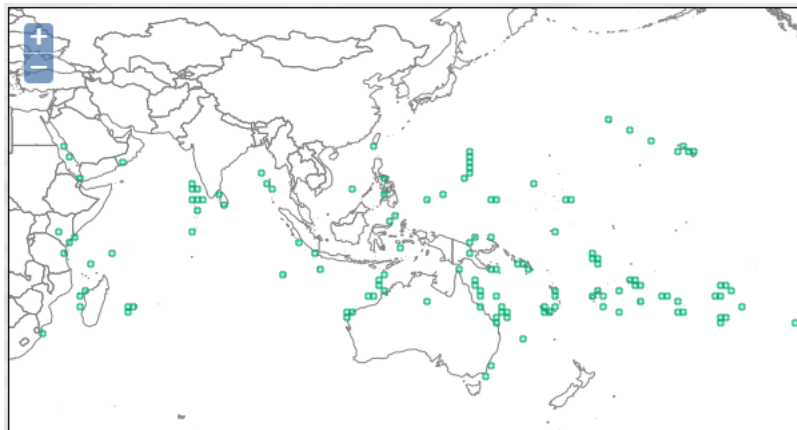


Figure 2. Occurrence of *A. mauritiana*, extracted from World Register of Marine Species. Available from <https://www.marinespecies.org> at VLIZ. Accessed 2025-08-21. doi:10.14284/170

Actinopyga varians

Found in the Pacific Ocean (formerly considered the Pacific range of *A. mauritiana* before its taxonomic revision): American Samoa (USA); Australia; Brunei Darussalam; Cambodia; China; Comoros; Cook Islands; Fiji; French Polynesia (France); Guam (USA); Indonesia; Japan; Kiribati; Malaysia; Federated States of Micronesia; Myanmar; Nauru; New Caledonia (France); Niue; Commonwealth of the Northern Mariana Islands (USA); Palau; Papua New Guinea; Philippines; Samoa; Singapore; Solomon Islands; Thailand; Tonga; USA (Hawaiian Islands); Vanuatu; Viet Nam; Wallis and Futuna Islands (France).



Figure 3. Occurrence of *A. varians*, extracted from World Register of Marine Species. Available from <https://www.marinespecies.org> at VLIZ. Accessed 2025-08-21. doi:10.14284/170

Actinopyga miliaris

Widely distributed throughout the Indo-Pacific and is reported to occur in: American Samoa (USA); Australia; Brunei Darussalam; Cambodia; China; Comoros; Cook Islands; Egypt; Eritrea; Fiji; French Polynesia (France); Guam (USA); India; Indonesia; Israel; Japan; Jordan; Kenya; Kiribati; Madagascar; Malaysia; Maldives; Marshall Islands; Mauritius; Mayotte (France); Federated States of Micronesia; Mozambique; Myanmar; Nauru; New Caledonia (France); Commonwealth of the Northern Mariana Islands (USA); Palau; Papua New Guinea; Philippines; Réunion (France); Saudi Arabia; Seychelles; Singapore; Solomon Islands; Somalia; Sri Lanka; Sudan; Taiwan Province of China; United Republic of Tanzania; Thailand; Tokelau; Tonga; Tuvalu; Vanuatu; Viet Nam; Wallis and Futuna Islands (France); Yemen.

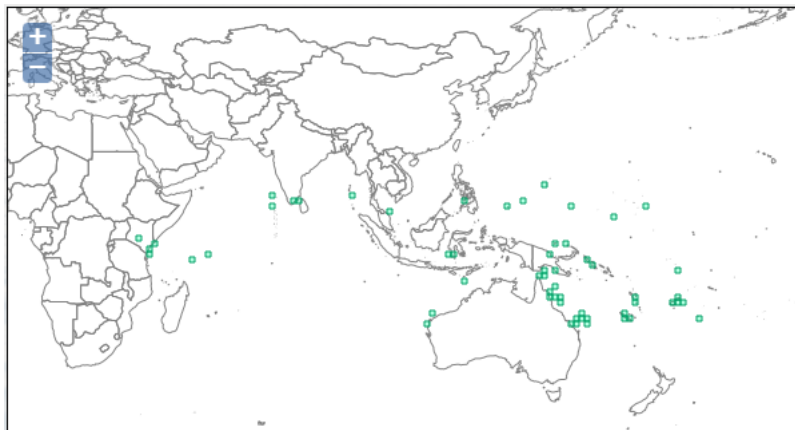


Figure 4. Occurrence of *A. miliaris*, extracted from World Register of Marine Species. Available from <https://www.marinespecies.org> at VLIZ. Accessed 2025-08-21. doi:10.14284/170

Actinopyga lecanora

Found in: American Samoa (USA); Cambodia; China; Djibouti; Fiji; Guam (USA); Indonesia; Japan; Kenya; Madagascar; Malaysia; Mauritius; Mozambique; Myanmar; New Caledonia (France); Commonwealth of the Northern Mariana Islands (USA); Papua New Guinea; Philippines; Réunion (France); Seychelles; Solomon Islands; Somalia; Taiwan; United Republic of Tanzania; Thailand; Tonga; Vanuatu; Viet Nam. Also distributed in SE Arabia, Sri Lanka, Bay of Bengal, East Indies, north Australia, Philippine, China, south Japan and South Pacific Islands.

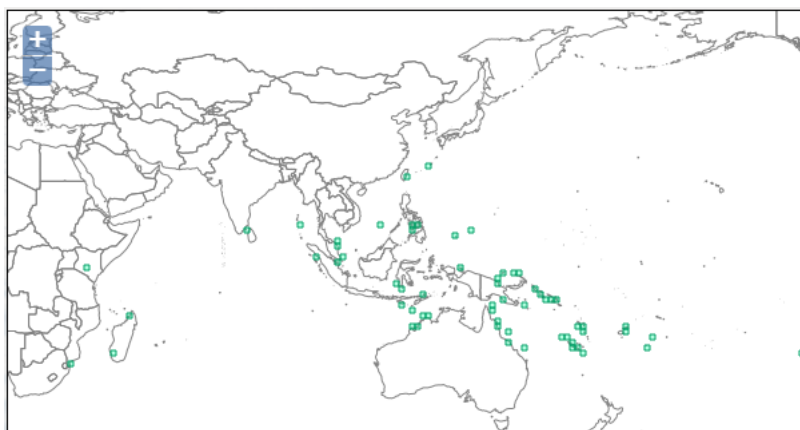


Figure 5. Occurrence of *A. lecanora*, extracted from World Register of Marine Species. Available from <https://www.marinespecies.org> at VLIZ. Accessed 2025-08-21. doi:10.14284/170

***Actinopyga palauensis*:** Australia; Fiji; Indonesia; Malaysia; New Caledonia (France); Niue (non-Party); Palau; Solomon Islands; Papua New Guinea; Philippines; Viet Nam; Thailand; Tonga.

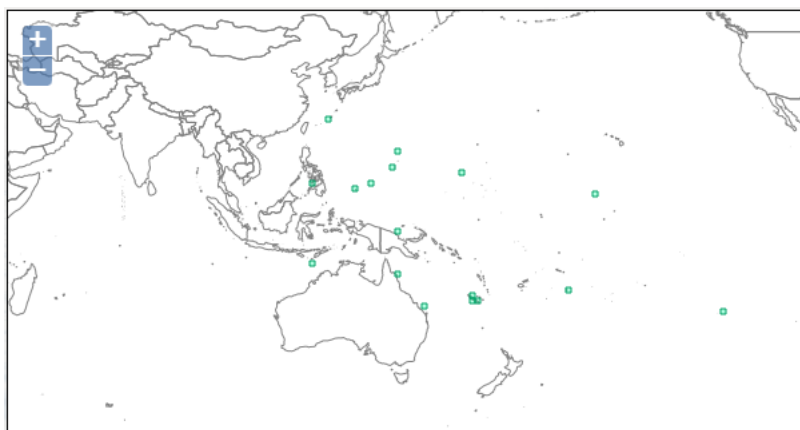


Figure 6. Occurrence of *A. palauensis*, extracted from World Register of Marine Species. Available from <https://www.marinespecies.org> at VLIZ. Accessed 2025-08-21. doi:10.14284/170

IUCN Global Category

Actinopyga echinites - Vulnerable A2bd (assessed 2010, ver. 3.1)

Actinopyga mauritiana - Vulnerable A2bd (assessed 2010, ver. 3.1)

Actinopyga miliaris - Vulnerable A2bd (assessed 2010, ver. 3.1)

Actinopyga varians – Data Deficient (assessed 2012, ver. 3.1)

Actinopyga lecanora – Data Deficient (assessed 2010, ver. 3.1)

Actinopyga palauensis – Least Concern (assessed 2010, ver. 3.1)

Biological and trade criteria for inclusion in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)

A) Trade regulation needed to prevent future inclusion in Appendix I

Actinopyga echinites, *A. mauritiana*, and *A. miliaris* were classified as Vulnerable on the IUCN Red List in 2010, all with decreasing population trends driven by overharvesting. Global population decreases were estimated at 30-40% based on estimates of depletion and over-exploitation across their ranges. Long-term, standardised surveys and comprehensive monitoring are largely absent across their ranges, so inferences of population trends often rely on site-specific surveys. Evidence of recovery is variable, despite fishery closures in several countries. A 2013 study reported 20% of global sea cucumber fisheries as depleted, 38% over-exploited, 14% fully exploited, and 27% underexploited including temperate deeper-water fisheries, moratorium-protected, or nascent fisheries.

Actinopyga echinites

According to the 2010 IUCN Red List assessments, *A. echinites* was classified as Vulnerable having decreased by more than 60-90% since the 1960s in at least 50% of its range. *A. echinites* was considered overexploited in at least 40% of its range, though exact declines are difficult to estimate (Conand et al., 2013). Site specific surveys have revealed local declines.

Across the 53 documented range States for *A. echinites*, population trend data is sparse. Only 12 countries (23%) have specific information available regarding population trends, while the remaining 41 countries (77%) lack such data, according to the data below only. The available data, where it exists, often points to significant declines, including cases of commercial extinction, overfishing, and reduced densities across surveyed habitats.

Indian Ocean

- **Mauritius and the outer island lagoons:** relative abundance fell from 3.45% (as a percentage of species composition in 1998) to 1.08% (2011), and 0.10% (2016).
- **Seychelles:** combined density estimates for *Actinopyga echinites*, *A. miliaris*, and *A. palauensis* (due to identification issues) based on a sea cucumber fishery stock survey showed declined from ~3.5 individuals/hectare, ind./ha (2004) to ~2.5 ind./ha (2021-2022).
- **Mozambique:** Southern fisheries (Maputo and Inhambane) experienced a “virtual collapse.” The fishery shifted to northern regions, where fishers report declining catches. Although not species-specific, *A. echinites* and *A. mauritiana* are among the most harvested species.
- **Madagascar:** No species-specific population trend data available, but declining export volumes may indicate population decreases. A density of 25 ind./ha was recorded for *A. echinites* in 2008 (CITES Secretariat, 2025).
- **Réunion:** The mean density of the species was 290 ± 94.1 ind./ha in 2014 (CITES Secretariat, 2025).
- **Egypt:** Several studies have recorded significant population declines of *Actinopyga* species in the Red Sea over recent decades due to overfishing, and sea cucumbers in general are considered overexploited in the region. *A. echinites* densities dropped from 2,450 to 240 ind./ha between 2000 and 2016.

Pacific Ocean

- **North Sulawesi, Indonesia (Bunaken National Marine Park):** only a single *A. echinites* was observed in 1994 over a 17-year survey period (1993–2010); species absent in subsequent surveys, suggesting it may have been extirpated.
- **South Sumatra, Indonesia (Lampung):** *A. echinites* and *A. varians* were found at “low densities” in 2016; fishermen report sea cucumbers were harder to collect, and dives need to be deeper than in previous decades.
- **New Caledonia (La Grande Terre):** Average density in reef flat habitats was 9.35 ind./ha across 50 reef sites in 2007.
- **New Caledonia (La Foa, South Province):** In 2021, densities of *A. echinites* averaged 16.14 ind./ha, ranging from 6.64 to 25.11 ind./ha.
- **Papua New Guinea:** 2015– 2016 national stock assessments indicate overfishing of *A. miliaris* and *A. echinites* in several locations, with limited recovery even during periods of fishery closure (Hair et al., 2016).
- **Fiji:** high densities of *A. echinites* in 2003 declined to commercial extinction at certain sites (Muaivuso) by 2009, due to intense fishing pressure.
- **Philippines:** While *A. echinites* is common in all the marine biogeographic regions (Cañada et al., 2020; De Guzman and Quiñones, 2021; Escobedo et al., 2025), populations are reportedly decreasing and further field assessments are needed to determine their status in some localities. Sea cucumber researchers have indicated that the species is undergoing 25% - 75% population reduction in some regions. Mean population densities vary widely: 0.75-4 mature ind./ha catch in the West and Northern Philippine Seas (Gamboa et al., 2025), 11 ind./ha in one Palawan Marine Protected Area (Jontila et al., 2018), 1-2 individuals per 300 m² in Anda, Pangasinan (Bautista et al., 2017), and 567–2,567 ind./ha on one island in Surigao del Sur (Sornito et al., 2022).

Actinopyga miliaris

According to the IUCN’s 2010 Red List assessment, *A. miliaris* was classified as Vulnerable having decreased by more than 60-90% since the 1960s in at least 50% of its range.

Across the 52 documented range States for *A. miliaris*, population trend data is available for 14 countries, representing 27% of the total. The remaining 38 countries, or 73%, lack specific data on population trends, according to the data below only. Where data is available, trends vary widely.

Site specific surveys have also revealed local declines.

Indian Ocean

- **Seychelles:** A stock survey combining *A. echinites*, *A. miliaris*, and *A. palauensis* found densities declined from approximately 3.5 ind./ha in 2004 to around 2.5 ind./ha in 2021–2022

- **Kenya:** In 2007, in southern Kenya, *A. miliaris* was estimated to have a density of 0.75 ± 0.5 ind./ha.
- **Mozambique:** Southern fisheries experienced a "virtual collapse", shifting effort northward where catches are reportedly declining. *A. miliaris* is among the most targeted commercial species in this fishery.
- **Egypt:** Densities dropped from 2,160 ind./ha (2000) to 370 ind./ha (2016). A 2022 study reported very low densities (80 ind./ha) in northern Red Sea seagrass beds, attributed to overfishing.
- **Eritrea:** A population density of 157.5 ind./ha of *A. miliaris* was estimated in near shore waters (Kalaeb et al., 2008).

Pacific Ocean

- **Fiji:** high densities recorded in 2003 declined to commercial extinction at certain sites (Muaivuso) by 2009, due to intense fishing pressure.
- **Tonga:** across three island groups, densities were "well below" healthy reference levels of 200 ind./ha. In Ha'apai, densities dropped from 16.37 ± 11.80 ind./ha (2014) to no individuals (2016), based on a 2016 survey of reef benthos transect stations.
- **New Caledonia** (La Grande Terre): In 2007, average density of *A. miliaris* in lagoon habitats was 3.78 ind./ha.
- **Palau:** A joint CPUE for *A. miliaris* and *A. echinites* in Palau showed 68.2 individuals per diver/hour.
- **Papua New Guinea:** Density estimates ranged from 0.1 ind./ha in Milne Bay Province to 57 ind./ha in Oro.
- **Vanuatu:** There was an estimated 785 ind./ha for this species in 1987, indicating a relatively high abundance (CITES Secretariat, 2025).

Actinopyga mauritiana

According to IUCN's 2010 Red List assessment, *A. mauritiana* was classified as Vulnerable having decreased by more than 60-90% since the 1960s in at least 60% of its range (e.g. South Pacific, Philippines, Indonesia, Papua New Guinea). However, *A. mauritiana* was considered overexploited in at least 25% of its range (e.g. India) (Conand et al., 2013). At the time of the assessment, *A. varians* was not yet separated from *A. mauritiana*.

Across the 15 documented range States for *A. mauritiana*, population trend data is available for five countries, representing 33% of the total, according to the data below only. The remaining 10 countries, or 67%, lack specific data on population trends. Where data is available, it indicates significant declines in population densities and signs of overexploitation.

- **Egyptian Red Sea:** densities dropped from 2,610 ind./ha (2000) to zero individuals at Abu Ghosoun (2016). At six coastal sites, densities fell from 612 ind./ha (2004) to 7.5 ind./ha (2014), just 1.2% of the original stock. A 2022 study reported very low densities (160 ind./ha) in northern Red Sea seagrass beds, attributed to overfishing.
- **Mozambique:** the intensification of the sea cucumber fishery resulted in a "virtual collapse" in southern fisheries. Fishers in northern Mozambique have also reported decreased catches, indicating a decline in populations. While no species-specific data are available, *A. echinites* and *A. mauritiana* are among the main commercial species harvested in this fishery.
- **Madagascar:** Although the country is a major exporter of sea cucumbers, no population trend data are available for this species. A decline in export volumes despite high demand may suggest declining populations.
- **Eritrea:** transect data used to calculate a population density of 35 ind./ha in near shore waters in 2008 (Kalaeb et al., 2008).

Actinopyga varians

According to the IUCN's 2012 Red List assessment, *A. varians* was classified as Data Deficient, due to limited information available on its population status, habitat, ecology, major threats or conservation measures (Samyn, 2013). The 2012 assessment predated *A. varians* formal recognition as a valid species separate from *A. mauritiana* by Michonneau et al. (2013) and *A. varians* added as a valid species in WoRMS in 2013. References to *A. mauritiana* in the Pacific are now assumed to pertain to *A. varians*, given that *A. mauritiana* has been shown to occur solely in the Indian Ocean region (Purcell et al., 2023).

Across the 32 documented range States for *A. varians*, population trend data is available for 7 countries, representing 22% of the total, according to the data below only. The remaining 25 countries, or 78%, lack specific data on population trends. Among the countries with data, trends vary.

- **Saipan (Commonwealth of the Northern Mariana Islands):** populations demonstrated recovery following a fishing moratorium, increasing from ~32,977 (1997) to 250,578 (2006), with size structures indicating a fully recovered population.

- **Indonesia** (South Sumatra, Lampung): *A. varians* was recorded at low densities in 2016. Fishers reported that sea cucumbers in general had become harder to collect, requiring deeper dives.
- **Tonga**: *A. varians* densities across three island groups (Tongatapu, Ha'apai, and Vava'u) were well below healthy reference densities of 200 ind./ha. Long-term population declines were documented, particularly in Ha'apai.
- **New Caledonia** (La Grande Terre): In 2007, average density of *A. varians* in crest habitats across 50 reef sites was 8.12 ind./ha.
- **Papua New Guinea**: *A. mauritiana* (*A. varians*) was found in moderate densities on deeper outer reef slopes but increasingly targeted as shallow stocks declined (Hair et al., 2016).

Regional survey data for *Actinopyga*:

Indonesia (2016-2018): across 23 localities (east to west Indonesia), moderate to low occurrences of *Actinopyga* were recorded: *A. echinites* (21), *A. lecanora* (6), *A. mauritiana* (6), and *A. miliaris* (5) compared to much higher counts of *Holothuria atra* (3326), *Holothuria hilla* (194), and *Holothuria scabra* (159) among others (Setyastuti et al., 2024). However, results cannot be extrapolated to describe the national population status, considering Indonesia is comprised of more than 17,000 islands. Export statistic data released by the Indonesian Ministry of Marine Affairs and Fisheries indicate that sea cucumber export volume from 2000 onwards tended to decrease (KKP, 2022).

Philippines: According to Junio-Menez in litt. (2025) *A. echinites*, *A. mauritiana* and *A. miliaris* were assessed as near threatened in the national red list assessment (2020) (Gamboa et al., 2025), noting decreasing populations, though species remain widespread in many locations across all six biogeographic regions. Species-specific density estimates are limited and based on one-time assessments. There is also evidence that there are still areas where no commercial harvesting takes place aside from those in effective marine protected areas (Junio-Menez in litt., 2025). *A. lecanora* densities were recorded at 155 ind./ha in Arrecife Island, Palawan, but none in other areas (Jontilla et al., 2018). The volume of traded dried *A. lecanora* in a warehouse was recorded to be around 900 kg in 2018; and only 71 kg and 30 kg in 2022 in Tawi-Tawi (Halun, pers comm. to Junio-Menez in litt., 2025).

Queensland, Australia: FAO expert panel report includes unpublished catch weight data recorded over a 30-year period for *A. mauritiana* and *A. miliaris* by the Queensland Sea Cucumber Association, noted it does not show declines in population trends (FAO, 2025).

B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences

Sea cucumbers are sedentary animals that are particularly vulnerable to overexploitation because they are easy to collect in their shallow area of occurrence, and do not require sophisticated fishing techniques. Specimens in shallow water can be collected by free diving, while deeper water individuals require SCUBA or hookah equipment to access. *Actinopyga* species are among the most conspicuous sea cucumbers in the tropics due to their relatively large size and epibenthic lifestyle - exposed all day in shallow water (Netchy et al., 2025). Despite a probable moderate-high productivity rate and a lifespan likely covering decades, populations have been slow to recover after depletion, as fertilisation success is density dependent, meaning low density populations struggle to replenish. Strong fishing pressure causes a decrease in species biomass density and populations are unable to replenish once they have fallen below critical mass (Toral-Granda, 2006). *A. mauritiana* is more difficult to collect as it hides in crevices and is found on the reef crest. Juveniles, being cryptic, are generally protected from exploitation (Purcell et al., 2023).

International trade of sea cucumbers often lacks proper taxonomic identification at a genus or species level, making it difficult to assess the impact on wild sea cucumber species (Toral-Granda in litt., 2022). Overfishing to meet the demands of the bêche-de-mer trade has caused depleted populations of *A. echinites*, *A. mauritiana* and *A. miliaris* across their respective ranges. Such declines in *Actinopyga* species - which are medium-value species, demonstrate the "boom and bust" cycles of sea cucumber harvest trade; depletion of more valuable species such as *Holothuria scabra* and *H. nobilis*, and continuing demand - has led to a shift in harvesting pressure to medium-value species including *Actinopyga* species. Aquaculture for *Actinopyga* species remains relatively unexplored. Small-scale aquaculture *Actinopyga* initiatives reported in Pacific Island countries, such as the Federated States of Micronesia and Palau, have reportedly been largely abandoned in recent years.

Several sources indicate that *A. echinites*, *A. mauritiana*, *A. miliaris* and *A. varians* represent a significant proportion of global holothurian exports. Indeed, the vast majority of sea cucumber harvest is for export, with only four of 77 fisheries in major sea cucumber fishing regions solely for subsistence use according to a 2013 review. In a global survey conducted by Purcell et al. (2012) for the FAO, *A. miliaris*, *A. echinites*, and *A. mauritiana* were identified by traders as routine components of shipments primarily processed into dried form and exported to Chinese markets. FAO global capture production of sea cucumbers peaked at 62,000 tonnes in 2018, dropping to ~45,000–50,000 tonnes in 2020-2021, suspected to be due to COVID-19 impacts. Self-reported harvest data are likely to be based on processed dried weights and therefore likely underestimate the true scale of global sea cucumber fisheries, as it would be better indicated using live wet weight. Although FAO 3-alpha species codes

exist for all six proposed *Actinopyga* species, no captures have been reported to FAO under these codes in the Aquatic Sciences and Fisheries Information System (ASFIS) List of Species for Fishery Statistic Purposes.

Average prices reported for *Actinopyga* species vary widely by market and year. For *A. echinites*, retail surveys in Guangzhou in 2016 recorded USD69/kg. *A. mauritiana* was slightly higher at USD72/kg in the same market, with no more recent benchmarks available. *A. lecanora* was reported at USD76/kg in Guangzhou in 2016 and much higher at USD166/kg in Hong Kong SAR of China, while Solomon Islands benchmark prices in 2025 set its value at USD120/kg for local buying and USD161/kg for export. *A. palauensis* sold for USD77/kg in Guangzhou and USD145/kg in Hong Kong SAR of China in 2016, with 2025 Solomon Islands benchmarks at USD78/kg locally and USD116/kg for export. For *A. varians*, only Solomon Islands values were available, at USD96/kg for local purchase and USD121/kg for export, while *A. miliaris* was priced at USD96/kg locally and USD137/kg for export. In addition, online advertisements in 2018 across Malaysia and Singapore suggested an average of USD70/kg for *Actinopyga* spp.

Actinopyga echinites

According to LEMIS, between 2009-2024, the USA imported a total of 4,597 kg of *A. echinites* meat, mostly from Taiwan Province of China (87%). Of this trade, 4,000 kg were reported in 2010 and the remainder in 2017. Additional imports included 120 kg of bodies from Fiji, with half in 2015, and the remainder between 2022-2023. Between 2019-2020, the USA imported a total of 8,266 kg bodies, mostly re-exported via Hong Kong SAR of China, with reported origins from Mozambique (53%) and Japan (46%).

According to Volza (2025), a commercial trade information platform aggregating global shipping manifests and customs records, dried exports of *A. echinites* (under the general HS code for dried sea cucumbers **0308**) included 221 kg reported as “dried sea cucumber” (*Actinopyga echinites* - HSN product 03081920 code) exported from Indonesia to Malaysia in August-September 2024. A separate 18 kg shipment of *A. echinites* also exported from Indonesia to Malaysia was recorded in April 2025 priced at USD300 (according to the bill of lading). Volza is not an official customs database, but aggregates and interprets trade data from multiple sources, providing insights into imports and exports by product and country.

- **Philippines:** *A. echinites* was considered to be among the most exploited sea cucumber species for export, in a value chain analysis of sea cucumbers in the Philippines in 2010.
- **Mozambique:** According to Conand et al. (2022), *A. echinites* was among the main commercial species harvested (the remainder being *A. mauritiana* and *Holothuria* species).
- **Australia:** *A. echinites* (and *A. mauritiana* i.e. *A. varians*) were the main species harvested, reported by the CITES Management Authority.
- **Solomon Islands:** a market survey indicated both *A. echinites* and *A. miliaris* were regularly among the top ten species in export volumes and value, out of ~20 commercially harvested species. Traders and middlemen operating in provincial areas reported consistent demand from Chinese buyers, particularly in Hong Kong SAR of China (Ramofafia et al., 2007).

Actinopyga mauritiana

According to LEMIS data spanning 2009-2024, the USA imported a total of 4,889 kg of *A. mauritiana* sea cucumber meat, all between 2022–2024. A further 470 kg of unspecified products were reported in 2009. However, most of these imports were reported from non-range States, namely Haiti, Jamaica and Nicaragua, therefore these imports may represent a different species. An exception was imports of 357 kg *A. mauritiana* bodies, primarily from Fiji (92%) and these were largely reported between 2022-2023.

According to the shipping manifests and customs data available in Volza, documented dried sea cucumber imports for October 2024 from Indonesia to Thailand included 189 kg of *A. mauritiana*. Reported values for these dried sea cucumbers ranged widely, from tens to several hundred USD per kilogram, reflecting both market demand and variability in product quality (Volza database, 2025).

- **Mozambique:** *A. mauritiana* was also among the main commercial species harvested (the remainder being *A. echinites* and *Holothuria* species), according to Conand et al. (2022).

Actinopyga miliaris

According to LEMIS, from 2009-2024, the USA imported a total of 2,946 kg of *A. miliaris* sea cucumber bodies, 99% from Fiji. Of this trade, 2,556 kg were reported between 2015-2017 and the remainder 2022-2023. A further 291 kg of meat was imported to the USA (69% from Indonesia), of which 241 kg were reported between 2022–2024 and the remainder in 2013.

According to the Volza database (2025), documented dried sea cucumber imports for October 2024 from Indonesia to Malaysia included approximately 786 kg of *A. miliaris*. Specific export data mentioning *A. miliaris* also include

four shipments labelled dried sea cucumbers (*Actinopyga miliaris* - HSN product code 03081920) totalling 756 kg exported from Indonesia to Malaysia in April 2025, total price (according to the bill of lading) of almost USD20,000.

- **Seychelles:** in 2017, the Seychelles fishing authority noted that *A. miliaris* was among the main species caught in its sea cucumber fishery, primarily export-driven.
- **Madagascar:** *A. miliaris* (along with *A. lecanora*) were among the most commonly collected species for export in the Masoala Peninsula, based on household interviews conducted between 2015 – 2020 by Rothamel et al. (2023). Madagascar is considered one of the key exporters of sea cucumbers in the Indian Ocean. It is possible that Madagascar's large volumes of sea cucumber exports are due to the development of aquaculture initiatives over the past few decades.
- **Kenya:** *A. miliaris* is actively fished, representing 17% of total catches (Purcell et al., 2023). The fishers sell the sea cucumbers to local middlemen who then process and sell them to exporters (Ochiewo et al., 2010).
- **China, Indonesia and the Philippines:** *A. miliaris* is a heavily fished species (Purcell et al., 2023).

Actinopyga varians

A. varians is commercially important in southern China, Japan, Malaysia, Indonesia and the Philippines (Purcell et al., 2023). *A. varians* was not reported in the LEMIS USA data, potentially as it is still recorded as *A. mauritiana*.

Trade was also reported in the species proposed for inclusion as lookalikes:

- ***A. palauensis*:** according to LEMIS data, 99 kg of meat and 35 kg bodies imported from **Australia** in 2022 and 2024.
- ***A. lecanora*:** According to the Volza database, documented dried sea cucumber exports in 2024 included 428 kg of *A. lecanora* from Indonesia to Thailand, and 94 kg from Indonesia to Hong Kong SAR of China. More recent shipments specifically mentioning *A. lecanora* in May 2025 include 637 kg (~USD9,000) exported from Indonesia to Hong Kong SAR of China, and labelled as dried sea cucumber (*Actinopyga licanora*) HSN product code 03081920 330 kg (~USD51,000) of dried boiled *A. lecanora* using HSN code 0308192000 was also exported from Philippines to Malaysia.

A study sampled 38 retail shops in Hong Kong SAR of China in 2022 and found that *A. lecanora* and *A. palauensis* were not present in any of these outlets despite previous records indicating their presence in the market (Purcell et al. 2025). This finding could indicate a decrease in trade availability, going undetected due to processing or mislabelling.

According to LEMIS, from 2009 to 2020, the USA also imported over 15,000 kg of *A. agassizii*, 180 kg of *A. spinea*, and 687 kg of unspecified *Actinopyga* spp. *A. agassizii* occurs in the tropical western Atlantic and Caribbean and therefore does not overlap in range with the six Indo-Pacific *Actinopyga* species considered in this proposal. In contrast, *A. spinea* is distributed in New Caledonia and Australia, overlapping with the ranges of the species proposed for inclusion in the Appendices. Identification guides note that *A. spinea* is frequently grouped with *A. miliaris* by fishers and processors and is probably misidentified as that species.

General

- **New Caledonia:** landing surveys in 2007 recorded *A. miliaris* as the second most observed species (13.7% of total individuals measured), after *Holothuria whitmae* (26%), with *A. palauensis* (7%), *A. varians* (formerly *A. mauritiana*- 5.5%), and *A. echinites* (4.4%) also among the main species (Purcell et al. 2009).
- **Papua New Guinea:** *A. mauritiana*, *A. miliaris*, and *A. echinites* consistently ranked among the top five in terms of catch volume and market value, alongside high-value species like *Holothuria scabra* and *Thelenota ananas*. Dried *Actinopyga* species (especially *A. miliaris* and *A. mauritiana*) commanded some of the highest prices paid to local fishers, making them priority targets despite regulatory measures (Hair et al., 2016).
- In contrast, sea cucumbers are generally not targeted by fisheries in countries such as Thailand, where they are considered bycatch, and no commercial fishery exists in Reunion.
- **Indonesia:** *Actinopyga* species are not among the primary targeted sea cucumbers in trade but have been more recently identified as commercially exploited, potentially due to their medium to low economic value (Setyastuti et al., 2019; Setyastuti et al., 2024).

Illegal trade

A 2025 study revealed that over 50 countries and territories experience illegal fishing and poaching of their sea cucumber resources, likely an underestimate (Mercier et al., 2025). Furthermore, between 2015 and 2021, at least 92 tonnes of sea cucumbers were seized globally from 204 incidents. Although no studies focus solely on the illegal trade of *Actinopyga* spp, illegal activity persists even in areas with export bans, such as Fiji, where sales continue despite a 2017 ban. Illegal fishing within Marine Protected Areas appears to be widespread, and the effectiveness of spatial and temporal sea cucumber fishery closures remains unclear.

Conand (2018) emphasised that illegal sea cucumber fishing is often driven by local actors and mobile foreign poachers using SCUBA, seabed hiding techniques, and nighttime raids, threatening both wild populations and mariculture, as seen in Madagascar, where security costs for enclosures have become prohibitively high as such illegal activity is so intense.

Despite a ban on sea cucumber harvesting in Egypt since 2004, densities of *A. echinites* and *A. miliaris* have significantly decreased between 2000 – 2016, and *A. mauritania* fell to 1.2% of its historic stock between 2004 and 2014, reflecting either illegal fishing and the negative impact on populations or the slow recovery rate of *Actinopyga* populations.

A TRAFFIC report documented the seizure of at least 101 tonnes of sea cucumbers and 6,976 individual specimens across a total of 163 incidents related to India between 2010 and 2021. The top three destinations for sea cucumbers trafficked from India were Sri Lanka, China, and Southeast Asia. While nearly 200 species of sea cucumber have been reported in India, nearly all the incidents referred to sea cucumbers rather than species-specific identification (Fernandes et al., 2022).

Inclusion in Appendix II to improve control of other listed species

A) Specimens in trade resemble those of species listed in Appendix II under Res. Conf. 9.24 (Rev. CoP17) Annex 2 a or listed in Appendix I

A. lecanora and *A. palauensis* are proposed on the basis that specimens in trade resemble those of the other four species proposed for listing. While some morphological distinctions exist, the six species considered in this proposal share considerable overlapping characteristics. For example, the small, morphologically simple ossicles that define the genus are often similar between species, making species differentiation challenging especially in traded form (Netchy et al., 2025). Colour and pattern also vary within species. There are many variants (morphs) of the different species, and different local (trade) names usually consist of similar looking species (in terms of form and size) (Juinio-Menez in litt., 2025). This further complicating identification.

Due to identification issues, *A. echinites*, *A. miliaris*, and *A. palauensis* were lumped together as “spork” in a survey of the sea cucumber population in the Seychelles’ fishery between 2021–2022. *A. varians* is frequently identified as *A. mauritiana*, as the designation of the Pacific variant of *A. mauritiana* as *A. varians* is not universal in the scientific literature or in management plans. Visual surveys “probably” misidentified *A. palauensis* as *A. miliaris*. Murphy et al. (2019) species identification guide for Australia’s Torres Strait fishery indicates that *A. echinites* can be confused with *A. mauritiana* and *A. miliaris*, and that in turn, *A. miliaris* can be confused with *A. palauensis* and *A. spinea* (the latter species not part of this proposal). In the Philippines, Sea Cucumber researchers noted that while some dried products can be distinguished (e.g. *A. lecanora*), small and medium-low economic value species are often lumped together in trade; sorted by size and processing quality rather than species (Juinio-Menez in litt., 2025). The species *A. cf. flammea* (Cherbonnier, 1979) was also noted in Purcell et al. (2023) guide on commercially important sea cucumbers of the world as a similar processed shape to *A. palauensis*, though the body is covered in papillae. It is distributed in the Central Western Pacific, namely New Caledonia, Fiji and Tonga, somewhat overlapping with some of the proposed species.

Additional Information

Threats

The primary threat to sea cucumber populations is overfishing, driven by the high international demand for bêche-de-mer. While traditionally traded as dried products, some countries have shifted toward fresh, frozen and salted products, which are many times heavier (Mercier et al., 2025). Sea cucumber harvest originated in Southeast Asia and tropical Pacific countries, before expanding to the Indian Ocean following “boom-and-bust” cycles of intense harvesting and stock collapse.

Beyond culinary uses, sea cucumbers are highly sought-after for their perceived biological and pharmaceutical benefits, and for the cosmetics industry. Since a paper came out in 2020 claiming that sulphated polysaccharides from sea cucumbers could inhibit the activity of the COVID-19 virus, (Son et al., 2020), sea cucumber consumption in Shanghai, Wuhan, Xi’an, Chengdu, Changsha and other places has apparently shown explosive growth (Son et al., 2020, Sea Cucumber Industry Branch of China Fisheries Association, 2022).

Sea cucumbers are also threatened by environmental threats including climate-change induced ocean warming and declining pH, which impairs sea cucumber physiology and immune function, with a 0.5 pH unit decline reducing larval survival by 49%. Pollution, particularly microplastics and heavy metals, poses an additional risk to sea cucumbers due to their sensitive feeding and respiratory systems.

Conservation, management and legislation

Numerous domestic restrictions exist throughout the species' ranges, though few legal instruments are specific to the *Actinopyga* species. Various countries have instituted fishing moratoria and area closures in response to overexploitation:

- **India:** prohibited sea cucumber fishing since 2001, under the Wildlife Protection Act.
- **Tonga:** lifted and then reinstituted a moratorium since 1996, instituted due to continued overfishing.
- **Guam:** no commercial export is allowed, and local harvests are restricted to a daily limit.
- **American Samoa:** subsistence harvests permitted but commercial fishing and trade prohibited.
- **Northern Mariana Islands:** moratorium on fishing sea cucumbers in place since 1998.
- **Philippines:** there is a size limit for harvesting sea cucumbers.
- **Federated States of Micronesia:** a sea cucumber management plan sets weight and export volume restrictions.
- **Madagascar, Fiji, Kenya, Seychelles, Solomon Islands, and Vanuatu:** set gear restrictions, such as prohibiting the use of SCUBA for harvesting sea cucumbers.
- **Papua New Guinea:** set a species-specific Total Allowable Catch (TAC), requires licensing, and prohibits the use of SCUBA.
- **Seychelles:** requires licensing and compliance with management measures.
- **Egypt:** banned the harvest of sea cucumbers in 2003, improving diversity and density of several commercial species of sea cucumbers.
- **Australia:** all five commercial sea cucumber fisheries employ a variety of measures, including effort and gear restrictions, rotational fishing zones, minimum size limits, TAC limits, ecological risk assessments, and catch and effort monitoring. *The Western Australian sea cucumber fishery (targeting A. echinites and Holothuria scabra) became the world's first to earn Marine Stewardship Council (MSC) certification, setting a standard for sustainable management via independent auditing and assurance. Products will be harvested, processed and marketed by Tasmanian Seafoods to Singapore, mainland China, Hong Kong SAR and Taiwan Province of China (MSC, 2019).*

Captive breeding

Aquaculture for these species is very limited, meaning that trade continues to rely on wild populations, with few prospects for restocking through captive-release programmes. *The successful application of aquaculture in rebuilding wild populations has only been documented in a few species, mostly high grade like Holothuria scabra (Mercier et al., 2025).*

Implementation challenges (including similar species)

Processed sea cucumbers are frequently traded in forms that obscure species identification (e.g., dried, gutted), posing identification challenges for customs officers and enforcement agencies. While some training materials and identification guides exist, more comprehensive resources are needed (Zhong et al., 2021).

Potential risk(s) of a listing

A potential risk of listing Actinopyga species in CITES Appendix II is the displacement of fishing pressure onto non-listed species, as serial depletion along the value chain has been documented for other sea cucumbers (Conand et al., 2014). Sea cucumbers comprise approximately 1,800 species, of which approximately 80 sea cucumber species are fished in more than 100 countries (Mercier et al., 2025). Purcell et al. (2025) noted that an increase in retail prices of sea cucumber species was potentially exacerbated by trade restrictions from previous CITES listings of the teatfish group and Thelenota spp.

Regarding the effectiveness of a CITES listing, countries have experienced mixed success in conducting and approving Non-Detriment Findings (NDFs) and Legal Acquisition Findings (LAFs) for the teatfish species (Holothuria nobilis, H. fuscogilva and H. whitmaei) following their CITES listing in Appendix II in 2019. This has highlighted the need for strengthened technical support to scientific authorities for preparing NDFs. Equally important is targeted support at the grassroots level to inform fishery stakeholders including fishers, traders, and exporters about the requirements and potential benefits of CITES protocols (Mercier et al., 2025). In many cases, coastal communities, particularly marginalised artisanal fishers, are heavily dependent on sea cucumber harvests for their livelihoods, often operating within weak governance systems or informal economies and with limited alternative income. The ability of Parties to implement CITES regulations has translated into extended periods before listings enter into force, as exemplified by the listings of three teatfish species and the Thelenota genus (Mercier et al., 2025).

Initiatives to improve management and recover stocks may help towards conservation and could be preferable to international trade restrictions. However, without strong enforcement and other regulatory measures, the effectiveness of size limits and restocking is rather questionable (Purcell et al., 2013).

Potential benefit(s) of listing for trade regulation

A CITES listing would strengthen monitoring and regulation of international trade in sea cucumbers, helping to reduce overexploitation. It would also support sustainable use through NDFs, raise awareness among stakeholders, and provide opportunities for technical assistance, research, and capacity building. Furthermore, it would improve trade reporting, promote coordinated management at national and regional levels, and contribute to the long-term socio-economic benefits for fishers (Bruckner, 2006; Toral-Granda, 2006).

Other comments

Trade chains for sea cucumbers can be reportedly complicated, often passed through multiple intermediaries and transformations before reaching markets (Pedroza-Gutiérrez and López-Rocha, 2021). Therefore, strong traceability systems are encouraged. One effective model is the labelling scheme used in the caviar industry or the tagging system used for crocodile skins (Mundy and Sant, 2015).

If the proposal to list these species were to be accepted, consideration might be given in the future to including *A. spinea* in Appendix II (not included in this proposal), given that it shares morphological features with *A. miliaris* and is frequently misidentified as that species by fishers and processors. This would help to avoid enforcement difficulties and ensure consistent regulation of trade.

References

- Bruckner, A. (editor) (2006) The Proceedings of the Technical workshop on the conservation of sea cucumbers in the families Holothuridae and Stichopodidae. NOAA Technical Memorandum NMFSOPR 44, Silver Spring, MD.
- CITES Secretariat (2025) Notification to the Parties No. 2025/102: Proposals to amend Appendices I and II. Provisional assessments by the Secretariat. Convention on International Trade in Endangered Species of Wild Fauna and Flora. Available at: <https://cites.org/sites/default/files/notifications/E-Notif-2025-102.pdf>
- Conand, C., Purcell, S. and Gamboa, R. (2013) *Actinopyga echinites*. The IUCN Red List of Threatened Species 2013: e.T180518A1642310. <http://dx.doi.org/10.2305/IUCN.UK.2013-1.RLTS.T180518A1642310.en>
- Conand, C., Purcell, S. and Gamboa, R. (2013) *Actinopyga mauritiana*. The IUCN Red List of Threatened Species 2013: e.T180337A1616879. <http://dx.doi.org/10.2305/IUCN.UK.2013-1.RLTS.T180337A1616879.en>
- Conand, C., Polidoro B., Mercier A., Gamboa R., Hamel J.F. and Purcell, S. (2014) The IUCN Red List assessment of aspidochirotid sea cucumbers and its implications. *SPC Beche-de-mer Information Bulletin* 34.
- FAO (2025) Report of the Eighth FAO Expert Advisory Expert Panel for the Assessment of Proposals to Amend Appendices I and II of CITES Concerning Commercially Exploited Aquatic Species – Bangkok, 7–11 July 2025 and Rome, 21–25 July 2025. *FAO Fisheries and Aquaculture Report* 1482. Rome. <https://doi.org/10.4060/cd6542en>
- Fernandes M., Chopra M., Gautam A. and Badola, S. (2022) In Deep Water: India's Sea Cucumber in illegal wildlife trade. TRAFFIC India, New Delhi. Available at: https://www.traffic.org/site/assets/files/19807/sea_cucumbers_in_illegal_trade_report_nov_2022.pdf
- Gamboa, R.U., Meñez, A.J., Abesamis, N.P., Sabanal, B.T., Florece, C. and Tan, J. (2025) First National Workshop: Marine Invertebrates of the Philippines Aquatic Redlist Committee (Aspidochirotida). Project Terminal Reprot. University of the Philippines Mindanao, University of the Philippines Marine Science Institute and Silliman University.
- Halun, Z. per comm. to Juinio-Menez, A. (2025) *in litt.* to IUCN/SSC Sea Cucumber Specialist Group.
- Juinio-Menez, A. (2025) *in litt.* to IUCN/SSC Sea Cucumber Specialist Group.
- Jontila, J.B.S., Monteclaro, H., Quintio, G.F., Santander-de Leon, S.M. and Altamirano, J.P. (2018) Status of sea cucumber fishery and populations across sites with different levels of management in Palawan, Philippines. *Ocean and Coastal Management* 165:225–234.
- Kalaeb, T., Ghirmay, D., Semere, Y. and Yohannes, F. (2008) Status and preliminary assessment of the sea cucumber fishery in Eritrea. *SPC Beche de Mer Information Bulletin* 27.
- KKP (2022) Statistik Ekspor Hasil Perikanan Tahun (2017 – 2021)
- Marine Stewardship Council (2019) World's first sea cucumber fishery proves it's sustainable. Marine Stewardship Council Website. Available at: https://www.msc.org/media-centre/press-releases/press-release/world-s-first-sea-cucumber-fishery-proves-it-s-sustainable?utm_source=chatgpt.com
- Mercier, A., Purcell, S.W., Montgomery, E.M., Kinch, J., Byrne, M. and Hamel, J.F. (2025) Revered and Reviled: The Plight of the Vanishing Sea Cucumbers. *Annual Review of Marine Science* 17(1):115–142. <https://doi.org/10.1146/annurev-marine-032123-025441>
- Moore, B., Bosserelle, P., Mailau, S., Vi, S., Fonua, S., Havea, T. and Malimali, S. (2017) The status of sea cucumbers in the Kingdom of Tonga in 2016. Noumea, New Caledonia. Pacific Community. Available at: <https://www.spc.int/digitalibrary/get/jipgt>

- Mundy, V. and Sant, G. (2015) Traceability Systems in the CITES Context: A review of experiences, best practices and lessons learned for the traceability of commodities of CITES-listed shark species. Available at: <https://cites.org/sites/default/files/eng/prog/shark/docs/BodyofInf12.pdf>
- Netchy, K., Samyn, Y. and Paulay, G. (2025) Dividing the monopoly: the most broadly distributed *Actinopyga* is two species (Holothuroidea: Holothuriidae). *Zootaxa* 5673(1):27–44. <https://doi.org/10.11646/zootaxa.5673.1.2>
- Ochiewo, J., de la Torre-Castro, M., Muthama, C., Munyi, F. and Nthuta, J.M. (2010) Socio-economic features of sea cucumber fisheries in southern coast of Kenya. *Ocean & Coastal Management* 53(3):192–202. <https://doi.org/10.1016/j.ocecoaman.2010.01.010>
- Pedroza-Gutiérrez, C. and López-Rocha, J.A. (2021) Ungovernable systems: The strength of informal institutions in the sea cucumber fishery in Yucatan, Mexico. *PLoS One* 16(3):e0249132. <https://doi.org/10.1371/journal.pone.0249132>
- Purcell, S.W., Hair, C.A. and Mills, D.J. (2012) Sea cucumber fisheries: global analysis of stocks, management measures and drivers of overfishing. *FAO Fisheries and Aquaculture Technical Paper* 516. Rome. Available at: <https://www.fao.org/3/i2728e/i2728e.pdf>
- Purcell, S.W., Mercier, A., Conand, C., Hamel, J.-F., Toral-Grande, M.V., Lovatelli, A. and Uthicke, S. (2013) Sea cucumber fisheries: Global analysis of stocks, management measures and drivers of overfishing. *Fish and Fisheries* 14(1):34–59. <https://doi.org/10.1111/j.1467-2979.2011.00443.x>
- Purcell, S.W., Lovatelli, A., González-Wangüemert, M., Solís-Marín, F.A., Samyn, Y. and Conand, C. (2023) Commercially Important Sea Cucumbers of the World. *FAO Species Catalogue for Fishery Purposes* 6 (Rev. 1). Rome. 2nd ed. Available at: <https://openknowledge.fao.org/server/api/core/bitstreams/aa81298b-5d93-42d8-a0fa-b35172c3404b/content>
- Ramofafia, C., Lincoln-Smith, M.P. and Lane, I. (2007) Managing sea cucumber fisheries with an ecosystem approach in the Solomon Islands. *SPC Beche-de-mer Information Bulletin* 25:11–19.
- Setyastuti, A., Wirawati, I., Hadiyanto, H., Nurjain, N., Permadi, S. Hadi, T.A. and Sjafrie, N.D.M. (2024) New insight into the diversity, biometric distribution, and relationships of commercial sea cucumber species from Indonesia. *Fisheries Research* 279:107124.
- Setyastuti, A., Wirawati, I., Permadi, S. and Vimono, I.B. (2019) Teripang Indonesia: Jenis, Sebaran dan Status Nilai Ekonomis. *PT. Media Sains Nasional*. Available at: https://www.researchgate.net/publication/334557606_Teripang_Indonesia_Jenis_Sebaran_dan_Status_Nilai_Ekonomis
- Toral-Granda, V. (2006) Situation biologique et commerciale des concombres de mer des familles Holothuriidae et Stichopodidae. AC22 Doc. 16 Annexe. Available at: <https://cites.org/sites/default/files/fra/com/ac/22/F22-16.pdf>
- Volza (2025) Dried Sea Cucumber Imports From World Market Size & Demand Based On Export Trade Data. Available at: <https://www.volza.com/p/dried-sea-cucumber/export/>. Accessed on 1 August 2025.
- Zhong, S., Qiao, Y., Zhao, L., Huang, G., Liu, Y. and Huang, L. (2021) Characterization and phylogenetic analysis of the complete mitochondrial genome of *Actinopyga lecanora* (Jaeger, 1833) (Holothuriida: Holothuriidae). *Mitochondrial DNA. Part B, Resources* 6(10):2801–2802. <https://doi.org/10.1080/23802359.2021.1970641>

Inclusion of Golden Sandfish *Holothuria lessoni* in Appendix II

Proponent: European Union

Summary: Sea cucumbers (Holothuroidea) are marine echinoderms comprising approximately 1,800 species worldwide. The Golden Sandfish *Holothuria lessoni* is a large tropical species typically reaching lengths of 25-35 cm, with some individuals growing up to 46 cm. The species inhabits shallow waters (up to 30 metres), with its range extending from Africa and the Indian Ocean to Asia and the Western Central Pacific.

The species is in demand for luxury food and traditional medicine, particularly in many Asian countries where consumption of bêche-de-mer, dried and cooked sea cucumbers, is popular. It is reported as the highest-priced tropical sea cucumber in market surveys in 2022 in Hong Kong SAR of China but was scarce in those markets.

Golden Sandfish is harvested throughout its range. Once common in parts of its range, exploitation has led to observed density reductions and in some places, local extirpations (e.g. Australia, specifically Ashmore Reef, the north and east of Fiji, and Vanuatu, Efate). No global population estimates are available for *H. lessoni*, but it was listed as Endangered on the IUCN Red List in 2010, an assessment now noted to need updating. This assessment estimated the species' global population had decreased by at least 50% over the previous 30 - 50 years, primarily due to overfishing and loss of mature individuals. These estimates were inferred in part based on comparison with another species, *H. scabra*, with which it is commonly traded, taking into account exploitation levels and biological vulnerability, as well as, observed population decreases across much of its range.

H. lessoni is considered intrinsically vulnerable to overharvesting, due to its sedentary behaviour, late maturity (2-5 years), density-dependent reproduction, low recruitment rates, as well as accessibility to harvest. Its generation length is thought to span several decades, and despite probable medium productivity, it is known to recover slowly after depletion, especially when population densities are too low for successful spawning, since reproduction relies on external fertilisation and close proximity of individuals. Although females produce several million eggs, survival rates are low due to the vulnerability of eggs and larvae to predation and environmental conditions.

Sea cucumber fisheries have developed in many parts of the world and have followed boom-and-bust patterns and most have declined as a result of overfishing. Whilst aquaculture development has focused mainly on *H. scabra*, *H. lessoni* has been explored for cultivation due to its high market value. though current production remains small in scale, and any benefits to wild populations are likely limited to local contexts.

Trade in *H. lessoni* is widespread but poorly tracked, as species-specific reporting is generally lacking. *H. lessoni* was exported from Indonesia in significant volumes from 2017–2020 (totalling 93,050 kg). In the Solomon Islands, the Ministry of Fisheries and Marine Resources (MFMR) reported export volumes of 925 kg in 2021 and 2,095 kg in 2022. According to the UN Comtrade database, between 2015 and 2024, Indonesia, Malaysia, and mainland China were among the leading exporters of sea cucumbers under general sea cucumber HS codes (exports from *H. lessoni* range States). Saudi Arabia, Hong Kong SAR of China, and mainland China were the top importers. Market prices for *H. lessoni* in Hong Kong SAR of China reached USD849/kg in 2016, and USD635/kg in 2022. Local prices also reflect high value in several locations. Illegal trade involving this species has been reported in several countries.

Identification challenges arise as *H. lessoni* is often sold together with *H. scabra* under the local name “putian,” for example in the Philippines. Morphological variation within the species in colour and body shape also contributes to misidentification risks. Enforcement challenges include misreporting, fraud, and difficulty in species-level identification, especially in dried or processed forms.

Few regulatory instruments are specific to *H. lessoni*, although it is managed under general sea cucumber frameworks in several range States. In Australia, the species is commercially harvested in

regulated fisheries, with seasonal closures, limited entry, licensing, and Total Allowable Catch (TAC) in place. Papua New Guinea also includes the species in its national sea cucumber management plan, which prohibits SCUBA use, requires licensing, and sets TACs. Fiji, Vanuatu, and Kiribati have similar national plans or general restrictions that apply to *H. lessoni*. Tonga, after a long moratorium, excluded the species from catch quota allocations when the fishery reopened in 2021. In the Seychelles, *H. lessoni* was authorised for exploitation under a TAC of 100,000 individuals in 2022-2023, but just under 9,000 were caught.

Holothuria lessoni is proposed for listing in CITES Appendix II, in accordance with Article II, Paragraph 2 (a) of the Convention and satisfying Criterion B of Annex 2a of Resolution Conf. 9.24 (Rev. CoP17). Three other species in the genus are already listed in Appendix II (*Holothuria fuscogilva*, *H. nobilis* and *H. whitmaei*) as well as the genus *Thelenota*.

Analysis: *Holothuria lessoni* is widely reported as over-exploited throughout much of its range, driven primarily by international trade for bêche-de-mer. Anecdotal and recorded evidence suggests that this species commands high market prices, driving targeted exploitation even in remote areas. The species was classified as Endangered in 2010, having undergone an estimated decrease of over 50% globally over the previous 30 – 50 years, with a then decreasing population trend. It is likely that fishing pressure has increased since then and population decreases may have accelerated, with more recent evidence of localised extirpation or severe depletion in specific locations. Considering the species' biological vulnerabilities that hinder population recovery, it is likely that trade regulation is required to ensure that harvesting does not reduce wild populations to a level that threatens their survival. Therefore, the species **meets the criteria in Annex 2aB** of Res. Conf 9.24 (Rev CoP17).

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

Holothuria lessoni (Golden Sandfish) (Massin et al., 2009). *H. lessoni* was previously considered as *H. scabra* var. *versicolor* (Conand, 1989) but was identified as a distinct species on the basis of a taxonomic study by Massin et al. (2009) and previously, Uthicke et al. (2005) concluded that *H. scabra* and *H. scabra* var. *versicolor* were distinct based on allozyme and mtDNA sequence analysis.

The proposal utilises CITES Standard Reference for sea cucumbers: *Taxonomic Checklist of selected Sea Cucumber taxa relevant to CITES following CoP19* extracted from the World Register of Marine Species (WoRMS) accessed 5 June 2025, with edits from the Nomenclature Specialist of the CITES Animals Committee.

Range

Australia (including Cocos (Keeling) Islands and Christmas Island); Bangladesh; territory formerly known as British Indian Ocean Territory; Cambodia; China; Comoros; Fiji; Indonesia; Kenya; Madagascar; Malaysia; Maldives; Mauritius; Mayotte (France); Mozambique; Myanmar; New Caledonia (France); Niue; Palau; Papua New Guinea; Philippines; Réunion (France); Samoa; Seychelles; Singapore; Solomon Islands; Somalia; Sri Lanka, United Republic of Tanzania; Thailand; Tonga; Vanuatu; Viet Nam; Wallis and Futuna (France).

American Samoa and India are included in the distribution list of IUCN, the CITES Scientific Authorities of the USA (*in litt.* to European Commission, 2024) and India (*in litt.* to European Commission, 2025) confirmed that *H. lessoni* does not occur in their waters. According to Dissanayake (*in litt.* to UNEP-WCMC, 2025), the species does not occur in Sri Lanka.



Figure 1. Distribution of *H. lessoni*, IUCN (2012). The IUCN Red List of Threatened Species. Version 2025-1

IUCN Global Category

Endangered A2bd (assessed 2010, ver. 3.1). Needs updating.

Biological and trade criteria for inclusion in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)

A) Trade regulation needed to prevent future inclusion in Appendix I

Holothuria lessoni typically measures 25–35 cm (up to 46 cm), with an average length of around 30 cm reported from populations in Australia, Madagascar and New Caledonia. *H. lessoni* reaches maturity typically at 22 cm and a weight of ~ 480 grams, between two and five years. While the species' exact generation length is unknown, it is thought to span several decades longer than its estimated lifespan of approximately 15 years, due to reproductive and environmental factors. Reproduction depends on external fertilisation at high densities, making recovery from depletion slow. Although females produce millions of eggs, survival rates are low due to the vulnerability of eggs and larvae to predation and environmental conditions. Natural mortality (M) was estimated at 0.6 per year in 2005, suggesting high productivity, but more recent unpublished data (0.4/year) place it in medium productivity species range.

No global population size estimates for *H. lessoni* exist. It was considered a common species in 2010, though abundance varies significantly by location.

H. lessoni was assessed as Endangered on the IUCN Red List in 2010 and noted to have a decreasing population trend. In 2010, the decline was estimated to be at least 50% throughout the global range over the previous 30–50 years. *The estimated decline in H. lessoni populations is based on the assumption that it is similarly affected by overfishing as H. scabra, which has experienced an estimated 50% decline over the past 50 years prior to 2010. This assumption is supported by the fact that H. lessoni is targeted in similar ways, is larger and rarer, may be more valuable, and is often misidentified or grouped with H. scabra in catch statistics across much of its range (Conand et al., 2013).* There is also evidence of *H. lessoni* depletions in some areas, as well as being targeted as a commercially valuable species and having not recovered from a fishing closure in one part of the range. Such depleted populations demonstrate the pattern of a “boom” of intense captures that many sea cucumber fisheries experience, followed by fishery collapse due to overfishing.

Temporal declines of *H. lessoni* populations (or recorded under its former taxonomic classification *H. scabra* var. *versicolor*) within individual range States were observed across numerous studies:

Pacific Ocean:

- **New Caledonia:** densities dropped from an average of 82 ind./ha (1980s) to an overall density of 1.5 ind./ha (2021–2022).
- **Tonga:** In Ha'apai and Vava'u, density fell to 0 ind./ha in 2016, from 0.12 ind./ha (Ha'apai, 2014) and 2 ind./ha (Vava'u, 2010–11); despite a 7–10 year fishing moratorium, no strong recovery. In Tongatapu the species had low densities but did not show decreases (3 ind./ha in 2010–2011 and 5 ind./ha in 2016). A 2019 survey across 90,000 m² of reef habitat recorded just eight *H. lessoni* individuals, only three above the size at maturity of 22 cm.
- **Fiji:** no individuals found in 2012–2015 surveys across multiple provinces, since it was last recorded in Muaivuso in 2009 in very low numbers. Fishers (2014) indicated it was the least observed and captured sea cucumber out of 23 species. Preston (2023) indicated these collective studies suggested the species may have become locally extirpated.
- **Vanuatu:** only one specimen was found across multiple surveyed sites in 2003. In 2019–2020, a comprehensive survey around Efate Island (50+ dives across seasons, day and night) found zero

individuals, suggesting local extirpation. However, it was observed in low numbers in the Maskelyne Islands in 2011 and 2023 surveys.

- **Samoa:** it was considered rare in 2008 surveys, not reported in 2009 and 2013 surveys, and only observed at one of 125 sites in 2019 (density of 0.7 ind./ha).
- **Solomon Islands:** described as “rare” across eight provinces surveyed (2011-2012). Most sites reported low density values of 1.2–6.9 ind./ha, though specific densities ranged from 0.5 to 333 ind./ha, in Taro and Star Harbour respectively. At Lord Howe Atoll in 2021, densities were low and sample sizes insufficient for analysis of size distribution.
- **Papua New Guinea:** in 2006, three specimens of *H. scabra versicolor* were observed out of ~21,000 individuals surveyed in the New Ireland Province. In 2017, post-moratorium, the species comprised only 3.3% of 35,000 sea cucumbers caught in Tigak Islands (6.3% by volume).
- **Australia:** observed at 7 of 50 sites in the eastern Torres Strait (2007), but later only observed at one site (east of Ugar Island) out of 289 (2019–2020). No sightings at Ashmore Reef since 1987, despite 13 surveys covering 107 ha. This indicates local extirpation due to overharvesting.
- **Philippines:** *H. lessoni* is rare with very limited distribution in the Philippine archipelago, and reported in only a few locations in Mindanao in the southern Philippines (Halun per. comm to Juinio-Menez in litt., 2025).
- **Indonesia:** 14 individuals across 23 sites (2016-2018), representing 0.3% of the 4,553 sea cucumbers recorded. The species was considered to have medium-low abundance in the wild. However, it is noted this result cannot be extrapolated to describe the national population status, considering Indonesia is comprised of more than 17,000 islands.

African region of Indian Ocean:

- **Seychelles:** absent from all sites exposed to fishing pressures in 2004; only found in a marine park at very low densities. Still low numbers in 2021-2022 surveys, with uncertain population status and likely not commercially viable.
- **Madagascar:** 17 ind./ha in Beankiho (2008), absent elsewhere. By 2022, *H. lessoni* had very low representation among sea cucumber species in monitored catch: Sarodrano (4.4%), Ankiembe (1.2%), Andrevo (0.6%).
- **Mauritius (Chagos Archipelago):** general sea cucumber decreases were reported in 2014 compared to previous surveys in 2006 and 2010, but no specific data for *H. lessoni*.
- **Comoros:** sea cucumbers are generally considered to be endangered according to a 2024 ministerial decree; the specific status of *H. lessoni* is unclear.
- **United Republic of Tanzania:** in 2011, *H. lessoni* was reported as having a high per piece price, but had a low mean catch per unit effort, indicating a low abundance.

Gulf of Thailand & Andaman Sea:

- No individuals recorded in 2022 surveys; the species' presence is uncertain.

B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences

The main threat to sea cucumber populations is overfishing driven by the high international demand for bêche-de-mer, in particular Asian markets. Global demand is also expected to increase. *Sea cucumbers comprise approximately 1,800 species, of which approximately 80 species are known to be fished in more than 100 countries (Mercier et al., 2025). Purcell et al. (2025) noted that an increase in retail prices of sea cucumber species was potentially exacerbated by trade restrictions from the listings of Holothuria fuscogilva, H. nobilis and H. whitmaei, at CoP18 (effective 28/08/2020) and Thelenota spp. at CoP19 (effective 25/05/2024).*

H. lessoni is one of the highest commercially valued species, prized as a cultural delicacy, and used to treat a wide range of ailments in traditional medicine, as well as for the production of cosmetic products. *H. lessoni* is collected throughout its range and widely reported as over-exploited. It is often confused with *H. scabra* in trade.

UN Comtrade data (2015-2024) indicate a total of 44,135 metric tonnes of aggregated sea cucumber (re-)exports from *H. lessoni* range States under HS codes 030811, 030812, 030819 and 160561. *Disaggregated data to species level is not possible given there is no specific HS code for H. lessoni.* The top exporters of sea cucumbers were Indonesia (29%), Malaysia (25%), mainland China (10%), Thailand (7%), Australia (5%), Sri Lanka (5%), Singapore (5%), Philippines (3%), Fiji (2%) and the Maldives (2%). Top importers of sea cucumbers from *H. lessoni* range States were Saudi Arabia (37%), Hong Kong SAR of China (23%) and mainland China (20%), with mainland China (33%), Hong Kong SAR of China (28%) and Viet Nam (11%) as the top destinations. *From 2015-2023, the importer-reported UN Comtrade data trade in sea cucumbers grew at a compound annual growth rate of 4.56%. UN Comtrade data from 2024 was not included in this calculation because it was incomplete.*

Anderson (2011) estimated global catch and production of sea cucumber fisheries, including from aquaculture had increased 13–16-fold over the past two to three decades, as market prices rose in line with heightened demand; *with capture production alone rising 4–5 fold from ~5,000 (1975) to ~22,500 tonnes (2005).*

According to FAO, global capture production of sea cucumbers in 2011–2021 peaked at 62,000 t in 2018, then dropped to ~45,000–50,000 t in 2020–2021 (likely due to the impacts of the COVID-19 pandemic).

Legal trade

There are no reliable estimates of the total volume of *H. lessoni* in international trade. Most sea cucumber catch and international trade statistics are not reported to the species level. However, several sources indicate *H. lessoni* is heavily exploited in parts of its range, consistently targeted for its high commercial value, and actively traded, particularly in Southeast Asia and the Pacific, despite signs of population depletion and absence from many market and export records. This self-reported harvest data is likely to be based on processed dried weights, which may underestimate the true scale of global sea cucumber fisheries that could be better determined using live, wet weights. According to LEMIS, 70 kg of wild-sourced *H. lessoni* meat was commercially imported into the USA since 2019, all exported from Indonesia (USFWS *in litt.*, to UNEP-WCMC, 2025).

Country-level trade and exploitation of *H. lessoni*:

- **Indonesia:** *H. lessoni* (*H. scabra* var. *versicolor* at the time) is regarded as “heavily exploited” (Tuwo, 2004). According to Wirawati *et al.* (2021), *H. lessoni* was exported from Indonesia in significant volumes totalling 93,050 kg (2017–2020), with main destinations including Hong Kong SAR of China, mainland China, and Singapore. Exports declined sharply from 30,613 kg (2017) to 9,100 kg (2020). Similar reductions in export volumes were not observed over the same period for the CITES Appendix II listed species: *H. fuscogilva*, *H. nobilis* and *H. whitmaei*.
- **Papua New Guinea:** *H. lessoni* considered a commercially important species, but export data are limited. Following a national moratorium, 1,141 individuals were recorded in a 10-day monitoring period in 2017. The species continues to be harvested in small quantities in various provinces.
- **Solomon Islands:** the MFMR reported export volumes of 925 kg in 2021 and 2,095 kg in 2022. In 2025, of the 27 sea cucumber species included in the published benchmark prices for buying and export of sea cucumbers, it was assigned the highest national buying price of all species (USD270 per piece).
- **Australia:** harvested in three fisheries, though in relatively small volumes. Annual catch ranged between 0–8 tonnes each year between 2013–2022 in the Queensland Sea Cucumber Fishery. In the Coral Sea Fishery, the Australian Fisheries Management Authority reported catch of *H. lessoni* under the ‘any other sea cucumber species’ category so *disaggregation of H. lessoni was not possible*, but the harvest of *H. lessoni* did occur in small volumes.
- **Fiji:** historically “very high value”, one of 13 species of high commercial importance to Fiji, and last recorded at Muaivuso, Viti Levu in 2009 but has not been recorded in export data for the past decade. It remains on the buyer price list in 2013 but is now likely commercially unviable in many areas.
- **New Caledonia:** not recently listed among major export species, although data before 2020 grouped catches under “miscellaneous” species. Its wet price was reported as USD3.57/kg in 2020.
- **Seychelles:** after no authorisation between 2020–2022, *H. lessoni* was authorised by the Fisheries Authority for legal exploitation under a TAC of 100,000 pieces for the 2022–2023 fishing season. Only 8,933 were caught, CPUE was 63 pieces per diver per day, and quota usage for the species among fishing vessels decreased from 12% (2022–23 season) to 5% (2023–24 season).
- **Madagascar:** regarded as a heavily exploited species and by 2018, among the highest-value sea cucumber species sold in three southwest coastal communities, with prices up to USD15.60 per processed piece. *H. lessoni* represented 0.6% to 4.4% of catches. It is actively fished in both the southwest and northeast Masoala Peninsula. *H. lessoni* was also targeted for aquaculture development, particularly in the southwest Toliara Province, contributing to high national export volumes.
- **Tonga:** not included in catch quotas allocation when the fishery reopened in 2021 (after closure since 2013).
- **Vanuatu:** the species was listed in the national sea cucumber management plan (2019–2024) with fixed prices (USD20.88/kg wet, USD52.20/kg dry). No export volumes could be found.
- **Philippines:** regarded as a heavily exploited species; recorded in catches at three out of five Mindanao sites (2019–2021). However, most catches consisted of medium to lower-value species, suggesting *H. lessoni* stocks had already been depleted and shifted to lower value species due to overexploitation of high value sea cucumbers. *H. lessoni is a high-value species, like H. scabra, and priced according to size. There are more H. scabra specimens than H. lessoni being traded in Tawi-Tawi, but they are sold together as trepang because they are priced the same (Halun pers. comm to Juinio-Menez in litt., 2025).*
- **Maldives:** 13,039 kg of various *Holothuria* species were exported in 2024, but no exports were recorded specifically as *H. lessoni*. The CITES Management Authority noted that 165 kg of unidentified *Holothuria* exported could potentially have included *H. lessoni*.

- **Mauritius:** considered part of the sea cucumber export trade, though no species-specific export data are available.
- **Hong Kong SAR of China:** among the highest-priced and scarcest tropical sea cucumbers. In 2022, retail prices averaged USD503/kg (maximum USD635/kg). Only four batches were found across 38 market shops, reflecting low availability despite high demand.

Illegal trade:

The illegal harvest and trade of sea cucumbers, driven by the high demand for bêche-de-mer is pervasive, widespread and affects all stages of the supply chain. This includes *pre-fishing activities such as forged licenses, corruption, and labour exploitation; fishing practices (illegal methods, poaching, overharvesting), post-harvest activities (armed robbery, smuggling, money laundering), processing (which involves illegal firewood harvesting for drying), and in sale and trade (through misreporting, fraud, and tax evasion)* (Phelps and Morrow, 2024). Although no seizures specifically involving *H. lessoni* were identified in EU-TWIX, WiTIS, or other enforcement databases, the species high value, frequent misidentification and presence in poorly monitored supply chains strongly suggest it is involved in illegal trade.

Additional Information

Threats

Beyond overfishing, this species is also threatened by environmental threats including destructive fishing methods like bottom trawling, which damages seabed habitats, and climate change causing rising ocean temperatures, acidification, and salinity changes. This impairs sea cucumber physiology and immune function. Pollution, particularly microplastics and heavy metals, poses an additional risk, as sea cucumbers are highly sensitive to contaminants due to their unique feeding and respiratory systems.

Conservation, management and legislation

Numerous domestic restrictions exist throughout the species' range, though few legal instruments are specific to *H. lessoni*. Nonetheless, several countries have implemented measures in response to overfishing and stock depletion.

- **Australia:** *H. lessoni* is commercially harvested in Queensland under a regulated fishery, with limited entry, seasonal closures, and a TAC, and formal stock assessments. The species is listed as a priority for management due to its high value.
- **Fiji:** the species is in the national sea cucumbers management plan, which imposes seasonal closures, size limits, and bans on the use of SCUBA for harvesting.
- **Papua New Guinea:** sea cucumber fisheries, including *H. lessoni*, are subject to a national management plan that includes licensing, TACs, and restrictions on the use of SCUBA diving gear.
- **Vanuatu:** imposed periodic moratoria on sea cucumber fishing, including *H. lessoni*, to allow for stock recovery; the species is covered under national fisheries regulations that control harvesting and trade.
- **Kiribati:** indirectly regulated through general sea cucumber measures, including size limits and bans on the use of SCUBA.
- **Tonga:** implemented a moratorium on all sea cucumber harvesting due to widespread overexploitation, which includes *H. lessoni*.

Artificial Propagation/captive breeding

Of the few tropical species to have been cultured, only the sandfish H. scabra has been bred extensively (Purcell et al., 2012). H. lessoni has seen few attempts at aquaculture (Purcell et al., 2012). The species shows promise as an alternative aquaculture commodity alongside H. scabra, and Isostichopus fuscus (Purcell et al., 2014), though production is currently small scale and any benefits to wild stocks would be very localised. The FAO expert panel report noted ongoing mariculture of H. lessoni in Tonga and Indonesia (FAO, 2025).

Implementation challenges (including similar species)

H. lessoni juveniles are often mistaken for H. scabra. Distinguishing features of H. lessoni from its close relative include a golden/ bronze coloration, an elongated and ventrally flattened body, and the dorsal surface features stubby black papillae whereas H. scabra tends to be duller (usually greyish or dark brown coloration), stouter and more robust, with deep wrinkles in the dorsal surface and a whitish ventral side (possibly the reason why it is locally called 'putian') (Halun pers. comm to Juinio-Menez in litt., 2025). While distinction may be possible with live/fresh H. lessoni, the distinguishing features described may be affected by processing. Coloration within H. lessoni varies from a dark greyish-black to beige with or without black blotches. The beige, black, and blotched varieties of H. lessoni were categorised as of high (commercial) value in Mindanao, the Philippines (through informant surveys and focus group discussions between 2019 and 2020 across 32 municipalities). The species is locally referred to as "big putian". Putian is the common name for H. scabra (Arriesgato et al., 2022). Hybridisation between H. scabra and H. lessoni further complicates morphological identification (Brown et al., 2025), and trade in processed forms (dried, gutted) obscure morphological features, creating enforcement challenges despite the availability of some training materials and identification guides.

Other comments

A potential impact of listing individual sea cucumber species in CITES Appendix II is the displacement of fishing pressure onto non-listed species, as serial depletion along the value chain has been documented for other sea cucumbers (Conand et al., 2014). Purcell et al. (2025) noted that an increase in retail prices of sea cucumber species was potentially exacerbated by trade restrictions from previous CITES listings of the teatfish group and *Thelenota* spp. However, CITES Appendix II regulation need not necessarily add significantly to the existing costs of compliance with national fisheries and export regulation.

Some countries have experienced mixed success in conducting and approving NDFs and legal acquisition findings for teatfish species (*Holothuria nobilis*, *H. fuscogilva* and *H. whitmaei*) following their CITES listing in Appendix II in 2019. This has highlighted the need for strengthened technical support to scientific authorities for preparing NDFs. Equally important is targeted support at the grassroots level to inform fishery stakeholders including fishers, traders, and exporters about the requirements and potential benefits of CITES (Mercier et al., 2025). In many cases, coastal communities, particularly marginalised artisanal fishers, are heavily dependent on sea cucumber harvests for their livelihoods, often operating within weak governance systems or informal economies and with limited alternative income.

References

- Arriessgato, N.C., Cruz, J.A.C., Rosales, A.C., Lim, C.E. and Detalla, A.L. (2022) Status of Sea Cucumber Fisheries in Mindanao: Harnessing Local Knowledge in Determining Diversity and Production Values. Department of Agriculture–Bureau of Fisheries and Aquatic Resources – National Fisheries Research and Development Institute. Philippines.
- Brown, K.T., Southgate, P.C., Duy, N.D.Q., Havimana, L., Mmochi, A.J., Basita, R.K., Delghandi, M., Stockwell, B. and Lal, M.M. (2025) The sandfish identity: phylogeographic reconstructions uncover a species complex within the Indo-Pacific distribution of *Holothuria* (*Metriatyla*) *scabra*. *BMC Ecology and Evolution* 25(78). <https://doi.org/10.1186/s12862-025-02421-8>
- Conand, C., Purcell, S. and Gamboa, R. (2013) *Holothuria lessona*. The IUCN Red List of Threatened Species 2013: e.T180275A1609567. <http://dx.doi.org/10.2305/IUCN.UK.2013-1.RLTS.T180275A1609567.en>
- Conand, C., Polidoro, B., Mercier, A., Gamboa, R., Hamel, J.F. and Purcell, S. (2014) The IUCN Red List assessment of aspidochirotid sea cucumbers and its implications. SPC Beche-de-mer Information Bulletin #34
- FAO (2025) Report of the Eighth FAO Expert Advisory Panel for the Assessment of Proposals to Amend Appendices I and II of CITES Concerning Commercially-Exploited Aquatic Species – Bangkok, 7–11 July 2025 and Rome, 21–25 July 2025. FAO Fisheries and Aquaculture Report, No. 1482. Rome. <https://doi.org/10.4060/cd6542en>
- Halun, Z per comm. to Juinio-Menez, A. (2025) *in litt.* IUCN/SSC Sea Cucumber Specialist Group
- Mercier, A., Purcell, S.W., Montgomery, E.M., Kinch, J., Byrne, M. and Hamel, J.F. (2025) Revered and Reviled: The Plight of the Vanishing Sea Cucumbers. *Annual Review of Marine Science* 17(1):115-142. doi: 10.1146/annurev-marine-032123-025441
- Phelps Bondaroff, T.N. and Morrow, F. (2024) Chapter 13: Sea cucumber crime. In: Mercier, A., Hamel, J.-F., Suhrbier, A. and Pearce, C. (eds), *The World of Sea Cucumbers*. Academic Press. <https://doi.org/10.1016/B978-0-323-95377-1.00009-6>
- Purcell, S.W., Hair, C.A. and Mills, D.J. (2012) Sea cucumber culture, farming and sea ranching in the tropics: progress, problems and opportunities. *Aquaculture* 368–369:68–81. <https://doi.org/10.1016/j.aquaculture.2012.08.053>
- Purcell, S.W., Polidoro, B.A., Hamel, J.F., Gamboa, R.U. and Mercier, A. (2014) The cost of being valuable: predictors of extinction risk in marine invertebrates exploited as luxury seafood. *Proceedings of the Royal Society B* 281(1781):20133296. <https://doi.org/10.1098/rspb.2013.3296>

Inclusion of the following fifteen tarantula species in Appendix II: *Grammostola rosea*, *Acanthoscurria chacoana*, *Acanthoscurria insubtilis*, *Acanthoscurria musculosa*, *Acanthoscurria theraphosoides*, *Avicularia hirschii*, *Avicularia rufa*, *Avicularia avicularia*, *Catumiri argentinense*, *Cyriocosmus berate*, *Cyriocosmus perezmilei*, *Haplotremus albipes*, *Holothele longipes*, *Pamphobeteus antinous*, and *Umbyquyra acuminatum*

Proponents: Argentina, Plurinational State of Bolivia, Panama

Summary: Fifteen recognised species of tarantulas, from nine genera, are proposed for inclusion in Appendix II. *Grammostola rosea* is proposed for inclusion under Article II, paragraph 2(a) (lead species), while the 14 other species from the genera *Acanthoscurria*, *Avicularia*, *Catumiri*, *Cyriocosmus*, *Haplotremus*, *Holothele*, *Pamphobeteus*, and *Umbyquyra* are proposed as 'lookalikes'. These species are distributed across a range of South and Central American countries, with Bolivia (Plurinational State of, hereafter referred to as Bolivia) representing a common range State for all fifteen, including two endemics (*Cyriocosmus perezmilei* and *Haplotremus albipes*). None of the species has been assessed for the IUCN Red List, although *Grammostola rosea* and *Pamphobeteus antinous* are listed as vulnerable in national assessments in Chile and Peru, respectively. Currently 37 tarantula species are listed in CITES, with 36 listed in Appendix II and the Antilles pinktoe tarantula *Caribena versicolor* listed in Appendix III.

Current and historical population estimates for all the proposed species are lacking, and population trends unknown. These tarantulas inhabit a variety of ecosystems including deserts, scrublands, deciduous forests, and grasslands. They are typically found in burrows or natural cavities and are sensitive to habitat disturbance. The primary threat to these species appears to be habitat loss and fragmentation, driven by agricultural expansion, urban development, fire, and logging. Collection for the international pet trade is also reported as a secondary threat affecting all fifteen species.

Many of these species, particularly *Grammostola rosea*, are reported in the pet trade, with the USA and Europe being the main destinations. Between 2012 and 2021, over 74,000 live *Grammostola rosea* individuals were imported into the USA, 80% of which were wild-sourced, predominantly from Chile; this trade virtually ceased in 2015 when wild harvest of all tarantulas for commercial export was prohibited in Chile. In 2020, Bolivia prohibited the export of all wildlife for the pet trade. Trade in other species has also been reported in lower numbers, including wild-sourced individuals of *Pamphobeteus antinous*, *Cyriocosmus bertae*, *Catumiri argentinense*, and *Avicularia avicularia*. In other range States, regulations vary, and captive breeding is limited or absent.

Illegal trade, species misidentification and misreporting are ongoing concerns. Seizure records confirm the presence of low numbers of these species in unregulated and illegal trade, including incidents in Brazil, Colombia, the Philippines, China, and Europe. Experts report that juveniles and hatchlings are particularly difficult to identify to species level, potentially posing challenges for enforcement. However most juvenile spiders in trade are already legally captive-sourced. In contrast, adult specimens from captive sources are not commonly found in the legal trade, as long maturation times and limited profitability make large-scale raising of mature individuals for sale impractical.

Analysis: Comprehensive data on the historical and current wild population sizes of the fifteen tarantula species proposed for inclusion in Appendix II are lacking, and population trends remain unknown. Several of the proposed species appear in international trade, with *Grammostola rosea* reportedly among the most commonly traded tarantulas globally, the majority reported from Chile, which banned the export of all tarantula species in 2015. Reported trade in other species is low and largely reported as captive-bred. Misidentification and misreporting of trade appear to be common issues, with species often reported under synonyms or traded from non-range States, complicating regulation.

The impact of trade on wild populations remains unclear for *Grammostola rosea* or any of the other fourteen species, with little population information available. Given recent regulations in Chile, and the

absence of significant trade from 2017 onwards, there is no evidence to indicate that regulation of trade in any of these fifteen species is required to ensure that the harvest of specimens from the wild is not reducing the wild populations to a level at which their survival might be threatened; **none appears to meet the criteria for inclusion in Appendix II in Annex 2a or 2b** of Res. Conf. 9.24 (Rev CoP17).

If the Parties did consider that any species met the criteria in Annex 2a, it is not clear that other species would meet the criteria in Annex 2bA. Adult specimens of the species under consideration are reportedly distinguishable. Juveniles and hatchlings are difficult to differentiate and could present identification challenges for enforcement authorities; although tarantulas are frequently traded as juveniles and hatchlings, these life stages are predominantly sourced from captive breeding operations. No implementation challenges in currently listed species have been raised.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

No CITES Standard Reference covers these species of tarantula and no standard reference is recommended in the proposal for adoption. All fifteen tarantula species are recognised as valid in the World Spider Catalog (2025), with each of the genera containing additional species beyond those proposed for inclusion in Appendix II.

Species and description	Range	IUCN Global Category	National assessments
<i>Acanthoscurria chacoana</i> Brèthes, 1909	Argentina, Bolivia, Brazil and Paraguay	Not Assessed	
<i>Acanthoscurria insubtilis</i> Simon, 1892	Bolivia and Brazil	Not Assessed	
<i>Acanthoscurria musculosa</i> Simon, 1892	Argentina, Bolivia and Paraguay	Not Assessed	
<i>Acanthoscurria theraphosoides</i> Doleschall, 1871	Bolivia, Brazil, French Guiana and Peru	Not Assessed	
<i>Avicularia hirschii</i> Bullmer, Thierer-Lutz & Schmidt, 2006	Bolivia*, Brazil, Ecuador and Peru	Not Assessed	
<i>Avicularia rufa</i> Schiapelli & Gerschman, 1945	Brazil, Bolivia, Ecuador and Peru	Not Assessed	
<i>Avicularia avicularia</i> Linnaeus, 1758	Bolivia, Brazil, French Guiana, Guyana, Peru, Suriname, Trinidad and Tobago, Venezuela	Not Assessed	
<i>Catumiri argentinense</i> Mello-Leitão, 1941	Argentina, Bolivia* and Chile	Not Assessed	
<i>Cyriocosmus bertae</i> Pérez-Miles, 1998	Bolivia*, Brazil	Not Assessed	
<i>Cyriocosmus perezmilei</i> Kaderka, 2007	Bolivia	Not Assessed	
<i>Grammostola rosea</i> Walckenaer, 1837	Argentina, Bolivia and Chile	Not Assessed	Assessed as vulnerable in Chile.
<i>Haplotremus albipes</i> Simon, 1903	Bolivia	Not Assessed	
<i>Holothele longipes</i> Koch, 1875	Bolivia, Brazil, **, French Guiana, Guyana, Panama, Peru, Suriname, Trinidad and Tobago, Venezuela	Not Assessed	
<i>Pamphobeteus antinous</i> Pocock, 1903	Bolivia and Peru	Not Assessed	Assessed as vulnerable in Peru.
<i>Umbyquyra acuminatum</i> Schmidt & Tesmoingt, 2005	Bolivia and Brazil	Not Assessed	

* These countries were not considered a range State for the species in the World Spider Catalog (2025).

** Colombia was also included as a range State for *Holothele longipes* in the World Spider Catalog (2025).

IUCN Global Category

None of the 15 tarantula species are assessed on the IUCN Red List of Threatened Species.

Biological and trade criteria for inclusion in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)**A) Trade regulation needed to prevent future inclusion in Appendix I**

No quantitative population size or trend data is available for any of the fifteen species. *Experts report there is no national-scale population assessment for Grammostola rosea, only localized studies (Noboa in litt., 2025). In Chile in 2016, a study reported that G. rosea occurred in lower densities in areas of high anthropogenic disturbance compared with areas of low anthropogenic disturbance (e.g., 750 individuals per hectare in Cerro Huechuraba and 1,350 individuals per hectare in Panul Park respectively (Aguilera et al., 2019)).*

There is a lack of detailed data on the distribution and biology of the fifteen species.

Grammostola rosea

This species occurs along the Andes in northern and central Chile, Bolivia and northwestern Argentina. In Bolivia, the species has been observed in the two southwestern departments of Potosí and Oruro Departments; and in Chile the regions of Atacama, Coquimbo, Valparaíso, Metropolitan and Biobío. *Grammostola rosea* occurs from 0 to 3,700 metres above sea level (m a.s.l.) in Bolivia, and 0 to 1,500 m a.s.l. in Chile, where its abundance was highest at low altitudes and decreased above 700 m a.s.l.

Grammostola rosea inhabits a wide range of habitats from deserts and scrublands to deciduous forests and grasslands. Tarantulas within the genus *Grammostola* have slow life strategies with females recorded to live around 15 to 20 years, and males from two to 10 years. Fecundity data was not available for this species but noted among tarantulas (Theraphosidae) to vary greatly from dozens to over a thousand eggs.

The species *Grammostola rosea* was reported to be affected by the loss, fragmentation and declining quality of its habitat, driven by agricultural and urban expansion, fire and logging. *The proponents state that in Chile, the species' declining habitat extent and quality were projected to decrease by more than 30% over the next 18 years, representing three generations. In Chile, collection of Grammostola rosea from the wild for the international pet trade was reported to have led to local extirpations. Experts report there is no evidence that population decreases can be specifically attributed to trade (Noboa in litt., 2025).*

B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences

All fifteen of the proposed species were reported in international trade, observed on online wildlife trade platforms. In a global study of tarantula trade using LEMIS data, from 2000 to 2021, it was reported that *Grammostola* species were among the most highly traded tarantulas to and from the USA, with a high proportion of this trade in wild-sourced individuals. However, it was also noted that species had been reported under synonyms, such as *Grammostola spathulate* (a synonym of *Grammostola rosea*), and that wild-sourced individuals were mostly reported from non-range States. This indicates a high degree of misreporting of tarantula species in international trade is taking place and that effective identification and regulatory mechanisms to prevent unregulated and illegal trade are lacking.

Experts report that the USA represents the largest global market for tarantula species (Cooper in litt., 2025). According to LEMIS database, over a ten-year period, from 2012 to 2021, a total of 74,222 live individuals of Grammostola rosea (including the G. rosea synonym G. porteri [29,970, 40%]) were imported by the USA from range States (Argentina, Bolivia and Chile) for commercial purposes. This trade predominantly occurred from 2012-2015, markedly decreasing in 2016 and no trade reported from 2017 to 2021.

Virtually all of these live individuals originated from Chile (>99%, 74,182), with only 40 individuals originating from Bolivia (<1%) and none from Argentina. Most of this live trade was reported as wild-sourced individuals (82%, 60,595), with the rest in captive-bred individuals (18%, 13,627).

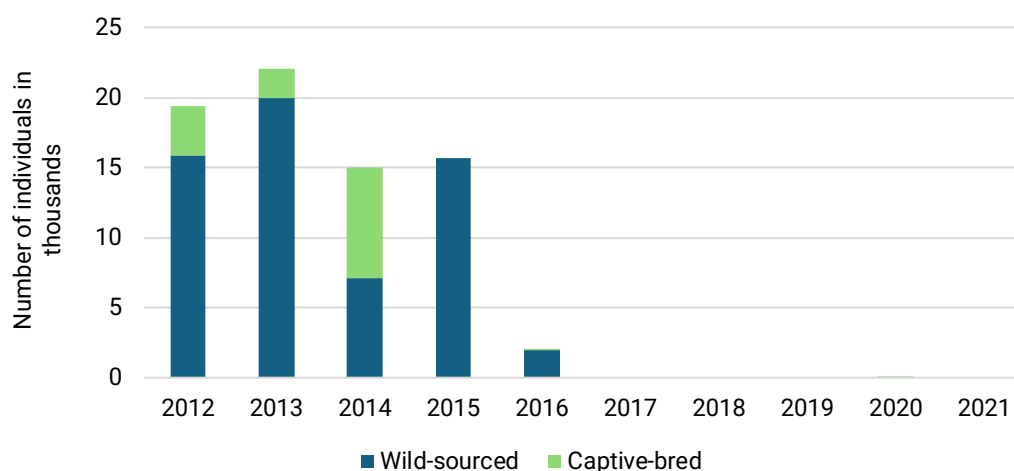


Figure 1: Imports of live *Grammostola rosea* (including *G. porteri*) into the USA between 2012 and 2021. Source: LEMIS database.

Over the same period, in addition to *G. rosea* (and *G. porteri*), a total of 69,358 live *Grammostola* spp. were reported as imported by the USA for commercial purposes from *G. rosea* range States (Argentina, Bolivia and Chile). Just over half of this trade was in wild-sourced individuals (37,061, 53%). No trade occurred from 2017 to 2021.

Trade in the other 14 tarantula species over the same period totalled 80 live wild-sourced individuals of *Pamphobeteus antinous* (50), *Cyriocosmus bertae* (16), *Catumiri argentinense* (12) and *Avicularia avicularia* (2) reported as imported by the USA for commercial purposes from the species' range States. No trade occurred from 2016 to 2021.

The proponents state that in Peru, in 2023, 1,787 tarantulas from 11 unspecified species were exported to Europe, the United Kingdom of Great Britain and Northern Ireland (UK) and the USA.

Illegal trade

The proponents state that illegal and unregulated trade represent a significant threat to many tarantulas (*species unspecified*). Evidence from current trade indicates that misidentification and misreporting of tarantula species in trade, such as *G. rosea*, is resulting in illegal and unsustainable trade.

No seizure data specific to the fifteen tarantula species was reported in the SS. However, general seizure data was presented for tarantula species, including seizures potentially involving the fifteen species sourced from the wild. These seizures included 20 individuals (*species unspecified*) seized in the USA originating from Bolivia in 2005, and two instances of tarantula seizures in Bolivia in 2019 and 2022. In Brazil, in 2021, 19 *Acanthoscurria* spp. were seized. In Colombia, 188 tarantulas were seized between 2022-2024 (*species unspecified*). In Paraguay, 291 tarantulas from 7 species were seized in 2013, allegedly with false permits; some species (*unspecified*) reportedly occurring in Bolivia. In Peru, two seizures were reported in 2024, including six and 320 specimens (*species unspecified*).

In Bolivia, the lack of specific regulations for *Grammostola rosea* and the absence of authorised breeding facilities, indicate that many individuals exported or sold online are likely sourced illegally through unregulated wild collection.

No seizure records were identified in EU-TWIX.

WiTIS contained three seizure records for the 15 tarantula species. Two seizures in the Philippines included one live *Grammostola rosea* individual in 2017, and 11 tarantula individuals comprising *Grammostola rosea* and *Acanthoscurria chacoana* in 2019. A seizure in Brazil in 2019 included two live *Pamphobeteus antinous* individuals.

Further seizure records provided by TRAFFIC on *Grammostola rosea* included 700 individuals seized in China in 2014 (Des Bois, 2015b), one individual seized in France in 2014 (Des Bois, 2015a), one individual seized in the Canary Islands in 2022 (Des Bois, 2022), and one individual seized in French Guiana in 2022 (Des Bois, 2024). Seizures involving the other 14 species included: a seizure in Brazil of a *Acanthoscurria* spp. in 2024 (Des Bois, 2024), 25 *Avicularia avicularia* individuals seized in Brazil in 2024 (Des Bois, 2024), one *Avicularia avicularia* individual discovered in the UK in 2022 (Des Bois, 2022), 33 *Avicularia avicularia* individuals seized in French Guiana in 2022 (Des Bois, 2024), and 23 individuals belonging to the *Xenesthis* and *Pamphobeteus* genera seized in Colombia in 2018 (Des Bois, 2019).

Inclusion in Appendix II to improve control of other listed species**A) Specimens in trade resemble those of species listed in Appendix II under Res. Conf. 9.24 (Rev. CoP17) Annex 2 a or listed in Appendix I**

Currently 37 tarantula species are listed in CITES, with 36 listed in Appendix II and only the Antilles pinktoe tarantula *Caribena versicolor* listed in Appendix III. The species listed in Appendix II include the five genera: *Aphonopelma* (1), *Brachypelma* (11), *Poecilotheria* (15), *Sericopelma* (2) and *Tliltocatl* (7). According to the CITES Trade Database, direct trade in *Theraphosidae* over the most recent ten years, 2015-2024, comprised almost entirely of just under 300,000 live captive-bred individuals for commercial purposes. Predominantly in *Tliltocatl albopilosum* (78,504, 27%), *Brachypelma smithi* (31,461, 11%) and *Brachypelma boehmei* (30,096, 10%). Direct trade in live wild-sourced tarantula species for commercial purposes as reported by exporters, involved 50 *Tliltocatl albopilosum* individuals exported by Nicaragua to USA in 2017. Importers reported a total of 1,032 individuals across five species (*T. albopilosum*, *B. boehmei*, *B. auratum*, *B. baumgarteni*, *B. hamorii* and *B. emilia*).

Identifying tarantula species is difficult for non-experts, especially when dealing with juveniles and hatchlings. *Grammostola rosea* was reported to be easily confused with other *Grammostola* species (22 recognised species in World Spider Catalog), especially *G. pulchra*. However, these life stages are predominantly sourced from captive breeding operations.

Additional Information**Threats**

The primary threats to *Grammostola rosea* are habitat loss, degradation and fragmentation. In Chile, where *G. rosea* is associated with sclerophyllous forests and hawthorn patches, its area of occupancy was projected to decrease by more than 30% over the next 18 years. A 2020 study noted the *G. rosea* population decreases were due to expanding real estate and agricultural development, which degrade habitat quality and isolate populations. In Bolivia, similar pressures affect this species, with extensive deforestation driven by agricultural expansion, fires, and land degradation affecting key ecosystems such as the dry and desert puna ecoregions. Experts report the species' intrinsic traits such as slow sexual maturation, low fecundity, and long-life cycles make it vulnerable to overexploitation over the medium to long term (Noboa in litt., 2025).

In general, arachnids such as tarantulas, are reported to face combined risks from intrinsic factors such as slow life histories and limited dispersal ability, and extrinsic factors including sensitivity to degraded natural habitats. Collection for the international pet trade was reported to be a secondary threat affecting virtually all fifteen species.

Conservation, management and legislation

In Bolivia, the Bolivian Ministerial Resolution (No. 042/2020 Article 19) stipulates that the collection of tarantula species and their export for the commercial pet trade is prohibited (as is the export of any wildlife species for the pet trade).

In Chile, *G. rosea* has been included in the hunting law (No 19.473) since 2015, and is classified as vulnerable in the national Red List. Experts report that following the inclusion of tarantulas in the Chile Hunting Law (No 19.473) their capture, and export was prohibited (Noboa in litt., 2025). This corresponds with LEMIS data showing a notable drop and cessation in trade into and from the USA of wild-sourced *Grammostola rosea*, including synonyms, from 2015 onward (see Figure 1).

In Peru, it was reported that there are regulations governing the trade of *P. antinous* from zoos, and in the Red Book of Threatened Fauna of Peru the species is listed as vulnerable due to population reductions caused by habitat fragmentation and illegal trade.

Artificial Propagation/captive breeding

There are no official records of authorised captive-breeding centres for commercial or conservation purposes for these tarantula species in Bolivia. No information is provided on captive-breeding facilities in other countries. Experts report that breeding tarantulas is not profitable, as they can take up to ten years to reach a marketable size (Noboa in litt., 2025).

Implementation challenges (including similar species)

Identification of tarantula species by non-experts, especially juvenile and hatchling specimens, represents a significant challenge to effective regulation and enforcement. Due to taxonomic uncertainty, implementation likely requires adequate identification materials for customs officials to avoid misidentification.

The proponents list a number of potential lookalike tarantula species for the fifteen species included in the proposal. As these species are not included in the proposal, it is unclear how these challenges will be mitigated.

All tarantula species included in the proposal appear to be those occurring in Bolivia. It is therefore unclear whether tarantula species in other range States also resemble those included in the proposal, and what the impact of listing only the species in the proposal would have on their trade.

Other Considerations

Should the proposal not be adopted, the range States may consider listing their respective species in Appendix III. In which case Bolivia could stipulate a zero-export quota for wild specimens, which would reflect the current national prohibition on commercial export of tarantulas for the pet trade.

As stated in CoP20 Inf. 6, no CITES Standard Reference exists for tarantulas in these genera and no standard reference has been proposed for adoption for the species included in this listing. Each of the genera contains additional species beyond those proposed for inclusion in Appendix II. As such, a standard reference for the Theraphosid tarantulas appears to be a task for the Animals Committee following CoP20 if Proposal 38 is adopted.

References

- Aguilera, M.A., Montenegro, V.R. and Casanueva, M.E. (2019) Impact of disturbed areas on Theraphosidae spiders diversity (Araneae) and first population data of *Grammostola rosea* (Walckenaer) in Panul Park. *Ecology and Evolution* 9(10):5802–5809.
- Cooper, E. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Des Bois, R. (2015a) On the Trail: Information and analysis bulletin on animal poaching and smuggling n°6 / 1 July – 30 September 2014. With the encouragement and financial support of the Fondation Brigitte Bardot, the Ligue pour la Protection des Oiseaux and the Fondation Franz Weber.
- Des Bois, R., (2015b) On the Trail: Information and analysis bulletin on animal poaching and smuggling n°7 / 1 October – 31 December 2014. With the encouragement and financial support of the Fondation Brigitte Bardot, the Ligue pour la Protection des Oiseaux and the Fondation Franz Weber.
- Des Bois, R., (2019) On the Trail: The defaunation bulletin Quarterly information and analysis report on animal poaching and smuggling n°23 / 1 October 2018 – 31 January 2019. With the encouragement and financial support of the Fondation Brigitte Bardot, the Ligue pour la Protection des Oiseaux and the Fondation Franz Weber.
- Des Bois, R., (2022) On the Trail: The defaunation bulletin n°35 / 1 January – 31 March 2022. With the encouragement and financial support of the Fondation Brigitte Bardot, the Ligue pour la Protection des Oiseaux and the Fondation Franz Weber.
- Des Bois, R., (2024) On the Trail: The defaunation bulletin n°45 / 1 April – 31 May 2024. With the encouragement and financial support of the Fondation Brigitte Bardot, the Ligue pour la Protection des Oiseaux and the Fondation Franz Weber.
- Noboa, K. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Resolución Ministerial (2020) N° 042/2020 de 10/02/2020 que aprueba el "Reglamento de Restricción y Control al Comercio de Vida Silvestre". Gobierno del Estado Plurinacional de Bolivia. Available at: <https://faolex.fao.org/docs/pdf/bol217993.pdf>
- World Spider Catalog (2025) World Spider Catalog. Version 26. Natural History Museum Bern. Available at <http://wsc.nmbe.ch>. Accessed on 14 July 2025.

Inclusion of South African Abalone *Haliotis midae* in Appendix II with an annotation “dried form only”

Proponent: South Africa

Summary: South African Abalone *Haliotis midae*, locally known as “perlemoen”, is a large marine snail that is a highly sought after seafood delicacy. The species is endemic to South Africa where it is found in kelp forests along rocky coasts. The proposal is for inclusion in Appendix II with an annotation limiting application of CITES to specimens traded in the “dried form only”. The proponent explains that this is the predominant form that illegally taken wild abalone is traded, in contrast to legal exports that mostly derive from a large, regulated commercial abalone farming/ranching model, the products of which are exported in fresh, frozen, dried or canned form. The proponent believes that it is unnecessary to regulate exports of fresh, frozen or canned form under CITES since the impact on the wild stock is negligible and the administrative burden on South Africa would be significant as would be the socio-economic and livelihood impact.

The species is slow growing, has a late maturity and can live for over 30 years. It was globally assessed as Endangered on the IUCN Red List in 2020 with a continuing population reduction and local extirpations noted due to overexploitation for trade. There are no estimates of its population size. In one of its main conservation areas in the Western Cape, Betty’s Bay Marine Protected Area, by 2012, densities of adult members of the species had declined to 1% of the level recorded in the 1990s. Over a similar period, Dyer Island, another conservation area in the Western Cape, reported high densities in 1990, with 40% of individuals being above legal size (4.5 inches or 11.4 cm) between 0-5 m depth; by 2013, only 4% of the population was above legal size. Catch Per Unit Effort (CPUE), where it has been recorded, has significantly declined since first recorded in 1983. In some fishing zones, the spawning biomass is estimated to have dropped below 20% of its estimated pre-fished biomass. The latest status review of marine fishery resources in South Africa concluded that the *Haliotis midae* remains in a depleted state, with the collective fishing pressure from both the legal commercial fishery and illegal harvesting being higher than is sustainable, and likely to worsen in the future.

The Eastern Cape, although not historically part of the commercial fishery, has become a major source for the illegal *H. midae* trade since substantial stocks were found in 1997, with observed declines at Cape Recife and Bird Island. The elevated level of illegal fishing effort was determined to have had a negative effect on stocks with symptoms of decline and local collapse. More recent research suggests that wild spawner biomass may have persisted despite heavy poaching.

Haliotis midae is primarily traded legally from farms (3000 tonnes/year) and ranches (200 tonnes/year), sold in live, frozen, canned and dried forms. There is also a small commercial wild fishery with a Total Allowable Catch (TAC) of 41.6 tonnes in 2024. Commercial divers are allocated an annual quota in fishing zones, the catch from which they often sell on to the companies running the farms who will export the product. To export the species the exporter needs to be able to demonstrate its origin and the holder of the quota. The species is also traded illegally from wild harvest at an estimated 2000-2500 tonnes/year predominantly in dried form, with a small quantity traded as frozen. Data show that nearly 60% of dried imports into Hong Kong SAR of China were exported by South Africa, with the remainder reported as exports from non-range States suggesting illegal harvest is being moved across borders for export. Overall, the greatest threat to the species is the illegal harvest of wild populations. There is however an economically important legal fishery.

Analysis: South African Abalone *Haliotis midae* is traded both legally and illegally. Live, frozen, canned and dried forms are traded legally internationally, mostly from regulated farming and ranching operations with relatively small wild TAC; dried forms are also traded illegally from wild harvested stock, with a small quantity traded in frozen form. Harvest for illegal trade is having a major impact on wild populations. Where data are available, reductions in densities are significant and historical data indicate reductions in CPUE. In some key conservation areas, the species has undergone a marked historical decline to an extent where it may already meet the biological criteria for inclusion in

Appendix I in Res. Conf. 9.24 (Rev. CoP17) as some populations have seen spawning biomass drop below 20% of its estimated pre-fished biomass. Considering the species' biological vulnerability, clearly trade regulation is needed to ensure that harvesting does not reduce wild populations to a level that threatens their survival and therefore **the species meets the criteria for inclusion in Appendix II** set out in Annex 2a of the Resolution.

However, **it is not clear that the Parties could adopt this proposal as currently presented, since the proposed annotation appears to be inconsistent with the text of the Convention.** Definitions set out in Article I of the Convention text do not provide for specification of which parts and derivatives of an animal species listed in Appendix I or II will be subject to CITES trade controls. Article I(b)(2) specifies that "in the case of an animal: for species included in Appendices I and II", specimen "means any animal, whether alive or dead, and any readily recognizable part or derivative thereof". This is further emphasised in Res. Conf. 11.21 (Rev.CoP19) which states that "RECALLING that an annotated listing of an animal or plant species in any of the three Appendices always includes the whole live or dead animal or plant, as well as any specimen specified in the annotation".

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

In 2007, Haliotis midae was listed by South Africa in CITES Appendix III, but this was withdrawn in 2010 as it was considered too difficult to implement (Peters, 2021).

Taxonomy

Haliotis midae (Linnaeus, 1758)

Range

South Africa

IUCN Global Category

Endangered A2ad (assessed 2020, ver 3.1)

Biological and trade criteria for inclusion in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)

Haliotis midae is endemic to South Africa and the only commercially traded South African *Haliotis* species. *Haliotis midae* are slow growing, reaching sexual maturity at around seven years of age, and take approximately 8-9 years to reach the minimum legal size of 11.4 cm shell breadth (SB). They reach a maximum size of 18 cm SB and are believed to live to an age of greater than 30 years (DFFE, 2025). The SS states that a three-generation period for *Haliotis midae* would be 42 years.

This species of abalone occurs off the coast of South Africa from St. Helena Bay in the Western Cape to just north of Port St Johns on the East Coast with some specimens reported as far north as Durban. This species lives from the intertidal to 25 m depth. The highest densities occur in waters less than 5 m depth (DFFE, 2025).

Population

In the 2020 IUCN Red List Assessment of Haliotis midae it was noted that owing to the precipitous decline in populations of Haliotis midae, the primary conservation effort is best focussed on restructuring fisheries to offer security and economic return to the fishers through long-term exclusive fishing rights that incentivise stock protection while reducing levels of poaching through robust enforcement. The current policy of promoting stock rebuilding by means of Haliotis midae ranching and stock enhancement is working in areas where it is being implemented and needs wider encouragement. In the absence of this approach, the decline in population of this species would continue. Owing to some resilience in recruitment and the increasing use of stock enhancement, this species was assessed as Endangered (EN A2ad), however, failure to extend recovery programmes including legal enforcement and anti-poaching measures will inevitably result in this species likely becoming Critically Endangered in the near future (Peters, 2021).

In the Western Cape at Dyer Island, transect data in 1990 showed high densities, with 41% of the species population occurring between 0 and 5 m depth being above legal size. However, by 2013, its density had collapsed and only 4% of the population was above minimum legal size. As the Betty's Bay MPA no longer functions as a closed system (see below), it is reported that Dyer Island should continue as a closed area (DFFE, 2025). DFFE (2025) reported that the Haliotis midae resource remains in a depleted state, with the collective fishing

pressure from both the legal commercial fishery and illegal harvesting being higher than is sustainable, and therefore likely to worsen the resource status in the future. With heavy fishing pressure being noted and the stock status being described as Depleted/Heavily depleted. There are probably some populations that have been extirpated, due to intensive illegal fishing giving rise to fragmentation across its distribution (Peters, 2021).

Population trends

Based on Commercial catch per unit effort (CPUE) (DFFE, 2025) in the Western Cape, the commercial fishery is managed across seven zones, A-G, with A in the south and then progressing north (Peters, 2021).

Zones A and B

An inspection of the nominal CPUE data shows no marked changes in the CPUE from Zone A over the past three seasons. The apparent slight increase in the nominal CPUE in Zone B over the past three seasons must however be weighed against the consideration that these are nominal and not standardised CPUE data, and concerns that have been raised about the accuracy of CPUE data-reporting in Zones A and B in recent years. One of these is the concern that *Haliotis midae* commercial fishing vessels have been observed around the Dyer Island closed area, so that the recent CPUE values for Zones A and B may have been artificially inflated by catches off Dyer Island.

Zones C and D

Spawning biomass projections in 2016 showed continuing declines in resource abundance in Zones C and D at recent estimated levels of illegal take. The resource has been severely reduced by the lobster-urchin effect on recruitment of abalone (see Threats), in addition to the effects of illegal fishing. Populations in these two Zones were also estimated to be below the 20% limit reference point set out in the management objectives (suggesting populations had depleted by 80%). No new data have become available to suggest a change in the previously estimated status of the resource in these Zones.

In the Betty's Bay MPA (situated in Zone D) which is the species main conservation area, in 2012, the mean density had been reduced to 1% of former pristine levels, recorded in the 1990s (Peters, 2021).

Zones E to G

Inspection of the nominal CPUE data shows a decline over the past three seasons (2020/21 to 2022/23) in Zones E, F and G. Under-reporting of catch data has been a challenge.

Eastern Cape

Although the Eastern Cape has not historically been part of the commercial fishery and was considered less intensively exploited, since 1997 substantial stocks of abalone have been found and it has become a major source for the illegal abalone trade. The failure of the state to issue fishing rights and neglect enforcement, allowed the emergence of a full-scale illegal fishery which, by 2005, was operating a fleet of 30 purpose-built vessels extracting 1,000–2,000 t of abalone with an export value of USD35–70 million per year. Surveys in the southern section of this region suggest over-fishing has had an impact, with total density and mean size significantly negatively correlated to exploitation indicators, and because of the high commercial value of abalone, illegal fishing has spread into remote and secluded localities. Sexually mature individuals are scarce at most sites and in areas of high exploitation pressure, abundance of subadults has been reduced, with abalone populations generally deteriorating. During surveys at the remote Great Fish Point, infralittoral abalone densities had fallen by almost 60% since initial sampling, with a further estimated 6000 individuals taken during a single poaching event in 2004. Further research in the Cape Recife area of the Eastern Cape, describes recruitment remaining consistent with no trends discernible between historical survey estimates and the results obtained. Although there was wide variation in the cryptic abalone abundance, the consistency in recruitment suggests that wild spawner biomass has persisted despite heavy poaching, and this was likely owing to the small size at sexual maturity of *Haliotis midae* in the area and/or viable pockets of older spawner stock not taken by the illegal fishery (Peters, 2021).

History of exploitation.

The commercial (diver) fishery for *Haliotis midae* started in the late 1940s and was dominated by five large *Haliotis midae* processing plants. Catches were initially unregulated and in 1965 catches reached a peak of nearly 3,000 t. By 1970, catches had rapidly declined but the fishery remained stable with a Total Annual Catch of around 600-700 t until the mid-1990s. However, after this period there were continuous reductions in commercial catches (DFFE, 2025).

In the 1990's, the recreational fishery was extensive and there was a significant increase in illegal fishing activities which continued, resulting in declines in *Haliotis midae* and in 2003/04, the recreational fishery was closed. In the late 1990's, to increase participation in the fishery, particularly for people who had been previously marginalised, Subsistence Rights were introduced in 1998/99, and were replaced by two-year medium-term Rights. In 2003/04, 10-year long-term rights were allocated, broadening participation in the fishery. At the same time, management zones were replaced with Territorial Use Rights in Fisheries (TURFs), which aimed to create a sense of ownership of the

resource, introducing co-management of the resource and improving compliance with regards to illegal activities. Despite all these efforts, illegal fishing remained high (DFFE, 2025).

Since the early 1990s, poaching of this species has taken place on an extensive scale, resulting in a rapid reduction in average size and densities of individuals throughout the species' range, including marine protected areas (MPAs). The SS reports that wild stocks in accessible sites throughout the range have been depleted. *Illegal fishing occurs in all areas of Haliotis midae range but moves around areas depending on resource abundance and law enforcement presence. In February 2008, diving prohibitions were introduced, and the commercial fishery was closed. In 2010, the fishery was opened again, conditional on a 15% per annum reduction in poaching. The reduction in illegal harvesting has not been met (DFFE, 2025).*

Recent Trade

Legal trade

Most of the legal exports of *Haliotis midae* originate from abalone farms and ranches. A limited wild catch is permitted, the Total Allowable Catch in 2024 was just 41 tonnes (in 1995/96 the TAC was 615 tonnes and in 2007/08 it was 75 tonnes) which is divided between 302 commercial divers. Most of the wild catch quota is sold to some of the companies who own abalone farms. *Haliotis midae* produced legally is exported in fresh, frozen, dried or canned form.

Confiscations of illegally harvested Haliotis midae are also sold through a tender process to processing facilities which then export the Haliotis midae to the legal market. Income generated from these sales go to the Marine Living Resources Fund to aid research, management and conservation of all marine resources in the country, as stipulated by the Marine Living Resources Act (Okes et al., 2018).

Whilst Haliotis midae was listed in CITES Appendix III between 2007 and 2010, CITES Trade Database (2025) records were available for 2008 and 2009. In 2008, it was reported that ~77 tonnes of wild-sourced Haliotis midae were exported by South Africa for commercial purposes, ~52 tonnes of meat (exported to Hong Kong SAR of China), ~8.4 tonnes in the live form (primarily exported to Hong Kong SAR of China) and ~16 tonnes of shells (exported to Germany, India, Netherlands, Poland, and the United States of America). In 2008, ~3,115 tonnes of captive-bred Haliotis midae were exported by South Africa for commercial purposes, ~1,929 tonnes in the live form, ~1,164 tonnes of meat, and ~22 tonnes of shells. In 2009 similar volumes were exported for commercial purposes, ~3,166 tonnes of captive-bred Haliotis midae were exported by South Africa, ~1,494 tonnes in the live form, ~1,570 tonnes of meat and ~103 tonnes of shells.

Abalone farms

The SS states there are 13 abalone farms in South Africa, 12 in the Western Cape and one in the Eastern Cape. The South African abalone farms produce 3000 tonnes/year. One Namibian farm produces 90 tonnes/year. The Abalone Farmers Association of South Africa (AFASA) is an important membership organization for the South African farms and the South African Abalone Export Council (SAAEC) has been constituted within the AFASA group (although not all members of AFASA are members of the SAAEC). The highest volumes of *Haliotis midae* are exported by four large companies (Abagold, I&J (formerly Irvin & Johnson are now owned by The J. R. Simplot company in the USA), Sea Harvest/Aquinion and HK).

Abalone ranching

Rights for abalone ranching (valid for 15 years) are granted by the Department of Forestry, Fisheries and the Environment (DFFE) and are seen as an important part of the government initiative Operation Phakisa. There are currently five abalone ranches in South Africa, producing 200 tonnes of *Haliotis midae*/year (AFASA, 2025). If successfully implemented, commercial scale abalone ranches could enhance Territorial User Rights Fisheries (TURF). *Wild caught abalone are used in all of the farms for brood stock to maintain a strong gene pool (Burgener in litt., 2025) and the young hatchery produced Haliotis midae are seeded into the wild in kelp beds, making a significant contribution to the total abundance, with increases ranging between 7% and 20% dependent on quality of habitat, increasing to as high as 80%. Private security companies are then contracted to protect the ranches and farms (Peters, 2021). It should be noted that the term "ranching" used in the SS does not meet the CITES definition of "ranching".*

Between 2000 and 2016, Hong Kong SAR of China imported 90% of *Haliotis midae*. The rest was imported by Japan (3%), Singapore (2%), Taiwan Province of China (2%), Macau SAR of China (1%), and the remaining 2% by mainland China, Malaysia, the USA and Canada. The only legal production of the endemic *Haliotis midae* outside of South Africa, is from a single abalone farm near Ludertiz, Namibia which reportedly produced 90 tonnes of *Haliotis midae* in 2023, also mainly for export to Hong Kong SAR of China. *Namibia's farm is land based, established in 2003, using an initial stock of wild-caught juvenile abalone (Okes et al., 2018).*

Illegal fishing and trade

The SS states that illegal exploitation of wild *Haliotis midae* populations is widely recognized as an extreme case of an IUU (illegal, unregulated or unreported) fishery. Between 1994 and 2006, there was a ten-fold increase in the illegal *Haliotis midae* trade. Since then, it has been estimated that over 20,500 tonnes of *Haliotis midae* had been poached and illegally traded over a 10-year period, and this situation has worsened since 2014. The illegal trade in *Haliotis midae*, at about 2000 – 2500 tonnes/year (estimated between 2000-2016) is over 50 times greater than the current Total Allowable Catch (41.6 tonnes in 2024). *The total mass of imports was 55,863 t, while only 18,905 t was legally produced, suggesting 36,958 t was illegally harvested, representing an average of 2,174 t per annum, and equating to over 96 million individual abalone illegally harvested over this period (Okes et al., 2018).*

The SS notes that most of the illegal trade is of abalone in a dried form, apart from a small quantity traded in frozen form. Cross border trade in illegally harvested *Haliotis midae* represents an estimated 43% of total exports. Data from the Hong Kong Census and Statistics Department showed that only 57% of dried *Haliotis midae* imports in Hong Kong SAR of China were exported by South Africa. The rest were exported by countries where *Haliotis midae* does not occur naturally: Mozambique (21%), Namibia (7%), Zimbabwe (7%) and Zambia (6%), with the remaining 2% coming from Angola, eSwatini, the Congo and Kenya combined. Namibia, with one abalone farm, is an exception, but at the time of the TRAFFIC report by Okes et al. (2018) was producing far less than Hong Kong SAR of China's import data showed. Since then, Lesotho has also been flagged as a transshipment country for poached *Haliotis midae*, dried and in bags disguised as bags of potatoes, then transported to Asia via Zimbabwe.

The SS states that there are no recorded incidents of illegal trade in live *Haliotis midae* and there are no known incidents of illegally sourced *Haliotis midae* being used by legal canned producers.

The management objectives for Haliotis midae to ensure sustainable utilisation and recovery have been to prevent the Haliotis midae spawning biomass dropping below 20% of its estimated pre-fished biomass and see it recover to 40% of that level within 15 years of reopening the commercial fishery in 2009/10 (DFFE, 2025). It was noted that to achieve this, illegal harvesting must be substantially reduced.

Additional Information

Threats

Predation

Urchins (Parechinus angulosus) provide protection to juvenile Haliotis midae, which hide beneath their spines away from predation. In some zones (C and D) rock lobsters Jasus lalandii are encroaching onto abalone territory where they are predating on the urchins which in turn is leaving Haliotis midae exposed to predation. In those areas where rock lobster intrusion has been reported there is total recruitment failure of Haliotis midae, however, this is probably a temporary effect and not an indication of imminent biological population collapse, since the lobster population has its own overfishing problems. Although the importance of urchins as shelter to young Haliotis midae is not in doubt, studies in the Eastern Cape show that newly settled abalone do not have an obligate relationship with urchins and that 50% counted in a recruitment survey were not under urchins (Peters, 2021).

The ecosystem state in Zones A and B is currently different to Zones C and D, with very few lobsters present, a lower biomass of seaweeds and sessile species, more encrusting corallines, and urchins and grazers still present in relatively high abundance (DFFE, 2025).

Allee effect

Close proximity of sexes is necessary for success in broadcast spawning. Where individuals become too widely dispersed, as little as more than 2 m apart, recruitment failure may follow, leading to shrinking populations and eventual local extinction. In addition, the larvae of haliotids rapidly sink and become immediately benthic with limited dispersal ability, severely limiting opportunities for recovery in the event of stock depletion. There is some defence in that young Haliotis midae, up to 80 mm length, live in a cryptic state offering inbuilt resilience, allowing a reservoir of individuals to emerge. Furthermore, with populations on reefs highly heterogenous, small pockets of inaccessible abalone in exposed open coast are likely to seed larvae over wide areas (Peters, 2021).

Disease

Abalone diseases originating from pathogens occurring in the natural environment and discharged into marine habitats from aquaculture facilities have plagued the industry for decades. However, the abalone aquaculture industry in South Africa has a very strict protocol which forbids translocation of abalone between regions, minimising the threat (Peters, 2021).

Conservation, management and legislation

The SS states that the main national legal framework governing fisheries management in South Africa is the Marine Living Resources Act (MLRA), 1998 (Act No. 18 of 1998). Although the MLRA is based on fisheries

management principles of optimum use, conservation and a precautionary approach, these have failed for a variety of reasons.

The SS states that in 2003, the Hermanus "Green Court" was a first for South Africa, launched as a pilot project to deal with prosecutions of marine environmental crime. Functioning as a Regional Court, by 2005 it had an 85% conviction rate, but in 2006 the Department of Justice decided to close it. Abalone poaching has worsened immensely since then.

The SS states that abalone aquaculture and abalone ranching are both highly regulated in South Africa, with the legal requirements including permits under the MLRA for grow-out facilities, to possess brood-stock and to operate a hatchery. Ranching is also affected by provincial conservation legislation (such as the Northern Cape Nature Conservation Act (Act 9 of 2009), the Nature and Environmental Conservation Ordinance 19 of 1974 (Cape) and the Northern Cape Planning and Development Act No. 7 of 1998).

As *Haliotis midae* is an Endangered species, its trade has been regulated in the European Community since 2008 triggered by the CITES Appendix-III listing and following the earlier EC Council Regulation. *It is now listed in Annex D of the EC Council Regulation to allow the trade to be monitored more closely, requiring an import notification to be submitted by the importer (UNEP, 2025).* Although the USA only imports a fraction (< 1%) of *Haliotis midae* exports, two USA laws that apply to endangered species in fisheries (the Lacey Act and the Pelly Amendment) may also apply to wild caught *Haliotis midae*.

The SS states that the Territorial User Rights in Fisheries (TURF) policy, introduced in 2003 in the hope that coastal communities would play a significant role in combatting illegal fishing, has failed. However, a blend between TURF and abalone ranching may have potential. At present, in response to the poaching situation, most of the fisheries law enforcement budget is spent on Fisheries Protection vessels, but this has been largely ineffective in the management and conservation of *Haliotis midae* populations. Although regular abalone stock assessments are necessary to determine TACs, the last assessment of the *Haliotis midae* fishery was undertaken in 2016. Since then, TACs have been based on outdated data: 96 tonnes in 2017/18 and 2018/19, 50.5 tonnes in 2019/20, 2020/21 and 2021/22 and according to a legal quota holder, 41.6 tonnes in 2024.

Artificial Propagation/captive breeding

The SS states that only one *Haliotis* species is farmed in South Africa (*Haliotis midae*). Since the 1990s, South Africa has become a major producer from land-based farms and, to a lesser extent, from ranching. In total, 13 South African abalone farms produce 3000 tonnes of *Haliotis midae* with a further 200 tonnes from five ranches. While South Africa is the largest non-Asian abalone producer from aquaculture, current production is about 1.3% of global production.

The SS states that in 2010, the South African government simultaneously identified areas for abalone ranching and published guidelines for marine ranching and stock enhancement. The guidelines stressed that stock enhancement will not be considered in areas where "stocks have not been depleted below 20% of pre-exploitation levels". The challenge from a conservation genetics perspective is that due to decades of uncontrolled decimation of wild *Haliotis midae* stocks, this applies to the entire former geographic range of *Haliotis midae*.

Selective breeding of *Haliotis midae* for aquaculture is well advanced in South Africa and regulated through the Genetically Modified Organisms Act 15 of 1997 and the Animal Improvement Act 62 of 1998. Concerns about the risks posed to the genetic integrity of wild *Haliotis midae* stocks if commercially grown abalone are released into the wild are known and there are widely recognised, globally applicable guidelines for responsible stock enhancement.

Other comments

There would be the potential for illegal traders to adapt the form they trade the species in and launder the illegally sourced *Haliotis midae* in the frozen form which could not be detected. However, the frozen form poses challenges as it would require freezer capacity for storage and transport, increasing costs. Shifting to trade the species in live or canned form would not be a viable option for illegal traders.

Definitions set out in Article I of the Convention text do not provide for specification of which parts and derivatives of an animal species listed in Appendix I or II will be subject to CITES trade controls. Article I(b)(2) specifies that "in the case of an animal: for species included in Appendices I and II", specimen "means any animal, whether alive or dead, and any readily recognizable part or derivative thereof". This is further emphasised in Conf. 11.21 (Rev.CoP19) which states that "RECALLING that an annotated listing of an animal or plant species in any of the three Appendices always includes **the whole** (*emphasised for effect*) live or dead animal or plant, as well as any specimen specified in the annotation".

References

- Burgener, M. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- DFFE (Department of Forestry, Fisheries and the Environment) (2025) Status of the South African marine fishery resources 2025. Cape Town: DFFE
- Okes, N., Bürgener, M., Moneron, S. and Rademeyer, J. (2018) Empty Shells - An assessment of abalone poaching and trade from southern Africa. TRAFFIC, Cambridge, UK.
- Peters, H. (2021) *Haliotis midae*. *The IUCN Red List of Threatened Species* 2021: e.T78771094A78772518. <https://dx.doi.org/10.2305/IUCN.UK.2021-3.RLTS.T78771094A78772518.en>. Accessed on 01 August 2025.
- UNEP (2025) *Haliotis midae*. Species+ Website. Compiled by UNEP-WCMC, Cambridge, UK. Available at: https://www.speciesplus.net/species#/taxon_concepts/9001/legal Accessed on 15 August 2025

Amend annotation #3 to exempt finished products packaged and ready for retail trade of thin-sliced roots derived from artificially propagated plants of *Panax quinquefolius*

Proponent: United States of America

Summary: This proposal concerns the exemption of finished products packaged and ready for retail trade of roots sliced 1-3 mm thick derived from artificially propagated plants of American Ginseng *Panax quinquefolius*. This amendment would add to the current exemption, under annotation #3, of manufactured parts or derivatives of powders, pills, extracts, tonics, teas and confectionery that apply to whole and sliced roots and parts of roots, and would apply exclusively to *P. quinquefolius*.

American Ginseng is native to the USA and Canada, with less than 1% of the global population occurring in Canada. It is listed as endangered nationally within Canada and vulnerable in the USA but has no global IUCN Red List Assessment. American Ginseng is in demand in international trade for use in herbal supplements, traditional medicine, and wellness products, particularly in Asian countries. Cultivated (e.g. artificially propagated) American Ginseng has become an increasingly significant portion of legal trade and, between 2014 and 2023, accounted for more than 99% of all CITES global exports. The USA was the only direct commercial exporter of wild-sourced roots since 2014, which totalled approximately 135,000 kg. Wild harvest and sale (even domestically) of American Ginseng is not permitted in Canada but between 2014 and 2023, Canada was the main global exporter of *Panax quinquefolius* roots from artificial propagation, with 98% of all exports totalling around 30 million kg. The USA reported exports of ~1.5 million kg of artificially propagated roots in the same period.

The proponent states that the intention of proposed amendment is to reduce the administrative burden associated with trade in artificially propagated roots. Currently, individuals purchasing legally cultivated packaged and sliced American Ginseng, often bought in small amounts by tourists, require CITES Appendix II export documents. They often also need to obtain import permits from countries implementing stricter domestic measures (including China, the main importing country for this product between 2020 and 2023). The USA reports that the process of obtaining such documentation can be costly and burdensome, with the potential to discourage lawful trade. It is argued that the exemption would streamline legal trade and reduce permitting requirements for this low-risk product, allowing enforcement agencies to focus instead on those products with a higher conservation risk.

The USA states that to meet the requirements for the proposed exemption, producers would have to package sliced, artificially propagated roots with clear visual identification, such as transparent windows or clear panels. Such packaging is already available in the USA. If the proposal were accepted its use would become mandatory to support effective enforcement. The proposal does not state the extent to which packaged sliced roots make up a significant proportion of artificially propagated American Ginseng exported from Canada, or whether Canada has the capacity to produce similar packaging. A targeted search for sliced American Ginseng for sale in Canada identified five online platforms based in Canada offering sliced roots of American Ginseng (with prices and ages indicating these were artificially propagated roots), with international shipping available.

Canada, which holds a very small proportion of the wild population of *P. quinquefolius*, has noted the potential for wild roots - particularly of lower grades - to be laundered as sliced roots, with some instances of illegal wild harvest having occurred in Canada within the last ten years. Information provided in the proposal, and corroborated by industry and plant trade experts, indicates that this is however unlikely. The market for wild and cultivated roots of American Ginseng differs markedly, with the value of wild roots associated with their age, size, shape, and surface texture (e.g., such as having twisted shapes), which are all features which slicing roots can remove. Wild roots are therefore typically sold whole, and reportedly rarely sliced before export. There is no clear economic incentive for slicing wild sourced American Ginseng and hence a low risk of wild harvested roots being laundered in this form. Artificially propagated roots are typically smoother and more uniform in appearance, harvested at a younger age (e.g., at three to four years compared to over ten years for wild roots), and sold in higher volumes, or sliced. Wild roots typically sell for USD250-850 a pound but can sometimes be worth up to USD1000, whilst artificially propagated roots typically sell for USD10-

25 a pound.

A rapid online survey conducted for this analysis in 2025 did identify sliced American Ginseng labelled as wild for sale on five online platforms, with prices ranging from USD16 to USD560 for less than a pound. One online platform offering sliced ginseng labelled as wild for USD560 was the only platform with a company seemingly based in the USA. One other additional online platform based in the USA and shipping internationally offered a custom slicing service for wild roots prior to export. The sliced roots offered for sale in each instance were all clearly labelled as wild-sourced, and the probability of laundering wild roots into finished products advertised as artificially propagated still appears to be unlikely.

Analysis: There are clearly differentiated markets for wild sourced and cultivated American Ginseng. Given the lack of economic incentives to slice wild roots, which are valued internationally for their shape, size, and other morphological features, there appears to be a low risk of wild roots being sliced and laundered as finished products derived from artificially propagated plants. The **amendment to annotation #3 would therefore appear not to pose a threat** to the conservation of wild populations of this species. This amendment is **in line with Res. Conf. 11.21 (Rev. CoP18)** on the *Use of annotations in Appendices I and II* and Res. Conf. 9.24 (Rev. CoP17), which recommend Parties ensure that annotations to listings in the Appendices include those specimens that first appear in international trade as exports from range States and that dominate the trade and the demand from the wild. A possible enhancement of the annotation could be to include the requirement that the packaging used facilitates clear visual identification.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

Panax quinquefolius L. 1753. Synonyms *Aralia canadensis* Tourn., *Aralia quinquefolia* Decne. & Planch., *Aureliana canadensis* Lafit, *Ginseng quinquefolium* Alph. Wood, *Panax americanum* (Raf.) Raf.

Range

Canada, USA

IUCN Global Category

Not assessed.

CITES listing history

Panax quinquefolius has been included in CITES Appendix II since 1975.

Discussion

The proposal concerns an amendment to CITES Annotation #3, which applies to American Ginseng *Panax quinquefolius* and Asian ginseng *Panax ginseng* (Russian populations only). The amendment would amend Annotation #3 for *Panax quinquefolius* only, to exempt finished, retail-packaged, sliced roots (1–3 mm thick) derived from artificially propagated plants. This would change annotation #3 from:

- Whole and sliced roots and parts of roots, excluding:
 - a) manufactured parts or derivatives such as powders, pills, extracts, tonics, teas and confectionery'

To (see addition in bold)

- Whole and sliced roots and parts of roots, excluding:
 - a) manufactured parts or derivatives such as powders, pills, extracts, tonics, teas and confectionery; and
 - b) **finished products packaged and ready for retail trade of roots sliced 1-3 mm thick derived from artificially propagated plants of *Panax quinquefolius***

Panax quinquefolius is native to the USA (occurring across Eastern and Central states) and Canada (where it occurs in Ontario and Quebec only) (NatureServe, 2025). Less than 1% of the global population occurs in Canada (Carignan and Branchaud, 2018). Under CITES, “finished products packaged and ready for retail trade” are defined in the Interpretation to the CITES Appendices¹⁸, to mean “Products, shipped singly or in bulk, requiring no further processing, packaged, labelled for final use or the retail trade in a state fit for being sold to or used by the general public.”

Panax quinquefolius is widely used internationally in herbal supplements, traditional medicine, and wellness products. Trade involves both wild-harvested and artificially propagated roots, with the USA serving as a major exporter primarily to East Asian markets where the species holds high medicinal value. Whole roots - both wild and artificially propagated - are commonly exported in bulk, often packed in 100-pound (~45 kg) barrels, primarily for large-scale distribution to East Asian markets. In these markets, whole roots are frequently further processed for retail sale or additional distribution. Sliced roots are typically derived from artificially propagated plants, often field-grown under shade in tilled soil. These sliced products are packaged for retail sale and are widely used in traditional herbal preparations or purchased by tourists and taken to their home countries.

The markets for wild and artificially propagated *P. quinquefolius* roots are fundamentally distinct. Wild roots, usually over 10 years old, are prized for their age, rarity, and unique morphological characteristics - such as twisted shapes, pronounced growth rings, and darker, corkier skin- which are culturally and commercially important in East Asian traditional medicine markets. Wild roots are typically sold whole and command premium prices ranging from USD250 to USD850 per dry pound (~450 g), with exceptional specimens and higher demand fetching prices as high as USD1,000 per pound. In contrast, artificially propagated roots are harvested much younger, generally after 3 to 4 years, resulting in smoother, more uniform roots. These are sold in higher volumes, either whole or sliced, at substantially lower prices, commonly between USD10 and USD25 per dry pound. This segment primarily supplies mainstream retail and wellness product manufacturers.

Cultivated American Ginseng has become an increasingly significant portion of legal trade, supplying a more stable and consistent source supported by best management practices. Between 2014 and 2023, artificially propagated American Ginseng has accounted for over 99% of direct commercial global exports of American Ginseng as reported in CITES Trade Database.

The proposal states that between 2020 and 2023, Canada and the USA exported more than 35,000,000 pounds (around 15.9 million kg) of dried artificially propagated roots, accounting for 99% of American Ginseng exports (UNEP-WCMC, 2025). Canada was the top exporter of artificially propagated roots for commercial purposes in recent years between 2014 and 2023 according to the CITES Trade Database, exporting 98% of artificially propagated roots globally (~29.7 million kg out of ~31.5 million kg) (Figure 1). The USA reported around 1.5 million kg, with the remaining exported mostly from Mainland China (~177,000 kg) and Hong Kong SAR of China (~166,000 kg).

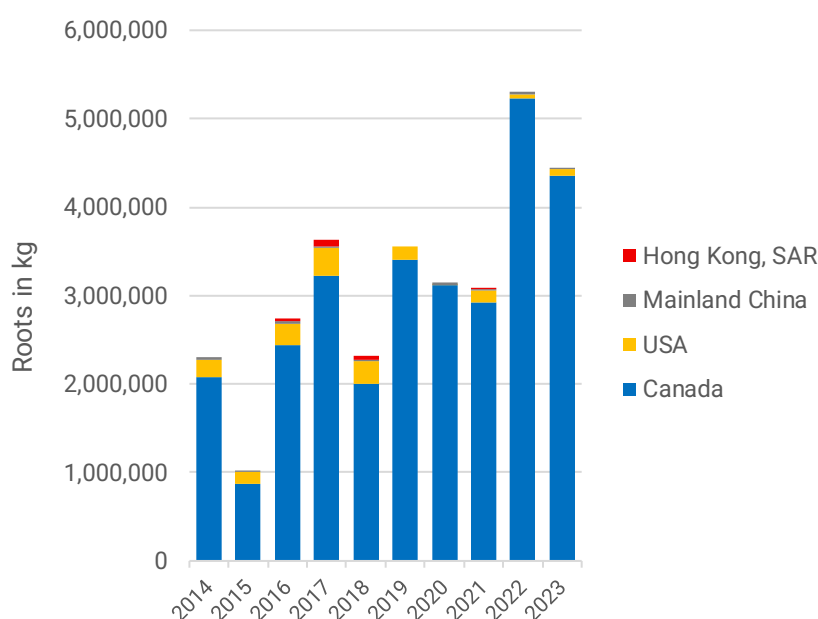


Figure 1. Mass in kg of direct commercial exports of artificially propagated roots of *Panax quinquefolius* as reported by exporting countries between 2013 and 2024. Source CITES Trade Database.

¹⁸ <https://cites.org/eng/app/appendices.php>

*Mainland China was the main importer of artificially propagated roots between 2020 and 2023 (72% of all imports), with the remaining imported mostly by Hong Kong SAR of China (25%). The USA has been the only exporter of wild harvested *P. quinquefolius* for commercial purposes since 2015, although Canada reported exports of 11 kg of wild sourced roots for personal use between 2017 and 2022.*

Trade is currently complex due to differing domestic measures and regulatory frameworks. Some countries do not recognize the CITES personal effects exemption, requiring permits for even small quantities, creating barriers for tourists purchasing artificially propagated *P. quinquefolius* as souvenirs. For example, Mainland China, the largest consumer and importer, requires both export and import permits for Appendix-II species, including artificially propagated American Ginseng, which complicates tourism-related purchases and other low-risk trade in certain artificially propagated American Ginseng products. Individuals purchasing legally cultivated, packaged American Ginseng for personal use at present must obtain CITES export permits or risk confiscation. This process is often costly and burdensome, discouraging lawful trade. By exempting this clearly defined, low-risk product, the amendment would streamline legal trade, reduce administrative burdens, and support the sustainable management of *P. quinquefolius* under CITES, consistent with the guidance and principles for annotations in paragraph 6 of Res. Conf. 11.21 (Rev. CoP19) on *Use of annotations in Appendices I and II*.

Additional Information

Conservation, management and legislation

In Canada, *Panax quinquefolius* is listed as endangered at the federal level under the Species at Risk Act and is therefore afforded protection on federal land. Wild harvest is prohibited for the purposes of sale or distribution, and *only cultivated American Ginseng can be legally traded in Canada, with even the possession of wild American Ginseng collected in Canada prohibited within the country* (Ontario Ministry of Natural Resources and Forestry, 2019). *The species is also protected at a state level. In Québec, the Act Respecting Threatened or Vulnerable Species prohibits any activity that harms or is likely to harm a specimen of *Panax quinquefolius* from a wild population in the province. In Ontario, *Panax quinquefolius* is protected through the Ontario Endangered Species Act, 2007 (ESA), with harvest of roots from wild populations prohibited under this Act (CITES Scientific and Management Authorities in Canada in litt., 2025). As it is deemed to not pose a risk to the province's wild population, the sale of field-cultivated *Panax quinquefolius* is permitted under Ontario's ESA, but other methods of commercial cultivation of *Panax quinquefolius* are not currently permitted (CITES Scientific and Management Authorities in Canada in litt., 2025).*

In the USA, the species has been nationally assessed as vulnerable (NatureServe, 2025). At the federal level, the U.S. Fish and Wildlife Service oversees the US CITES Export Program (CEP) for American Ginseng, under which 22 States and one Tribe have codified regulations concerning the harvest and sale of *P. quinquefolius*. These regulations include designated harvest and buying seasons; minimum plant harvest size and/or age limits; mandatory seed replanting at or near the site of harvest; licensing requirements for dealers; and root certification prior to export. All harvested roots intended for international trade must carry a State- or Tribe-issued certificate along with a valid CITES export permit, both presented to officials at the port of export.

The USA states that to meet the proposed exemption criteria for excluding certain American Ginseng specimens from CITES permitting requirements, producers would have to package sliced, artificially propagated roots with clear visual identification, such as transparent windows or clear panels. While this packaging is currently available for voluntary use, it would become mandatory to support effective enforcement. The proposed exemption applies exclusively to finished products—specifically, sliced roots (1–3 mm thick) of artificially propagated American Ginseng—that are packaged, labelled, and ready for retail sale. According to the CITES definition, these products must be shipped singly or in bulk, require no further processing, and be fit for sale or use by the general public. This packaging would facilitate non-invasive inspection by enforcement officers, ensuring only compliant, traceable products qualify for the exemption and helping prevent laundering of wild harvested roots. *It is not clear whether Canada would have the capacity to introduce such packaging for these products, or whether such sliced roots are currently a large proportion of their export market, as the term for 'roots' when reported in the CITES Trade Database does not differentiate between sliced and whole roots. A rapid online review did identify five online platforms based in Canada selling sliced American Ginseng roots internationally, two shipping to the USA and three globally. None of the five online platforms stated the roots were wild-sourced, with ages stated (4-5 years) and prices (USD11-92 for less than a pound) compatible with artificially propagated roots.*

Artificial Propagation

Panax quinquefolius is widely cultivated in the USA through two main artificial propagation methods: field cultivation under artificial shade, and woods-grown cultivation under natural forest canopy. The US Fish and Wildlife Service recognises American Ginseng produced through these methods as artificially propagated under Res. Conf. 11.11 (Rev. CoP18) on *Regulation of trade in plants*. Cultivation occurs under controlled and verifiable conditions, employing cultivated seed and established horticultural techniques. These practices enhance traceability and regulatory oversight. Roots produced through wild-simulated methods do not meet the CITES definition of artificial propagation and therefore remain subject to wild harvest regulations.

Implementation challenges (including similar species)

Panax ginseng (Korean or Asian ginseng) is included in CITES Appendix II with Annotation #3 applying specifically to populations in the Russian Federation. *Based on the CITES Trade Database, there have been no legal exports of Panax quinquefolius from the Russian Federation since 1994 (CITES, 2025).* The proposed amendment applies exclusively to *P. quinquefolius* and does not extend to *Panax ginseng* or any other species. This focus ensures clarity, maintains regulatory specificity, and minimizes potential enforcement challenges or broader trade implications.

Potential risk(s) of the proposed amendment

Although laundering wild roots as cultivated material could be a potential concern, the proposal states it is highly unlikely given the proposed exemption specifically for sliced artificially propagated roots as a distinct commodity. Wild roots by contrast are typically traded whole *and at much higher prices than artificially propagated roots*. Slicing wild roots removes key morphological features -such as size, shape, and surface texture- that buyers of whole wild roots highly value, significantly lowering their market value and eliminating premium traits. Market prices reflect this distinction: field-grown roots typically sell for USD10–25 per dry pound, whereas wild or wild-simulated roots command much higher prices ranging from USD250–850 per pound. Substituting wild roots into the sliced root market is economically disincentivised and unlikely. Sliced artificially propagated roots primarily serve a different market - tourists, wellness consumers, and retail herbal buyers. To ensure the exemption supports efficient enforcement and traceability, products must be packaged to allow clear visual identification by enforcement officers, such as packaging with transparent windows or clear panels. Although such packaging currently exists, making this a mandatory requirement will help prevent substitution or laundering of wild roots, strengthening regulatory oversight. *Whilst this clear packaging would ensure that packets do not contain whole wild roots, rapid online surveys (see below) have indicated that there are sliced roots of American Ginseng labelled as wild being sold online and shipped internationally (including some from the USA). It is not clear whether these could be visually differentiated from artificially propagated sliced roots, although it was also not possible to verify whether these products were truly products from wild-sourced ginseng or simply labelled as such to enable a higher sale price. Therefore, clear labelling indicating artificial propagation would be desirable. The proposed annotation does not specify the type of packaging required for the exemption to apply, the USA could consider amending the proposal to explicitly include such a requirement.*

The supporting statement notes that Canada, as a co-range State with a small endangered wild population of *P. quinquefolius*, has expressed concern over the exemption's potential impact on their wild populations. *Consultations with the CITES Scientific and Management Authorities in Canada identified that Canada's 2018 federal recovery strategy for American Ginseng lists illegal root harvest as one of four threats that are of a "high level of concern", with the authorities noting that wild roots of lower grade or value that are intended to be cut anyway for consumption as a remedy, could potentially be sliced (before export) to try to evade the export control if the proposed amendment were to be successful (CITES Scientific and Management Authorities in Canada in litt., 2025).*

There is some evidence of current illegal trade in P. quinquefolius from Canada in trade databases in recent years, with reports of six seizures of P. quinquefolius from Canada in EU-TWIX between 2017 and 2023. The total quantities seized were fairly small (27 roots and ~1.5 kg of dried plants) and the source of all specimens seized was unknown. Two seizures of wild-sourced roots imported into the USA from Canada were also reported in LEMIS; 10,000 roots in 2017 and 300 g of roots in 2019. Two other shipments of wild sourced roots totalling 8 kg from Canada have been refused entry since, both in 2022 and reportedly imported for personal use. Ginseng surveillance by federal enforcement officers on federal lands in Canada has not detected illegal harvest in recent years but such activity may nevertheless occur as poachers make efforts to hide their traces (CITES Scientific and Management Authorities in Canada in litt., 2025). A media report in 2017 stated conservation officers in Canada were utilising video surveillance, and marking ginseng roots with ultraviolet dyes (e.g. to remove their value), to deter poachers (Crawford, 2017).

The Ministry of Environment, Fight Against Climate Change, Wildlife and Parks in Quebec (one of the two States in which American Ginseng occurs in Canada) did not have any recent, documented information on the presence of illegal activities involving Panax quinquefolius in Québec but informal communications with various local organizations and partners suggest that this threat is, however, very real in the province, mainly in the border regions of Outaouais, Montérégie, and Estrie (CITES Scientific and Management Authorities in Canada in litt., 2025). In Ontario (the other Canadian state in which American Ginseng occurs), one threat to the species is reported to be unregulated commercial cultivation on forested land, especially if it occurs on sites near wild populations, with wild Panax quinquefolius reported to be highly sought after for the medicinal value of its root in Ontario and commonly poached (CITES Scientific and Management Authorities in Canada in litt., 2025). Three media reports outlined instances of illegal poaching of American Ginseng in Ontario in the last ten years, with one stating 41 live American Ginseng plants were seized (some reported to be propagated, but some harvested from the wild between 2018 and 2020), and two others outlining instances in which individuals were fined for selling unstated quantities of wild American Ginseng in 2015 (two individuals) and 2016 (one individual).

Consultations with multiple industry and medicinal plant experts confirmed that the laundering of wild American Ginseng as sliced roots was unlikely and low risk (Commender; Filyaw; Fletcher; Timoshyna in litt, 2025). Consultations with one industry expert confirmed the price differences for wild and cultivated roots in the proposal, with the expert stating that wild ginseng is valued at USD600-900 per pound (with prices for the last season ranging from USD600-700/pound for wild dried whole/semi whole roots), but in the high tens to low twenties per pound for artificially propagated specimens, giving no monetary incentive for laundering of wild specimens as artificially propagated (Fletcher in litt., 2025).

A rapid online survey identified sliced roots labelled as wild for sale online on five platforms with prices as follows:

- USD223 (SGD288) for 37.5 g (less than a pound)
- USD400 (MYR1,7000) for 30 g (less than a pound)
- USD560 for 4 oz (113 g or a quarter of a pound)
- USD23 for 60 g (less than a pound)
- USD16 for 30 g (less than a pound)

There was no clear indication of the quality of the wild roots on the online advertisements, which makes it difficult to ascertain whether the sliced roots labelled as wild came from lower grade wild roots, and it was not possible to verify whether the products being offered were truly wild sourced. The online platforms offering wild sliced roots for sale for USD560 appeared to be hosted by a company based in the USA, with the remaining overseas. One other US platform offered a slicing service of whole wild roots (prices ranging USD83-230) which was advertised for USD10. This US-based platform also sold a CITES permit with the products offered. All products of wild sliced roots identified in the snapshot online survey were clearly labelled as wild sourced, and some of the prices are higher than would be expected for the sliced product based on the evidence provided by experts and in the proposal. An industry expert noted that there is no clear incentive to label 'wild' as cultivated, although mislabelling cultivated to 'wild' does have economic incentives, pointing out that anytime a ginseng root is altered from its natural state/form, it is more difficult to determine its source and/or how it was grown (Fletcher in litt., 2025).

Potential benefit(s) of listing for trade regulation

This targeted exemption would reduce permitting burdens for growers, exporters, and small-scale buyers. The lack of international recognition of CITES personal effects exemptions, including in major markets, complicates trade in legally cultivated, low-risk products and can deter legitimate commerce. The proposal states that the increasing role of artificially propagated American Ginseng in global trade supports conservation by helping to meet demand while easing pressure on wild populations. Facilitating efficient trade in clearly distinguishable, lower-risk products strengthens legal supply chains, discourages illegal harvest, and enhances transparency, further alleviating pressure on wild plants. One expert noted that as the markets for wild and artificially propagated roots differ, it is not clear to what extent the market for artificially propagated products from American Ginseng relieves pressure on wild populations (Timoshyna in litt., 2025).

Key outcomes of the proposed exemption stated by the proponents include:

- Facilitates compliance with CITES by clarifying treatment of low-risk products;
- Supports enforcement by allowing authorities to focus on higher-risk items;
- Promotes sustainable trade alternatives aligned with conservation goals;
- Reduces permitting requirements for exporters and consumers, particularly tourists;
- Preserves enforcement authority to inspect and seize non-compliant materials;
- Improves trade efficiency for American ginseng exporters and importing Parties.

References

- Commender, K. (2025) in litt. to the IUCN/TRAFFIC Analyses Team, Cambridge, UK
- Crawford (2017) *Canada's endangered wild ginseng under threat from poachers*. Available at <https://www.cbc.ca/news/politics/wild-ginseng-poaching-endangered-plants-1.4212920>. Accessed August 2025.
- Filyaw, T. (2025) in litt. to the IUCN/TRAFFIC Analyses Team, Cambridge, UK
- Fletcher, E. (2025) in litt. to the IUCN/TRAFFIC Analyses Team, Cambridge, UK
- Ontario Ministry of Natural Resources and Forestry (2019) *Recovery Strategy for American Ginseng (Panax quinquefolius) in Ontario*. Available at <https://www.ontario.ca/page/american-ginseng-recovery-strategy>. Accessed August 2025.
- Timoshyna, A. (2025) in litt. to the IUCN/TRAFFIC Analyses Team, Cambridge, UK
- NatureServe (2025) *Panax quinquefolius*. NatureServe Explorer. Accessed 25 June 2025. <https://explorer.natureserve.org/>

Inclusion of Chilean Palm *Jubaea chilensis* in Appendix I

Proponent: Chile

Summary: The Chilean Palm *Jubaea chilensis* is a large, slow growing palm endemic to the mediterranean region of Chile. It is a very long-lived species that matures at 75-150 years and can live to over 700 years. It is the only native palm in mainland Chile and has a restricted and highly fragmented distribution. Its extent of occurrence is estimated at ~24,000 km² with an area of occupancy of 1,600 km². The population has been estimated at ~120,000 mature individuals, having decreased by more than 50% over the last 300 years (three generations), and possibly by 98% in the last 500 years. Population declines have been driven by habitat loss from land-use changes and fire damage, felling of palms, the harvesting of its fruits and seeds and their predation, particularly by introduced rabbits and livestock. The population is largely composed of older individuals, with limited recruitment of younger plants. Due to this demographic imbalance and multiple threats to its habitat, *Jubaea chilensis* was globally assessed as Endangered on the IUCN Red List in 2021, as well as being nationally assessed as endangered in Chile in 2020.

In the past, young stems were eaten as hearts of palm. Trees were cut to extract sap used to make palm syrup “miel de palma”, which was also fermented to make palm wine. Felling in Chile has been restricted by law since 1941. The fruits of *Jubaea chilensis* have been harvested for centuries; the nuts are edible once the hard shells are removed and are known in South America as “coco chileno” or “coquito”. Collection of seed for domestic use and export is believed likely to have a significant impact on recruitment of wild plants. Nuts or coquitos, are offered for sale in specialized shops in Europe and on the internet. The high price that the Chilean Palm seed has reached in the international market, mainly Asian, has generated an unusual rise in the collection of seeds/nuts, leading to this practice being suspended in areas where collection by local communities has traditionally been allowed. Information provided by Chilean Customs show exports primarily of the seeds/nuts as well as plants to Europe, Hong Kong SAR of China and Singapore.

Currently around 90% of trees are found in three areas (Ocoa, Cocolán and Las Siete Hermanas), all protected, although illicit seed collection is believed to take place in some of these. The species has been widely propagated and planted as an ornamental in Chile, often outside its indigenous range, and in other countries, including the USA and Europe. One foundation in Chile, set up in 1997 specifically to address conservation of the Chilean Palm, has raised over two million plants since its establishment, many planted elsewhere in Chile.

Various conservation measures are in place for *Jubaea chilensis* including through legislation, *in-situ* protection and implementation of a National Conservation Plan since 2005. The general objective of the Plan is to develop conservation actions for the species by reestablishing sustainable natural populations and promoting cultivation of the species in areas of its natural distribution, within a framework of environmentally sustainable social and economic development.

Analysis: *Jubaea chilensis* is a late-maturing long-lived palm with a relatively restricted range in central Chile, whose seeds and sap have been widely exploited for food. Its habitat has been affected by land-use change, deforestation and fire and it is believed to have undergone significant historic population declines. With over 100,000 individuals (mostly mature) estimated in the wild, most in protected areas, it does not currently have a small population. At present the population appears stable, although recruitment is low and fire in particular remains an ongoing threat. Seeds have traditionally been collected by local communities (sometimes illegally) with an unknown proportion of these being exported. Economic benefits derived from this export are unclear, although prices of seeds in overseas markets have increased markedly in recent years. The species is widely grown as an ornamental within Chile and elsewhere. Although historical declines are believed to have occurred, current protection (both through the majority of its wild population now occurring in protected areas and by regulation of harvest) appear to have stabilised the wild population. It is therefore **unclear whether *Jubaea chilensis* meets the criteria for inclusion in Appendix I** set out in Annex 1 of Res. Conf. 9.24 (Rev. CoP17) or requires regulation in Appendix I.

Inclusion in Appendix II would ensure that any trade was regulated and non-detrimental, with an option to establish a zero quota for wild-sourced seeds. Furthermore, nurseries intending to export plants would not require registration in accordance with Res. Conf. 9.19 (Rev. CoP15) *Registration of nurseries that artificially propagate specimens of Appendix-I plant species for export purposes.*

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS

Taxonomy

Jubaea chilensis is the only species in the monotypic genus *Jubaea*.

No standard reference is recommended in proposal CoP20. Prop. 41. In accordance with CoP20 Inf. 6 a time-stamped extract from POWO prepared by the flora nomenclature specialist is proposed for adoption as a standard nomenclature reference (see CoP20 Doc. 110 Add., Annex 1):

POWO. (2025). *Jubaea*. World Checklist of Vascular Plants. Facilitated by the Royal Botanic Gardens, Kew. Published on the Internet; <https://powo.science.kew.org/> Retrieved 28 July 2025.

Range

Chile

IUCN Global Category

Endangered A2c (assessed 2021, ver. 3.1)

Biological criteria for inclusion in Appendix I

A) Small wild population

The remaining Jubaea chilensis trees are distributed across eleven subpopulations, with the largest found in Parque Nacional La Campana, previously estimated at around 60,000 individuals. Other significant populations include Palmeras de Cocalán with approximately 35,000 trees, and El Salto (Las Siete Hermanas) with about 6,500. However, a more recent survey conducted by CONAF in La Campana reported a revised count of 40,405 individuals (CONAF, 2016). These three sites represent around 90% of the total population. There are other localities with less than 1,000 J. chilensis individuals. Overall, the total population is estimated not to exceed 120,000 individuals (González et al., 2009).

Plants at Las Siete Hermanas were affected by fire in 2022 and 2024 although most mature individuals are likely to have survived (Gardner and Hepp, 2025).

B) Restricted area of distribution

The current geographic distribution of the Chilean palm is restricted, with an estimated extent of occurrence of 23,836 km² and area of occupancy of 1,612 km² (Chaparro, 2021). It is endemic to the Mediterranean region of Chile, occurring only in the central zone and distributed from Coquimbo to Maule. Its presence extends from sea level on the coast up to 2,000 m above sea level, mainly associated with sclerophyllous forest and scrubland. Currently, its populations are small and highly fragmented, mainly in dry areas on the slopes of the Coastal Range, with some populations also found in the central valley. The known northernmost populations are at Hacienda Las Palmas in Coquimbo, while the southernmost ones are in the locality of Tapihue, in the Maule region.

C) Decline in number of wild individuals

The species had a suspected past reduction of 50% of population size in the last 300 years (three generations); past changes in land use and logging of individuals in combination with current substitution for cultivation, browsing on juveniles by rodents, livestock, wildfires, as well as intensive exploitation of its fruits has led to a decline of 50% of area of occupancy (Chaparro, 2021). Most populations present an age structure with a large percentage of adult individuals and a low percentage of juvenile and infant individuals suggesting continuing decline in the future. Most seedlings are lost to exotic herbivores such as rabbits and hares.

Trade criteria for inclusion in Appendix I

The species is or may be affected by trade

The high price that the Chilean Palm seed has reached in the international market, mainly Asian, has generated an unusual rise in the collection of seeds.

International trade is recorded for products of this species, including, fruits, plants, and seeds. Data compiled by the Chilean Customs Service for the period 2018-2023 identified trade in species products from the regions of Valparaíso, Metropolitana, O'Higgins, and Maule to five destination countries and territories: Germany, the Netherlands, Singapore, Ireland, and Hong Kong SAR of China, for a total Free On Board (FOB, i.e. seller loads goods onto ship and pays all related costs) amount of USD225,372 (see Figure 1).

The main exported product corresponds to the nuts or "coquitos de palma," which represent 88% of the total exported during the period 2018-2023. The value of trade (USD FOB) for this period is shown in Figure 1.

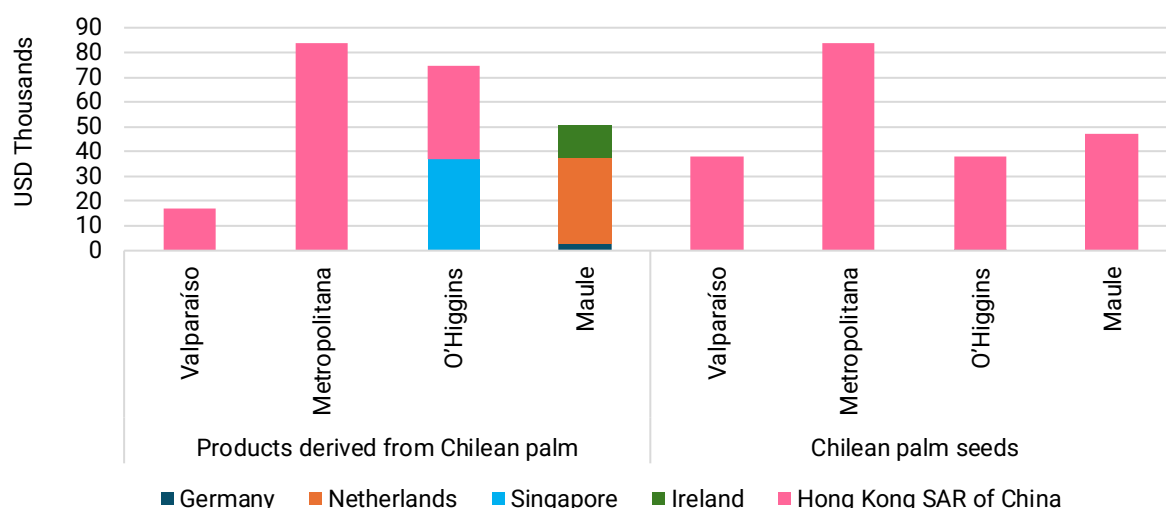


Figure 1. Export of *Jubaea chilensis* products and seeds by total USD FOB from four Chilean states to importing countries, 2018-2023.

Harvesting agreements were made between Corporación Nacional Forestal (CONAF) and local communities surrounding the palm grove in La Campana National Park (with the largest population of *Jubaea chilensis*) from 1986 until 2017. The agreements ended after a significant demand from international markets led to a drastic rise in seed prices, and increased levels of seed harvesting. Harvesting was completely banned inside the National Park. Nonetheless, the pressures associated with international demand were not regulated at that time, and the responsibility for the conflict fell mainly on local harvesters (Urresty-Vargas et al., 2024). Seed harvesting in Las Siete Hermanas has been estimated at just over 12 tonnes per season representing 94% of total production in that population (Cordero in litt., 2025).

During the final ripening stage of Chilean Palm seeds, illegal harvesting, primarily for sale at fairs or to candy producers, is frequently reported within protected areas such as national parks and reserves. This activity poses a direct threat to the species' conservation and is legally classified as theft or robbery, punishable by imprisonment and fines, with penalties also applicable to individuals involved in the trade or commercial use of the seeds.

The proportion of Chilean Palm products exported rather than traded domestically within Chile is unrecorded. However, the scale of the domestic market in seeds, fruits and plants of *J. chilensis* is likely to be larger than the export trade.

Additional Information

Threats

The main threats to this species are habitat loss from forest fires and land-use change, illegal harvesting of fruits for commercial purposes (which have high demand on the international market), seed predation (by introduced and native rodents) and browsing by animals, water extraction in its natural distribution areas, drought phenomena over the last 14 years, the introduction of invasive alien species that compete for habitat, urban expansion, and deforestation. The high fragmentation of its populations and the high degree of threat to its habitat have resulted in a significant spatial retreat and decline of its populations, mainly due to past anthropogenic activities and the almost non-existent regeneration of current populations.

All studied populations of *J. chilensis* have low genetic diversity, high inbreeding, and no evidence of isolation by distance. Habitat fragmentation, degradation of the sclerophyllous forest, lack of long-distance dispersers for the seeds, and a deficit of natural regeneration have contributed to promoting inbreeding and low levels of genetic diversity in the palm groves of *J. chilensis*. Although existing populations are not at imminent risk of extinction due

to the species longevity, the inbreeding rate could increase rapidly because of climate change and human impact. Therefore, taking these factors into account, it is suggested that this palm species should be considered as critically endangered by international conservation organizations (Jara Arancio et al., 2022). Other experts agree that the species may qualify for the highest category of risk (Cordero et al., 2021; Ovalle et al., 2024).

The mediterranean vegetation in which *J. chilensis* grows is itself threatened, which intensifies the threat to the species as it functions as shelter vegetation for the recruitment of seedling (Fleury et al., 2015).

Viña del Mar city had the third largest population of Chilean Palm in the country with 6,061 adult specimens in 2008. In February 2022, a large-scale fire devastated 1,661 palms, mostly individuals over 500 years old, on 27% of the palm grove's surface area. A major fire in the first week of February 2024 swept away most of the remaining population in this area (Ovalle et al., 2024).

Conservation, management and legislation

In situ conservation: An important part of the total population of Chilean palm is located in the Ocoa sector inside La Campana National Park, in the Valparaíso region. This wild area was created on October 17, 1967, with an area of 8,000 hectares. There are also small, protected populations inside sectors of the Lago Peñuelas National Reserve also in the Valparaíso region, created in 1952. Both La Campana National Park and Lago Peñuelas National Reserve fall within the "La Campana Peñuelas" Biosphere Reserve. The second important concentration of its population is located in the locality of Cocalán, in the Palmas de Cocalán National Park, located in the O'Higgins region, commune of Las Cabras. It was created in 1989, with an area of 3,709 hectares, which are privately owned.

Management and legislation: The species is under legal protection and regulation through various instruments, including D.N° 908 of July 3, 1941, of the Ministry of Lands and Colonization; Decree Law 701 of 1974 (Forestry Promotion Law), Law 19.561 on Forestry Promotion and its modifications introduced in D.L. No. 2,565, and Law 20.283 on Native Forest and Forestry Promotion. Furthermore, this species was categorized as "endangered" according to the 16th classification process of wild species established in the General Environmental Framework Law (Law 19.300). A permit from the Agricultural and Livestock Service (SAG) is required to carry out any type of cutting, exploitation, or transfer of Chilean palm.

Since 2005, CONAF has been implementing the National Conservation Plan for the Chilean Palm, *Jubaea chilensis* in accordance with Decree Law 701 of 1974. The general objective of the Plan is to develop conservation actions for the species by reestablishing sustainable natural population structures, and to promote the cultivation of the species in all areas of its natural distribution, within the framework of an environmentally sustainable social and economic development.

In 2021 a proposal was made declaring Chilean palm as a "Natural Heritage" species under national legislation (CDD, 2021). Approval of this is recommended by Ovalle et al. (2024).

Artificial Propagation/captive breeding

Jubaea chilensis has been grown in botanic gardens for over a century and is currently recorded in 118 ex situ collections (BGCI, 2024). It is a hardy species tolerating temperatures down to about -12°C and it is popular in cooler temperate areas with a winter-wet mediterranean climate. It is often planted as an ornamental and street tree in Mediterranean regions and is widely grown in California, for example.

The slow growth (particularly when young) and limited availability in the past mean that *J. chilensis* is not common in cultivation. Older cultivated trees often produce abundant fruit, so seedlings are becoming more widely available (Palmpedia, 2025). Plants and seeds are available from specialist nurseries in Europe and the USA and from online sources. One website supplying palm seeds and plants to the international market notes that *J. chilensis* is "our bestselling palm" (Rare Palm Seeds, 2025).

In Chile, Fundación PALMA was established in 1997 with legal status granted by Supreme Decree No. 211 of 6 March 1998 of the Ministry of Justice. Since its constitution, this Foundation has reproduced more than two million specimens of *Jubaea chilensis* in its nurseries, a figure that compares with the total population of 120,000 specimens existing at the time of the beginning of its work. The requirement for a suitable ecosystem to achieve its natural reproduction, has been successfully tested on the Foundation's grounds and specimens have been planted in many places within Chile.

Jubaea chilensis is produced by commercial nurseries within Chile for the ornamental horticulture trade. Further commercial production may help reduce trade pressures on wild populations.

Implementation challenges (including similar species)**Other Considerations:**

Jubaea chilensis is also widely produced by nurseries in other countries. There is an established international trade in propagated plants of this species for ornamental horticulture that will be impacted by a CITES Appendix-I listing.

As cultivars of this species are currently commercially traded, inclusion of Jubaea chilensis in Appendix I may increase the administrative burden on countries currently trading in cultivated plants and seeds of the species and its inter-generic hybrids.

Potential risk(s) of a listing

Listing Jubaea chilensis on CITES Appendix I may have substantial implications for the global ornamental palm trade. While much of this trade may rely on cultivated rather than wild-sourced material, further investigation is needed to clarify the origin of seeds used in international commerce.

An Appendix-I listing may also constrain Chile's ability to use regulated trade as a conservation tool. Legal and sustainable harvesting or export of seeds and plants can incentivize local stewardship and enhance the species' value within Chile. Restricting such mechanisms could reduce opportunities for conservation-linked economic benefits (Baker in litt., 2025).

Additionally, limitations on the export of coquitos may adversely affect local livelihoods, particularly in communities that depend on this resource for income.

Other comments

According to León-Lobos et al. (2022), four parts of Jubaea chilensis palms have been consumed as food. As well as the fruits and seeds, eaten fresh and in sweet dishes, and the sap which was boiled to produce syrup, young stems were consumed in the past as hearts of palm or "palmitos".

The harvesting of Jubaea chilensis nuts (commonly known as coquitos) continues to represent an important source of income for local communities. However, this practice is currently at odds with biodiversity conservation objectives established by Chile's National Forestry Corporation (CONAF) for La Campana National Park (LCNP).

Since 1986, CONAF had formal agreements with surrounding communities to allow seasonal harvesting within the park. These agreements were discontinued in 2017, following a rise in international demand for coquitos, and harvesting was subsequently prohibited within LCNP.

Despite the ban, many local harvesters continue to collect seeds in the Ocoa sector each autumn. For these individuals, the activity is not only a livelihood but also a cultural tradition passed down through generations. They generally do not perceive their actions as harmful to the forest ecosystem—a view that contrasts with CONAF's conservation stance. Some community members have suggested that seed predation by rodents may play a more significant role in limiting palm regeneration than human harvesting (Urresty-Vargas et al., 2024).

References

- Baker, W. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- BGCI (2024) GlobalTreePortal. Botanic Gardens Conservation International. Richmond, UK. Available at: <https://www.bgci.org/resources/bgci-databases/globaltree-portal/species-search/?species=Jubaea+chilensis> Accessed 7 July 2025.
- CDD (Cámara de Diputados y Diputadas de Chile) (2021) Proyecto de Ley que Declara a la Palma Chilena como Patrimonio Natural de Chile, Santiago, Chile. Available at: <https://www.camara.cl/verDoc.aspx?prmTipo=SIAL&prmID=63120&formato=pdf>
- Chaparro, C.B. (2021) *Jubaea chilensis*. The IUCN Red List of Threatened Species 2021: e.T38586A2880356. <https://dx.doi.org/10.2305/IUCN.UK.2021-3.RLTS.T38586A2880356.en>
- Chile (1941) Decreto Supremo N° 908 de 1941. Diario Oficial de la República de Chile.
- CONAF (2016) Plan de manejo del Parque Nacional La Campana (versión completa). Corporación Nacional Forestal, Región de Valparaíso. Available at: <https://www.conaf.cl/centro-documental/plan-de-manejo-del-pn-la-campana-version-completa/>
- Cordero, S., Gálvez, F. and Fontúrbel, F.E. (2021) Multiple Anthropogenic Pressures Lead to Seed Dispersal Collapse of the Southernmost Palm *Jubaea chilensis*. *Frontiers in Ecology and Evolution* 9:719566. <https://doi.org/10.3389/fevo.2021.719566>
- Cordero, S. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Fleury, M., Wara, M., Vásquez, R., González, L. and Bustamante, R. (2015) Recruitment dynamics of the relict palm, *Jubaea chilensis*: Intricate and pervasive effects of invasive herbivores and nurse shrubs in Central Chile. *PLoS ONE* 10(7):e0133559. <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0133559>

- Fundación Palma Chilena (2025) Inicio. Palma Chilena Website. Available at: <https://www.palmachilena.cl>
Accessed on 25 July 2025.
- Gardner, M.F. and Hepp, J. (2025) *Jubaea chilensis*. Trees and Shrubs Online Website. Available at: <https://treesandshrubsonline.org/articles/jubaea/jubaea-chilensis/> Accessed on 13 September 2025.
- González, L.A., Bustamante, R., Navarro, C., Rafael, M., Herrera, M. and Toral Ibáñez, M.A. (2009) Ecology and Management of the Chilean Palm (*Jubaea chilensis*): History, Current Situation and Perspectives. *PALMS* 53(2):68–74.
- Jara-Arancio, P., Carvalho, C.S., Carmona-Ortiz, M.R., Bustamante, R.O., Schmidt Villela, P.M, Andrade, S.C.S., Peña-Gómez, F.T., González, L.A. and Fleury, M. (2022) Genetic Diversity and Population Structure of *Jubaea chilensis*, an Endemic and Monotype Gender from Chile, Based on SNP Markers. *Plants* 11:1959. <https://doi.org/10.3390/plants11151959>
- León-Lobos, P., Díaz-Forestier, J., Díaz, R., Celis-Diez, J.L., Diazgranados, M. and Ulian, T. (2022) Patterns of Traditional and Modern Uses of Wild Edible Native Plants of Chile: Challenges and Future Perspectives. *Plants* 11:744. <https://doi.org/10.3390/plants11060744>
- Ovalle, J., Pauchard, A., Scherson, R. and Aranson, J. (2024) The iconic *Jubaea chilensis* teeters on the edge of local extinction: a plea for enhanced conservation policies. *Biodiversity and Conservation* 33:3923–3927.
- Palmpedia (2025) *Jubaea chilensis* – Palm Grower's Guide. Available at: https://www.palmpedia.net/wiki/Jubaea_chilensis Accessed on 7 July 2025.
- Rare Palm Seeds (2025) *Jubaea chilensis* – Chilean Wine Palm – Buy Seeds. Available at: <https://www.rarepalmseeds.com/jubaea-chilensis> Accessed on 8 August 2025.
- Urresty-Vargas, C., Catalán, E., Razeto, J. and Sarmiento, F.O. (2024) Towards Biocultural Conservation of Chilean Palm Landscapes: Expanding Perspectives from Historical Ecology. *Land* 13:2206. <https://doi.org/10.3390/land13122206>

Inclusion of species *Beaucarnea hookeri* and *Beaucarnea glassiana* in Appendix II, as part of the listing of the *Beaucarnea* spp.

Proponents: Depositary Government (Switzerland) and Mexico

Summary: The genus *Beaucarnea*, known as Ponytail Palms, consists of plants native to countries in Central America (Belize, El Salvador, Guatemala, Honduras) and Mexico, traded primarily for use as ornamentals. The genus was included in Appendix II in 2017 after Mexico submitted a proposal to list one species (*B. recurvata*) based on criterion B of Annex 2a of Res. Conf. 9.24 (Rev. CoP16) and the other 10 species then recognised in the genus as lookalikes, given their similarity in seedlings and seeds (reportedly the main specimens entering international trade). At that time Mexico considered illegal harvesting of wild plants for ornamental purposes to be one of the main threats to *B. recurvata*.

At the time of the genus listing, no standard reference for the genus was adopted. However, two species formerly included in *Calibanus* have since been transferred into the genus *Beaucarnea*, (now *Beaucarnea hookeri* and *Beaucarnea glassiana*), and are recognised in the proposed standard reference (Rojas-Piña *et al.*, 2014) to be considered for adoption at CoP20 under Nomenclature matters (see CoP20 Doc. 110). As *B. hookeri* and *B. glassiana* were not considered part of *Beaucarnea* by the proponent at the time of the original listing in 2017, the Plants Committee agreed that their inclusion would constitute a substantive amendment to the current listing. At its 27th meeting (PC27) the Plants Committee invited the Depositary Government, in consultation with Mexico, to assess both species against the listing criteria in accordance with Res. Conf. 9.24 (Rev. CoP17) and to submit proposals for inclusion in Appendix II to CoP20 in accordance with paragraph 2f) of Res. Conf. 12.11 (Rev. CoP19) on *Standard nomenclature*.

According to the proposal, adult forms of *Beaucarnea* species (including *B. hookeri* and *B. glassiana*) display characteristics that can differentiate them from each other. However, the seeds and seedlings are reported to be indistinguishable to non-specialists and constitute the main specimens of *Beaucarnea* entering international trade. This proposal is therefore to include *B. hookeri* and *B. glassiana* in Appendix II under the look-alike criterion for inclusion in Appendix II based on criterion A of Annex 2b in Res. Conf. 9.24 (Rev. CoP17).

Beaucarnea hookeri and *Beaucarnea glassiana* are both endemic to Mexico and reportedly found in relatively few locations. *B. hookeri* was reported to be locally relatively common in the late 1960s. Information on its current status, and of *B. glassiana*, is lacking. There is some evidence of trade in seedlings and seeds of *B. hookeri* on online platforms in countries outside Mexico although according to one source all plants in trade derive from collections made in 1968. Of the nine online platforms found to be offering *B. hookeri* specimens for sale, all but two offered international shipping from countries including China, Czechia, Germany, Italy, the Netherlands and South Africa.

According to the CITES Trade Database, from 2017 to 2023, exporters have reported trade in *Beaucarnea* spp. to be mostly in artificially propagated live specimens and seeds, with most of these coming from the species *B. guatemalensis* and smaller quantities from *B. recurvata*. Most exports of seeds are reported to be from the USA, and most live specimens from China, Guatemala and Thailand. Mexico has reported very few exports of specimens from *Beaucarnea* species since the listing in 2017 (around 200 specimens and 1,150 kg of seeds), and all specimens are reported to be from artificial propagation.

In Mexico, harvest of terrestrial and native wildlife species is permitted under certain circumstances: it must be proven that harvest rates are lower than the natural renewal of the populations subject to use and that harvest will not have a detrimental effect on populations. Based on reports of seizures in EU-TWIX between 2017 and 2023 there are some indications of illegal trade in *B. recurvata* but none of these have reported Mexico as a country of origin, with most originating from the Netherlands. A high proportion of these seizures was reported as in artificially propagated plants with none reported to be of wild origin; it is probable that the remaining seizures were also of artificially propagated plants that lacked appropriate CITES certificates.

The genus listing proposal submitted at CoP17 in 2016 for *Beaucarnea* species stated artificial propagation of *Beaucarnea recurvata* did occur in nurseries in Mexico but that this did not satisfy the demand of the ornamental plant market at the national and international levels.

Analysis: There is evidence of some international trade in *Beaucarnea hookeri* seeds and seedlings (e.g. in the form that is reported to be difficult to distinguish for *Beaucarnea* species), but there is no evidence that *B. glassiana* is in international trade. It is likely that all *B. hookeri* specimens in trade originate in plants long-established in cultivation. Small quantities of species in the genus *Beaucarnea* being exported from Mexico in recent years (all reportedly being from artificial propagation) with no indication that illegal trade is still a threat to the wild species. Given that almost all global trade since the listing in 2017 has also been reported to be in artificially propagated specimens, regulation of trade in these two endemic species is unlikely to be necessary to bring trade under effective control for any of the *Beaucarnea* species currently in Appendix II. The **species do not therefore appear to meet the criteria in 2bA**. Nevertheless, this proposal has been submitted after consideration by the Plants Committee **following the guidance of Res. Conf. 12.11** (Rev. CoP19) on *Standard nomenclature* and **adding these species under the genus listing of *Beaucarnea* may be a more pragmatic solution than adding an annotation to clarify that they are excluded from the Appendices.**

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

Beaucarnea spp. were first listed in CITES Appendix II in 2017 (UNEP, 2025).

Taxonomy

Beaucarnea hookeri (Lem.) Baker (1872), *Beaucarnea glassiana* (L.Hern. & Zamudio) Rojas-Piña *et al.*, (2014). Both species were previously in the genus *Calibanus*.

Range

Mexico (endemic)

IUCN Global Category

Not assessed

Inclusion in Appendix II to improve control of other listed species

A) Specimens in trade resemble those of species listed in Appendix II under Res. Conf. 9.24 (Rev. CoP17) Annex 2 a or listed in Appendix I

The genus *Beaucarnea* consists of plants native to countries in Central America (Belize, El Salvador, Guatemala, Honduras) and Mexico, which are traded for use as ornamental plants. The 17th meeting of the Conference of the Parties in 2016 adopted proposal CoP17 Prop. 50, submitted by Mexico, to include *Beaucarnea* spp. in CITES Appendix II, based on compliance with criterion B of Annex 2a of Res. Conf. 9.24 (Rev. CoP16) for *Beaucarnea recurvata* and with criterion A of Annex 2b for the other 10 species in the genus that were recognized at the time of listing. The latter were included based on reports that although adult forms of the species in the genus display characteristics that can help differentiate the species (e.g., leaf anatomy, trunk shape, foliage density), the seeds and seedlings are indistinguishable to non-specialists and constitute the main specimens entering international trade.

At its 27th meeting (PC27; Geneva, Switzerland, 2024), the Plants Committee agreed to include the Mexican endemic species, *Beaucarnea hookeri* and *B. glassiana*, in a standard nomenclature reference for the genus (Rojas-Piña *et al.*, 2014). Because these species were not part of the genus *Beaucarnea* at the time of their inclusion in Appendix II and therefore their international trade is not regulated by CITES, the nomenclature change constitutes a substantive amendment to the current listing of the genus. Taking this into account, PC27 invited the Depositary Government, in consultation with Mexico, to assess both species against the listing criteria in accordance with Res. Conf. 9.24 (Rev. CoP17) and to submit proposals for inclusion in Appendix II to CoP20 in accordance with paragraph 2f) of Resolution 12.11 (Rev. CoP19) on Standard nomenclature.

The proposal states that *Beaucarnea hookeri* and *B. glassiana* display the typical morphological characteristics of the genus in their adult stage (a thick stem, broadened at the base that serves as water storage, as well as linear,

elongated leaves), which allow them to be distinguished from *B. recurvata*. However, it is not easy for non-experts to recognise them from each other or from other *Beaucarnea* species, particularly regarding juvenile specimens (seedlings) and seeds. In species of this genus, the seeds are spherical to ellipsoid, brown, reddish, or yellowish in color, all with three lobes: two smaller ones and a larger lobe that generally contains the embryo. Considering the morphological similarities of seeds and seedlings of *Beaucarnea hookeri* and *B. glassiana* to *B. recurvata* and other species of the genus listed in Appendix II, the proposal states that these two species meet the look-alike criterion for inclusion in Appendix II (criterion A of Annex 2b of Res. Conf. 9.24 (Rev. CoP17)), and it is therefore proposed that the species *Beaucarnea hookeri* and *B. glassiana* be included in the genus-level listing of *Beaucarnea* spp. in Appendix II. *This criterion states that 'species may be included in Appendix II if specimens of the species in the form in which they are traded resemble specimens of a species included in Appendix II under the provisions of Article II, paragraph 2 (a), or in Appendix I, so that enforcement officers who encounter specimens of CITES-listed species are unlikely to be able to distinguish between them'.*

Beaucarnea hookeri is distributed in the states of Guanajuato, Hidalgo, Querétaro, San Luis Potosí, and Tamaulipas. Following the discovery of *B. hookeri* in the mid-19th century, the species was little known outside its range until populations were discovered in 1968 in San Luis Potosí. The species was reportedly relatively common in mountainous areas where it was called *Sacamecate* by local people, its leaves being used for thatching roofs and scouring pots as they contain a soaplike substance. Plants and seeds were collected at that time and introduced into cultivation. It is believed that all plants commercially available are descended from this collection (smgrowers.com). *B. glassianus*, a similar second species, was discovered in Guanajuato State in 1995 and described in 2003 (smgrowers.com). It is not known to be in cultivation. There is little recent information on the status of either species.

Beaucarnea hookeri is listed as "threatened" on the national list of endangered species in Mexico (which is said to be equivalent to an IUCN Red List Assessment of Vulnerable or Near Threatened (MacipRíos et al., 2015)) and *B. glassiana* has not yet been evaluated.

A rapid survey of online trade in *B. hookeri* and *B. glassiana* found some offers for sale for *B. hookeri* (under its' synonym *Calibanus hookeri*) only. The species was found on eight online platforms and one Facebook group, with prices ranging from USD10-110 for seedlings, and around USD100 each for two instances in which the photos of the plant being offered for sale appeared to be an adult. Only one platform offered seeds, with no prices stated and the product out of stock. None of the online advertisements were for companies or platforms based in Mexico (they were based in China, Czechia, Germany, Italy, the Netherlands and South Africa), but almost all offered international shipping. No evidence of international trade in the two species (including under names from their previous genus *Calibanus*) was found in US trade data in LEMIS between 2008 and 2021, or in seizure records in EU-TWIX or WiTIS based on available data (from 2014-2023 for EU-TWIX and 1993-2025 for WiTIS).

Since the listing of *Beaucarnea* species in 2017, Mexico has reported very few direct exports: five live specimens of *B. gracilis*, 200 *Beaucarnea* spp., and ~1,150 kg of seeds from *B. recurvata*, with all specimens reported to be from artificial propagation. There have also been reports of 24 seizures of *B. recurvata* in EU-TWIX between 2017 and 2023, although none of these reported Mexico as an origin country. One seizure was for three seeds, with the remaining for over 7,500 live plants. Two seizures, totalling around 5,900 live plants, stated the source of plants seized was artificial propagation, but the remaining 22 seizures stated the source of the plants were unknown. The Netherlands was the country of origin reported for most seizures (17 out of 24) with the rest from Japan, the United Kingdom of Great Britain and Northern Ireland, Thailand, and China, and in three seizures, the country of origin was unknown. The listing proposal for the genus at CoP17 stated that despite the production of *B. recurvata* in intensive Wildlife Management and Conservation Units (UMAs) and Properties and Facilities that Manage Wildlife in a Confined Manner Outside its Natural Habitat (PIMVS), it does not satisfy the demand of the ornamental plant market at the national and international levels, and therefore, the seeds are constantly subject to illegal overcollection (CoP17 Prop 50).

Additional Information

Other comments

In Mexico, Mexico's General Wildlife Law (2000) Article 84 states that any harvest of terrestrial and native wildlife species (e.g., including threatened species) is permissible under certain circumstances, including that it must be proven that harvest rates are lower than the natural renewal of the populations subject to use and that harvest will not have a detrimental effect on populations.

The listing proposal for the genus at CoP17 in 2016 stated that at the time, no export permits had been granted for *B. recurvata* since 2009 (CoP17 Prop 50). It also stated that foreign buyers often visited sites where the species is distributed and asked farmers to collect as many juvenile plants as possible, in exchange for a daily payment. The wild plants were reportedly then taken to 'clandestine' nurseries to acclimatize them and subsequently be placed in pots and offered to legally registered nurseries, before being prepared for sale abroad.

The inclusion of so-called 'lookalike' species in the Appendices is the subject of Agenda item 102 at CoP20 where Parties are asked to consider document CoP20 Doc. 102 'Considering the 'lookalike' criterion Annex 2B A of Res. Conf. 9.24 (Rev. CoP17) on Criteria for amendment of Appendices I and II.' The document draws attention to the regulatory burden created by the increasing numbers of such listings, stating that implementation may divert resources from regulation of species that are genuinely affected by international trade, and proposes a series of actions to be undertaken by the Animals and Plants Committees, the Secretariat and the Standing Committee with a view to making recommendations for consideration at CoP21.

References

- CoP17 Prop. 50 (2016) Inclusion of the genus *Beaucarnea* (Lemaire, 1861) in Appendix II. (Proponent Mexico).
- MacipRíos, R., Ontiveros, R., LópezAlcaide, S., and CasasAndreu, G. (2015) The conservation status of the freshwater and terrestrial turtles of Mexico: A critical review of biodiversity conservation strategies. *Revista Mexicana de Biodiversidad* 86(4):1048–1057. <https://doi.org/10.1016/j.rmb.2015.09.013>
- Rojas-Piña, V., Olson, M.E., Alvarado-Cárdenas, L.O. and Eguiarte, L.E. (2014) Molecular phylogenetics and morphology of *Beaucarnea* (Ruscaceae) as distinct from *Nolina*, and the submersion of *Calibanus* into *Beaucarnea*. *Taxon* 63(6):1193–1211. <https://doi.org/10.12705/636.31>
- SanMarcos Growers (https://www.smgrowers.com/products/plants/plantdisplay.asp?plant_id=289) Accessed on 30 August 25.
- UNEP (2025). The Species+ Website. Nairobi, Kenya. Compiled by UNEP-WCMC, Cambridge, UK. Available at: www.speciesplus.net. Accessed on 16 July 2025.

Inclusion of *Commiphora wightii* in Appendix II

Proponent: European Union

Summary: *Commiphora wightii* is a slow-growing shrub or small tree that occurs in arid areas of India, Pakistan and Oman. The stems and branches produce an oleoresin or gum, known commonly as guggul or Indian bdellium, which has been harvested from the wild for medicinal use for centuries. The yields obtained by traditional tapping methods are in the order of 250-500 g of dry guggul per tree each season. Over time the tapping methods used generally lead to the death of individual trees.

Commiphora wightii faces numerous threats, the most serious of which are unsustainable harvesting practices, habitat degradation, overgrazing, and limited natural regeneration. The species was assessed as Critically Endangered on the IUCN Red List in 2014 due to an estimated global population decline of >80% over the three generations; this assessment needs updating. There is little more recent population information available; one recent study in Pakistan found that the average height of shrubs had noticeably reduced as mature plants are tapped for the resin; the populations in Oman are reported to be stable and in India local studies indicate that healthy populations of the species exist in some areas so that decline is not consistent throughout the range.

The export of wild-harvested guggul resin from India has been banned for over 30 years. The Indian state of Gujarat was historically the key producer and exporter. However, decreasing wild populations in India and increased state level regulation of wild-harvested gum have increasingly shifted sourcing of wild-harvested resin to Pakistan. In the early 2000s, an estimated quantity of 450 tonnes of *C. wightii* gum was exported from Pakistan to India each year for the industrial pharmaceutical industry. Finished products are processed in and exported from India, apparently now mainly produced from imports of resin from Pakistan. However, wild-harvested resin from India can also be processed for export. The wide availability of *C. wightii* products for sale on online marketplaces demonstrates ongoing worldwide commercial demand. Exported material can be in the form of bulk extract, or as powders, capsules and tinctures.

Techniques to cultivate the species and improve tapping methods are being developed in India but remain small-scale and insufficient to meet commercial demand. The success of cultivation efforts will not be known for years to come.

Extract (resin), powders and finished products all appear to be products first seen in trade. It may be necessary to ensure that these are specified as included as part of the listing through the inclusion of an annotation. According to the definition of "Extract" in Interpretation section of the Appendices this would cover "Any substance obtained directly from plant material by physical or chemical means regardless of the manufacturing process. An extract may be solid (e.g. crystals, resin, fine or coarse particles), semi-solid (e.g. gums, waxes) or liquid (e.g. solutions, tinctures, oil and essential oils)."

Analysis: *Commiphora wightii* is a relatively widespread shrub which occurs at moderate density in arid regions of western India, Pakistan and Oman. It has been extensively harvested for its gum and there has reportedly been a marked decline in the wild population size, expected to continue with a decrease in area and quality of habitat, and significant levels of exploitation and an inherent biological vulnerability. High levels of wild harvesting of *C. wightii* gum for international trade, with very limited, if any, cultivated alternatives, contributed to the species being assessed as Critically Endangered on the IUCN Red List assessment in 2014 based on population declines. Recent information on the population size at a global level is scarce although the species is reported to be locally abundant with healthy populations in parts of India (e.g. the Kachchh region of Gujarat) and in Oman. With depletion of accessible resources in India, most resin processed in India is imported from Pakistan, with some wild harvested resin also contributing to the exports of processed products from India. *Commiphora wightii* may meet criterion 2aB for inclusion in Appendix II Res. Conf. 9.24 (Rev. CoP17).

No annotation was proposed specifying parts and derivatives to be covered by the Appendix-II listing. Extract (resin), powders and finished products all appear to be products first seen in trade. Therefore,

if the Parties considered an annotation was necessary #1 would cover the most relevant products in trade.

Summary of Available Information

Text in non-italics is based on information in the Supporting Statement (SS); text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

No standard reference is recommended in proposal CoP20 Prop. 43. In accordance with the guidance provided in CoP20 Inf 6 a time-stamped extract from POWO prepared by the flora nomenclature specialist is proposed for adoption as a standard nomenclature reference (see CoP20 Doc. 110 Add., Annex 2):

POWO (2025). *Commiphora wightii*. World Checklist of Vascular Plants. Facilitated by the Royal Botanic Gardens, Kew. Published on the Internet; <https://powo.science.kew.org/> Retrieved 29 July 2025.

The genus *Commiphora* includes 181 accepted species (POWO, 2025).

The species was formerly known as *Commiphora mukul* or *Balsamodendron mukul* and was renamed as *C. roxburghii* in 1962. However, the name *C. wightii* (Arn.) Bhandari had been published prior to *C. roxburghii*, and therefore *C. wightii* (Arn.) Bhandari is recognised as the valid binomial name. This is the name currently accepted by POWO (2025) and *World Flora Online* (WFO, 2025). *C. mukul* continues to be frequently used in the pharmacological scientific literature.

Range

Native: India, Pakistan, Oman

Presence unconfirmed: Bangladesh and Nepal

IUCN Global Category

Critically Endangered A2cd (assessed 2014, needs updating, ver.3.1)

Biological and trade criteria for inclusion in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)

A) Trade regulation needed to prevent future inclusion in Appendix I

Small wild population

No global population estimate is available for *C. wightii* but, based on density and range extrapolations provided below, it is unlikely that the population size is small.

In India, data are available on population density for C. wightii in Rajasthan and Gujarat, the states where the species is most commonly recorded. Average population density for C. wightii across Rajasthan has been estimated as 21.9 plants/ha, however, forest blocks were selected as they were known to be areas where C. wightii occurred. The highest densities were reported from the district of Sawai Madhopur (105.9 plants/ha) and the lowest in Jhalawar (1.6 plants/ha) (Tomar et al., 2021). Another study focusing on C. wightii densities in desert and mountainous areas of Rajasthan estimated an average density of 40 plants/ha in the mountainous zone compared to 25 plants/ha in the desert zone based on 52 sampling plots measuring 0.04 ha (400 m²) each over 26 sites (Reddy et al., 2012). In Gujarat, C. wightii is most abundant in Kachchh and Junnagar districts. Density of plants has been recorded as up to 46 plants/ha in one area of Kachchh (Patel, 2018).

In Pakistan, a recent baseline study in the Tharparkar District surveyed 64 distinct locations using a standardized method of ten quadrats per site (IUCN, 2023). Based on this extensive field survey, the estimated density of C. wightii was approximately 13.78 plants/ha. Field observations further revealed that the average size of the C. wightii shrubs was notably reduced, with an average height of 1.3 ± 0.4 meters. This is significantly smaller than the species' potential maximum height, which may exceed four meters. The diminished stature of the shrubs is likely attributable to anthropogenic pressure, particularly the harvesting of mature plants for gum tapping. Such exploitation typically results in the eventual death of the tapped individuals, leading to the rarity of fully grown, mature specimens in the surveyed area.

Populations of C. wightii in the Eastern Hajar mountains of Oman are large and healthy (Patzelt, 2015).

Restricted area of distribution

In India and Pakistan, C. wightii is associated with semi-arid to arid open and hilly habitats. The IUCN Red List assessment for the species gives an estimated extent of occurrence (EOO) of 20,000-22,000 km² and an estimated area of occupancy (AOO) of just 2,000-2,200 km² (Ved et al., 2015); however, this estimate does not consider the species' range in Oman.

In India, *C. wightii* is most commonly recorded in the states of Rajasthan and Gujarat and is also found in Haryana, northern Madhya Pradesh, Maharashtra and Karnataka. *Presence in Maharashtra is uncertain with reliable records limited to only four Indian states – Gujarat, Haryana, Madhya Pradesh and Rajasthan (Brindavanam in litt., 2025). Within Madhya Pradesh the extensive ravine system known as the Chambal ravines is an important area for the species (Thomas et al., 2022).*

In Pakistan, *C. wightii* occurs in the provinces of Sindh and Balochistan. *The majority of the Commiphora wightii subpopulation is concentrated in the Tharparkar District (Thar Desert of Pakistan, area about 22,000 km²).*

In Oman, C. wightii occurs in the north of the country, on the lower slopes of the Eastern and Western Hajar mountains with Vachellia tortilis and Euphorbia larica at 50–1000 m altitude. There is an overlap in distribution of C. kua and C. wightii in the eastern Hajar mountains and a few locations in the foothills of the western Hajar (Ghazanfar, 2007). It is also found in the coastal areas of Dhofar, in southern Oman (Miller and Morris, 1988).

Decline in number of wild individuals

There has been a marked overall decline in the number of wild individuals of the species which is ongoing in parts of its range and could warrant Appendix-II listing in the near future. A decline in the number of wild individuals was reported in 2014, but this has not been quantified on a global scale. The species was listed as Critically Endangered on the IUCN Red List in 2014, on the basis of an estimated global population decline of over 80% over the past three generations (84 years, 1930-2014) *as a result of habitat loss and degradation, coupled with unregulated harvesting and tapping of oleo-gum resin (Ved et al., 2015). Biological characteristics of the species including slow growth, very low regeneration with poor germination rates make it inherently vulnerable to over-exploitation.* The IUCN Red List assessment is based on information for India and Pakistan only. More recent information on the species' overall population status has not been found.

Local studies indicate that healthy populations of the species exist in some areas so that decline is not consistent throughout the species' range. For example, extensive field studies at different times indicated that the natural population of C. wightii in Kachchh region of Gujarat in India were healthy and largely undisturbed (Brindavanam in litt., 2025).

The populations of *C. wightii* in Oman are reported to be stable.

B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences

Harvesting from the wild using destructive tapping techniques is threatening the survival of the species in significant parts of its range. Harvesting is both for traditional use and for international trade. *C. wightii* is inherently vulnerable to overexploitation as a result of its slow growth rate, relatively low dispersal ability, and poor germination rates. Unregulated commercial exploitation is therefore expected to impact the recruitment and regeneration capacity of *C. wightii*, which is already heavily depleted.

Harvest is focused on individuals that are at least 5 years old (and 8-10 years old in India). *Commiphora wightii* trees are tapped for their dark green or pale brown oleoresin, an exudate produced from the branch and stems of the plant upon injury. Gum oozing from the stems and branches of *C. wightii* is triggered by infection from pathogenic *Xanthomonas* bacteria. While this occurs naturally in the wild, a gum suspension containing this bacterium is often applied to tapping cuts, alongside other chemicals such as ethephon to increase gum yields. *More traditional tapping methods involve making several deep incisions on the stem to extract the maximum amount of gum. A paste is applied around the incision consisting of horse or wild ass urine, gum resin and copper sulphate. This crude method increases the amount of gum three to four times over that obtained under normal tapping procedures, but the shrub subsequently becomes unfit for tapping for several years and ultimately plants may die due to the injurious effect of copper sulphate (Oldfield and Jenkins, 2012).* Seasonal average yield per tree is 250-500 grams.

The level of trade is difficult to determine as there appears to be no national or global monitoring of trade in the species. Estimates do however suggest that international trade is considerable. According to the National Medicinal Plants Board of India, estimated annual trade in *C. wightii* oleoresin, stems and roots was 1000-2000 metric tonnes (t) *for the financial year 2014-15 (Goraya and Ved, 2017). This quantity includes domestic retail use, and industrial demand for the domestic and export market. Domestic production accounts for only a fraction of the total annual demand with imports in large quantities supplying the demand (Goraya and Ved, 2017).* Currently, there is no commercial scale cultivation of *C. wightii*, and destructive methods of harvesting gum produce significantly higher gum yields. Considering the level of trade in *C. wightii* and the high level of demand that cannot be met from cultivation, destructive harvest and population declines will likely continue within wild populations without the introduction of measures to regulate harvest and international trade.

C. wightii products are traded under several different harmonized system tariff codes (HS Codes), which are not specific to genus or species. For this reason, trade volumes for *C. wightii* are uncertain; however, available estimates indicate that the amount of material traded internationally is considerable. The CITES MA of Oman reported no significant exports in the species. The Indian state of Gujarat was historically the key producer and exporter of *C. wightii* gum. However, over recent decades, gum was reported to be increasingly exported from Pakistan to India, where it is then processed and re-exported. In 1999, 90% of the estimated 500 tons (508 t) consumed by the Indian pharmaceutical industry was met through imports from Pakistan, equating to 450 tons (457 t) imported by India from Pakistan per year (*this would require an estimated ~0.9 to 1.8 million trees based on production figures per tree*). The situation was similar in 2006-2007, where a nation-wide study conducted by the National Medicinal Plants Board of India found that the majority of *C. wightii* raw material in India was sourced through imports.

The estimated consumption by Indian herbal manufacturing units for the period 2005-2006 comprised 909 t (dry weight) with the major proportion of domestic requirements met through imports from Pakistan (Ravikumar et al., 2018). More recently, the estimated domestic demand in India is about 1,610 t whereas the availability is nearly 10 t, annually (Thomas et al., 2022). During the 2024 season about 15 tons (15 t) of guggul gum was harvested from plantation sources in the Morena area (Brindavanam in litt., 2024 based on pers. comm Mr Zakir Hussain).

According to Indian export data from Zaub Technologies & Data Services Pvt Ltd, which included shipment descriptions that could be used to determine species in trade, India's export of *C. wightii* oleoresin extract was estimated at 27.5 t in 2014; as described above, it is possible a proportion of this trade represents re-exports. Using a conversion factor of 7:1 (w/w3) based on an average of reported drug-to-extract ratios in the wider literature, this was suggested to represent approximately 192.5 t raw oleoresin. Extract was reportedly exported to Brazil, Canada, Costa Rica, the European Union (Austria, Croatia, Czechia, Italy, Finland, France, Germany, Netherlands, Romania, Slovakia, Spain), Indonesia, Japan, Malaysia, Saudi Arabia, South Africa, the UK and the USA.

Table 1: Export of Guggul gum and its products from India in 2023 including HS code, importing country, quantity in kilograms (kg) and price in USD in integers.

HSN	Product Description Amended	Destination	Sum of Qty (kg)	Sum of Price (USD)
13021919	Guggul extract 2.5% sterones	Germany	700	15519
		Italy	589	16368
		USA	425	91060
		Ireland	400	31895
		South Africa	200	8539
		Mexico	200	10353
		Indonesia	175	11138
	Guggul extract	France	661	11866
		USA	558	31972
		Netherlands	107	4993
		Japan	100	12917
		Russia	75	1797
		Belarus	25	586
		Brazil	25	2030
		Austria	20	798
		Australia	10	108
	Guggul dry extract	USA	500	10648
		Australia	75	1639
	Guggul oil	United Arab Emirates	500	9945
	Guggul extract 10% sterones	Italy	404	28263
		Belgium	50	2197
		Netherlands	10	361
	Guggul extract 5% sterones	Germany	100	5347
		UK	50	2655
	Guggul extract capsules	United Arab Emirates	40 (Number)	179

HSN	Product Description Amended	Destination	Sum of Qty (kg)	Sum of Price (USD)
13021990	Guggul extract 2.5% sterones	France	200	3982
		Indonesia	150	10948
		Germany	75	1466
	Guggul extract 20% sterones	Belgium	300	19229
	Guggul extract	USA	230	7526
		Indonesia	25	1700
	Guggul extract 10% sterones	France	150	6180
13021999	Guggul extract capsules	Hungary	3	157
	Guggul dry extract	Brazil	3	60
13021999	Guggul extract	Mexico	100	5500
	Guggul extract - food supplement	Netherlands		
13021911			5	133
Grand Total			7200	476853

Source: Volza. (2023). Guggul extract Exports from India - Export data with price, buyer, supplier, HSN code. Retrieved from <https://www.volza.com/p/commiphora-mukul-or-guggul-extract/export/export-from-india>

Very limited information is currently available on exports of guggul from Pakistan.

Populations of C. wightii in Oman are not exploited commercially but are collected for local use.

Local inhabitants readily distinguish two species, as "dhee" (C. wightii) and "mukul" (C. kua), with separate uses. The resin has been used for medicinal and other purposes: in the eastern Hajar region of northern Oman, the gum-resin is used for chest and liver ailments, and as soap for the final bath given to the dead (Ghazanfar in litt., 2025).

Extract (resin), powders and finished products all appear to be products first seen in trade. It may be necessary to ensure that these are specified as included as part of the listing through the inclusion of an annotation. According to the definition of "Extract" in Interpretation section of the Appendices this would cover "Any substance obtained directly from plant material by physical or chemical means regardless of the manufacturing process. An extract may be solid (e.g. crystals, resin, fine or coarse particles), semi-solid (e.g. gums, waxes) or liquid (e.g. solutions, tinctures, oil and essential oils)". Annotation #1 (All parts and derivatives, except:

- a) seeds, spores and pollen (including pollinia);*
- b) seedling or tissue cultures obtained in vitro transported in sterile containers;*
- c) cut flowers of artificially propagated plants; and*
- d) fruits, and parts and derivatives thereof, of artificially propagated plants of the genus Vanilla.)*

appears to be the most appropriate to cover products first seen in trade.

Illegal trade

In June 2024 an illegal consignment containing over 17,000 kg of guggul was confiscated at Karachi Gateway Terminal in a joint operation by the provincial wildlife department and the Anti-Narcotics Force. The provincial law places a ban on the extraction of the plant's secretion, its trade, export, and commercial-scale transportation. The consignment was destined for UAE [Huge shipment of 'protected' gum seized at Karachi port - Newspaper - DAWN.COM](#).

No records for this species were reported in the WITIS or EU-TWIX seizure databases.

Additional Information

Threats

*C. wightii is threatened by habitat loss and fragmentation driven by agriculture, mining, and urban development. The species' naturally low regeneration rates, coupled with overgrazing by sheep and collection of branches as firewood in the rainy season, competition with invasive species (e.g. *Prosopis juliflora*) and highly fragmented populations, are considered to amplify the impact of the species' exploitation for international trade in parts and derivatives for medicinal use.*

In Oman, the threats to the species' include habitat loss, degradation and fragmentation, caused by infrastructure development such as road building, urban growth, and agricultural expansion. Furthermore, increasingly extended periods of drought in the region are an additional stressor on the population. The population of C. wightii is stable

on the Hajar mountains of northern Oman. The population is not extensive, nor continuous, but fragmented, as are most other populations of species on the mountains of northern Oman. Shrubs of *C. wightii* in very good form were seen during a visit in 2022, growing on steep rocky slopes in an area close to urban development and consequently without goats or camels (Ghazanfar in litt., 2025). In 2025, some populations close to the Capital area have further decreased in size and are more isolated, due to development. But overall, and in particular in the foothills of the Eastern Hajar, the populations remain large and healthy (Patzelt in litt., 2025).

Conservation, management and legislation

In **India**, the management of harvest and cultivation of medicinal plants varies from state to state. While tapping of *C. wightii* appears to be limited or prohibited in Rajasthan, in Gujarat, *C. wightii* gum is categorized as a nationalized item, meaning that collection, sale and marketing is administered through the Gujarat State Forest Development Corporation Ltd. The state of Gujarat has prohibited harvesting wild populations of the species (Brindavanam in litt., 2025).

A Public Notice published in 1994, which revised the 1992-1997 Export and Import Policy, prohibited the export of *C. wightii* obtained from the wild (including its derivatives and extracts) (Public Notice No. 47 (PN)/92-97). In 2005, Public Notice No. 47(RE-2005)/2004-2009 included *C. wightii* in a list of species for which a new scheme to promote export of minor forest products did NOT apply. It is unclear whether the export ban on wild products is enforced. The CITES MA of India (in litt. to European Commission, 2024) stated that while the export of raw *C. wightii* resin was prohibited, export of processed products was permitted and “encouraged”. It should be noted that raw guggul resin that is reported to originate from India appears to be available for sale on international markets.

The CITES MA of India reported that initiatives were underway to develop non-lethal resin extraction methods, with one method successfully developed by the Arid Forest Research Institute (AFRI); however, the extent of practical implementation of these methods is unclear. *C. wightii* has also been prioritized for conservation of seed germplasm as part of an initiative to conserve the forest genetic resources of Rajasthan and Gujarat. Previous conservation initiatives include the launch in 2007 of a Save Guggul Movement, working with local communities in the Aravali Hills in Rajasthan State, building awareness and encouraging sustainable management of the resource (Soni, 2010). The following year a project was launched by the Indian National Medicinal Plants Board to undertake enrichment planting of guggul over several hundred hectares in the Kutch region (Oldfield and Jenkins, 2012).

The State Government of Gujarat established a network of Medicinal Plant Conservation Area (MPCA) sites in Kachchh region during 2007-2008 for conservation of the species. A study to assess the outcomes of conserving Guggul in two of the MPCA sites has been carried out using regeneration trends and resource health as criteria (Brindavanam et al., 2022).

CCRAS (Central Council for Research in Ayurvedic Sciences) under Ministry of Ayush has a Guggul herbal farm at Mangliawas (near Ajmer), Rajasthan, which focuses on the conservation and research of *Commiphora wightii* in India. This farm has been working for many decades (Uniyal in litt., 2025).

DMAPR-ICAR (Directorate of Medicinal and Aromatic Plants Research) based in Anand, Gujarat, has developed guidelines for sustainable harvesting of gum from the Guggul plant. However, implementation of these practices remains limited, and available evidence suggests that adoption across harvesting communities has been minimal, with few measurable outcomes to date (Uniyal in litt., 2025).

In **Pakistan**, *C. wightii* was declared a protected species in 2024 under the Sindh Wildlife Protection, Preservation, Conservation and Management Act of 2020. It is unclear if this measure prohibits or simply regulates the harvest of *C. wightii* gum.

Following the recent survey undertaken (IUCN, 2023), IUCN Pakistan has announced the establishment of a Thar Flora Conservation Station in the Tharparkar region. This initiative aims to prioritize the conservation and propagation of endangered plant species including *C. wightii* within their natural ecological range. The conservation station is expected to serve as a critical facility for research, restoration, and community-based conservation efforts focused on the unique and vulnerable flora of the Thar Desert (Sirohi in litt., 2025.)

At a national level in Oman *C. wightii* has been assessed as least concern (Patzelt, 2014). There is no legislation or management plan for *C. wightii* in Oman. The species occurs in the As Sareen Nature Reserve and is highly likely to also occur in the Ras Al Sajar Nature Reserve (Patzelt in litt., 2025).

Artificial Propagation/captive breeding

Cultivation of *C. wightii* has been recognized as essential for the conservation of the species for around 50 years and the urgent need for large scale cultivation to fulfil demand continues to be highlighted by experts as well as

the need for development of *ex situ* conservation methods such as artificial propagation of high yielding genotypes. However, cultivation of *C. wightii* is challenging, with a lack of quality planting material as a result of slow and unpredictable germination acting as a key barrier. Vegetative propagation by stem cutting is considered the most effective method. *The seeds of C. wightii exhibit poor germination but techniques are being developed to enhance the process.*

In India the National Medicinal Plants Board has promoted resource augmentation projects in Gujarat and Rajasthan with development of propagation techniques and nurseries (Patel, 2018). There is also local cultivation of C. wightii in the Chambal ravines of Madhya Pradesh. Local farmers have been growing C. wightii on their bunds and unproductive patches of land for over 20 years with about 15 tons (15 mt) harvested from this source in 2024 (Brindavanam in litt., 2025).

While some cultivation initiatives exist, they are limited in scale and not yet sufficient to meet commercial demand or relieve pressure from wild populations (Dhyani in litt., 2025).

This species is recorded in nine ex situ collections (BGCI, 2024). Cultivation at Oman Botanic Garden showed very limited success with poor results from cuttings and low seed germination (Patzelt in litt., 2025).

Implementation challenges (including similar species)

Identification of products in trade may represent a significant challenge. Morphologically, *C. wightii* gum is reported to be difficult to distinguish from oleoresins of other species of *Commiphora* (myrrh) and species of *Boswellia* (Frankincense). Neither of these genera are currently listed in the CITES Appendices. Resins from species of the two genera can be mixed together or used as substitutes. *Boswellia ovalifoliolata* has for example been reported as an adulterant of *Commiphora wightii* in trade (Cunningham et al., 2018). Identification of oleoresins generally requires the use of chromatography, e.g. high-performance liquid chromatography (HPLC) and thin-layer chromatography (TLC). The presence of guggulsterones (a plant steroid) in *C. wightii* distinguishes the gum of this species from others in the genus and from species of other genera producing characteristic peaks in chromatograms. DNA-based methods of identification have also been developed. However, it is unclear whether identification is feasible for non-experts or trained customs officers.

The nomenclature applied to *C. wightii*, with the name *C. mukul* also used (although generally considered a synonym) may also be a source of implementation difficulties.

As noted by Yang et al. (2024), there have always been problems in identifying the origin of medicinal products derived from Commiphora. Taking the plants C. mukul and C. wightii as examples, they are the same tree species, Commiphora wightii (Arnott) Bhandari syn. Commiphora mukul (Hook. ex Stocks) Engl, but because of their different names in circulation, the resins they produce are often considered to be two different myrrh. This misunderstanding not only affects the proper use of the medicine but also prevents research and development of its medicinal value.

References

- Brindavanam, N.B., Goraya, G.S., Singh, S.P, Bhatt J., Paradva, B., Poptani, R., Vadodaria, V., Mahato, A.K.R. and Vijaykumar, V. (2022) A study on *in situ* conservation of *Commiphora wightii* (Arn.) Bhandari in MPCA model. *Medicinal Plants* 14(4):648–657.
- Brindavanam, N.B. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- CITES Secretariat (2025) Appendices of the Convention. Nomenclature matters. Standard nomenclature. Addendum. CoP20 Doc. 110 Add.
- Cunningham, A.B., Brinckmann, J., Kulloli, R.N. and Schippmann, U. (2018) Rising trade, declining stocks: the global guggul (*Commiphora wightii*) trade. *Journal of Ethnopharmacology* 223:22–32.
- Dhyani, A. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Ghazanfar, S.A. (2007) Flora of the Sultanate of Oman, Vol. 2, Crassulaceae-Apiaceae (Text + photo CD-ROM). Scripta Botanica Belgica. Belgium: National Botanic Garden of Belgium, Meise. 220 pp.
- Ghazanfar, S.A. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Goraya, G.S. and Ved, D.K. (2017) Medicinal Plants in India: An Assessment of their Demand and Supply. National Medicinal Plants Board, Ministry of AYUSH, Government of India and Indian Council of Forestry Research and Education.
- Miller, A.G. and Morris, M. (1988) Plants of Dhofar. The Office of the Adviser for Conservation of the Environment, Diwan of Royal Court, Sultanate of Oman.
- Oldfield, S. and Jenkins, M. (2012). Wild flora for improved rural livelihoods. Case studies from Brazil, China, India and Mexico. Botanic Gardens Conservation International, Richmond, U.K.
- Patel, A.C. (2018) Conservation & resource augmentation of Guggulu (*Commiphora wightii*) in Gujarat. Presentation made at International Seminar on Medicinal Plants: An Associated event of VIII Word Ayurveda Congress, 14–17 December 2018.
- Patzelt, A. (2014) Oman Plant Red Data Book. Diwan of Royal Court, Oman Botanic Garden.

- Patzelt, A. (2015) Synopsis of the Flora and Vegetation of Oman, with Special Emphasis on Patterns of Plant Endemism. Jahrbuch 2015 der Braunschweigischen Wissenschaftlichen Gesellschaft, Tagung 2014, 282–317. Available at: <http://www.digibib.tu-bs.de/?docid=00060240>.
- Patzelt, A. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- POWO (2025) *Commiphora wightii*. Plants of the World Online Website. Available at: <https://powo.science.kew.org/results?q=Commiphora%20wightii> Accessed on 8 August 2025.
- Ravikumar, K., Begum, S.N., Ved, D.K., Bhatt, J.R. and Goraya, G.S. (2018) Compendium of traded Indian medicinal plants. FRLHT.
- Reddy, C.S., Meena, S.L., Krishna, P.H., Charan, P.D. and Sharma, K.C. (2012) Conservation Threat Assessment of *Commiphora wightii* (Arn.) Bhandari—An Economically Important Species. *Taiwania* 57(3):288–293.
- Sirohi, H.M. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Soni, V. (2010) Conservation of *Commiphora wightii*, an endangered medicinal shrub, through propagation and planting, and education awareness programs in the Aravali Hills of Rajasthan, India. *Conservation Evidence* 7:27–31.
- Soomro, N.A., Sirohi, M.H., Malik, I.N. and Hasaan, F. (2023) Ecological baseline survey report Tharparkar Sind. IUCN.
- Thomas, M., Prajapati, U., Tripathi, N., Upadhyay, K., Batham, L., Pahalwan, D.K. and Koutu, G.K. (2022) Survey of data deficient endangered Guggul plants (*Commiphora wightii*) in District Morena, Madhya Pradesh. Directorate of Research Services, JNKVV, Jabalpur, 34p.
- Tomar, U.K., Singhal, H., Gaur, A. and Saini, L.S. (2021) Population density, genetic diversity and hot spots of *Commiphora wightii* (Arnott) Bhandari in Rajasthan State. *Journal of Applied Research on Medicinal and Aromatic Plants* 25:100323. <https://doi.org/10.1016/j.jarmap.2021.100323>
- Uniyal, R. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Ved, D., Saha, D., Ravikumar, K. and Haridasan, K. (2015) *Commiphora wightii*. *The IUCN Red List of Threatened Species* 2015: e.T31231A50131117. <http://dx.doi.org/10.2305/IUCN.UK.20152.RLTS.T31231A50131117.en>
- WFO (2025) *Commiphora wightii* (Arn.) Bhandari. Available at: <http://www.worldfloraonline.org/taxon/wfo-0000617524>. Accessed on 14 July 2025.
- Yang, Y., Sun, X., Peng, C., Wei, J. and Yang, X. (2024) The Genus *Commiphora*: An Overview of Its Traditional Uses, Phytochemistry, Pharmacology, and Quality Control. *Pharmaceuticals* 17:1524. <https://doi.org/10.3390/ph17111524>
- Thomas, M., Prajapati, U., Tripathi, N., Upadhyay, K., Batham, L., Pahalwan, D.K. and Koutu, G.K. (2022) Survey of data deficient endangered Guggul plants (*Commiphora wightii*) in District Morena, Madhya Pradesh. Directorate of Research Services, JNKVV, Jabalpur, 34p.
- Volza (2023) Guggul extract Exports from India – Export data with price, buyer, supplier, HSN code. Available at: <https://www.volza.com/p/commiphora-mukul-or-guggul-extract/export/export-from-india>

Transfer of Cycad Spurge *Euphorbia bupleurifolia* from Appendix II to Appendix I

Proponent: South Africa

Summary: The Cycad Spurge *Euphorbia bupleurifolia* is a very slow-growing succulent species endemic to South Africa. *Euphorbia bupleurifolia* has not been assessed by IUCN but was assessed as critically endangered in a national red list assessment in 2022. It is popular in international trade as an ornamental plant and has been included in Appendix II since 1975 under the listing of the genus *Euphorbia* (currently with annotation #4).

Historically the species was found from inland areas around Pietermaritzburg in the province of KwaZulu-Natal, down along the coast, and then further west throughout the eastern region of the Eastern Cape up to Makhanda but most of the remaining populations are now restricted to the Eastern Cape province with a restricted area of distribution with an estimated area of occupancy of around 130 km², and an extent of occurrence of around 65,000 km². Subpopulations are small or very small and fragmented, varying in size from one to 675 individuals and with distances between the nearest neighbouring populations often around 8 km. Around 40% of suitable habitat is estimated to have been lost due to infrastructure development, mining and agriculture. Field surveys of *Euphorbia bupleurifolia* in 2018 across most historic localities identified just over 1,700 individual plants with a total population size estimated at around 2,500 individuals. Plants were only present in nine of the 31 historic subpopulations surveyed. The national red list assessment inferred a marked decline of 97% since 1975 based on estimates of the quantities of plants with a suspected wild origin in trade relative to the number of plants remaining in the wild from the 2018 field surveys.

Euphorbia bupleurifolia is a protected indigenous plant in South Africa; collection within state and private nature reserves requires a permit from the relevant conservation authority, whilst landowner permission is required to gather plants on privately owned land.

Historically there have been large quantities of *Euphorbia bupleurifolia* directly exported by South Africa, with over 63,000 plants reported since 1980. Between 2014 and 2018, South Africa exported just under 4000 plants per year; the only recorded exports since then are of 200 in 2021. Almost all exports have been declared as artificially propagated, but it is suspected that a large proportion were wild-collected, with many specimens observed in nurseries showing characteristics of wild-collected plants and two nurseries admitting to having sourced large plants from the wild without the permits required. One of the nurseries that had been the source of over 94% of ~ 30,000 *E. bupleurifolia* plants exported from South Africa between 2010 and 2018, has since been implicated in laundering of wild plants and has been closed, which is likely the main reason for a decline in reported *E. bupleurifolia* exports since 2019.

Although the species occurs in accessible areas and is legally protected where it grows, current enforcement is believed to be insufficient to prevent illegal harvesting. The species is reportedly moderately difficult to grow when propagated from seed, which would appear to make it challenging to produce large quantities in cultivation.

Additional pressures on the species include wild harvesting for local use as well as ongoing habitat loss and degradation.

Analysis: The overall wild population of *Euphorbia bupleurifolia* is small with some very small subpopulations. It has a restricted area of occupancy within a wider extent of occurrence and occurs at very few locations, with evidence of a decrease of 70% in the number of known subpopulations.

There is evidence that export in wild specimens falsely declared as artificially propagated has previously impacted populations of the species and may have been the main cause of marked recent population declines. Recent enforcement efforts appear to have stopped, or at least reduced, the illegal wild harvest and trade but there is evidence of some continued demand for the species, including exports of live plants from South Africa in small numbers in the last five years. The species

therefore **appears to meet the biological and trade criteria for inclusion in Appendix I** in Res. Conf. 9.24 (Rev. CoP17).

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

Listed in CITES Appendix II under a genus listing for *Euphorbia* spp. in 1975. The current listing of this genus is under annotation #4 which for this species exempts seeds, seedlings or tissue cultures obtained *in vitro* transported in sterile containers, and cut flowers of artificially propagated plants.

Taxonomy

Euphorbia bupleurifolia Jacq.

Range

South Africa (endemic)

IUCN Global Category

Not assessed.

Biological criteria for inclusion in Appendix I

A) Small wild population

The proposal states that the current known population (based on the best available information) is estimated to comprise less than 2,500 mature individuals, noting that this figure is likely an underestimate of the national abundance. *The national red list assessment states that field surveys in 2018 indicate that between 2,460 and 2,658 mature individuals are likely extant (Mhlongo and Pfab, 2022).* A total of just 1,724 plants was counted during systematic field surveys in 2018 in 31 known historical localities across the species' distribution. There is anecdotal information to suggest that the plants are more common than ascertained during these surveys, with several additional subpopulation localities recently identified and at least one extant subpopulation not yet surveyed. Nevertheless, plants were only found in nine of the 31 known localities surveyed, with subpopulations varying in size from one plant to 675 individuals. *Around two thirds (67%) of the known remaining subpopulations are very small, comprising less than 100 individuals each, with the most central subpopulations having a substantially larger number of individuals compared to the peripheral northerly and the westernmost subpopulations.*

B) Restricted area of distribution

Euphorbia bupleurifolia is endemic to South Africa. The area of occupancy (AOO) for *E. bupleurifolia* according to the national red list assessment is estimated to be 132 km² with an extent of occurrence (EOO) of around 64,496 km². *A research paper published in 2023 by the authors of the national red list assessment estimated a smaller area of occupancy of 36 km² and extent of occurrence of 22,200 km² based on surveys at 31 historical localities in 2018 but noted that two additional localities were identified after the field surveys were conducted, (Mhlongo et al., 2023), which may account for the slightly larger area of occupancy and extent of occurrence stated in the red list assessment.*

The species' historical range is reported to have stretched from the inland areas around Pietermaritzburg in the province of KwaZulu-Natal, down along the coast, and then further west throughout the eastern region of the Eastern Cape up to Makhanda (formerly known as Grahamstown). Distances between the nearest neighbouring populations are 8 km, thus, the distribution is considered to be highly fragmented. Searches of the 31 previously recorded localities revealed that only nine populations (29%) still had plants in 2018, indicating a contraction of the distribution. The species is thought to occur in several additional localities, but its distribution can no longer be considered as widespread, with most of the remaining populations now restricted to the Eastern Cape province. *The national red list assessment stated that between 10 and 36 subpopulations are likely to be extant based on extensive surveys undertaken across this species range in 2018 (Mhlongo and Pfab, 2022).*

C) Decline in number of wild individuals

This formerly widespread and common species has suffered a significant loss in the number and size of subpopulations owing to ongoing removals and habitat loss across its distribution range. The species is claimed to have experienced a reduction of at least 97% since 1975 due to illegal collecting for the international specialist succulent trade and the local traditional medicine trade. This decline is estimated based on the reported number of (suspected wild origin) plants exported for the horticultural trade in addition to those suspected to have been harvested for trade in local traditional medicine markets, relative to the number of plants remaining in the wild as

determined by extensive surveys across this species' distribution in 2018. Further evidence of this decline is that less than 30% of the historical localities recorded for *E. bupleurifolia* were found to have plants during the 2018 field surveys. Less than 5% of the remaining subpopulations occur within some type of protected area, and ongoing loss of individuals due to trade and habitat loss is likely to cause at least a further 10% decline in the population by 2035. About 40% of the suitable habitat for the species has also been lost, mostly due to infrastructure developments, mining, and agriculture.

Trade criteria for inclusion in Appendix I

The species is or may be affected by trade

Euphorbia bupleurifolia is a popular plant in the international ornamental trade. This species has been listed in CITES Appendix II under a genus listing for *Euphorbia* spp. since 1975 (UNEP, 2025). Several parts and derivatives of the species, including seeds, are currently exempt from regulation under CITES as it is currently listed with annotation #4, but would no longer be exempt from regulation under CITES Appendix I.

According to CITES Trade Data, South Africa has reported over 63,800 live plants of *Euphorbia bupleurifolia* in direct exports for commercial purposes since 1980. Only seven live plants (two in 1997 and 5 in 2007) were reported to be wild-sourced, with the remaining sourced from artificial propagation. Since 2019, there has been an overall decline in the number of direct commercial exports of *E. bupleurifolia* reported by South Africa, from an average of ~3,840 live plants a year from 2014-2018 to just 200 since 2019 (all in 2021 and imported by Japan) (see Figure 1).

Direct exports by other countries (mostly the Netherlands, with smaller amounts from Japan, Czechia and Thailand) have also declined in recent years, with the highest quantity of exports reported between 2015 and 2017 (~1,990 a year) and an average of 225 a year between 2018 and 2023, with recent exports almost all reported to be from the Netherlands. Very few exports of commodities other than live plants were reported over the ten years (just ten stems from South Africa to Thailand in 2020). All direct exports between 2014 and 2023 were reported to be in artificially propagated specimens.

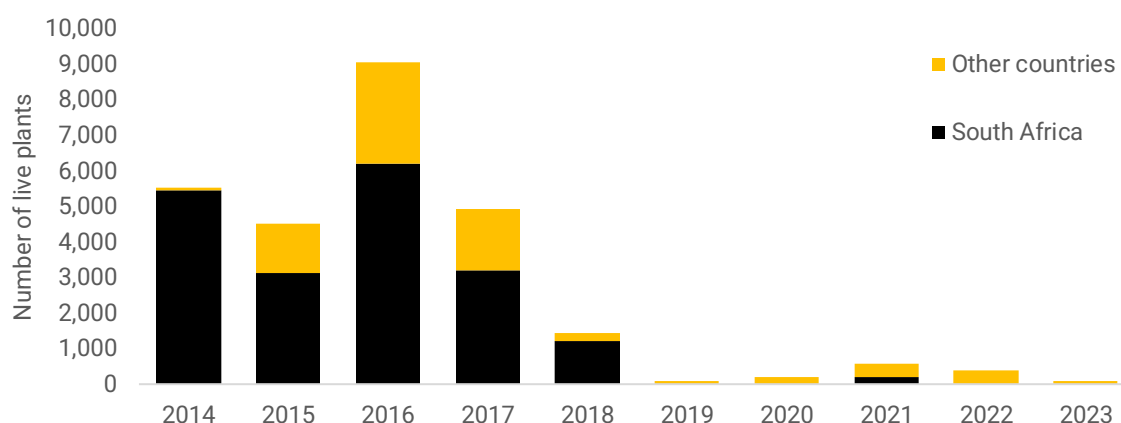


Figure 1. Direct commercial exports of live plants of *Euphorbia bupleurifolia* by South Africa and other countries between 2014-2023. All plants were reportedly artificially propagated. Source: CITES Trade Database.

The proposal states that whilst most of the specimens exported were reported as artificially propagated, recent investigations have revealed evidence of previous and continued large-scale removals and laundering of wild plants into the legal trade. A 2018 audit of the five nurseries known to be cultivating the species found that 98% of the ~8,700 *E. bupleurifolia* plants under cultivation at the time showed strong indications of being wild collected. None of the facilities were able to provide proof of legal acquisition of their parental stock, while two facilities admitted to sourcing large plants from the wild without the necessary permits. The nursery that had exported more than 94% of *E. bupleurifolia* plants between 2010 and 2018 (~28,400 out of 30,354 plants) has since been implicated in laundering of wild plants and 3,556 wild *E. bupleurifolia* plants were seized from this nursery in 2019. It is suspected that many of the plants that have previously been exported by this facility over the years may have been illegally obtained from the wild population. This nursery has since had their stock confiscated and closed down (Newton in litt., 2025), which is likely the main reason for a decline in exports of *E. bupleurifolia* plants exported from South Africa since 2019.

Euphorbia bupleurifolia is also traded locally for traditional medicine use and trade. The white milky sap/latex of the species has purported applications in the treatment of various skin ailments and is used in both the Zulu and Xhosa cultures. Extensive levels of local harvesting have previously been reported across the Eastern Cape with wild-sourced plants commonly traded in large volumes in traditional medicine markets and shops. More recent surveys of local trade areas, as well as interviews with traditional healers, herbalists and traders revealed that at least 1,721 wild *E. bupleurifolia* individuals may have been harvested from the wild in 2018. A large proportion of

harvesters had reported collecting up to 20 plants at a time, although three individuals claimed to have harvested more than 100 plants each over one year. Given that the species has been observed for sale in local markets since the early 1990s, the continued annual removals of wild plants for the local trade would have contributed significantly to the decline of *E. bupleurifolia*. The scale of local trade for horticultural purposes has not been fully assessed but the species is not commonly seen for sale locally and nurseries growing the species largely supply international markets.

Euphorbia bupleurifolia was not found in seizure data from EU TWIX and WiTIS based on available data (from 2014-2023 for EU TWIX and 1993-2025 for WiTIS).

Additional Information

Threats

Euphorbia bupleurifolia was assessed on the national red list in South Africa as critically endangered in 2019 due to factors including a restricted distribution, habitat loss (estimated at 40% of suitable habitat), and inferred population declines of at least 97% since 1975 based on estimates of quantities illegally collected for trade relative to the number of plants remaining in the wild based on population surveys in 2018. The species is also long-lived and extremely slow growing, with a generation length estimated as 20 years (Mhlongo and Pfab, 2022).

Subpopulations of the species are generally dominated by medium-to-large adults, and seedlings/juveniles are scarce (observed in low numbers of <4% in 2018 in just three of the subpopulations), indicating that regeneration by seedling recruitment is slow. *It's not clear how much of a threat to the population this is*- one study postulated that low or episodic recruitment is not a cause for concern for long-lived species as the persistence of abundant mature individuals can sustain populations. The national population shows variable regeneration potential with flowering observed in 27% of all individuals (n= 1724) surveyed in 2018 and the percentage of flowering individuals in subpopulations ranging from 5% to 85%.

Habitat degradation and disturbance is a growing threat to populations surveyed across the Eastern Cape, particularly those in easily accessible areas. Trampling (by both humans and cattle) was noted in all but one of the surveyed populations. Other disturbances observed included illegal dumping, grazing, vehicle damage and recreational activities including quad-biking, group training activities and cycling. The damage to plants and habitat from these disturbances is postulated to have negative implications for the reproductive output and recruitment of the species. Most of the preferred vegetation types are sensitive to change, and many have undergone observed high percentage losses, particularly the coastal grassland ecosystems in KwaZulu-Natal that have been completely lost to development or modified and degraded by agricultural practices.

Conservation, management and legislation

Euphorbia bupleurifolia is a protected plant in both the Eastern Cape and KwaZulu-Natal. Collection of such plants within state and private nature reserves requires a permit from the relevant conservation authority, whilst landowner permission is required to gather plants on privately owned land. It is unlikely that authorities would issue collection permits for protected species within protected areas but the number of *E. bupleurifolia* plants has been severely reduced to less than 30 individuals in the three surveyed subpopulations that occur within protected areas. Access to two of the reserves is somewhat restrictive but the subpopulation in the third reserve is easily accessible. There is furthermore no confidence in landowner permission as a control mechanism on privately owned land. At present, resource limitations with respect to the capacity (including relevant skillsets) and budget within both provinces renders existing harvest and management controls ineffective, not only hindering active field monitoring of the species' populations but also the ability to detect any illegal activities.

Updates to national and provincial legislation that will increase legislative protection for the species are progressing slowly, although implementation of existing legislation is inadequate and hampered by inadequate conservation resources within both the Eastern Cape and KwaZulu-Natal. Officials are presently ill-equipped to deal with the evolving pressures on the country's biodiversity. For example, it is increasingly difficult to detect instances of poaching, smuggling and laundering of succulent plants facilitated largely by the internet and rapidly developing social media platforms. It is also challenging to keep track of privately owned collections. Due to ease of access, many subpopulations are at an increased risk from illegal harvest.

There is currently no field monitoring programme for the species and monitoring of harvest for the local trade does not take place due to limited capacity and resources within the provinces. Records of exports of *E. bupleurifolia* from South Africa extracted from the CITES Trade Database are analysed by the South African National Biodiversity Institute (SANBI) on behalf of South Africa's Scientific Authority. The proposal states that the quality of this data is not always as good as it should be, owing to errors and inconsistencies in reporting by both exporting and importing countries. Lack of monitoring across local nursery facilities also undermines the accuracy of the trade data.

Artificial Propagation

There are at least five nurseries in South Africa known to be involved in the cultivation and trade of *E. bupleurifolia* and although four of the five facilities were found to be compliant with the CITES definition of “under controlled conditions” during an audit in 2018, none of the nurseries were able to produce clearly marked, legally acquired parental stock. Whilst three nurseries had only juvenile plants on site, *E. bupleurifolia* plants at two of the nurseries (which had most of the plants) were mostly large with distinctive wild features, including high degrees of variation in sizes, irregularly shaped stems and growth forms, evident animal damage as well as lichen growth along the stems. Propagated exclusively from seed, the species is not the easiest to grow and is considered a moderate-care plant suited for succulent enthusiasts with some experience. *Euphorbia bupleurifolia* may grow rapidly during the juvenile phase, with growth gradually slowing once plants begin to flower.

Implementation challenges (including similar species)

The species is not easily confused with any other natural species within the genus *Euphorbia*. It can be easy to mistake *E. bupleurifolia* for the commonly grown and more widely available artificially produced hybrid species, *E. japonica*, although *Euphorbia bupleurifolia* can be distinguished from *E. japonica* by its typically unbranched, pinecone-like, narrow and cylindrical stem (up to 20 cm tall) and its narrow, strap-shaped leaves. The stems of *E. japonica* in contrast are often flatter or more globular (lower and wider) and more freely branching, whilst the leaves are small, more clustered and with a more compact appearance.

Potential benefit(s) for trade regulation of a transfer from Appendix II to I

The species has been included in Appendix II since 1975 under the genus listing of *Euphorbia*. However, the proposal states this has had a limited effect on preventing the unsustainable trade in wild-harvested plants.

The NDF for *E. bupleurifolia* concluded that trade in the species is detrimental to its survival in the wild and should any trade be considered in the future, it ought to be restricted to strictly artificially propagated specimens (consistent with the requirements of CITES Res. Conf. 11.11 (Rev. CoP18)) and linked to a restoration programme for the species. The NDF further recommended that significant improvements to management, control, monitoring and protection measures are essential to support a sustainable trade in the species.

With an Appendix I-listing, international exports could be limited to CITES-registered nurseries that satisfy all conditions for artificial propagation. More resources would be deployed to regularly inspect these nurseries and the parental stock and to monitor the artificial propagation of the plants. Importing CITES Parties would pay more attention to the specimens imported to ensure that they are artificially propagated, and they would monitor in-country plant traders that stock the species.

References

- Mhlongo, N.N. and Pfab, M.F. (2022) *Euphorbia bupleurifolia* Jacq. National Assessment: Red List of South African Plants version. Accessed on 29 July 2025
- Mhlongo, N.N., Pfab, M.F. and Witkowski, E.T.F. (2023) Distribution, population structure and microhabitat profile of *Euphorbia bupleurifolia*. *South African Journal of Botany* 163:401-411.
- Newton, D. (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- South Africa, National Biodiversity Institute (SANBI) (2025) Red List of South African Plants. Available at: <https://redlist.sanbi.org/> Accessed on 16 July 2025.
- UNEP (2025) The Species+ Website. Nairobi, Kenya. Compiled by UNEP-WCMC, Cambridge, UK. Available at: www.speciesplus.net. Accessed on 16 July 2025.

Deletion from Appendix II of *Afzelia bipindensis* populations located in Central Africa: Cameroon, Central Africa Republic, Congo, Democratic Republic of the Congo, Equatorial Guinea and Gabon

Proponents: Burundi, Cameroon, Central African Republic, Congo, Democratic Republic of Congo, Equatorial Guinea, Gabon

Summary: *Afzelia bipindensis* is an African tree species that provides commercially important timber known generally as Afzelia, pod mahogany, African Mahogany or Doussie. It has been one of the most traded African *Afzelia* species and has a widespread distribution occurring at generally low density in evergreen and semi-deciduous rainforest in Angola, Cameroon, Central African Republic, Congo, Côte d'Ivoire, Democratic Republic of the Congo (DRC), Equatorial Guinea, Gabon, Nigeria, Uganda and Zambia. Commercial harvesting for timber mainly takes place in Central Africa. Three other species of *Afzelia* also occur in the same region: *A. africana*, *A. bella*, and *A. pachyloba*. *Afzelia* timber has a wide range of uses for decorative veneer, flooring, door frames, staircases, docks, boatbuilding, exterior millwork and construction, furniture, musical instruments, turned objects, inlays, and other small speciality wood items.

Afzelia bipindensis and the other African species in the same genus were included in Appendix II in 2023 after CoP19 with annotation #17, which includes logs, sawn wood, veneer sheets, plywood, and transformed wood.

Since listing Cameroon, Congo, DRC and Equatorial Guinea have reported exports to European and Asian countries and imports to Belgium are reported from Gabon. Implementation of the Appendix-II listing has included the development and publication of Non-Detriment Findings (NDFs) and declaration of associated export quotas for *A. bipindensis* by Cameroon and DRC. Export quotas for 2025 have also been set by Congo and Equatorial Guinea although the non-detriment basis for these quotas is not known. The development of NDFs for *Afzelia* spp. was recommended as a priority by the 26th Meeting of the Plants Committee in 2023.

Illegal logging is reported to be the main threat to the species alongside other threats such as deforestation resulting from agriculture, urbanisation and mining. The extent to which illegal logging of *A. bipindensis* occurs is unknown but, in the Central African region, a large proportion of total log production is believed to be of illegal origin.

The proposal is to delete the populations of *Afzelia bipindensis* that occur in six central African countries (Cameroon, Central African Republic, Congo, DRC, Equatorial Guinea and Gabon) from Appendix II. The proponents indicate that with strict management measures and robust traceability systems for the species in place within these countries, and with satisfactory levels of regeneration, international trade regulation is not necessary for the species in this part of its range. Populations of *A. bipindensis* in Angola, Côte d'Ivoire, Nigeria, Uganda and Zambia would remain in the Appendices. No exports nor imports of *Afzelia* species have been reported from Angola, Côte d'Ivoire, Nigeria and Zambia since the CITES listing. Uganda has re-exported significant quantities of *Afzelia* reported as *A. africana* and originating in South Sudan.

Available information, based largely on forest inventories in forest management units (FMUs) in Central Africa, indicates moderate abundance of *A. bipindensis* among commercially exploited timber species, with potentially a large population of mature trees showing generally good regeneration.

Five *Afzelia* species are currently identified as of major commercial importance: *Afzelia africana*, *A. bella*, *A. bipindensis*, *A. pachyloba* and *A. quanzensis*. The timber of different species of *Afzelia* has been difficult to distinguish and is commonly marketed under the same commercial names. The proponents note that there has now been a clarification of preferred names for different species by the timber industry. Furthermore, they note that the processed products from *A. bipindensis* can be differentiated from those of other species, notably by anatomical and technological characteristics

and with the help of identification tools. They consider that Appendix-II listing for the species for lookalike reasons is not required.

Analysis: Available information, based largely on forest inventory in central African FMUs indicates that *Afzelia bipindensis* remains a widespread species occurring at moderate density with a large population of mature trees. Measures are reportedly in place to ensure the sustainable and legal harvesting and trade of timber of the species in Cameroon, Congo, DRC, Central African Republic, Equatorial Guinea and Gabon, and regeneration appears to be good within managed concessions. Non-detriment Findings (NDFs) have been published for Cameroon and DRC and export quotas published for Congo and Equatorial Guinea although it is not known if these are based on NDFs. Based on this information the species in these range States **may not meet the criteria for inclusion in Appendix II** in Annex 2a of Res. Conf. 9.24 (Rev. CoP17).

However, it appears that **the species still meets the lookalike criteria for inclusion in Appendix II** (Criterion A of Annex 2b of Res. Conf. 9.24 (Rev. CoP17)). Action has been taken to clarify and standardise trade names for timber of different species of African *Afzelia*, and techniques are available to differentiate the wood. Nevertheless, standardised names may not be universally applied and enforcement officers without access to appropriate identification techniques or tools would generally be unlikely to be able to distinguish specimens of *A. bipindensis* in the form in which they are traded from those of other species in the genus listed in Appendix II. This applies to the species, *A. africana*, *A. bella*, and *A. pachyloba* which also occur within Central African countries and are in international trade, as well as to other species in the genus. Furthermore, timber products from the populations of *A. bipindensis* would only be differentiated from those of populations of the same species in other countries covered by the Appendix-II listing by their stated origin. This may lead to laundering given that timber consignments of species for export from Central African countries are known to include timber consolidated from different countries and may include a substantial proportion of illegally harvested timber.

If adopted this proposal would result in a split-listing of *Afzelia bipindensis* by geographical populations. Some populations of the species would remain in the Appendices, and the populations covered by the proposal would not. Annex 3 of Res Conf 9.24 (Rev. CoP17) notes that “*split-listings that place some populations of a species in the Appendices, and the rest outside the Appendices, should normally not be permitted*” in view of enforcement problems it might create, although there are a few precedents for such split-listing in the Appendices.

Summary of Available Information

Text in non-italics is based on information in the Supporting Statement (SS); text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

Afzelia bipindensis (Harms). There is no CITES Standard Reference for this species. In accordance with guidance provided in CoP20 Inf. 6, *Afzelia* spp. is amongst the group of taxa indicated as priorities for production of standard nomenclature references (see CoP20 Doc. 110, paragraph 82). In this light, a time-stamped extract from POWO prepared by the flora nomenclature specialist is proposed for adoption as a standard nomenclature reference (see CoP20 Doc. 110 Add., Annex 3):

POWO. (2025). African populations of *Afzelia* spp. World Checklist of Vascular Plants. Facilitated by the Royal Botanic Gardens, Kew. Published on the Internet; <https://powo.science.kew.org/> Retrieved 7 August 2025.

Afzelia bipindensis is one of seven African *Afzelia* spp. currently recognised, the others being *Afzelia africana*, *A. bella*, *A. pachyloba*, *A. parviflora*, *A. peturei* and *A. quanzensis*. Globally 12 species are recognised (POWO, 2025).

Range (populations proposed for deletion from the Appendices are underlined)

Angola, Cameroon, Central African Republic, Congo, Democratic Republic of Congo, Equatorial Guinea, Gabon, Nigeria, Uganda. Possibly also *Zambia*, *Côte d'Ivoire* (see below).

It has also been reported in Zambia. In Equatorial Guinea, A. bipindensis occurs in the continental region (Rio Muni). It has been poorly documented within the country and, for example, is not recorded in the country by de la Estrella et

al. (2010) or by BGCI (2025). POWO (2025) does, however list Equatorial Guinea in the countries of distribution. The species is also recorded in Côte d'Ivoire by POWO (2025).

IUCN Global Category

Vulnerable A1cd (assessed 1998 [needs updating], ver. 2.3).

Biological and trade criteria for retention in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)

A) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences

In the Central African region, the relatively low natural density of this species is offset by efficient natural regeneration. Inventories have been made in Forest Management Units of Cameroon, Central African Republic, Congo, DRC and Gabon as a basis for sustainable forest management and harvesting. Most of the harvest of *A. bipindensis* takes place under regulated forest concessions, with annual volumes defined by the competent national authorities (Donkpegan, 2017). At a workshop held in Libreville in December 2022, the species was globally assessed as least concern based on projections of low future population decline (aligned with IUCN Red List criterion A3) in both concession and non-concession areas (Loubota Panzou et al., 2023). This assessment was conducted independently of the formal IUCN Red List process.

Management of harvesting and trade in Central African countries where *Afzelia bipindensis* is commercially exploited are designed to maintain populations at sustainable levels. Nevertheless, concerns remain about population reduction in *A. bipindensis* in parts of its range. The species has recently been evaluated as vulnerable in Nigeria, where timber production and habitat destruction are cited as threats (Bamigboye et al., 2022). Conservation policies to prevent the indiscriminate harvesting of this and other species of *Afzelia* that occur in Nigeria are recommended (Bamigboye et al., 2022). It has been noted that *A. bipindensis* is declining in Nigeria due to intensive exploitation for timber, firewood and charcoal. Regeneration is poor in areas where few large trees remain (Ajayi, 2025). Furthermore, a recent study has also highlighted the need for conservation through regulated land use in Dibombari highland, Cameroon Mountain (Cameroon) and the forests around Oban hills in Nigeria due to projected loss of *A. bipindensis* because of climate change (Salako et al., 2021).

A. bipindensis is considered to be vulnerable in Uganda, where it is rarely seen and is harvested for timber. Previously, *A. bipindensis* was categorised as nationally vulnerable in Cameroon in 2011, due to heavy exploitation for the international timber trade. The species was listed as nationally endangered in DRC in 2014, because of unsustainable logging, deforestation caused by urbanisation and agriculture, and tree bark damage by forest elephants. Overexploitation of the species in DRC has led to a "continued and accelerated decline" considered "irreversible" due to illegal logging, poor forest governance, and slash and burn agriculture negatively impacting regeneration. It is also considered to be threatened in Zambia (GlobalTreePortal, 2025).

Legal international trade

In a recent survey of timber legality verification in China, *Afzelia bipindensis* and *A. pachyloba* were reported as being amongst the main timber species imported from African countries by the companies interviewed (Samejima, 2023). According to CITES Trade Database direct exports of *A. bipindensis* comprised ~5,670 m³ of pre-convention and ~2,580 m³ wild-sourced logs, sawn wood and timber for commercial purposes from Central African Countries between 2020 and 2024 (see Table 1). No direct exports of *A. bipindensis* were reported by other countries. Congo also reported direct commercial export of ~1,670 m of wild-sourced logs and sawn wood from *Afzelia* species in 2023. The main exporting countries of *A. bipindensis* are Cameroon, Congo and DRC, with sawn wood and logs destined for Asian and European markets. No exports nor imports of *Afzelia* species have been reported from Angola, Nigeria and Zambia since the CITES listing. The availability of Nigerian Doussie, *Afzelia* spp. is however promoted on various websites (for example: Renamyglobal.com; Alibaba.com; globalsuppliersonline.com; highviewuk.com). Uganda has re-exported significant quantities of *Afzelia* that originated in South Sudan (all reported as *A. africana*).

Table 1. Quantities of logs, sawn wood and timber of *A. bipindensis* reported in direct commercial exports by Central African countries between 2023 and 2024. Source codes: Wild (W); Pre-Convention specimens (O). Data are reported by exporters aside from those marked with an *, which indicate importer reported data. This was due to exporters not yet providing CITES annual trade reports for these years. Source: CITES Trade Database.

Exporter	2023	2024
Cameroon	4,921 m ³ (O)	557 m ³ * (O)
Equatorial Guinea	1,232 m ³ (W)	-
Congo	178 m ³ (source not stated)	889 m ³ (W)
DRC	-	451 m ³ (W)
Gabon	189 m ³ * (O)	-

The 2024 NDF for *A. bipindensis* in Cameroon allocated a national cutting quota of 22,872 m³ for the 2024 financial year, representing 28% of the exploitable volume and 40% of the harvesting potential of all titles. This quota is distributed as follows: for the Permanent Forest Estate: 13,337 m³ or 58% of national production and for the Non-Permanent Forest Estate (which includes Community Forests and other categories of forest management): 9,536 m³ or 42% of national production (Mbock et al., 2024). The 2024 NDF for *A. bipindensis* in DRC states that export quotas for *A. bipindensis* will be set based directly on the harvest forecasts from logging companies, resulting from annual exploitation inventories conducted over the area constituting an Annual Cutting Area (ACA). These ACAs are defined based on the Forest Management Plans (FMP) resulting from the implementation of validated Forest Management Agreements (FMA). The results of the exploitation inventories conducted on each ACA will be presented in the corresponding Annual Operation Plans (AOP), which will be verified and validated by the forestry administration in accordance with legal provisions (Anon, 2024). Export quotas have also been set by Congo and Equatorial Guinea.

Illegal trade

Afzelia bipindensis is reported to be mainly threatened by illegal logging for its wood. It is not known how significant an impact this has on the overall population. In Central Africa almost 46% of total log production is said to come from illegal logging, particularly in non-permanent forests. These volumes are, however, mainly destined for local markets which have little interest in *A. bipindensis* because of the difficulty of working the timber.

Despite considerable efforts to improve the enforcement of forestry laws, governance, and trade in the Congo Basin countries, illegality still occurs throughout the timber supply chain. Some reports have asserted that 90 percent of the timber from the Congo Basin rainforests may be illegally sourced and facilitated by high-level crime (Mahonghol, 2024).

WiTIS recorded four seizures totalling 39 logs reported in Cameroon between 2016 and 2021. No seizure records for *Afzelia bipindensis* (or other *Afzelia* species) could be found in EU TWIX between 2014-2023.

In general, illegality in the forestry sector remains an issue for Central African countries. In Cameroon for example, Kamkuimo, et al. (2025) note that illegal exploitation is still a major concern in the country and frequently documented in various stages of the local supply chain. The risks are enormous and appear in all categories of law. High levels of illegal timber trade between DRC and Uganda have been reported although not specifically in *Afzelia* spp. (Blanshe, 2024). The Congo also faces challenges related to forest governance and illegal logging (WorldTradePro, 2023) and, despite a strong legislative framework for forest management, illegal logging is still considered widespread in Gabon (Forest Trends, 2024a).

Cameroon's ports act as a central shipping hub for timber-producing countries across West and Central Africa, complicating supply chains and origin labelling (Forest Trends, 2024b). Challenges remain concerning the management of trade in CITES-listed timber from third countries in transit through Cameroon. As a transit country, Cameroon is not able to guarantee the durability, legality or traceability of the timber. It has no power to verify that the conditions of issuance of CITES permits – particularly the making of NDFs and legal acquisition findings – were respected in the countries of origin. It has been reported that a high proportion of CITES-listed species exported from the ports of Cameroon come in fact from the Central African Republic and from the Congo. Moreover, specimens from neighbouring countries are not subject to the same control procedures as timber from Cameroon. The authorities of Cameroon only verify the presence of the required documents, but the procedure of entry into the port, customs clearance and export is managed directly by the authorities of the countries concerned, which have offices in the port of Douala and cooperate with the customs of Cameroon. It is therefore not possible to thoroughly monitor the exports made by the countries of the region that transit through Cameroon (CITES Secretariat, 2025b).

Retention in Appendix II to improve control of other listed species

A) Specimens in trade resemble those of species listed in Appendix II under Res. Conf. 9.24 (Rev. CoP17) Annex 2a or listed in Appendix I

The SS also states that the commercial classification of African species in the *Afzelia* genus has been clarified, with *A. bipindensis* designated as “doussie” and *A. pachyloba* as *Pachyloba* and *A. africana* as “lingue”, and that this change, endorsed by industry professionals, reflects the technological differences observed between them. It is not known however the endorsed names are widely used in a standard way.

The taxonomic and commercial similarity between African *Afzelia* spp. has long led to confusion during forest inventories, particularly in the field, where morphological distinctions between species can appear subtle. Timber of *Afzelia bipindensis* resembles that of other species in the same genus that are currently listed in Appendix II.

In the Central African region, *A. bipindensis* coexists with several other morphologically related species of the same genus: *A. africana*, *A. bella* and *A. pachyloba*, which may be more severely affected by international trade. All are commonly marketed under the generic name of doussie. CITES Trade Data show that Central African countries

exporting *A. bipindensis* (see Table 1) have also reported direct exports of commodities from these *A. africana* and *A. pachyloba* between 2023 and 2024:

- Cameroon reported direct commercial exports of ~41,700 m³ pre-convention sawn wood from *A. pachyloba* in 2023.
- Congo reported direct commercial exports of 400 m³ wild-sourced logs and sawn wood and 187 m³ logs (source not stated) from *A. pachyloba* between 2023 and 2024.
- DRC reported direct commercial exports of 35 m³ wild-sourced sawn wood from *A. africana* in 2024.
- Equatorial Guinea reported direct commercial exports of ~ 610 m³ wild-sourced timber from *A. pachyloba* in 2023.

The IUCN Red List global assessment for *A. africana* is Vulnerable (VU A2cd). The rationale for the assessment states that despite being widespread with an extent of occurrence (EOO) estimated to be 4,850,397 km², intensive and unsustainable harvesting of this species has resulted in a population reduction of at least 30% over the past three generations (150 years) (Hills, 2020). *Afzelia pachyloba* was globally assessed as Vulnerable (VU A1d) in 1998 and annotated as 'needs updating'. *Afzelia bella* was globally assessed as Least Concern in 2019 (LC) (Hills, 2019).

The SS notes that recent studies have shown that *A. bipindensis* can be clearly differentiated both by specific morphological characters (leaves, seeds and bark) and by distinct chemical signatures detected by mass spectrometry. It is not clear, however, how readily available these techniques are and whether they are appropriate for enforcement officers.

Additional Information

Threats

The main threat to the species is logging due to the value of its timber, especially illegal logging in non-permanent forest areas. In addition, *A. bipindensis* is threatened by agriculture (small and large scale), habitat degradation and destruction, urbanisation and mining activities.

Conservation, management and legislation

Afzelia bipindensis occurs in various protected areas. Management of harvesting takes place in Central African areas designated for timber production.

Artificial Propagation/captive breeding

Afzelia bipindensis is recorded in four ex-situ collections (BGCI, 2025). The species is included in a propagation manual for Nigerian trees of conservation concern (Ajayi, 2025).

The species is not well adapted to artificial plantations. Young stands of *A. bipindensis* are often invaded by faster-growing trees and vines, resulting in slow growth or even death of *A. bipindensis*. Plantations in Gabon show that survival and growth are greater on cleared land and open areas than in the forest understory.

Other comments

The SS states that despite developing NDFs, producing countries are facing lengthy administrative delays, particularly with regard to the issuance of import permits by member countries of the European Union, the main destination market, which has had a severe impact on legal international trade in the species. *Should the delisting proposal be accepted, it is not clear whether all populations of the species would remain in Annex B of the EU Regulation and still require NDFs under stricter domestic measures.*

References

- Ajayi, T. (2025) *Afzelia bipindensis*. Forest Center Website. Available at: <https://forestcenter.iita.org/afzelia-bipindensis/>. Accessed on 9 August 2025.
- Anon (2024) Avis de Commerce Non Préjudiciable pour l'Exploitation et le Commerce de *Afzelia bipindensis* Harms en République Démocratique du Congo. Institut Congolais pour la Conservation de la Nature.
- Bamigboye, S.O., Odewa, S.A., Jimoh, M.O., Lassi, I.A. and Momo, B.O. (2022) Studies on extinction risk of genus *Afzelia* in Nigeria from biogeographical perspective. *Nigerian Annals of Pure and Applied Sciences* 5(1):191–199.
- Blanshe, M. (2024) Timber Smuggling: From DRC, Through Uganda, to the Rest of the World. Pulitzer Center Website. Available at: <https://pulitzercenter.org/projects/timber-smuggling-drc-through-uganda-rest-world>
- BGCI (2025) Botanic Gardens Conservation International Species Search Website. Available at: <https://www.bgci.org/resources/bgci-databases/globaltree-portal/species-search/?species=Afzelia+bipindensis>
- CITES Secretariat (2025a) Appendices of the Convention. Nomenclature matters. Standard nomenclature. Addendum. CoP20 Doc. 110 Add.
- CITES Secretariat (2025b) Application of Article XIII in Cameroon. SC78 Doc. 33.4.

- de la Estrella, M., Cabezas, F.J., Aedo, C. and Velayos, M. (2010) The Papilionoideae (Leguminosae) of Equatorial Guinea (Annobón, Bioko and Río Muni). *Folia Geobot* 45:1–57.
- Donkpegan, A.S.L. (2017) Histoire évolutive du complexe *Afzelia smith* (Leguminosae -Caesalpinioideae) dans les écosystèmes forestiers et savanicoles d'Afrique tropicale. Université de Liège - Gembloux Agro-Bio tech.
- Forest Trends (2024a) Timber Legality Risk Dashboard: Gabon. Available at: https://www.forest-trends.org/wp-content/uploads/2022/01/Dashboard-Gabon_Sept-2024_v2-1.pdf
- Forest Trends (2024b) Timber Legality Risk Dashboard: Cameroon. Available at: https://www.forest-trends.org/wp-content/uploads/2022/01/Dashboard-Cameroon_August2024.pdf
- Hills, R. (2020) *Afzelia africana*. *The IUCN Red List of Threatened Species* 2020: e.T33032A67742420. <https://dx.doi.org/10.2305/IUCN.UK.2020-3.RLTS.T33032A67742420.en>
- Hills, R. (2019) *Afzelia bella*. *The IUCN Red List of Threatened Species* 2019: e.T60757654A60757657. <http://dx.doi.org/10.2305/IUCN.UK.2019-3.RLTS.T60757654A60757657.en>
- ITTO (2025) *Tropical Timber Market Report* 29(4):24–25.
- Kamkuimo, P., Guiny, J., Kamga, J. and Modjo, A. (2025) A Guide to documentary analysis for EUTR/EUDR stakeholders within the context of carrying out due diligence. TRAFFIC & FODER.
- Loubota Panzou, G.J., Monthe, F., Ligot, G., Julve, C. and Doucet, J.L. (2023) Actualisation du statut de vulnérabilité des espèces ligneuses exploitées en Afrique centrale. Available at: http://www.ppecf-comifac.com/tableau_recapitulatif.html Accessed on 2 July 2025.
- Mbock, G., Tabue Mbobda, R.B., Ngnibi Lounse, E.X., Ndaora Saiba, A., Adamou, A. and Betti, J.L. (2024) Avis De Commerce Non Préjudiciable de *Afzelia bipendensis* (Doussié rouge) au Cameroun.
- Mahonghol, D. (2024) The Regional Security Imperative to Protect the Congo Basin. *Africa Security Brief* 44. Africa Center for Strategic Studies. Available at: www.africacenter.org
- Plants of the World Online (POWO) (2025) Plants of the World Online Website. Kew Royal Botanic Gardens. Available at: <https://powo.science.kew.org/> Accessed on 13 August 2025.
- Salako, G., Oyebanji, O.O., Olagunju, T.E. and Howe, G.T. (2021) Potential impact of climate change on the distribution of some selected legumes in Cameroon and adjoining Nigeria border. *African Journal of Ecology* 59(4):959–975. <https://doi.org/10.1111/aje.12915>
- Samejima, H. (2023) Status of Timber Legality Verification by Business Entities in China. Institute for Global Environmental Strategies (IGES) on behalf of the International Tropical Timber Organization (ITTO), Yokohama, Japan.
- WorldTradePro (2023) Who is the largest exporter of timber in Africa? – World Trade Pro – Navigating the World of Commodities: Discuss, Share, Trade. Available at: <https://worldtradepro.com/forum/topic/who-is-the-largest-exporter-of-timber-in-africa/>

Transfer of Brazilwood *Paubrasilia echinata* from Appendix II to Appendix I

Proponent: Brazil

Summary: *Paubrasilia echinata*, commonly known as Brazilwood, Pernambuco or Pau-brasil, is a slow-growing leguminous tree, reaching around 15 m in height with a maximum trunk diameter of around 70 cm. It is endemic to the Mata Atlântica (Atlantic Coastal Forest) biodiversity hotspot in Brazil. Many aspects of the biology of Brazilwood and the composition and structure of the plant communities in which it occurs are poorly known. The wood is currently used worldwide for the manufacture of high-quality bows for musical instruments, for which the species has been exploited for over 200 years. *Paubrasilia echinata* was globally assessed as Endangered on the IUCN Red List in 1998 and was assessed in 2024 as critically endangered by Brazil's National Centre for Flora Conservation.

Originally included in Appendix II with Annotation #10 at CoP14 (2007) as *Caesalpinia echinata*, that name becoming a synonym of *Paubrasilia echinata* in 2019, following taxonomic changes adopted at CoP18. Annotation #10 initially included logs, sawn wood, veneer sheets and unfinished wood articles used for the fabrication of bows for stringed musical instruments. The annotation exempted finished bows until February 2023 when an amendment to the Annotation adopted at CoP19 came into force including "all parts, derivatives and finished products, except re-export of finished musical instruments, finished musical instrument accessories and finished musical instrument parts". This is the only species subject to this annotation. In response to ongoing trade pressure on wild populations of the species, Brazil now proposes to transfer it to Appendix I, a listing that would apply to all parts and derivatives (in accord with Article 1b(iii) of the Convention).

No empirical estimates of the natural populations of *P. echinata* across the Atlantic Forest are currently available. The species is fragmented between forest remnants and localised extinctions of subpopulations have been observed. The deforestation of the Mata Atlântica has been intensifying in recent years with almost 50,000 ha lost to deforestation between 2021 and 2024. Although reliable data on the size of natural populations of the species in the remaining forest fragments is lacking, estimates by CNCFlora in 2024 suggest there are around 10,000 mature trees. It is also estimated that the population of this species has declined by more than 80% over the past three generations. The species is widely cultivated as an ornamental and has been extensively planted in Brazil and elsewhere. Small-scale plantations have been established but none is known to produce export-grade wood.

The species has been heavily traded for over 500 years, initially as a source of red dye (brazilein) and more recently as timber. Since the early 1800s, the species' wood, which is highly valued for its combination of durability, flexibility and resonance, has been used to produce bows for musical instruments such as violins, violas, cellos, and double basses. It has been estimated that over half a million mature trees were removed over the last five centuries. Trade in finished bows is predominantly international. Based on available invoices from the past 25 years, Brazilian companies, where the vast majority of bows are manufactured, sold about 8,000 finished bows in the domestic market (6%) and more than 130,000 abroad (94%). In contrast, bow blanks are mainly traded at the national level, with around 460,000 units sold within Brazil (91%) and only around 40,000 exported (9%). More than half of the finished bow exports are destined for the USA, with the remainder going primarily to Europe and smaller volumes to Asia. In contrast, over half of the bow blank exports are to Japan, followed by the USA, and to a lesser extent, Europe.

Domestic trade in *P. echinata* wood between companies and bow makers within Brazil has been regulated and controlled under the Document of Forest Origin (DOF) system since 2006, but there appears to be some uncertainty about the accuracy of records of existing stocks registered in Brazil at the time the Appendix-II listing came into force. The DOF system did not regulate domestic or international trade in bows as finished products until June 2022, when a domestic permit requirement for the export of finished bows became effective. Since then, and the amendment to Annotation #10, no CITES Permits have been issued by Brazil for international bow trade.

According to an international survey of bow makers conducted in July 2022, the average annual global production of bows made from *P. echinata* is approximately 25,000. A conservative estimate suggests that 144 trees are required each year to produce 700 high-quality violin bows, based on an average wood yield of one cubic metre per tree, implying that around 5,000 trees would need to be used annually to sustain a production of 25,000 high-quality bows, excluding existing stockpiles. However, other estimates differ, with one stating that a single mature tree could produce up to 500 bows with 25-50 of high-quality.

Harvesting of *P. echinata* in its natural habitat and export is prohibited under Brazilian law, and only wood from planted trees that are registered with the environmental agency (none of which are currently registered according to the Brazilian Management Authority), or pre-Convention specimens can be traded. Its inclusion on the official list of Brazilian flora threatened with extinction, means collection, harvesting, transportation, storage, handling, processing, and commercialisation of wild-sourced specimens is prohibited. However, recent investigations in Brazil have shown that illegal trade in *Paubrasilia echinata* timber for bow-making has been, and remains, significant.

The proponent seeks listing of the species in Appendix I, which would result in additional administrative procedures for companies trading products derived from the species outside Brazil; particularly in transactions involving finished musical instruments, their accessories, and parts, which are currently exempt from CITES permit requirements under Annotation #10.

The proponent suggests that, according to Res. Conf. 16.8 (Rev. CoP17), frequent cross-border, non-commercial movements of musical instruments may be facilitated by Musical Instrument Certificates (MICs). These certificates would enable musicians and orchestras to transport Brazilwood instruments across borders. However, according to this Resolution as it currently stands, MICs only apply to specimens listed in Appendix I acquired before the species was included in the Appendices, as well as to species listed in Appendices II and III. In the case of *Paubrasilia echinata*, if it were to be listed in Appendix I, MICs could only apply to instruments acquired before 2007 (when the species was listed after CoP14). Any cross-border movement of musical instruments and their accessories and parts acquired after 2007 would need to be permitted on a case-by-case basis in compliance with Articles III and VII of the Convention. Such authorisation would be limited to personal and household instruments, pre-Convention instruments, and instruments, their accessories and parts recognised as derived from artificially propagated trees grown in CITES-registered nurseries.

Analysis: *Paubrasilia echinata* has been subject to extensive historical exploitation for international trade. Its natural population is small and fragmented, with records of extirpation of some subpopulations. Based on rates of deforestation within its natural range (Atlantic Coastal Forest) the wild population is likely to have declined by more than 80% over the last three generations. There is evidence of ongoing international demand in the USA, Europe, and Asia, with instances of significant illegal trade reported and it is likely that illicit felling to produce timber for export is continuing to have an impact on the wild population. *P. echinata* therefore **appears to meet the biological and trade criteria for inclusion in Appendix I** under Criterion C of Annex 1 of Res. Conf. 9.24 (Rev. CoP17).

The proposed transfer of this species from Appendix II to Appendix I would effectively ban all commercial trade in wild-sourced Brazilwood specimens. Although the proponent has expressed the expectation that the use of Musical Instrument Certificates could facilitate the cross-border movement of musical instruments and their parts by musicians and orchestras, this would not be universally applicable as such documents cannot be issued for Appendix-I species acquired after their inclusion in the Appendices, in this case 2007.

If there are concerns about practical application of an Appendix-I listing, **Brazil could alternatively propose retaining the species in Appendix II with an amended Annotation #10 to cover “all parts, derivatives and finished products” (removing the current exemptions for re-export of finished musical instruments, their accessories and parts), combined with a zero-export quota for wild specimens of *P. echinata* traded for commercial purposes.** This option would strengthen controls on international trade while still allowing the use of MICs for non-commercial cross-border movements.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

Paubrasilia echinata was first listed in Appendix II in 2007 under the name *Caesalpinia echinata*. Following a taxonomic revision in 2016, which was adopted at CoP18, this name was subsequently recognised as a synonym of *Paubrasilia echinata* in 2019. This listing included Annotation #10, which covered “Logs, sawn wood, veneer sheets, including unfinished wood articles used for the fabrication of bows for stringed musical instruments”. In 2023, Annotation #10 was amended to include “All parts, derivatives and finished products, except re-export of finished musical instruments, finished musical instrument accessories and finished musical instrument parts”.

Taxonomy

Paubrasilia echinata (Lam.) Gagnon, H.C.Lima & G.P.Lewis

Range

Brazil

IUCN Global Category

Endangered A1acd (assessed 1998 [needs updating], ver. 2.3)

Biological criteria for inclusion in Appendix I

A) Small wild population

The Atlantic Forest is one of the most threatened biomes in the world. Brazilwood occurs in environments that have been severely degraded from colonization to the present, leading to a sharp decline in habitat quality and the extirpation of local subpopulations. Natural subpopulations are no longer found in Sergipe. A single forest fragment with a native population of Brazilwood is situated in the state of Espírito Santo in Aracruz, and a new population has recently been discovered in another municipality of the same state. Natural populations of Brazilwood are found in forest fragments of Rio de Janeiro, where urban sprawl poses a significant threat to these populations.

Although there is a lack of reliable data on the size of natural populations in the remaining species forest fragments, estimates from the National Centre for Flora Conservation (CNCFlora), which is linked to the Rio de Janeiro Botanical Garden (JBRJ), suggest that there are around 10,000 adult individuals. Consequently, Brazil has evaluated the species as critically endangered in 2024; it was globally assessed as Endangered on the IUCN Red List in 1998 (*an assessment that needs updating*).

B) Restricted area of distribution

Brazilwood occurs exclusively in Brazilian territory, only in the Atlantic Forest biome, in the coastal strip between the states of Rio de Janeiro and Rio Grande do Norte. The Atlantic Forest currently covers only 12.4% of the area that existed 500 years ago. The original geographical distribution of Brazilwood and the size of its native populations have been reduced by selective logging, caused by the exploitation of its wood and by the expansion of urban centres.

The species inhabits inland moderately moist forests and in a range of much drier habitats, including disturbed dry coastal cactus scrub rocky outcrops, seasonally semideciduous forest (tabuleiro forest), and restinga, a type of coastal forest with well drained sandy soil. Isolated trees have been cultivated as ornamental plants in streets and parks throughout the country and sometimes can be found in commercial plantations too.

The historical geographical distribution of Brazilwood has severely decreased due to deforestation, logging, exploitation for its wood, the development and expansion of urban centres, and changes in land-use for agriculture and forestry operations. At present, the largest populations of Pau-brasil are situated in forested areas within fully protected conservation units and in cocoa-cabruca agroforestry systems in southern Bahia, where they are used as shade trees for growing crops of *Theobroma cacao*. These areas are declining as the cocoa plantations are being rapidly converted into pastures.

C) Decline in number of wild individuals

The Atlantic Forest is one of the most threatened biomes in the world, and Brazilwood occurs in environments that have been severely degraded from colonisation to the present. This long-standing degradation has led to a sharp decline in habitat quality and the extinction of local subpopulations, fragmenting the species' distribution, reducing genetic variability, and limiting gene flow between remaining populations. Natural populations no longer exist in Sergipe, and in Espírito Santo only one native population remains in a forest fragment in Aracruz, with a second recently discovered in Vila Velha. In Rio de Janeiro, urban sprawl poses a major threat to populations

persisting in forest fragments. The decrease of cocoa production areas has further aggravated the situation, as *P. echinata* has traditionally been maintained to provide shade in cocoa-cabruca agroforestry systems.

As previously indicated, only 12.4% of the Atlantic Forest's original native vegetation remains. Between 2021 and 2022, deforestation led to the loss of approximately 20,100 hectares of forest, marking the highest annual reduction in recent years. This was followed by a slightly lower but still substantial loss of 14,700 hectares between 2022 and 2023. Most recently, from 2023 to 2024, an additional 14,400 hectares were cleared. Although the rate of forest loss appears to be gradually declining, the cumulative impact over these three years remains substantial. These ongoing losses, combined with the intensive exploitation of Brazilwood for international trade, continue to drive the decline of native populations throughout the species' range.

Using multiple criteria from the IUCN Red List and analysing millions of herbaria and forest inventory records, De Lima et al. (2024) produced automated conservation assessments for all tree species in the Atlantic Forest biodiversity hotspot. Their findings estimate an 84% decline in the population size of Paubrasilia echinata over the past three generations (i.e. 1874-2024).

Trade criteria for inclusion in Appendix I

The species is or may be affected by trade

Paubrasilia echinata was initially overexploited for the extraction of red dyes (brazilein) during the Portuguese occupation. *Intense harvesting continued until synthetic dyes became available in 1875, by which time extensive population declines had already taken place (Varty, 1998).* Starting from the mid-18th century, the species' wood has been used to build stringed musical instrument bows and is still internationally sought after due to its high levels of resonance, density and durability. *According to Rocha (2010), approximately 527,000 mature specimens have been harvested over the last 500 years.*

Using invoices from the past two decades as a basis, Guimarães (2025) estimated that 94.2% of all the bows for violins, violas, cellos and double basses produced in Brazil are exported. The international market primarily demands professional-grade bows—classified as gold, silver, and nickel—whose production requires exceptionally high-quality wood. It is estimated that more than 90% of harvested Brazilwood is discarded as waste during the production process, deemed unsuitable for professional bow-making due to stringent quality standards. *It is estimated that 144 trees are needed to produce 700 violin bows (i.e., almost five violins per tree), considering an average wood production rate of one cubic meter per tree (Espey, 2010, cited in CNCFlora, 2024).* Estimates by the International Pernambuco Conservation Initiative (IPCI), however, indicate that a single mature tree may yield as many as 500 bows, though only 5-10% (25-50 bows) would meet the highest quality suitable for soloists (IPCI in litt., 2025).

After analysing 346 cases over the past 25 years, Guimarães (2025) found that Brazilian companies, where the vast majority of finished bows are manufactured, sold 7,986 finished bows in the domestic market (6%) and 131,232 abroad (94%). In contrast, bow blanks were mainly traded nationally, with 464,515 units sold within Brazil (91%) and only 45,163 exported (9%).

The main international consumers of finished bows were the USA (55%), Belgium (16%), and Japan (10%). Approximately 20% were distributed among other countries and territories, including Germany, the Netherlands, Italy, Portugal, the United Kingdom of Great Britain and Northern Ireland (UK), Taiwan Province of China, and Canada. For bow blanks, Japan and the USA were also the principal importers, accounting for 58% and 24% of exports respectively, followed by Germany, Italy, and Portugal, which together received around 16% of the total.

The USA, Japan, Belgium, Germany, the Netherlands, Portugal, Italy, and France currently represent the largest consumers of raw *P. echinata*. *USA data report a total of 452 pieces being imported between 2009–2018 into Belize (70%), Brazil (20%), Mexico (5%), Austria (4%), Republic of Korea (0.7%) and Canada (0.2%). Over 95% of the pieces were labelled as wild-sourced and the rest as unknown. However, only about 24% were registered as having originated from Brazil. One instance of 316 pieces (approximately 70% of total pieces traded) being traded for commercial purposes was reported to have originated in Belize. No international trade of seedlings, seeds or bark is known. According to Cumine (2007), the term "Brazilwood" has also been used in trade to describe lower quality timber that is not *P. echinata*, but probably *Massaranduba* (*Manilkara* spp.).*

*Following the Appendix-II listing in 2007 and until 2020, CITES data include a total of 129 instances of *P. echinata* trade, totalling more than 15,000 kg and approximately 35 m³ of sawn wood reported as being exported and over 25,500 kg and more than 15 m³ reported as being imported. The top importer countries included the USA (17% of total imports), Japan (12%), China (11%), Switzerland (10%), Germany (9%), Hong Kong SAR of China (5%), Republic of Korea (5%), France (4%), Italy (4%), and Canada (4%). The top exporters were the USA (34% of total exports), Brazil (20%), France (9%), Germany (8%), Italy (6%), and the UK (4%). From 2021 to 2023 trade records in the CITES Trade Database consist of only two direct exports, one from Brazil to the USA for scientific purposes and the other being in of two wood products for personal purposes, while records show 22 re-export events (and 11 import events), the*

great majority consisting of pre-Convention specimens exported mainly by the USA who exported around 5 cm³ of plywood for commercial purposes, followed by France which exported wood products mostly for personal purposes.

A July 2022 survey by the IPCI sent to international professional organisations and bow makers recorded the average number of bows produced per year by each bow manufacturer. Approximately 25,000 bows are manufactured every year worldwide by the 337 different bow makers who responded to the survey. Among them, approximately 91% produce fewer than 50 bows per year and nearly 44% produce fewer than ten (IPCI in litt., 2025).

Harvesting of *P. echinata* in its natural habitat and export is prohibited under Brazilian law since 2006 (Federal Law No. 11,428/2006 and Federal Decree No. 6,660/2008) and only wood from planted trees that are registered with the environmental agency or pre-Convention specimens can be traded. Its inclusion on the official list of Brazilian flora threatened with extinction (since 1992), means that collection, harvesting, transportation, storage, handling, processing, and commercialisation of wild-sourced specimens is prohibited.

Through CITES Notification 2025/076, IBAMA, Brazil's CITES Management Authority, reported that it is currently assessing the declared stockpiles of *Paubrasilia echinata* wood held by bow-making companies to verify their legal origin. To date, none of the companies evaluated has been found to possess stockpiles deemed eligible for commercialization under current regulatory and legal requirements (CITES, 2025).

Illegal trade

Recent investigations in Brazil have shown that illegal trade in *P. echinata* timber for bow-making has been, and remains, significant. Since 2002, more than 1,000 one-metre logs, the standard size for bow making, and over 200 m³ of timber have been confiscated across several states, primarily Bahia and Espírito Santo. Much of this timber was laundered through fraudulent documents, including the misuse of pre-Convention certificates and permits for plantation-grown trees. Virtual credits from the DOF system were also manipulated to disguise the origin of illegally harvested wood, mainly sourced from Bahia, home to Pau-Brasil National Park, the largest remaining native population of *P. echinata*.

Over the past five years, investigations by Brazilian authorities have revealed ongoing illegal harvesting of *P. echinata* to supply the international bow-making industry. Operation "Dó-Ré-Mi" in 2018 exposed widespread illegalities in Espírito Santo, resulting in the seizure of over 292,000 bows and bow blanks and fines to around 45 individuals and companies. Operation Dó Ré Mi has exposed a sophisticated criminal network that fraudulently acquired permits, manipulated the DOF system, and exported bows abroad, particularly to Europe and the USA.

Between 2021 and 2022, additional seizures and police operations at nearly 60 locations uncovered a criminal network that generated over USD46 million through illegal exports. Investigations found that many companies and independent bow makers submitted false declarations to legitimize illegal timber, mixing defective scrap wood with high-quality stock to meet export demands.

According to Guimarães (2025), the majority of bows and bow blanks sold by Brazilian companies over the past 25 years likely originated from illegal sources concealed by fraudulent reporting, document misuse, and deceptive laundering schemes. *In Brazil, enforcement actions by IBAMA, in collaboration with the Federal Police, have led to significant wood seizures. However, challenges remain, such as identifying illegal stocks, which underscores the need for more robust control systems. Parties, including Canada and Germany, are implementing stock controls and inventories to ensure compliance with CITES regulations (CITES Secretariat, 2025).*

Between 2009 and 2018, the LEMIS database recorded 24 shipments involving Paubrasilia echinata (reported as Caesalpinia echinata), mostly for personal, scientific, or commercial purposes. There seem to be no records in LEMIS for 2019 or 2020. Of these shipments, 15 (62.5%) were cleared, while 9 (37.5%) were refused, indicating that a significant portion faced issues with compliance or documentation. Refusals included personal imports of worked wood items from Brazil and Mexico (2009, 2014), and commercial shipments such as 65 worked wood pieces from Brazil in 2016 and 316 logs from Belize in 2018. Some exports of pre-Convention Brazilwood bows to the UK in 2015 were cleared, as were a few items associated with scientific use or travelling exhibitions. Most of the specimens were wild-sourced. While some shipments were successfully authorised, the relatively high refusal rate suggests challenges in meeting the established trade requirements.

Between 2018 and 2022, WITIS recorded four seizures of Paubrasilia echinata in Brazil. Two of these occurred in 2018: one involved a large quantity of worked wood comprising 20,804 items, while the volume of the other was not specified. In 2022, authorities seized 120 units of sawn wood and an additional 114 items of worked wood. All incidents were recorded as domestic seizures within Brazil, with no reported trade routes outside the country.

From 2014 to 2023, EU-TWIX records only three seizures involving Brazilwood. These include the export of 1 kg of a "Pernambuco bow instrument" from the USA to the UK in 2021; the export of 0.5 m³ of sawn wood from the USA to Sweden in 2023; and the import of 15 wood products by a tourist travelling from China to Poland in 2023.

Additional information

Threats

In addition to large volumes of illegal harvest and trade, with substantial quantities of timber being discarded as waste by manufacturers, *P. echinata* is threatened by habitat loss and fragmentation due to continued deforestation, urban development, tourist development, and agriculture. Fragmented populations are causing a decrease in genetic variation of the species. Further pressures on the species include ongoing illegal logging activities.

Conservation, management and legislation

The Atlantic Coastal Forest encompasses at least 28 Private Natural Heritage Reserves (RPPN), as well as many protected areas. Landowners within the Reserves guarantee protection of local animals and plants in exchange for tax reductions.

Specific national regulations and governance measures are in force to protect *P. echinata* (Table 1).

Table 1. List of national legal instruments that govern the exploitation of native Brazilian plants (including *P. echinata*).

Legal or Regulatory Instrument	Description
Federal Law No. 6,607/1978	Declares <i>Brazilwood Paubrasilia echinata</i> as the national tree and establishes a public awareness campaign on its historical importance, as well as the creation of tree nurseries throughout the country for conservation and civic distribution purposes.
Federal Decree No. 750/1993	Prohibits the harvesting of native vegetation in the Mata Atlântica (Atlantic Forest) biome.
CONAMA Resolution No. 278/2001	Authorizes IBAMA to suspend authorizations—granted either directly or through delegation—for the harvesting of endangered species in the Atlantic Forest, until science-based technical criteria are established to ensure sustainable use and genetic conservation.
CONAMA Resolution No. 300/2002	Supplements the cases eligible for harvesting authorization under Article 2 of Resolution No. 278/2001.
CONAMA Resolution No. 317/2002	Establishes the necessary criteria for the genetic conservation and sustainable use of threatened plant species in the Atlantic Forest. These must be included in State Plans for Conservation and Use, based on technical and scientific studies.
Federal Law No. 11,428/2006	Provides for the use and protection of native vegetation in the Atlantic Forest biome, including prohibitions on the exploitation of species listed as threatened in the official national flora list.
Federal Decree No. 6,660/2008	Regulates Federal Law No. 11,428/2006, detailing mechanisms for the protection of native vegetation in the Atlantic Forest.
Federal Law No. 12,651/2012	Establishes the Brazilian Forest Code , which allows federal and state authorities to prohibit or restrict the cutting of endangered species and provides a general framework for the protection of native vegetation.
MMA Ordinance No. 320/2012	Establishes the National Program for the Conservation of Brazilwood .
MMA Ordinance No. 443/2014	Lists <i>P. echinata</i> as threatened with extinction in Brazil. It imposes full protection for species in the categories Extinct in the Wild (EW), Critically Endangered (CR), Endangered (EN), and Vulnerable (VU), including prohibition of collection, harvesting, transportation, storage, handling, processing, and commercialisation, among others.
MMA Ordinance No. 148/2022	Publishes the updated national list of threatened species , reaffirming the protected status of <i>P. echinata</i> .
MMA Normative Instruction No. 01/2015	Establishes procedures for the approval of Sustainable Forest Management Plans (SFMPs) for species listed as VU, and sets timeframes for use restrictions on CR and EN species.
IBAMA Normative Instruction No. 21/2014	Establishes the National System for the Control of the Origin of Forest Products (SISFLORA) , enabling traceability and legality verification of forest products.
IBAMA Normative Instruction No. 8/2022	Establishes procedures for the export of wood products and by-products from native species originating from natural forests or plantations.

Bow makers represent the community most involved in current conservation and plantation efforts, despite most of them living outside Brazil (Lichtenberg et al., 2022). Since 2000, the IPCI has partnered with government agencies

from Brazil, non-governmental organisations, several universities, local cacao farmers and regional science and conservation organisations among others to carry out activities aimed at the conservation of *P. echinata*. These include growing and planting of seedlings, developing and implementing replanting strategies in farm and forest locations, funding inventories and the study of the distribution and dynamics of the species, mapping forest fragments and investing in studies into the genetic, anatomy and growth of the species, as well as silviculture and pernambuco propagation methods (IPCI in litt., 2025). Over 340,000 *P. echinata* seedlings had been planted in five states (Bahia, Espírito Santo, Pernambuco, Rio Grande do Norte, and Paraíba) by 2025 (IPCI in litt., 2025). Approximately half of all replanting occurred in natural preserves and urban areas, whereas the remainder were planted on farms and small-scale plantations for potential commercial use once the trees reach maturity at 35–40 years (IPCI in litt., 2025).

Various other international conservation collaborations have been undertaken over the past thirty years. Recently the species was part of a project running from 2019 to 2022 led by RPPN Sítio Duas Cachoeiras & Jardim Botânico Araribá in Brazil. It is also part of a conservation project led by the New York Botanic Garden in Southern Bahia, Brazil. The project aims to enrich and enlarge the population of the species through replanting with seedlings grown from locally collected seeds; and to increase the skills for in-situ conservation. Additionally, the Ecological Reserve of Guapiaçu will join forces with local tree nurseries to add seedlings of this species to the restoration programme of degraded pastures on the Atlantic coast of Brazil. The species is also covered by the project: 'Restoration of threatened tree populations in the Pedra Talhada Biological Reserve in northeast Brazil' led by NORDESTA Reforestation & Education (BGCI GlobalTree Portal, 2025).

No monitoring programme for natural populations of *P. echinata* is currently in force. According to Santana et al. (2020), several plantations in the northeastern states of Pernambuco, Rio Grande do Norte, and Alagoas are being monitored by a Brazilian non-governmental organisation called Associação Plantas do Nordeste.

Artificial propagation

No large-scale commercial plantations of *P. echinata* exist. Wood from small commercial plantations and conservation initiatives is not traded. Brazil indicates that none of the plantations in the country currently meet the requirements established by national laws, which determine that: a) existing plantations must be registered in a timely manner with the environmental agencies, and b) technical projects prepared by legally qualified professionals must be submitted.

In a CITES Notification published at Brazil's request in June 2025 (CITES Notification 2025/076), IBAMA stated that no plantations are registered in Brazil's federal control system (SINAFLO). While some state environmental authorities maintain records of plantations, these remain outside the federal registration system. Brazil indicates that the available plantations are technically inadequate and have been misused to launder illegal timber and concludes that *Paubrasilia echinata* plantations in Brazil are currently not suitable for supplying bows, whether for the domestic or international market.

In a 2025 report, the CITES Secretariat expressed concern about the lack of communication between state environmental authorities and IBAMA. While the federal agency claims to be unaware of formal plantation registrations, some producers have indicated that they had already completed this process with state authorities in regions such as Bahia and Espírito Santo. The CITES Secretariat suggests that closer collaboration between state agencies and IBAMA would facilitate the regularisation of these plantations by providing rural producers with the technical support they need to complete the federal-level registration process (CITES Secretariat, 2025).

Little information on the success or failure of propagation efforts or forestry experiments is available. Several partnerships carry out planting of Brazilwood seedlings in privately owned areas for future commercial exploitation. However, the ideal age for harvesting planted *P. echinata* trees for the production of bows remains largely unknown. A study by Rolim and Piotto (2018) on a 24-year-old plantation found that the species needs long cycles (40–50 years) to reach at least 30 cm dbh. According to the IPCI, the tree begins to produce the dark heartwood for which it is renowned at 10 years old, and the heartwood becomes dominant after 25–30 years (IPCI in litt., 2025). Moreover, timber traders prefer wild-grown wood as it is considered to be superior to that grown in plantations. A paper by Lichtenberg et al. (2022) reports that the timber quality of planted trees compared to high-quality timber from trees in natural habitats has been seriously questioned, as have regulations for their commercial use.

There is no available information on plantations of *P. echinata* outside of Brazil.

Implementation challenges (including similar species)

Paubrasilia echinata has no similar species listed in CITES. The wood can be easily identified by its orange/reddish colouration, storied rays on the tangential face, and the presence of brazilin, which appears as a reddish dye when in contact with a basic solution. The general aspect and colour help distinguish *P. echinata* from similar species (such as *Brosimum rubescens*, *Centrolobium* spp. and *Manilkara* spp.). *Handroanthus* spp.,

which are listed in Appendix II, and *Dialium guianense*, which is not listed in CITES, are also harvested for the production of bows for musical instruments; these may be distinguished from *P. echinata* by the deposits in heartwood vessels (*Handroanthus* spp.) and by axial parenchyma in narrow bands (*Dialium guianense*).

Other considerations:

A complete ban on Brazilwood commercial trade could inadvertently increase illegal trade given the high demand from the musical instrument industry. There is a need to strengthen traceability and enforcement mechanisms to protect this species. Implementing a robust traceability system is essential to ensure that only wood of verified legal origin enters the market. The ultimate goal is to establish a well-regulated market that reduces the appeal of the black market, thereby encouraging sustainable harvesting from commercial plantations and legalized stocks.

References

- Botanic Gardens Conservation International (BGCI) (2025) GlobalTree Portal – *Paubrasilia echinata*. Available at: <https://www.bgci.org/resources/bgci-databases/globaltree-portal/species-search/?species=Paubrasilia+echinata>
- CITES (2025) Notification to the Parties 2025/076 requested by Brazil: *Paubrasilia echinata*: updated information including on plantation growth and stockpiles. <https://cites.org/sites/default/files/notifications/E-Notif-2025-076.pdf>
- CITES Secretariat (2025) *Paubrasilia echinata* bows: Fine tuning traceability solutions. <https://cites.org/sites/default/files/documents/E-SC78-73-A2.pdf>
- Cumine, T. (2007) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- CNCFlora. (2024) Painel de dados do Centro Nacional de Conservação da Flora. Available at: <https://cncflora.jbrj.gov.br/ficha/602728>.
- De Lima, R.A.F., Dauby, G., Gasper, A.L., Fernandez, E.P., Vibrans, A.C., Oliveira, A.A., Prado, P.I., Souza, V.C., Siqueira, M.F. and Steege, H. (2024) Comprehensive conservation assessments reveal high extinctions risks across Atlantic Forest trees. *Science* 383:219-225. DOI: 10.1126/science.abq509
- IPCI (International Pernambuco Conservation Initiative) (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Lichtenberg, S., Huber-Sannwald, E., Reyes-Agüero, J.A., Anhuf, D. and Nehren, U. (2022) Pau-brasil and string instrument bows telecouple nature, art, and heritage. *Ecology and Society* 27(1):32. <https://doi.org/10.5751/ES-13047-270132>
- Rocha, Y.T. (2010) Distribuição geográfica e época de florescimento do Pau-brasil (*Caesalpinia echinata* Lam. – Leguminosae). *Revista do Departamento de Geografia* 20:23–36.
- Varty, N. (1998) *Caesalpinia echinata*. The IUCN Red List of Threatened Species 1998: e.T33974A9818224. <http://dx.doi.org/10.2305/IUCN.UK.1998.RLTS.T33974A9818224.en>.

Deletion from Appendix II of populations of *Pterocarpus soyauxii* located in Central Africa: Angola, Cameroon, the Central African Republic, Congo, the Democratic Republic of the Congo, Equatorial Guinea and Gabon

Proponents: Burundi, Cameroon, Central African Republic, Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon

Summary: *Pterocarpus soyauxii* is an African tree species that provides commercially important timber known generally as mukula, rosewood, African Padauk or African Padouk. It is one of the most exploited timber species in tropical Africa with a wide distribution in lowland forests of Angola, Cameroon, Central African Republic, Congo, Democratic Republic of the Congo (DRC), Equatorial Guinea, Gabon and Nigeria. Eleven other species of *Pterocarpus* occur in Africa, with distributions at country-level overlapping. The species was included in Appendix II in 2023 along with other unlisted African *Pterocarpus* populations following CoP19. Deletion from the Appendices of the regional populations from seven central African countries, that is all range States except Nigeria, is now proposed.

Globally there are around 40 species of *Pterocarpus* native to tropical and subtropical regions, many of which produce valuable hardwoods. China has been a major importer of *Pterocarpus* timber for use in traditional Hongmu and other rosewood furniture. Significant growth in the consumption of Hongmu and other “rosewoods” from 2010 led to dramatically increased levels of exploitation of *Pterocarpus* along with *Dalbergia* spp. and other genera producing timber with similar properties. The increase in trade in the African species *P. erinaceus* was of particular concern leading to its inclusion in Appendix II in 2017 following CoP17. A second species, *P. tinctorius* was included in Appendix II in 2019 following CoP18. Subsequently there were reports of increased pressure on other non-CITES African *Pterocarpus* species. Trade has generally shifted between African *Pterocarpus* spp. depending on availability with multiple species traded under the same names. *P. soyauxii* has been one of the main African species of *Pterocarpus* recorded in trade within China in recent years. It also comprised around 80% of recorded African rosewood imports to Viet Nam during 2018-2022.

Concern about the level of exploitation and trade in timber of *Pterocarpus* species in Africa, including in *P. soyauxii*, led to the inclusion of all African populations of *Pterocarpus* species in Appendix II in 2023 after CoP19 with annotation #17, which includes logs, sawn wood, veneer sheets, plywood and transformed wood. This species has not been globally assessed on the IUCN Red List and no overall population estimates are available although at least one recent assessment has concluded the species is not currently of conservation concern. It is widely distributed in suitable habitat, occurring as individual trees or small groups in forested areas. Forest inventories indicate that the population may be very large – comprising millions of mature trees – and is likely to be able to sustain current harvest levels.

The proposal seeks to delete *Pterocarpus soyauxii* populations from all range States except Nigeria on the basis that they do not meet the “lookalike” criteria for inclusion in Appendix II, specifically criterion A of Annex 2b of Res. Conf. 9.24 (Rev. CoP17). The proponents consider that transformed products of *P. soyauxii* (e.g., sawn wood, parquet, deck boards) can now be reliably distinguished from those of the other species of the genus with available scientific techniques. Timber of *P. soyauxii* may be difficult to distinguish from that of the other CITES Appendix-II listed rosewoods, *Dalbergia* (genus listing at CoP18) and *Guibourtia* (three African species listed at CoP18). According to the supporting statement Central African countries are implementing sustainable forest management and rigorous traceability mechanisms that make it possible to guarantee the origin and identity of the species in trade.

CITES trade data show ~69,500 m³ of logs, sawn wood and timber of *P. soyauxii* reported in direct commercial exports by the seven Central African countries in 2023 and 2024, most (70%) reported either as wild-sourced or of remaining pre-Convention stock. Angola, Cameroon, Congo, DRC, and Equatorial Guinea have all reported some quantities of logs, sawn wood and timber from *P. soyauxii* in direct commercial exports, and although Central African Republic and Gabon have not yet submitted

CITES trade data for 2023-2024, importers have reported imports of *P. soyauxii* in those years. For three of these countries (Angola, Cameroon and DRC) CITES trade data show direct commercial exports of logs and sawn wood in other *Pterocarpus* species also took place between 2020 and 2023. Despite considerable efforts to improve the enforcement of forestry laws, governance, and trade in the Congo Basin countries, large quantities of timber in general are believed to be illegally sourced.

Implementation of CITES for *P. soyauxii* has included development of Non-Detriment Findings (NDFs) and annual quotas for the species by Cameroon, DRC and Gabon. Annual quotas have also been developed by Congo and Equatorial Guinea.

The supporting statement recommends the establishment of a monitoring mechanism if the populations are removed from the Appendices to prevent any drift towards future over-exploitation, although presumably this would no longer be under the auspices of CITES.

Analysis: Available information indicates that the populations of *P. soyauxii* located in Angola, Cameroon, Congo, DRC, Central African Republic, Equatorial Guinea and Gabon **may not meet Criterion B of Annex 2a** of Res. Conf. 9.24 (Rev. CoP17). Measures are reportedly in place to ensure the sustainable and legal harvesting and trade of timber of the species in the countries for which the delisting is proposed, and regeneration appears to be good within managed concessions. The extent and impact of illegal logging are currently difficult to quantify, but illegal logging is believed to be the main ongoing threat. Uncertainties remain regarding the sustainability of harvesting and trade in *P. soyauxii* and the development of NDFs for the species in some exporting range States.

Although available scientific techniques may exist to distinguish *P. soyauxii* from those of other species in the genus, enforcement officers may not have access to these. Of particular concern may be distinguishing specimens of *P. soyauxii* from those of the Endangered *P. erinaceus*, which is currently subject to a range of CITES trade suspensions. Furthermore, timber products from the populations of *P. soyauxii* proposed for delisting would only be differentiated from products from populations covered by Appendix II by their stated origin. Evidence indicates that timber consignments exported from Central Africa are frequently consolidated, often in Cameroon, from multiple countries such as Central African Republic, Congo and Cameroon itself. Due to weak enforcement and porosity in supply chains, these consignments incorporate timber of uncertain or illegal origin. **These populations of *P. soyauxii* thus appear to continue to meet the 'lookalike' criteria in Annex 2b** of Res. Conf. 9.24 (Rev. CoP17).

If adopted this proposal would result in a split-listing of *P. soyauxii* by geographical populations, with the population of the species in Nigeria remaining in the Appendices, and the populations in countries covered by the proposal outside the Appendices. Annex 3 of Res Conf 9.24 (Rev. CoP17) notes that "*split-listings that place some populations of a species in the Appendices, and the rest outside the Appendices, should normally not be permitted*" in view of enforcement problems it might create, although there are a few precedents for such split-listing in the Appendices.

Summary of Available Information

Text in non-italics is based on information in the Supporting Statement (SS); text in italics is based on additional information and/or assessment of information in the SS.

Taxonomy

Pterocarpus soyauxii Taub. (1895)

Range (populations proposed for deletion from the Appendices are underlined)

Angola, Cameroon, Central African Republic, Congo, DRC, Equatorial Guinea, Gabon and Nigeria.

IUCN Global Category

Not assessed.

Biological and trade criteria for retention in Appendix II (Res. Conf. 9.24 (Rev. CoP 17) Annex 2a)**B) Regulation of trade required to ensure that harvest from the wild is not reducing population to level where survival might be threatened by continued harvest or other influences**

P. soyauxii has a widespread distribution occurring in Angola, Cameroon, Central African Republic, Congo, DRC, Equatorial Guinea, Gabon and Nigeria. It occurs in lowland dense evergreen and semi-deciduous wet forests. In January 2020, the total area of evergreen and semi-deciduous forests in Central Africa countries was estimated at around 200 million hectares, of which 185 million hectares showed no visible signs of disturbance.

The Central African forest region is broadly subdivided into two major types of land tenure: (i) the permanent forest estate with areas designated for timber production and also designated as formal protected areas; and (ii) the non-permanent forest domain consisting of agricultural landscapes and young or mature secondary forests. Within areas designated for timber production, management is generally through the development of mandatory forest management plans in forest concession areas. The implementation of forest management plans provides data on the spatial distribution of commercial species, including *P. soyauxii*, contributing to a better understanding of the structure and dynamics of harvested stands.

There is no information on the overall population size of P. soyauxii. It generally occurs as scattered individuals or small groups within the forest (Jansen, 2005). However, the SS indicates that, considering the average density of mature stems and the total area of production forests in Central Africa, the total population of mature *Pterocarpus soyauxii* was estimated at more than 18,883,000 individuals within its regional range by 2023. It is also stated that the species is abundant and shows good natural regeneration, with no signs of overall decline, and is well-represented in protected areas.

Natural regeneration is currently considered to support sustainable levels of trade. It has been noted that 88% of studied population structures for P. soyauxii in Central Africa were good with no major regeneration problems noticed throughout its natural distribution area. This species occurs in many forests where logging companies operate, meaning that pre-harvest inventories data are available and consequently population structures are incorporated into the management plans (Mousset Moumbolou, 2019).

Although a global IUCN Red List assessment is not yet available for P. soyauxii it is unlikely that the criteria for assessment within a threat category would be met in terms of distribution, population decline and other factors. At a workshop held in Libreville in December 2022 the species was assessed as least concern based on estimates of low population decline in the future (IUCN Red List criterion A3) both in concession and non-concession areas (Loubota Panzou et al., 2023). At a national level, P. soyauxii has previously been assessed as threatened in DRC and not threatened in Cameroon (BGCI, 2025).

Data on population densities, regional population estimates and presence in protected areas supports the proposition that current trade is not driving extinction risk. However, data appear to derive mainly from concessions and managed areas (potential sampling bias); non-negligible illegal harvest is noted in non-permanent forest domains (40–50% of timber may enter the informal economy regionally - see African Development Bank, 2018); and regional heterogeneity may mask local declines. More independent, pan-range monitoring evidence and an assessment of resilience to changes in demand is required (Mohonghol in litt., 2025). It has also been noted that trees do not coppice well, stump regrowth is weak and uneconomic for wood production (Jansen, 2005).

P. soyauxii has been traded internationally for many years and was among the first timber species to be exported from Gabon. Between 2000–2003, Gabon exported 120,000 m³ of P. soyauxii logs annually. Since 2010, the export of logs has been banned by Gabon. Padouk was the second most important timber exported from Gabon by volume 2007–2017 (all commodities combined) with a volume of almost 1,195,000 m³ (Mahonghol et al., 2020). Much of this is presumably P. soyauxii, which is listed as one of the 24 species currently harvested (Timber Trade Portal, 2025). Pterocarpus soyauxii is also reported to be one of eleven species that account for about 80% of timber production in DRC (Timber Trade Portal, 2025). CITES Trade Data between 2017 and 2023 show that P. soyauxii accounts for most direct commercial exports of Pterocarpus species from Central African countries in recent years between 2023 and 2024, with P. erinaceus being the species most exported prior to this (Figure 1). Overall ~69,500 m³ of logs, sawn wood and timber from the species have been reported in direct commercial exports by Central African countries between 2023 and 2024.

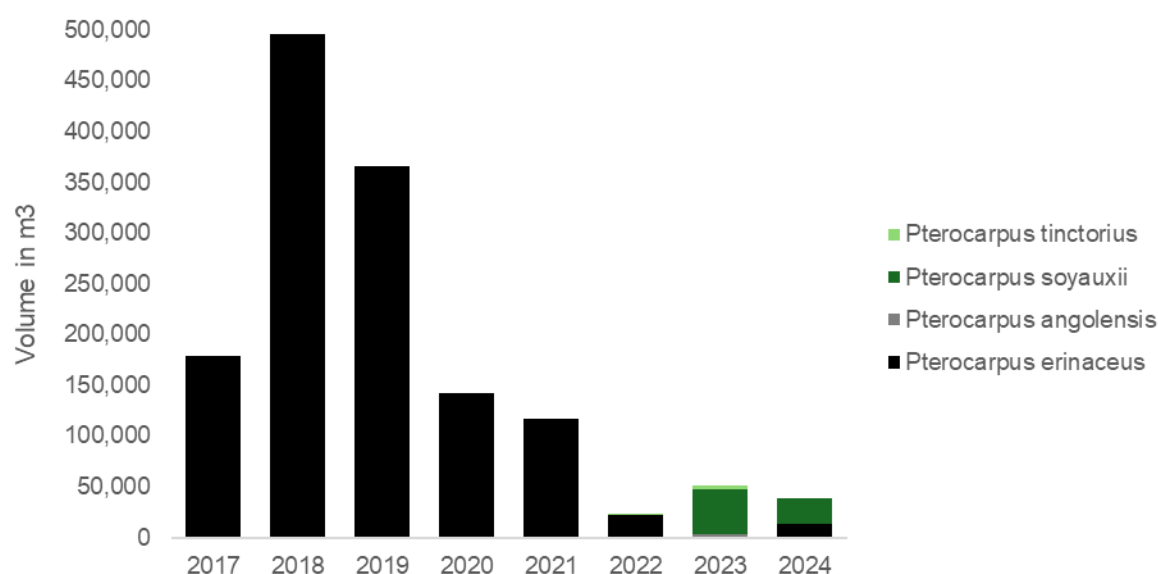


Figure 1. Volume in cubic metres of direct commercial exports of *Pterocarpus* species from Central African countries between 2017 and 2023. Source CITES Trade Database.

Since *P. soyauxii* was listed in CITES, exporters Angola, Cameroon, Congo, DRC, and Equatorial Guinea have reported notable quantities of logs, sawn wood and timber in direct commercial exports (Table 1). Two other countries- Central African Republic and Gabon- have not yet reported CITES Trade Data for 2023-2024, but importers have reported trade in *Pterocarpus soyauxii* from these countries in those years.

Most trade reported by exporters is in wild-sourced specimens (~41,200 m³; 68% of direct commercial exports) with the remaining pre-Convention. Congo also exported ~8,500 m³ of logs and sawn wood with no source stated in 2023. Top importers were China (70% of all imports), Viet Nam (16%) and Belgium (7%), with the rest reported to be imported in small quantities by 17 other countries.

In addition to the data presented in Table 1 for trade reported by volume in cubic metres, France reported importing ~45,000 kg of pre-Convention wood products from Gabon (who has not yet reported trade data for 2023), and India reported importing ~51,000 kg of wild sourced logs, with the exporting country not stated. Trade data discrepancies are often due to errors or differences in reporting (CITES Secretariat and UNEP-WCMC, 2022).

Table 1. Direct commercial exports of *Pterocarpus soyauxii* logs, sawn wood and timber reported by exporters and importers in cubic metres between 2023 and 2024 (for which data is incomplete). Source CITES Trade Database.

Country	Exporter-reported quantity (m³)			Importer-reported quantity (m³)		
	Pre-convention	Wild	Source not stated	Pre-convention	Wild	Assisted production
Angola	625	307		850	64	24
Cameroon	18,668	0		10,714	32	0
Central African Republic*	0	0		375	5,913	0
Congo		25,750	8,468	17,687	3,707	0
DRC		5,463		9,666	497	0
Equatorial Guinea		10,154		1,401	8,083	0
Gabon*	0	0	0	15,981	557	0
Total	19,293	41,674	8,468	56,674	18,853	24

*Have not yet submitted CITES annual trade reports for 2023-2024

Three countries have produced NDFs for *P. soyauxii* but NDF processes in other countries are unclear. Cameroon's NDF for *P. soyauxii* established a national cutting quota of 31,907 m³ representing 29% of the exploitable standing volume and 40% of the harvesting potential. For 2024, Gabon has an NDF with a national harvest quota of 345,028 m³ and quota of 158,713 m³ for processed products out of a harvest potential of 363,187 m³. DRC has set a

national export quota for *P. soyauxii* of 48,934 m³ for 2025. Quotas have also been set by Congo and Equatorial Guinea.

In a CITES study of rosewoods, *P. soyauxii* was categorised as medium priority in terms of NDF development (CITES Secretariat, 2024) but this category was assigned for all new rosewood species listed at CoP19 in 2022 as a precautionary measure due to no trade data yet being available for these species.

Central African countries have put in place a range of national legal instruments to regulate the sustainable management of forest resources and the protection of biodiversity. Most of the countries have legislation which provides for responsible harvesting and trade of forest resources including *P. soyauxii*. The policy and legal frameworks provide a good basis for the development of NDFs.

Despite sustainable management policies and practice for *P. soyauxii* there is no guarantee of enforcement and effectiveness across all range States and along the supply chain. More information is needed on consistent, verifiable effectiveness from for example audits, compliance rates and seizure data (Mahonghol in litt., 2025). *P. soyauxii* is mainly threatened by the illegal logging of its wood. The extent of illegal trade in *P. soyauxii* across its range is largely unknown, although the SS states that in Central Africa almost 46% of total log production comes from illegal logs. In general illegality in the forestry sector remains an issue for Central African countries. As noted by Mahonghol (2024) despite considerable efforts to improve the enforcement of forestry laws, governance, and trade in the Congo Basin countries, illegality still occurs throughout the timber supply chain with some reports asserting that 90% of the timber from the Congo Basin rainforests may be illegally sourced and facilitated by high-level crime. In Cameroon for example, Kamkuimo et al. (2025) note that illegal exploitation of timber is still a major concern in the country and frequently documented in various stages of the local supply chain. Since the species listing in 2022, WiTIS has also recorded six seizures totalling 25 logs and 46 pieces of sawn wood from *P. soyauxii*, with all of these reported to have originated in Cameroon and were seized within the country, with all stated to be wild-sourced. Prior to the listing and between 2016 and 2021, around 35 seizures totalling around 500 wild-sourced sawn wood/logs were also reported seized in Cameroon (again with origin reported to be Cameroon).

High levels of illegal timber trade between DRC and Uganda have been reported (Blanshe, 2024) although not specifically in *Pterocarpus* spp.. The Congo also faces challenges related to forest governance and illegal logging (WorldTradePro, 2023) and, despite a strong legislative framework for forest management, illegal logging is still considered widespread in Gabon (Forest Trends, 2024a).

Cameroon's ports act as a central shipping hub for timber-producing countries across West and Central Africa, complicating supply chains and origin labelling. The port of Douala has also been identified as a centre for organized crime and illicit exports (Forest Trends, 2024b). Challenges remain concerning the management of trade in CITES-listed timber from third countries in transit through Cameroon. As a transit country, Cameroon is not able to guarantee their durability, legality or traceability. Cameroon has no powers to verify that the conditions of issuance of CITES permits – particularly the making of NDFs and legal acquisition findings – were respected in the countries of origin. It has been reported that a high proportion of CITES-listed species exported from the ports of Cameroon come in fact from the Central African Republic and from the Congo. Moreover, specimens from neighbouring countries are not subject to the same control procedures as timber from Cameroon. The authorities of Cameroon only verify the presence of the required documents but the procedure of entry into the port, customs clearance and export is managed directly by the authorities of the countries concerned, which have offices in the port of Douala and cooperate with the customs of Cameroon. It is therefore not possible to thoroughly monitor the exports made by the countries of the region that transit through Cameroon (CITES Secretariat, 2025).

Timber such as rosewood is amongst the high-value items that are poached and smuggled between Nigeria and Cameroon (TRAFFIC, 2024). Nigeria has, in the past, exported large quantities of illegally harvested rosewood from Cameroon. In 2024, an MoU was signed by Nigeria's Environment Minister and Cameroon's Minister for Forestry and Wildlife, to address wildlife poaching and illegal trade through a framework to jointly implement transboundary wildlife programmes, intelligence sharing protocols, and enhanced enforcement cooperation (TRAFFIC, 2024).

Retention in Appendix II to improve control of other listed species

A) Specimens in trade resemble those of species listed in Appendix II under Res. Conf. 9.24 (Rev. CoP17) Annex 2a or listed in Appendix I

The proposal states that this species was listed on Appendix II in accordance with Res. Conf. 9.24 (Rev. CoP17) Annex 2(b), Criterion A due to its similarity to other listed African species of the same genus and the presumed difficulty for control agencies to distinguish commercial products from these different species. However new evidence shows that processed products from *P. soyauxii* can be reliably distinguished from those of other species in the genus using available scientific techniques.

Despite progress in timber identification methodologies these may not be widely available to enforcement officers. Research on timber supply chains in Cameroon and the Republic of Congo, for example recorded challenges

including the limited availability of specialists in timber species identification among customs officials and limited knowledge of forestry laws and regulations (Mahonghol, 2022).

Currently all African populations of *Pterocarpus* species are included in Appendix II of CITES. Species of particular conservation concern include *P. erinaceus* listed as *Endangered* by IUCN and the very rare Cameroonian endemic *P. zenkeri* also listed as *Endangered*. *P. tessmannii* is listed as *Near Threatened* by IUCN. *P. erinaceus* has been subject to a trade suspension from Nigeria and Liberia since 2018; from Cameroon, Central African Republic, Chad, Gambia, Guinea-Bissau, Mali and Togo since 2022 and Guinea since 2023.

Other *Pterocarpus* spp. occurring in Central African countries, and the *Pterocarpus* spp. which have been reported in direct commercial exports from each country between 2017 (e.g., the genus listing) and 2023 are summarised in Table 2. Exporter reported data were used in most instances unless otherwise specified but importer reported data were used for Central African Republic and Gabon as they have not yet reported CITES Trade Data for all years between 2017 and 2023.

The SS indicates that *P. soyauxii* occurs in geographical areas that are well differentiated from those of other CITES-listed *Pterocarpus* species. However, there may be mixing of wood of different species in trade consignments.

Table 2. *Pterocarpus* spp. occurring in Central African countries, and the *Pterocarpus* spp. which have been reported in direct commercial exports from each country between 2017 and 2024. Source: CITES Trade Database; de la Estrella et al., 2010; BGCI, 2025.

Central African range State	Other <i>Pterocarpus</i> spp. occurring in the country*	Other <i>Pterocarpus</i> spp. exported from the country
Angola	<i>P. angolensis</i> , <i>P. lucens</i> , <i>P. rotundifolius</i> , <i>P. tinctorius</i>	<i>P. angolensis</i> (3,065 m³ wild sourced sawn wood in 2023) <i>P. tinctorius</i> (~3,940 m³ wild sourced sawn wood between 2022-2023)
Cameroon	<i>P. erinaceus</i> , <i>P. mildbraedii</i> , <i>P. osun</i> , <i>P. santalinoides</i> , <i>P. zenkeri</i>	<i>Pterocarpus</i> spp. (60 m³ wild sourced sawn wood in 2023)*
Central African Republic	<i>P. erinaceus</i> , <i>P. mildbraedii</i> (?)	None
Congo	<i>P. angolensis</i> , <i>P. lucens</i>	None
DRC	<i>P. angolensis</i> , <i>P. lucens</i> , <i>P. rotundifolius</i> , <i>P. tessmannii</i> , <i>P. tinctorius</i>	<i>P. tinctorius</i> (~75,500 m³ pre-convention and 900 m³ wild sourced logs between 2020 and 2023)**
Equatorial Guinea	<i>P. mildbraedii</i> , <i>P. osun</i> (?) <i>P. santalinoides</i> (?), <i>P. mildbraedii</i> , <i>P. tessmannii</i>	None
Gabon	<i>P. erinaceus</i> (?), <i>P. mildbraedii</i> , <i>P. tessmannii</i>	None
Other range States		
Nigeria	<i>P. erinaceus</i> , <i>P. lucens</i> , <i>P. mildbraedii</i> , <i>P. osun</i>	<i>P. erinaceus</i> (~251,000 m³ sawn wood with no source stated in 2018)

*This trade was reported by the importer USA. Cameroon did not report any exports.

**This trade was reported by the importers China (>99%) and Viet Nam. DRC did not report any exports.

It has been noted that alternatives to *P. soyauxii* include the wood of *Pterocarpus osun* from southern Nigeria and Cameroon, *Pterocarpus tessmannii* from Equatorial Guinea, Gabon and DRC, and *Pterocarpus tinctorius* from DRC and Angola which are also marketed as African padauk (Jansen, 2005). The SS states that *Pterocarpus tinctorius* is mainly harvested in south-eastern DRC (Katanga) and transits to Zambia via uncontrolled cross-border networks, but it is not exported from the same ports as those used for Padauk (i.e., Gabon, Congo, Cameroon). The extent of illegal trade in timber of *Pterocarpus* spp. exported through Zambia is highlighted by Forest Trends (2021). CITES Trade Data shows that there have been direct commercial exports of wild-sourced *Pterocarpus tinctorius* reported from Angola and DRC between 2020 and 2023. There are no reports of significant trafficking of *P. erinaceus* from Central African countries. Documented seizures mainly concern West Africa, in the context of intense trade with Asia. The marketing channels are clearly distinct from those of *P. soyauxii*.

Other genera traded with the name rosewood include species of *Dalbergia* (genus listing at CoP18) and *Guibourtia* (three African species listed at CoP18). *The timber of P. soyauxii is one of the main African species of Pterocarpus currently recorded in trade within China. In Kunming for example, it is recorded with other rosewood species such as P. erinaceus, P. tinctorius and Guibourtia spp. in markets of Kunming, traded as hewn logs and squared timber having been brought into China mainly through Zhangjiagang (Zhang and Chen, 2022b). This species also comprises around 80% of African rosewood imports to Vietnam during 2018-2022 with a value of USD62million (Panjiva, 2022).*

Additional Information

Threats

The main threat to this species is reported to be illegal logging. *Pterocarpus soyauxii* is among the most exploited species in Central Africa. Studies on the impact of logging on the species and habitats are needed. Other threats include the expansion of agriculture and urbanisation (Loubota Panzou *et al.*, 2023).

Conservation, management and legislation

Central African countries have put in place a set of national legal instruments aimed at regulating the sustainable management of forest resources and the protection of biodiversity. In Central Africa, most of the countries with *Pterocarpus soyauxii* populations have a regulatory framework, based on existing forestry laws, that is aimed at ensuring the sustainable management of forests and the responsible exploitation and trade of forest resources, including *P. soyauxii*. Traceability systems for commercial timber have been developed.

Artificial Propagation/captive breeding

*Silvicultural experiments and plantation trials have been undertaken in Congo, Cote d'Ivoire and Nigeria. Propagation by seed is effective. Seeds will germinate in the shade, but seedlings are light demanding, requiring abundant light to recruit adequately. In trial plantations annual height growth varied between 1.6 m and 2.7 m in the first seven years. The mean annual volume growth, including thinning, was 20–30 m³/ha over a period of 15 years (Jansen, 2005). In some forest concessions, active planting initiatives for *P. soyauxii* have been implemented, reinforcing the potential for assisted regeneration of the species.*

P. soyauxii is recorded in five *ex situ* collections (BGCI, 2025)

Other comments

The SS states that the compulsory preparation of NDFs in producer countries with their attendant administrative delays as well as delays in the issue of import permits by member countries of the European Union, have had a severe impact on legal international trade in the species. *Should the delisting proposal be accepted it is not clear whether the EU would retain the species as a whole in Annex B. and therefore still require NDFs under stricter domestic measures regardless of the origin of the species.*

References

- African Development Bank (2018) Développement intégré et durable de la filière bois dans le bassin du Congo: Opportunités, défis et recommandations opérationnelles.
- Blanshe, M. (2024) Timber Smuggling: From DRC, Through Uganda, to the Rest of the World. Pulitzer Center Website. Available at: <https://pulitzercenter.org/projects/timber-smuggling-drc-through-uganda-rest-world>
- BGCI (2025) Botanic Gardens Conservation International Species Search Website. Available at: <https://www.bgci.org/resources/bgci-databases/globaltree-portal/species-search/?species=Afzelia+bipindensis>
- CITES Secretariat (2024) Report on the conservation and trade of CITES-listed rosewood species [Leguminosae (Fabaceae)]. PC27 Doc. 27. Available at: <https://cites.org/sites/default/files/documents/E-PC27-27.pdf>
- CITES Secretariat (2025) Application of Article XIII in Cameroon. SC78 Doc. 33.4.
- CITES Secretariat and UNEP-WCMC (2022) A guide to using the CITES Trade Database. Version 9. Geneva, Switzerland, and Cambridge, UK.
- de la Estrella, M., Cabezas, F.J., Aedo, C. and Velayos, M. (2010) The Papilionoideae (Leguminosae) of Equatorial Guinea (Annobón, Bioko and Río Muni). *Folia Geobot* 45:1–57.
- Forest Trends (2024a) Timber Legality Risk Dashboard: Gabon. Available at: https://www.forest-trends.org/wp-content/uploads/2022/01/Dashboard-Gabon_Sept-2024.pdf
- Forest Trends (2024b) Timber Legality Risk Dashboard: Cameroon. Available at: https://www.forest-trends.org/wp-content/uploads/2022/01/Dashboard-Cameroon_August2024.pdf
- Forest Trends (2021) Timber Legality Risk Dashboard: Zambia. Available at: <https://www.forest-trends.org/wp-content/uploads/2022/01/Zambia-Timber-Legality-Risk-Dashboard-IDAT-Risk.pdf>
- ITTO (2025) Tropical Timber Market Report.
- Jansen, P.C.M. (2005) *Pterocarpus soyauxii* Taub. PROTA (Plant Resources of Tropical Africa / Ressources végétales de l'Afrique tropicale) Website. Available at: <https://prota.prota4u.org/protav8.asp?q=pe&p=Pterocarpus+soyauxii> Accessed on 12 August 2025

- Kamkuimo, P., Guiny, J., Kamga, J. and Modjo, A. (2025) A Guide to documentary analysis for EUTR/EUDR stakeholders within the context of carrying out due diligence. TRAFFIC and FODER.
- Loubota Panzou, G.J., Monthe, F., Ligot, G., Julve, C. and Doucet, J.L. (2023) Actualisation du statut de vulnérabilité des espèces ligneuses exploitées en Afrique centrale. Available at: http://www.ppecf-comifac.com/tableau_recapitulatif.html. Accessed on 2 July 2025
- Mahonghol, D. (2022) Improving timber supply chains will help unlock sustainable production and consumption this International Day of Forests. TRAFFIC Website. Available at: <https://www.traffic.org/news/improving-timber-supply-chains-will-help-unlock-sustainable-production-and-consumption-this-international-day-of-forests/>
- Mahonghol, D., Outhwaite, W. and Momballa, M.C. (2020) Timber trade trends in Central Africa 2007-2017: Cameroon, Democratic Republic of Congo, Gabon and Republic of Congo. June 2020, Yaoundé Cameroon. TRAFFIC Edition. Yaoundé, Cameroon and Cambridge, UK.
- Mahonghol, D. (2024) The Regional Security Imperative to Protect the Congo Basin. *Africa Security Brief* 44
- Mahonghol, D. (2025) *in litt.* to IUCN/TRAFFIC Analyses Team, Cambridge, UK.
- Momballa-Mbun C., Mgaza A., Floros C. and Chen H.K. (2023) An overview of the timber traceability systems in the Congo Basin countries. TRAFFIC.
- Mousset Moumbolou, C.L. (2019) Étude du statut de conservation de trois espèces de bois d'œuvre d'Afrique Centrale : *Millettia laurentii* de Wild., *Pterocarpus soyauxii* Taub. et *Triplochiton scleroxylon* K. Schum. MSc thesis, Universidad Internacional de Andalucía: 50 p.
- Panjiva (2002) Panjiva Database. Downloaded by Forest Trends August 2022.
- Timber Trade Portal (2025) Country Profiles available at: <https://www.timbertradeportal.com/en/p/5/country-profiles>
- TRAFFIC (2024) Historic agreement between Nigeria and Cameroon to tackle wildlife crime. TRAFFIC Website. Available at: <https://www.traffic.org/news/historic-agreement-between-nigeria-and-cameroon-to-tackle-wildlife-crime/>
- WorldTradePro (2023) Who is the largest exporter of timber in Africa?. World Trade Pro – Navigating the World of Commodities: Discuss, Share, Trade. Available at: <https://worldtradepro.com/forum/topic/who-is-the-largest-exporter-of-timber-in-africa/>
- Zhang, K. and Chen, H.K. (2022b) China's rosewood market survey. TRAFFIC.

Inclusion of species *Aloe bergeriana*, *Aloe jeppeae*, *Aloe subspicata*, and *Aloe welwitschii* in Appendix II, as part of the listing of the *Aloe* spp.

Proponents: Depositary Government (Switzerland), South Africa and Zimbabwe

Summary: The genus *Aloe* consists of over 600 species, most native to Africa. Aloe species have a variety of uses and are mostly traded as live specimens, stems, extracts, flowers, dried plants, and leaves and have a variety of uses. The genus *Aloe* has been included in CITES Appendix II since 1975, with 21 of these species now included in Appendix I.

Four additional species *Aloe bergeriana*, *Aloe jeppeae*, *Aloe subspicata*, and *Aloe welwitschia*, previously comprising the genus *Chortolirion* and now included within *Aloe*, are proposed for inclusion in the genus listing for *Aloe*. This classification is also followed in the updated CITES Checklist for *Aloe* species, which is proposed for adoption and inclusion as standard nomenclature reference at CoP20 (see document CoP20 Doc.110). The change in taxonomy was previously discussed by the Nomenclature Working Group at the 27th meeting of the CITES Plants Committee in 2024. It was concluded that any change to the appendices would be substantive and that the best course of action would be to invite the Depositary Government to work with South Africa on the present proposal to include the four *Aloe* species previously treated as *Chortolirion* in Appendix II in accordance with paragraph 2f) of Res. Conf. 12.11 (Rev. CoP19) on *Standard nomenclature*.

Inclusion of these species in Appendix II is claimed to be justified under criterion A of Annex 2b in Res. Conf. 9.24 (Rev. CoP17). The proposal states that although their flowers distinguish these four species from other aloes, the bulbous base of the four species and thin grass-like leaves are very similar to other small bulbous grass aloe species, with the bulblike stem bases being the form in which they - and the currently listed aloes - appear in trade. It would therefore be difficult to separate the four species proposed for inclusion with certainty and ease from those of other small bulbous grass-aloes already listed in the genus.

Aloe bergeriana is native to Namibia, South Africa, and Zimbabwe, *A. subspicata* to Botswana, Eswatini, Lesotho, Namibia, and South Africa, *A. jeppeae* is endemic to South Africa, and *A. welwitschii* endemic to Angola. Three of the four species (excepting *A. welwitschii*) have been assessed within South Africa as not of conservation concern. There is no evidence of international trade in any of the species. According to CITES trade data (2019-2023), out of the seven countries in which these four species are native, exports of wild sourced specimens of any *Aloe* species were only reported from South Africa (mostly extracts, derivatives and powder of *Aloe ferox* and *Aloe arborescens*) and Zimbabwe (mostly stems of *Aloe arborescens*) for commercial purposes.

Analysis: Of the four *Aloe* species included in the proposal, three have been assessed as not threatened (the fourth is unassessed) and none is known to be in international trade, nor is there likely to be demand for wild-collected specimens of any of them. Therefore, these **species do not meet the criteria for inclusion in Appendix II in Annex 2a** of Res. Conf. 9.24 (Rev. CoP17). Their similarity to other small, bulbous grass aloe species means there is **some possibility of their meeting the criteria under Annex 2b** of the Resolution, but none of these other bulbous grass aloe species are known currently to be in trade as wild-collected plants. Nevertheless, this proposal has been submitted after consideration by the Plants Committee **following the guidance of Res. Conf. 12.11** (Rev. CoP19) on *Standard nomenclature* and **adding these species under the genus listing of *Aloe* may be a more pragmatic solution than adding an annotation to clarify that they are excluded from the Appendices.**

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

The Aloe genus was first listed in 1975 (UNEP, 2025).

Taxonomy

Aloe bergeriana (Dinter) Boatwr. & J.C.Manning (2014), *Aloe jeppeae* Klopper & Gideon F.Sm. (2013), *Aloe subspicata* (Baker) Boatwr. & J.C.Manning (2013) and *Aloe welwitschii* Klopper & Gideon F.Sm. (2013). All four species were previously in the genus *Chortolirion*, which was transferred to the genus *Aloe*.

Range

The distribution of the four species is:

- *Aloe bergeriana* – Namibia, South Africa, Zimbabwe
- *Aloe jeppeae* – South Africa
- *Aloe subspicata* – Botswana, Eswatini, Lesotho, Namibia, South Africa
- *Aloe welwitschii* – Angola

IUCN Global Category

Not assessed.

Inclusion in Appendix II to improve control of other listed species

A) Specimens in trade resemble those of species listed in Appendix II under Res. Conf. 9.24 (Rev. CoP17) Annex 2 a or listed in Appendix I

The *Aloe* genus was first listed in 1975 (UNEP, 2025) and currently the genus of *Aloe*, except for *Aloe vera* (= *Aloe barbadensis*), is listed in the CITES Appendices. There are over 600 species listed overall, with 21 species listed in Appendix I and the remainder in Appendix II. Of the 21 Appendix-I listed species, 17 are endemic to Madagascar with 4 occurring in southern Africa: *Aloe albida* in Eswatini and South Africa, *Aloe pillansii* in Namibia and South Africa, *Aloe polyphylla* in Lesotho, and *Aloe vossii* in South Africa.

All aloes were traditionally included in a broadly circumscribed single genus, *Aloe*. Based on evidence from molecular and phylogenetic studies on aloes and their relatives, the genus *Aloe* (in the broad sense) has been divided into segregate genera, namely the true aloes in the genus *Aloe*, as well as *Aloestrela* (ancient aloe), *Aloiampelos* (rambling aloes), *Aloidendron* (tree aloes), *Aristaloe* (awn-leaf aloe), *Gonialoe* (kanniedood aloes), and *Kumara* (fan aloes). Results from this research also confirmed that the genus *Chortolirion*, comprising four species (now *Aloe bergeriana*, *Aloe jeppeae*, *Aloe subspicata*, and *Aloe welwitschia*) is embedded within *Aloe* in the phylogeny for this group of plants. This classification is also followed in the updated CITES Checklist for aloe species that will be proposed at CoP20 for adoption and inclusion as standard nomenclature reference in Res. Conf. 12.11 (Rev. CoP19). None of the four species have been assessed on the IUCN Red List, and all are native to countries within Southern Africa. *Aloe bergeriana* (native to Namibia, South Africa and Zimbabwe) and *Aloe subspicata* (Botswana, Eswatini, Lesotho, Namibia, and South Africa) were listed as least concern on the national list of endangered species in South Africa in 2018 (*A. subspicata* assessed 2014 (SANBI, 2025)), and *Aloe jeppeae* (endemic to South Africa) was listed as least concern nationally in 2014 (SANBI, 2025). *A. welwitschia* has not been assessed (SANBI, 2025).

The genus *Chortolirion* has not been included in the Appendices to CITES. Including these four (now *Aloe*) species in the genus-level listing in Appendix II would thus constitute a substantive change as it alters the scope of protection under the Convention in accordance with paragraph 2f) of Resolution Conf. 12.11 (Rev. CoP19) on *Standard nomenclature*. This matter was discussed by the Nomenclature Working Group at the 27th meeting of the CITES Plants Committee (July 2024). It was decided that the best course of action would be to invite the Depositary Government to work with South Africa on a proposal to include the four *Aloe* species previously treated in the genus *Chortolirion* in Appendix II, for submission to CoP20, in accordance with paragraph 2f) of Res. Conf. 12.11 (Rev. CoP19)¹⁹ [see recommendation 5b) on p. 10 of the Nomenclature Working Group report (PC27 Com. 4)].

The proposal states that the four species *Aloe bergeriana*, *Aloe jeppeae*, *Aloe subspicata*, and *Aloe welwitschia* previously in the *Chortolirion* genus are small bulbous aloes with thin, non-succulent, grasslike leaves that are variously spirally twisted, with minute marginal teeth. Although the haworthia-like flowers will distinguish these four species from other aloes, the bulbous base of the plants formed by slightly fleshy leaf-bases, and the thin grass-like leaves are very similar to other small bulbous grass aloe species. Individual, non-flowering bulbs of

¹⁹ Paragraph 2f of Resolution 12.11 (Rev. CoP19) states 'whenever a change in the name of a taxon included in the Appendices is proposed, the Secretariat, in consultation with the Animals or Plants Committee, determine whether this change would alter the scope of protection for fauna or flora under the Convention. In the case where the scope of a taxon is redefined, the Animals or Plants Committee shall evaluate whether acceptance of the taxonomic change would cause additional species to be included in the Appendices or listed species to be deleted from the Appendices and, if that is the case, the Depositary Government should be requested to submit a proposal to amend the Appendices in accordance with the recommendation of the Animals or Plants Committee, so that the original intent of the listing is retained. Such proposals should be submitted for consideration at the next regular meeting of the Conference of the Parties, at which the recommendations of the Animals and Plants Committees will be considered'.

members of *Aloe* section *Chortolirion*, as they would appear in trade, would be difficult to separate with certainty and ease from those of other small bulbous grass-aloes. The proposal states the look-alike criterion is thus very much applicable to these four species and including them in Appendix II is therefore warranted based on criterion A of Annex 2b in Resolution Conf. 9.24 (Rev. CoP17). This criterion states that 'species may be included in Appendix II if specimens of the species in the form in which they are traded resemble specimens of a species included in Appendix II under the provisions of Article II, paragraph 2(a), or in Appendix I, so that enforcement officers who encounter specimens of CITES-listed species are unlikely to be able to distinguish between them'.

According to the CITES Trade Database, between 2019 and 2023 there have been notable quantities of commercial exports of products from *Aloe* species including ~ 64.5 million live specimens and large quantities of parts and derivatives (~45.5 million specimens, 2.2 million kg, and ~140,000 litres of commodities such as stems, extracts, flowers, dried plants, leaves). At least 140 different *Aloe* species (out of around 600 currently listed) have been reported in direct exports by over 25 countries. Most of these exports are reported to be in artificially propagated specimens, but South Africa is the top exporter of specimens from wild sources, reporting direct exports of 1,000 live specimens as well as large quantities of parts and derivatives of *Aloe* species (~140,000 litres, 1.5 million kg, and ~18,000 specimens).

Expert consultations on evidence of trade in the species proposed for inclusion identified little evidence of trade in the four species. Online searches – including of the RHS Plant Finder website – did not identify the species in trade in the United Kingdom of Great Britain and Northern Ireland (UK CITES Scientific Authority in litt., 2025). An expert in South Africa stated they were aware of a few nurseries/online plant traders that do sell these species but most likely not in large quantities (Klopper in litt., 2025). This expert was not able to state whether these plants for sale were cultivated or wild-collected and added that because these species do not have spectacular flowers, they have likely not been that sought after, noting, however, that since they are now aloes, their popularity might increase, especially for those collectors who aims to get each species of aloe in their collections. This expert also stated that these plants are traded under *Chortolirion angolense* or more recently *Aloe welwitschia*, with these names likely used in a broad concept that includes all four species. A snapshot online survey for the four species – including these two names – identified just one online advertisement for a live plant under the name *Chortolirion angolense* on a platform based in South Africa, which offered international shipping, with no price stated. No evidence of international trade in the four species was found in US trade data in LEMIS between 2008 and 2021, or in seizure records available from 2014-2023 for EU-TWIX and 1993-2025 for WITIS.

According to CITES trade data, out of the seven range States for the four species, only South Africa and Zimbabwe currently commercially export specimens from *Aloe* species already listed in the CITES Appendices, and only commodities from *Aloe ferox* and *Aloe arborescens* are reported to be wild sourced (CITES, 2025) (Table 1). Based on these data, *Aloe bergeriana*, *Aloe jeppeae*, *Aloe subspicata* could meet criterion A of Annex 2b in Res. Conf. 9.24 (Rev. CoP17) if they could be mistakenly traded in products containing extract, powder, and derivatives of *Aloe ferox* and *Aloe arborescens*, and *Aloe bergeriana* could also meet the criteria if stems of the species resemble those of *Aloe arborescens*. Given *Aloe welwitschii* is endemic to Angola, and Angola has not exported products from *Aloe* species in recent years, it appears unlikely this species could be mistaken in trade with *Aloe* species listed in CITES.

Table 1. Commercial exports of CITES listed *Aloe* species from range States of the four species proposed for inclusion in CITES Appendix II under criterion A of Annex 2b in Resolution Conf. 9.24 (Rev. CoP17), and *Aloe* species in direct exports 2019-2023. Source: CITES Trade Database

Range State	<i>Aloe</i> species proposed for inclusion occurring in this range State	Commercial exports of wild-sourced <i>Aloe</i> species 2019-2023
South Africa	<i>Aloe bergeriana</i> , <i>Aloe jeppeae</i> , <i>Aloe subspicata</i>	<i>Aloe ferox</i> (~2 million kg, 1.9 million specimens, and 44,000 litres- mostly extract with some powder and derivatives). <i>Aloe arborescens</i> (13,500 kg and 14,000 specimens - mostly powder with some extract and leaves).
Zimbabwe	<i>Aloe bergeriana</i>	<i>Aloe arborescens</i> (6,000 stems)
Angola	<i>Aloe welwitschii</i>	
Botswana	<i>Aloe subspicata</i>	
Eswatini	<i>Aloe subspicata</i>	No commercial exports of wild-sourced <i>Aloe</i> species
Lesotho	<i>Aloe subspicata</i>	
Namibia	<i>Aloe bergeriana</i> , <i>Aloe subspicata</i>	

Other considerations:

The inclusion of so-called 'lookalike' species in the Appendices is the subject of Agenda item 102 at CoP20 where Parties are asked to consider document CoP20 Doc. 102 'Considering the 'lookalike' criterion Annex 2bA of Res. Conf. 9.24 (Rev. CoP17) on *Criteria for amendment of Appendices I and II.*' The document draws attention to the regulatory burden created by the increasing numbers of such listings, stating that implementation may divert resources from regulation of species that are genuinely affected by international trade, and proposes a series of actions to be undertaken by the Animals and Plants Committees, the Secretariat and the Standing Committee with a view to making recommendations for consideration at CoP21.

References

South African National Biodiversity Institute (SANBI) (2025) Red List of South African Plants. Available at:

<https://redlist.sanbi.org/> Accessed on 16 July 2025.

UK CITES Scientific Authority (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.

Klopper, R. (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK.

UNEP (2025) The Species+ Website. Nairobi, Kenya. Compiled by UNEP-WCMC, Cambridge, UK. Available at: www.speciesplus.net. Accessed on 16 July 2025.

UNEP-WCMC (Comps.) (2025) The Checklist of CITES Species Website. CITES Secretariat, Geneva, Switzerland. Compiled by UNEP-WCMC, Cambridge, UK. Available at: <http://checklist.cites.org>. Accessed on 16 July 2025.

Transfer of Parlatore's Podocarp *Podocarpus parlatorei* from Appendix I to Appendix II

Proponent: Argentina

Summary: *Podocarpus parlatorei* is a conifer endemic to the montane Yungas forests of Argentina and Plurinational State of Bolivia (Bolivia). Heavily exploited between the 1950s and mid-1970s, it was included in Appendix I when the Convention came into force in 1975. The species has recently been globally assessed as Vulnerable on the IUCN Red List, 2025 in press, due to its restricted and fragmented distribution, ongoing habitat decline from climate change and selective logging, and a small area of occupancy.

Podocarpus parlatorei is relatively abundant in certain high-elevation forest patches (above 1,800 m), with densities of up to 300 individuals per hectare reported in both long-term monitoring plots and national forest inventories in Argentina; and even higher densities up to 1,700 regenerating individuals per hectare in Bolivia. However, these values reflect local abundance within a naturally fragmented range restricted to mountain slopes in the eastern Andes, with an extent of occurrence limited by elevation and climate tolerance and area of occupancy limited to ~280 km². *P. parlatorei* depends on a relatively narrow range of warm-season temperatures and precipitation levels, making it particularly sensitive to climatic changes. Modelling predicts substantial loss of suitable habitat within its range by 2050 due to climate change.

Historically, *P. parlatorei* was intensively logged for its lightweight, workable wood, highly prized in fine carpentry and furniture making. Since 1975, however, there have been only three legal international transactions reported to CITES, involving timber pieces and leaves for scientific purposes. Nevertheless, there appears to be some potential interest in the species as an ornamental tree in international horticulture, as it has been observed in public gardens in Europe and is available commercially in countries such as New Zealand. Ornamental plants in cultivation are considered to derive from cultivated stock, rather than from recent wild collection.

Conservation and management measures are well established in Argentina and Bolivia through national forestry legislation and land-use planning frameworks. Argentina also employs remote sensing and traceability systems for native forest species, although *P. parlatorei* is not currently harvested from the wild. Artificial propagation of *P. parlatorei* has been trialled in Argentina mainly for restoration and conservation purposes, but not on a commercial scale. Experimental nurseries have been established in the Yungas, but no commercial plantations are known to exist, and any potential trade in timber would therefore be of wild origin. While the species is present in some protected areas in Argentina and Bolivia, studies indicate that population structure and regeneration in Bolivia do not reflect a healthy conservation status. In addition to traditional pressures such as agricultural deforestation, expanding real estate development on mountain slopes—particularly around Salta, Argentina—is having a growing impact on the species. Moreover, the most genetically diverse subpopulations in the southern range lie outside protected zones.

Analysis: Information available for *Podocarpus parlatorei* does not indicate a small wild population and it can be locally abundant. It has a restricted area of occupancy, which is impacted by development, but the forests in which it occurs have suffered little deforestation. While the species is subject to climate-related risks, projected population decreases would not appear to justify its retention in Appendix I. Therefore, ***P. parlatorei* does not appear to meet the biological criteria** outlined in Res. Conf. 9.24 (Rev. CoP17) for inclusion in Appendix I.

No significant international trade has occurred for nearly 50 years, and there is no indication that future demand would threaten its survival in the wild. There is some potential interest in *P. parlatorei* as an ornamental tree in temperate horticulture, which would remain regulated and subject to Non-detriment Findings. Therefore, **a transfer to Appendix II, would be consistent with the precautionary measures set out in Annex 4, section A.2 of that Resolution.**

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

Podocarpus parlatorei was listed in Appendix I since the Convention entered into force in 1975.

Taxonomy

Podocarpus parlatorei (Pilg. in Engler)

Synonyms: *Podocarpus angustifolius* (Parl. non Griseb) and *Nageia angustifolia* ((Parl.) Kuntze)

There is no CITES Standard Reference for this species.

Range

Argentina and Bolivia (Plurinational State of)

IUCN Global Category

Vulnerable B2ab (i,ii,iii,iv,v) (assessed 2025, in press, ver 3.1) (Quiroga and Premoli, in press)

Biological criteria for inclusion in Appendix I

A) Small wild population

Locally abundant with densities reaching approximately 300 individuals per hectare. This species tends to occur above 1,800 metres above sea level, forming dense patches on steep mountain slopes. Two data sources were used to assess the abundance and structure of *Podocarpus parlatorei* in Argentina: the Subtropical Network of permanent plots and the 2nd National Inventory of Native Forests (INBN2). The permanent plot network, managed by ProYungas and the National Institute of Forests and Conservation (IER-CONICET), comprises over 50 sites located in the Yungas forests of Salta, Jujuy, and Tucumán. With more than 30 years of tree remeasurement data, the network offers detailed information on forest dynamics, growth, and species composition in relatively well-conserved forest areas. In contrast, the INBN2 represents a systematic sampling of Argentina's forests conducted in 2015, covering a range of forest conditions and management regimes.

However, the plot network is concentrated in relatively undisturbed areas, and little information exists for genetically distinct populations further south in Catamarca Province, which lie outside protected areas and may be subject to different pressures (Quiroga and Premoli, 2012).

It is also considered an effective pioneer species, often displaying strong regeneration after large-scale disturbances of both natural and anthropogenic origin (Quiroga and Premoli, in press).

A complementary study in Bolivia, based on 81 plots across 16 localities in Santa Cruz Department, found that both adult density and regeneration of *P. parlatorei* varied widely across sites. While some areas showed high values (e.g., up to 650 adults/ha in Vallegrande and 1,700 regenerating individuals/ha in Pucará), overall densities were lower at the landscape level (200 and 1,100 individuals/ha, respectively). The results highlight spatial variability and underline the importance of local environmental factors and land-use pressures influencing population viability (Nina Churqui et al., 2022).

B) Restricted area of distribution

Podocarpus parlatorei has a broad north–south distribution of approximately 1,000 km but a narrow west–east range of about 100 km on the eastern-most flanks of the Andes of northwestern Argentina and Bolivia. It is a conspicuous and numerically well-represented element in montane forests across a latitudinal band of more than 1000 m above sea level (asl), from Catamarca (southernmost limit) and Tucumán to 1,900-2,000 m asl towards the north in Jujuy and Salta in Argentina, to Cochabamba and La Paz in Bolivia. This elevation-dependent distribution results in naturally fragmented populations, limited to the peaks of the mountains of the eastern Andean range, preferentially on the eastern and northeastern flanks, where they are exposed to prevailing orographic rainfall (Quiroga and Premoli, 2012). In Argentina, Parlatore's Podocarp covers an area of almost 20,000 km².

Despite the species' large extent of occurrence, its area of occupancy is limited to 276 km². Between six and fifteen distinct locations have been identified as threatened due to ongoing and projected declines in habitat suitability, particularly affecting peripheral and outlying subpopulations. Furthermore, distribution models indicate that some locations will undergo broad projected changes in suitable habitat in the near future in north, central, and southern areas.

According to Quiroga et. al. (2018), *Podocarpus parlatorei* is sensitive to a relatively narrow range of both warm season temperature and precipitation. By the mid-late twenty-first century (within one generation), areas of high suitability for *P. parlatorei* are not expected to shift geographically but overall suitability across its range is projected to decline significantly. An emerging threat relates to climate change; recent research indicates a significant decline

in the optimum habitat throughout its range, with many parts of its range becoming completely unsuitable by 2050 even under relatively conservative climate change scenarios (Quiroga et al., 2018).

C) Decline in number of wild individuals

Historically the species' pattern of distribution was reported to be stable with regard to the number of subpopulations and the range in 2008, and at local level it was observed that many of the subpopulations were starting to expand into grazing areas (Quiroga and Gardner, 2013). There are no available figures for the historic loss through logging, therefore, it is not possible to estimate the reduction in the species population size. However, Quiroga and Premoli (in press) estimate that it is reasonable to suspect that there has been a past decline approaching 30%.

According to the 2025 in press global IUCN Red List assessment, the population of mature *P. parlatoarei* individuals is decreasing, attributed to the inferred selective felling of mature trees and increased fragmentation in parts of its range (Quiroga and Premoli, in press).

Trade criteria for inclusion in Appendix I

The species is or may be affected by trade

In past decades, this species was heavily exploited due to the high quality of its wood, which is lightweight, easy to work with, and suitable for peeling, making it valuable for fine carpentry and furniture production (Novara, 2012). According to Romel et al. (2022), such overexploitation led to its inclusion in CITES Appendix I in 1975, in order to ensure its conservation by prohibiting international trade.

Selective logging is limited to local use; many of the locations are on very steep slopes, often in remote forests where weather conditions are such, that it is difficult for commercial forestry activities. It is used by local rural communities as a source of firewood, to make wooden posts, utensils or housing, as living fencing around houses and pastures (Quiroga and Premoli, in press).

Since 1975, the CITES Trade Database has included only four records of movements for this species, which appear to refer to three transactions: two for scientific purposes involving three pieces of wood in 1998 and three kilograms of pieces of wood in 2004, and one for personal use involving eight 'bone carvings' in 2020 (quite likely an erroneous record).

From 2008 to 2023, LEMIS data only includes a record of refused import of eight 'bone carvings', which is also included in the CITES Trade Database. No trade records were found in either the EU-TWIX database or data from TRAFFIC's WITIS dataset.

There is currently no recorded international demand for products derived from *Podocarpus parlatoarei*. For many years, this demand has been met by timber from commercial plantations of exotic species of the genera *Pinus* and *Eucalyptus*. There has never been any international trade in products from this species since the CITES Convention came into force.

However, there seems to be potential interest in using the species as an ornamental plant in international horticulture, as it has been observed in several public gardens in the UK. So far such specimens have been grown from cultivated stock (Christian, 2023). *Podocarpus parlatoarei* is also available commercially as a garden plant; for example in New Zealand, through an online retailer that specifies the trees are field-grown in open fields (The Plant Store, 2025). Although *Podocarpus parlatoarei* shows some potential in international horticulture as an ornamental species (Christian, 2023) and may attract limited trade aimed at broadening the genetic base of artificially propagated specimens, such trade is not expected to be significant.

Precautionary Measures

Species not in demand for trade; transfer to Appendix II unlikely to stimulate trade in, or cause enforcement problems for, any other species included in Appendix I

There is some interest in *Podocarpus parlatoarei* as an ornamental plant in international horticulture, but there is no known demand for products derived from it, and international trade has remained minimal since the implementation of CITES.

The proposal to transfer the species to Appendix II is in accordance with the precautionary measures outlined in Annex 4 of Resolution Conf. 9.24 (Rev. CoP17), paragraph A.2.a.i, as there is currently no demand for this species in international trade and such a transfer is not expected to stimulate trade or create enforcement challenges for any species listed in Appendix I.

Species likely to be in demand for trade, but its management is such that the CoP will be satisfied with:

- A) Implementation by the range States of the requirements of the Convention, in particular Article IV; and**
- B) appropriate enforcement controls and compliance with the requirements of the Convention**

Export quota or other special measure

No export quotas are included and the species is not anticipated to be in great demand.

Additional Information

Threats

The forest types where Parlatore's Podocarp is found are, proportionally, the least affected by the forest loss occurring in the Yungas region. Of a total of 1,417 km² of forest loss identified between 2006 and 2022 in the Yungas, only 6% (85 km²) occurred in areas where the species is present. The annual rate of forest loss in the Selva and Montane Forests—its primary habitat—is significantly lower (0.03%) than both the regional average (0.21%) and the national rate (0.55%) during the same period. The steep slopes and altitude above 1,100 m where this species is found limit agricultural expansion and access, reducing the risk of deforestation.

However, as well as expansion of the agricultural frontier current real estate developments on mountain slopes are threatening P. parlatorei populations. In Salta near the capital city many urban developments are being undertaken within the habitat of P. parlatorei (Premolin and Quiroga in litt., 2025).

Historically, *Podocarpus parlatorei* was heavily logged for its timber, leading to its inclusion in CITES Appendix I in 1975. Although commercial exploitation has declined, it remains an important timber species, and selective logging continues for national use in many parts of its range outside protected areas. However, the Second National Inventory of Native Forests, completed in 2019, and long-term monitoring indicate that the species maintains a high relative abundance with no significant differences between managed and conserved areas where logging of the species is prohibited, suggesting that logging has not affected its density.

Additionally, human-set fires used to manage grasslands for livestock ranching contribute to habitat loss. In some regions, despite reduced grazing and increased rainfall, forests are not recolonizing degraded grasslands—a phenomenon attributed to recurrent fires that inhibit succession by killing seedlings and altering soil and microclimate conditions.

Climate change is an emerging and significant threat (Quiroga et al., 2018). Research indicates a substantial decline in suitable habitat across the species' range, with many areas projected to become completely unsuitable by 2050—even under conservative climate scenarios (Quiroga et al., 2018). Podocarpus parlatorei is sensitive to a narrow range of warm-season temperatures and summer precipitation. Its habitat suitability is expected to decline without shifting upslope, due to limited availability of higher elevation areas, leading to increased fragmentation of subpopulations (Quiroga et al., 2018). Much of its range was predicted to become completely unsuitable by 2050 even under relatively conservative climate change scenarios (Quiroga et al., 2018).

Conservation, management and legislation

In Argentina, Parlatore's Podocarp is protected under National Law 26.331, which sets minimum standards for the conservation and sustainable use of native forests through land-use planning (OTBN). This law requires provinces to classify forest areas based on conservation value and regulate activities accordingly, with stricter controls on high-value areas. Provincial laws in Salta, Jujuy, Tucumán, Catamarca, and La Rioja have implemented the OTBN framework in regions where the species occurs. In Bolivia, the species is protected under the general framework of Forestry Law No. 1700, which regulates the sustainable use of forest resources, including high-mountain forests where this species is found.

Argentina has established tools to monitor native forests and ensure traceability of forest species listed in CITES Appendix II, including a national remote sensing and GIS-based monitoring system and the Forest Administration, Control and Verification System for permit management and control of forest product movements. All international trade in wild species requires prior approval, and currently, there is no wild collection of Parlatore's Podocarp.

Some parts of the species' population are found within protected areas in Argentina and Bolivia, but the most genetically diverse subpopulations in the southern region remain outside these zones, highlighting the need for targeted conservation efforts due to their ecological and evolutionary importance (Quiroga and Premoli, in press).

Between 2006 and 2009, Argentina conducted a Periodic Review of *Podocarpus parlatorei* under CITES, concluding that the species should remain in Appendix I due to insufficient information to justify its transfer. This precautionary decision was documented in PC18 Doc.26. At PC27 (Geneva, 2024), the species was again selected for review at the 27th Meeting of the Plants Committee (PC27; Geneva, July 2024), and Argentina assumed the responsibility of conducting the review in response to Notification No. 084/2024.

Artificial Propagation/captive breeding

The artificial propagation of Parlatores' Podocarp in Argentina has been the subject of study due to its ecological value. Institutions such as the National Institute of Agricultural Technology (INTA) and universities (such as the National University of Tucumán) have conducted propagation trials. Experimental forest nurseries have been established in areas of the Yungas to produce seedlings for reforestation and conservation. There is no record of the existence of any commercial plantation of this species.

In 2015, Kew reported that there were no known commercial plantations of this species, indicating that any trade in timber would be wild in origin (Groves and Rutherford, 2015). However, Christian (2023) claims that ornamental plants being grown are so far from cultivated stock (likely from long past collection of wild material).

Implementation challenges (including similar species)

Even though the genus *Podocarpus* has approximately 94 species, no particular challenges are foreseen as the species covered by the proposal is not in trade.

References

- Christian, T. (2023) *Podocarpus parlatores* Pilg. Trees and Shrubs Online Website. International Dendrology Society. Available at: <https://www.treesandshrubsonline.org/articles/podocarpus/podocarpus-parlatores/>
- Groves, M. and Rutherford, K. (2025) CITES and Timber: A guide to CITES-listed tree species. Kew, Royal Botanic Gardens.
- Nina Churqui, R.E., Pinto Viveros, M.A., Sánchez, J., Arroyo Herbas, L., Espinoza, D., Banegas, E., Pantoja, S. and Toledo, M. (2022) Estructura forestal y regeneración natural de poblaciones del pino de monte (*Podocarpus parlatores* Pilg.) en el Departamento de Santa Cruz, Bolivia. *Ecología Austral* 32:307–318. <https://doi.org/10.25260/EA.22.32.2.0.1873>
- Novara, L.J. (2012) Podocarpaceae Endl. *Aportes Botánicos de Salta – Ser Flora* 12:1–6.
- Premoli, A.C. and Quiroga, M.P. (2025) *in litt.* to IUCN/TRAFFIC Analysis Team, Cambridge, UK.
- Quiroga, M.P., Premoli, A.C. and Kitzberger, T. (2018) Niche squeeze induced by climate change of the cold-tolerant subtropical montane *Podocarpus parlatores*. *Royal Society Open Science* 5(180513). <http://dx.doi.org/10.1098/rsos.180513>.
- Quiroga, P. and Gardner, M. (2013) *Podocarpus parlatores*. *The IUCN Red List of Threatened Species* 2013: e.T32019A2808150. <http://dx.doi.org/10.2305/IUCN.UK.2013-1.RLTS.T32019A2808150.en>
- Quiroga, M.P., Pacheco, S., Malizia, L.R. and Premoli, A.C. (2012) Shrinking forests under warming: evidence of *Podocarpus parlatores* (pino del cerro) from the subtropical Andes. *Journal of Heredity* 103(5):682–691. <https://doi.org/10.1093/jhered/ess031>
- Quiroga, P. and Premoli, A. (in press) *Podocarpus parlatores*. *The IUCN Red List of Threatened Species* 2025: e.T32019A150296054
- The Plant Store (2025) *Podocarpus parlatores* - Field-Grown. The Plant Store Website. Available at: <https://www.theplantstore.co.nz/products/field-grown/podocarpus-parlatores-fg/>

Transfer of *Avonia quinaria* from Appendix II to Appendix I

Proponent: South Africa

Summary: *Avonia quinaria* is a slow-growing succulent plant with attractive flowers native to South Africa and Namibia. The species occurs in specific habitats within select areas of the Namaqualand Hardeveld and Richtersveld Bioregions, mostly on quartz gravel plains and some mountain-top plateaus. *Avonia quinaria* has not been assessed globally by IUCN, but was assessed as endangered in South Africa's National Red List in 2022 and has not yet been assessed nationally in Namibia.

In South Africa, the species has a restricted and fragmented distribution occurring across an estimated 17,200 km² with an estimated area of occurrence of between 84 and 120 km². Population surveys in South Africa between 2019 and 2023 found that 86% of surveyed populations had densities of less than 1 plant/m², with estimates based on this that fewer than 500,000 plants exist in the wild. There are no available data for the area or extent of occurrence of *Avonia quinaria* in Namibia, but it is reported to occur only marginally in the far southeast of the country. All populations of the species are reported to be patchily distributed. Anecdotal information indicates that the species is extremely long-lived, with large individuals possibly exceeding 100 years, and that growth is exceptionally slow, with plants requiring decades to attain even minimal size. Regeneration is infrequent and often dependent on rare recruitment events linked to sustained high rainfall.

Avonia quinaria is popular in international trade as an ornamental plant and has been included in Appendix II since 1975 under a genus listing of *Avonia* species (with annotation #4).

Large quantities of live *A. quinaria* plants have been reported in direct exports from South Africa in recent years (~43,000 between 2014 and 2019, with the highest quantities in 2015 and 2017). All direct exports by both South Africa and other countries have been reported as artificially propagated, but there are clear indications of wild collected plants entering trade. Observations during a law enforcement operation at one nursery (now closed) known to have been granted CITES export permits for over 33,000 live plants between 2010 and 2018 resulted in seizures of over 3,000 plants with wild characteristics. In 2019, the South African Police Service initiated "project Crassula" to investigate succulent plant trade, and since then, a further 21,000 live plants have been seized. There has been a recent notable decline in exports reported in the CITES Trade Database since 2019, likely as a result of increased enforcement efforts and closure of the nursery concerned. South Africa last reported direct commercial exports of 312 live plants in 2019, with none reported in CITES trade data between 2020 and 2023.

The species reportedly exists in accessible areas within South Africa, and although it is a protected plant and requires a permit for collection, enforcement at present is apparently still not adequate to protect it from illegal harvest.

Illegal wild harvest of the species is thought to have led to population declines in South Africa, and larger adult size classes (the size targeted for trade), are reported to be entirely absent or drastically reduced in almost all populations, with some now regarded as non-viable. The National Red List assessment for South Africa in 2022 assessed the species as endangered based on inferred population reductions exceeding 50% over three generations (two past and one future). No evidence of trade in Namibia was found although there are anecdotal reports of collection there.

It is thought that artificial propagation of this species is unlikely to be able to produce the large quantities of individuals exported from South Africa in recent years given their extremely slow growth rates, with the caudex taking at least 8-12 years to enlarge even under ideal conditions. Other factors affecting the population are slow regeneration rates, habitat degradation, the likely impact of climate change (including increased droughts), and the potential for mining in known habitats.

Analysis: *Avonia quinaria* has a restricted area of distribution, largely in South Africa, with fragmented populations believed to have decreased significantly. It is intrinsically vulnerable due to its slow

growth. Little information is available for the species in Namibia, although it is reported as occurring only marginally in the country. The species has been affected by collection for the ornamental plant trade at least in part for export, with many populations reportedly largely or entirely depleted of mature plants. It therefore **appears to meet the criteria for inclusion in Appendix I** under Res. Conf. 9.24 (Rev. CoP17) based on criterion B.

If adopted, currently exempted products including seeds and seedlings would also be subject to the Appendix-I listing; nurseries artificially propagating the species for commercial purposes would need to be registered with the CITES Secretariat in accordance with Res. Conf. 9.19 (Rev. CoP15) on *Registration of nurseries that artificially propagate specimens of Appendix-I plant species for export purposes*.

The transfer of the genus *Avonia* to *Anacampseros* has been proposed in the revision to Standard nomenclature at CoP20 (see document CoP20 Doc. 110). If accepted this species would be referred to as *Anacampseros quinaria*.

Summary of Available Information

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

CITES Background

Listed in CITES Appendix II under a genus listing for *Avonia* spp. in 1975 (UNEP, 2025). The current listing of this genus is under annotation #4 which for this species exempts seeds, seedlings or tissue cultures obtained *in vitro* transported in sterile containers, and cut flowers of artificially propagated plants.

Taxonomy

Avonia quinaria (E.Mey. ex Fenzl) G.D.Rowley
Scientific synonyms *Anacampseros quinaria* E.Mey. ex Fenzl,

A new standard reference for the species has been proposed in CoP20 Doc 110. If adopted, *Avonia quinaria* would be listed in the Appendices as *Anacampseros quinaria*.

Range

South Africa, Namibia

IUCN Global Category

Not assessed.

Biological criteria for inclusion in Appendix I

A) Small wild population

Population size estimates are exceedingly difficult to establish given the cryptic nature of the plants and as such, density has been used as a proxy for abundance. Population surveys were carried out in 2019, 2022 and 2023 during the reproductive period of the species (between October and December). A total of 22 populations were surveyed, which is estimated to represent around 85% of the national population. Plant density in most of the populations was found to be low, with 86% of the surveyed populations having a density of less than 1 plant/m² (varying between 0.01 – 0.89 plants/m²). The highest plant density recorded in the field between 2022 and 2023 was 2.68 plants/m². Extrapolating these field data to account for populations not surveyed, the estimated population size is unlikely to exceed 500,000 individuals in total.

B) Restricted area of distribution

Avonia quinaria occurs in South Africa and Namibia, though only marginally occurring in Southeastern Namibia. An expert on the species in Namibia stated that the plants only occur in the far south-east known as the Gariep Centre of endemism, and that in some populations they seem to be relatively common but overall are very localised and confined to specific habitats (Loots in litt., 2025). The Global Biodiversity Information Facility has just two occurrence records for the species in Namibia since 1990, and 46 for South Africa (see Figure 1).

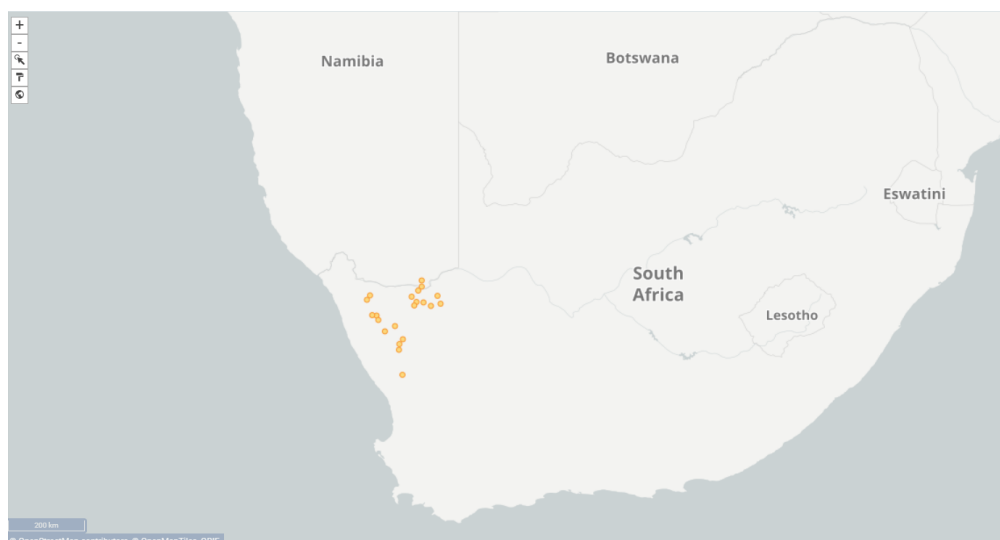


Figure 1. Known occurrence records for *Avonia quinaria* in Namibia (2) and South Africa (46) since 1990 based on records in The Global Biodiversity Information Facility²⁰.

Estimates for the extent of occurrence or area of occupancy were not available in Namibia, but within South Africa, the species has a restricted and fragmented distribution occurring across an estimated 17,206 km² within the arid north-western regions of the Succulent Karoo Biome, predominantly in the Northern Cape province (Figure 1). One population was recently found on the border with the Western Cape province. Although the distribution range is relatively large, the area of occupancy (AOO) is quite small (estimated at 120 km² nationally, though this is likely to be an overestimate of the actual area occupied by the species in South Africa). Populations tend to be confined to small areas of between 0.002 km² and 0.507 km² and are patchily distributed. *The 2022 national red list assessment for the species in South Africa estimated a lower area of occupancy for the species, reporting it to be 84 km² (Mhlongo et al., 2022).*

Avonia quinaria occurs only in select areas of the Namaqualand Hardeveld and Richtersveld Bioregions, mostly on quartz gravel plains and some mountain-top plateaus. These areas are characterized by shallow and skeletal soils and low soil pH and are thus similar to that of rocky habitats and outcrops. The species appears to prefer growing amongst fine to medium-grain gravel and plants are usually absent from areas where the grain is coarse or large.

C) Decline in number of wild individuals

Whilst the status of the seven distinct ecosystem/vegetation types that the species inhabits are least concern, they are all either poorly protected or completely unprotected and are thus highly susceptible to degradation and loss. Analysis of the size class distributions of surveyed populations indicates that the larger adult size classes (>33 cm²) are entirely absent or drastically reduced in almost all populations. Although formerly abundant in suitable habitat, most populations have been significantly reduced by illegal harvest over an extended period and this threat is ongoing. Since 2022, the species is currently assessed as endangered (A4bd) on the Red List of South African Plants, indicating a population reduction of >50%. Although no evidence of local extinction has been documented for any known *A. quinaria* population, field surveys established that several populations have been harvested to the point of non-viability, some comprising as little as fewer than five individuals. *The national red list assessment from South Africa states that reductions in the population of 50% over three generations are based on two generations in the past (e.g., the last 60 years), and one generation into the future (30 years) due to declines from illegal collection of mature individuals, and states that populations in Namibia are also suspected to be declining due to illegal collection (Mhlongo et al., 2022). The source of evidence for these estimates of decline is not stated in the assessment.*

Trade criteria for inclusion in Appendix I

The species is or may be affected by trade

Avonia quinaria has been listed in CITES Appendix II under a genus listing for *Avonia* spp. since 1975. According to CITES trade records, South Africa is by far the largest exporter of this species, accounting for approximately 98% of all reported global exports in live plants. The proposal states that an estimated 38,647 live plants have been permitted for export from the country since 1996, the vast majority (97%) between 2011 and 2021.

²⁰ GBIF.org (30 July 2025) GBIF Occurrence Download <https://doi.org/10.15468/dl.26fuw9>

According to the CITES Trade Database, direct commercial exports reported by South Africa were in higher quantities than this, with ~43,000 live artificially propagated *A. quinaria* exported in recent years between 2014 and 2019. Far fewer were reported by importers- around 28,000 in the same period (Figure 2). The highest exports were reported in 2015 (~15,000) and 2017 (~12,600), with just over 300 in 2019. Since 2019, however, there have been no direct commercial exports of live *A. quinaria* plants reported by South Africa. Although Japan did report 280 live plants imported from South Africa in 2021. No exports of wild specimens of *A. quinaria* have been reported by South Africa (or Namibia) since the species was listed in 1975, with the first trade records reported to CITES in 1996.

Very few exports were reported by other countries in the last ten years of available data between 2014 and 2023 (around 740 artificially propagated live individuals from China, Czechia, and Thailand), and none from Namibia. However, there is anecdotal evidence of illegal collection of the species in Namibia (Loots in litt., 2025). The species was not found in seizure data in EU-TWIX and WiTIS based on available data (from 2014-2023 for EU-TWIX and 1993-2025 for WiTIS).

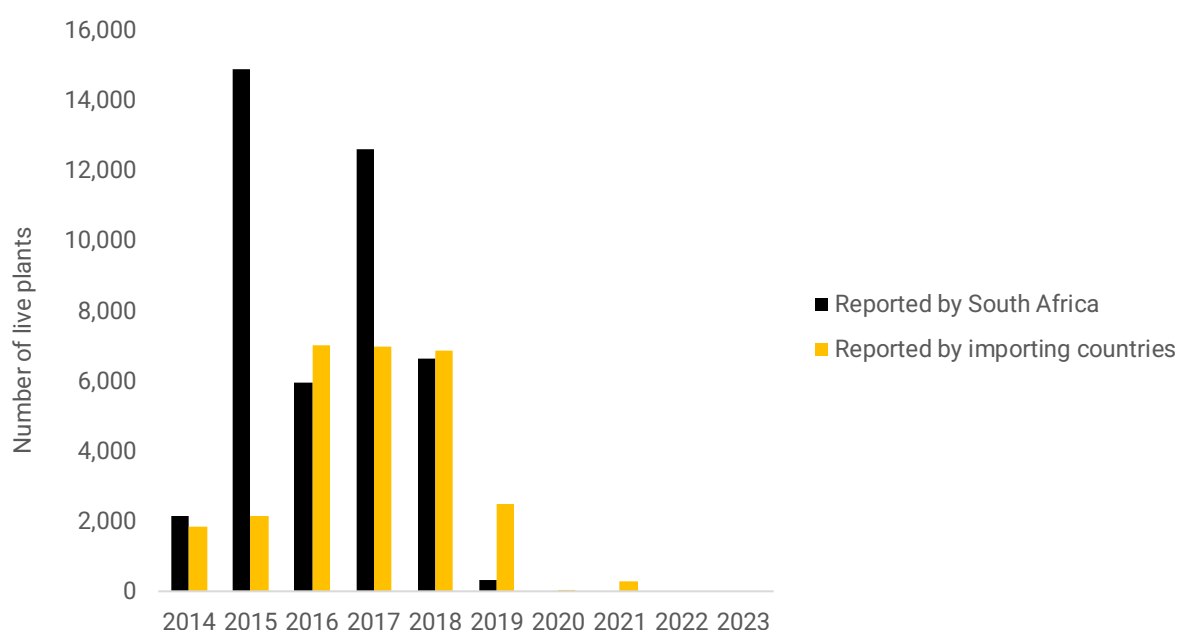


Figure 2. Direct exports of live *Avonia quinaria* plants reported in direct commercial exports by South Africa and importing Parties between 2014 and 2023. Source CITES Trade Database

Despite exports of *Avonia quinaria* from South Africa being reported as from artificially propagated specimens in the CITES Trade Database, it is feared that large numbers of wild-collected plants were laundered into the legal trade during the 2010s, with details and numbers below:

- During a law enforcement operation conducted in May 2019 at the only South African nursery known to be exporting the species at the time, 3,326 *A. quinaria* plants displaying wild characteristics were seized. Many of the plants in the nursery had a distinctly oblong shape, a consequence of having grown between parallel layers of rock in habitat. Some blackening of the stems from exposure to harsh sunlight was also evident, something that would not generally feature on the caudices of artificially propagated plants. Some cultivated seedlings were also present but comprised a relatively small proportion of the total collection and their sizes and shapes stood in stark contrast to the varying sizes and irregular shapes of the wild specimens.
- CITES permits had been issued for the export of 33,237 “artificially propagated” *A. quinaria* plants from this facility between 2010 and 2018 and it is reasonable to suspect that all or most of this material was of wild origin. *This nursery has since been closed down (Uys in litt., 2025). CITES trade data shows that only ~26,900 live Avonia quinaria plants were reported in commercial exports from South Africa in this period. This indicates that not all permitted plants were exported, but that the facility likely accounted for a significant share of South Africa’s reported legal trade during that period.*
- Over the period January 2019 to May 2024, a further 21,240 wild *A. quinaria* were seized by law enforcement agencies (orders of magnitude more than any other seized CITES Appendix II plant species during this period), and many more plants than this have likely left the country illegally.

- *Illegal harvest in South Africa was also reported in 2023 where landowners in the Northern Cape (in the Northeastern parts Avonia quinaria's range) indicated that plants were removed from their properties (Uys in litt., 2025).*

In May 2019, the South African Police Service (SAPS) initiated "project Crassula" to investigate succulent plant trade through the use of informants and setting up "busts". The various engagements of numerous stakeholders involved (SAPS Stock Theft and Endangered Species Units, South African National Biodiversity Institute, local Nature Conservation officials and various NGO's) since 2019, created awareness amongst local law enforcement (SAPS) of this "new" environmental crime, which was previously only dealt with by conservation officials. More attention has been given to these types of crimes (e.g., illegal succulent trade) since 2019 (Uys in litt., 2025). These increased enforcement efforts, in addition to the closure of the large nursery found laundering wild specimens, may be the cause of the reduction in exports of live Avonia quinaria plants from South Africa reported in the CITES Trade Database since 2019.

As well as local and international nurseries, wild specimens of *A. quinaria* have been observed for sale on various online social media and trading platforms. The proposal states the large and unique shape of plants offered for sale in online advertisements on social media platforms indicates their wild origin. The proposal states that ongoing harvest of wild plants is uncontrolled and illegal (with no legal harvest permitted as present) and is likely driven by targeted operations where large specimens are poached for short-term economic yields in a way that is likely to cause the near extinction of the species. Access to populations is unrestricted as most of the populations fall outside of any kind of formally or informally protected area.

Impacts of continued harvest have been noted in the field in South Africa with some populations being depleted to the point of complete non-viability and others having had the population structure severely impacted leading to instability and possibly affecting the regenerative potential of these populations. Due to the methods employed during harvesting (i.e., sweeping away layers of quartz and digging holes in delicate soils), habitat degradation has been observed, while other threats to the habitat include overgrazing (particularly on communal land), and mining. The severity of these threats is made worse by more frequent and prolonged droughts. The slow-growing nature of the species means that obtaining sizable specimens in cultivation is a very long process and adult individuals have continually been targeted by harvesters on behalf of collectors and traders. Discrepancies between sizes of seized plants and plants remaining in the wild further signifies that large adult plants have historically been targeted by harvesters, which is extremely detrimental to such long-lived, slow-growing taxa.

Local trade in *A. quinaria* is minimal. Owing to the species slow-growing nature, wild specimens are sought after by collectors and hobbyists.

The current listing of this genus is under annotation #4 which for this species exempts seeds, seedlings or tissue cultures obtained *in vitro* transported in sterile containers, and cut flowers of artificially propagated plants. Under an Appendix-I listing, parts and derivatives, including seeds, would no longer be exempt from regulation under CITES.

Additional Information

Threats

The species is a long-lived, dwarf caudiciform plant with an extremely slow growth rate. The species regenerates sexually from seed and although large quantities of seed may be produced, safe sites/microhabitats for germination and conditions conducive to persistence of plants to adulthood are not abundant within the arid environment the species occupies. Regeneration in arid regions is rare and often relies on specific microhabitats and sustained high rainfall, meaning recruitment events may be spaced decades or centuries apart.

Habitats for the species are fragile and highly sensitive to changes brought on by threats such as climate change, mining, and sustained overgrazing. Degradation caused by these threats generally leads to changes in plant demographics, followed by plant and animal losses, and ultimately, changes in soil condition and processes that can be difficult to reverse or remedy. While the species is able to withstand some minor level of disturbance (displaying resilience in terms of grazing impact), activities like mining, illegal harvesting and soil disturbance severely threaten the specialized quartz fields it depends on. Effective restoration of these sensitive areas (that are home to many local endemics) would require large-scale, intensive intervention focused on restoring ecological processes, soil structure, and microhabitat suitability—efforts that can be expensive and slow, with passive recovery possibly taking centuries in highly degraded areas.

Mining exploration and prospecting has taken place in the vicinity of several *A. quinaria* populations, potentially putting them at risk should mining in these areas continue. Due to its slow growing nature, the species is seldom affected directly by grazing but associated impacts on the habitat can affect the species' ability to recruit and survive. The severity of these threats is made worse by more frequent and prolonged droughts which are

becoming more common within the Northern Cape province. The province has recently gone through an eight-year period of drought.

One expert stated a key threat for A. quinaria in Namibia was climate change, which could result in a loss of at least 40% of the population within one to two generations, with the impact on A. quinaria likely greater than for other species due to their highly specialised habitat (Loots in litt., 2025).

Conservation, management and legislation

Avonia quinaria is legally protected under the Northern Cape Nature Conservation Act No. 9 of 2009 in South Africa. In line with the terms set out in this legislation, no person may pick, import, export, transport, cultivate or trade in a specimen of a protected plant without a permit. No formal management or harvest strategy exists for the species and there has been no formal management of wild harvesting of the species to date. The species remains poorly protected with just one population known to occur in a formally protected area. Access to most populations remains unrestricted, monitoring and enforcement is weak, and resource shortages in staffing, infrastructure, and funding make effective regulation and protection nearly impossible. As there is no legal harvest permitted at present, potential conservation incentives (for the species and its habitat) tied to managed harvesting are also lost. The Northern Cape, South Africa's largest and most biodiverse province, faces growing pressure from the illegal trade of endemic plants like *A. quinaria* and unless significant investment is made in both in-situ and enforcement capabilities the survival of *A. quinaria* and other threatened plants will remain at risk despite their protected statuses.

Of 22 recently surveyed populations of the species, nine occurred on communal land, eight on property belonging to private landowners, and the remaining five on state-owned land including just two (small) *A. quinaria* populations (<1% of the national population) that occur within a Provincial Nature Reserve. The presence of the species on private lands and in reserves does not preclude possible harvest of the species in these areas as there have been several incursions at these localities where the illegal harvest of other succulent species has taken place. Whilst provincial legislation also makes provisions for the management and protection of the species, enforcement of and compliance with the regulations is challenging to guarantee and the regulations governing harvest thus does not afford any actual, practical level of protection on the ground. The distribution of the species across remote areas coupled with the lack of resources in terms of sufficient budget and staff, presents no confidence in any type of protection afforded to the species at present. Although each of the ecosystems/vegetation types occupied by the species have been assessed as least concern, the protection of these habitats across state, private and communal lands is weak/lacking, leaving the species and its habitat vulnerable to disturbance and change.

There is currently no dedicated field monitoring programme for *A. quinaria*, though recent baseline population data collated between 2019 and 2023 could inform future monitoring efforts.

Artificial Propagation

Although there are a handful of South African nurseries (<5) who may offer limited quantities of artificially propagated plants for sale to the South African public, there is no evidence that any of the nurseries have the capacity to produce the large quantities of plants that have been exported from the country over the years. There are currently only two nurseries licensed to trade in this species internationally.

The species is reportedly relatively easy to propagate from seed, but plants grow very slowly. It is therefore not common to see artificially propagated large plants in cultivation and plants are incredibly sensitive to overwatering and prone to rot in cultivation.

Anecdotal information from experienced growers suggests that the lifespan of large individuals may easily exceed a century and even in ideal conditions under cultivation, the caudex (*stem*) takes many years to enlarge. Information from growers suggests that seed loses viability within a year making long-term seed banking unlikely. Seeds germinate best during the winter rainy season at mild temperatures but lose viability quickly.

Implementation challenges (including similar species)

Whilst there are species that share similar features with *A. quinaria*, particularly those in the genus that are small in size with cylindrical stems and papery white coverings (stipules), the species is quite distinctive in its appearance and not easily confused even with its closest relatives. The most similar species is likely to be *Avonia papyracea* which has notably larger and elongated stems (5-10 mm in diameter, up to 15 cm long) compared with other members of the genus. *Avonia ustulata* is another closely related dwarf species with tufted silvery-grey worm-like branches closer in size to *A. quinaria*, however, *A. ustulata* does not form as robust a caudex as *A. quinaria*. *Avonia quinaria* is also said to have the showiest flowers of all species in the genus.

References

Mhlongo, N.N., Rodgers, C. and Dreher, H. (2022) *Anacampseros quinaria* E.Mey. ex Fenzl. National

Assessment: Red List of South African Plants version. Accessed on 30 July 2025
Loots, S. (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK
UNEP (2025) The Species+ Website. Nairobi, Kenya. Compiled by UNEP-WCMC, Cambridge, UK. Available
at: www.speciesplus.net. Accessed on 16 July 2025.
Uys, N. (2025) *in litt.* to the IUCN/TRAFFIC Analyses Team, Cambridge, UK

Amend annotation #4

Proponent: United Kingdom of Great Britain and Northern Ireland (as Chair of the Working Group on Annotations)

Summary: The proposal seeks to amend Annotation #4 paragraph f) in line with the consensus recommendation made by the Standing Committee at its 78th meeting (Geneva, 2025), following discussion in the Working Group on Annotations. The proposed amendments are as follows (with added text shown underlined and deleted text shown in ~~strike through~~):

f) finished products packaged and ready for retail trade of *Aloe ferox* and *Euphorbia antisyphilitica* ~~packaged and ready for retail trade~~;

The proposed amendment addresses a minor ambiguity in the existing wording, it being unclear whether it is currently intended to apply to all finished products of *Aloe ferox* or those specifically packaged and ready for retail trade. It also harmonises use of this language with its use in paragraph g of Annotation #4. The term “finished ‘products packages and ready for retail trade” is defined in paragraph 8 of the interpretation section of the Appendices as:

“Products, shipped singly or in bulk, requiring no further processing, packaged, labelled for final use or the retail trade in a state fit for being sold to or used by the general public.”

The proposed amendment will have no substantive impact on the trade in the species referred to in the annotation (*Aloe ferox* and *Euphorbia antisyphilitica*). Use of the term would also be in line with its use in annotations #2 and #13.

Although the proposed amendment would make no substantive difference to the listing, at the 78th meeting of the Standing Committee it was considered necessary to submit a proposal in line with Resolution Conf. 11.21 (Rev. CoP19) on *Use of Annotations in Appendices I and II*, as annotations to species listed in the Appendices are considered substantive annotations that are integral parts of species listings and they may be adopted, deleted or amended only by the Conference of the Parties.

Analysis: The proposed amendment has been discussed and agreed by the Annotations Working Group of the Standing Committee. Its **adoption would make no substantive change to the listings** of the species to which it applies and **clarifies and harmonises its use** within the annotation and with the definition of the term in the interpretation section of the Appendices.

Discussion

Text in non-italics is based on information in the Proposal and Supporting Statement (SS), text in italics is based on additional information and/or assessment of information in the SS.

The proposal seeks to amend Annotation #4 paragraph f) in line with the consensus recommendation made by the Standing Committee at its 78th meeting (Geneva, 2025), following discussion in the Working Group on Annotations. The proposed amendments are as follows (with added text shown underlined and deleted text shown in ~~strike through~~):

Annotation #4 All parts and derivatives, except:

- a) seeds (including seedpods of Orchidaceae), spores and pollen (including pollinia). The exemption does not apply to seeds from Cactaceae spp. exported from Mexico, and to seeds from *Beccariophoenix madagascariensis* and *Dypsis decaryi* exported from Madagascar;
- b) seedling or tissue cultures obtained in vitro transported in sterile containers;
- c) cut flowers of artificially propagated plants;
- d) fruits, and parts and derivatives thereof, of naturalized or artificially propagated plants of the genus *Vanilla* (Orchidaceae) and of the family Cactaceae;
- e) stems, flowers, and parts and derivatives thereof, of naturalized or artificially propagated plants of the

- genera *Opuntia* subgenus *Opuntia* and *Selenicereus* (Cactaceae);
- f) finished products packaged and ready for retail trade of *Aloe ferox* and *Euphorbia antisyphilitica*
~~packaged and ready for retail trade;~~
 - g) finished products derived from artificial propagation, packaged and ready for retail trade of cosmetics containing parts and derivatives of *Bletilla striata*, *Cynoches cooperi*, *Gastrodia elata*, *Phalaenopsis amabilis* or *Phalaenopsis lobbii*.

Although the Annotation itself applies to a large number of taxa, the proposed amendment to paragraph f relates specifically to trade in Aloe ferox and Euphorbia antisyphilitica.

This reorganisation of the existing wording in paragraph f clarifies and harmonises use of the phrases "finished products" and "packaged and ready for retail trade" with their use in paragraph g of the same annotation. It would also be in line with the use of this phrase in other Annotations, namely #2 and #13, and with the term "Finished products packaged and ready for retail trade" defined in paragraph 8 of the interpretation section of the Appendices as:

"Products, shipped singly or in bulk, requiring no further processing, packaged, labelled for final use or the retail trade in a state fit for being sold to or used by the general public."

Decisions 18.323-18.326 (rev CoP19) directed the Secretariat, Plants Committee and Parties to engage in a process to review the impact of the amendment annotation #4 (that exempted finished products of *Aloe ferox* at CoP18) on international trade and on the populations size, distribution, status and harvest of *Aloe ferox*. The conclusion of that process was summarised in CoP Doc 109. Some Parties indicated that they have to invest less resources to checking small consignments, reported trade volumes have not decreased, and overall may be increasing, and that the species seemed to be well-managed and South Africa appears to be taking steps to ensure trade remains non-detrimental to wild populations (CoP19 Doc 109).

The Standing Committee at its 78th meeting considered that it was necessary to submit a proposal in line with Resolution Conf. 11.21 (Rev. CoP19) on *Use of annotations in Appendices I and II*, annotations to species listed in the Appendices are considered substantive annotations that are integral parts of species listings, they may be adopted, deleted or amended only by the Conference of the Parties, in accordance with the provisions of Article XV of the Convention. As such, in line with Rules 24 and 25 of its Rules of Procedure, the proposed amendments endorsed by the Standing Committee require that a proposal amending an Annotation (in this case Annotation #4) be submitted by one or more Parties to the Convention for consideration by the Conference of the Parties at CoP20.

The proposed amendment, which is submitted in line with Article XV, makes no substantive change to the listing of these species and therefore there is no risk to the species of accepting this amendment.

The International Union for Conservation of Nature (IUCN) is a membership Union uniquely composed of both government and civil society organisations. By harnessing the experience, resources and reach of its more than 1,400 Member organisations and the input of some 17,000 experts, IUCN is the global authority on the status of the natural world and the measures needed to safeguard it.

The IUCN Species Survival Commission (SSC), the largest of IUCN's six commissions, has over 10,500 species experts recruited through its network of over 168 groups (Specialist Groups, Task Forces and groups focusing solely on Red List assessments). Biodiversity loss is one of the world's most pressing crises, with many species' populations declining to critical levels. SSC is dedicated to halting this decline in biodiversity and to provide an unmatched source of information and advice to influence conservation outcomes, as well as contribute to international conventions and agreements dealing with biodiversity conservation.

TRAFFIC works to ensure that trade in wild species is legal and sustainable, for the benefit of people and the planet. With 50 years of expertise, TRAFFIC is a trusted advisor and partner to governments, businesses, law enforcement, and communities worldwide—tackling the root causes of illegal trade in wild species.

For more information visit:

www.iucn.org
www.traffic.org